

The Grammar of Salish Reduplication

by

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A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF

Doctor of Philosophy

in

THE FACULTY OF GRADUATE AND POSTDOCTORAL
STUDIES

(Linguistics)

The University of British Columbia

(Vancouver)

April 2025

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Abstract

This dissertation proposes a model of the spell-out component of the grammar that accounts for the form and position of reduplicative morphemes across all twenty-three Salish languages. The model integrates components of Prosodic Morphology and Moraic Theory, Generalised Nonlinear Affixation, and Stratal Optimality Theory. Its empirical basis is a comprehensive survey of reduplication across the Salish language family, including both published and archival sources. The survey is supplemented by original fieldwork on two languages (Comox-Sliammon and Lillooet) which allows specific predictions of the model to be tested pertaining to the relevant domain(s) for reduplication, the type of phonological material present in lexical entries, and which aspects of the phonological grammar determine if, where, and how a reduplicative morpheme is realised.

With respect to domain, a stratum-based model is required to account for the realisation of reduplicative morphemes, including patterns of allomorphy, variation in form and position, and systematic gaps in multiple reduplication patterns. Strata in the model developed here are defined purely on a phonological basis, without reference to morphological domains. This approach also accounts for the form of reduplicative morphemes that include either linear (segmental) or non-linear (prosodic) representations, or a combination of both. To predict the correct surface position of reduplicative morphemes, a two-step model of lexical insertion is adopted. Morphemes are linearised at the first step, prior to the insertion of phonological content; it is only at the second step that differences between linear and non-linear representations emerge. These differences, including the shape and position of suprasegmental and segmental content, are determined by the phonological grammar at the relevant stratum; the phonology determines when and

if segments are reduplicated, and if they will be positioned as prefixes, suffixes, or infixes.

This dissertation is the first formal phonological analysis of reduplication that considers the full range of reduplication patterns across the entire Salish language family. Aside from its theoretical contribution, it serves as a comprehensive and accurate documentary source for further theoretical and typological investigation.

Lay Summary

Reduplication is a word-building process where meaning is added by doubling certain sounds. This dissertation describes types of reduplication found across all twenty-three Salish languages (a family of Indigenous languages spoken in the Pacific Northwest); it is informed by a combination of linguistic fieldwork, documentation, and archival materials. Similarities and differences between types of reduplication in closely related languages are significant for understanding how speakers represent their languages in their “mental grammars”; for example, how they decide what sounds like a “good” word in their language. Three main findings include: (i) reduplication does not happen “all at once” when building a word (some types happen later than others); (ii) different Salish languages order types of reduplication in different ways; and (iii) speakers generally only tolerate one type of reduplication at a time (i.e., words with two types are only allowed if one type happens before the other).

Preface

The phonological analysis presented in this dissertation is my own work, though certain aspects are guided by earlier collaboration. The unpublished fieldwork data included in this dissertation represents ongoing collaboration. Below, I detail papers and collaborations that have informed the analysis, generalisations, and data included in this dissertation.

In Chapter 4, I include an analysis of ʔayʔaʃuθəm (Comox-Sliammon) C₁ reduplication that is partly based on generalisations first given in the following conference papers:

- Mellesmoen, G. (2017). All the Small Things: Diminutive Reduplication as Infixation in ʔayʔaʃuθəm. In A. Anghelescu, M. Fry, M. Huijsmans, & D. Reisinger (Eds.), *Papers for the 52nd International Conference on Salish and Neighbouring Languages (ICSNL 52)* (pp. 151–169). Vancouver: University of British Columbia Working Papers in Linguistics.
- Mellesmoen, G. (2020). A Reanalysis of CV- Reduplication in Comox-Sliammon. In *Proceedings of the 35th Annual Northwest Linguistics Conference (NWLC 35)* (pp. 1–10). Victoria: Working Papers of the Linguistics Circle of the University of Victoria

The analysis in this dissertation makes different theoretical claims; it replaces these earlier papers. I have a forthcoming conference paper presenting this updated analysis, including the discussion of syllable structure in Chapter 3:

- Mellesmoen, G. (to appear). Monomoraic syllables with non-moraic vowels in Salish: The structure of a monomoraic syllable in Comox-Sliammon

reduplication. In *Proceedings of the 41st West Coast Conference on Formal Linguistics (WCCFL) 41*. Somerville: Cascadilla Press.

Analysis of St'át'imcets (Lillooet) C₁ reduplication in Chapters 4–5 is informed by joint work with Henry Davis, which appeared as:

- Davis, H., & Mellesmoen, G. A Newly Discovered Reduplication Pattern in St'át'imcets and its Implications. In D. K. E. Reisinger, L. Griffin, G. Mellesmoen, S. Nederveen, J. Schillo, & B. Trotter (Eds.), *Papers for the 58th International Conference on Salish and Neighbouring Languages (ICSNL) 58* (pp.10–46). Vancouver: University of British Columbia Working Papers in Linguistics.

Henry Davis provided the fieldwork data and description in the first half of the paper; I contributed the formal phonological analysis and theoretical implications in the second half of the paper. While the paper includes much of the same analysis I give in this dissertation, some elements have since been revised (e.g., abandoning the “stem” as a phonological domain).

For the analysis of Hul'q'umi'num' (Halkomelem) in Chapter 5, description and generalisations are informed by collaborative work with Suzanne Urbanczyk. This collaboration includes the following conference papers:

- Mellesmoen, G., & Urbanczyk, S. (2020). Mora Affixation and Halkomelem Imperfective Allomorphy. In D. K. E. Reisinger, H. Green, M. Huijsmans, G. Mellesmoen, & B. Trotter (Eds.), *Papers for the 55th International Conference on Salish and Neighbouring Languages (ICSNL 55)* (pp. 238–256). Vancouver: University of British Columbia Working Papers in Linguistics.
- Mellesmoen, G., & Urbanczyk, S. (2021) Avoiding Multiple Reduplication without INTEGRITY. In D. K. E. Reisinger & M. Huijsmans (Eds.), *Proceedings of the 37th West Coast Conference on Formal Linguistics (WCCFL 37)* (pp. 169–178). Somerville: Cascadilla Proceedings Project.
- Mellesmoen, G. & Urbanczyk, S. (2021). Binariness in Prosodic Morphology and elsewhere. In R. Bennett et al. (Eds.), *Supplemental proceedings of the*

2020 Annual Meeting on Phonology (AMP). Washington: Linguistic Society of America.

- Mellesmoen, G. (2024). Reduplicative morphemes and their non-reduplicative allomorphs in Stratal OT: Stem-level and word-level reduplication in Hul’q’umi’num’. In R. Autry et al. (Eds.), *Proceedings of the West Coast Conference on Formal Linguistics (WCCFL) 39* (pp. 346–351). Somerville: Cascadia Press.

In Chapter 5, I present a critique of this earlier work based on independent research. Elements of the formal analysis (applied to St’át’imcets) will be publishing in an upcoming conference paper:

- Mellesmoen, G. (to appear). Limited by number of morphemes, not copied segments multiple reduplication and triplication in St’át’imcets. In *Proceedings of the 2024 Annual Meeting on Phonology (AMP)*. Washington: Linguistic Society of America.

In Chapter 5, discussion of an ungrammatical combination of plural C_1C_2 and imperfective C_1 builds on an observation made in the following paper:

- Huijsmans, M., & Mellesmoen, G. (2021). How to distribute events: ?ay?ajuθəm pluractionals. *International Journal of American Linguistics (IJAL)* 87(3), 339–368.

The data for this section and conclusions about the semantics associated with this combination are informed by continued collaboration with Marianne Huijsmans. The phonological description and analysis are my own.

Analysis of certain cross-Salish patterns in Chapter 6 comes from more in-depth analyses developed in the following papers:

- Mellesmoen, G. (2021). Syllables and reduplication in Bella Coola (Nuxalk). In D. K. E. Reisinger, L. G. Hannah Green, M. Huijsmans, G. Mellesmoen, & B. Trotter (Eds.), *Papers for the 56th International Conference on Salish and Neighbouring Languages (ICSNL 56)* (pp. 226–243). Vancouver: University of British Columbia Working Papers in Linguistics

- Mellesmoen, G. (2022). DIG reconstruction and the secret life of Proto-Salish diminutives: An amphichronic analysis of diminutives in Upper Chehalis. In D. K. E. Reisinger, L. Griffin, G. Mellesmoen, S. Nederveen, B. Oliver, & B. Trotter (Eds.), *Papers for the 57th International Conference on Salish and Neighbouring Languages (ICSNL 57)* (pp. 67–86). Vancouver: University of British Columbia Working Papers in Linguistics

Each of the above papers is cited where relevant throughout the dissertation; otherwise, the content of the dissertation represents original research that has not appeared elsewhere.

Fieldwork for this dissertation on ʔayʔajuθəm (Comox-Sliammon) and Secwempemčtsín (Shuswap) was conducted as part of the ʔayʔajuθəm Lab and Secwempemčtsín Working Groups at UBC. Fieldwork data from these two languages was elicited by me, either independently or in collaboration with another lab member. Some examples in Chapter 5 were elicited by Marianne Huijsmans on my behalf; I use footnotes to identify the relevant examples. All fieldwork data on St’át’imcets comes from elicitation conducted by Henry Davis.

This research was conducted with approval from the UBC Research Ethics Board, under the titles Dictionary Project (H15-01026) and BC FNL Grammar Project (H20-02664-A002), both with Henry Davis as principal investigator.

Generative AI was not used for any aspect of this dissertation.

Table of Contents

Abstract	iii
Lay Summary	v
Preface	vi
Table of Contents	x
List of Tables	xxix
List of Figures	xxxv
Acknowledgments	.xxxvi
Dedication	.xxxix
1 Introduction	1
1.1 Focus and Goals of this Dissertation	2
1.1.1 Reduplication and Non-Concatenative Morphology	2
1.1.2 Main Contributions and Claims	4
1.1.2.1 Architecture of the Grammar	5
1.1.2.2 Main Theoretical Claims	7
1.1.2.3 Contributions to Description and Documentation	7
1.2 Structure of this Dissertation	8
1.3 Salish Languages	10
1.3.1 Endonyms vs. Exonyms	10

1.3.2	Geographical Distribution and Language Classification . .	12
1.3.3	Language Status	14
1.4	Methodology	16
1.4.1	Fieldwork	16
1.4.2	Published and Archival Sources	18
1.4.3	Methodological Considerations for Reduplication	18
1.4.3.1	Eliciting Reduplication	19
1.4.3.2	Phonologically Regular and Active Patterns . .	20
1.4.3.3	Productivity and Reduplication	22
1.4.4	Transcription and Glossing	23
1.4.4.1	Segments	24
1.4.4.2	Suprasegmental Elements	25
1.4.4.3	Modifications to Glossing and Translations . . .	25
1.4.4.4	Unreduplicated Forms for Comparison	26
1.5	Sources and Documentation of Salish Reduplication	26
1.5.1	Central Salish	27
1.5.1.1	Lushootseed	28
1.5.1.2	Twana	29
1.5.1.3	Nooksack	30
1.5.1.4	Straits (Northern Straits and Klallam)	30
1.5.1.5	Halkomelem	31
1.5.1.6	Squamish	33
1.5.1.7	Comox-Sliammon	33
1.5.1.8	Sechelt	34
1.5.1.9	Pentlatch	34
1.5.1.10	Summary: Central Salish	34
1.5.2	Tsamosan	35
1.5.2.1	Upper Chehalis	35
1.5.2.2	Cowlitz	36
1.5.2.3	Lower Chehalis	36
1.5.2.4	Quinault	36
1.5.2.5	Summary: Tsamosan	37
1.5.3	Tillamook	37

1.5.4	Interior Salish	37
1.5.4.1	Lillooet	38
1.5.4.2	Thompson River Salish	39
1.5.4.3	Shuswap	40
1.5.4.4	Montana Salish-Spokane-Kalispel	40
1.5.4.5	Moses Columbia	41
1.5.4.6	Coeur d'Alene	41
1.5.4.7	Okanagan	42
1.5.4.8	Summary: Interior Salish	42
1.5.5	Bella Coola	42
2	Survey of Reduplication in Salish	44
2.1	Examples and Sources	44
2.2	Terminology and Glossing Conventions	45
2.2.1	Morpheme Breaks and Glossing	46
2.2.2	What Counts as Reduplication?	48
2.2.2.1	Reduplicative Morphemes	49
2.2.2.2	Phonological Criteria for Reduplicative Morphemes	50
2.2.3	Defining Core Reduplication Patterns	52
2.2.3.1	Core Reduplication Patterns	52
2.2.3.2	Peripheral Reduplication Patterns	53
2.2.4	Labels for Reduplication Patterns	55
2.2.4.1	Form-Based Primary Classification	56
2.2.4.2	Meaning-Based Secondary Classification	58
2.2.5	Additional Terminology	60
2.2.5.1	Pivots: Edge vs. Prominence Alignment	60
2.2.5.2	Non-Reduplicated Segments: Fixed vs. Epenthetic	61
2.2.6	Classifying Patterns	64
2.3	Core Reduplication Patterns across Salish	65
2.4	C ₁ C ₂ Reduplication	67
2.4.1	Position of C ₁ C ₂ Reduplication	68
2.4.2	Vowels in C ₁ C ₂ Reduplication	69

2.4.3	Meanings Associated with C ₁ C ₂ Reduplication	70
2.4.3.1	Plural C ₁ C ₂ Reduplication	71
2.4.3.2	Property-Denoting C ₁ C ₂ Reduplication	72
2.4.3.3	Distribution of C ₁ C ₂ Reduplicative Morphemes	75
2.4.4	Summary: C ₁ C ₂ Reduplication	77
2.5	C ₁ Reduplication	77
2.5.1	Position of C ₁ Reduplication	77
2.5.2	Vowels in C ₁ Reduplication	79
2.5.3	Meanings Associated with C ₁ Reduplication	80
2.5.3.1	Diminutive C ₁ Reduplication	81
2.5.3.2	Aspectual C ₁ Reduplication	82
2.5.3.3	Plural C ₁ Reduplication	83
2.5.3.4	Person-Linked C ₁ Reduplication	84
2.5.3.5	Distinguishing Types of C ₁ Reduplication	85
2.5.3.6	Distribution of C ₁ Reduplicative Morphemes	87
2.5.4	Summary: C ₁ Reduplication	89
2.6	C ₂ Reduplication	90
2.6.1	Position of C ₂ Reduplication	90
2.6.2	Vowels in C ₂ Reduplication	92
2.6.3	Meanings Associated with C ₂ Reduplication	93
2.6.4	“Triplication” (C ₂ ² Reduplication)	95
2.6.5	Summary: C ₂ Reduplication	96
2.7	V ₁ Reduplication	97
2.7.1	Central Salish V ₁ Reduplication	97
2.7.2	Interior Salish V ₁ Reduplication	98
2.7.3	Tsamosan V ₁ Reduplication	99
2.7.4	Summary: V ₁ Reduplication	100
2.8	Multiple Reduplication	101
2.9	Generalisations and Gaps in Salish Reduplication	102
2.9.1	Generalisations about Core Types of Reduplication	102
2.9.2	Generalisations about Form	103
2.9.3	Generalisations about Meaning	104
2.9.4	Generalisations about Syntactic Distribution	104

3	Phonological Theory and Background	106
3.1	Salish Reduplication and Phonological Theory	107
3.2	The Mora, Nucleus, and the Salish Syllable	109
3.2.1	Moraic and Non-Moraic Consonants	110
3.2.2	Moraic and Non-Moraic Vowels	110
3.2.3	Two Types of Monomoraic Syllables in Salish	111
3.3	Prosodic Morphology	112
3.3.1	Background: Prosodic Morphology Hypothesis	112
3.3.2	Modifications: The Mora as a Prosodic Unit	114
3.4	Generalized Nonlinear Affixation	115
3.4.1	Underlying Forms/Affixes in GNLA	117
3.5	Architecture of the Grammar	118
3.5.1	Spans and Linearisation	119
3.5.1.1	Domain: w-span	119
3.5.1.2	Linearisation: Prefix Feature	121
3.5.2	Stratification: Phonological Levels	123
3.5.2.1	Stratal OT	123
3.5.2.2	Stratification and the Grammar	125
3.5.2.3	Level Diacritics: Classifying Morphemes by Stratum	127
3.5.2.4	Summarising Constraint Rankings	129
3.6	Predictions of the Phonological Theory	131
4	Single Reduplication	133
4.1	Case Study I: Comox-Sliammon	134
4.1.1	Comox-Sliammon Phonology	134
4.1.2	Preview of Comox-Sliammon Reduplication	136
4.1.3	Level 1	136
4.1.3.1	Level 1 C ₁ Reduplication	137
4.1.3.2	Level 1 C ₁ C ₂ Reduplication	155
4.1.3.3	Summary of Level 1 Reduplication	162
4.1.4	Level 2	163
4.1.4.1	Level 2 C ₁ Reduplication	164

4.1.4.2	Level 2 C ₁ C ₂ Reduplication	171
4.1.4.3	Summary of Level 2 Grammar	174
4.1.5	Level 3	176
4.1.5.1	Level 3 C ₂ Reduplication	177
4.1.5.2	Level 3 V ₁ Reduplication	180
4.1.5.3	Summary of Level 3 Grammar	187
4.1.6	Summary of Reduplication in Comox-Sliammon	189
4.2	Case Study II: Lillooet	193
4.2.1	Lillooet Phonology	194
4.2.2	Preview of Lillooet Reduplication	194
4.2.3	Level 1	195
4.2.3.1	Level 1 C ₁ Reduplication	195
4.2.3.2	Level 1 C ₁ C ₂ Reduplication	200
4.2.3.3	Summary of Level 1 Grammar	207
4.2.4	Level 2	208
4.2.4.1	Level 2 C ₂ Reduplication	208
4.2.4.2	Level 2 C ₁ Reduplication	221
4.2.4.3	Glottalisation and C ₁ Reduplication	237
4.2.4.4	Summary of Level 2 Grammar	252
4.2.5	C ₂ ² Reduplication (Triplication)	255
4.2.5.1	Form of C ₂ ² Reduplication (Triplication)	255
4.2.5.2	Input for C ₂ ² Reduplication (Triplication)	257
4.2.5.3	C ₂ ² Reduplication (Triplication) at a Later Stratum	258
4.2.6	Summary of Reduplication in Lillooet	259
4.3	Conclusion	264
5	Multiple Reduplication	265
5.1	Introduction	265
5.2	Background	266
5.3	Case Study I: Halkomelem (Allomorphy)	267
5.3.1	Single Reduplication in Halkomelem	268
5.3.1.1	Level 1: Imperfective	269
5.3.1.2	Level 2: Diminutive	276

5.3.1.3	Level 2: Plural	279
5.3.2	Multiple Reduplication	281
5.3.2.1	Attested: Diminutive Imperfective	281
5.3.2.2	Attested: Plural Imperfective	282
5.3.2.3	Unattested: Diminutive Plural	284
5.3.3	Restrictions on Multiple Reduplication	285
5.3.3.1	Multiple Reduplication is Limited by Morphological Affiliation	286
5.3.3.2	Alternate Approach: Multiple Reduplication is Limited by Segments	291
5.3.3.3	Summary of Approaches	294
5.4	Case Study II: Lillooet (Multiple Reduplication Gaps)	295
5.4.1	Predictions for Multiple Reduplication in Lillooet	295
5.4.2	Multiple Reduplication in Lillooet	297
5.4.2.1	Attested: Level 1 C ₁ C ₂ and Level 2 C ₁ Reduplication	297
5.4.2.2	Attested: Level 1 C ₁ C ₂ and Level 2 C ₂ Reduplication	299
5.4.2.3	Attested: Level 1 C ₁ and Level 2 C ₁ Reduplication	300
5.4.2.4	Attested: Level 1 C ₁ and Level 2 C ₂ Reduplication	301
5.4.2.5	Ungrammatical: Level 1 C ₁ C ₂ and Level 1 C ₁ Reduplication	301
5.4.2.6	Ungrammatical: Level 2 C ₁ and Level 2 C ₂ Reduplication	303
5.4.3	Summary: Multiple Reduplication in Lillooet	304
5.4.4	Analysis: Multiple Reduplication in Lillooet	305
5.4.5	Evaluating Alternative Approaches: Multiple Reduplication	308
5.4.6	Evaluating Alternative Approaches: Triplication	310
5.4.7	Summary: Morpheme vs. Segment Restrictions	311
5.5	Case Study III: Comox-Sliammon	312
5.5.1	Predictions for Multiple Reduplication in Comox-Sliammon	312
5.5.2	Combinations Excluded	314
5.5.3	L1-L1 Combinations	315

5.5.3.1	Attested (Allomorphy): Level 1 C ₁ and Level 1 C ₁ C ₂ Reduplication	315
5.5.3.2	Attested (Allomorphy): Level 1 C ₁ and Level 1 C ₁ Reduplication	316
5.5.4	L1-L2 Combinations	317
5.5.4.1	Attested: Level 1 C ₁ and Level 2 C ₁ C ₂ Reduplication	317
5.5.5	L1-L3 Combinations	318
5.5.5.1	Attested: Level 1 C ₁ and Level 3 V ₁ Reduplication	318
5.5.6	L2-L2 Combinations	320
5.5.6.1	Ungrammatical: Level 2 C ₁ and Level 2 C ₁ C ₂ Reduplication	320
5.5.7	L2-L3 Combinations	322
5.5.7.1	Attested: Level 2 C ₁ C ₂ and Level 3 V ₁ Reduplication	322
5.5.7.2	Attested: Level 2 C ₁ and Level 3 C ₂ Reduplication	323
5.5.7.3	Attested: Level 2 C ₁ and Level 3 V ₁ Reduplication	323
5.5.8	L3-L3 Combinations	324
5.5.8.1	Ungrammatical: Level 3 C ₂ and Level 3 V ₁ Reduplication	324
5.5.9	L1-L2-L3 Combinations	325
5.5.9.1	Attested: Level 1 C ₁ , Level 2 C ₁ , and Level 3 V ₁ Reduplication	325
5.5.10	Analysis	326
5.5.10.1	Diminutive and Plural Properties	326
5.5.10.2	Double C ₁ Diminutive Plurals at Level 1	329
5.5.10.3	Diminutive Plurals	331
5.5.11	Summary	332
5.6	Conclusion	334

6	A Model of Reduplication: Cross-Salish Variation and Implications for Theory	335
6.1	Architecture of the Grammar: Overview	336

6.2	Step 1: Structure Defines Span (Syntax)	338
6.3	Step 2: Lexical Insertion, Step I (Morphosyntax)	339
6.4	Step 3: Linearisation (Morphology)	339
6.4.1	Feature-Based Linearisation	339
6.4.2	Cross-Salish Variation: [prefix] vs. [suffix]	340
6.5	Step 4: Lexical Insertion, Step II (Morphophonology)	344
6.5.1	Allomorphy and Lexical Insertion	344
6.5.2	Lexical Entries	345
6.5.2.1	Suprasegmental: Prosodic Units	345
6.5.2.2	Suprasegmental: Features	352
6.5.2.3	Segmental: Fixed Segments	353
6.5.3	Cross-Salish Variation: Underlying Forms	354
6.5.3.1	Surface Bimoraic Syllable: Lushootseed, Tsamosan, Bella Coola	355
6.5.3.2	Single Fixed Segmental Allomorphs: Montana Salish-Spokane-Kalispel and Moses Columbia Repetitive	356
6.5.3.3	“Onset” Reduplication	358
6.5.3.4	Segmental (Consonant-Only): Inchoative in Southern Interior Salish	363
6.5.3.5	Summary: Lexical Entries for Reduplicative Morphemes	364
6.6	Step 5: Phonological Derivation (Phonology)	365
6.6.1	Stratified Word-Level Grammar	365
6.6.1.1	Phonologically-Defined Levels	366
6.6.1.2	Reduplication is Fission	366
6.6.1.3	Different Constraint Rankings at Each Strata	366
6.6.1.4	Number of Segments Associated with Reduplicative Morphemes	367
6.6.1.5	Linearisation in the Phonological Derivation: Prefix Constraints	368
6.6.1.6	Restrictions on Multiple Reduplication	369
6.6.2	Cross-Salish Variation - Word-Level Grammar	370

6.6.2.1	Cross-Salish Variation: Summary of Case Studies	371
6.6.2.2	Cross-Salish Variation: Extending the Analysis .	376
6.6.2.3	Cross-Salish Variation: Morpheme to Level Mappings and Number of Relevant Levels . . .	383
6.7	Step 6: Post-Reduplication Phonological Processes and Phonetic Implementation (Phonology/Phonetics)	390
6.7.1	Post-Reduplication Phonology	391
6.7.1.1	Post-Lexical Phonology: Schwa Deletion in Lillooet	392
6.7.1.2	Word-Level Post-Reduplication Phonology: Compensatory Lengthening in Bella Coola and Upriver Halkomelem	392
6.7.2	Phonetic Implementation	395
6.8	Implications of the Model	396
6.8.1	Modularity and the Interface between Morphology and Phonology	396
6.8.2	Non-Concatenative Morphology	397
6.8.2.1	Linearisation	398
6.8.2.2	Allomorph Selection	402
6.8.3	Phonological, not Morphological Domains	403
6.9	Conclusion	405
References		406
A Survey of Reduplication Patterns		441
A.1	Survey Methods, Sources, and Transcription	441
A.2	Interior Salish Reduplication	444
A.2.1	Coeur d'Alene Reduplication	444
A.2.1.1	Coeur d'Alene Sources and Transcription	444
A.2.1.2	Coeur d'Alene C ₁ C ₂ Plural/Pluractional Reduplication	445
A.2.1.3	Coeur d'Alene C ₁ C ₂ Property-Denoting Reduplication	447

A.2.1.4	Coeur d'Alene Diminutive C ₁ Reduplication . .	448
A.2.1.5	Coeur d'Alene Change-of-State C ₂ Reduplication	449
A.2.1.6	Coeur d'Alene V ₁ Reduplication	450
A.2.1.7	Other Reduplication in Coeur d'Alene	451
A.2.2	Moses Columbia Reduplication	452
A.2.2.1	Moses Columbia Sources and Transcription . .	452
A.2.2.2	Moses Columbia Plural/Pluractional C ₁ C ₂ Reduplication	452
A.2.2.3	Moses Columbia Property-Denoting C ₁ C ₂ Reduplication	453
A.2.2.4	Moses Columbia Diminutive C ₁ Reduplication .	454
A.2.2.5	Moses Columbia Repetitive C ₁ Reduplication .	455
A.2.2.6	Moses Columbia Change-of-State C ₂ Reduplication	456
A.2.2.7	Repetition in Moses Columbia	457
A.2.3	Okanagan Reduplication	459
A.2.3.1	Okanagan Sources and Transcription	459
A.2.3.2	Okanagan C ₁ C ₂ Plural/Pluractional Reduplication	459
A.2.3.3	Okanagan C ₁ C ₂ Property-Denoting Reduplication	460
A.2.3.4	Okanagan Diminutive C ₁ Reduplication	461
A.2.3.5	Okanagan Change-of-State C ₂ Reduplication . .	463
A.2.3.6	Okanagan V ₁ Reduplication	464
A.2.3.7	Other Reduplication in Okanagan	464
A.2.4	Montana Salish-Spokane-Kalispel Reduplication	466
A.2.4.1	Montana Salish-Spokane-Kalispel Sources and Transcription	466
A.2.4.2	Montana Salish-Spokane-Kalispel Plural/Pluractional C ₁ C ₂ Reduplication	467
A.2.4.3	Montana Salish-Spokane-Kalispel Diminutive C ₁ Reduplication	468
A.2.4.4	Montana Salish-Spokane-Kalispel Repetitive C ₁ Reduplication	470

A.2.4.5	Montana Salish-Spokane-Kalispel Change-of-State C ₂ Reduplication	472
A.2.4.6	Montana Salish-Spokane-Kalispel V ₁ Reduplication	473
A.2.4.7	Repetition in Montana Salish-Spokane-Kalispel	475
A.2.5	Lillooet Reduplication	477
A.2.5.1	Lillooet Sources and Transcription	477
A.2.5.2	Lillooet Plural/Pluractional C ₁ C ₂ Reduplication	477
A.2.5.3	Lillooet Property-Denoting C ₁ C ₂ Reduplication	479
A.2.5.4	Lillooet Diminutive C ₁ Reduplication	480
A.2.5.5	Lillooet Pluractional C ₁ Reduplication	482
A.2.5.6	Lillooet Plural (Initial) C ₁ Reduplication	483
A.2.5.7	Lillooet Change-of-State C ₂ Reduplication	484
A.2.5.8	Other Reduplication in Lillooet	486
A.2.6	Shuswap Reduplication	487
A.2.6.1	Shuswap Sources and Transcription	487
A.2.6.2	Shuswap Plural/Pluractional C ₁ C ₂ Reduplication	487
A.2.6.3	Shuswap Property-Denoting C ₁ C ₂ Reduplication	489
A.2.6.4	Shuswap Diminutive C ₁ Reduplication	490
A.2.6.5	Shuswap First Person C ₁ Reduplication	491
A.2.6.6	Shuswap Change-of-State C ₂ Reduplication	492
A.2.6.7	Other Reduplication in Shuswap	493
A.2.7	Thompson River Salish Reduplication	494
A.2.7.1	Thompson River Salish Sources and Transcription	494
A.2.7.2	Thompson River Salish Plural/Pluractional C ₁ C ₂ Reduplication	494
A.2.7.3	Thompson River Salish Property-Denoting C ₁ C ₂ Reduplication	496
A.2.7.4	Thompson River Salish Diminutive C ₁ Reduplication	497
A.2.7.5	Thompson River Salish Affective C ₁ Reduplication	499

A.2.7.6	Thompson River Salish Change-of-State C_2	
	Reduplication	500
A.3	Central Salish Reduplication	502
A.3.1	Comox-Sliammon Reduplication	502
A.3.1.1	Comox-Sliammon Sources and Transcription . .	502
A.3.1.2	Comox-Sliammon Plural/Pluractional C_1C_2	
	Reduplication	502
A.3.1.3	Comox-Sliammon Property-Denoting C_1C_2	
	Reduplication	504
A.3.1.4	Comox-Sliammon Diminutive C_1 Reduplication	505
A.3.1.5	Comox-Sliammon Plural C_1 Reduplication . . .	508
A.3.1.6	Comox-Sliammon Progressive C_1 Reduplication	509
A.3.1.7	Comox-Sliammon Change-of-State C_2	
	Reduplication	511
A.3.1.8	Comox-Sliammon V_1 Reduplication	511
A.3.2	Halkomelem Reduplication	514
A.3.2.1	Halkomelem Sources and Transcription	514
A.3.2.2	Halkomelem Plural/Pluractional C_1C_2	
	Reduplication	514
A.3.2.3	Halkomelem Property-Denoting C_1C_2	
	Reduplication	517
A.3.2.4	Halkomelem Imperfective C_1 Reduplication . .	518
A.3.2.5	Halkomelem Diminutive C_1 Reduplication . . .	521
A.3.2.6	Other Reduplication in Halkomelem	524
A.3.3	Lushootseed Reduplication	525
A.3.3.1	Lushootseed Sources and Transcription	525
A.3.3.2	Lushootseed Plural/Pluractional C_1C_2	
	Reduplication	525
A.3.3.3	Lushootseed Diminutive C_1 Reduplication . . .	527
A.3.3.4	Lushootseed Plural C_1 Reduplication	530
A.3.3.5	Lushootseed C_2 Reduplication	530
A.3.3.6	Lushootseed V_1 Reduplication	532
A.3.4	Nooksack Reduplication	534

A.3.4.1	Nooksack Sources and Transcription	534
A.3.4.2	Nooksack Plural/Pluractional C ₁ C ₂ Reduplication	534
A.3.4.3	Nooksack Property-Denoting C ₁ C ₂ Reduplication	536
A.3.4.4	Nooksack Diminutive C ₁ Reduplication	536
A.3.4.5	Other Reduplication in Nooksack	537
A.3.5	Northern Straits and Klallam Reduplication	539
A.3.5.1	Northern Straits and Klallam Sources and Transcription	539
A.3.5.2	Northern Straits and Klallam Plural/Pluractional C ₁ C ₂ Reduplication	539
A.3.5.3	Northern Straits and Klallam Property-Denoting C ₁ C ₂ Reduplication	542
A.3.5.4	Northern Straits and Klallam Diminutive C ₁ Reduplication	543
A.3.5.5	Northern Straits and Klallam Imperfective C ₁ Reduplication	546
A.3.5.6	Other Reduplication in Northern Straits and Klallam	548
A.3.6	Pentlatch Reduplication	549
A.3.6.1	Pentlatch Sources and Transcription	549
A.3.6.2	Pentlatch C ₁ C ₂ Reduplication	549
A.3.6.3	Pentlatch C ₁ Reduplication	550
A.3.7	Sechelt Reduplication	553
A.3.7.1	Sechelt Sources and Transcription	553
A.3.7.2	Sechelt Plural/Pluractional C ₁ C ₂ Reduplication	553
A.3.7.3	Sechelt Property-Denoting C ₁ C ₂ Reduplication	555
A.3.7.4	Sechelt Diminutive C ₁ Reduplication	556
A.3.7.5	Sechelt Imperfective C ₁ Reduplication	559
A.3.7.6	Sechelt Plural C ₁ Reduplication	560
A.3.7.7	Sechelt Change-of-State C ₂ Reduplication . . .	561
A.3.8	Squamish Reduplication	563
A.3.8.1	Squamish Sources and Transcription	563
A.3.8.2	Squamish Plural/Pluractional C ₁ C ₂ Reduplication	563

A.3.8.3	Squamish Property-Denoting C ₁ C ₂ Reduplication	565
A.3.8.4	Squamish Progressive C ₁ Reduplication	566
A.3.8.5	Squamish Diminutive C ₁ Reduplication	566
A.3.9	Twana Reduplication	568
A.3.9.1	Twana Sources and Transcription	568
A.3.9.2	Twana Plural C ₁ C ₂ Reduplication	568
A.3.9.3	Twana Diminutive C ₁ Reduplication	570
A.3.9.4	Twana Diminutive V ₁ Reduplication	571
A.3.9.5	Other Reduplication in Twana	572
A.4	Tsamosan Reduplication	573
A.4.1	Cowlitz Reduplication	573
A.4.1.1	Cowlitz Sources and Transcription	573
A.4.1.2	Cowlitz Pluractional C ₁ C ₂ Reduplication	573
A.4.1.3	Other Reduplication in Cowlitz	574
A.4.2	Lower Chehalis Reduplication	576
A.4.2.1	Lower Chehalis Sources and Transcription	576
A.4.2.2	Lower Chehalis C ₁ C ₂ Pluractional Reduplication	576
A.4.2.3	Other Reduplication in Lower Chehalis	577
A.4.3	Quinalt Reduplication	579
A.4.3.1	Quinalt Sources and Transcription	579
A.4.3.2	Quinalt Pluractional C ₁ C ₂ Reduplication	579
A.4.3.3	Quinalt V ₁ Reduplication	580
A.4.3.4	Other Reduplication in Quinalt	581
A.4.4	Upper Chehalis Reduplication	582
A.4.4.1	Upper Chehalis Sources and Transcription	582
A.4.4.2	Upper Chehalis Pluractional C ₁ C ₂ Reduplication	582
A.4.4.3	Upper Chehalis Change-of-State (“Slow”) C ₂ Reduplication	584
A.4.4.4	Upper Chehalis Superlative C ₂ Reduplication	584
A.5	Tillamook Reduplication	586
A.5.0.1	Tillamook Sources and Transcription	586
A.5.1	Tillamook Plural/Pluractional C ₁ C ₂ Reduplication	586
A.5.2	Tillamook Diminutive C ₁ Reduplication	587

A.5.3	Tillamook Pluractional and Continuative C ₁ Reduplication	588
A.5.4	Tillamook Change-of-State C ₂ Reduplication	589
A.6	Bella Coola Reduplication	591
A.6.1	Bella Coola Sources and Transcription	591
A.6.2	Bella Coola Plural C ₁ C ₂ Reduplication	591
A.6.3	Bella Coola Diminutive C ₁ C ₂ and C ₁ Reduplication . . .	593
A.6.3.1	Bella Coola Diminutive C ₁ Reduplication . . .	595
A.6.3.2	Bella Coola Diminutive C ₁ C ₂ Reduplication . .	596
A.6.4	Phonological Patterns Shared between Different Types of Reduplication in Bella Coola	597
B	Survey of Multiple Reduplication Patterns	598
B.1	Interior Salish Multiple Reduplication	600
B.1.1	Coeur d'Alene Multiple Reduplication	600
B.1.1.1	Diminutive Plural	600
B.1.1.2	Diminutive and Property-Denoting	601
B.1.1.3	Plural Change of State	601
B.1.1.4	Diminutive Change of State	602
B.1.2	Moses Columbia Multiple Reduplication	604
B.1.2.1	Diminutive Plural	604
B.1.2.2	Plural Change of State	604
B.1.2.3	Diminutive Change of State	605
B.1.3	Okanagan Multiple Reduplication	607
B.1.3.1	Diminutive Plural	607
B.1.3.2	Diminutive and Property-Denoting	608
B.1.3.3	Plural Change of State	608
B.1.3.4	Diminutive Change of State	609
B.1.4	Montana Salish-Spokane-Kalispel Multiple Reduplication	610
B.1.4.1	Diminutive Plural	610
B.1.4.2	Diminutive Repetitive	611
B.1.4.3	Diminutive Change of State	611
B.1.4.4	Diminutive, Plural, and Change of State	612
B.1.5	Lillooet Multiple Reduplication	613

	B.1.5.1	Diminutive Plurals	613
	B.1.5.2	Diminutive and Property-Denoting	614
	B.1.5.3	Plural Change of State	615
	B.1.5.4	Diminutive and Initial	616
	B.1.5.5	Change of State and Initial	616
	B.1.5.6	Triplication + Change-of-State	617
B.1.6	Shuswap	Multiple Reduplication	618
	B.1.6.1	Diminutive Plural	618
	B.1.6.2	Diminutive and Property-Denoting	619
	B.1.6.3	First Person and Plural	620
	B.1.6.4	First Person Properties	621
	B.1.6.5	Comparatives	621
B.1.7	Thompson River Salish	Multiple Reduplication	622
	B.1.7.1	Diminutive Plural	622
	B.1.7.2	Diminutive and Property-Denoting	623
	B.1.7.3	Plural Change of State	624
	B.1.7.4	Diminutive Change of State	625
	B.1.7.5	Plural and Property-Denoting	626
	B.1.7.6	Affective Change of State	627
	B.1.7.7	Affective Diminutive	627
	B.1.7.8	Plural, Diminutive, and Change of State	628
	B.1.7.9	Diminutive, Plural, and Property-Denoting	629
	B.1.7.10	Affective, Diminutive, and Property-Denoting	629
B.2	Central Salish	Multiple Reduplication	630
	B.2.1	Comox-Sliammon Multiple Reduplication	630
	B.2.1.1	Diminutive and Property-Denoting	630
	B.2.1.2	C ₁ Plural and Property-Denoting	631
	B.2.1.3	Diminutive and C ₁ Plural	632
	B.2.1.4	Diminutive and C ₁ C ₂ Plural	632
	B.2.1.5	Diminutive and V ₁ Plural	633
	B.2.1.6	C ₁ C ₁ Plural and V ₁ Plural	634
	B.2.1.7	Progressive Change of State	635
	B.2.1.8	Progressive and V ₁ Plural	636

B.2.1.9	Diminutive, Progressive, and V ₁ Plural	636
B.2.1.10	Diminutive, V ₁ Plural, and C ₁ C ₂ Plural	637
B.2.2	Halkomelem Multiple Reduplication	638
B.2.2.1	Diminutive Imperfective	638
B.2.2.2	Plural Imperfective	639
B.2.3	Lushootseed Multiple Reduplication	641
B.2.3.1	Diminutive and C ₁ C ₂ Plural	641
B.2.3.2	Diminutive and V ₁ Plural	643
B.2.3.3	C ₁ C ₁ Plural and C ₂ Reduplication	643
B.2.4	Sechelt Multiple Reduplication	644
B.2.4.1	Diminutive and C ₁ C ₂ Plural	644
B.2.4.2	Imperfective Change of State	645
B.2.4.3	C ₁ Plural and Change of State	645
B.2.4.4	C ₁ C ₂ Plural and Change of State	646
B.2.5	Northern Straits and Klallam Multiple Reduplication . . .	647
B.2.5.1	Diminutive and Property-Denoting	647
B.2.5.2	Imperfective Diminutives	648
B.2.6	Twana Multiple Reduplication	649
B.2.6.1	Double Diminutive	649
B.3	Tillamook Multiple Reduplication	651
B.3.1	Plural Change of State	651
B.3.2	Plural + Unclear C ₁ Reduplication	651
C	Index of Constraints and Summary of Rankings	653
C.1	Index of Constraints	653
C.2	Summary of Linear Rankings	660
D	Sources for Salish Reduplication	662
D.1	Bella Coola	663
D.2	Coeur d'Alene	664
D.3	Comox-Sliammon	664
D.4	Cowlitz	666
D.5	Halkomelem	666

D.6	Lillooet	668
D.7	Lower Chehalis	669
D.8	Lushootseed	669
D.9	Montana Salish-Spokane-Kalispel	671
D.10	Moses Columbia	672
D.11	Nooksack	673
D.12	Okanagan	673
D.13	Pentlatch	674
D.14	Quinault	674
D.15	Sechelt	674
D.16	Shuswap	675
D.17	Squamish	675
D.18	Straits (Northern Straits and Klallam)	676
D.19	Thompson	678
D.20	Tillamook	679
D.21	Twana	679
D.22	Upper Chehalis	680
E	NAPA to IPA Conversion Chart	681
F	Fission and Alternate Analyses of V₁ Reduplication	686

List of Tables

Table 1.1	Salish Exonyms and Endonyms	11
Table 1.2	Salish Languages	15
Table 2.1	Summary: Core Reduplication in Salish	66
Table 2.2	Core Reduplication Patterns in Salish Languages	67
Table 2.3	C ₁ C ₂ Reduplication in Salish Languages by Branch	75
Table 2.4	C ₁ C ₂ Reduplication in Salish Languages	76
Table 2.5	Number of Languages with C ₁ Reduplication by Branch	87
Table 2.6	Types of C ₁ Reduplication	88
Table 2.7	C ₁ Reduplication: Form-to-Meaning	89
Table 2.8	C ₂ Reduplication in Salish Languages by Branch	96
Table 2.9	V ₁ Reduplication in Salish Languages by Branch	100
Table 2.10	Meanings of V ₁ Reduplication	100
Table 2.11	Multiple Reduplication in Salish Languages by Branch	101
Table 3.1	Example Summary of Constraint Rankings at a Stratum	130
Table 4.1	Preview of Comox-Sliammon Reduplicative Affixes	136
Table 4.2	Level 1 Reduplicative Affixes	137
Table 4.3	Preliminary Level 1 Grammar: Diminutive	152
Table 4.4	Level 1 Reduplicative Affixes	155
Table 4.5	Preliminary Level 1 Ranking from Section 4.1.3.1.2	159
Table 4.6	Comox-Sliammon Level 1 Grammar	163
Table 4.7	Affix Constraints at Level 1	163
Table 4.8	Level 2 Reduplicative Affixes	164

Table 4.9	Number of Underlying and Surface Moras (CəSC Base)	170
Table 4.10	Comox-Sliammon Level 2 Grammar	171
Table 4.11	Affix Constraints at Level 1 vs. Level 2	176
Table 4.12	Level 3 Reduplication	176
Table 4.13	Comox-Sliammon Level 3 Grammar (Preliminary)	180
Table 4.14	Place Features	183
Table 4.15	Comox-Sliammon Level 3 Grammar (Updated)	186
Table 4.16	Comox-Sliammon Affix Constraints	189
Table 4.17	Constraints Motivating Parsing and Filling	190
Table 4.18	Constraints Motivating Fission and Epenthesis	191
Table 4.19	Constraints Related to Consonant Features	191
Table 4.20	Constraints Related to Alignment and Position	192
Table 4.21	Allomorph Selection and Vowel/Syllable Constraints	193
Table 4.22	Lexical Entries in Comox-Sliammon	193
Table 4.23	Preview of Lillooet Reduplicative Affixes	195
Table 4.24	Level 1 Reduplicative Affixes	195
Table 4.25	Lillooet Level 1 Grammar (Preliminary)	200
Table 4.26	Lillooet Level 1 Grammar (Updated)	207
Table 4.27	Lillooet Affix Constraints	207
Table 4.28	Level 2 Reduplicative Affixes	208
Table 4.29	Stress Assignment	214
Table 4.30	Stress and Reduplication: <i>qax^wx^wáwł</i> ‘break down’	214
Table 4.31	Stress and Reduplication: <i>nləpəpánaʔ</i> ‘get completely buried’	215
Table 4.32	Lillooet Level 2 Grammar (Preliminary)	221
Table 4.33	Lillooet Level 2 Grammar (Preliminary - Second Version)	228
Table 4.34	Lillooet Level 2 Grammar (Preliminary - Version 3)	237
Table 4.35	Level 2 Affix Constraints	254
Table 4.36	Constraints Motivating Parsing and Filling	260
Table 4.37	Fission and Epenthesis Constraints	260
Table 4.38	Position of Fissioned Consonants	261
Table 4.39	Building Prosodic Structure	262
Table 4.40	Constraints Dealing with Lexical Stress and Phonotactics	262
Table 4.41	Glottalisation and [c.g.] Constraints	263

Table 4.42	Lexical Entries in Lillooet	263
Table 5.1	Phonologically Conditioned Allomorphs of the Imperfective (Halkomelem)	272
Table 5.2	INT _S Violations for Plural	292
Table 5.3	INT _S Violations for Diminutive Plural Imperfective (Unattested)	292
Table 5.4	INT _S Violations for Plural Imperfective	293
Table 5.5	Inventory of Lillooet Reduplication	296
Table 5.6	Predictions for Possible Lillooet Multiple Reduplication	296
Table 5.7	Multiple Reduplication with C ₁ C ₂ (Level 1) and C ₁ (Level 2) .	299
Table 5.8	Lillooet Multiple Reduplication Inventory	305
Table 5.9	Inventory of Comox-Sliammon Reduplication	312
Table 5.10	Predictions for Comox-Sliammon Multiple Reduplication . . .	313
Table 5.11	Excluded Pairs	314
Table 5.12	Box - Predicted Derivation (Ungrammatical Output)	331
Table 5.13	Comox-Sliammon System: Revised	333
Table 5.14	Comox-Sliammon Multiple Reduplication	333
Table 6.1	Mora in Case Studies	346
Table 6.2	Syllable in Case Studies	348
Table 6.3	Syllable with Nucleus and Mora in Case Studies	350
Table 6.4	Summary: Foot	352
Table 6.5	Underlying Forms in Comox-Sliammon, Lillooet, and Halkomelem	355
Table 6.6	Prosodic and Segmental Content in Lexical Entries	365
Table 6.7	Epenthesis vs. Fission: Constraint Rankings	367
Table 6.8	Ranking of Prefix Constraints	369
Table 6.9	Reduplicative Morphemes in Comox-Sliammon, Lillooet, and Halkomelem	371
Table 6.10	Levels in Comox-Sliammon, Lillooet, and Halkomelem	372
Table 6.11	Levels in Comox-Sliammon, Lillooet, and Halkomelem	373
Table 6.12	Lengthening vs. Reduplicating Grammars	378
Table 6.13	Mora Constraints	382

Table 6.14	Number of Reduplication Relevant Strata	384
Table 6.15	Reduplication in Thompson River Salish	389
Table 6.16	Levels in Thompson River Salish	390
Table 6.17	Post-Reduplication Compensatory Lengthening	395
Table A.1	Reduplication in Coeur d'Alene	444
Table A.2	Reduplication in Moses Columbia	452
Table A.3	Reduplication in Okanagan	459
Table A.4	Reduplication in Montana Salish-Spokane-Kalispel	466
Table A.5	Reduplication in Lillooet	477
Table A.6	Reduplication in Shuswap	487
Table A.7	Reduplication in Thompson River Salish	494
Table A.8	Reduplication in Comox-Sliammon	502
Table A.9	Reduplication in Halkomelem	514
Table A.10	Reduplication in Lushootseed	525
Table A.11	Reduplication in Nooksack	534
Table A.12	Reduplication in Northern Straits and Klallam	539
Table A.13	Reduplication in Pentlatch	549
Table A.14	Reduplication in Sechelt	553
Table A.15	Reduplication in Squamish	563
Table A.16	Reduplication in Twana	568
Table A.17	Reduplication in Cowlitz	573
Table A.18	Reduplication in Lower Chehalis	576
Table A.19	Reduplication in Quinault	579
Table A.20	Reduplication in Upper Chehalis	582
Table A.21	Reduplication in Tillamook	586
Table A.22	Reduplication in Bella Coola	591
Table B.1	Multiple Reduplication in Salish Languages by Branch	598
Table B.2	Multiple Reduplication in Coeur d'Alene	600
Table B.3	Multiple Reduplication in Moses Columbia	604
Table B.4	Multiple Reduplication in Okanagan	607
Table B.5	Multiple Reduplication in Montana Salish-Spokane-Kalispel	610

Table B.6	Multiple Reduplication in Lillooet	613
Table B.7	Multiple Reduplication in Shuswap	618
Table B.8	Multiple Reduplication in Thompson River Salish	622
Table B.9	Multiple Reduplication in Comox-Sliammon	630
Table B.10	Multiple Reduplication in Halkomelem	638
Table B.11	Multiple Reduplication in Lushootseed	641
Table B.12	Multiple Reduplication in Sechelt	644
Table B.13	Multiple Reduplication in Northern Straits and Klallam	647
Table B.14	Multiple Reduplication in Twana	649
Table B.15	Multiple Reduplication in Tillamook	651
Table C.1	Sections for Constraint Definitions	653
Table C.2	Constraint Definitions	654
Table D.1	Sources for Bella Coola Reduplication	663
Table D.2	Sources for Coeur d'Alene Reduplication	664
Table D.3	Sources for Comox-Sliammon Reduplication	664
Table D.4	Sources for Cowlitz Reduplication	666
Table D.5	Sources for Halkomelem Reduplication	666
Table D.6	Sources for Lillooet Reduplication	668
Table D.7	Sources for Lower Chehalis Reduplication	669
Table D.8	Sources for Lushootseed Reduplication	669
Table D.9	Sources for Montana Salish-Spokane-Kalispel Reduplication	671
Table D.10	Sources for Moses Columbia Reduplication	672
Table D.11	Sources for Nooksack Reduplication	673
Table D.12	Sources for Okanagan Reduplication	673
Table D.13	Sources for Pentlatch Reduplication	674
Table D.14	Sources for Quinault Reduplication	674
Table D.15	Sources for Sechelt Reduplication	674
Table D.16	Sources for Shuswap Reduplication	675
Table D.17	Sources for Squamish Reduplication	675
Table D.18	Sources for Northern Straits and Klallam Reduplication	676
Table D.19	Sources for Thompson Reduplication	678

Table D.20	Sources for Tillamook Reduplication	679
Table D.21	Sources for Twana Reduplication	679
Table D.22	Sources for Upper Chehalis Reduplication	680
Table E.1	NAPA Conversion Chart: Obstruents	681
Table E.2	NAPA Conversion Chart: Sonorant Consonants	683
Table F.1	Defining Reduplication - Inchoatives	689
Table F.2	Defining Reduplication - Link Vowels	690

List of Figures

Figure 1.1	Architecture of the Grammar (PF)	5
Figure 1.2	Salish Languages (Green) (Suttles, n.d.)	12
Figure 1.3	Salish Branches (Orange) and Languages (Pink)	13
Figure 1.4	Central Salish Languages (Dark Pink), and Dialects (Light Pink)	28
Figure 1.5	Tsamosan (Dark Pink) and Dialects (Light Pink)	35
Figure 1.6	Interior Salish Languages (Dark Pink) and Dialects (Light Pink)	38
Figure 6.1	Architecture of the PF Component of the Grammar	336
Figure 6.2	Example Tree	338
Figure 6.3	Example Tree	340

Acknowledgments

First and foremost, I want to express my gratitude and respect to the knowledge keepers, language teachers, communities, and linguists whose commitment to Salish languages has resulted in a rich documentary record. It would not be possible to discuss reduplication in all twenty-three Salish languages if not for their contributions to the field. However, the true heart of this dissertation is the major empirical contributions of the speakers of ʔayʔaʃuθəm (Betty Wilson, Elsie Paul, the late Freddie Louie, the late Joanne Francis, the late Karen Galligos, the late Marion Harry, the late Maggie Wilson, Phyllis Dominic, Margaret Vivier, and Jerry Francis), St'át'imcets (Carl Alexander - Qwaʔyán'ak, through his work with Henry Davis), and Secwepemctsín (Garlene Dodson). Their contributions have filled gaps in previous documentation, which I hope will be valuable for their communities, future language learners, and linguists alike. I would also like to thank the members of ʔayʔaʃuθəm-speaking communities who have shared their knowledge with me, including Betty Wilson, Gail Blaney, Jacqueline Mathieu, Koosen Pielle, and Hoss Timothy.

I could not have undertaken this dissertation without the mentorship of Henry Davis. I am greatly indebted to him for introducing me to the Salish language family, conducting fieldwork on St'át'imcets for my dissertation, and getting me started on my own fieldwork with speakers of ʔayʔaʃuθəm and Secwepemctsín. Henry challenged me to go beyond a phonological analysis and consider non-concatenative morphology with respect to the larger architecture of the grammar. His generosity with time and his (*very* honest but thoughtful) feedback is what enabled me to write the best possible version of this dissertation. He has always

seen the best in me and my work; I not have asked for a better supervisor.¹

I am very grateful to my committee for the helpful feedback, discussions, and encouragement. I appreciate the time and care Lisa Matthewson put into reading every draft I sent her: she played a huge role in the ~200 pages on the semantics of reduplication that did not end up in my dissertation. I left every meeting with her feeling excited about semantics and fieldwork. I also want to thank Gunnar Hansson for providing the phonology expertise with respect to my analysis: he pointed out many of the potential problems that ended up having the most interesting solutions (including a particularly challenging problem with my analysis of multiple reduplication).

I would like to extend my sincere gratitude to Peter Svenonius, my external examiner, for his insightful comments and questions on this work. I also appreciate the thought-provoking questions that Douglas Pulleyblank and Stefan Dollinger asked as my university examiners. The questions raised at my defence give me directions to pursue in future work. Thank you to Bruce Miller for chairing my defence.

Much of my understanding of reduplication in specific languages has been shaped by conversation with others through Salish Working Group and the International Conference on Salish Languages. In particular, I thank the following people for sharing insight into particular languages at various points, including Mat Andreatta, Donna Gerdts, Marianne Ignace, John Lyon, and Susan Blake. I also thank past and present members of the ʔayʔaʃuθəm Lab (Marianne, Daniel, Henry, Bailey, Laura, Bruno, Kaining, Roger, and Jillian) and Secwepemctsin Working Group (Sander, Bruce, Julia, Mitchi, Henry, Celina, and Hannah) for many helpful conversations.

I have been very lucky to have incredible mentors during my graduate studies. I am extremely grateful to Suzanne Urbanczyk, Amanda Cardoso, and Marianne Huijsmans, who I consider dear friends, co-authors, and role models. I am also thankful for the support and thoughtful professional advice that Molly Babel,

¹Marion Caldecott, who first introduced me to phonology, thanks Henry in her MA thesis acknowledgments for being on her committee despite his “abhorrence of phonology”. I would like to provide a very necessary update after 26 years: I have decided to bestow Henry with the title of “Honorary (albeit reluctant) Expert on Salish Phonology”.

Anne-Michelle Tessier, Carla Hudson Kam, and Patti Janssen have shared with me. I would also be remiss not to mention the incredible classmates that I have had, who are too numerous to list here, with special mention to those in AMT's Lab (Kaili, Alex, Sijia, and Gazelle), who never fail to make me laugh, as well as my cohort (Khia, Daniel, Roger, and Bruno), who were there on the very first day of this journey.

I greatly appreciate the emotional support and endless distractions provided by Michael Schwan and Shannon Arsenault when I needed it the most. I also want to thank my family for their encouragement and support. Thank you to my mom, dad, Fred, Torie, Dennis, Felicity, Hadley, Anne, and Papa. In particular, I would like to thank to my mom and Fred for always being happy to do ferry pick up and drop off for ʔayʔajuθəm Lab field trips.

Finally, I want to acknowledge the funding that has supported the fieldwork undertaken for this dissertation: this work was made possible by a SSHRC grant awarded to Henry Davis (435-2020-1293) and several grants from the Jacobs Research Fund.

Dedication

This dissertation is dedicated to the late Freddie Louie, late Joanne Francis, and late Karen Galligos. I am so grateful to have known them. They taught me so much about ʔayʔajuθəm. Their kindness and laughter during elicitation sessions is something I will always cherish.

Chapter 1

Introduction

Non-concatenative morphology, including reduplication, offers insight into the interface between morphology and phonology, including the architecture of the grammar and division of labour between different components. In this dissertation, I propose a model of reduplication that integrates components of Prosodic Morphology and Moraic Theory (McCarthy & Prince, 1986; Hayes 1989), Generalised Nonlinear Affixation (Bermúdez-Otero, 2012; Zimmermann, 2017), and Stratal Optimality Theory (Kiparsky 2015; Bermúdez-Otero 2017).

Each component of the model developed here is motivated through case studies of individual Salish languages (Comox-Sliammon, Lillooet, and Halkomelem); predictions of the model are further considered in the context of the Salish language family as a whole.¹ Salish languages provide the ideal conditions for a typological examination and phonological analysis of reduplicative patterns within and across languages; they have a number of productive types of reduplication that apply across different lexical categories (see, e.g., Jelinek & Demers, 1997), as well as participating in multiple reduplication (see, e.g., Broselow, 1983). The data in this dissertation come from fieldwork, published sources, and archival materials.

In this introductory chapter, I first introduce the goals of the dissertation, including my main claims and contributions (Section 1.1), as well as outlining

¹When referring to the languages, I use exonyms, which are traditionally used in diachronic and comparative treatments of the language family, rather than endonyms; this decision is motivated in Section 1.3.1.

the structure of the dissertation (Section 1.2). This is followed by an overview of the Salish language family (Section 1.3), which covers language names, major branches, and dialects. Section 1.4 provides an overview of my methodology, including discussion of different types of data, as well as methodological decisions related to transcription and glossing. Section 1.5 concludes this chapter with a summary of the state of documentation for each Salish language, highlighting existing descriptions and analyses that contain either examples or description of reduplication.

1.1 Focus and Goals of this Dissertation

1.1.1 Reduplication and Non-Concatenative Morphology

Morphology can be *concatenative* or *non-concatenative*; concatenative morphemes “involve linear affixation of material to a base which remains relatively unperturbed”, while non-concatenative morphemes “involve morphological material from both the base and affix interleaved” (Kastner & Tucker, 2019, p.1).

The primary focus of this dissertation is on a specific type of non-concatenative morphology: *reduplication*. Descriptively, reduplication is the repetition of phonological material; repeated segments are used to mark the meaning associated with a reduplicative morpheme. While reduplication is the most common term for this type of non-concatenative morphology, other terms are sometimes used in the literature, such as “copying” or “doubling”.² However, given the range of allomorphy associated with certain reduplicative morphemes, such as aspectual marking in Halkomelem (see Section 5.3) or vowel lengthening in Upper Chehalis (see Section 6.6.2.2.1), it is not possible to talk about reduplication in Salish without considering other non-concatenative processes, such as metathesis and ablaut. Non-concatenative processes other than reduplication are therefore included where relevant.

While the use of reduplication is relatively peripheral to the grammar of some

²Hess (1967) labels reduplicative morphemes in Lushootseed as “chameleon morphemes”, which is a name chosen to characterise their flexibility of form and dependence on their environment to inform the choice of segmental content. This is an apt label because reduplication differs from other affixes with segmental content, which have stable or “fixed” forms across words.

European languages, being used for example in colloquial English only to mark a particular kind of contrastive focus (Ghomeshi, Jackendoff, Rosen, & Russell, 2004), it is attested more robustly in the grammar of many other languages. In fact, reduplication is widespread geographically: 278 of 368 languages surveyed in the World Atlas of Linguistic Structures (WALS) use reduplication, including both full and partial patterns (Rubino, 2013). Full reduplication refers to the doubling of a full word, root, or stem; partial reduplication refers to doubling a smaller portion of phonological word, such as a consonant or a syllable. The focus of my dissertation is partial reduplication.³

In languages where reduplication is prominent, there may be multiple reduplicative processes, which take different forms and serve different functions. These can be combined to create words with *multiple reduplication*. I investigate both single and multiple reduplication patterns in this dissertation.

Reduplication raises significant questions for theories of morphophonology. While the meaning of a given reduplicative morpheme may be the same across words, its phonological form is dependent on the identity and form of other morphemes in the word. This means that reduplication provides insight into the order in which morphemes are added, because the segments targeted for copying must already be present at the point in the derivation when reduplication occurs. Furthermore, there are phonological restrictions on the number and types of reduplicative morphemes that can co-occur within a word in multiple reduplication (see, e.g., Mellesmoen & Urbanczyk, 2021a; Zimmermann, 2021b).

Reduplication in Salish languages has been central to argumentation in phonological theory. However, most phonological analyses of reduplication in Salish only focus on a single type of reduplication in a single language (e.g., C_1 diminutive in Shuswap in Bell 1983 and Lushootseed in Mellesmoen & Urbanczyk, 2024), or on well-attested combinations (e.g., diminutive C_1 and plural C_1C_2 reduplication across multiple languages in Broselow 1983), without considering the full inventory of reduplicative morphemes in a language. This

³One of the terms for reduplication used in Salish linguistics is “total reduplication” (see, e.g., Kuipers 1967 or Van Eijk 1997); despite the label, this type of reduplication is an example of partial reduplication, not full reduplication. See discussion of C_1C_2 (= “total”) reduplication in Chapter 2 (Section 2.4).

leaves a number of questions unanswered, such as those pertaining to how and whether the analysis of one reduplicative morpheme is compatible with the others, or with the phonological grammar as a whole; exceptions that include detailed discussions of reduplication in a single language with formal phonological analysis are found in B. Carlson (1989) for Spokane, Jimmie (1994) for Thompson River Salish, Urbanczyk (2001) and Kirkham (1992) for Lushootseed, and Crawford (2015) for Klallam.

This dissertation addresses a gap in the description and formal analysis of Salish reduplication by proposing a theoretical approach informed by a survey of reduplication in all 23 Salish languages based on published and archival sources; the survey is supplemented by fieldwork on two languages (Comox-Sliammon and Lillooet), which allows specific predictions of the model to be tested.

1.1.2 Main Contributions and Claims

The primary goal of this dissertation is to propose a model of the spell-out component of the grammar that can account for the form and position of reduplicative morphemes across Salish, with a specific focus on the phonological derivation. In particular, I provide a formal analysis that answers the following questions:

- What is/are the relevant *domain(s)* for reduplication?
- What type of phonological material is present in the *lexical entries* of reduplicative morphemes?
- What aspects of the phonological *grammar* determine how, where, and if a reduplicative morpheme is realised?

With respect to the domain, I argue that a *stratum-based* model of phonology is required to account for the morphophonological grammar of reduplication, including attested patterns of allomorphy, variation in the form and position of reduplicative morphemes, and systematic gaps in multiple reduplication patterns. I propose a modified version of Stratal Optimality Theory where strata are defined purely on a phonological basis, without resorting to morphological domains; this

represents a departure from previous work in Stratal Phonology (Kiparsky, 2015; Bermúdez-Otero, 2017), as well as other frameworks in the tradition of Lexical Phonology (Mohanan, 1982; Kiparsky, 1985).

1.1.2.1 Architecture of the Grammar

Figure 1.1 shows the architecture of grammar proposed in this dissertation.

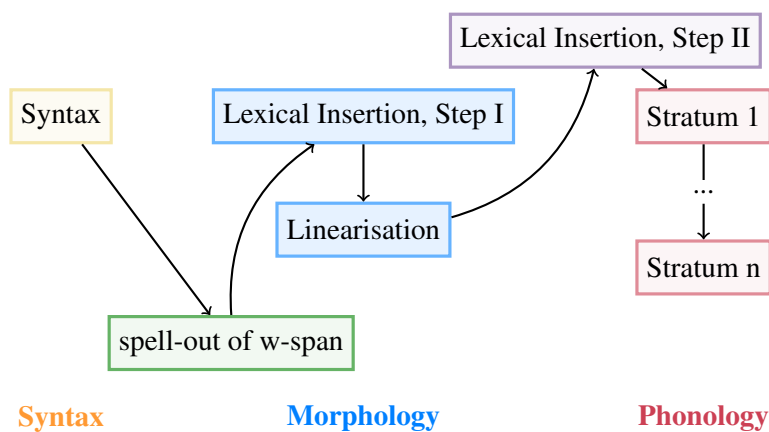


Figure 1.1: Architecture of the Grammar (PF)

1.1.2.1.1 Morphosyntax

The only morphosyntactic domain relevant for reduplicative morphemes in Salish is the morphological word, which corresponds to a w-span. Following Svenonius (2016), a w-span is a sequence of heads in a head-complement relationship with the feature [w] marking the highest head in the span that corresponds to a word. Lexical insertion takes place in two steps, following Svenonius and Bye (2012) and Svenonius (2012). The first step of lexical insertion, called *L-Match*, follows the spell-out of a w-span; syntactic heads are matched with morphosyntactic feature bundles.

I assume that lexical entries contain features for linearisation, following work on clitics by Huijsmans (2023a) and H. Davis and Huijsmans (2024). Reduplicative morphemes contain either a [prefix] or [suffix] feature, which provides instructions for linearisation in the morphological component of the grammar. Linearisation

acts on morphosyntactic feature bundles prior to the insertion of phonological content.

1.1.2.1.2 Morphophonology

Phonological content is inserted at the second stage of lexical insertion, called *Insert*. For the phonology of non-concatenative morphemes, I adopt the model of *Generalised Nonlinear Affixation*, which assumes that non-concatenative morphology involves the affixation of non-linear phonological content, which can include floating elements of prosodic structure or features (Bermúdez-Otero, 2012). I adapt the prosodic hierarchy proposed by Zec (1988), which assumes moraic theory (Hyman, 1985; Hayes, 1989); the mora and the nucleus are crucial with respect to Salish reduplication.

I show that reduplication can be triggered by linear (segmental) affixes, as well as non-linear (prosodic) affixes. The phonological content supplied at the second step of lexical insertion can therefore include both segmental and suprasegmental content. All affixes are linearised in the same way by the morphology (prior to the second stage of lexical insertion), it is only following the second stage of lexical insertion that differences between non-linear and linear representations emerge.

Differences in the realisation of suprasegmental and segmental content are determined by the phonological grammar at the relevant stratum; the phonology determines when and if segments are reduplicated, and further if the reduplicative morpheme will be prefixed, infix, or suffix. A consequence of the model in Figure 1.1 is that linear order is determined across different components of the grammar (syntax, morphology, phonology).

The derivation of a phonological word takes place across a number of ordered strata, which each have their own grammar (i.e., constraint ranking). While the phonological derivation of a given word involves evaluation at each stratum, affixes are lexically specified for a level; this specification determines when the affix is parsed into prosodic structure.

Given that the only relevant morphosyntactic unit is the w-span that determines which morphemes will be present in the input to the phonology at the first stratum (Level 1), evidence for strata is phonological in nature. Morpheme-to-stratum mapping determines the shape and position of a reduplicative morpheme

(determined by the constraint ranking), whether it is parsed into the phonological word early or late in the derivation, and whether it may participate in multiple reduplication (given restrictions on multiple reduplication at the same stratum).

1.1.2.2 Main Theoretical Claims

The main theoretical claims of this dissertation include the following:

1. *Domain:*

- (a) Phonological and morphosyntactic domains beneath the level of the word are not isomorphic.
- (b) The smallest morphosyntactic domain relevant for the phonology is the word; there is no evidence for a stem.
- (c) Strata are phonologically defined.

2. *Lexical Entries:*

- (a) Underlying forms of reduplicative morphemes can contain linear and non-linear phonological content.
- (b) Concatenative and non-concatenative morphemes are not inherently different; they are treated the same with respect to linearisation and allomorphy prior to being parsed into the phonological word.

3. *Grammar:*

- (a) The form and position of reduplication depends on the constraint rankings across different strata.
- (b) Only one prosodic affix can be parsed into the word per stratum.

1.1.2.3 Contributions to Description and Documentation

A second goal of this dissertation is to serve as a reference for and overview of Salish reduplication, contributing to the documentation and description of the phenomenon. The survey in Chapter 2 and the data provided in Appendix A (types of reduplication) and Appendix B (attested combinations of reduplicative

morphemes) present an overview of Salish reduplication in an accessible format that can be used as the empirical basis for further theoretical and typological investigation. Chapters 4 and 5 include fieldwork data, crucially including negative judgments where relevant, from my own fieldwork (Comox-Sliammon and Shuswap), as well as by Henry Davis (Lillooet).

1.2 Structure of this Dissertation

This dissertation combines a survey of reduplicative patterns across all 23 Salish languages with more in-depth examination and analysis of three specific languages through case studies.

In the remainder of this chapter, I provide background information about the Salish language family (Section 1.3), including language names, major branches, and dialects. I also provide an overview of my methodology (Section 1.4), including discussion of sources, transcription, and glossing. The chapter concludes by reviewing the available documentation for each Salish language (Section 1.5), highlighting existing descriptions and analyses that contain either examples or description of reduplication.

Chapters 2 and 3 establish the foundation for the dissertation. In Chapter 2, I provide a set of phonological criteria used to identify reduplicative morphemes and define key terminology used to classify and describe reduplication in Salish (Section 2.2). I define reduplication as *fission*, which is a 1:2(+) mapping between input and output segments. The rest of Chapter 2 describes all the reduplication patterns found in Salish (Sections 2.3–2.9), which establishes the range of empirical variation that any phonological analysis must capture.

Key assumptions about phonological theory needed for the analysis of Salish reduplication are introduced in Chapter 3, including those pertaining to prosodic structure (Section 3.2), reduplicative affixes (Sections 3.3–3.4), and the architecture of the grammar (Section 3.5). In this chapter, I explain how underlying representations of non-concatenative morphemes are “defective” in ways that require some kind of phonological repair (see, e.g., Zimmermann, 2017). I also propose a new version of stratum-based phonological derivation where the strata (levels) are divorced from morphosyntactic domains.

Chapters 4 and 5 constitute the empirical and analytical core of the dissertation; they further develop and test predictions of the theoretical approach to reduplication proposed in Chapter 3. Chapter 4 presents detailed case studies of Comox-Sliammon (Section 4.1) and Lillooet (Section 4.2) that consider the full inventory of reduplicative morphemes in each language. An analysis is proposed that can account for all productive or phonologically regular patterns and is compatible with the overall phonological grammar needed for each language; its empirical predictions are tested on the two languages.

Chapter 5 considers multiple reduplication patterns, where reduplicative morphemes co-occur in the same word. A theory of multiple reduplication, restricting the number of reduplicative morphemes that may occur at a given phonological level, is first proposed following a case study of Halkomelem (Section 5.3), based on secondary sources. Predictions of the one-per-level restriction on multiple reduplication are then tested through case studies of Lillooet (Section 5.4) and Comox-Sliammon (Section 5.5), which include negative data; negative data is crucial because my analysis in this chapter predicts that combinations of two prosodic affixes at the same stratum are impossible, which cannot be conclusively determined based on positive data alone.

Chapter 6 concludes the dissertation by returning to the cross-Salish variation laid out in Chapter 2; this chapter asks how well the model of reduplication built over Chapters 3–5 captures and predicts variation across the Salish language family. The main theoretical claims and predictions are evaluated and discussed.

Supplementary material to the dissertation can be found in the appendices. Appendix A provides examples of reduplication for all 23 Salish languages, while Appendix B provides examples of multiple reduplication. Appendix C is a list of phonological constraints with definitions. Appendices A and B can be read as companions to Chapters 2 and 6. Appendix C contains reference material to accompany Chapters 3–6. Appendix D is a finders' guide to existing descriptions and analyses of core patterns, organised by language; it is a supplementary bibliography of Salish reduplication that readers can consult for further information about a particular type of reduplication or language. Appendix E provides a conversion between phonetic alphabets. Appendix F discusses alternate analyses of V_1 (vowel) reduplication.

1.3 Salish Languages

This section provides an overview of the Salish language family. Endonyms and exonyms are discussed in Section 1.3.1. The geographical distribution and linguistic grouping of languages are introduced in Section 1.3.2. Current language status is discussed in Section 1.3.3.

1.3.1 Endonyms vs. Exonyms

Given that the scope of this dissertation covers the whole Salish language family and that the choice of names used for a single language may vary (e.g., Sliammon, Comox, Comox-Sliammon, and ʔayʔaʃuθəm are all used to describe the same language), I first establish a standardized set of labels to allow for comparison across different languages and dialects.

Potential labels include exonyms, which are traditionally used in diachronic and comparative treatments of the language family, and endonyms, which are used by the speech communities. I have chosen to adopt the exonyms for practical reasons. First, endonyms often come from specific communities, which may not represent the language as a whole. For example, the endonym used by speakers of the Mainland dialect of Comox-Sliammon is ʔayʔaʃuθəm, (literally: “speaking well”). However, [θ] in the Mainland dialect corresponds to [s] in the Island dialect, which means the endonym is ʔayʔaʃusəm in the Island dialect; neither endonym ʔayʔaʃuθəm can reflect pronunciation in both varieties.⁴ Halkomelem is another language where endonyms are insufficient for cross-language comparisons: there is no single endonym for the language because each dialect has a different pronunciation and orthography (i.e., Hul’q’umi’num’ for the Island dialect, həŋqəminəŋm for the Downriver dialect, and halqemeylem for the Upriver dialect). I choose therefore to use exonyms as they are best suited for cross-family comparisons between languages, rather than specific dialects.

Table 1.1 shows how exonyms used in this dissertation correspond to endonyms. Dialects are only listed where different endonyms are used.

⁴There are no L1 speakers of the Island dialect. However, there are language learners in the K’ómoks community, which traditionally spoke this variety.

Table 1.1: Salish Exonyms and Endonyms

Exonym	Endonym
Bella Coola	Nuxalk
Tillamook	hutyéyu
Central Salish	
Comox-Sliammon	
Mainland Comox	ʔayʔajuθəm
Island Comox	ʔayʔajusəm
Halkomelem	
Island Halkomelem	Hul'q'umi'num'
Downriver Halkomelem	hənqəmínəm
Upriver Halkomelem	halqemeylem
Klallam	nəx ^w sʔəyəmúcən
Lushootseed	dəx ^w ləšucid, x ^w lələšucid
Nooksack	Lhechalosem, Lhechelesem
Northern Straits	
Sooke	T'sou-ke
Saanich	SENĆOŦEN
Songish	Lekwungen
Samish	Xws7ámeshqen
Lummi	Xwlemi' Chosen
Pentlatch	pentl'ach
Sechelt	she shashishalhem
Squamish	Skwxwú7mesh sníchim
Twana	tuwaduq
Tsamosan	
Cowlitz	sʔpúlmiq
Lower Chehalis	ʔəwálməš
Quinault	K ^w ínaył
Upper Chehalis	ʔ ^w áyáytł
Interior Salish	
Coeur d'Alene	snchitsu'umshtsn
Lillooet	St'át'imcets
Montana Salish-Spokane-Kalispel	
Montana Salish (aka Flathead)	Séliš
Kalispel	Qlispé
Spokane	Npoqínišcn
Moses Columbia	nxaʔamxcín
Okanagan	nsyilxcn
Shuswap	Secwepemctsin
Thompson River Salish	nłeʔkepmxcín

1.3.2 Geographical Distribution and Language Classification

There are 23 Salish languages; they are spoken in the Pacific Northwest of North America. The geographical distribution of the family includes coastal British Columbia, the southern interior of British Columbia (up to the southern Cariboo region), Washington, Idaho, Montana, and Oregon. For general surveys of the family see Czaykowska-Higgins and Kinkade (1998), H. Davis (2019), and Watanabe (2024). A map is given in Figure 1.2.^{5,6}

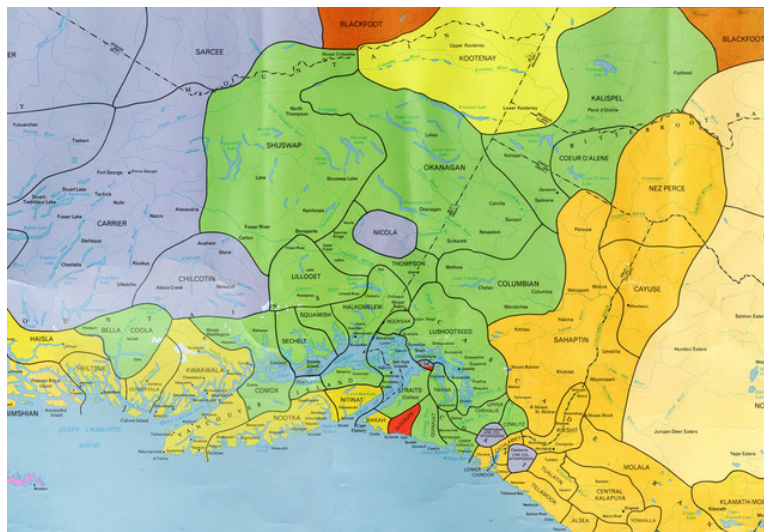


Figure 1.2: Salish Languages (Green) (Suttles, n.d.)

Salish languages are spoken in two main cultural areas (Czaykowska-Higgins & Kinkade, 1998, p.1): a Plateau culture area (e.g., Walker, 1998), which includes the Interior Salish languages, and a Northwest Coast culture area (e.g., Suttles, 1990), which includes the Central Salish and Tsamosan languages, as well as Bella Coola and Tillamook.

Salish languages have been grouped and classified in different ways. The language groupings shown in Figure 1.3 follow H. Davis (2019), who divides Salish languages into three major groups: Interior Salish, Coast Salish, and

⁵The orientation of this map reflects Salish conceptions of geography, which reference downriver/towards the coast in opposition to upriver/away from the coast, rather than north and south.

⁶Tillamook is not correctly marked: it should be coloured green because it is a Salish language.

Bella Coola (consisting of a single language). Interior Salish further divides into Northern and Southern branches; Coast Salish divides into three: Tsamosan, Central Salish, and Tillamook (consisting of a single language); and Tsamosan divides into two branches: Maritime and Inland.

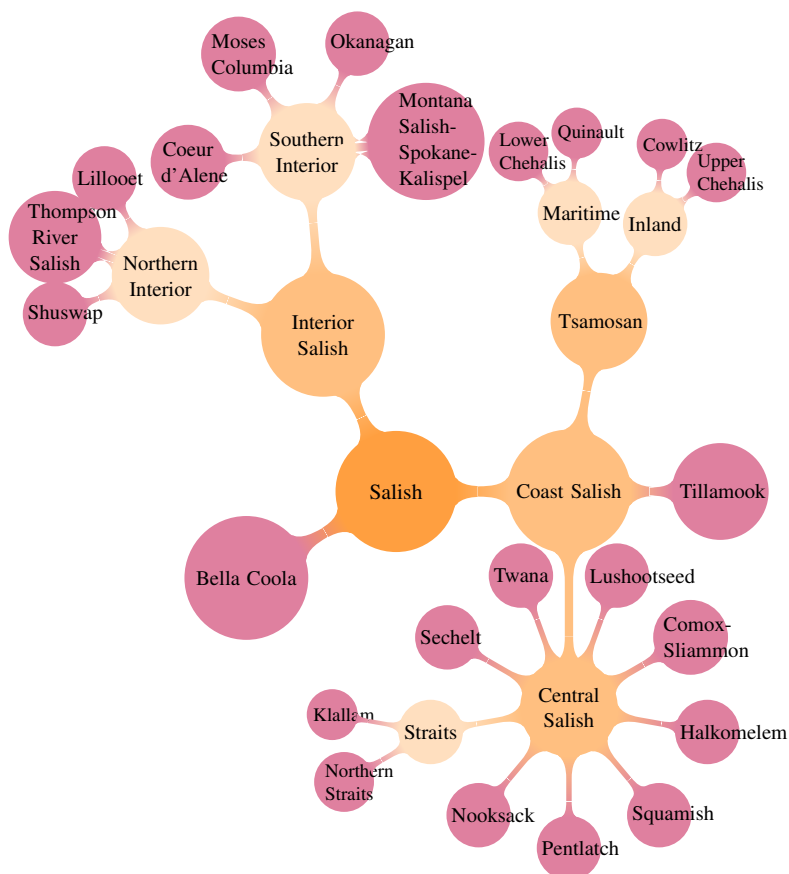


Figure 1.3: Salish Branches (Orange) and Languages (Pink)

The divisions shown in Figure 1.3 differ slightly from other classifications. Both Czaykowska-Higgins and Kinkade (1998) and Kroeber (1999) divide Salish into five co-equal groups: Interior Salish, Central Salish, Tsamosan, Tillamook, and Bella Coola. In contrast, Kuipers (2002) divides Salish into two: Interior Salish and Coast Salish (which includes Bella Coola). While there is debate with respect to the existence and composition of an umbrella Coast Salish grouping,

all classifications converge on Interior Salish (7 languages), Central Salish (10 languages), and Tsamosan (4 languages) as established branches. Tillamook has more in common with Central Salish than with Interior Salish (Czaykowska-Higgins & Kinkade, 1998, p.4); Bella Coola shares traits with both Central and Interior Salish (Kroeber, 1999, p.6).

I adopt a five-way division between Interior Salish, Central Salish, Tsamosan, Bella Coola, and Tillamook for my survey of Salish reduplication. These divisions represent the largest groupings that are informative for synchronic and typological work; this is not a complete diachronic classification.

1.3.3 Language Status

Languages, linguistic groupings, and estimated age of remaining first language speakers as of 2018 are given in Table 1.2, which is adapted from H. Davis (2019, pp.452–453).⁷

⁷Numbers reported here include revised estimates from Henry Davis (p.c.).

Table 1.2: Salish Languages

Branch	Language	Estimated L1 Speaker Age in 2018 (Dialect)
	Bella Coola	80+
Central Salish	Comox-Sliammon	65+ (Mainland)
	Pentlatch	N/A
	Sechelt	90+
	Squamish	90+
	Halkomelem	75+ (Island), 90+ (Upriver)
	Nooksack	N/A
	Twana	N/A
	Lushootseed	90+? (Northern)
	Northern Straits	80+ (Saanich), 80+? (Samish)
	Klallam	N/A
Interior Salish	Moses Columbia	90+
	Coeur D'Alene	90+
	Shuswap	70+ (Western), 80+? (Eastern)
	Montana Salish-	90+ (Spokane), 75+ (Kalispel),
	Spokane-Kalispel	80+ (Montana Salish/Flathead)
	Okanagan	60+ (Northern/Lakes),
	Thompson River Salish	70+ (Southern/Colville)
Tsamosan	Lillooet	70+ (Lytton/Canyon),
		70+ (Nicola Valley)
		70+ (Upper/Fountain/Fraser River),
		65+ (Lower/Mount Currie)
	Upper Chehalis	N/A
	Cowlitz	N/A
	Lower Chehalis	N/A
	Quinault	N/A
	Tillamook	N/A

All Salish languages are at a critical point with respect to language status. Many languages do not currently have speakers who learned the language from birth; of those who did, most were over 65 years of age in 2018. The current status of Salish languages in Canada is a direct result of the residential school system, which disrupted the transmission of language and culture (see, e.g., Truth &

Reconciliation Commission of Canada, 2015; First Peoples' Cultural Council, 2018). However, despite the challenges associated with a small, aging population of first language speakers, language revitalisation and documentation activities are ongoing across different language communities. These activities include creating dictionaries and grammars, university degree programs, immersion programs, and pedagogical materials in the languages.

1.4 Methodology

Data in this dissertation come from fieldwork, archival materials, and published sources. I discuss fieldwork elicitation and other methodological aspects in Section 1.4.1. Limitations and considerations relevant to using published and archival materials are covered in Section 1.4.2. Issues pertaining to productivity and phonological regularity are addressed in Section 1.4.3. The section concludes with a discussion of transcription and orthographic considerations in Section 1.4.4.

1.4.1 Fieldwork

Given that there are still L1 speakers of about half a dozen Salish languages, the investigation of Salish reduplication is not limited to previous documentation: empirical predictions may be tested explicitly in elicitation where it is possible to work with L1 speakers.⁸ Elicitation not only allows for systematic documentation of reduplicative patterns, but also crucially includes negative evidence. Fieldwork data may thus be used to confirm that gaps are systematic (ungrammatical), rather than accidental. It may also provide information about productivity, discussed further in Section 1.4.3, as well as insight into whether attested restrictions on the distribution of reduplicative morphemes are phonological, as opposed to lexical, syntactic, or semantic.

Fieldwork in this dissertation consists of elicitation with language consultants from Comox-Sliammon, Lillooet, and Shuswap. Elicited data from Comox-Sliammon comes from fieldwork I have conducted from 2016 onwards.⁹ I

⁸H. Davis (2019) states that fieldwork with L1 speakers is practically possible for 6/23 languages.

⁹This work is in collaboration with the ʔayʔaʃuθəm Lab, which includes Marianne Huijsmans, Daniel Reisinger, Henry Davis, Bailey Trotter, and Laura Griffin

also provide some Shuswap examples from my fieldwork conducted from 2020 onwards.¹⁰ Fieldwork for Lillooet was conducted by Henry Davis.

Elicitation took place over many different sessions and combined a range of common fieldwork methods, including translations from the target language and from English of individual words or sentences, describing pictures or storyboards, elicitation production tasks with specific contexts, forced choice/matching, and acceptability judgments. Key negative data comes from elicitation where speakers were given potential reduplicated forms and asked if they were acceptable. If the form was acceptable, the speaker was asked to provide a (volunteered) translation to confirm its acceptability and meaning.

The type of elicitation varies between different languages and with different speakers. Elicitation for Comox-Sliammon was conducted exclusively in person until 2020, with a mix of in person and online sessions (conducted via Zoom) from 2020 onward. Elicitation for Shuswap was mostly online, with one in-person session. Elicitation that Henry Davis conducted for Lillooet was predominantly in person, with a handful of online sessions.

I provide speaker initials alongside elicited data; initials also distinguish fieldwork data from published or archival materials, which are indicated with a citation to a source and page number. Examples and grammaticality judgments given throughout this dissertation come from the following consultants:

- ʔayʔajuθəm (Comox-Sliammon)
 - BW: Betty Wilson
 - EP: Elsie Paul
 - FL: Freddie Louie (†)
 - JF: Joanne Francis (†)
 - KG: Karen Galligos (†)
 - MH: Marion Harry (†)
 - PD: Phyllis Dominic

¹⁰This work is in collaboration with the Secwepemetsín Working Group, which includes Sander Nederveen, Bruce Oliver, Julia Schillo, Henry Davis, and Michelle Kamigaki-Baron.

- MV: Margaret Vivier
- MW: Maggie Wilson (†)
- WJF: Jerry Francis
- St’át’imcets (Lillooet)
 - CA: Carl Alexander
- Secwepemctsín (Shuswap):
 - GD: Garlene Dodson

1.4.2 Published and Archival Sources

Aside from original fieldwork, this dissertation uses extensive examples and description from both published and archival materials. The published sources are grammars, dictionaries, published texts, and papers (or dissertations and theses). Archival materials include field notes, unpublished papers, as well as texts or word-lists.¹¹

Archival materials offer documentation for languages with little published material. With archival materials, source interpretability is often contingent on the reader’s familiarity with a language, the Salish language family, and the source data itself (e.g., whether a fieldworker was consistent in their transcription and description of various patterns). One of the goals of this dissertation is to provide a synthesis of data for all 23 Salish languages that is presented in a manner that is accessible and consistent, such that it can be used by theorists, typologists, and those generally interested in learning about reduplication in Salish.

1.4.3 Methodological Considerations for Reduplication

There are a number of methodological issues that must be considered when dealing with reduplication. In particular, there are a number of different judgments a speaker might make when faced with a reduplicated word they have never

¹¹Materials were accessed from the University of Washington archives, BC Archives, the UBC Library, and the UBC Department of Linguistics. Other materials are freely available online in archives with digital collections, such as the California Language Archive.

heard before. Reduplication and fieldwork are discussed in Section 1.4.3.1. Considerations that arise while conducting fieldwork are also relevant when deciding which examples of reduplication to include and exclude from published and archival sources. I define phonologically regular and active patterns in Section 1.4.3.2; issues related to productivity are discussed in Section 1.4.3.3.

1.4.3.1 Eliciting Reduplication

Eliciting reduplication in Salish languages poses a methodological challenge; this is summarised by H. Davis and Mellesmoen (2023, p.17):

“Eliciting reduplicated forms is tricky, because judgments are never straightforward unless the form is in common circulation in the language. For less common / unattested forms, at least the following levels of acceptability must be distinguished: (i) the speaker knows and uses the form readily with the expected meaning; (ii) the speaker has heard the form and can ascribe it the expected meaning, but doesn’t use it; (iii) the speaker has never heard the form but can still identify its expected meaning; and (iv) the speaker has never heard the form and doesn’t know what it would mean.”

For some lexical items, there is a general preference for using a non-reduplicative strategy instead of reduplication. For example, a periphrastic phrase with an independent word meaning “small” or “a little bit” is often offered before reduplication when speakers are prompted to produce a diminutive form.¹² Examples from Shuswap are given in (1). The consultant first volunteered the sentence in (1a) with *kʷoyʔsə* ‘small/little bit’, and then accepted the suggested form with diminutive reduplication in (1b); she confirmed that (1b) could be used to mean the same thing and in the same context as the original sentence.

¹²A preference for periphrastic diminutives alone is not reflective of language loss: periphrastic diminutives are also found in languages with diminutive affixes, such as Greek (Vassilaki, 2017).

- (1) a. $\gamma\partial=s-\Gamma^w\partial s\sim\Gamma^w\acute{e}s-t$ $m<\text{?}>\acute{i}\check{x}^w-\partial n-s$
 DET=NMLZ-PDEN~sunny-CHAR melt<INCH>-CTR-3ERG
 $\gamma\partial=s-w\acute{u}x^wt$ $k^w\text{oy}\acute{i}\text{?s}\partial\acute{\lambda}ux^w$
 DET=NMLZ-snow small EXCL
 ‘The sun was able to melt the snow a little bit.’ (GD, vf, Shuswap)
- b. $\gamma\partial=s-\Gamma^w\partial s\sim\Gamma^w\acute{e}s-t$ $m<\text{?}>\acute{i}\check{x}^w-\partial n\acute{w}\acute{e}<w>\partial\acute{n}-s$
 DET=NMLZ-PDEN~sunny-CHAR melt<INCH>-NTR<DIM>-3ERG
 $\gamma\partial=s-w\acute{u}x^wt$
 DET=NMLZ-snow
 ‘The sun was able to melt the snow a little bit.’ (GD, sf, Shuswap)

A combination of volunteered and suggested forms are used throughout the dissertation. I mark volunteered forms as vf and suggested forms as sf, as in the examples above. Any suggested form that I include must (i) be accepted by the speaker as a possible form, and (ii) must be either accepted within a specific context, accepted with an explicit suggested translation, or given a translation by the speaker (marked as vt).

Given that not all types of reduplication are volunteered equally often, speaker judgments provide essential negative data; without these judgments, it would not be possible to know if gaps reflect forms that are not possible for phonological reasons, or if they instead reflect facts related to speaker preference or particular lexical items.

1.4.3.2 Phonologically Regular and Active Patterns

As my goal is to present a phonological analysis of Salish reduplication, my focus is on patterns where reduplication is both phonologically active and regular, even where the pattern may not be fully productive. I use the term *phonologically regular* to describe patterns that can be identified with a consistent form across multiple lexical items (i.e., can be grouped with reduplication found elsewhere in the language); the term *phonologically active* is used when alternations between reduplicated and non-reduplicated forms of the same lexical item are attested.

Though the unreduplicated form of a word is typically the base for a phonologically active process, there are cases where the simplest attested form of the word is a reduplicated one; however, even in these cases, sometimes

reduplication can be “undone” under particular circumstances, which means it is phonologically active. For example, in Shuswap, reduplication is used when counting people or animals (e.g., Anderson, 1999). The word *səséle* ‘two’ has lexical reduplication, as shown in (2a); while it may undergo further reduplication when counting animals, as shown in (2b), this lexicalised reduplication is removed when counting people, as shown in (2c).

- (2) a. *sə~séle*
RED~two
‘two’
b. *sə~sé<s>le*
RED~two<RED>
‘two animals’
c. *tk-séle*
LOC-two
‘two persons’

(Kuipers, 1974a, pp.55–56, Shuswap)

I also consider reduplication to be phonologically active in words without a non-reduplicated counterpart if there is evidence that the grammar treats a seemingly lexicalised type of reduplication like any other productive type. Examples of phonologically active reduplication with no unreduplicated counterpart include any pattern where active (reduplication-specific) allomorphy is found, as well as any situation where seemingly lexicalised reduplication shows reduplication-specific phonological behaviour (e.g., blocking the addition of a productive type of reduplication due to limitations on multiple reduplication).

An example of the first type is provided in (3): despite the fact that the root *tul* never occurs without diminutive C_1 reduplication in Comox-Sliammon (i.e., *titul* ‘small’), the reduplication is phonologically active, because it displays the exact same pattern of phonologically conditioned allomorphy as other regularly reduplicated diminutives in the language, namely a prefixed C_i -reduplicant with monosyllabic bases and an infixed -C- reduplicant with polysyllabic ones (see Section 4.1.3.1.1 for further description and analysis).

- (3) a. **ti**~tul
 DIM~small
 ‘small’
 b. tu<**t
 small<DIM>-head
 ‘small head’ (KG, vf, Comox-Sliammon)**

I use the term “lexicalised reduplication” to refer to patterns that are not phonologically active nor regular; the term “lexically-specified” (or “lexical reduplication”) is used to describe patterns where a type of reduplication is part of the lexical entry for a given word or root, but is both phonologically active and regular. The lexical item is stored with some amount of prosodic structure that triggers reduplication as an active and regular phonological process.

1.4.3.3 Productivity and Reduplication

Phonologically regular and active reduplication encompasses patterns that vary greatly with respect to productivity. Productivity refers to whether a given type of reduplication can be freely added to new words, provided the reduplicative morpheme and the rest of the word are phonologically and semantically compatible. It is important to note that the total number of examples across the whole lexicon is not necessarily a useful indicator of productivity. For example, Kroeber (1988, pp.142–143) comments on the “low frequency in both discourse and the dictionary” of a reduplicative pattern marking a change-of-state in Comox-Sliammon, noting that it was deemed “marginal and no longer productive” by Harris (1981) and Hagège (1981). However, Kroeber (1988, p.143) further observes that this type of reduplication only occurs with adjectives; my fieldwork confirms that this type of reduplication is frequently offered and accepted with adjectival roots, which suggests that it is productive within its limited domain. In general, therefore, productivity must be estimated relative to the particular syntactic or semantic domains where a reduplicative morpheme applies: these may be quite restricted.

Estimations of productivity based on attested forms are also limited by the scope and genre of the documentation. For example, most of the reduplicated

forms in Nooksack given in Galloway (1984b, 1993b) come from the work of Paul Fetzer, who recorded “several thousand examples of plurals, about 100 diminutives, and about 100 diminutive plurals” (Galloway, 1993b, p.104). Fetzer notes that the consultant, George Swanaset, showed some reluctance when asked about diminutives and would only provide reduplicated diminutives when speaking about non-living items (unless in the context of a story). His preferred strategy for forming diminutives involved periphrastic expressions (using the word for ‘small’), or using a non-reduplicative affix (meaning ‘baby’). This example is relevant for the challenges for eliciting reduplication introduced in Section 1.4.3.1. Frequency of reduplication in documentation cannot be taken as evidence for productivity. If speakers prefer periphrastic expressions to reduplication, productive reduplication processes may be underrepresented. However, targeted elicitation may lead one to overestimate the frequency of the reduplicative pattern.

Given that I rely on existing documentation for languages other than Comox-Sliammon, Lillooet, and Shuswap, it is not possible to comment on the productivity of every type of reduplication in Salish. However, where possible, I cite discussions of productivity for specific patterns from the literature and include examples with loanwords where possible; if a loanword is attested with reduplication, this means that the pattern was phonologically active, if not productive, at some point between when the word was borrowed and when the form was recorded.

1.4.4 Transcription and Glossing

Some researchers on Salish languages use transcription systems that are opaque to those unfamiliar with the fieldworker and their work. I have adapted examples given in these lesser known transcription systems, as well as examples given in practical orthographies, to the Northern Americanist Phonetic Alphabet (NAPA), as typically employed in the Salish literature. For example, J. Davis (1970b, p.29) uses [č̣] for what would be transcribed as [t̪ʰ] in the International Phonetic Alphabet (IPA), while Galloway (1990, p.1) uses [θ’]; I use [t̪ʰ] for this sound, following the NAPA conventions developed in the Salish comparative literature.

1.4.4.1 Segments

The NAPA representations I adopt do not always match the surface realisation of a sound. I follow Kroeber (1999) in using “s” and “c” for the alveolo-palatal sounds [š] and [č] in five of the Interior Salish languages (Lillooet, Thompson River Salish, Shuswap, Moses Columbia, and Okanagan); while his motivation for this is “orthographic simplicity”, given a lack of contrast with alveolar [c] and [s] (Kroeber, 1999, p.xxix), it also allows for clearer cross-Salish comparisons and identification of cognates (H. Davis, p.c.). For example, “s” ([š]) in Interior Salish languages corresponds to “s” ([s]) in Central Salish, which both come from Proto-Salish *s, whereas [š] in Central Salish corresponds to [x] in most of Interior Salish (and to *x in Proto-Salish).

Furthermore, I abstract away from surface variation and phonetic conditioning of vowels for languages where I am familiar with the transcription system or where the original source provides underlying or intermediate forms. The representations I give for Comox-Sliammon and Lillooet, where I provide examples from fieldwork, are intermediate phonological representations that are prior to phonetic implementation. However, given that there is considerable variability in transcription systems with respect to vowel quality and archival sources may use different conventions, I leave vowels as-is for languages I am less familiar with, such as Quinault, unless the vowel can be identified as an epenthetic non-moraic vowel ([ə]). See Chapter 3 (Section 3.2.2) for the non-moraic status of [ə] in Salish; identifying non-moraic vowels is relevant for the phonological analysis of Salish reduplication given in Chapters 4–6.

Instances of epenthetic [ə] may not be transcribed in published or archival sources. Both optional and obligatory schwas may be missing, so when vowels are omitted in the source materials, it is sometimes unclear when [ə] should be posited. For example, Doak (2023) gives *tčén̩m* ‘take hold of’ in Coeur d’Alene: it is unclear whether the /n̩m/ cluster should have a vowel (schwa) or a sonorant consonant as its nucleus (i.e., [tčén̩əm] or [tčén̩m]). In cases of uncertainty, I do not include vowels that are absent in the source material. However, I include regularly epenthetic [ə] in my transcriptions wherever possible, and the same goes for [ə] due to reduction of a full (moraic) vowel; these are highly relevant

for reduplication, because the surface distribution of [ə] is sensitive to prosodic structure and sonority.¹³ Therefore, if I can be confident in the position of [ə] based on the phonology of a language or transcription in a corroborating source, I add [ə].

1.4.4.2 Suprasegmental Elements

I include both glottalisation and primary stress where present in the original sources, except I omit stress on monosyllabic words and where it falls in a fixed position (i.e., in Comox-Sliammon). Secondary stress is rarely marked in source materials; its status is unclear and so I omit it.

For fieldwork data, glottalisation is marked when it was recorded in elicitation. It should be noted, however, that there is speaker variation and glottalisation is falling out of use in Comox-Sliammon and is absent in places it would be expected based on older documentation (see discussion of Comox-Sliammon glottalisation in Watanabe, 1994a; Caldecott & Mellesmoen, 2018). Glottalisation is more robust in Interior Salish languages (e.g., glottalisation is frequent in the data from Lillooet elicited by Henry Davis).

1.4.4.3 Modifications to Glossing and Translations

Where I can be confident, I make minor adaptations to glosses and morpheme breaks for consistency and clarity. For example, I replace the label “progressive” in Downriver Halkomelem examples from Suttles (2004) with “imperfective” to reflect its semantic contribution. For places where only a partial translation is given, I only make changes where the correspondence between reduplication and meaning is straightforward. For example, I leave (dim.) in *titést* ‘approaching it (dim.)’, as given in Suttles (2004, p.173), because there are a number of different interpretations possible for a diminutive verb (e.g., small event, low intensity, small participant), but I give *qəqiləm* ‘eyes’ in lieu of ‘eye (pl.)’, as given in Suttles (2004,

¹³The inclusion of [ə] is a question of debate in the descriptive literature. For example, Kuipers (1974a) and Nater (1984) do not assume underlying schwas in their descriptions of Shuswap and Bella Coola, respectively, and only represent epenthetic or derived schwas in limited circumstances. In contrast, L. C. Thompson and Thompson (1992) include schwas in proposed underlying forms for Thompson River Salish. My assumptions about underlying vowels are given in Chapter 3 (Section 3.2).

p.216), because this translation can be ascertained from this gloss.

Another place where I have modified glosses from original sources is in the translation of third person pronouns, which are gender neutral in Salish; it is therefore more accurate to translate example words and sentences using the gender neutral pronoun ‘they’. Where this might contribute to ambiguity between singular and plural in discussion of pluractionality, the clarification “sg.” is included. It should be noted that many of the examples I provide from my fieldwork do include translations with ‘he’ or ‘she’. I keep most of these with the original translation, as they were generally elicited with a specific referent who the consultant knows or elicited with a storyboard about specific characters.

1.4.4.4 Unreduplicated Forms for Comparison

Wherever possible, when I present examples of reduplication, I present an unreduplicated word based on the same root for comparison; these comparison forms are introduced with “cf.”. Where possible, the unreduplicated forms given for comparison come from the same source as the example. If the unreduplicated form was not provided in the original and a suitable form could not be located elsewhere in the source, a comparison form was located in the primary dictionary or grammar for the language. Where no form could be located, only a presumed root is given. No comparison form is provided for unreduplicated words, including those with lexicalised reduplication that is not phonologically active.

1.5 Sources and Documentation of Salish Reduplication

General sources and work specifically on reduplication for each Salish language are discussed in this section. For each language, I identify the primary documentation that exists for the language as a whole, as well as subsequent work that provides an analysis or reinterpretation of reduplication on the basis of either primary or secondary data. I cover Central Salish in Section 1.5.1, Tsamosan in Section 1.5.2, Tillamook in Section 1.5.3, Interior Salish in Section 1.5.4, and Bella Coola in Section 1.5.5.

This is not a comprehensive overview of all existing materials, particularly texts and pedagogical materials. Grammatical descriptions and dissertations

for each language include more comprehensive overviews of existing materials. Some general bibliographies for or including Salish languages include Pilling (1893), Butler (1941), Wolff (1947), Adler (1961), Lorient (1964), Martin (1975), A. Mattina (1989), and in particular Van Eijk (2008), which incorporates previous bibliographies.¹⁴ Many papers including examples of reduplication in Salish can be found in the archive of the International Conference on Salish and Neighbouring Languages (ICSNL).¹⁵

Papers which draw on multiple sources to discuss reduplication in a cross-Salish context include Haeberlin (1918), Reichard (1959), B. Carlson and Thompson (1982), Van Eijk (1990b), Bessell (1993b), Demers and Jelinek (1996), Jelinek and Demers (1997), Anderson (1997), Haag (1998), Van Eijk (1998a), Van Eijk (1998b), Urbanczyk (2004b), Urbanczyk (2005), Yu (2007), Urbanczyk (2019), Mellesmoen and Urbanczyk (2021c), and Mellesmoen (2024a).

1.5.1 Central Salish

There are ten Central Salish languages, as shown in Figure 1.5. It should be noted that Figure 1.5 omits the follow major groupings for reasons of space: North Georgia languages (Comox-Sliammon, Pentlatch, and Sechelt), South Georgia languages (Nooksack, Halkomelem, Squamish, Northern Straits, and Klallam), and Puget languages (Lushootseed and Twana).

¹⁴A list of early linguists studying Salish languages, which can be used for seeking further materials, can be found in Kinkade (1971).

¹⁵The ICSNL archive can be found here: <https://lingpapers.sites.olt.ubc.ca/>.

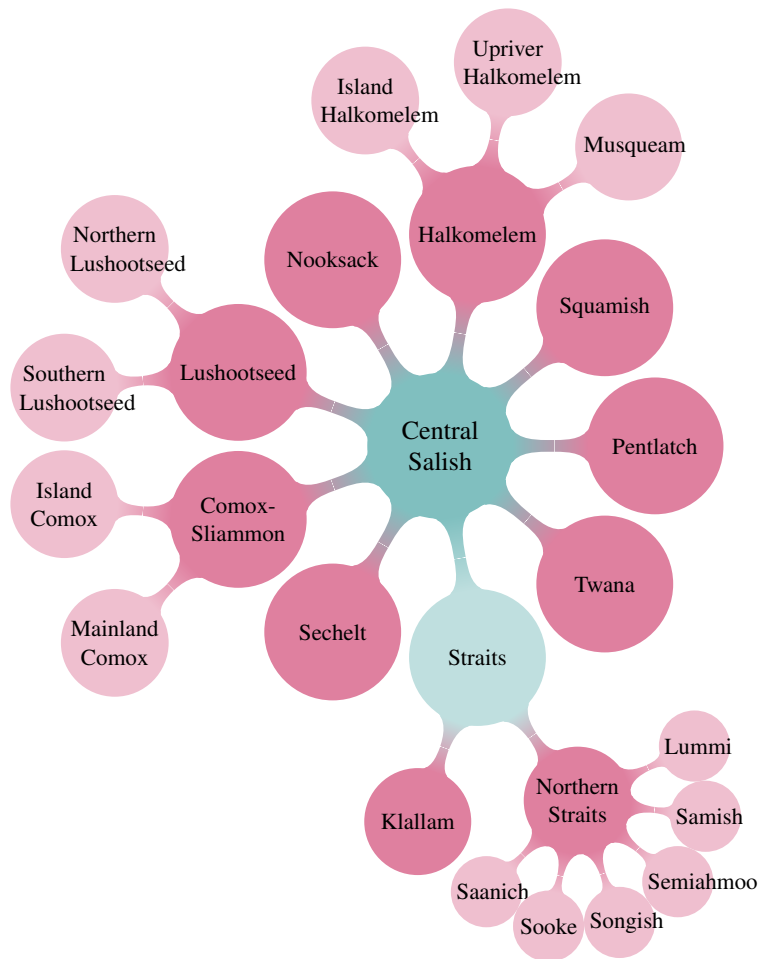


Figure 1.4: Central Salish Languages (Dark Pink), and Dialects (Light Pink)

1.5.1.1 Lushootseed

Lushootseed is the language with the most thorough documentation and description of reduplication in Salish. Papers containing primary data include Hess (1966), Van Eijk and Hess (1986), Bates (1986), and Bates (1990). The main resources for Lushootseed are the grammatical descriptions in Hess (1967), Hess (1995), Hess (1998), and Hess (2006), as well as a dictionary (Bates, Hess, & Hilbert, 1994).

Earlier documentation with examples of reduplication include a Lushootseed dictionary (Gibbs, 1877), which has been reproduced with analysis and

commentary in Zahir (2009). Other primary documentation includes sketches of the phonology and morphology based on field notes and/or recordings in Ransom (1945), Tweddell (1950), and Snyder (1968).

The most extensive phonological treatment of reduplication in any Central Salish language is Urbanczyk (2001), which uses Lushootseed data from Bates et al. (1994). Other works that draw data from Hess (1967), Bates (1986), or Bates et al. (1994) for data include Broselow (1983), Kirkham (1991), Kirkham (1992), Kim (1992), Urbanczyk (1996a), Urbanczyk (1996b), Park (1999), Alderete et al. (1999), Urbanczyk (1999), Urbanczyk (2000), Struijke (2000), Lonsdale (2001), Fitzpatrick and Nevins (2003), Fitzpatrick (2006), Urbanczyk (2006), Mellesmoen and Urbanczyk (2021b), and Mellesmoen and Urbanczyk (2024).

Many other phonological analyses of Lushootseed draw upon sources which are themselves based on secondary data. For example, Park (2000) cites Broselow (1983) and Urbanczyk (2001) as sources. Samuels (2010) uses data from Fitzpatrick (2006). Benua (1997), Frampton (2009), Zimmermann (2021a), and Zimmermann (2021b) draw on various papers written by Urbanczyk, which are mentioned above. Data in S. Davis (1988) comes via McCarthy and Broselow (1983). Fidelholtz (1978) also covers plural and diminutive reduplication in Lushootseed, though the source of data is unclear.

The Metcalf recordings (part of the Burke Museum's Special Collections at the University of Washington) are another primary data source (see, e.g., Hilbert, 1974); Kye (2023) presents a grammatical sketch of Lushootseed based on these recordings. Van Eijk (2008) also notes that unpublished materials of Haeberlin's are present in the University of Washington Libraries.

1.5.1.2 Twana

Drachman (1969) is a comprehensive dissertation on Twana phonology based on fieldwork with Louise Pulsifer, said to be the last fluent L1 speaker of the language, including description and analysis of reduplication; Skokomish Indian Tribe (2020) is a dictionary of Twana compiled from the recordings and documentation made by Drachman. Kim (2017a) and Kim (2017b) are phonological analyses of reduplication that are based on secondary data from

Drachman (1969).

N. Thompson (1979) is another dictionary of the language that draws on archival sources. N. Thompson (1985) is a dissertation on Twana baby talk, which includes discussion of reduplication.

1.5.1.3 Nooksack

Galloway (1984b, 1993b) explores reduplication in Nooksack, including numerous examples; his description of Nooksack is drawn from fieldwork undertaken by Pamela Amoss, Barbara Efrat, Laurence Thompson, and Paul Fetzner. A brief sketch of the phonology of the language can be found in Amoss (1961).

1.5.1.4 Straits (Northern Straits and Klallam)

Given similarities in the reduplicative systems of the two Straits languages (Northern Straits and Klallam), I group them together under the label of Straits throughout most of this dissertation. The choice to group them together is motivated by the fact that there is no clean distinction between the reduplication systems of Northern Straits and Klallam; in fact, reduplication patterns distinguish the Saanich dialect of Northern Straits from the other Northern Straits dialects and Klallam (see discussion in Section A.3.5.2 of the appendix). I note whether each Straits example comes from Klallam or from a dialect of Northern Straits. I highlight places where the specific language or dialect is relevant to the discussion.

The description of the Samish dialect of Northern Straits in Galloway (1986, 1990) relies mostly on data from George Gibbs (e.g., Gibbs, 1863), but also includes some primary fieldwork with Lena Daniels.

Reduplication in the Sooke dialect of Northern Straits is included in description of the dialect based on primary fieldwork data (Efrat, 1969).

Patterns of reduplication in the Songish dialect of Northern Straits are included as part of a description of the phonology and morphology of the dialect in Raffo (1972), as well as in a paper on aspect (Raffo, 1970). M. Mitchell (1975) is an MA thesis based on primary data that also includes a brief discussion of reduplication in Songish.

Reduplication in the Lummi dialect of Northern Straits is mentioned in Demers

(1974), Demers and Jelinek (1984), Demers and Jelinek (1996), and Jelinek and Demers (1997). Demers and Jelinek conducted fieldwork with L1 speakers of the language.

There is a grammar and dictionary for Klallam (Montler, 2012, 2015) based on primary fieldwork. L. C. Thompson and Thompson (1968, 1971) also conducted fieldwork on Klallam, and further made use of recordings made by Leon Metcalf. Reduplicated diminutives and plurals in Klallam were also recorded by Leo Frachtenberg, his fieldnotes are available through the American Philosophical Society. Additional discussion of Klallam including primary data can be found in Montler (1989).

There is a grammar and dictionary for the Saanich dialect of Northern Straits (Montler, 1986, 2018), which draws on primary fieldwork. Additional discussion of Saanich reduplication including primary data can be found in Montler (1988) and Montler (1989). Aspectual reduplication in the Saanich variety of Northern Straits is discussed in Turner (2005), Turner (2006), Turner (2007), Kiyota (2008), Leonard and Turner (2010), and Turner (2011); all of these are based on primary fieldwork. Reduplication in Saanich is also mentioned in Leonard (2009) and Leonard (2019), which uses a combination of data from Montler and original fieldwork.

Primary data from Montler (e.g., Montler, 1986) is used to inform analyses of Saanich reduplication in Stonham (1994), Kurisu (2001), S. Davis and Ueda (2006), Stonham (2007), and Kirchner (2010). Svenonius and Bye (2012) base their analysis of Saanich reduplication on secondary data in S. Davis and Ueda (2006). Zimmermann and Trommer (2013) and Mellesmoen (2023) each draw on multiple secondary sources. Primary data from Montler (2012) is used to inform an analysis of Klallam reduplication in Crawford (2015).

1.5.1.5 Halkomelem

There are three major dialects of Halkomelem: Downriver Halkomelem, Upriver Halkomelem, and Island Halkomelem. Examples are identified by dialect and dialect differences are discussed where relevant.

Suttles (2004) is the primary source for Downriver Halkomelem (Musqueam). Shaw (2008) draws on Suttles (2004), as well as additional materials with an unclear source. The discussion of Halkomelem in Inkelas (2014) cites Shaw (2008) as a data source. Suttles (2004) is the data source for Shaw (2004) and H. Davis (2024). The data source for Shaw (2002) is not clear.

Galloway (1993a, 2009) are the primary sources for Upriver Halkomelem.¹⁶ Additional discussion of reduplication can be found in Galloway (1973), Galloway (1984a), and Galloway (1991). J. Thompson (2005a, 2009) discusses plural reduplication in Upriver Halkomelem using data from primary fieldwork. Kurisu (2001) and Wiltschko (2005, 2008) use data from Galloway (1980, 1993a). The source of data in Wiltschko and Steriopolo (2007) is not clear. Urbanczyk (1998), Monahan (2002), Brown (2004), Zimmermann (2009), Zimmermann (2013), and Zimmermann and Trommer (2013) all use data from Galloway (1993a). The source of Upriver Halkomelem data for Marinakis (2004), which mentions reduplication, is unclear.

The primary data source for Island Halkomelem is the dictionary (Hukari & Peter, 1995). Reduplication is discussed across various papers drawing on primary data, including Hukari (1976, 1977, 1978, 1981a, 1981b, 1984). Diminutive reduplication is also briefly mentioned in Gerdtz and Peter (2021). Other primary documentation for Island Halkomelem can be found in Leslie (1979) and Jones (1976).

Bianco (1996) uses Hukari and Peter (1995) as a corpus, but also conducted fieldwork to check stress placement; while reduplication is not a focus, the attention to vowels, prosodic structure, and sonority in her dissertation is highly relevant. Hukari and Peter (1995) is also the source for a number of papers discussing reduplication including Urbanczyk (2004a), Urbanczyk (2010), Urbanczyk (2011), Mellesmoen and Urbanczyk (2020), Mellesmoen and Urbanczyk (2021b), Mellesmoen and Urbanczyk (2021a), and Mellesmoen (2024b).

¹⁶I cite Galloway (1977) at various points throughout the dissertation, which is an earlier version of Galloway (1993a).

1.5.1.6 Squamish

Kuipers (1967) is a grammar of Squamish based on primary fieldwork; it is used as a source of data for Demers and Horn (1978), S. Davis (1984), Dyck (2004), and Lǐ and Ponsford (2018). There is also a dictionary available for Squamish (Squamish Nation Education Department, 2011).

Work explicitly discussing reduplication with data from primary fieldwork includes Bar-el and Watt (1998), Bar-el (2000), Bar-el and Watt (2001), Bar-el (2006), Bar-el (2008), and Jacobs (2012). Downing (2008) cites Bar-el (2000) as a data source.

1.5.1.7 Comox-Sliammon

Early documentation of Comox-Sliammon includes Sapir (1915), which focuses on noun reduplication.¹⁷ Wilbur (1973) uses data from Sapir (1915).

Harris (1981) and Hagège (1981) are grammatical sketches of the Island dialect based on primary fieldwork; both include discussion of reduplicative patterns.

Fieldwork with speakers of the Mainland dialect informed the work of J. Davis (1970b), J. Davis (1970a), J. Davis (1971), and J. Davis (2019), which all discuss reduplication.¹⁸ Kroeber (1988) also includes fieldwork data on change-of-state reduplication in Comox-Sliammon.

The main documentation for the Mainland dialect of the language includes a grammar (Watanabe, 2003) and a dissertation on the phonology of the language (Blake, 2000). Reduplication is also discussed in Blake (1992), Watanabe (1994a), Watanabe (1994b), Watanabe (1997), and Watanabe (2010). Czaykowska-Higgins (2006) draws on Urbanczyk (2005) for secondary data, which in turn comes from Watanabe (1994a).

Further discussion of reduplication patterns based on primary data collection can be found in Mellesmoen (2017), Mellesmoen (2018a), Huijsmans, Mellesmoen, and Urbanczyk (2018), Mellesmoen and Huijsmans (2019c), Mellesmoen, Davis, and Matthewson (2020), Huijsmans and Mellesmoen (2021), Huijsmans (2023b), Reisinger (2024), Huijsmans and McCarthy (2024), and

¹⁷See Reisinger and Griffin (2022) for further discussion of early documentation and materials for Comox-Sliammon.

¹⁸John Davis has archived his recordings and field notes through the California Language Archive.

Mellesmoen (in press). Mellesmoen (2021a) and Mellesmoen (2018b) use a combination of primary data and secondary data from multiple sources; Mellesmoen (2020) uses Watanabe (2003) as a data source.

The description and documentation of Comox-Sliammon used in this dissertation is largely informed by original fieldwork on the Mainland dialect (as detailed in Section 1.4.1).

1.5.1.8 Sechelt

The discussion of Sechelt reduplication in this dissertation is based on the grammatical description in Beaumont (1985) and data drawn from the single major dictionary of the language (Beaumont, 2011). The dictionary contains very extensive exemplification of reduplication, which has facilitated identification of previously undescribed reduplication patterns. Some examples and discussion of reduplication can also be found in Hill-Tout (1904).

1.5.1.9 Pentlatch

There is very little documentation of Pentlatch.¹⁹ Data used for this dissertation originally come from Boas (1886); I also use Kinkade's retranscriptions and notes on the Boas material, which are available through the archives at the University of Washington. This documentation is also used as a secondary source by Pulhug (2023), which discusses reduplication.

1.5.1.10 Summary: Central Salish

Aside from Pentlatch, Twana, and Nooksack, reduplication is relatively well-described for at least one dialect of each Central Salish language. Conclusions based on Central Salish languages are therefore based on a relatively large corpus of data.

¹⁹For discussion of the available sources and language context, see Andreatta, Recalma, Qualicum First Nation, and Urbanczyk (2023).

1.5.2 Tsamosan

There are four Tsamosan languages, as shown in Figure 1.5. The Tsamosan languages can be divided into Maritime and Inland languages. None have been used as first languages since the 1960s.

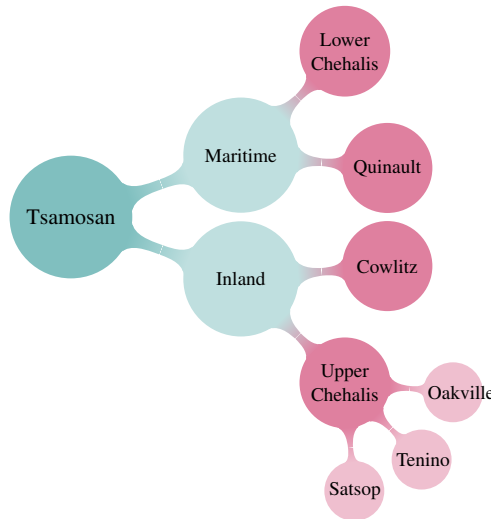


Figure 1.5: Tsamosan (Dark Pink) and Dialects (Light Pink)

Reduplication across Tsamosan languages is briefly discussed in Kinkade (1993); data comes from a mix of sources.

1.5.2.1 Upper Chehalis

Kinkade (1995, p.347) notes that Franz Boas amassed a large body of texts and field notes between 1927 and 1935, while his own work commenced in 1960. There are a grammatical description and a dictionary of Upper Chehalis (Kinkade, 1963, 1991). Reduplication is included in a sketch of the phonology and morphology of the language (Kinkade, 1963), as well as in Kinkade (1975, 1985, 1995).

Mellesmoen (2022) is an analysis considering reduplication in Upper Chehalis using data from Kinkade (1991).

1.5.2.2 Cowlitz

There is very little documentation of Cowlitz. I refer exclusively to Kinkade (2004) in my survey, which is informed by fieldwork conducted by M. Dale Kinkade in 1967; Kinkade (2004) also draws on earlier documentation by John Harrington, who was interested in the phonetics and lexicon than the grammar (Kinkade, 1993, p.176).²⁰

1.5.2.3 Lower Chehalis

Documentation of Lower Chehalis is sparse and therefore very little is known about processes of reduplication in the language. Robertson (2014) notes that the earliest documentation consists of brief word-lists (Curtis, 1907–1930; Hale, 1846; Scouler, 1841; Swan, 1857).

Reduplication is briefly discussed in Snow (1969), which includes data from primary fieldwork. Robertson (2014) is an overview of Lower Chehalis morphosyntax based on Snow (1969) and notes on the morphology from M. Dale Kinkade, as well as unpublished materials collected by Franz Boas, Myron Eells, John Harrington, Leon Metcalf and members of the Lower Chehalis speech community.

1.5.2.4 Quinault

There is very little existing documentation for Quinault. The only explicit discussion of reduplication in Quinault is found in the work of Ruth Modrow; she conducted fieldwork and produced a full dictionary including a brief description of the grammar and phonology (Modrow, 1971), as well as some teaching materials (e.g., Modrow, 1969); however, there are some issues with transcription in these sources. Kinkade (1993, p.176) states that the best documentation of Quinault is in the form of unpublished field notes by J. Gibson (1964), which include examples of reduplication.

²⁰See Kinkade (2004, pp.xii–xiv) for an overview of existing Cowlitz documentation.

1.5.2.5 Summary: Tsamosan

The amount and quality of available documentation for the Tsamosan languages is quite sparse; there are also very few explicit descriptions of reduplicative patterns compared to other branches of the family. Upper Chehalis has the most extensive documentation.

1.5.3 Tillamook

Tillamook is geographically separated from the rest of Salish by Chinookan and Athabaskan languages, though it bears similarity to Central Salish languages (Kinkade, 1990, p.10).

Tillamook is one of the languages where documentation of reduplication is limited; this is because Tillamook ceased being spoken as a first language earlier than most other Salish languages. An overview of the Tillamook language, including reduplication, can be found in Edel (1939), which draws on field notes from primary fieldwork conducted by Boas (1890). A combined vocabulary including work by Boas, Edel, and Melville can be accessed through the American Philosophical Society (Edel, 1935).

Egesdal and Thompson (1996, 1998) also draw on the unpublished field notes of Laurence C. Thompson and M. Terry Thompson in a discussion of Tillamook inflectional morphology, which includes brief discussion of reduplication. Reduplication is also discussed in L. C. Thompson and Thompson (1985). Kim and Gardiner (2016) draw on Edel (1939).

1.5.4 Interior Salish

There are seven Interior languages, as shown in Figure 1.6. The Interior Salish languages split into two sub-branches: Northern Interior Salish and Southern Interior Salish languages. The Northern Interior branch includes Shuswap, Lillooet, and Thompson River Salish. The Southern Interior branch includes Okanagan, Moses Columbia, Montana Salish-Kalispel-Spokane, and Coeur d'Alene.

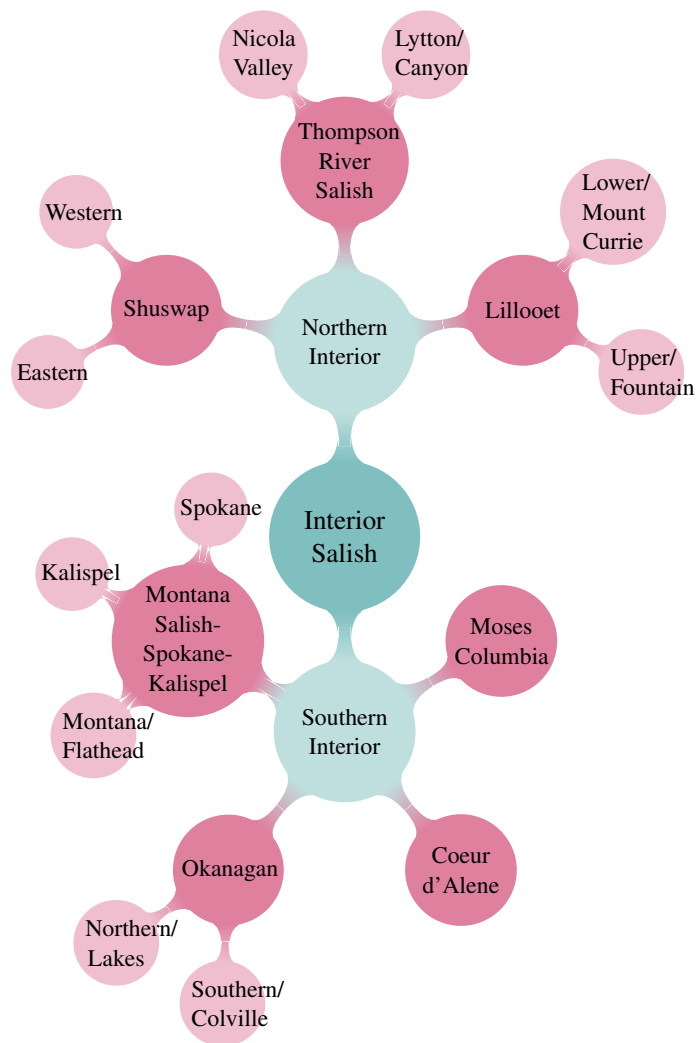


Figure 1.6: Interior Salish Languages (Dark Pink) and Dialects (Light Pink)

1.5.4.1 Lillooet

Primary documentation of Lillooet includes a dictionary and grammar (Van Eijk, 1997, 2013), which contain many examples of reduplication. I have also drawn examples from an unpublished grammar and dictionary by H. Davis, which are both based on original fieldwork. Papers using primary data discussing at least one reduplicative morpheme include Van Eijk (1981a), Van Eijk and Hess (1986),

Van Eijk (1993), Matthewson (2000), H. Davis (2011), Van Eijk (2011), H. Davis (2014), and H. Davis and Mellesmoen (2023).

Nelson (1998), Bird (2000), and Shaw (2005) use secondary data from Van Eijk's work. Downing and Inkela (2015) also use data from Van Eijk (1997). Discussion of reduplication in Caldecott (2009) uses data and description from Van Eijk (1997) and H. Davis (in prep).

Examples from and generalisations about Lillooet in this dissertation are largely based on fieldwork conducted by Henry Davis (as detailed in Section 1.4.1).

1.5.4.2 Thompson River Salish

Primary documentation of Thompson River Salish includes a grammar (L. C. Thompson & Thompson, 1992), a dictionary (L. C. Thompson & Thompson, 1996), and a brief grammatical sketch which includes discussion of reduplication (L. C. Thompson, Thompson, & Egesdal, 1996). Different types of reduplication in Thompson River Salish are discussed in L. C. Thompson and Thompson (1981), B. Carlson and Thompson (1982), and L. C. Thompson (1985).

The most comprehensive examination and analysis of Thompson River Salish reduplication is Jimmie (1994), which uses data from L. C. Thompson and Thompson (1992, 1996); Jimmie (1994) also includes some data from original fieldwork. Discussion of glottalisation associated with Thompson River Salish reduplication can be found in Luntzlara (in prep), which is also informed by primary data.

Papers using secondary data include Coelho (1999) and Shaw (2005), who use examples from the work of Laurence C. Thompson and M. Terry Thompson, and Broselow (1983), who uses Haeberlin (1918). Data in Zimmermann (2021b) also comes from Haeberlin (1918) by way of Broselow (1983). Kroeber (1988) also discusses reduplication in Thompson River Salish, drawing on data from B. Carlson and Thompson (1982).

1.5.4.3 Shuswap

Shuswap data used in this dissertation largely comes from Kuipers (1974a, 1989), which is a grammatical description based on primary fieldwork; reduplication is also discussed in Kuipers (1974b).

Discussions of Shuswap reduplication using Kuipers (1974a, 1989) as a source include Broselow (1983), Van Eijk (1990a), Taylor (1996), Hendricks (1998, 1999), Nakamura (2000), Bjorkman (2006), and Kim and Gardiner (2016). Bell (1983) mostly draws on examples from Kuipers (1974a), but some were checked through fieldwork. McCarthy and Broselow (1983) cite Bell (1983) as a data source.

In addition to Kuipers (1974a), there is another grammatical description in J. Gibson (1973), which is based on primary fieldwork. Reduplication is briefly summarised in M. Ignace and Ignace (2017).

1.5.4.4 Montana Salish-Spokane-Kalispel

There is no single term for the language consisting of the three major dialects Montana Salish (Flathead), Spokane, and Kalispel. All examples from Montana Salish-Spokane-Kalispel are annotated for specific dialect throughout the dissertation and appendices.

Vogt (1940) is a grammar of Kalispel, based on primary fieldwork. Camp (2007) provides an interlinear analysis of textual material from Vogt (1940). In his edition of a Kalispel grammar (Post, 1904), Speck (1977) includes discussion of reduplication. Idris (1981) uses data from Haeberlin (1918), which originally came from Giorda (1879).

The main source of primary data for Spokane is B. Carlson (1972a), a grammar, and B. Carlson and Flett (1989), a dictionary. Papers including discussion of at least one reduplicative morpheme include B. Carlson (1972b), B. Carlson (1980), B. Carlson and Thompson (1982), B. Carlson (1989), Bates and Carlson (1989), Bates (1990), B. Carlson and Bates (1990), B. Carlson (1996), and Bates and Carlson (1998). Kroeber (1988), Yu (2007), and Vaux and Wolfe (2009) discuss reduplication in Spokane, using secondary data from B. Carlson and Thompson (1982).

Primary fieldwork and a survey of the literature inform Black (1996), which summarises processes of reduplication in Spokane. Inkelas (2008) uses secondary data from Black (1996).

There is a dictionary of Montana Salish (Pete, 2010). Thomason (1997, 1999) contain discussion of reduplication that includes primary data from Montana Salish; this is compared to data from other varieties, including Spokane, which is drawn from B. Carlson and Flett (1989).

1.5.4.5 Moses Columbia

Reduplication data from primary fieldwork on Moses Columbia is discussed in Kinkade (1982) and Kinkade (1989). Some non-reduplicated forms given for comparison come from Kinkade (1981), which is a word-list.

Reduplication is also discussed in Willett and Czaykowska-Higgins (1995), Czaykowska-Higgins (1993a), Czaykowska-Higgins and Willett (1997), Czaykowska-Higgins (1998), and Willett (2003); they acknowledge Kinkade's fieldnotes as an important source for their work, though they also carried out primary fieldwork. Czaykowska-Higgins (2006) uses examples of reduplication from Willett's fieldwork. Czaykowska-Higgins (1985) and Czaykowska-Higgins (1993b) use data sourced from Kinkade's fieldnotes. Bessell (1993a) provides a few reduplicated examples that come from Bessell and Czaykowska-Higgins (1991). Raimy (2009) uses data from Czaykowska-Higgins and Willett (1997). Urbanczyk (2005) uses data from Czaykowska-Higgins (1993b).

1.5.4.6 Coeur d'Alene

Early documentation of Coeur d'Alene includes a grammar (Reichard, 1938) and a cross-Salish comparison of five languages (Reichard, 1959). Barthmaier (1996) is a dictionary of Coeur d'Alene based on file slips from Reichard (1938).

There is also a dictionary and a teaching course written by Lawrence Nicodemus (Nicodemus, 1975), who was a speaker and language consultant to others who worked on the language, including Reichard. Lyon and Greene-Wood (2007) and Lyon (2010) are compilations of lexical information transcribed from Nicodemus.

Patterns of reduplication are discussed in a grammatical overview (Doak, 1997) and a paper on reduplication (Doak, 2004), based on both original fieldwork and data from Reichard (1938) and Nicodemus (1975). Fieldwork also informs a discussion of Coeur d'Alene aspect, including discussion of relevant processes of reduplication, in Vincent (2014).

1.5.4.7 Okanagan

Primary documentation for Okanagan includes a dictionary (A. Mattina, 1987), a grammatical description (A. Mattina, 1973), and papers discussing reduplication (A. Mattina, 1979, 1993; A. Mattina & Peterson, 1997). Primary data on reduplication is also included in Watkins (1970) and Pattison (1978). Watkins (1974) also discusses reduplication as part of a paper considering documentation from Franz Boas.

1.5.4.8 Summary: Interior Salish

Reduplication in Northern Interior Salish languages is described more thoroughly than for most other Salish languages; reduplication in Southern Interior languages is not as well described, though the Spokane dialect of Montana Salish-Spokane-Kalispel is an exception.

1.5.5 Bella Coola

Czaykowska-Higgins and Kinkade (1998, p.4) state that Bella Coola is the “most divergent of the Salish languages, showing influence of its Wakashan neighbours, as well as a number of innovations specific to it”. As a result of these differences, Bella Coola cannot be easily grouped with any of the other Salish languages; Kinkade (1990, p.10) considers it equally distant from the others.

Reduplication is discussed in Newman (1937, 1947, 1971), which are based on original fieldwork. Ingram (1973) and Brown (2004) use data from Newman (1971). Primary fieldwork on Bella Coola forms the empirical basis for a grammar (Nater, 1984) and dictionary (Nater, 1990). Papers focused on reduplication which present data from primary fieldwork include Saunders and Davis (1972) and Nater (1978).

Bagemihl (1991) is an influential paper on syllables and reduplication in Bella Coola based on a combination of primary fieldwork and secondary sources. Work using Bagemihl (1991) as a source includes Cook (1994), M. Gibson (1995), K. Carlson (1997), Raimy and Idsardi (1997), Nelson (2003), Marlo (2004), Shaw (2005), Topintzi (2005), Topintzi (2006), Frampton (2009), Vaux and Wolfe (2009), and Kirchner (2010). Angermeyer (2003) uses data from Newman (1971), Nater (1984), and Bagemihl (1991). The source of Bella Coola data is not clear in Nelson (2003), but likely can be traced back to Bagemihl (1991). Mellesmoen (2021b) uses secondary data sourced from Nater (1990).

Chapter 2

Survey of Reduplication in Salish

Reduplication is found in every Salish language, though the number of reduplicative morphemes in each language varies. The generalisations presented in this chapter, and throughout the dissertation, are based on a survey of existing documentation for all twenty-three Salish languages. I describe my approach to compiling the data from these sources in Section 2.1. Terminology, definitions, and glossing conventions related to reduplication and reduplicative morphemes are provided in Section 2.2.

Identification of a particular morpheme as reduplicative in this survey is based on a set of phonological criteria, which I establish in Section 2.2. I present my survey of reduplication grouped by major pattern (Sections 2.3–2.8) with illustrative examples; a more comprehensive set of data for each language is provided in Appendices A and B. The chapter concludes with a discussion of generalisations emerging from the cross-Salish survey (Section 2.9).

2.1 Examples and Sources

Sources for the survey include available grammatical descriptions for each language, papers specifically about at least one type of Salish reduplication, and

papers about reduplication which include data from Salish languages.¹ I use dictionaries, archival materials, and other papers to supplement the survey where needed, particularly where they provide different descriptions or examples of a specific type of reduplication, or where reduplication has not previously been discussed in detail.

Illustrative examples of reduplication for each language can be found in Appendix A; examples of multiple reduplication are provided in Appendix B. A summary of types of reduplication present in sources consulted can be found in Appendix D. See Section 1.4 for more about methods, including transcription and standardisation across sources. Discussion of which sources are available for each language and the type of data (primary or secondary) can be found in Section 1.5.

2.2 Terminology and Glossing Conventions

In the literature on Salish languages, a single morphophonological pattern may be labelled and described in different ways across different sources, even for a single language. Variation in nomenclature may hinder cross-linguistic comparisons. Furthermore, labels are not innocent: they often contain implicit analyses. It is particularly important to choose labels carefully when covering reduplicative patterns across an entire language family, as in this dissertation.

In this section, I establish and define the terminology and glossing conventions I use throughout this dissertation. In Section 2.2.1, I explain the glossing conventions used to mark reduplication, before defining reduplicative morphemes and core types of reduplication in Sections 2.2.2 and 2.2.3 respectively. I introduce form-based and meaning-based classification labels in Section 2.2.4 and additional terminology in Section 2.2.5.

¹As my classification is primarily form-based, I focus on papers which address the morphology or phonology of reduplication. Further, using “reduplication” as a main search term means papers with a primarily semantic focus are under-represented in the survey; for example, papers with “aspect” in the title without mention of reduplication were not included in the initial list, even where reduplication plays an important aspectual role. Some of these papers were added to the list later.

2.2.1 Morpheme Breaks and Glossing

Reduplication is marked throughout this dissertation using either ~ or <>. This notation is derived from a combination of Rule 9 and Rule 10 of the *Leipzig Glossing Rules*, which pertain to reduplication and infixes, respectively (Comrie, Haspelmath, & Bickel, 2008, pp.7–8).

I use ~ where the content associated with reduplicative morphemes can be linearly separated from adjacent morphemes (i.e., where reduplication can be segmented as a prefix or suffix); <> is used where this is not possible (i.e., reduplication surfaces as an infix). These glossing conventions are illustrated with the Sechelt examples in (4).^{2,3} Aspectual (imperfective) reduplication in (4a) can be delineated from the root (RED~root); in contrast, the segments associated with the diminutive morpheme in (4b) appear in the middle of the root (root<RED>).

- (4) a. ʔə~ʔɪtən (cf. ʔɪtən ‘eat’)
RED~eat
‘eating’ (Beaumont, 1985, p.18, Sechelt)
- b. ká<kə>n-át-cut (cf. kánátcut ‘lower one’s voice’)
lower.voice<RED>-CTR-REFL
‘lower voice a bit, speak a bit more softly’ (Beaumont, 2011, p.272, Sechelt)

While the examples in (4) correspond to two different types of reduplication, the same type of reduplication may be marked with ~ in some words and with <> in others. The pair of examples from Sechelt in (5) shows the same type of reduplication (change-of-state), which is positioned after the second consonant of a root in both cases. The segments associated with the reduplicative morpheme follow the root when the root has the shape CVC (CVC~VC); the segments occur in the middle of the root when the root has the shape CVCC (CVC<VC>C).

²Labels for reduplication are introduced and motivated in Section 2.2.4. Until then, I use RED as a provisional label for any morpheme realised as reduplication; labels in the gloss line will be replaced with meaning-based labels wherever possible from Section 2.2.4.2 onward.

³Beaumont (2011, pp.1–2) marks emphasis on multiple vowels with the diacritic ˈ; it is not clear which of these correspond to primary (or secondary) stress. I choose to leave these diacritics in Sechelt examples (see Section A.3.7).

- (5) a. $k^w\acute{o}m\sim\acute{o}m$ (cf. $k^w\acute{o}m\acute{o}lqín$ ‘have red hair’)
 red~RED
 ‘getting red’ (Beaumont, 2011, p.607, Sechelt)
- b. $qíl<\acute{i}l>t$ (cf. $qílt$ ‘bright’)
 light<RED>
 ‘daybreak, dawn, getting light’ (Beaumont, 2011, p.645, Sechelt)

Readers familiar with Salish languages are cautioned that the glossing may sometimes look unfamiliar when informed by a particular analysis. For example, the glossing of the Comox-Sliammon examples in (6) is informed by the analysis in Chapter 4 (Section 4.1.3.1.1): segments associated with the diminutive morpheme may occur to the left of the root (RED~root) or within it (root<RED>), as shown in (6a) and (6b), respectively. This diverges from previous analyses of diminutive reduplication in Comox-Sliammon, which treat it as either exclusively prefixing (Blake, 2000; Watanabe, 2003) or infixing (Mellesmoen, 2017).

- (6) a. $\acute{t}i\sim\acute{t}aqt$ (cf. $\acute{t}aqt$ ‘mountain’)
 RED~mountain
 ‘small mountain’ (JF, Comox-Sliammon)
- b. $\acute{t}i<\acute{?}>m\acute{o}n$ (cf. $\acute{t}im\acute{o}n$ ‘road, door’)
 road<RED>
 ‘narrow walkway (small trail)’ (JF, Comox-Sliammon)

The glossing I use corresponds to the output of the phonology (i.e., the surface form), rather than the input. Most reduplicative morphemes in Salish are prefixes in the input, even when they surface as infixes. The choice between ~ and <> should not be taken as a complete phonological analysis. Where there is ambiguity as to which segments are associated with a particular reduplicative morpheme, I chose between ~ and <> depending on which is best supported by the language and/or used in previous literature. For example, it is not clear whether the segments associated with the reduplicative morpheme in the Lushootseed word in (7) are in the first or second syllable. I use ~ for this type of reduplication in Lushootseed, which is consistent with previous analyses of the language (e.g., Urbanczyk, 2001).

- (7) **síd**~síd[˙]-cut *or* síd<**síd**>[˙]-cut (cf. *sídqcut* ‘move over’)
RED~spin-REFL spin<RED>-REFL
‘spin self around’ (Bates et al., 1994, p.203, Lushootseed)

The position of morpheme breaks and which segments to gloss as the reduplicative morpheme is informed by:

- assumptions necessary to derive the correct surface forms in the phonological analyses given in the case studies in this dissertation (Comox-Sliammon, Lillooet, Halkomelem)
- absence of stress and/or presence of [ə], rather than an underlying vowel (Coeur d’Alene, Moses Columbia, Okanagan, Montana Salish-Spokane-Kalispel, Shuswap, Thompson River Salish, Northern Straits and Klallam, Sechelt, Cowlitz, Quinault, Upper Chehalis, Bella Coola)
- comparison with closely related languages and/or consideration of previous descriptions (Lushootseed, Nooksack, Pentlatch, Squamish, Twana, Lower Chehalis, and Tillamook)

2.2.2 What Counts as Reduplication?

Descriptively, reduplication refers to the repetition or doubling of segments. Formally, I follow Pulleyblank (2008) and Kirchner (2010), among others, and analyse reduplication as fission, rather than copying: an input segment is “split” into two output segments. Reduplication thus yields a 1:2 mapping between an input and an output. The phonological framework adopted in this dissertation, discussed further in Chapter 3 (Section 3.4), assumes that reduplication results from the affixation of a morpheme which lacks segmental content, following the work of Svenonius and Bye (2012) and Zimmermann and Trommer (2013), among others.

Within this framework, reduplication occurs when the phonological grammar (i.e., constraint ranking) favours fission over other potential strategies to provide segmental content to a prosodic unit which must be associated with segments in the output. The main prosodic units relevant for reduplication are the mora (μ)

and syllable (σ), though the nucleus (N) and foot (Ft) are also important (see the prosodic hierarchy introduced in Section 3.2). Morphemes with segmental content may also trigger reduplication to repair a marked structure (e.g., to provide an onset for a syllable).

2.2.2.1 Reduplicative Morphemes

The definition of reduplication adopted here includes patterns that would be labelled as “phonological doubling” according to the Dual Theory of reduplication proposed by Inkelas (2008): I will not consider morphological doubling, which is also referred to as syntactic reduplication in some sources (e.g., Kirchner, 2010).⁴ Phonological doubling for Inkelas (2008, p.367) is when “a consonant or vowel is required to be present in the output but is not present, or if present is not featurally specified” in the input, or “if a particular morphological construction imposes a template requiring a consonant, vowel, or mora in a position where the input has none to provide”.⁵

Since reduplication is a phonological process under the framework I assume, not an inherent property of particular morphemes, I use the term “reduplicative morpheme” to refer to morphemes with at least one allomorph expressed with reduplication. This approach to reduplication is well-suited to a discussion of Salish languages, where the same morpheme may not always be realised with reduplication. For example, reduplication is only one possible way to realise the imperfective morpheme in Halkomelem (see Section 5.3.1.1.1 in Chapter 5 for a discussion of the full range of allomorphs). The set of examples in (8) shows that the same morpheme is associated with consonant reduplication (8a) and ablaut (8b). The Halkomelem imperfective morpheme is considered reduplicative because at least one allomorph is realised as reduplication.

- (8) a. $\text{c}\text{ə}\sim\text{c}\acute{\text{e}}\text{ʔ-t}$ (cf. ceʔt ‘turn it over’)
 RED~turn.over-TR
 ‘turning it over’ (Hukari, 1984, p.147, Island Halkomelem)

⁴An example of possible morphological doubling would be the repetition of a root in $k\acute{\text{a}}x^w k\acute{\text{a}}x^w k\acute{\text{a}}x^w$ ‘knock, knock, knock’ in Moses Columbia (Van Eijk, 1998a, p.465); this pattern is discussed and set aside in Section 2.2.3.2.2.

⁵This definition includes compensatory reduplication, as described by Yu (2005).

- b. $\dot{c} < e > k^w \dot{x} - t$ (cf. $\dot{c} \dot{\partial} k^w \dot{x} t$ ‘fry it’)
 fry <RED>-TR
 ‘frying it’ (Hukari, 1984, p.148, Island Halkomelem)

2.2.2.2 Phonological Criteria for Reduplicative Morphemes

The criteria I use to identify reduplicative morphemes are the following:

1. Reduplication occurs as a direct result of the addition of a morpheme, where fission provides segmental content necessary for a well-formed output.
2. At least one allomorph of the morpheme is expressed with reduplication.

These criteria provide a phonological definition of reduplication and reduplicative morphemes that I apply throughout this chapter. Lexical entries for reduplicative morphemes comprise two types: (i) those with prosodic units which require reduplication to provide segmental content (e.g., an affixed syllable), and (ii) those with insufficient segmental content to meet the requirements for a well-formed output (e.g., they otherwise yield a syllable without an onset).

The examples provided in (9) show the repetitive morpheme in Montana Salish-Spokane-Kalispel, which qualifies as a reduplicative morpheme under my criteria. The repetitive morpheme may be realised as *-e-*, as in (9a), or *-eC-*, where C stands for a reduplicated consonant, as in (9b).⁶ This type of reduplication meets the criteria because a morpheme is added and at least one of its realisations involves fission. Fission occurs to provide an onset, which is necessary for a well-formed syllable (Bates & Carlson, 1998, p.106).

- (9) a. $\dot{l} < e > k^w - p - \text{min}$ (cf. $\dot{l} k^w \dot{p} i m$ ‘He speared it.’)
 pierce/spear <REP>-INCH-instrument
 ‘pitchfork’ (Bates & Carlson, 1998, p.105, Spokane)
- b. $\dot{c} e \sim \dot{c} i p - \dot{s}$ (cf. $\dot{c} i \dot{p} \dot{p}$ ‘It accidentally got pinched.’)
 REP~pinch-PL
 ‘scissors, shears’ (Bates & Carlson, 1998, p.104, Spokane)

⁶The repetitive surfaces as either *-a-* or *-aC-* in an environment that triggers vowel lowering (Bates & Carlson, 1998, p.104).

It is harder to classify vowel reduplication than consonant reduplication, as there are more marginal cases where vowels from different sources may appear to exhibit a 1:2 mapping between input and output. I do not consider vowel lengthening as reduplication unless there is a good argument that it results from a 1:2 mapping, such that one input vowel has two output correspondents at some point in the phonological derivation (even if the VV sequence is realised phonetically as a single long vowel). Vowel length that corresponds to a 1:1 mapping where an input vowel corresponds to a single output linked to two moras does not count as reduplication.⁷ In other words, there must be a phonological motivation for fission: adding a morpheme must create the circumstances to trigger a 1:2 input-output mapping. In the case of vowel reduplication, this must be either triggered by the necessity of a nucleus for prosodic well-formedness, or the presence of a mora or syllable in the input that may be filled by a vowel.

The examples in (10) show a type of vowel reduplication found in Montana Salish-Spokane-Kalispel and Okanagan that meet my criteria for inclusion. The inchoative marker *-ʔ-* is typically only considered a reduplicative morpheme in older sources, such as Vogt (1940) and Reichard (1959). However, there is evidence of a 1:2 mapping for at least some forms containing the inchoative marker. Whether fission of a vowel results from the presence of an underlying mora associated the inchoative morpheme, or alternatively represents a place where the grammar requires the addition of a vowel for prosodic reasons, either is consistent with my criteria for reduplication.

- (10) a. $q^w <uʔ> úʔ$ (cf. *hi q^w uʔ* ‘It is dusty.’)⁸
 dusty<INCH>
 ‘It gets dusty.’ (Vogt, 1940, p.64, Kalispel)

⁷Lengthening a segment may be used instead of reduplication to fill a prosodic unit in the phonology or can be a process that happens following reduplication (see Sections 6.6.2.2.1 and 6.7.1.2); either way, lengthening a segment does not count as fission.

⁸The Spokane form *q^w ʔuʔ* is given in B. Carlson and Flett (1989, p.176). It is possible that there is idiolectal or dialectal variation with respect to the presence of a doubled vowel with the inchoative. See Mellesmoen (2024a) for further discussion of V₁ reduplication across different Salish languages with multiple sources of data.

- b. kən ɬ<aʔ>át (cf. *cʰat* ‘be/get wet’)⁹
 1SG.SUBJ wet<INCH>
 ‘I got wet.’ (Watkins, 1970, p.107, Okanagan)

Further discussion of the inchoative forms in (10) and other possible analyses can be found in Appendix F.¹⁰

2.2.3 Defining Core Reduplication Patterns

In this section, I make a phonological distinction between “core” and “peripheral” reduplication patterns, which are defined in Section 2.2.3.1 and Section 2.2.3.2, respectively. I use this distinction to define the inclusion criteria for the cross-Salish survey in this chapter, as well as for the case studies in Chapter 4.

2.2.3.1 Core Reduplication Patterns

I define *core* reduplication patterns as those with shared phonological properties which are widely attested across the family. The relevant phonological properties include the number and identity of segments copied. Core patterns include those where two consonants are copied, those where one consonant is copied (divided into two based on whether the first or second consonant of a given phonological string is copied), and those where a vowel is copied without copying a consonant. The resulting core reduplication patterns and the labels I use (C_1C_2 , C_1 , C_2 , and V_1) are introduced and defined in Section 2.2.4.1.

Identifying core patterns makes it possible to conduct a typological survey of reduplication in Salish because they define types of reduplication based on form, which allows classification of reduplication types even where little is known about the meaning associated with reduplication or the phonological grammar generally (e.g., for Pentlatch or Quinault). Particular reduplication patterns across languages can then be compared within each category on the basis of position and meaning.

The core reduplication patterns are the smallest reduplicative units that can be associated with distinct meanings. They form the basic set of reduplication

⁹The underlying /ʃ/ in the root *ʃat* ‘wet’ (10b) is absent in forms with -ʔ-, based on the examples provided in A. Mattina (1987, p.88).

¹⁰Discussion of “link vowels” can also be found in Appendix F.

patterns that can then potentially be combined to yield multiple reduplication. Combinations of reduplicative morphemes (multiple reduplication) are discussed further in the case studies in Chapter 5 and across Salish in Chapter 6 (Section 6.6.2.3).

2.2.3.2 Peripheral Reduplication Patterns

There are a handful of reduplication patterns across the family that do not fit within the four core types. These patterns are either found with limited productivity in a single language ($C_1C_2C_3$ reduplication in Lower Chehalis, discussed in Section 2.2.3.2.1), or appear to involve multiple iterations of reduplicative morphemes ($C_1C_2^2$ reduplication in Interior Salish, discussed in Section 2.2.3.2.2). I call these patterns *peripheral* reduplication patterns.¹¹ I summarise peripheral patterns in this section before setting them aside.

2.2.3.2.1 Peripheral Pattern I: $C_1C_2C_3$ Reduplication

Lower Chehalis is the only Salish language that has a reduplication process where three consonants are copied. Robertson (2014, p.135) notes that “the meaning conveyed seems to be an aspect-like ‘intermittent action’”. Examples are given in (11).

- (11) a. čəpáʔx̣ ~ čəpəx̣-ŵ-n
lightning ~ PL-INTR-3.SUBJ.IPFV
‘It is lightening off and on.’ (Robertson, 2014, p.136, Lower Chehalis)
- b. ʔi-xəpəq̣ ~ xəpəq̣-ŵ-n
IPFV-strike ~ PL-INTR-3.SUBJ.IPFV
‘They (stones) always strike together.’ (Robertson, 2014, p.136, Lower Chehalis)

Robertson (2014, p.135) notes that this reduplication process is limited in use: it is restricted to CVCVC-shaped roots, which are less common than CVC and CVCC roots in Lower Chehalis and across Salish. He suggests that both the reduplication and the words where it has been found may have been borrowed from Chinuk

¹¹Repeated C_2 reduplication (“triplication” or C_2^2 reduplication) is discussed in Section 2.6.4 and as part of the case study of Lillooet reduplication in Chapter 4 (Section 4.2.5).

Wawa.¹² There is no evidence from other Salish languages to support positing a core reduplication process that copies three consonants.¹³

2.2.3.2.2 *Peripheral Pattern II: Repetition*

A second peripheral pattern is the apparent repetition of a core pattern. Examples from Lillooet and Moses Columbia with two repetitions are provided in (12). Van Eijk (1998a, p.465) observes that all syllables in these repetitions “tend to receive equal stress”.¹⁴ Examples from Montana Salish-Spokane-Kalispel (from the Montana Salish and Spokane dialects) and Okanagan with three repetitions are given in (13).

- (12) a. kóx^w kóx^w kóx^w
 ‘knock, knock, knock’ (Van Eijk, 1998a, p.465, Moses Columbia)
- b. ka-múlmúlmúl-a
 ‘repeatedly got dipped in water’ (Van Eijk, 1998a, p.466, Lillooet)
- (13) a. cá cá cá cá
 ‘the hollering noise one someone makes when getting beaten up’
 (Thomason, 1999, p.255, Montana Salish)
- b. líw líw líw líw
 ‘sound of continuous ringing bells’ (Thomason, 1999, p.255, Spokane)
- c. ʔáp ʔáp ʔáp ʔáp
 ‘sound of feet slapping on the floor’ (Thomason, 1999, p.255, Montana Salish)
- d. pəs pəs pəs pəs
 ‘[Coyote’s call to his helpers]’ (Thomason, 1999, p.255, Okanagan)

¹²Chinuk Wawa, also known as Chinook Jargon, is a pidgin language that arose from contact and trade; it has influence from varieties of Chinookan, Salish, and Wakashan languages, as well as English and French (Robertson, 2011, pp.5–6).

¹³While Galloway (1993b) does describe a pattern that looks like C₁C₂C₃ (CVCC) reduplication in Nooksack, this pattern reflects his decision to analyse glottalised sonorants as a sonorant-glottal stop sequence instead of a single segment. This reduplication pattern can be classified as C₁C₂ reduplication.

¹⁴Primary stress in the Lillooet forms is likely final (H. Davis, p.c.).

Van Eijk (1998a, p.465) notes that repetitions of reduplication, such as those in (12) and (13), are “sound-imitative” or “refer to movements”. Czaykowska-Higgins and Kinkade (1998) also observe that repetition itself may be used to yield an imitative effect. This is known in the literature on sound symbolism as “gestalt iconicity” (see, e.g., H. Davis & Trotter, 2023, and references therein). Further, repetition in (13) above is not clearly phonological reduplication; it is also consistent with morphological doubling (see, e.g., Inkelas, 2008), such that there may be multiple instances of the root at the point of lexical insertion.

Another reported instance of repetition is found with a final (C₂) consonant in Montana Salish-Spokane-Kalispel, in both the Spokane and the Montana Salish dialects. B. Carlson and Flett (1989) and Thomason (1999) document three repetitions, as shown in (14); these forms are onomatopoeic.

- (14) a. cússss
 ‘rustling noise when you walk in dry brush or pushing through brush’
 (Thomason, 1999, p.253, Montana Salish)
- b. i cáxxxx
 ‘the sound of frying food’ (Thomason, 1999, p.253, Montana Salish)
- c. mallll
 ‘sound of a gurgling stomach or brook’ (B. Carlson & Flett, 1989, p.50, Spokane)

Repetition in (14) does not clearly correspond to a prosodic unit nor a copy-repair associated with a segmental affix, which means that none of the fission-based analyses from Section 2.2.2.2 are plausible explanations. I set these cases aside because they are better analysed as repetition than reduplication. The translations suggest an onomatopoeic quality; it is possible that these forms are used stylistically, similar to how vowels may be drawn out for emphasis when telling a story.

2.2.4 Labels for Reduplication Patterns

A single reduplication pattern either in the same language or across the family may be called different things across different sources. Variation in nomenclature

between sources and languages can hinder cross-linguistic and comparative analysis even in the best case scenario; in the worst case scenario, labels may suggest or introduce analyses that are misleading. Labeling inconsistency is particularly an issue when those with less familiarity with the languages or associated naming conventions draw on descriptions of Salish languages. In order to discuss all twenty-three languages, it is therefore necessary to unify the disparate labels under headings that allow for comparison across the family.

Labels for types of reduplication may be based on phonological form (FORM-BASED) or draw on meaning (MEANING-BASED). An example of a form-based label employed in the Salish literature is “total” reduplication (e.g., Kuipers, 1967); an example of a meaning-based label is “augmentative” (e.g., Van Eijk, 1998a). Each of these labels may be used to describe the same reduplication pattern, which I categorise as C_1C_2 reduplication (see the following section). I adopt a form-based classification system, introduced in Section 2.2.4.1; this classification system uses additional meaning-based labels to distinguish between semantic subtypes of the same form, as discussion in Section 2.2.4.2.

2.2.4.1 Form-Based Primary Classification

Using a form-based classification, I identify four core reduplication types: C_1C_2 reduplication, C_1 reduplication, C_2 reduplication, and V_1 reduplication. These labels refer to the defining characteristics of reduplicant shape (i.e., number and type of segments copied).

An example of each type is provided in (15). C_1C_2 reduplication is repetition of the first and second consonant of a root, as shown in (15a).¹⁵ C_1 reduplication is the repetition of a consonant either at the beginning of a root or immediately before a stressed vowel, as shown in (15b). C_2 reduplication involves repeating a consonant (either the second consonant of the root or the consonant directly after a stressed vowel), as shown in (15c). V_1 reduplication is the repetition of a vowel, as shown in (15d).

(15) a. C_1C_2 Reduplication

¹⁵Except in the case of Bella Coola, where the position of C_1C_2 reduplication is relative to the first sonorant segment in the word: see Section 2.4.1.

s-cúy<cuý>qs-mit-n (cf. cúyqs-m ‘stumble’)

IPFV-stumble<PL>-MID-3SG.SUBJ

‘They (sg.) are stumbling.’ (Kinkade, 2004, p.259, Cowlitz)

b. *C₁ Reduplication*

s-tə~tákən (cf. stákən ‘sock’)

NMLZ-DIM~sock

‘little sock’ (Montler, 1986, p.42, Saanich - Northern Straits)

c. *C₂ Reduplication*

łışáw~aw-m (cf. łışáwm ‘get cold’)

cold-INCH~C₂-MID

‘It got cold slowly.’ (Kinkade, 1985, p.191, Upper Chehalis)

d. *V₁ Reduplication*

q<ił>i~qp-a?am (cf. qəp?əm ‘touch (s.t.)’)

DIM<PL>~touch-A.INTR

‘touch repeatedly’ (Watanabe, 2003, p.403, Comox-Sliammon)

The division of Salish reduplication into these four core types follows the description of Lushootseed reduplication in Kirkham (1991), who classifies seven reduplicative patterns into four categories: CVC, CV, V, and VC. However, my modified notation (C₁C₂, C₁, V₁, and C₂) abstracts over differences related to vowels. For example, C₁C₂ reduplication patterns are defined by the fact that two different consonants undergo fission. A vowel is not included in the C₁C₂ label because the presence of a vowel and its source varies across the family (see Section 2.4); the label C₁C₂ abstracts away from individual patterns of fission or epenthesis of a vowel, which allows reduplication patterns to be grouped to facilitate comparison.

Excluding variable segments is also motivated by patterns in individual languages. For example, both -C- and Ci- are surface realisations of diminutive C₁ reduplication in Comox-Sliammon, as was shown in (6) earlier in this chapter (see Section 4.1.3.1.1 for discussion of this pattern). The two allomorphs can be grouped together as C₁ reduplication because they both reduplicate a consonant.

2.2.4.2 Meaning-Based Secondary Classification

I use secondary meaning-based labels in order to differentiate between sub-types of reduplication. These additional labels are necessary because there may be different meanings associated with the same reduplicative shape. For example, diminutive, plural, and progressive meanings are all marked with C₁ reduplication in Comox-Sliammon, as shown in (16).

(16) a. *Comox-Sliammon Diminutive C₁ Reduplication*

- i. **pi**~**piq** (cf. **piq** ‘wide’)
DIM~wide
‘It is a little bit wide.’ (BW)
- ii. **k^wi**~**k^was** (cf. **k^was** ‘hot’)
DIM~hot
‘It is warm.’ (BW)

b. *Comox-Sliammon Plural C₁ Reduplication*

- i. **ʔa**<**ʔ**>**y-at=č** (cf. **ʔayatč** ‘I am holding it.’)
hold<PL>-CTR+STAT=1 SG.SUBJ
‘I am holding them.’ (BW)
- ii. **ʔi**<**ʔ**>**k^w-it** (cf. **ʔik^wit** ‘It is sewn.’)
sew<PL>-STAT
‘They are sewn.’ (BW)

c. *Comox-Sliammon Progressive C₁ Reduplication*

- i. **ʔi**~**ʔik^w-it-as** (cf. **ʔik^wit** ‘It is sewn.’)
PROG~sew-CTR-3ERG
‘She is sewing it.’ (BW)
- ii. **ʔu**~**ʔuk^w-uʔ** Fred (cf. **ʔuk^w** ‘fly’)
PROG~fly-PST Fred
‘Fred was flying.’ (BW)
- iii. **čə**<**č**>**ʔ** (cf. **čəʔ** ‘rain’)
rain<PROG>
‘It’s raining.’ (BW)

There are three different reduplicative morphemes in (16), but all are categorised as types of C_1 reduplication. They can be differentiated through the addition of a semantic label (e.g., “diminutive”). The labels used in interlinear glossing are always based on meaning, unless a meaning is too unclear to label. For example, I use the label C_2 for (15c) above to gloss C_2 reduplication in Upper Chehalis which is associated with change of state or “slowness” (Kinkade, 1985).

The choice of secondary label is informed by the literature on particular languages and cross-Salish patterns. Most reduplicative morphemes fall into three categories when classified by meaning: plural, diminutive, and aspectual. Where little is known about the meaning of the reduplicative morpheme, I default to more general labels. For example, I use the gloss ASP to refer to the type of C_1 reduplication in Pentlatch shown in (17), which resembles progressive reduplication in Comox-Sliammon and Squamish.

(17) *Pentlatch Aspectual C_1 Reduplication*

- a. s-**ta**~**táyaq**=čì (cf. *táyaq* ‘angry’)
STAT-ASP~angry=1SG.SUBJ
‘I am angry.’ (Boas, 1886)
- b. **kʷa**~**kʷáy**=čì (root: *kʷáy*- ‘hungry’)
ASP~hungry=1SG.SUBJ
‘I am hungry’ (Boas, 1886)

An example from Squamish that is comparable to (17) is provided in (18a). Given that there is semantic work on progressive C_1 reduplication in Squamish (Bar-el, 2008), in addition to previous descriptive work on the language (Kuipers, 1967), I gloss this morpheme as PROG following the literature.

(18) *Squamish Progressive C_1 Reduplication*

- a. čən wa **tá**~**tayaq** (cf. *táyaq* ‘get angry’)
1SG.SUBJ IPFV PROG~angry
‘I am angry’ (Bar-el, 2006, p.245, Squamish)
- b. na **pá**~**payaq**-ant-as ta=s-nəxʷit-s (cf. *páyaq* ‘be fixed’)
RL PROG~fix-TR-3ERG DET=NMLZ-canoe-3POSS
‘He’s in the process of fixing the canoe.’ (Bar-el, 2008, p.47, Squamish)

2.2.5 Additional Terminology

In addition to the four main form-based labels, there are two additional types of terminology related to form: the first concerns *pivots* (where copied segments are positioned); the second concerns non-reduplicated segments sometimes associated with reduplicative morphemes. These are introduced in Section 2.2.5.1 and Section 2.2.5.2, respectively.

2.2.5.1 Pivots: Edge vs. Prominence Alignment

The possible positions where copied material can be placed are called *pivots* (Yu, 2007). Yu (2007) distinguishes *edge-oriented* from *prominence-oriented* pivots for affixation. Edge-oriented affixes are positioned with respect to a word or morpheme edge, and prominence-oriented affixes are positioned relative to prosodic prominence (e.g., they follow a stressed vowel). Examples of edge-aligned and prominence-aligned C_1 reduplication in Lillooet are provided in (19). The reduplicative morpheme is aligned to the left-edge of the root in (19a), while it follows the stressed vowel in (19b).

- (19) a. *Edge-Aligned: Initial C_1 Reduplication*
s-**q**^wə~q^wlíp (cf. *sq^wlip* ‘black tree moss’)
NMLZ- C_1 .INIT~black.tree.moss
‘lots of black tree moss’ (Van Eijk, 1997, p.57, Lillooet)
- b. *Prominence-Aligned: Diminutive C_1 Reduplication*
kapó<**p**>uh (cf. *kapúh* ‘coat’)
coat<DIM>
‘small coat’ (Van Eijk, 1997, p.257, Lillooet)

The position of a reduplicative morpheme in the input does not necessarily match its position in the output. For example, I analyse forms like (19b) above as containing a reduplicative prefix that is realised as an infix, such that the reduplicated consonant is adjacent to the stressed vowel (see Section 4.2.4.2 in Chapter 4). In this chapter, I use the labels prefix, infix, and suffix descriptively to refer to the position of a reduplicant in the surface form; the relationship between input affix position and linearisation is discussed further in Chapter 3 (Section

3.5.1.2).

2.2.5.2 Non-Reduplicated Segments: Fixed vs. Epenthetic

Segments associated with reduplicative morphemes may arise from (i) reduplication itself, (ii) fixed segments, or (iii) epenthesis. Fixed and epenthetic segments are not copies of existing material (i.e., they do not involve fission). I define fixed segments in Section 2.2.5.2.1 and epenthetic segments in Section 2.2.5.2.2, immediately below.

2.2.5.2.1 Fixed Segments

Fixed segments, such as the [i] in Sechelt diminutive reduplication, are part of a lexical entry (and therefore present in the input to the phonology). Examples of fixed segments are bolded in (20).¹⁶

- (20) a. s-**lí**~ləm-ay (cf. *slómay* ‘branch’)
NMLZ-DIM~?-tree
‘small branch’ (Beaumont, 2011, p.771, Sechelt)
- b. ?íl~**al** (root: ?il- ‘cry’)
cry~C₂
‘sing’ (Drachman, 1969, p.279, Twana)
- c. **dú**~tuqusu (cf. *tuqusu* ‘beaver’)
DIM~beaver
‘little beaver’ (Edel, 1939, p.14, Tillamook)

Lexical entries for some reduplicative morphemes consist only of a fixed segment. For example, repetitive C₁ reduplication in Salish-Spokane-Kalispel corresponds to an underlying /e/; allomorphy is conditioned by syllable structure (Bates & Carlson, 1992). The repetitive affix surfaces as an infix -e- when the unreduplicated word starts with a consonant cluster, as shown in (21a); reduplication occurs when the unreduplicated word starts with a CV- sequence, as shown in (21b). Non-

¹⁶I am unsure if (20c) should have [q] or [q^w], due to conditioning from the following rounded vowel [u]. I leave it as [q] because my best guess is that it is related to Proto-Salish *təq, which is found in words relating to (beaver) dams (Kuipers, 2002, p.106).

reduplicative prefixes, such as the nominaliser *s-* in (21a), do not factor into the choice between *-e-* and *-eC-*.

- (21) a. $s\text{-}\dot{m}\langle e\rangle\dot{l}k^w\text{-}\dot{e}ye?$ (cf. $mlk^w\dot{e}lx^w$ ‘a fur rug’)
 NMLZ-solid<REP>-seem
 ‘shadow’ (Bates & Carlson, 1998, p.105, Spokane)
- b. $\dot{c}\langle e\dot{c}\rangle ip\text{-}\dot{s}$ (cf. $\dot{c}ip\dot{p}$ ‘It accidentally got pinched.’)
 pinch<REP>-PL
 ‘scissors, shears’ (Bates & Carlson, 1998, p.104, Spokane)

2.2.5.2.2 Epenthetic Segments

Epenthetic segments are not present in the input, in contrast to fissioned and fixed segments, which correspond to segments present in the input.

An epenthetic segment may accompany a reduplicative morpheme, such as in C_1C_2 reduplication in Halkomelem. A schwa is epenthesized between the doubled consonants, as shown in (22), because the reduplicative morpheme requires a vocalic nucleus.¹⁷

- (22) a. $\text{ʔ}\dot{a}m\sim\text{ʔi:m}\dot{a}\theta$ (cf. $\text{ʔim}\dot{a}\theta$ ‘grandchild’)
 PL~grandchild
 ‘grandchildren’ (Galloway, 1977, p.108, Upriver Halkomelem)
- b. $s\text{-}\dot{t}\dot{a}n\sim\dot{t}\dot{e}ni?$ (cf. $s\dot{t}\dot{e}ni?$ ‘woman’)
 NMLZ-PL~woman
 ‘women’ (Leslie, 1979, p.49, Island Halkomelem)

While epenthetic segments are typically vowels (and more likely to be [ə] than any other vowel), there are also cases of epenthetic consonants accompanying reduplication. For example, V_1 reduplication in Comox-Sliammon, analysed in Section 4.1.5.2, has a floating [c.g.] (constricted glottis) feature in the input alongside a prosodic unit (μ). A glottal stop is inserted and a vowel is reduplicated to provide segmental content for the feature and the prosodic unit, respectively.

¹⁷In Chapter 6 (Section 6.5.2.1.3), I argue that $\sigma_{N+\mu}$ is the input for this type of C_1C_2 reduplication: epenthetic [ə] associates with the nucleus node of the affixed prosodic unit.

- (23) a. $\text{ɬ} < \mathbf{a?} > \mathbf{ak}^w \sim \text{ɬuk}^w$ (cf. ɬuk^w ‘fly’)
 $C_1 C_2 . \text{PL} < V_1 . \text{PL} > \sim \text{fly}$
 ‘flying around’ (BW, vf, Comox-Sliammon)
- b. $p < \mathbf{a?} > a < p > \text{ya-sx}^w - \text{?ut}$ (cf. $pa?a$ ‘one’)
 $\text{one} < V_1 . \text{PL} > < \text{DIM} > - \text{CAUS-EXCL}$
 ‘She did them one by one.’ (MV, vf, Comox-Sliammon)

Epenthesis also accompanies reduplication as a repair, in which case it follows reduplication in order to ensure the well-formedness of a word where reduplication creates an impossible or dispreferred sequence. For example, an additional [ə] is inserted to break up clusters created by diminutive C_1 reduplication in Lillooet, as shown in (24a–b). The position of [ə] is sensitive to the overall shape of the word, rather than the position of consonants associated with the diminutive morpheme.¹⁸ A schwa inserted following C_1 reduplication does not have a fixed position: compare (24a) and (24b). Additionally, [ə] is not obligatory as part of the reduplication: the diminutive form in (24b) requires an additional epenthetic [ə] (the second) to meet phonotactic constraints, while a diminutive form with the same root in (24c) does not.

- (24) a. $\text{xz} \acute{\text{ə}} < \mathbf{z} > \mathbf{\acute{ə}m}$ (cf. xzum ‘big’)
 $\text{big} < \text{DIM} >$
 ‘a bit bigger’
- b. $\text{z} \acute{\text{ə}} < \mathbf{z} > \mathbf{x\acute{ə}t}$ (cf. zaxt ‘long’)
 $\text{long} < \text{DIM} >$
 ‘a bit longer’
- c. $\text{z\acute{ə}} < \mathbf{z} > \mathbf{xt-\acute{a}n}$ (cf. $\text{z\acute{a}xtan}$ ‘lengthen s.t.’)
 $\text{long} < \text{DIM} > - \text{CTR}$
 ‘lengthen s.t. a bit’

(H. Davis & Mellesmoen, 2023, p.13, Lillooet)

Where epenthesis varies depending on word shape and is not required by the reduplicative morpheme (i.e., to provide a nucleus for an affixed syllable), the

¹⁸See Chapter 4 (Section 4.2.4.2.3) for further description and an analysis of diminutive C_1 reduplication in Lillooet.

segment is not glossed as part of the reduplicative morpheme (see e.g., the Lillooet examples in (24) above).

2.2.6 Classifying Patterns

I have standardised the descriptions from previous work using the form-based classification system described in Section 2.2.4 above, which sorts patterns into C_1C_2 , C_1 , C_2 , and V_1 reduplication. In general, there is high consistency across sources with respect to the presence of the three consonant-based reduplication patterns (C_1 , C_1C_2 , and C_2). However, I diverge from previous descriptions where motivated by the criteria described in Section 2.2.2, as stated in (25).

- (25)
1. Reduplication (a 1:2(+) input-output mapping) occurs as a direct result of the addition of the morpheme, where reduplication provides segmental content necessary for a well-formed output.
 2. At least one allomorph is expressed with reduplication.

I include putative patterns of reduplication only if they can be analysed in a manner consistent with how I have defined reduplication, based on existing description or what can be reasonably be extracted from available data. I set aside two possible cases of C_2 reduplication, which are found in Bella Coola (Newman, 1947) and Quinault (Modrow, 1971), as well as possible cases of V_1 reduplication, which are found in Lower Chehalis (Robertson, 2014) and Bella Coola (Angermeyer, 2003). These are discussed in for each language in Appendix A.

Cases where the data are too scarce or variable to determine if a pattern can be include: V_1 and C_2 reduplication in Nooksack (Galloway, 1984b), V_1 reduplication in Tillamook (Edel, 1939), V_1 reduplication in the Island dialect of Halkomelem (Leslie, 1979), V_1 reduplication in the Songish dialect of Straits (Raffo, 1972), and C_1 and C_2 reduplication in Lower Chehalis (Robertson, 2014). If additional data were to support the inclusion of these patterns as reduplication, they would not be phonologically exceptional; they can easily be accounted for with the phonological theory described in Chapter 3.

Given that the survey undertaken here largely focuses on attested forms in existing documentary sources, the conclusions I draw are necessarily based on

the *presence* of reduplication. The absence of a particular type of reduplication in the survey does not rule out the possibility that the reduplication pattern might exist, especially in languages with limited description. For example, it would not be surprising if Pentlatch has C_2 reduplication, given that it is present in both of the other languages of the North Georgia sub-branch of Salish (Sechelt and Comox-Sliammon). This cannot be concluded from the available documentation, however.

Finally, I set aside reduplicated numerals used to count animals or humans. Anderson (1997) surveys these forms and finds that counting forms typically employ one of the core reduplication patterns (C_1C_2 and C_1), but their use with numerals is semantically distinct and varies in arbitrary ways across the languages. The examples in (26) compare reduplicated numerals for ‘six people’ in Interior Salish. The reduplication in the Shuswap word is not edge-aligned, unlike regular C_1C_2 reduplication in Interior Salish (see Section 2.4.1). In Lillooet, both C_1C_2 and C_1 reduplication are attested for the same numeral.

- (26) a. $təq̣méḳməkst$ (cf. $təq̣méḳst$ ‘six’)
 ‘six people’ (Kuipers, 1974a, p.56, Shuswap)
 b. $ʔáq̣ʔəq̣mekst$ (cf. $ʔáq̣mekst$ ‘six’)
 ‘six people’ (Anderson, 1997, p.3, Thompson River Salish)
 c. $n-ʔáq̣ʔəq̣əmkst$ or $n-ʔáq̣əmkst$ (cf. $ʔáq̣əmkst$ ‘six’)
 ‘six people’ (Van Eijk, 1997, p.129–130, Lillooet)

2.3 Core Reduplication Patterns across Salish

Table 2.1 shows how many languages in each branch of the family exhibit each of the four core reduplication patterns identified in the survey.¹⁹

¹⁹I discuss Northern Straits and Klallam together under the label Straits for reasons given in Section 1.5.1.4; however I tally them separately because they are different languages and otherwise the total number of Salish languages is incorrect (i.e., 22 instead of 23).

Table 2.1: Summary: Core Reduplication in Salish

Branch	C_1C_2	C_1	C_2	V_1
Central Salish (n=10)	10	10	4	3
Interior Salish (n=7)	7	7	7	3
Tsamosan (n=4)	4		1	1
Bella Coola (n=1)	1	1		
Tillamook (n=1)	1	1	1	
Total (n=23)	23	19	13	7

As the table shows, C_1C_2 and C_1 reduplication are widespread in Salish. All languages have C_1C_2 reduplication; C_1 reduplication is represented in every language outside of the Tsamosan branch. In contrast, C_2 reduplication is found in 13 out of 23 languages; V_1 reduplication is only found in seven. An inventory of the core types of reduplication by language is given in Table 2.2.

Table 2.2: Core Reduplication Patterns in Salish Languages

Language	C ₁ C ₂	C ₁	C ₂	V ₁
Comox-Sliammon	✓	✓	✓	✓
Sechelt	✓	✓	✓	
Pentlatch	✓	✓		
Squamish	✓	✓		
Nooksack	✓	✓		
Halkomelem	✓	✓		
Straits <i>Northern Straits and Klallam</i>	✓	✓		
Lushootseed	✓	✓	✓	✓
Twana	✓	✓	✓	✓
Tillamook	✓	✓	✓	
Upper Chehalis	✓		✓	
Cowlitz	✓			
Lower Chehalis	✓			
Quinault	✓			✓
Lillooet	✓	✓	✓	
Thompson River Salish	✓	✓	✓	
Shuswap	✓	✓	✓	
Okanagan	✓	✓	✓	✓
Montana Salish-Spokane-Kalispel	✓	✓	✓	✓
Moses Columbia	✓	✓	✓	
Coeur d'Alene	✓	✓	✓	✓
Bella Coola	✓	✓		

I discuss each core type of reduplication in the following sections: C₁C₂ in Section 2.4, C₁ in Section 2.5, C₂ in Section 2.6, and V₁ in Section 2.7. This is followed by a discussion of multiple reduplication patterns in Section 2.8.

2.4 C₁C₂ Reduplication

Two consonants are copied in C₁C₂ reduplication, typically with an epenthetic or reduplicated vowel placed between the copied consonants (i.e., resulting in a

surface CVC shape). The surface positions of segments associated with C₁C₂ reduplication are compared in Section 2.4.1. The presence and quality of vowels in C₁C₂ reduplication are considered in Section 2.4.2. The meanings associated with and number of C₁C₂ reduplication processes are covered in Section 2.4.3.

2.4.1 Position of C₁C₂ Reduplication

Segments associated with C₁C₂ reduplication are positioned relative to the edge of a domain (e.g., word) for all languages except Bella Coola. The segments associated with the reduplicative morpheme may surface in a prefixed or a suffixed position (or an infix position for CVCC roots). For example, the plural C₁C₂ reduplicative morpheme in Comox-Sliammon is the first syllable in (27a); in contrast, the property-denoting C₁C₂ reduplicative morpheme in Moses Columbia is the second syllable in (27b).²⁰

- (27) a. *Edge-aligned: Realised as First Syllable*
 čəy~čuý (cf. čuy ‘child’)
 PL~child
 ‘children’ (Watanabe, 2003, p.327, Comox-Sliammon)
- b. *Edge-Aligned: Realised as Second Syllable*
 cál~cál-t (cf. cáll ‘get shaded’)
 shade~PDEN-CHAR
 ‘shady’ (Willett, 2003, p.243, Moses Columbia)

In Bella Coola, the position of the reduplicative morpheme depends on prosodic structure, rather than the edge of a stem or word.²¹ C₁C₂ reduplication targets the first syllable with a moraic (non-schwa) vowel or sonorant.^{22,23} Concatenative prefixes do not factor into the position of reduplication, as shown in (28a), nor are they copied (28b).

²⁰Stress is used to differentiate between plural (first syllable) and property-denoting (second syllable) C₁C₂ reduplication in Moses Columbia (see Czaykowska-Higgins, 1993).

²¹Prosody-aligned reduplication can be identified by the fact that its position is variable; it is sensitive to prosodic structure, not aligned to an edge.

²²While Bella Coola allows words (and syllables) without vowels (e.g., Bagemihl, 1991), reduplication does not apply to these words without epenthesis of a moraic vowel (Mellesmoen, 2021b).

²³See Chapter 3 (Section 3.2) for more on moraic segments.

(28) *Prosody-Aligned*

- a. s-tan~**tan**-mx (cf. *stan* ‘mother’)
NMLZ-mother~PL-person
‘female ancestors’ (Nater, 1978, p.2, Bella Coola)
- b. ʔa-sc̣-ał~**čł** (cf. *ʔascał* ‘waterfall’)
LOC-spray-precipitation~PL
‘heavy rain’ (Nater, 1984, p.118, Bella Coola)

The generalisation that prefixes are never copied extends across the family (Van Eijk, 1998a). Examples from each of the major branches of reduplication with words including prefixes are provided in (29). The prefix is never copied.

- (29) a. s-**nəx**^w~nəx^wł (cf. *snəx^wł* ‘canoe’)
NMLZ-PL~canoe
‘canoes’ (Kuipers, 1967, p.100, Squamish)
- b. t-čén-**čən**-əm (cf. *tčénm* ‘take hold of’)
LOC-hold~PL-MID
‘They held on with both hands.’ (Reichard, 1938, p.635, Coeur d’Alene)
- c. s-cúy<**cuy**>qs-mit-n (cf. *cúyqs-m* ‘stumble’)
IPFV-stumble<PL>-MID-3SG.SUBJ
‘They (sg.) are stumbling.’ (Kinkade, 2004, p.259, Cowlitz)

2.4.2 Vowels in C₁C₂ Reduplication

The presence and quality of a vowel in C₁C₂ reduplication can vary. In most languages, the vowel is an epenthetic [ə], as in Comox-Sliammon (30a); however, there are exceptions, such as a fixed segment /i/ in Tillamook (30b), or a faithful copy in Lushootseed (30c).²⁴

²⁴Edel (1939, p.11) notes alternations in Tillamook between stem vowels that vary across recordings with “no apparent change in context or meaning”, including [a] (“a”) and [ə] (“u”); the example in (30b) appears to be an example of this. However, [ə] before a uvular sound is realised lower (i.e., closer to [a]) in many languages, so it is not clear that the pronunciation of these would be very different in this word.

- (30) a. $k^wəs \sim k^wusən$ (cf. $k^wusən$ ‘star’)
 PL $\sim$ star
 ‘stars’ (Watanabe, 2003, p.372, Comox-Sliammon)
- b. $g^wəʔ ʔiq \sim ʔəqí-i$ (cf. $ʔaqi$ ‘to run away’)
 FUT PL $\sim$ run.away-1 SG.SUBJ
 ‘I will skip around.’ (Edel, 1939, p.14, Tillamook)
- c. $húd \sim hud$ (cf. $hudčup$ ‘light fire, burn wood’)
 PL $\sim$ burn
 ‘burning here or there’ (Hess, 1967, p.65, Lushootseed)

A vowel may also be absent in the surface form of a word with C_1C_2 reduplication, as shown in the Bella Coola example in (31), repeated from (28b). The absence of a vowel is dictated by the general phonological grammar of the language.

- (31) $ʔa-sč-aʔ \sim čʔ$ (cf. $ʔasčaʔ$ ‘waterfall’)
 LOC-spray-precipitation $\sim$ PL
 ‘heavy rain’ (Nater, 1984, p.118, Bella Coola)

The presence of a vowel in the reduplicant in the Comox-Sliammon example in (30a) and the absence of a vowel in the reduplicant in the Bella Coola example in (31) are consistent with the phonological grammars of each language: Comox-Sliammon tolerates fewer consonant clusters than Bella Coola, forcing the presence of a vowel where Bella Coola lacks one. In general, the presence of a vowel is determined by the phonology of a particular language, and in particular whether a reduplication process would create phonotactically illicit sequences.²⁵

Further description and exemplification of C_1C_2 reduplication patterns in individual languages can be found in Appendix A.

2.4.3 Meanings Associated with C_1C_2 Reduplication

C_1C_2 reduplication marks plurality and/or pluractionality across all 23 languages. A secondary type of C_1C_2 reduplication is associated with properties/adjectives. These two types of C_1C_2 reduplication are discussed in Sections 2.4.3.1 and

²⁵See discussion of schwa in Chapter 3 (Section 3.2), as well as syllabification in Comox-Sliammon in Chapter 4 (Section 4.1.1).

2.4.3.2, respectively. The cross-Salish distribution of C₁C₂ reduplication by meaning is discussed in Section 2.4.3.3.

2.4.3.1 Plural C₁C₂ Reduplication

Plurality marked by C₁C₂ reduplication varies across languages, ranging from being restricted to a specific type of plurality (e.g, pluractional verbs in Tsamosan languages that refer to events distributed in either time or space) to marking general plurality of both events and entities (i.e., both plural and pluractional meanings).

C₁C₂ reduplication may be sensitive to lexical category and/or the event-entity distinction. When C₁C₂ reduplication acts as a marker of plurality in the verbal domain, events may be distributed across participants, times, and spaces. For example, Bar-el (2008) finds that C₁C₂ reduplication with verbs marks habitual or iterative events in Squamish. J. Thompson (2009) finds that plural morphology on verbs in Upriver Halkomelem can mark a plurality of times, space-times, and participants. Broadly, the use of terms such as “repetitive” by A. Mattina (1973) for Okanagan and “continuative” by Nater (1984) for Bella Coola suggest that there is a range of ways in which events can be distributed with pluractional uses of C₁C₂ reduplication.

Certain languages have stricter restrictions on the use of C₁C₂ reduplication than others. For example, plural C₁C₂ reduplication with verbs in Comox-Sliammon requires both spatial and temporal distribution (Mellesmoen & Huijsmans, 2019c). While Comox-Sliammon allows plural entity readings when C₁C₂ reduplication occurs with nouns, C₁C₂ reduplication in Upper Chehalis *only* allows for a distributive interpretation over events (Kinkade, 1995, p.357); Upper Chehalis does not use C₁C₂ reduplication to mark plural arguments on a verb or plural entities in the nominal domain.

Some languages allow a plural argument reading on adjectives with C₁C₂ reduplication, as shown for Comox-Sliammon and Sechelt in (32).

- (32) a. *piq~piq-ašin* (cf. *piq* ‘wide’)
PL~wide-foot
‘wide feet’ (FL, Comox-Sliammon)

- b. $\lambda\acute{a}q \sim \lambda\acute{a}qt\text{-}\acute{s}en$ (cf. $\lambda\acute{a}qt$ ‘long’)
 PL $\sim$ long-foot
 ‘long legs’ (FL, Comox-Sliammon)
- c. $\eta\acute{y} \sim \eta\acute{y}$ (cf. $\eta\acute{y}$ ‘good’)
 PL $\sim$ good
 ‘good (more than one person/thing)’ (Beaumont, 2011, p.192, Sechelt)

2.4.3.2 Property-Denoting C_1C_2 Reduplication

A second type of C_1C_2 reduplication is called “characteristic” in the Salish literature. As these often involve permanent (individual-level) properties, rather than temporary (stage-level) states, I use the label “property-denoting” for C_1C_2 reduplication that can be distinguished from plural/pluractional C_1C_2 reduplication either on the basis of form or meaning (i.e., where the form or meaning do not match that of plural C_1C_2 reduplication in a given language). Examples are given in (33): C_1C_2 reduplication denotes a individual-level property in (33a–b) and a stage-level one in (33c).

- (33) a. $g\acute{a}y \sim gay\text{-}at\text{-}t\acute{e}n$ (cf. *gayat* ‘ask s.o.’)
 PDEN $\sim$ ask-CONNECT-instrument
 ‘curious’ (PD, Comox-Sliammon)
- b. $k\acute{e}n \quad t\acute{t} \sim ta\acute{t}\text{-}t$ (cf. *ta\acute{t}* ‘straight, certain’)
 1SG.SUBJ PDEN $\sim$ straight-CHAR
 ‘I am honest.’ (A. Mattina, 1987, p.200, Okanagan)
- c. $\acute{c}\acute{a}q^w \sim \acute{c}q^w$ (cf. $\acute{c}\acute{a}q^w$ ‘dirt’)
 dirt $\sim$ PDEN
 ‘dirty’ (Montler, 2015, p.286, Klallam)

While previous analyses have proposed collapsing plural and property-denoting reduplication into a single “augmentative” category (see e.g., Van Eijk 1998a), there are both phonological and semantic reasons to keep them separate in at least some languages. For example, Czaykowska-Higgins (1993b) shows that C_1C_2 reduplication in Moses Columbia exhibits two distinct stress assignment patterns. She attributes this to a difference in position: plural reduplication is prefixed (stress

on the second syllable), while property-denoting reduplication is suffixed (stress on the first syllable). Examples of plural reduplication are shown in (34), while property-denoting forms are given in (35).²⁶ Forms with a clear property-denoting meaning across Interior Salish languages often have the affix *-t*, which I gloss as CHAR.

- (34) a. **kəł~káł-x̣t-n** l̥x
 PL~give-APP-1 SG.SUBJ PL
 ‘I gave them a lot of things.’ (cf. *t swát ci káłtus* ‘Who gave that to them (sg.)?’) (Czaykowska-Higgins, 1993b, p.68, Moses Columbia)
- b. **ṭəł~ṭəł-x̣n** (cf. *ṭinwí k^w ṭəłt* ‘Your face is dirty.’)
 PL~dirty-foot
 ‘Their (sg.) feet are dirty.’ (Czaykowska-Higgins, 1993b, p.70, Moses Columbia)
- c. **q̣aʔ~q̣aʔ-n** (cf. *q̣aʔaʔmínn* ‘It’s too small for me.’)
 PL~wedge.in-TR+1 SG.SUBJ
 ‘I put them away.’ (Czaykowska-Higgins, 1993b, p.70, Moses Columbia)
- (35) a. **x̣^wús~x̣^wus-cn** (cf. *x̣^wus* ‘hurry’)
 hurry~PDEN-mouth
 ‘talk fast’ (Czaykowska-Higgins, 1993b, p.70, Moses Columbia)
- b. **ṭəł~ṭəł-t** (cf. *ṭinwí k^w ṭəłt* ‘You are dirty.’)
 dirty~PDEN-CHAR
 ‘They (sg.) are dirty.’ (Czaykowska-Higgins, 1993b, p.70, Moses Columbia)
- c. **q̣aʔ~q̣aʔ-t** (cf. *q̣aʔaʔmínn* ‘It’s too small for me.’)
 wedge.in~PDEN-CHAR
 ‘crowded’ (Czaykowska-Higgins, 1993b, p.70, Moses Columbia)

The range of meanings associated with property-denoting reduplication in Moses Columbia is quite broad, particularly where the characteristic *-t* is not

²⁶Forms given for comparison in (34a–c) come from Willett (2003, pp.88,187,226). The comparison forms in (35b–c) come from Willett (2003, pp.187,226).

present, which may indicate that the property-denoting or adjectival semantics are contributed by that affix. This would be consistent with other analyses of property-denoting reduplication, both in Moses Columbia and more broadly across Interior Salish; Willett (2003, p.242) analyses property-denoting reduplication as a process of adjective formation in Moses Columbia, while H. Davis (2011) proposes that C_1C_2 forms an adjective in Lillooet. This type of meaning with C_1C_2 reduplication in Lillooet is sometimes found with a *-t* affix, which alternates with an *-m* form (see examples in Section 4.2.3.2.1).

Property-denoting C_1C_2 reduplication is also found in languages without productive use of the *-t* affix. See, for example, the non-plural use of C_1C_2 reduplication in Klallam and Comox-Sliammon in (33) above, or the Sechelt examples in (36b) and (37b) below. As in Moses Columbia, there are pairs in Sechelt where the same root allows for both plural and non-plural interpretations. Examples are provided in (36) and (37). Sechelt does not typically use two different forms to mark the distinction in meaning, though there is sometimes a difference in the reduplicant vowel, as shown in (36).

- (36) a. **ʔíy**~ʔiy-ʔál-um (cf. *ʔíyʔálum* ‘enjoy what one is eating’)
 PL~good-food-MID
 ‘enjoy what one is eating (more than one person)’ (Beaumont, 2011, p.983, Sechelt)
- b. **ʔáy**~ʔiy-ʔál-um (cf. *ʔíyʔálum* ‘enjoy what one is eating’)
 PDEN~good-food-MID
 ‘really enjoy what one is eating’ (Beaumont, 2011, p.967, Sechelt)
- (37) a. **máy**~má-y-əl (cf. *máyəl* ‘bad (rotten, spoiled)’)
 PL~bad-enter.state
 ‘bad (rotten, spoiled) (more than one thing)’ (Beaumont, 2011, p.33, Sechelt)
- b. s-**máy**~may-ít (cf. *smáy* ‘badness, evil’)
 STAT-PDEN~bad-STAT
 ‘feeble (frail, weak)’ (Beaumont, 2011, p.154, Sechelt)

There are phonological reasons to posit two separate reduplicative morphemes

in at least some languages (e.g., Moses Columbia). Furthermore, L. C. Thompson and Thompson (1992, p.89) provide the example in (38) with both types of C₁C₂ reduplication co-occurring in the same word in Thompson River Salish, with a meaning consistent with a compositional interpretation of both of them.

- (38) ?es-t-**paɬ**~páɬ~**paɬ**-t (cf. ?estpáɬpaɬt ‘grey, faded’)
 STAT-QUAL-PL~fade~PDEN-CHAR
 ‘several are grey, faded’ (L. C. Thompson & Thompson, 1992, p.89,
 Thompson River Salish)

Further work is required to determine the range of non-plural uses of C₁C₂ reduplication and how this differs across Salish languages.

2.4.3.3 Distribution of C₁C₂ Reduplicative Morphemes

The distribution of C₁C₂ reduplication is summarised in Table 2.3 for each branch of the family. I use “plural” to refer to both general plurality and pluractionality unless otherwise specified.²⁷

Table 2.3: C₁C₂ Reduplication in Salish Languages by Branch

	Plural	Plural and Property-Denoting	Unclear
Central Salish (n=10)	1	7	2
Tillamook (n=1)	1	0	0
Tsamosan (n=4)	4	0	0
Interior Salish (n=7)	1	6	0
Bella Coola (n=1)	0	0	1
Total	7	12	3

Of the 12 languages with multiple meanings associated with C₁C₂ reduplication, five languages show differences in form that correspond to a semantic distinction (“different form” in Table 2.4 below), and seven do not (“same form” in Table 2.4). As these distinctions are based on existing documentation, the absence of

²⁷Note that Tsamosan languages only have pluractional, not plural, C₁C₂ reduplication.

evidence for a distinction may reflect gaps in the documentary record and should be interpreted accordingly.²⁸

Table 2.4: C₁C₂ Reduplication in Salish Languages

One Function	Plural
	Tillamook, Cowlitz, Lower Chehalis, Lushootseed, Upper Chehalis, Quinault, Montana Salish-Spokane-Kalispel
Two Functions	Property-Denoting and Plural
Same Form	Coeur d’Alene, Shuswap, Lillooet, Okanagan, Sechelt Squamish, Nooksack
Different Forms	Comox-Sliammon, Thompson River Salish, Halkomelem, Moses Columbia Northern Straits, Klallam
Unclear	
	Pentlatch, Twana, Bella Coola

Distinguishing types of C₁C₂ reduplication on the basis of form is often not possible, even for languages with “different” forms in Table 2.4 because the two types of reduplication are often homophonous. For example, in Comox-Sliammon, the difference is only reflected in the placement of sonorant glottalisation and therefore is obscured with obstruent-only roots. In Sechelt, the two forms are usually the same, except some minimal pairs are differentiated with an additional [a] in the reduplicant for the plural form, as in (36) above.

Languages listed as having the same form also may have multiple patterns for C₁C₂ reduplication. For example, variable stress placement is observed in both Moses Columbia and Lillooet C₁C₂ reduplication, but stress is not used to differentiate different types of C₁C₂ reduplication in Lillooet (see analysis of C₁C₂ reduplication in Section 4.2.3.2.1), unlike in Moses Columbia (Czaykowska-Higgins, 1993b).

²⁸Limitations in available evidence are more likely to result in under-determination than over-determination of semantic distinctions.

2.4.4 Summary: C₁C₂ Reduplication

There are two sub-types of C₁C₂ reduplication in Salish: plural and property-denoting. As all languages have plural C₁C₂ reduplication but not all have property-denoting reduplication, there is an implicational relation between them: if a language has property-denoting C₁C₂ reduplication, then it has plural C₁C₂ reduplication.

2.5 C₁ Reduplication

C₁ reduplication is widespread in Salish (Czaykowska-Higgins & Kinkade, 1998); it is found in all Salish languages excluding the Tsamosan branch. C₁ reduplication includes the most diverse set of reduplicative processes in both form and meaning. With respect to form, the key parameters for variation include the position of the reduplicant and vowel presence/quality. These are discussed in Section 2.5.1 and Section 2.5.2, respectively. The meanings associated with C₁ reduplication are discussed in Section 2.5.3.

2.5.1 Position of C₁ Reduplication

There are both edge-aligned and prominence-aligned C₁ reduplication patterns in Salish: the reduplicated consonant may be either the first in the relevant domain (prefixes are excluded), or the one before the stressed vowel.

The segment(s) copied in edge-aligned C₁ reduplication may occur in the first syllable or second syllable of a word, as shown in (39a–b) and (39c), respectively.

- (39) a. **cí**~cíč-it (cf. *cíčit* ‘fork something’)
IPFV~root-CTR
‘forking/spearing something’ (Beaumont, 2011, p.875, Sechelt)
- b. **qí**~qíx^w (cf. *dx^wqíx^w* ‘toward the head of an inlet’)
DIM~upstream
‘a little upstream’ (Bates et al., 1994, p.187, Lushootseed)
- c. **k^wá**<**k^w**>əl-ám (cf. *k^wálám* ‘glow(ing) from heat/light’)
glow<DIM>-MID
‘dim/weak (of light)’ (Beaumont, 2011, p.603, Sechelt)

The vowels in (39a–b) are copies of the root vowel; however, [ə] is epenthesized in (39c) following the infixed consonant; vowels associated with C₁ reduplication are discussed further in Section 2.5.2 immediately below.

In the prominence-aligned patterns, found in the Northern Interior languages, C₁ reduplication targets the first consonant before the stressed vowel, and the copied consonant is placed after the stressed vowel. Examples from the three Northern Interior Salish languages are given in (40).

- (40) a. ʔəqt-ʔú<ʔ>əy ʔə=lətép (cf. ʔəqtʔúy ‘It’s too wide.’)
 wide-real<DIM> DET=table
 ‘The table is a little too wide.’ (GD, Shuswap)
- b. qemú<ṁ>t (cf. qemút ‘hat’)
 hat<DIM>
 ‘small hat’ (Jimmie, 1994, p.73, Thompson River Salish)
- c. s-pəq-mó<ṁ>x (cf. spəqmúx ‘swan’)
 NMLZ-white-excess<DIM>
 ‘little swan’ (H. Davis & Mellesmoen, 2023, p.15, Lillooet)

The examples in (40) further show that while prefixes are not copied in C₁ reduplication, suffixes may be (40a,c).

Languages with prominence-aligned C₁ reduplication may also have edge-aligned C₁ reduplication. Both patterns are found in Lillooet, as shown in (41).²⁹

- (41) a. *Edge-Aligned: Initial C₁*
- i. n-~~ʔ~~ə~xʔúm-anaʔ (cf. xʔum ‘big’)
 LOC-C₁.INIT-big-ear
 ‘big ears’ (Van Eijk, 1997, p.57, Lillooet)
- ii. s-~~q~~^wə~q^wlíp (cf. sq^wlíp ‘black tree moss’)
 NMLZ- C₁.INIT~black.tree.moss
 ‘lots of black tree moss’ (Van Eijk, 1997, p.57, Lillooet)

²⁹The edge-aligned C₁ reduplication pattern in (41a.ii) is phonologically regular but marginal in occurrence: see Chapter 4 (Section 4.2.3.1) for more about this type of reduplication.

- b. *Prominence-Aligned: Diminutive C₁*
- i. kapó<p>uh (cf. *kapúh* ‘coat’)
coat<DIM>
‘small coat’ (Van Eijk, 1997, p.257, Lillooet)
 - ii. x^wó<x^w>ktən (cf. *x^wík-tən* ‘knife’)
knife<DIM>-instrument
‘small knife’ (Van Eijk, 1997, p.254, Lillooet)

Prominence-aligned C₁ reduplication is only found in Northern Interior Salish. However, even in languages with only edge-aligned reduplication, there may be an additional component that is infixated further into a stem or word. For example, a glottal stop occurs in certain diminutive forms in Halkomelem and Straits (42): [ʔ] may be discontinuous with the rest of the reduplicative morpheme (i.e., realised as Cə...ʔ), as shown in (42b).

- (42) a. s-k^wíʔ~k^wθeʔ (cf. *sk^wθeʔ* ‘island’)
NMLZ-DIM~island
‘little island’ (Leslie, 1979, p.53, Island Halkomelem)
- b. xə~xá<ʔ>čeʔ (cf. *xáčeʔ* ‘lake’)
DIM~lake<DIM>
‘small lake’ (Galloway, 1990, p.46, Samish - Northern Straits)

2.5.2 Vowels in C₁ Reduplication

C₁ reduplication can also involve a vowel, which may be either (i) reduplicated (43a), (ii) a fixed segment (43b), or (iii) an epenthetic segment (43c).

- (43) a. *C₁ Reduplication with Reduplicated Vowel*
ʔú~ʔuq^wu-d (cf. *ʔúq^wud* ‘pull out’)
DIM~pull.out-CTR
‘pull part way out’ (Urbanczyk, 2006, p.183, from Bates et al., 1994, Lushootseed)

- b. *C₁ Reduplication with Fixed Vowel*
 cí~cəqət (cf. *c(ə)qat* ‘tree’)
 DIM~tree
 ‘a little tree’ (Galloway, 1984b, p.86, Nooksack)
- c. *C₁ Reduplication with Epenthetic Vowel*
 cə~cítx^w (cf. *citx^w* ‘house’)
 C₁.INIT~house
 ‘group of houses’ (Van Eijk, 1997, p.58, Lillooet)

A reduplicated, fixed, or epenthetic vowel is not present in all C₁ reduplication instances, however. For example, there are no vowels associated with C₁ reduplication in the examples given in (44).

- (44) a. pu<p>t (cf. *put* ‘boat’)
 boat<DIM>
 ‘little boat’ (Suttles, 2004, p.212, Downriver Halkomelem)
- b. ʔ=n-ci<c>tx^w (cf. *citx^w* ‘house’)
 DET=1SG.POSS-house<DIM>
 ‘my house’ (Kuipers, 1974a, p.39, Shuswap)

The absence of a vowel is determined by (i) the source of the vowel when present (underlying, copy, or epenthesis) and (ii) the grammar of the language. A vowel is present if supplied in the underlying form (fixed segment) or if the affixed prosodic unit requires one; an affixed mora can be filled by a coda consonant alone. Vowels may be omitted later in the derivation if consistent with patterns of unstressed vowel deletion in the language. Conversely, if reduplication of a consonant creates an illicit cluster, phonological repairs later in the derivation can supply an epenthetic vowel.

2.5.3 Meanings Associated with C₁ Reduplication

The most common meaning of C₁ reduplication is diminutivity, which is found in the 19 languages in my survey with C₁ reduplication. C₁ reduplication is also associated with aspectual and plural meanings across Salish. Diminutive C₁

reduplication is discussed in Section 2.5.3.1, aspectual C₁ reduplication in Section 2.5.3.2, and plural C₁ reduplication in Section 2.5.3.3.

2.5.3.1 Diminutive C₁ Reduplication

Diminutive reduplication on nouns either marks a small entity (count nouns), or a small amount (mass nouns), as shown in (45a) and (45b), respectively. Diminutive nouns may also refer to the young of an animal (45c) or be used as terms of endearment or familiarity (45d).

- (45) a. cí<č>čómin (cf. *cíčómin* ‘fork’)
fork<DIM>
‘little fork’ (Beaumont, 2011, p.874, Sechelt)
- b. s-nú<ń>yeʔ (cf. *snúyeʔ* ‘money’)
NMLZ-money<DIM>
‘a small amount of money’ (Jimmie, 1994, p.73, Thompson River Salish)
- c. s-cí~cqay (cf. *scóqay* ‘sockeye’)
NMLZ-DIM-sockeye
‘young sockeye (salmon)’ (Beaumont, 2011, p.795, Sechelt)
- d. pí~puʔtəd (cf. *púʔtəd* ‘shirt’)
DIM~shirt
‘thin shirt’ or ‘favourite shirt’ (Hess, 1967, p.62, Lushootseed)

Diminutive C₁ reduplication on adjectives refers to a lesser degree of the property denoted by the adjective (46a), or a lesser degree of difference between the property and a standard, in cases of comparison (46b).

- (46) a. híʔ~łqʷ (cf. *łəqʷ* ‘wet’)
DIM~wet
‘damp’ (Hess, 1967, p.8, Lushootseed)
- b. xzə<z>əm (cf. *xzum* ‘big’)
big<DIM>
‘a little bit bigger’ (H. Davis & Mellesmoen, 2023, p. 13, Lillooet)

Diminutive C₁ reduplication has a range of meanings with verbs. It may denote the small size of an event participant (47a), indicate that the event is diminished in intensity (47b), or indicate that the event is restricted in some way (generally in space or time) (47c).

- (47) a. **x^wí~x^witil** (cf. *x^wítíl* ‘fall from height’)
 DIM~fall.from.height
 ‘small item(s) fall(s) from a height’ (Hess, 1967, p.62, Lushootseed)
- b. **ííʔ~lu:-d** (cf. *lu:d* ‘hear something’)
 DIM~hear-CTR
 ‘hear something a little’ (Urbanczyk, 2006, p.208, from Bates et al. 1994, Lushootseed)
- c. **k^wí<k>^wt-ús-əm** (cf. *k^wítúsəm* ‘look out window’)
 look<DIM>-face-MID
 ‘look briefly out of the window’ (Beaumont, 2011, p.621, Sechelt)

2.5.3.2 Aspectual C₁ Reduplication

Aspectual C₁ reduplication is found in Central Salish (Kroeber, 1999, p.10). It has imperfective or progressive semantics.³⁰ For example, Bar-el (2008) shows that aspectual C₁ reduplication in Squamish marks the progressive. H. Davis, Huijsmans, and Mellesmoen (2020, p.120) argue that aspectual C₁ reduplication in Comox-Sliammon is also progressive. Examples of progressive C₁ reduplication from Squamish and Comox-Sliammon are given in (48).

- (48) a. **tú~túk^w** (cf. *túk^ws* ‘take (something) home’)
 PROG~go.home
 ‘be going home’ (Dyck, 2004, p.200, Squamish)
- b. **θa~θap-iš** (cf. *θapiš* ‘bathe’)
 PROG~bathe-TR
 ‘be bathing’ (Watanabe, 2003, p.392, Comox-Sliammon)

³⁰Other labels have been used to describe imperfective C₁ reduplication. For example, Montler (1986) uses the label “actual” for the Saanich dialect of Northern Straits, which is equivalent to what Galloway (1984b) calls “continuative” in his description of the Samish dialect.

2.5.3.3 Plural C₁ Reduplication

A third type of C₁ reduplication marks plurality. In Comox-Sliammon, for example, C₁ reduplication is used to mark plurality of arguments when applied to adjectives or derived states, as shown in (49a). C₁ reduplication marks multiple repeated events in Montana Salish-Spokane-Kalispel and Moses Columbia, as shown in (49b). I gloss the latter as REP.

- (49) a. *?a<ʔ>mut* (cf. *?amut* ‘be home’)
 be.home<PL>
 ‘They are all home’ (Watanabe, 2003, p.376, Comox-Sliammon)
- b. *ka~kíc-stu-m-s* (cf. *kn kicx* ‘I arrive’)
 REP~arrive-TR-1SG.OBJ-3ERG
 ‘They came here (to see me) several times.’ (Willett, 2003, p.209, Moses Columbia)

Lushootseed uses C₁ reduplication to mark plurality on nouns and plural arguments on verbs, as shown in (50). The same is true of Lillooet, though two different types of reduplication are used, as shown in (51): a (marginal) initial C₁ reduplication marks plural entities and an infix type of C₁ reduplication marks plural events (H. Davis & Mellesmoen, 2023).³¹

- (50) a. *s-ďá~duuk^w* (cf. *sduuk^w* ‘knife’)
 NMLZ-PL~knife
 ‘knives’ (Kirkham, 1991, p.1, Lushootseed)
- b. *sá~sax^w-əb* (cf. *sáx^wəb* ‘run, jump’)
 PL~run-MID
 ‘many run away’ (Hess, 1967, p.64, Lushootseed)
- c. *sá~saq^w* (cf. *saq^w* ‘fly’)
 PL~fly
 ‘many fly up’ (Hess, 1967, p.64, Lushootseed)

³¹There is a limited, likely lexicalised, version of initial C₁ reduplication that marks aspect in Lillooet. Examples are given in Chapter 4 (Section 4.2.3.1).

- (51) a. s-q^wə~q^wlíp (cf. *sq^wlip* ‘black tree moss’)
 NMLZ- C₁.INIT~black.tree.moss
 ‘lots of black tree moss’ (Van Eijk, 1997, p.57, Lillooet)
- b. x^wəlp-í<p>ləx (cf. *x^wəlpilx* ‘turn around’)
 turn<C₁.PL>-AUT
 ‘turn around and around’ (H. Davis & Mellesmoen, 2023, p.21, Lillooet)

Thompson River Salish has two types of C₁ reduplication: it has an initial (edge-aligned) C₁ reduplication pattern, distinct from its infix (prominence-aligned) C₁ reduplication, marking diminutivity. The meaning and productivity of this edge-aligned C₁ reduplication are unclear, but it has been labelled “affective” by L. C. Thompson and Thompson (1996). I include it with C₁ plural reduplication here because I group it with Lillooet initial C₁ reduplication based on form, though little can be said about its meaning at present. Some examples are given in (52).

- (52) a. x̣^wə~x̣^wáx̣^w~ox̣^w (cf. *ṇx̣^wax̣^wáyeʔqn* ‘summer shelter’)
 AFF~stick~COS
 ‘have many shoots’ (L. C. Thompson & Thompson, 1996, p.436, Thompson River Salish)
- b. ʔ^wə~ʔ^wóyt (diminutive cf. *ʔ^wóʔ^wiʔt* ‘take a nap’)
 AFF~sleep
 ‘sleep erratically’ (L. C. Thompson & Thompson, 1996, p.480, Thompson River Salish)
- c. ʔes-cə~cəʔ^w (cf. *ʔescóʔ^w* ‘have a stripe, be striped’)
 STAT-AFF~stripe
 ‘(The pattern is) all striped.’ (L. C. Thompson & Thompson, 1996, p.37, Thompson River Salish)

2.5.3.4 Person-Linked C₁ Reduplication

Unique amongst Salish languages, Shuswap uses C₁ reduplication with a first person argument including possessor, subject, and object, as shown in (53).

- (53) a. $\gamma=n-ci<\mathbf{c}>tx^w$ (cf. *citx^w* ‘house’)
 DET=1 SG.POSS-house<RED>
 ‘my house’ (Kuipers, 1974a, p.39, Shuswap)
- b. $ki<\mathbf{k}>cx-kn$ (cf. *tkicx* ‘arrived there’)
 arrive<RED>-1 SG.SUBJ
 ‘I arrive.’ (Kuipers, 1974a, p.39, Shuswap)
- c. $m-wí<\mathbf{w}>k-t-səm-s$ $rə=Mary$
 COMPL-see<RED>-TR-1 SG.OBJ-3SUBJ DET=Mary
 ‘Mary saw me.’ (cf. *swétý kwikts* ‘Whom did they (sg.) see?’)
 (Gardiner, 1996, p.175, Shuswap)

C_1 reduplication marking first person is prominence-aligned and identical in form to diminutive C_1 reduplication. Examples of diminutive C_1 reduplication are provided in (54).

- (54) a. Sander $?əpək^wut-s$ $ʔət\sim ʔá<\mathbf{ʔ}>\dot{\lambda}$ (cf. *ʔa\dot{\lambda}* ‘wet’)
 Sander coat-3POSS PDEN~wet<DIM>
 ‘Sander’s coat is a little wet.’ (Shuswap: GD, vf)
- b. Sander $x^w?it$ $tə=s-təq\sim tí<\mathbf{\dot{\lambda}}>qt$ $\gammaə=s-wík-əm-s$
 Sander many DET=NMLZ-PL~cloud<DIM> DET=see-MID-3POSS
 ‘Sander saw a lot of small clouds.’ (cf. *s\dot{\lambda}iqt* ‘cloud’) (Shuswap: GD, vf)

2.5.3.5 Distinguishing Types of C_1 Reduplication

C_1 reduplication patterns may be distinguished by position. Thompson River Salish and Lillooet each have a marginal edge-aligned C_1 reduplication in addition to their productive prominence-aligned C_1 patterns. Recall that Lillooet has two prominence-aligned C_1 reduplication patterns: one that marks diminutivity and one that marks pluractionality.³²

³²Though they surface in the same position, pluractional and diminutive C_1 reduplication in Lillooet can be distinguished by the vowel (H. Davis & Mellesmoen, 2023). An analysis is given in Chapter 4 (§4.2.4.2).

- (55) a. **kə**~kláx^w (cf. *klax*^w)
 C₁.INIT~muskrat
 ‘muskrat(s)’ (Van Eijk, 1997, p.56, Lillooet)
- b. s-ɣə<ɣ>p (cf. *sɣap*)
 NMLZ-tree<DIM>
 ‘little tree, sapling’ (H. Davis & Mellesmoen, 2023, p.13, Lillooet)
- c. cá<č>q-ən (cáqan ‘pull s.t. out’)
 pull.out<C₁.PL>-CTR
 ‘pluck something (a bird)’ (H. Davis & Mellesmoen, 2023, p.24, Lillooet)

Different types of C₁ reduplication may also be distinguished by vowel quality. For example, in Lushootseed, a fixed segment /a/ is associated with the plural; the diminutive either copies a vowel or has the fixed segment /i/. Squamish patterns with Lushootseed in having /i/ associated with the diminutive, as shown in (57). However, the use of the diminutive in Squamish is very limited (Dyck, 2004). Aspectual C₁ reduplication, which copies a vowel, is much more common.

- (56) a. s-**da**~duuk^w (cf. *sduuk*^w ‘knife’)
 NMLZ-PL~knife
 ‘knives’ (Kirkham, 1991, p.1, Lushootseed)
- b. s-**dí**~duuk^w (cf. *sduuk*^w ‘knife’)
 NMLZ-DIM~knife
 ‘small knife’ (Urbanczyk, 2006, p.208, from Bates et al. 2004, Lushootseed)
- (57) a. lí~lam (cf. *lam* ‘house’)
 DIM~house
 ‘little house’ (Squamish Nation Education Department, 2011, p.103, Squamish)
- b. na **pá**~payaq-ant-as ta s-nəx^wit-s (cf. *páyaq* ‘be fixed’)
 RL PROG~fix-TR-3ERG DET canoe-3POSS
 ‘They (sg.) are in the process of fixing the canoe.’ (Bar-el, 2008, p.47, Squamish)

There is good evidence for at least three types of C₁ reduplication across Salish on the basis of both form and meaning: diminutive, plural, and aspectual.³³ Diminutive and plural C₁ reduplication frequently have a fixed vowel, while aspectual C₁ does not.

2.5.3.6 Distribution of C₁ Reduplicative Morphemes

Table 2.5 shows the number of languages with at least one C₁ reduplication process.

Table 2.5: Number of Languages with C₁ Reduplication by Branch

Branch	C ₁	No C ₁
Central Salish	10	0
Tillamook	1	0
Tsamosan	0	4
Interior Salish	7	0
Bella Coola	1	0

A summary of the meanings associated with C₁ reduplication in each language is given in the table below. I use (✓) to indicate the more marginal edge-aligned C₁ reduplication processes in Thompson River Salish (“affective”) and Lillooet (“initial”), where the languages have more productive prominence-aligned C₁ reduplication.

³³In Central Salish, aspectual C₁ reduplication marks imperfective aspect.

Table 2.6: Types of C₁ Reduplication

# of Types	Language	Diminutive	Plural	Aspect	Person-Linked
3	Comox-Sliammon, Sechelt, Halkomelem	✓	✓	✓	
3	Lillooet	✓	✓ (✓)		
2	Moses Columbia, Tillamook, Montana Salish-Spokane-Kalispel, Lushootseed	✓	✓		
2	Shuswap	✓			✓
2	Squamish, Pentlatch, Straits	✓		✓	
2	Thompson River Salish	✓	(✓)		
1	Coeur d'Alene, Okanagan, Twana, Nooksack	✓			
0	Upper Chehalis, Quinault, Lower Chehalis, Cowlitz				
Unclear	Bella Coola				

All languages with C₁ reduplication use it to mark diminutivity. Eight also use C₁ reduplication to mark plurality, and seven (all in Central Salish) use C₁ reduplication to mark an aspectual contrast (imperfectivity).

Unlike with C₁C₂ reduplication (Section 2.4.3.3), different types of C₁ reduplication can be distinguished on the basis of form in every language, aside from Shuswap, where both diminutive and first-person are marked with prominence-aligned C₁ reduplication.³⁴ This is summarised in Table 2.7, which shows whether there is a difference in phonological form between the different types of C₁ reduplication in each language.

³⁴There may also be an edge-aligned C₁ reduplication process in Shuswap, however the data are too sparse at this point for me to draw this conclusion.

Table 2.7: C₁ Reduplication: Form-to-Meaning

Three Meanings	
<i>Diminutive, Plural, and Aspectual</i>	
<i>Different Forms</i>	Comox-Sliammon, Sechelt, Halkomelem
<i>Diminutive, Plural (Initial), and Plural (Pluractional)</i>	
<i>Different Forms</i>	Lillooet
Two Meanings	
<i>Diminutive and Plural</i>	
<i>Different Forms</i>	Montana Salish-Spokane-Kalispel, Tillamook, Lushootseed, Moses Columbia
<i>Diminutive and First Person</i>	
<i>Same Form</i>	Shuswap
<i>Diminutive and Aspectual</i>	
<i>Different Forms</i>	Straits (Klallam and Northern Straits), Squamish, Pentlatch
<i>Diminutive and “Affective”</i>	
<i>Different Forms</i>	Thompson River Salish
One Meaning	
<i>Diminutive</i>	
Coeur d’Alene, Okanagan, Twana, Nooksack	
Unclear/None	
Bella Coola, Tsamosan Languages	

2.5.4 Summary: C₁ Reduplication

If a language only has a single C₁ reduplication pattern, it will be diminutivity. The only languages without diminutive reduplication are the Tsamosan languages, which lack any process of C₁ reduplication, regardless of meaning.

Plural C₁ reduplication is scattered throughout the family: it is found in Southern and Northern Interior Salish, as well as in Central Salish and Tillamook. It is likely that this type of reduplication is more widespread than previously thought. For example, “consonant reduplication” as described by Van Eijk (1997) actually contains two separate C₁ reduplicative processes, distinguished by vowel quality: a plural C₁ with a full vowel, and a diminutive C₁ with a schwa (H. Davis

& Mellesmoen, 2023). In the eight languages with plural C₁ reduplication, it is associated with pluractionality (e.g., repetition or plural participants), rather than marking plurality in the nominal domain.

While aspectual C₁ reduplication is limited to Central Salish, it is a core part of the aspectual system in these languages and therefore well-attested in existing documentation.

2.6 C₂ Reduplication

C₂ reduplication copies either the second consonant in the root (edge-aligned reduplication) or the consonant following a stressed vowel (prominence-aligned reduplication). The positions and vowel qualities associated with C₂ reduplication are summarised in Sections 2.6.1 and 2.6.2, respectively. The meanings associated with C₂ reduplication are covered in Section 2.6.3. Triplication is discussed in Section 2.6.4. I summarise the distribution of C₂ reduplication across Salish languages in Section 2.6.5.

2.6.1 Position of C₂ Reduplication

C₂ reduplication may be edge-aligned or prominence-aligned. Edge-aligned patterns are found in Central Salish, Southern Interior Salish, and Tillamook; prominence-aligned patterns are found in Northern Interior Salish and Tsamosan.

In edge-aligned C₂ reduplication, a copy of the second consonant of the root immediately follows the consonant it copies. Examples from Coeur d’Alene are given in (58).

- (58) a. $\dot{q}ey \sim \dot{\alpha}y - \acute{u}t - \acute{\alpha}m$ (cf. *sqéyám* ‘writing’)
 write~COS-STAT-INDEF.OBJ
 ‘It is possible to write.’ (Reichard, 1938, p.607, Coeur d’Alene)
- b. $ni\acute{c} \sim \acute{\alpha}c - \acute{u}t - \acute{\alpha}m$ (cf. *ci:ntcám* ‘S/he cut some.’)
 cut-COS-STAT-INDEF.OBJ
 ‘It is cut-able.’ (Reichard, 1938, p.607, Coeur d’Alene)

Edge-aligned C₂ reduplication is positioned relative to the left-edge of the word: where the root has more than two consonants, the reduplicated consonant is

between the second and third consonants (59).

- (59) a. $\acute{q}íl<\acute{il}>t$ (cf. *qílt* ‘bright (moon, sky)’)
 bright<COS>
 ‘daybreak, dawn, getting light’ (Beaumont, 2011, p.645, Sechelt)
- b. $g^w\acute{e}l<\acute{e}l>\acute{a}pu$ (cf. *g^welápu* ‘you folks’)
 2SG.INDEP<C₂>
 ‘just you folks’ (Hess, 1967, p.66, Lushootseed)

In contrast, C₂ reduplication in Lillooet and Thompson River Salish is prominence-aligned and positioned relative to stress, as shown in (60) and (61), respectively.

- (60) a. $p\acute{e}t\sim p\acute{u}<\acute{e}t>t$ (cf. *pútxal* ‘to boil smth.’)
 PL~boil<COS>
 ‘rapids’ (Van Eijk, 1997, p.57, Lillooet)
- b. $c\acute{e}q\sim c\acute{í}<q\acute{e}>q$ (cf. *cíqeq* ‘get stabbed’)
 PL~stab<COS>
 ‘to get stabbed all over’ (Van Eijk, 1997, p.57, Lillooet)
- (61) n-ceqé $\acute{m}\sim\acute{m}$ (cf. *nceqémix* ‘recline, lie on back’)
 LOC-lie.on.back~COS
 ‘fall backwards accidentally’ (L. C. Thompson & Thompson, 1996, pp.20–21, Thompson River Salish)

The pattern in Shuswap is harder to discern because there are fewer documented examples. Further documentation of longer words is needed.³⁵ However, even if C₂ reduplication is prominence-aligned in Shuswap, it differs from the other Northern Interior languages because the reduplication is positioned before the stressed syllable rather than afterwards, as shown in (62).

- (62) a. $\acute{x}<\acute{e}x^w>\acute{u}x^w$ (cf. *xux^w* ‘numb’)
 numb<COS>
 ‘(got) numb’ (Kuipers, 1974a, p.39, Shuswap)

³⁵In my own fieldwork, I was not successful in eliciting C₂ reduplication with any multisyllabic words with stress on a syllable beyond the first two, which could differentiate between edge-aligned and prominence-aligned reduplication.

- b. $\dot{\lambda} < \text{əs} > \acute{a}s$ (cf. $\dot{\lambda}as$ ‘make a splashing sound’)
 splash <COS>
 ‘(got) soggy’ (Kuipers, 1989, p.163, Shuswap)

2.6.2 Vowels in C₂ Reduplication

The vowel accompanying C₂ reduplication may be a faithful copy, a fixed segment, or an epenthetic vowel (/ə/).

Tsamosan and Central Salish languages other than Twana with C₂ reduplication have a faithful copy of the vowel. Examples from Lushootseed are given in (63).

- (63) a. $dáy \sim \acute{a}y$ (cf. $dáy$ ‘very, only’)
 only ~ C₂
 ‘alone’ (Hess, 1967, p.66, Lushootseed)
- b. $s\acute{t}ád < \text{ad} > \acute{a}y?$ (cf. $s\acute{t}ád\acute{a}y?$ ‘woman’)
 NMLZ-woman <C₂>
 ‘woman (living) alone’ (Hess, 1967, p.66, Lushootseed)

Twana has a fixed segment /a/ in C₂ reduplication.³⁶ Examples are provided in (64).

- (64) a. $s\text{?}áq^w \sim \acute{a}q^w$ (root: $\text{?}áq\text{-}$)
 NMLZ-snow ~ C₂
 ‘snow’ (Drachman, 1969, p.172, Twana)
- b. $bíx^w \sim \acute{a}x^w$ (root: $bíx^w\text{-}$)
 smile ~ C₂
 ‘smile’ (Drachman, 1969, p.279, Twana)

All of the Interior Salish languages as well as Tillamook have an epenthetic /ə/ in C₂ reduplication. An example from Lillooet is provided in (65).

³⁶There is also a C₂ process associated with distributive numerals, but it is not possible to discern the vowel source because there are too few examples to establish a generalisation.

- (65) pú<ɬə>ɬ (cf. *púɬxal* ‘to boil smth.’)
 boil<COS>
 ‘come to a boil’ (Van Eijk, 1997, p.56, Lillooet)

When the vowel associated with C₂ reduplication is /ə/, its presence is determined by the general phonology of the language.³⁷

2.6.3 Meanings Associated with C₂ Reduplication

Across the family, C₂ reduplication is primarily associated with a change-of-state meaning. Kroeber (1988) compares the meaning of C₂ reduplication in Interior Salish and Comox-Sliammon, contrasting its putative “out of control” meaning in the Interior languages with “inceptivity” in Comox-Sliammon.

There are a few other meanings associated with C₂ reduplication. In Lushootseed, Hess (1967) states that C₂ reduplication is used to mark something “isolated”/“restricted” (66a), or something “aimless” (66b).

- (66) a. s-ɬád<ad>əyʔ (cf. *sɬádəyʔ* ‘woman’)
 NMLZ-woman<C₂>
 ‘woman (living) alone’ (Hess, 1967, p.66, Lushootseed)
- b. sáx^w~ax^w-əb (cf. *sáx^wəb* ‘run, jump’)
 run~C₂-MID
 ‘scurrying about trying to get things done’ (Hess, 1967, p.66, Lushootseed)

Kinkade (1985) finds that the most frequent meaning associated with C₂ reduplication in Upper Chehalis is to indicate that a change-of-state is slow or gradual. He finds 41 roots which allow for this use. Examples are provided in (67a-b). The next most frequent meaning associated with C₂ reduplication is the superlative, shown in (67c), which Kinkade finds with 29 roots. However, all the superlative uses of C₂ reduplication he lists are provided in combination with the superlative *sx^w...-s(t)*.

³⁷See the analysis of C₂ reduplication in Lillooet in Chapter 4 (Section 4.2) for further discussion and analysis.

- (67) a. $\acute{\lambda}i\acute{s}\sim\acute{a}w\sim\mathbf{aw}\sim m$ (cf. $\acute{\lambda}i\acute{s}\sim\acute{a}w\sim m$ ‘get cold’)
cold-INCH $\sim$ C₂-MID
‘it got cold slowly’ (Kinkade, 1985, p.191, Upper Chehalis)
- b. $s\sim t\acute{i}p\sim\mathbf{ip}\sim i\sim w\sim n$ (cf. $t\acute{i}p\acute{t}$ ‘get dark’)
IPFV-dark $\sim$ C₂ $\sim$ -INCH-IPFV
‘It gets dark gradually.’ (Kinkade, 1985, p.193, Upper Chehalis)
- c. $sx^w\sim\acute{\lambda}u\acute{k}^w\sim\mathbf{k}^w\sim st$ (cf. $\acute{\lambda}u\acute{k}^w$ ‘high’)
SUPR-high $\sim$ C₂-SUPR
‘highest’ (Kinkade, 1985, p.193, Upper Chehalis)

The other categories Kinkade (1985, p.189) lists include “persistent, frequent” (9 roots), “while going” (3 roots), “become a colour” (4 roots), “by oneself” (limited to independent pronouns), counting forms (limited to numbers), and plural (only applies to the lexical suffix *-əlwas* meaning ‘middle’). C₂ reduplication occurs with an additional 12 roots without a clear meaning (Kinkade, 1985, p.190). The English translations given by Kinkade for examples in the “persistent, frequent”, “while going”, “become a colour”, and residual categories suggest that all might involve a basic change of state semantics. For example, contrast (67b) above, which has a separate inchoative marker, with the examples given in (68) below. Kinkade (1985) labels (68a) as “become a colour”, while (68b) is listed under “persistent, frequent”. I group these uses, and the superlative, shown in (67c) above, under change of state.

- (68) a. $s\sim t\acute{i}p\sim\mathbf{ip}\sim n$ (cf. $t\acute{i}p\acute{t}$ ‘get dark’)
IPFV-dark $\sim$ COS-IPFV
‘it is kind of dark’ (Kinkade, 1985, p.194, Upper Chehalis)
- b. $s\sim\acute{t}an\acute{a}p\sim\mathbf{ap}\sim n$ (root: $\acute{t}an\acute{a}p\sim$)
IPFV-dark $\sim$ COS-IPFV
‘It gets darker and darker.’ (Kinkade, 1985, p.194, Upper Chehalis)

In Twana, there is a type of C₂ reduplication that Kim (2017a) analyses as truncated C₁C₂ reduplication. The consonant in C₂ position is reduplicated and the copied segment is positioned on the left edge of the root (prior to C₁). This type of reduplication is also called “wrong side” reduplication by Kim and Gardiner

(2016), who find a similar pattern in Tillamook. An example from Twana is given in (69). The meaning is consistent with C₁C₂ reduplication (i.e., it marks plurality). This type of reduplication is analysed in Chapter 6 (Section 6.5.3.3.2).

- (69) s-**q**~téqaw (cf. *steqéw* ‘horse’)
 NMLZ-PL~horse
 ‘horses’ (Drachman, 1969, p.32, Twana)

2.6.4 “Triplication” (C₂² Reduplication)

There is another reduplication pattern that I have classified under C₂ reduplication on the basis of shape. This “triplication” pattern appears to involve two instances of C₂ reduplication. This C₂² reduplication refers to the continuation of an event in Lillooet (H. Davis, p.c.), as shown by the examples in (70). C₂² reduplication is also found in Shuswap and Okanagan, as shown in (71) and (72), respectively.

- (70) a. t<**əlal**>íl (cf. *ílin* ‘sprinkle something’)
 sprinkle<TRIP>
 ‘to keep sprinkling’ (Van Eijk, 1997, p.56, Lillooet)
 b. ^ˈχ<**ətəp**>śí (cf. *χətp* ‘shake, shiver’)
 shiver<TRIP>
 ‘to keep shivering’ (Van Eijk, 1997, p.56, Lillooet)
- (71) a. p<**ətət**>ét (cf. *snptóleʔx^w* ‘dumplings’)
 bubble/boil<TRIP>
 bubble, boil (Kuipers, 1989, p.22, Shuswap)
 b. t<**ək^wək^w**>ík^w (cf. *tʔik^w* ‘fire’)
 fire<TRIP>
 ‘throw off sparks’ (Kuipers, 1989, p.22, Shuswap)
- (72) a. tər<**qq**>áq (root: *tíq-*)
 jump/kick/dance<TRIP>
 ‘jump up and down’ (A. Mattina, 1993, p.221, Okanagan)
 b. t<**q^wq^w**>iq^w (cf. *tíq^wt* ‘burst, explode’)
 burst/explode<TRIP>
 ‘explosion’ (A. Mattina, 1993, p.221, Okanagan)

Though it has not been discussed explicitly in descriptions of Thompson River Salish, I have been able to locate the following form in L. C. Thompson and Thompson (1996, p.422), which they describe as an “unusual apparent double reduplication”; they do not identify the root.

- (73) a. $\text{xən} < \dot{\text{q}}\text{a}\dot{\text{q}} > \acute{\text{a}}\dot{\text{q}} \sim \text{xən} < \dot{\text{q}}\text{ə}\dot{\text{q}} > \acute{\text{a}}\dot{\text{q}}$
 ?<TRIP>
 ‘rumbling, snorting noise’ (L. C. Thompson & Thompson, 1996, p.422, Thompson River Salish)

C_2^2 reduplication patterns are notable because stress always falls on the final syllable, contra to standard patterns of stress assignment.

On the surface, these patterns could be analysed as the combination of two C_2 reduplicative morphemes (i.e., as multiple reduplication) or as one instance of C_2^2 reduplication (i.e., as triplication). I analyse this pattern as triplication in the case study of Lillooet reduplication presented in Chapter 4 (Section 4.2.5).

2.6.5 Summary: C_2 Reduplication

C_2 reduplication is found in all branches of the Salish family. However, it is only present in four out of ten Central Salish languages and in one of four Tsamosan languages. It occurs in all of the Interior Salish languages, and is also found in Tillamook. The table below summarises the distribution of C_2 reduplication across the family.

Table 2.8: C_2 Reduplication in Salish Languages by Branch

	C_2 Reduplication	None/ Unclear
Central Salish (n=10)	4	6
Tillamook (n=1)	1	0
Tsamosan (n=4)	1	3
Interior Salish (n=7)	7	0
Bella Coola (n=1)	0	1
Total (n=23)	13	10

2.7 V₁ Reduplication

Only a vowel is copied in V₁ reduplication. This is the least frequent of the four core types of reduplication; it is found in seven languages, including three from the Central branch, three from the Interior branch, and one from the Tsamosan branch. I discuss the form and meaning of V₁ reduplication by branch: Central Salish in Section 2.7.1, Interior Salish in Section 2.7.2, and Tsamosan in Section 2.7.3.

2.7.1 Central Salish V₁ Reduplication

Examples of V₁ reduplication in Central Salish are given in (74). The examples in (74a–b) show a frequent combination of V₁ reduplication with C₁ reduplication in Twana and Comox-Sliammon; there is an additional glottal stop, which may be analysed as a fixed or epenthetic segment, inserted to provide an onset for the syllable created by copying the vowel. In examples like (74c) from Lushootseed, it is not clear if vowel reduplication is realised as two adjacent vowels or a long vowel (i.e., VV vs. V:).

- (74) a. q<iʔ>i~qp-aʔam (cf. *qəpʔəm* ‘touch (s.t.)’)
DIM<PL>~touch-A.INTR
‘touch repeatedly’ (Watanabe, 2003, p.403, Comox-Sliammon)
- b. be-kʷ<aʔ>á<kʷ>šə-čəd (cf. *bekʷašáčəd* ‘I’m cutting things.’)
CONT-cut<DIM><DIM>-1SG.SUBJ
‘I’m cutting a little.’
‘I’m cutting things small.’ (Drachman, 1969, p.137, Twana)
- c. čá<a>gʷəs (cf. *čəgʷás* ‘wife’)
wife<PL>
‘wives’ (Hess, 1967, p.65, Lushootseed)

V₁ reduplication in Twana is associated with diminutivity. It is found in combination with diminutive C₁ reduplication, which can either involve the affixation of a glottal stop in addition to reduplication (shape: ʔC) or copy a vowel (shape: CV). For this reason, I analyse V₁ reduplication in Twana as a realisation of the ʔC diminutive: a consonant is copied and there is a floating constricted

glottis feature ([c.g.]). Vowel reduplication is triggered to avoid a complex onset (e.g., [qʔ] in (74a) above).

V₁ reduplication in Lushootseed and Comox-Sliammon marks plurality. According to Hess (1967, p.64), in Lushootseed it marks plural arguments with verbs and plurality with nouns. Hess (1967, p.65) notes that V₁ reduplication with nouns is not a productive form of the plural. In Comox-Sliammon, V₁ marks a pluractional that involves temporally distributed events (Mellesmoen et al., 2020). In Comox-Sliammon, V₁ reduplication typically co-occurs with another type of reduplication: see description and analysis in Chapter 4 (Section 4.1.5.2).

2.7.2 Interior Salish V₁ Reduplication

While there is no evidence for V₁ reduplication in the Northern Interior branch, it is attested in three of the Southern Interior languages: Coeur d’Alene (75a), Montana Salish-Spokane-Kalispel (75b-c), and Okanagan (75d).³⁸

- (75) a. hən-tá<ʔa>š-álp (cf. *taš* ‘be inherently sweet’)
 LOC-sweet<INCH>-feeling
 ‘I felt sweetness’ (Reichard, 1938, p.612, Coeur d’Alene)
- b. n<aʔ>ás (cf. *hi nas* ‘It is wet.’)
 wet<INCH>
 ‘They (sg.) get wet.’ (Vogt, 1940, p.64, Kalispel)
- c. q^w<uʔ>úł (cf. *hi q^wuł* ‘It is dusty.’)
 dusty<INCH>
 ‘It gets dusty’ (Vogt, 1940, p.64, Kalispel)
- d. q^w<aʔ>ác (cf. *q^waʔcínaʔ* ‘warm weather’)
 warm<INCH>
 ‘It is getting warm.’ (Watkins, 1970, p.100, Okanagan)

V₁ reduplication in the examples above accompanies the inchoative affix -ʔ-: see Section 2.2.2. I treat vowel reduplication in these cases as an allomorph of the

³⁸My choice for morpheme breaks is informed by the position of the inchoative marker -ʔ- in words without V₁ reduplication. CVʔC surfaces in Coeur d’Alene where it is CʔVC in Okanagan and Montana Salish-Spokane-Kalispel (A. Mattina, 1993, pp.231–232).

inchoative infix. As there is variation across different descriptions as to whether the vowel is present with these forms, it is possible that there are differences between dialects and speakers regarding whether and when the inchoative triggers reduplication.

In addition to inchoative V₁ reduplication, Montana Salish-Spokane-Kalispel marks plurality with V₁ reduplication. Examples are provided in (76). Plural V₁ reduplication behaves like the inchoative V₁ reduplication, except the copied vowel is positioned after the stressed vowel, rather than before.

- (76) a. cí<ʔi>tx^w-s (cf. *citx^w* ‘house’)
house<PL>-3POSS
‘their house’ (Thomason, 1997, p.354, Montana Salish)
- b. nunx^wé<ʔe>nʔ-em (cf. *nunx^wéneʔ* ‘believe’)
believe<PL>-MID?
‘They (sg.) believed.’ (Pete, 2010, p.51, Montana Salish)

2.7.3 Tsamosan V₁ Reduplication

V₁ reduplication is documented by Modrow (1971) for Quinault. The stressed vowel is copied and a glottal stop separates the identical vowels, as shown in (77). V₁ reduplication in Quinault is associated with comparatives.

- (77) a. q^wumí<ʔi>n (cf. *q^wumín* ‘slow’)
slow<COMP>
‘slower’ (Modrow, 1971, p.356, Quinault)
- b. ná<ʔa>w (cf. *náw* ‘big’)
big<COMP>
‘bigger’ (Modrow, 1971, p.356, Quinault)

While Upper Chehalis and Cowlitz do not have vowel reduplication, as defined by a 1:2 output mapping, vowel lengthening is used to mark the diminutive and results from the affixation of a mora (Kinkade 1998, p.204; Mellesmoen 2022, p.73). While I do not count vowel lengthening as a type of reduplication, I discuss its theoretical relevance to the model of non-concatenative morphology that I adopt in Chapter 6 (Section 6.6.2.2.1).

2.7.4 Summary: V₁ Reduplication

Based on available documentation, I count seven languages with V₁ reduplication. Its distribution across the family is summarised in Table 2.9.

Table 2.9: V₁ Reduplication in Salish Languages by Branch

	V ₁ Reduplication	None/ Unclear
Central Salish (n=10)	3	7
Tillamook (n=1)	0	1
Tsamosan (n=4)	1	3
Interior Salish (n=7)	3	4
Bella Coola (n=1)	0	1
Total (n=23)	7	16

The meanings of V₁ reduplication are summarised in the table below.

Table 2.10: Meanings of V₁ Reduplication

Meaning	Languages
Plural/Pluractional	Comox-Sliammon (Mellesmoen et al., 2020), Montana Salish-Spokane-Kalispel (Thomason, 1997), Lushootseed (Hess, 1967)
Diminutive	Twana (Drachman, 1969)
Comparative	Quinault (Modrow, 1971)
Inchoative	Coeur d'Alene (Reichard, 1938), Montana Salish-Spokane-Kalispel (Vogt, 1940), Okanagan (Watkins, 1970)

V₁ reduplication is used to mark plurality in three languages, and is associated with scalar functions (diminutive and comparative) in two. The change-of-state (inchoative) use is found in three languages, where vowel reduplication is triggered by the infixation of the inchoative marker (-ʔ-). Montana Salish-Spokane-Kalispel is the only language to have more than one V₁ reduplication process: vowel reduplication can mark plurality or surface as part of the inchoative.³⁹

³⁹There may be an underlying /ʔ/ in the plural as well.

2.8 Multiple Reduplication

Multiple reduplication, the combination of more than one reduplicative morpheme in a single word, is attested in 15 of the 23 Salish languages.

Table 2.11: Multiple Reduplication in Salish Languages by Branch

	Yes	No	Lacks Sufficient Data
Central Salish (n=10)	7	2	1
Tillamook (n=1)	1	0	0
Tsamosan (n=4)	0	3	1
Interior Salish (n=7)	7	0	0
Bella Coola (n=1)	0	1	0
Total (n=23)	15	6	2

The criterion for being classified as multiple reduplication is that each of the component reduplication processes is attested independently.

Not all possible combinations of reduplicative processes are attested within any given language; multiple reduplication has restrictions on which (and how many) reduplicative morphemes can be combined in the same word. Multiple reduplication typically allows for the combination of at most two types of reduplicative morphemes.

The most common type of multiple reduplication is the combination of C_1C_2 and C_1 reduplication, which is most frequently used to form diminutive plurals.

- (78) a. hí~həp~húpit (cf. *húpit* ‘deer’)
 DIM~PL~deer
 ‘young deer (more than one)’ (Beaumont, 2011, p.115, Sechelt)
- b. pəs~pé<p>sətk^we (cf. *pəsətk^we* ‘lake’)
 PL~lake<DIM>
 ‘small lakes’ (Kuipers, 1974a, p.39, Shuswap)
- c. bí~bəd~bədaʔ (cf. *bədaʔ* ‘child, offspring’)
 DIM~PL~child
 ‘dolls, litter (of animals)’ (Bates et al., 1994, p.306, Lushootseed)

The combination of C_1C_2 and C_1 reduplication is included in one of the earliest descriptions of reduplication in Salish, that of Haeberlin (1918), who documents this combination of reduplicative morphemes to mark diminutive plural in Lillooet, Thompson River Salish, Okanagan, Lushootseed, Klallam, Northern Straits (Songish), Shuswap, and Comox-Sliammon.⁴⁰ In languages with few reduplicative morphemes, multiple reduplication is not attested. The languages which lack multiple reduplication include the Tsamosan languages, Bella Coola, Nooksack, and Squamish. Pentlatch and Quinault do not appear to have productive multiple reduplication, but this conclusion is based on limited data.

Multiple reduplication is discussed in more detail in the case studies in Chapter 5 and in Chapter 6 (Section 6.6.2.3), since it can be used to test predictions of particular theoretical approaches and assumptions. The basic generalisation is that at least one type of multiple reduplication is permitted for most languages with more than one reduplicative morpheme.

2.9 Generalisations and Gaps in Salish Reduplication

Attested and unattested patterns in Salish reduplication allow for some generalisations about the possible forms and meanings of reduplicative morphemes in the inventory of a single language and across the family. Generalisations about the core types of reduplication and inventories of reduplicative morphemes are summarised in Section 2.9.1. Generalisations pertaining to form and meaning associated with reduplication across the family are provided in Sections 2.9.2 and 2.9.3, respectively. The distribution of reduplicative processes across lexical categories is discussed in Section 2.9.4.

2.9.1 Generalisations about Core Types of Reduplication

Generalisations about core types of reduplication in Salish languages are given in (79).

⁴⁰These languages represent the documentation available at the time Haeberlin (1918) was writing; subsequent research has revealed more languages that permit the combination of C_1C_2 and C_1 reduplication.

(79) *Generalizations about Core Reduplication Patterns:*

1. **C₁C₂ Reduplication:** All 23 Salish languages have a C₁C₂ reduplication process associated with plurality of entities and/or events.
2. **C₁ Reduplication:** The second most common type of reduplication is C₁ reduplication, which is found in all languages except those in the Tsamosan branch. If a language has C₁ reduplication, it will use C₁ reduplication to mark diminutivity, though there may be other types of C₁ reduplication in the language.
3. **C₂ Reduplication:** C₂ change-of-state reduplication is found in thirteen languages. C₂ reduplicative processes, if present, show less cross-linguistic variation in form than C₁ and V₁ reduplication.
4. **V₁ Reduplication:** V₁ reduplication is less common compared to other core types of reduplication; where present, it shows less cross-linguistic variation in its form than in its meaning. Nine Salish languages have a V₁ reduplication process, though its meaning varies cross-linguistically between plurality, diminutivity, comparison, and change-of-state.

2.9.2 Generalisations about Form

Generalisations about the form of Salish reduplication are given in (80).

(80) *Generalisations about Form:*

1. **Position of Segments:** Most reduplicative morphemes in Salish are positioned relative to the left edge of the word (ignoring non-reduplicative prefixes), but reduplication in Lillooet, Shuswap, and Thompson River Salish may be aligned to stress, and reduplication in Bella Coola may be aligned to the leftmost vowel or sonorant segment.
2. **Number of Segments:** The maximum number of segments copied in reduplication (setting aside cases of multiple reduplication) is three,

though the most common number is one. The number of reduplicative morphemes in a word with multiple reduplication is usually no more than two.

2.9.3 Generalisations about Meaning

Reduplicative morphemes in Salish can be classified into three major semantic categories: (i) aspectual, (ii) plural, and (iii) scalar (including diminutive, comparatives, and superlatives). Aspectual reduplication marks imperfectives, progressives, continuatives, inceptives/inchoatives, and changes of state. Plural reduplication marks plurality of entities or events (including arguments). Reduplicative morphemes with scalar properties include processes that mark diminutivity, intensity, or comparison.

Types of reduplication that do not fall into the three major categories include non-plural (“property-denoting”) C_1C_2 reduplication, found across the family, and first person C_1 reduplication in Shuswap.

Reduplication in Salish is not used to mark tense, mood, or clause-type.

2.9.4 Generalisations about Syntactic Distribution

Reduplication often applies across syntactic categories, but the resulting meanings show sensitivity to semantic types. For example, if a reduplicative morpheme applies to both nouns and verbs, it targets entities in the nominal domain (e.g., plural) and events or their arguments in the verbal domain (e.g., pluractional).

Despite earlier claims by Jelinek and Demers (1997, p.303) that reduplication is not confined to any given class of lexical items, distributional restrictions have been documented for a number of languages. Aspectual C_1 reduplication is limited to the verbal domain, for example. H. Davis (2011) argues that the non-plural use of C_1C_2 reduplication in Lillooet is semantically vacuous and derives an adjective from a category-neutral root. Another example of the limited application of reduplication to specific categories of roots is seen in Comox-Sliammon: change-of-state C_2 reduplication “is found with all roots of basically stative aspectual character” (Kroeger, 1988, p.143), which Huijsmans (2023b, p.180) argues are adjectives. The fact that C_2 reduplication cannot be used to derive a

verb from a nominal root provides evidence that certain types of reduplication cannot apply indiscriminately to any category.

The supposed cross-categorical nature of reduplication in Salish has featured in debates about whether a noun-verb distinction exists (e.g., Kinkade, 1983; Van Eijk & Hess, 1986). In an argument for acategorical roots, Wiltschko and Steriopolo (2007) state that diminutive reduplication can occur with nouns, verbs, and adjectives in Halkomelem because the diminutive merges with a pre-categorical root. This claim is challenged by H. Davis and Trotter (2023) who argue that the Halkomelem diminutive must apply to categorially specified roots. More specifically, the diminutive in Halkomelem only occurs with imperfective stems in the verbal domain (Suttles, 2004); it is never found with perfective verbs. It follows that the diminutive has a selectional restriction related to viewpoint aspect, it cannot be combined with a pre-categorical root. Other restrictions of this nature are attested: plural C₁ reduplication in Comox-Sliammon applies only to states and stative-marked verbs (Watanabe, 2003, p.376).

Chapter 3

Phonological Theory and Background

Reduplication in Salish has significance for phonological theory because it provides empirical support for certain frameworks while showing where other approaches are insufficient. Over the next four chapters, I argue that the reduplicative morphemes found across Salish and within the grammars of individual languages provide empirical evidence in support of three main claims:

1. Reduplication happens across a number of sub-word phonological domains (strata), which do not correspond to morphosyntactic categories.
2. Lexical entries for concatenative and non-concatenative morphemes are not inherently different.
3. The realisation of a reduplicative morpheme is based on the grammar (constraint ranking) of the stratum at which it is parsed.

The rest of the dissertation is divided into four chapters: an overview of previous work on reduplication in Salish and the theoretical assumptions I adopt for the analysis (Chapter 3); an analysis of single reduplication patterns in Comox-Sliammon and Lillooet (Chapter 4); an analysis of multiple reduplication patterns in Halkomelem, Lillooet, and Comox-Sliammon (Chapter 5); and a discussion of the cross-Salish typology of reduplication and implications for theory (Chapter 6).

In this chapter, I situate my analysis of Salish reduplication in the context of phonological theory, covering background and relevant assumptions about Salish syllabification, as well as explaining the theoretical framework I employ and my assumptions about the architecture of the grammar.

3.1 Salish Reduplication and Phonological Theory

Salish reduplication has been analysed within a number of different theoretical frameworks and has been used to argue for (and against) many existing phonological theories.

Many early phonological analyses follow Marantz (1982) in using a templatic approach to reduplication, including Bates (1986), S. Davis (1988), and Bates and Carlson (1998), among others. Salish reduplication has also been analysed using a-templatic approaches within the (Parallel) Optimality Theory framework, including, for example, in the work of Urbanczyk (1996a, 1999, 2001, 2006) and Coelho (1999). Others have analysed the same or similar data with reference to co-phonologies (e.g., Downing, 2008), contrast enhancement (e.g., Czaykowska-Higgins & Urbanczyk, 2001; Urbanczyk, 2005), phonological precedence and linearisation (e.g., Fitzpatrick & Nevins, 2003; Fitzpatrick, 2006), compression between morphemes and edges (e.g., Hendricks, 1999), transderivational correspondence (e.g., Benua, 1997), distributed reduplication (e.g., Frampton, 2009), contiguous articulatory gestures (e.g., Angermeyer, 2003), and word faithfulness (e.g., Struijke, 2000). Salish reduplication has also been used to argue for a subjacency condition in morphology (Broselow, 1983), a separation between phonology and morphology in reduplication (Alderete et al., 1999; Inkelas, 2008), a separation between morphology and syntax in reduplication (Kirchner, 2010), and for theories concerning alignment or anchoring (Nelson, 1998, 2003; Nakamura, 2000; Shaw, 2005).

While Salish reduplication has been analysed and used to support many different phonological approaches, previous work largely focuses on a single type of reduplication (e.g., the various reduplicative and non-reduplicative allomorphs of the Halkomelem imperfective, such as in Monahan, 2002, Zimmermann, 2013, and Mellesmoen & Urbanczyk, 2020) or a specific combination of two types of

reduplication (e.g., multiple reduplication involving the plural and diminutive in Lushootseed, such as in Broselow 1983 and Park 2000). Exceptions to this generalisation are where reduplication forms part of a larger phonological study (e.g., Willett, 2003), or in work focusing on the reduplicative system of a single language (e.g., Jimmie, 1994; Urbanczyk, 2001; Crawford, 2015). I aim to complement these studies by contributing a formal analysis of Salish reduplication in the broader context of the full inventory of reduplicative morphemes, both within languages and across the family.

Two components are necessary for a cross-Salish analysis of reduplication that can extend beyond a single language or type of reduplication. First, it is necessary to adopt a framework that is capable of deriving the surface form, including linear position, of the attested reduplicative morphemes. Second, the analysis must be able to predict the form of words with multiple reduplication, which requires an explanation of any relevant interactions with other (reduplicative and non-reduplicative) morphemes. Minimally, an appropriate phonological framework must account for the form and position of reduplicated segments across both single and multiple reduplication patterns for each individual language. For a complete analysis of Salish reduplication, the same theoretical assumptions established for any one language must also be compatible with the grammars and reduplicative inventories of the other 22 languages.

The present analysis adopts the assumptions of Prosodic Morphology and Generalized Non-Linear Affixation (GNLA), defined in Sections 3.3 and 3.4, respectively. Reduplication is defined as fission (introduced in Section 2.2.2), which is a phonological repair strategy that provides segmental content to ensure a well-formed output. Reduplicative morphemes often involve an “empty” prosodic unit, such as a mora (μ) or a syllable (σ). GNLA has been used previously in analyses of non-concatenative morphology in Salish (e.g., Svenonius & Bye, 2012; Zimmermann, 2013; Mellesmoen & Urbanczyk, 2020), though analysing reduplicative morphemes as prosodic affixes in Salish predates GNLA (see, e.g., Jimmie, 1994; Willett & Czaykowska-Higgins, 1995).

The core assumptions of Prosodic Morphology and GNLA are presented within a Stratal OT framework, defined in Section 3.5.2.1, which is well-suited to analysing reduplication in languages with several productive reduplicative

processes that interact with other morphology in a particular order.

3.2 The Mora, Nucleus, and the Salish Syllable

I adopt the prosodic hierarchy proposed by Zec (1988), which adds the mora to earlier work proposing that prosodic structure is comprised of constituents arranged in a hierarchy (e.g., Selkirk, 1978; Nespor & Vogel, 1986). I focus on prosodic structure beneath the level of the prosodic word (PrWd). This includes the mora (μ), syllable (σ), and foot (Ft), as shown in (81).¹



The mora is crucial for understanding prosodic structure and syllabification in Salish and will play a key role in the analysis of reduplication in the following chapters. I assume that the mora is key to syllabification in Salish, adopting the theoretical assumptions of moraic theory (e.g., Hyman, 1985; Hayes, 1989).

Moraic theory proposes that the mora is the unit counted in syllabification, rather than the segment (cf. segmental theories in Levin, 1985, among others). The mora is a prosodic unit that corresponds to a phonological position and to phonological weight: for example, a light syllable has one mora and a heavy syllable has two. Moras are often described as timing units. However Campos-Astorkiza (2004) distinguishes between two types of moras: positional moras (μ_p) license segments by virtue of their position in the syllabification process, while timing moras (μ_t) correspond to time-related properties. Moras described in this chapter are positional moras. Key assumptions about moraicity and syllable structure in Salish are given in this section.

¹This will later be revised to include the nucleus; see Section 3.2.3 and Section 6.5.2.1.3.

3.2.1 Moraic and Non-Moraic Consonants

The moraicity of consonants in Salish languages is determined by their position within a syllable. Following previous work on Salish (e.g., Blake, 2000), I assume that onsets are always non-moraic and that codas are typically moraic.² While Blake (2000) focuses primarily on evidence from Comox-Sliammon, there are arguments for moraic codas in other Salish languages, including Moses Columbia (Czaykowska-Higgins & Willett, 1997) and Halkomelem (Mellesmoen & Urbanczyk, 2020).

As a working assumption throughout this dissertation, I view coda consonants as moraic, unless language-specific patterns emerge that warrant further discussion. For instance, Mellesmoen (2021b) argues that only continuant obstruents are moraic in Bella Coola: non-continuant obstruents (stops) are not. Exceptions of this kind underscore the role of language-specific patterns in reduplication. The moraicity of different consonants is a language-specific difference in the phonological grammar, which may result in cross-linguistic variation in non-concatenative morphology, including reduplication.

Finally, not all analyses of Salish phonology assume that coda consonants are or can be moraic. For example, Leonard (2019) argues that consonants in the Saanich dialect of Northern Straits (Central Salish) are not moraic. However, it is important to note that her argument comes from *a lack* of evidence for the necessity of moraic codas rather than positive evidence for non-moraic codas. This suggests that either a moraic or non-moraic analysis is possible; I assume that codas in Northern Straits are moraic for consistency across the family.

3.2.2 Moraic and Non-Moraic Vowels

Previous work on Salish syllables and schwa has posited that schwa is nuclear but non-moraic (e.g., Shaw, 1993; Shaw, Blake, Campbell, & Shepherd, 1999; Blake, 2000). In other words, a schwa may occupy the peak position or head a syllable, but it may not bear phonological weight. All other vowels in Salish are moraic.

I assume that [ə] is a non-moraic vowel, following Blake (2000) in particular, who proposes that a moraic coda is required for a syllable with a stressed schwa

²Assumptions about onsets exclude potential moraic consonants which are ambisyllabic.

to ensure that any stressed syllable has weight.³ As a schwa cannot bear its own mora, the syllable must have a coda consonant, which are moraic in Salish. A coda consonant can therefore contribute weight to a stressed syllable, as shown in (82).



The leftmost mora in a stressed syllable is the *head mora*. Where relevant, I use the notation μ^+ to mark a head mora in the analysis. An example is provided in (83), which shows that the head mora is leftmost in the stressed syllable, as well as in the foot.



3.2.3 Two Types of Monomoraic Syllables in Salish

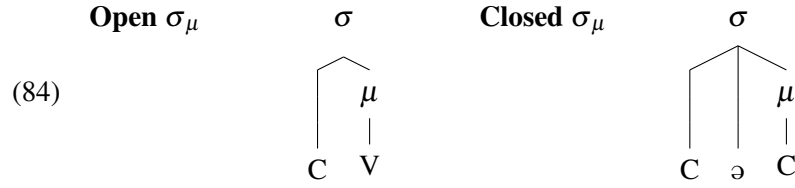
Given that [ə] is nuclear and non-moraic in Salish, there are two possible structures for monomoraic syllables: one with and one without a moraic vowel.⁴

A monomoraic syllable (σ_μ) in Salish can be closed (with a coda) or open (with no coda) depending on the identity of the nucleus (moraic vs. non-moraic), as shown in (84).⁵

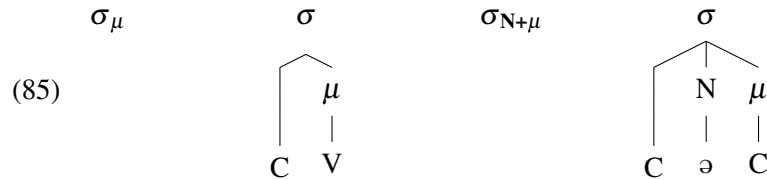
³It may be the case that schwa in a stressed syllable may share a mora with an adjacent consonant. This modification allows a stressed schwa to be associated with the head mora of a prosodic word, and therein bear acoustic correlates of stress despite being a non-moraic vowel.

⁴Minor or simple syllables without vocalic nuclei are not considered here. See Bagemihl (1991) for discussion of the Simple Syllable Hypothesis in relation to Bella Coola.

⁵I do not show bimoraic and trimoraic syllables here; I assume CəCC and CVC syllables are bimoraic, while CVCC syllables are trimoraic.



Given that a monomoraic syllable σ_μ corresponds to both configurations in (84), something additional is needed to differentiate between different syllable structures; I propose that a nucleus node is also a possible constituent of a syllable. The difference between an open monomoraic syllable and a closed monomoraic syllable is represented in (85). An open monomoraic syllable has a single dependant unit (the mora and the nucleus are one and the same), while the closed monomoraic syllable has two dependant units (the mora and the nucleus are separate).⁶



The syllable, mora, and nucleus will be shown in Chapter 4 to be relevant units for distinguishing different types of reduplication in Salish.

3.3 Prosodic Morphology

3.3.1 Background: Prosodic Morphology Hypothesis

The Prosodic Morphology Hypothesis, as laid out in McCarthy and Prince (1986), establishes prosodic units as the core of templatic morphology. While Prosodic Morphology was first fully articulated by McCarthy and Prince (1986, 1990),

⁶The nucleus in the open monomoraic syllable in (85) is not explicitly labeled. A question for future work is whether an explicit nucleus node is always present in syllables with a nucleus and, if so, where it sits in the prosodic structure relative to the mora for moraic vowels. At present, I do not assume that the mora dominates the nucleus, or vice versa. I treat the mora and the nucleus node as sharing an equal status as dependants of the syllable, regardless of whether they are both present in some capacity for moraic vowels (e.g., fused or combined). I discuss how reduplication motivates the nucleus as a part of the prosodic structure in Chapter 6 (Section 6.5.2.1.3).

Urbanczyk (2019) notes that their framework incorporates earlier insights from autosegmental approaches (e.g., Leben, 1973; Goldsmith, 1976). Autosegmental approaches were common in analyses of reduplication using skeletal templates (e.g., Marantz, 1982); Prosodic Morphology builds on these earlier insights and analyses.

The prosodic categories outlined by McCarthy and Prince (1986) are the prosodic word, foot, and syllable. They make explicit reference to reduplication to illustrate the difference between prosodic units and segments, and in fact make the case for prosodic units being better suited to the analysis of reduplication than segments. They argue that if a reduplicative morpheme could have the target shape XXX, where X is a segment (without any particular role in prosodic structure), a type of reduplication that copies the first three segments of a word (regardless of whether they are vowels or consonants) should be possible, such as the one in (86).

- (86) a. badupi - bad~badupi
 b. bladupi - bla~bladupi
 c. adupi - adu~adupi

(McCarthy & Prince, 1986, p.240)

They point out that processes such as the one in (86) are unattested. In fact, languages have a restricted set of reduplication patterns, often with the shapes CV or CVC (showing sensitivity to sonority and syllable structure); there is no evidence for reduplication counting a specific number of segments, and therefore there is no empirical motivation for segmental templates.

Prosodic Morphology captures the insight that reduplication and other non-concatenative morphological processes (e.g., ablaut and vowel lengthening) are sensitive to prosodic structure. The relevance of prosody is reflected by what is copied (e.g., CV reduplication matches a monomoraic syllable and CVC reduplication matches a bimoraic syllable), as well as by the position of the reduplicant (e.g., the diminutive in Shuswap is aligned relative to the stressed syllable).

Another component of Prosodic Morphology is the Satisfaction Condition,

given in (87), which states that every element in the template must be obligatorily satisfied (i.e., filled).

- (87) Satisfaction Condition: All elements in a template are obligatorily satisfied (McCarthy & Prince, 1986, p.243).

Under the Satisfaction Condition, the choice of segmental content to fill a prosodic template is determined by the phonological grammar.

McCarthy and Prince (1986, p.243) state that “the fact that the templates are bounded by a language’s prosody follows from their being literally built from that prosody”, which means that the realisation of reduplicative morphemes is shaped by what makes for a well-formed syllable, foot, and prosodic word in the language. Reduplication typically results in unmarked prosodic structures (e.g., McCarthy & Prince, 1994). In contrast, under a segmental approach, marked or atypical prosodic structure would be predicted because a segmental template (e.g., XXX) could result in unattested surface forms. Prosodic Morphology is therefore more appropriate for an analysis of reduplication patterns than a segment-based approach.

3.3.2 Modifications: The Mora as a Prosodic Unit

The centrality of the mora to Salish phonology was mentioned in Section 3.2. However, McCarthy and Prince (1986) do not consider the mora as an independent prosodic affix, despite its inclusion in the prosodic hierarchy. In other words, they distinguish between monomoraic and bimoraic syllables, but do not treat the mora as a stand-alone prosodic unit that may be affixed independent of a syllable.

In order to account for attested patterns in Salish, I include the mora as a possible prosodic affix. This addition does not pose any challenge to the core tenets of Prosodic Morphology, which already distinguishes syllables by moraic weight. Affixed moras are necessary for the analysis of C₁ reduplication in Salish, in particular (see discussion in Section 2.5). Examples of C₁ reduplication where a single consonant is copied are given in (88); these types of reduplication, most notably found in the Interior Salish languages, have been called “bare consonant reduplication” or “internal C reduplication” by Hendricks (1998) and Bell (1983), respectively.

- (88) a. $cq\acute{e}<\acute{q}>ɬp$ (cf. $cq\acute{e}ɬp$ ‘tree’)⁷
 tree<DIM>
 ‘small tree’ (Kuipers, 1974a, p.39, Shuswap)
- b. $x^w\acute{o}<x^w>kt\grave{a}n$ (from $x^w\acute{i}kt\grave{a}n$)
 knife<DIM>
 ‘small knife’ (Van Eijk, 1997, p.254, Lillooet)

If only syllables or feet could be affixed, the examples of single consonant reduplication in (88) would constitute an argument against Prosodic Morphology because there is no prosodic unit that corresponds to a single consonant. However, if the possible inventory of prosodic affixes includes moras and if coda consonants are analysed as moraic (see Section 3.2.1), then bare consonant reduplication patterns pose no problem for Prosodic Morphology because they correspond to a prosodic unit, namely a mora. A prediction of this approach is that restrictions on which segments can be moraic will match restrictions on which segments can be reduplicated if a morpheme contains an underlying mora; this will be revisited in Chapter 6 (see discussion of Bella Coola and Straits in Section 6.6.2.2.2).

3.4 Generalized Nonlinear Affixation

Generalized Nonlinear Affixation (GNLA) proposes that all non-concatenative morphology involves the affixation of non-linear (i.e., non-segmental) phonological content (Bermúdez-Otero, 2012).⁸ These “phonologically defective” representations may include affixed prosodic units, tones, or phonological features (Zimmermann, 2017). Reduplication in GNLA is typically triggered by the affixation of a floating prosodic unit.

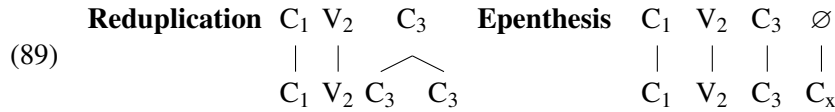
GNLA is implemented in Optimality Theory (Prince & Smolensky, 2004), a constraint-based grammatical model where potential output candidates are generated by GEN and evaluated by EVAL. Constraints are violable and the winning candidate is the one that best satisfies the highest ranked constraints. The surface form of a reduplicative morpheme thus follows from the constraint ranking

⁷One instance of / $\acute{q}$ / is deglottalised in (88a) because Shuswap only allows one ejective per word (see, e.g., L. C. Thompson and Thompson 1985).

⁸Generalized Nonlinear Affixation builds on earlier work by Lieber (1992) and Stonham (1994).

of the grammar, which determines the best way to provide segmental content to associate with (or “fill”) an affixed mora, syllable, or foot.

Reduplication is not an inherent property of any morpheme under GNLA, but instead reflects the phonological grammar (constraint ranking) of a language. Reduplication occurs when the grammar favours fission (where a single input segment corresponds to two segments in the output) over other possible repairs that would provide segmental content to the affixed (underspecified) prosodic unit, such as epenthesis. Schemas for reduplication and epenthesis are given in (89). The top row shows the input and the bottom row shows the output. In reduplication, one consonant in the input (C_3) is fissioned (split) into two segments in the output. Epenthesis inserts a new segment that was not present in the input (C_x).⁹



GNLA integrates the strengths of Prosodic Morphology and retains many of the desirable qualities of earlier template-based approaches, with a prosodic unit serving as a template, with the additional ability to account for allomorphy between non-concatenative processes; it can account for a range of non-concatenative morphological processes, including subtraction, mutation, lengthening and reduplication (Bermúdez-Otero, 2012). Under a GNLA approach, as defined by Bermúdez-Otero (2012), reduplication occurs following the affixation of an empty prosodic unit. Other non-concatenative processes can emerge if a given constraint ranking favours them over reduplication.

GNLA has been used previously to analyse non-concatenative morphology in Salish languages (e.g., Svenonius & Bye 2012; Mellesmoen & Urbanczyk 2020; Zimmermann 2021b). For example, C_1 reduplication is only one of the phonological processes used to mark imperfectivity in Halkomelem (compare reduplication in 90a to metathesis in 90b).¹⁰

⁹Fission violates the constraint IDENTITY, while epenthesis violates DEP; these constraints are defined in Chapter 4 in the first analysis section (Section 4.1.3.1.2).

¹⁰See Case Study #1 in Chapter 5 (Section 5.3) for discussion of reduplicative and non-reduplicative allomorphs.

- (90) a. $\text{h}^{\sim}\text{əc}\text{-ət}$ (from $\text{h}^{\sim}\text{cət}$)
 IPFV $\sim$ cut-CTR
 ‘cutting it’
- b. $\text{p} < \text{a} > \text{q}^{\text{w}}\text{-t}$ (from $\text{pq}^{\text{w}}\text{at}$)
 break < IPFV > -CTR
 ‘breaking it’

(Hukari & Peter, 1995, as cited in Mellesmoen & Urbanczyk, 2020, pp.240–241)

Zimmermann (2013), building on Urbanczyk (1998) and Kurisu (2001), analyses several allomorphs of Halkomelem imperfective C_1 reduplication within a GNLA approach using a segmentally empty prosodic foot, showing the strength of the approach for analysing non-concatenative allomorphy in Salish. Mellesmoen and Urbanczyk (2020) further show how a GNLA approach can be expanded to account for the remaining allomorphs of the Halkomelem imperfective morpheme (including schwa epenthesis and deletion) if the prosodic unit is analysed as a mora, rather than a foot. GNLA therefore allows for the unification of a number of distinct surface forms from a single input, with the choice of output predicted by the phonological grammar of the language. This reduces the amount of morpheme-specific information needed to correctly derive all possible surface forms.¹¹

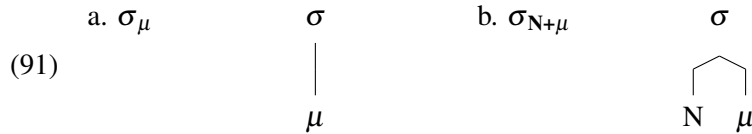
3.4.1 Underlying Forms/Affixes in GNLA

GNLA assumes that nonconcatenative processes, such as reduplication, follow insertion of “empty” (underspecified) phonological representations. These representations can be a single prosodic unit, such as a mora, or the lexical entry may include additional prosodic, segmental, or featural specifications.

Underlying forms may be minimally specified, or carry additional specifications that must also be satisfied by the grammar. For example, the underlying form of a morpheme may be a single syllable with no further

¹¹In fact, Svenonius and Bye (2012) argue against the need for *any* morphological component of the grammar, building on an earlier proposal by Lieber (1992), who claimed that syntax could linearise morphemes and that the phonological grammar can account for any alternations.

specification for number of moras or other prosodic structure. The grammar only needs to fill the syllable, which means that the least marked syllable for that context will emerge, regardless of whether it is weightless, monomoraic, or bimoraic. In contrast, another morpheme may also be an affixed syllable but carry additional structural specification, such as a mora (σ_μ) or a mora and nucleus ($\sigma_{N+\mu}$), as shown in (91). The basic monomoraic syllable σ_μ has a single mora attached to a syllable: it may be filled with either a consonant or vowel, which allows for either a closed or open monomoraic syllable to surface in the output. $\sigma_{N+\mu}$ is more specific than σ_μ ; it requires a closed syllable, though it may be either monomoraic (CəC) or bimoraic (CV $_\mu$ C).



Prosodic units may also be associated with features. The association of featural content with an affixed mora is proposed by Paschen (2018, p.215), who discusses a type of reduplication arising from the affixation of a mora and a floating [PHAR] (pharyngeal) feature in Fox (Algonquian). A minimally specified mora may be filled by any segment in a moraic position, while a mora with additional specification must be filled by a segment that also meets the requirements for that specification (e.g., epenthesis of a pharyngeal coda satisfies both the mora and an associated [PHAR] feature).

The range of possible affixed units (and any featural specifications) is constrained by the inventory of units and features present in the language.

3.5 Architecture of the Grammar

In this section, I outline my assumptions about the architecture of the grammar. The full model and how all the assumptions fit together are discussed in Chapter 6 (see Section 6.1 for a summary of the full model from spell-out through to phonetic implementation).

I adopt a span-based approach to the syntax-morphology interface (Section 3.5.1), but depart from previous work (e.g., Svenonius & Bye, 2012) in assuming

that the lexical entries of individual morphemes contain linearisation features, rather than linearisation being determined by a general algorithm (such as Mirror Theory). Most reduplicative morphemes in Salish are linearised as prefixes in the morphology; the phonological grammar then determines how the reduplicative morpheme is integrated into prosodic structure. I further assume stratification in the phonological grammar within a constraint-based framework: Stratal Optimality Theory (Section 3.5.2). All together, these elements permit a theoretical approach that can derive reduplication at the right time and place in the derivation.

3.5.1 Spans and Linearisation

The crucial morphosyntactic components of my analysis include a *w-span*, which defines a word, and linearisation features, which specify an affix as a prefix (linearised to the left) or a suffix (linearised to the right). The *w-span* is defined in Section 3.5.1.1; linearisation features are discussed in Section 3.5.1.2.

3.5.1.1 Domain: *w-span*

The relevant domain for reduplication in Salish is the word, which is defined morphosyntactically as a *w-span*, which I adopt as part of the span-based approach to spell-out proposed by Svenonius (2016, 2022). A span is a sequence of heads which stand in a head-complement relationship; the *w-span* thus includes all heads that are part of the word.

Svenonius (2022) uses the feature [w] to mark where spans are divided into words. The feature [w] sits on the highest head of the span that is part of a word. The definition of a lexical word, which is adapted from Svenonius (2016), is provided in (92). I replace “root” in the original definition with “minimal lexical head” to clarify where the span starts; it ends with the head marked with [w].

- (92) A minimal lexical head and the span dominating it up to the first [w] feature is a lexical word (Svenonius, 2016, p.12).

The *w-span* for a verb in Salish varies between languages: for example, imperfective aspect is marked by affixation in some languages, such as Comox-Sliammon (via progressive C₁ reduplication), while it is marked by an independent

word in others, such as Lillooet (e.g., via the imperfective auxiliary *wa?*). An example of each is provided in (93).

- (93) a. *One w-span: Comox-Sliammon C₁ Progressive*

qəʃji=ʔut ʔi~ʔim-it-Ø-as
still=EXCL PROG~walk-CTR-3OBJ-3ERG

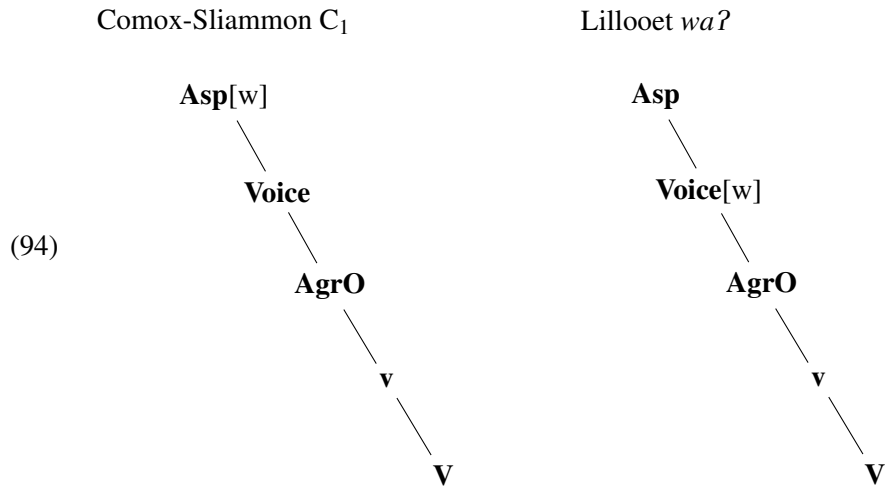
‘They (sg.) are still stepping on it.’ (KG)

- b. *Two w-spans: Lillooet wa? Imperfective*

wa? ʔiʔx̣c-am̃
IPFV scream-MID

‘They (sg.) are screaming.’ (H. Davis, Frank, Whitley, & The Upper St’át’imc Language Authority, in prep)

The relevant domain for reduplication can vary between languages: the w-span must include the relevant aspectual head in Comox-Sliammon, but not in Lillooet. Examples are shown in (94). These trees show the sequence of heads in the w-span. This style of tree follows Brody (2000) and Svenonius (2016): complements slope to the right, while specifiers slope to the left (if present).¹²



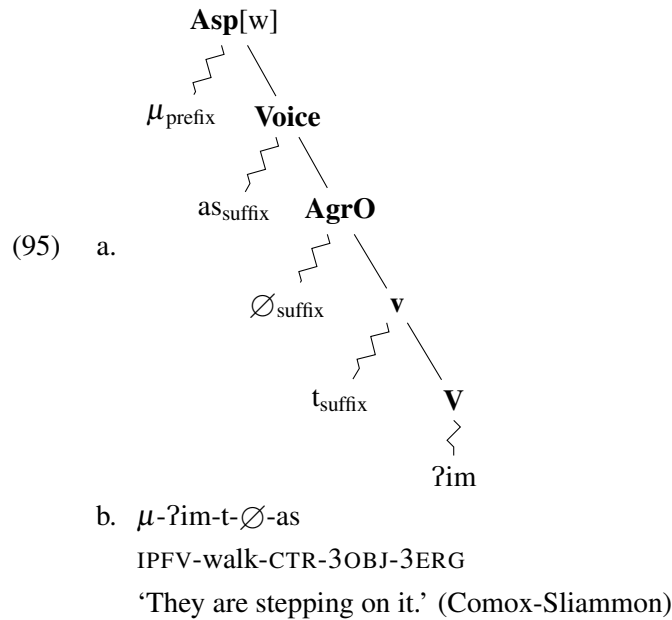
¹²This style of tree is particularly useful for seeing spans because they can quickly be identified as sequences of right sloping lines. See Section 6.2 for discussion of syntactic structure and the role of hierarchical structure in linearisation, which includes a more conventional tree.

3.5.1.2 Linearisation: Prefix Feature

I adopt an affix-driven model of linearisation following Huijsmans (2023a) and H. Davis and Huijsmans (2024). Lexical insertion occurs in two steps. The first step matches terminal elements in the syntax with lexical entries by their features; the second step inserts the phonological content. I follow Huijsmans (2023a) in assuming that linearisation also happens in two stages. Prior to the first stage of lexical insertion, sister nodes are unordered with respect to each other. After terminals are matched with lexical entries based on features, linear order can be established between the heads in the w-span. A further step of linearisation, which determines the correct order between pairs of heads, precedes the insertion of phonological content.

I treat all reduplication as prefixing in Salish, with the exception of Bella Coola (see Section 6.4.2), which is suffixing. The fact that most reduplication in Salish is prefixal is motivated by the fact that edge-aligned reduplication in Salish is always left-aligned (see Section 2.9.2). I assume all affixes carry a feature specifying their direction of attachment (i.e., [prefix] or [suffix]). This feature is available following the matching step of lexical insertion, which allows for the appropriate order of heads in a w-span to be established.

The [prefix] and [suffix] features are instructions for linearisation. Starting at the bottom of the tree, linearisation is determined based on the feature carried by whichever of the two heads is higher in the span. If it is a prefix, A will linearise to the left; if it is a suffix, A will linearise to the right. A w-span is shown in (95a) for the Comox-Sliammon verb given in (93a) in the previous section, with squiggly lines showing the lexical entries associated with heads. The linearisation feature associated with the lexical entry is marked with a subscript font next to the phonological material. The resulting linear order is shown in (95b).



All reduplicative morphemes in a word are spelled out and linearised as part of the same w-span. The input to the phonological component of the grammar contains all of the heads in a w-span in the order determined by linearisation; the phonological grammar can cause further changes, such as infixation.

For simplicity, I assume all affixes are marked for direction of attachment by a linearisation feature. The more standard approach would be to adopt Mirror Theory, which is based on the generalisation that “an exponent representing higher material in a span, if it is affixal, normally follows exponents of lower material” (Svenonius, 2016, p.13); this inverse ordering pattern is often taken as axiomatic (Svenonius, 2016, p.13). I do not adopt this approach as it establishes a fixed order between heads, such that a head is always linearised to the right of the one immediately beneath it; this would require proposing that reduplicative affixes are always suffixes in the morphosyntax while behaving as prefixes in the phonology. The simpler solution is to assume that all affixes, including non-concatenative ones, carry linearisation instructions.¹³

A two stage approach to linearisation incorporating linearisation features

¹³See Chapter 6 (Section 6.8.2.1) for further discussion of why an affix-driven approach to linearisation is preferable to Mirror Theory.

ensures that the reduplicative morpheme is correctly positioned as a prefix, prior to entering the phonological component of the grammar. Reduplicated segments may then surface in infixed positions if motivated by the phonological grammar.

3.5.2 Stratification: Phonological Levels

I assume that the phonological grammar is stratified: there are a number of different strata that correspond to phonological levels and have their own grammar (constraint ranking). I introduce Stratal Optimality Theory in Section 3.5.2.1. I further explain my assumptions about stratification and phonological levels in Section 3.5.2.2. Level diacritics, which specify when in the derivation (i.e., at which stratum) an affix is parsed into prosodic structure, are defined in Section 3.5.2.3. Finally, I explain the way I summarise constraint rankings in Section 3.5.2.4.

3.5.2.1 Stratal OT

Stratal Optimality Theory (Stratal OT) provides a phonological framework for analysing Salish reduplication, and makes testable predictions about possible patterns of allomorphy. The core components of Stratal OT are Optimality Theory and modularity, with the interleaving of morphology and phonology (Kiparsky, 2015).

Optimality Theory is a constraint-based framework that models phonological computation with the use of ranked and violable constraints. Stratal OT follows many of the same assumptions used in traditional Optimality Theory (Prince & Smolensky, 2008), though it is unlike traditional (parallel) Optimality Theory because rather than a single global evaluation, it employs multiple stages of derivation between lexical insertion and phonetic implementation.

Modularity is a crucial component of Stratal Phonology and refers to how the grammar is structured: one component interfaces with another where the output of the first component aligns with the input of the next one. Following from the tradition of Lexical Phonology (Kiparsky, 1982; Mohanan, 1982), the morphology and phonology are interleaved and stratified.

Kiparsky (2015) defines three principles central to Stratal OT. First, there is

stratification - phonology and morphology are divided into different levels called strata, each with their own constraint ranking. The standard number of strata is three, termed stem, word, and phrase. Second, levels (strata) are ordered and correspond to morphological domains. Third, Stratal OT assumes cyclicity, such that phonological content evaluated by constraints at the first stratum will also be evaluated at all subsequent strata as well.¹⁴

My analysis assumes stratification and modularity as properties of the phonological grammar. However, I do not link strata to morphological domains, as discussed in the following section (Section 3.5.2.2); this is a substantial divergence from previous work in Stratal OT, which assumes a correspondence between morphological and phonological domains. In my model, strata are labeled simply as ordered levels (i.e., Level 1, Level 2, Level 3), which correspond to steps of phonological computation in the derivation of the word. The phonological word and associated prosodic structure are built up across the levels, as shown in (96). Constraint rankings differ at each stratum, determining which phonological processes will apply to material parsed into the phonological word at a given point in the derivation.

(96) **Stratum-Based Derivation of the Word**

Input to Level 1

Level 1 Phonology

Output of Level 1 (and Input to Level 2)

Level 2 Phonology

Output of Level 2 (and Input to Level 3)

Level 3 Phonology

Output of Level 3

I follow Jaker and Kiparsky (2020) in assuming that the number of strata can vary across languages; polysynthetic languages typically need a greater number of levels to account for differences in allomorphy and phonological patterns

¹⁴Inkelas (2014, p.189) differentiates between “true cyclicity” (multiple derivations with the same grammar) and “layering” (multiple derivations with different grammars); the cyclicity discussed in this section is an example of the latter: there are multiple strata with different constraint rankings. In Stratal Phonology, however, all words must pass through all strata in sequence, regardless of which affixes are present (unlike co-phonologies, as in Downing, 2008).

associated with affixation.

3.5.2.2 Stratification and the Grammar

Following work on Stratal Phonology (e.g., Kiparsky, 2015; Bermúdez-Otero, 2017), I assume the phonological derivation occurs over a sequence of strata, such that particular phonological processes may apply at different levels. Within a constraint-based approach, stratification means that each stratum may have a different constraint ranking: the output of the previous stratum becomes the input to the next stratum. The grammar cannot access information about earlier stages in the derivation if it is not present in the output of the immediately previous stratum.

I diverge from previous work within Lexical Phonology and Stratal OT (Mohan, 1982; Kiparsky, 1982, 2015; Bermúdez-Otero, 2017) because my strata correspond to strictly phonological, not morphological, domains. All affixes in the phonological word (defined as the w-span in Section 3.5.1) are present in the input. Each affix carries a diacritic specifying a level, which is visible to the phonological grammar and determines when the affix is parsed into the prosodic structure.

Cross-Salish variation in aspectual reduplication provides support for a decoupling of morphosyntactic and phonological domains. For example, employing a traditional Stratal OT analysis, Mellesmoen and Urbanczyk (2021a) analyse the imperfective morpheme in Halkomelem as a stem-level affix, rather than a word-level affix. However, viewpoint aspect composes with the entire VP, and therefore forms an outer layer of the morphosyntax: how can outer morphology then be part of the stem domain? In Chapter 5 (Section 5.3), I argue that imperfective reduplication in Halkomelem happens at the earliest phonological stratum, Level 1. This analysis does not lead to any bracketing paradox or mismatch between morphology and phonology as long as phonological levels do not correspond to morphological domains. The imperfective is linearised to the left, as a prefix, and the phonological grammar can then build up the phonological word sequentially over the levels.

Assumptions needed to make morphosyntactically-defined strata work in an analysis of Salish reduplication have detrimental consequences for other parts of

grammar; this provides further motivation for phonologically-defined domains. Reduplication in Salish marks a number of common meanings, such as plurality and aspect. Where languages use reduplication to mark a similar meaning, there is little reason to suspect major differences in the morphosyntax. If reduplicative morphemes in two languages correspond to the same category of head or a head with a similar position in the verb span, an approach assuming morphosyntactic domains predicts that their phonological form should be computed at the same stratum; this prediction is not consistent with Salish reduplication. While imperfective reduplication in Halkomelem must happen at the first stratum, progressive reduplication happens at a later stratum in Comox-Sliammon (see Section 4.1). As progressive is a subtype of imperfective aspect, there is no reason to expect that the imperfective morpheme would be part of the stem domain in Halkomelem while the progressive would be part of the word domain in Comox-Sliammon. Either the languages have drastically different grammars with respect to the morphosyntax of aspect, or stages in the phonological derivation are not strictly defined by morphological domains; I adopt the latter view. The model proposed in this dissertation therefore assumes that morphosyntactic units are only relevant insofar as determining which morphemes are in the input to the phonological grammar (i.e., the w-span introduced in Section 3.5.1.1); morphosyntax is irrelevant for stratification. A consequence is that the output of the phonological grammar may be considered a word, but there is no need for a “stem”. This implication will be discussed briefly in Chapter 6 (Section 6.8.3).

Since phonological strata are not determined by morphological domains in my model, individual affixes must be associated with diacritics specifying the stratum at which they are parsed into the phonological word.¹⁵ The level that an affix is parsed into the word may be consistent with a morphological domain, such that “stem” affixes would be parsed at Level 1, but this is not a necessity. Level diacritics are discussed further in the following section.

While the specific implementation proposed in this dissertation is novel,

¹⁵Paschen (2018) observes similar challenges for the mapping between morphological domains and phonological strata in Stratal OT and proposes that diacritics on lexical entries may be needed to account for surface orders that are unexpected by the morphosyntax, though the details of such an analysis are not provided.

distinguishing phonological and morphological domains has precedents in Salish linguistics. Czaykowska-Higgins (1993a, p.268) argues that “the morphological properties of suffixes cannot predict their phonological behaviour” with respect to stress assignment in Moses Columbia. Czaykowska-Higgins argues that a stratum-ordering model is not sufficient because it would require at least ten strata to account for the patterns (pp.269–270); however, Trommer (2023) proposes a reanalysis within Stratal OT that captures some of these patterns with reference only to a stem and word domain. My model falls in the middle of these two approaches: I assume a stratum-ordering model as Trommer (2023) does, but I agree with Czaykowska-Higgins (1993a) that strata cannot be defined or limited in number on the basis of morphosyntactic domains.

3.5.2.3 Level Diacritics: Classifying Morphemes by Stratum

Recall from Section 3.5.1 that reduplicative morphemes are part of a w-span, which corresponds to a word. Following lexical insertion, the phonological content associated with the complete w-span is present in the input to the phonology. This means that all affixes will be in the input at the first stratum, even those that must be parsed into the prosodic structure later in the derivation (e.g., affixes that are positioned relative to stress).

A sample lexical entry for the (non-reduplicative) affix *-as* that marks third person subjects on transitive verbs in Comox-Sliammon is given in (97). The diacritic L2 marks this as an affix parsed into prosodic structure at Level 2.

(97) a. $as_{L2} \Leftrightarrow \langle 3ERG \rangle$

In contrast, the transitivity morpheme *-t* in (98) bears no diacritic.¹⁶ Morphemes without a diacritic, including roots, are parsed at Level 1. If the number of relevant strata in a language is n , then the number of possible diacritics is $n-1$.

(98) a. $as_{L2} \Leftrightarrow \langle 3ERG \rangle$
 b. $t \Leftrightarrow \langle CTR \rangle$

¹⁶I use the morphemes in (97) and (98) as examples of how the diacritics and levels could work, but this should not be taken as an analysis. The transitivity morpheme *-t* may actually be parsed at a later stratum than Level 1: see the discussion of link vowels and the transitivity morpheme in Appendix F.

The fact that unmarked affixes are parsed at Level 1 can be explained with the constraints $\text{PARSE}_{\text{Affix}}$ and AFFIX_{L2} , defined in (99). $\text{PARSE}_{\text{Affix}}$ is violated by any affix left unparsed; AFFIX_{L2} is violated by every affix with an L2 diacritic that is parsed into the phonological word.

- (99) a. $\text{PARSE}_{\text{AFFIX}}$: Assign a violation mark for any affix left unparsed.
b. AFFIX_{L2} : Assign a violation mark for every affix with an L2 diacritic that is parsed into the phonological word.

AFFIX_{L2} is ranked higher than $\text{PARSE}_{\text{Affix}}$ at Level 1, and the inverse is true at Level 2. These rankings are shown in (100), which shows a derivation for the Comox-Sliammon phrase *təqtas* ‘she closed it’.¹⁷ A foot is built at Level 1, which integrates the root and the control transitivizer *-t*; the latter is unmarked for phonological level. The third person suffix *-as* is left unparsed because it is marked with an L2 diacritic and any candidate that parses it would incur a fatal violation under AFFIX_{L2} . At Level 2, AFFIX_{L2} is lower than $\text{PARSE}_{\text{Affix}}$: the *-as* affix must be integrated into the phonological word. Brackets in (100) show parsed material.

- (100) a. Level 1: $\text{AFFIX}_{\text{L2}} \gg \text{PARSE}_{\text{Affix}}$

/tq + t + as _{L2} /	AFFIX_{L2}	$\text{PARSE}_{\text{AFFIX}}$
☞ a. (təqt) + as _{L2}		*
b. (təqtas)	*!	
c. (təq) + t + as _{L2}		**!

- b. Level 2: $\text{PARSE}_{\text{Affix}} \gg \text{AFFIX}_{\text{L2}}$

/(təqt) + as _{L2} /	$\text{PARSE}_{\text{AFFIX}}$	AFFIX_{L2}
☞ a. (təqtas)		*
b. (təqt) + as _{L2}	*!	

The diacritic specifying a stratum is only visible to the phonology until the affix is parsed into the phonological word. In the above tableau, *-as* bears a L2 diacritic in the input to Level 2, as well at Level 1; this specification only ceases to be accessible in the output of Level 2, when the affix has been parsed into the

¹⁷As laid out at the beginning of the Comox-Sliammon case study in Chapter 4 (Section 4.1.1), I assume the schwa in the root is epenthetic, following Blake (2000).

phonological word. In contrast, *-t* is unmarked and therefore parsed into the word at the first stratum. The Level 2 grammar cannot “see” that *-t* was an unmarked affix because it is fully integrated into the phonological word following Level 1.

I further assume that the classification of affixes for different levels is language-specific, and thus may vary between languages within the same family. Consider the imperfective in Halkomelem and the progressive in Comox-Sliammon, as mentioned in the previous section. In the morphology, both the morphemes correspond to an aspectual head and are linearised as prefixes; the difference between the two arises in the phonological derivation, because imperfective reduplication in Halkomelem occurs at Level 1, while the equivalent morpheme in Comox-Sliammon occurs at Level 2. Issues with defining strata by morphosyntactic domains are discussed further in Chapter 6 (Section 6.8.3).

3.5.2.4 Summarising Constraint Rankings

One of the key benefits of adopting the Stratal OT framework for an analysis of Salish reduplication is that it allows for different constraint rankings at different strata, which proves essential for the analysis of languages with a number of reduplicative morphemes with different forms. These rankings are not specific to reduplication or indexed to a particular morpheme (in contrast to co-phonologies, as elaborated in e.g., Downing, 2008): the derivation of even monomorphemic words will involve evaluation at each stratum in a fixed order. My analysis of reduplication is therefore constrained by the fact that affixes do not determine their own grammar, they just select for a given entry point in the derivation of a phonological word.

As the goal of the analyses in Chapter 4 is to present a working grammar for each stratum which can derive more than one type of reduplication, there are too many relevant constraints at each stratum to establish crucial ranking relations between every pair of constraints. Instead of presenting constraints in a single flat linear ranking, I therefore summarise constraint rankings in a manner modelled after constraint demotion and dominion hierarchies in work on phonological learning (e.g., Tesar and Smolensky, 1998). This provides a way of visualizing and discussing the rankings which is not misleading: the relative ranking for many

pairs of constraints cannot be determined and does not matter for the output.

An example of what the summary of constraint rankings look like in Chapter 4 for each stratum is given in Table 3.1. The first row of constraints includes the ones that are not crucially dominated by any of the other constraints which are relevant at a particular stratum in the analysis. The second row includes constraints that are crucially ranked below at least one of the constraints in the first row. Constraints given in subsequent rows are crucially ranked beneath at least one of the constraints in the second row. The columns show where there is a direct relationship between the constraints. For example, Constraint 6 in Table 3.1 is crucially ranked beneath Constraint 4, which is in turn crucially ranked beneath Constraint 1. The letters in the leftmost column show the relationship between constraints: a constraint in row B must be ranked beneath a constraint in row A of the same column.

Table 3.1: Example Summary of Constraint Rankings at a Stratum

A	Constraint 1	Constraint 2	Constraint 3
B	Constraint 4	Constraint 5	
C	Constraint 6		

A crucial ranking means that if one constraint (e.g., Constraint 2 in Table 3.1) was not ranked above another (e.g., Constraint 5), an unattested candidate would win over the attested form in some derivations. However, the representation says nothing about the pairwise ranking of a constraint with constraints in a higher row in *different* columns: while Constraint 5 may be given in a lower row than Constraint 1, their pair-wise ranking is not critical or not established in the present analysis. Constraint 3 is in the final column, which means it is either undominated, and must therefore dominate more than one of the other constraints in a lower row, or it is not crucially ranked relative to any other specific constraint.

The type of table exemplified by Table 3.1 is used to summarise crucial rankings in Chapter 4. Flat linear rankings used to create tableaux for each stratum can be found in Appendix C; however, many of the constraints are not crucially ranked relative to other constraints and so these flat rankings provide less information than the summary tables.

3.6 Predictions of the Phonological Theory

The phonological assumptions introduced in this chapter make predictions about reduplication that will be explored in subsequent chapters. The version of Stratal OT proposed in this dissertation assumes that strata are phonologically defined and mappings between morphemes and strata are lexically determined; lexical entries for reduplicative morphemes can therefore vary within and across languages with respect to their underlying content and their level diacritic, which determines the stratum at which they will be parsed. As all strata are ordered and have their own constraint ranking, level diacritics play an important role in determining the form and position of a reduplicative morpheme, as well as its potential interactions with other morphemes.

Regarding form and position, Stratal OT predicts that a language can fill the same prosodic unit (e.g., a mora) differently, depending on when it enters the derivation and the constraint ranking determining how that unit is filled. Specifically, the same input prosodic units may take on different forms depending on when they are parsed (e.g., at Level 1, Level 2, or Level 3). A type of reduplication (e.g., diminutive) may surface differently in one stratum compared to subsequent ones. This means that the total inventory of prosodic units available in affixation for a given language may be smaller than the total number of reduplicative processes, as two processes that surface differently may involve the same prosodic unit.

Modularity in Stratal OT limits possible interactions between reduplicative morphemes. The form and position of a reduplicative morpheme is not expected to be sensitive to the shape of affixes parsed at a later stratum than the one at which it is parsed. The influence of a morpheme at a prior stratum on the form of a reduplicative morpheme is limited, however: a reduplicative affix can only be sensitive to the full output of Level 1 (i.e., it cannot see individual morphemes).

The predictions in this section are tested through case studies of single reduplication patterns in Comox-Sliammon and Lillooet in Chapter 4. Multiple reduplication patterns in Halkomelem, Lillooet, and Comox-Sliammon are considered in Chapter 5. The case studies in Chapters 4 and 5 are followed by a broader consideration of the cross-Salish typology of reduplication in

Chapter 6 in the light of the theoretical model expounded in this chapter. The selection of case study languages is motivated by practical considerations, such as access to speakers (Comox-Sliammon and Lillooet) and the quality of existing documentation (Halkomelem), as well as by the relevance of these languages to the goals of the dissertation. The three languages exemplify a range of reduplication patterns, including allomorphy and multiple reduplication.

Chapter 4

Single Reduplication

Comox-Sliammon and Lillooet each have a rich inventory of reduplicative morphemes: Comox-Sliammon has seven types of reduplication, and Lillooet has six. These two languages are ideal for a case study because each represents a different major branch of the family (Central vs. Interior Salish). Further, it is possible to conduct fieldwork with speakers where data is sparse, meaning that analysis is not limited to existing documentation.

Reduplication patterns in Comox-Sliammon and Lillooet exhibit a combination of properties that can establish the necessary constraints (and rankings) needed to analyse reduplication across the family. Comox-Sliammon is particularly relevant for forming an analysis that may be adapted and extended to other Salish languages because it uses all four core types of reduplication (C_1C_2 , C_1 , C_2 , and V_1), introduced in Chapter 2 (Section 2.2.3). Lillooet is relevant for a cross-Salish analysis because it has both edge-aligned and prosody-aligned reduplicative patterns, the latter necessary for an analysis of diminutive reduplication in the Northern Interior Salish languages.

The first half of this chapter presents an analysis of reduplication in Comox-Sliammon (Section 4.1); the second half focuses on Lillooet (Section 4.2). Within each case study, the major subsections are structured by strata (*phonological domain*), with data and analysis presented in minor subsections for each reduplicative morpheme (*input/lexical entry*). At the end of each major subsection, a summary of the constraint ranking (*grammar*) is given.

4.1 Case Study I: Comox-Sliammon

The first case study in this chapter explores reduplication in Comox-Sliammon. An overview of relevant facts about the phonological grammar of the language is provided in Section 4.1.1, followed by a preview of the grammar of reduplication in Section 4.1.2. Description, data, and analysis are then presented for reduplication at Level 1 (Section 4.1.3), Level 2 (Section 4.1.4), and Level 3 (Section 4.1.5).

4.1.1 Comox-Sliammon Phonology

The phonology of Comox-Sliammon is influenced by contact with Kwak’wala, a neighbouring Northern Wakashan language (see Kroeber & Watanabe, 2004). Comox-Sliammon differs from other Salish languages in syllabification, prosody, and phonotactics in three main ways. First, stress always falls on the first syllable of a word, regardless of vowel quality, which is a departure from the rest of the language family, where stress is typically weight-sensitive or morphologically-governed (see e.g., Dyck, 2004, p.10).¹ Second, it has also lost all non-reduplicative prefixes. Third, complex onsets are not permitted.² This restriction on onsets rules out word-initial consonant clusters and limits the length of word-medial ones; epenthesis is required to repair illicit consonant clusters that would be tolerated elsewhere in the family.

Like other Salish languages, Comox-Sliammon differentiates moraic vowels (/i/, /a/, or /u/) from the non-moraic vowel (/ə/).³ The default epenthetic vowel is schwa, as in the rest of the family. Following Blake (2000), I assume that the distribution of schwa is fully predictable in Comox-Sliammon. This means that all instances of [ə] are epenthetic; schwa is never present in the input to the initial stratum.

A phonological pattern that is specific to Comox-Sliammon and relevant to reduplication concerns alternations between the voiced obstruents /j/ and /g/ and

¹Due to the predictability of stress assignment, I do not mark stress on any examples from Comox-Sliammon.

²There are very few exceptions where a fossilised prefix *s-* remains (Kroeber & Watanabe, 2004), such as *sjəsuʔ* ‘yesterday’. Further, some clitics are proclitic in Comox-Sliammon, which can yield complex onsets (e.g., Huijsmans, 2023a).

³My assumptions about moraic and non-moraic vowels are stated in Chapter 3 (Section 3.2.2).

the glides /y/ and /w/. Glide-stop alternations are conditioned by the position of the segment within a syllable, as discussed in Blake (1992) and Mellesmoen (2018b). The phonemes /j̥/ and /g/ surface as [j̥] and [g] in an onset, and [y] and [w] in a coda, respectively: examples are given in (101).⁴ This alternation can be characterised as a distinction between non-moraic (onset) and moraic (coda) allophones.

- (101) a. Root: *huj̥*- ‘finish’
- i. *huj̥*-it
finish-STAT
‘ready’
 - ii. *huy*
finish
‘stop, finish’ (EP/FL)
- b. Root: *tɔ̃*- ‘freeze’
- i. *taʔg*-it
freeze-STAT
‘frozen’
 - ii. *təw̃*
freeze
‘ice’ (FL)

Sequences of /əy/ or /əw/ within the same syllable are pronounced as [i] or [u], respectively, whether or not the glide is underlying, as in (102a), or comes from an underlying /g/ or /j̥/, as in (102b), where an underlying /g/ is reduplicated as part of C₁C₂ reduplication, yielding coda /w/; the resulting glide is subsequently realised as [u] because it follows /ə/.

⁴/g/ is realised as [ʔg] in (101b.i) because /g/ and /j̥/ pattern with sonorants; these glottalised sonorants may undergo glottal restructuring, such that an underlying /S’/ is realised as [ʔS] (Blake, 2000, pp.240–245). I argue in Mellesmoen (2018b) that the reason voiced obstruents pattern as obstruents is because /j̥/ and /g/ retain the feature [SV] (sonorant voice) from Proto-Salish *y and *w, which they developed from.

- (102) a. **čičayiš** (cf. *čayiš* ‘hand’)
 //čəy~čayiš//
 PL~hand
 ‘hands’ (JF)
- b. **huhigus** (cf. *higus* ‘chief’)
 //həw~higus//
 PL~chief
 ‘chiefs’ (EP/FL)

4.1.2 Preview of Comox-Sliammon Reduplication

The grammar of Comox-Sliammon contains two C_1C_2 reduplication processes (plural and property-denoting), three C_1 reduplication processes (diminutive, plural, and progressive), a C_2 reduplication process (change-of-state), and a V_1 reduplication process (plural).

Reduplication can occur at three different strata: Level 1, Level 2, and Level 3. The grammar of reduplication in Comox-Sliammon is summarised in Table 4.1.

Table 4.1: Preview of Comox-Sliammon Reduplicative Affixes

Stratum	Type	Affix	Section
Level 1	Diminutive C_1	μ or <i>i</i>	4.1.3.1.1
	Plural C_1	μ or <i>a</i>	4.1.3.1.3
	Property-Denoting C_1C_2	$\sigma_{N+\mu}$	4.1.3.2.1
Level 2	Progressive C_1	μ	4.1.4.1.1
	Plural C_1C_2	$\sigma_{N+\mu}$	4.1.4.2.1
Level 3	Change-of-State C_2	σ	4.1.5.1.1
	Plural V_1	μ [c.g.]	4.1.5.2.1

4.1.3 Level 1

I analyse diminutive C_1 , plural C_1 , and property-denoting C_1C_2 reduplication as Level 1 affixes. Diminutive C_1 and plural C_1 reduplication both involve the affixation of either a mora or a fixed segment allomorph (*i*- for diminutive and *a*- for plural); these morphemes are homophonous when they affix a mora.

Property-denoting C_1C_2 reduplication involves the affixation of a monomoraic syllable with a nucleus node that is not directly associated with the mora ($\sigma_{N+\mu}$).

The underlying forms and surface positions of the reduplicative morphemes are summarised in Table 4.2.

Table 4.2: Level 1 Reduplicative Affixes

	Underlying Form	Surface Position
Diminutive C_1	μ or i	Prefixed or Infix
Plural C_1	μ or a	Prefixed or Infix
Property-Denoting C_1C_2	$\sigma_{N+\mu}$	Prefixed

4.1.3.1 Level 1 C_1 Reduplication

C_1 reduplication at Level 1 marks either diminutivity or plurality, as shown in (103) and (104). The diminutive examples in (103a) and (103b) are homophonous with the plural examples in (104a) and (104b), respectively.

(103) *Diminutive C_1 Reduplication*

- a. lu<I>q^w-it (cf. *luq^wit* ‘something is stuck’)
get.stuck<DIM>-STAT
‘something is just a little stuck’ (PD)
- b. θ i< $\theta θ iqit ‘dug up’)
dig<DIM>-STAT
‘dug up (a little)’ (FL)$

(104) *Plural C_1 Reduplication*

- a. lu<I>q^w-it (cf. *luq^wit* ‘something is stuck’)
get.stuck< C_1 .PL>-STAT
‘more than two things are stuck’ (PD)
- b. θ i< $\theta θ iqit ‘dug up’)
dig< C_1 .PL>-STAT
‘dug up (multiple)’ (FL)$

Vowel quality is the only form-related clue which distinguishes diminutive and plural C_1 reduplication. When a non-reduplicated word has a non-moraic vowel (i.e., [ə]) in the first syllable or is monomoraic, both the diminutive and the plural will surface with a moraic vowel.⁵ The default patterns are that [i] in the reduplicant is associated with the diminutive and [a] is associated with the plural: compare (105) and (106). Instances of [i] in square brackets in (105) and (106) have a different source (either epenthetic or marking stativity); this is different than the one associated with the diminutive (e.g., the first [i] in 105).

(105) *Diminutive C_1 Reduplication*

ti~t[**i**]k (cf. tɔlk 'hole')

DIM~hole

'small hole' (BW)

(106) *Plural C_1 Reduplication*

ta~t[**i**]k (cf. tɔlk 'hole')

C_1 .PL~hole

'bunch of holes' (KG)

Identifying the type of C_1 reduplication on the basis of form can be challenging even when [i] or [a] are present, because both can occur for other reasons. For example, [i] can occur in addition to (or as part of) stative marking (see Watanabe, 2003, pp.376–384; Mellesmoen & Huijsmans, 2019b, pp.155–156). An identical pair of examples with stative marking, where both diminutive and plural forms have [i], is provided in (107) and (108).

(107) *Diminutive C_1 Reduplication*

ti~tq-it (cf. tɔqit 'closed')

DIM~close-STAT

'a little bit closed' (Watanabe, 2003, p.389)

⁵When I talk about words or roots with a non-moraic vowel, this refers to words or roots that surface with [ə] in the output; as stated in Section 4.1.2, I do not assume schwa is present in underlying forms.

(108) *Plural C₁ Reduplication*

ti~tq-it (cf. *təqit* ‘it is closed’)

C₁.PL~close-STAT

‘They are closed.’ (Watanabe, 2003, p.389)

However, even when the form of diminutive and plural C₁ reduplication is identical, it is usually possible to distinguish the two either by their meaning or by their distribution. For example, C₁ reduplication on a nouns always marks a diminutive.⁶ C₁ plural reduplication is semantically restricted to state-denoting predicates.⁷

Given the generalisations laid out in this section, I analyse /i/ as the fixed vowel associated with diminutive C₁ reduplication, and /a/ as the fixed vowel associated with plural C₁ reduplication. I use distribution and meaning to distinguish between forms where vowel quality does not provide evidence (i.e., words with a moraic vowel). I set aside forms with confounding factors (e.g., [i] with stative marking).

In the following subsections, I provide a description (Section 4.1.3.1.1) and

⁶One exception is the word *q^wəlay* ‘driftwood’, which has both a diminutive and plural form, as shown in (i) and (ii) respectively.

(i) *Diminutive C₁ Reduplication*

q^wi~q^wlay (cf. *q^wəlay* ‘driftwood’)

DIM~driftwood

‘little driftwood’ (FL)

(ii) *Plural C₁ Reduplication*

q^wa~q^wlay (cf. *q^wəlay* ‘driftwood’)

C₁.PL~driftwood

‘driftwood’ (FL)

⁷State-denoting predicates include adjectives, such as a colour in (i), and predicates derived with the stative marker, as shown in (ii). A plural state means that the state holds of multiple participants or that instances of the state are spatially distributed, which may involve a single participant.

(i) **pa**~puq^w məm~mimaw (cf. *puq^w* ‘grey’)

C₁.PL~grey PL~cat

‘light-coloured cats’ (FL)

(ii) **xa**~xəλ-it (cf. *xəλit* ‘it is broken’)

C₁.PL~break-STAT

‘They are broken.’

‘It is broken (in multiple places).’ (BW)

analysis (Section 4.1.3.1.2) of diminutive C₁ reduplication, which also accounts for homophonous plural C₁ forms. Further examples of plural C₁ reduplication are provided in Section 4.1.3.1.3, followed by an analysis showing the derivation of plural forms with [a] in Section 4.1.3.1.4.

4.1.3.1.1 Diminutive C₁ Reduplication

When the first vowel in the non-reduplicated form is [ə], the diminutive form has [i] between the matching consonants (109).⁸

- (109) a. **t̥i**~t̥xəməy̆ (cf. t̥əxəməy̆ ‘cedar’)
 DIM~cedar
 ‘small cedar’ (Watanabe, 2003, p.388)
- b. **q̣i**~q̣snaŋ̆ (cf. q̣əsnaŋ̆ ‘shirt’)
 DIM~shirt
 ‘small shirt’ (Watanabe, 2003, p.386)
- c. **t̥^θi**~t̥^θx̣^wuʔ (cf. t̥^θx̣^wu ‘cod’)
 DIM~cod
 ‘small cod’ (Watanabe, 2003, p.386)
- d. **θi**~θjapuq^w (cf. θəjapuq^w ‘hat’)
 DIM~hat
 ‘little hat’ (PD)

Diminutives built from monosyllabic words also have [i].⁹ Examples are shown in (110).

- (110) a. **t̥i**~t̥aqt̆ (cf. t̥aqt̆ ‘mountain’)
 DIM~mountain
 ‘small mountain’ (JF)

⁸There is additional glottalisation associated with the diminutive, as seen in (109a); however, this is variable across speakers and glottalisation of sonorants is falling out of use in Comox-Sliammon (Caldecott & Mellesmoen, 2018). I will not provide an analysis of reduplication-related glottalisation in Comox-Sliammon nor include it in the underlying form. However, the Proto-Salish diminutive was likely associated with glottalisation (Mellesmoen, 2022).

⁹While the choice of Ci- is robust for monosyllabic words, I have recorded some variation in choice between -C- and Ci- for younger speakers for multisyllabic words with [ə].

- b. **ti**~təw̄ (cf. təw̄ ‘ice’)
 DIM~ice
 ‘little bit of ice’ (JF)
- c. **pi**~puk^w (cf. puk^w ‘book’)
 DIM~book
 ‘little book’ (JF)

When the input is multisyllabic and has an underlying vowel, the first consonant of the root is doubled and positioned after the first vowel of the root, as shown in (111–113).

(111) *Root Vowel: /a/*

- a. **ʔa**<**ʔ**>ɬəm (cf. ʔaɬəm ‘salt’)
 salt<DIM>
 ‘little bit of salt’ (FL)
- b. sa<**s**>plin (cf. saplin ‘bread’)
 bread<DIM>
 ‘small piece of bread’ (JF)

(112) *Root Vowel: /i/*

- a. mi<**m**>ɬaɬ (cf. miɬaɬ ‘bear’)
 bear<DIM>
 ‘small bear’ (JF)
- b. ʔi<**ʔ**>mən (cf. ʔimən ‘road, door’)
 road/door<DIM>
 ‘narrow walkway (small trail)’ (JF)

(113) *Root Vowel: /u/*

- a. šu<**š**>k^wa (cf. šuk^wa ‘sugar’)
 sugar<DIM>
 ‘little bit of sugar’ (JF)
- b. su<**s**>payu (cf. supayu ‘axe’)
 axe<DIM>
 ‘small axe, hatchet’ (EP)

While I analyse the diminutive as a prefix in the input for all forms, copied segments may surface in an infix position, as in (111–113); infixation of the diminutive is motivated by the phonological grammar, as detailed in the analysis provided in Section 4.1.3.1.2 below.

Evidence of recent productivity of the diminutive is provided by the loanwords in (110c) and (111b) above (see discussion of productivity in Section 1.4.3.3). The diminutive form of the loanword *puk^w* ‘book’ in (110c), which is borrowed from English, shows that the Ci- form has been recently productive with monosyllabic inputs. The word for a ‘small piece of bread’ in (111b) is derived from the loanword *saplin* ‘bread’, which is borrowed from French via Chinuk Wawa; this shows that the -C- form was also recently productive. Another example of the -C- form of the diminutive with a loanword is *šuš^ka* in (111a); *šuk^wa* ‘sugar’ is borrowed from English.

I analyse the -C- form of the diminutive as the default; the Ci- form surfaces when the word would otherwise be monosyllabic or have [ə] as its first vowel.

4.1.3.1.2 Analysis of Diminutive Reduplication

I analyse the diminutive as either a mora or /i/; the constraint ranking leads to the reduplication of a single consonant regardless of allomorph choice. The lexical entry for the diminutive is given in (114). It is a Level 1 morpheme, so there is no diacritic on the lexical entry, as explained in Chapter 3 (Section 3.5.2.3).

- (114) *Comox-Sliammon Diminutive:*
 $\{\mu, i\} \Leftrightarrow \langle \text{DIM} \rangle$

Three constraints needed for the μ allomorph include MAX-FLOAT, *FLOAT, and REALIZEMORPHEME, which are defined in (115). MAX-FLOAT and *FLOAT, defined in (115a–b), motivate reduplication following affixation of a prosodic unit (i.e., filling a mora is preferred over deleting a floating mora or leaving a mora floating in the output). O in (115b) stands for output. REALIZEMORPHEME, defined in (115c), requires the surface form of $\alpha+\beta$ to differ from that of α alone; α stands for the phonological form a morpheme is added to and β stands for the morpheme added.

- (115) a. MAX-FLOAT (MAXFLT): All prosodic units or features that are floating in the input have output correspondents.

(modified from Wolf, 2005, p.370)

- b. *FLOAT: $\forall p \in O$, where p is a prosodic unit or phonological feature: $\exists s$, where s is a segment (i.e., phonological content on the segmental tier), and p dominates or is associated with s .

(modified from Saba Kirchner 2013, p.232)

- c. REALIZE MORPHEME (RM): Let α be a morphological form, β be a morphosyntactic category, and $F(\alpha)$ be the phonological form from which $F(\alpha+\beta)$ is derived to express a morphosyntactic category β . Then RM is satisfied with respect to β iff $F(\alpha+\beta) \neq F(\alpha)$ phonologically.

(Kurusu, 2001, p.39)

MAX-FLOAT is violated when floating elements in the input are not present in the output. *FLOAT bans floating prosodic units or phonological features in the output, which motivates phonological processes (e.g., fission, epenthesis, lengthening, etc.) to provide segmental content to fill a floating prosodic unit (Saba Kirchner, 2013, p.232). A violation mark is incurred under *FLOAT for each prosodic unit in the output that is not connected to segmental content. REALIZE MORPHEME requires every input morpheme to be expressed in the output; it is violated if no exponent of an input morpheme is present in the output, such that the output is identical to what it would be without the morpheme (Kurusu, 2001, p.39).

*FLOAT is dominated by $AFFIX_{L2}$ (and $AFFIX_{L3}$) in Comox-Sliammon at Level 1, such that floating prosodic units are only tolerated when they carry a L2 (Level 2) or L3 (Level 3) diacritic; they may remain floating until a later point in the derivation. The definition of $AFFIX_{L2}$, originally defined in Chapter 3 (Section 3.5.2.3) is restated in (116).

- (116) $AFFIX_{L2}$: Assign a violation mark for every affix with an L2 diacritic that is parsed into the phonological word.

The tableau in (117) summarises how the ranking of $AFFIX_{L2}$ and $PARSE_{AFFIX}$ relative to REALIZE MORPHEME, MAX-FLOAT, and *FLOAT achieves the desired

outcome at Level 1: a hypothetical mora with the L2 diacritic is left floating, while one with a L1 diacritic is filled in a manner that results in a different surface form than would have surfaced if it were not present. I use x to stand in for either a reduplicated or epenthetic segment in this tableau. I only mark moras where they correspond to a floating mora in the input unless otherwise stated. As laid out in Chapter 3 (Section 3.2), I assume that all non-schwa vowels and coda consonants are moraic.

(117)	$/\mu_{L2} + \mu + CVC/$	AFFIX _{L2}	MAXFLT	RM	*FLOAT	PARSE _{AFFIX}
	a. $\mu_{L2} CVx_{\mu}C$				*	*
	b. $CVx_{\mu}x_{\mu}C$	*!				
	c. $CVx_{\mu}C$		*!		*	
	d. $\mu_{L2} CV_{\mu}C$			*!	*	*
	e. $\mu_{L2} \mu CVC$				**!	**!

Candidate (117b), which fills the Level 2 mora, fatally violates AFFIX_{L2}. Candidate (117c) satisfies AFFIX_{L2} by deleting the Level 2 affix, which incurs a fatal violation under MAX-FLOAT because one of the input moras is absent. Candidate (117d) fatally violates REALIZEMORPHEME because the unmarked (i.e., non-L2) mora associates with a vowel, which would also be associated with a mora in the absence of the morpheme. Candidate (117e), which leaves both affixed moras floating, is ruled out because it violates the same constraints as the winning candidate but incurs two violations under each constraint.¹⁰

I exclude REALIZEMORPHEME and the affix constraints AFFIX_{L2} and PARSE_{AFFIX} from tableaux in the remainder of this chapter. I assume affix constraints are ranked as motivated in Chapter 3 (Section 3.5.2.3) and will only discuss their rankings as part of the final ranking given in summary sections. For brevity, I will only show affixes in the input which will be parsed at that particular stratum: a L2 or L3 diacritic will be provided alongside lexical entries in Sections 4.1.4 and 4.1.5, respectively, but will be omitted in the tableaux.

Reduplication is the fission of a segment, which creates a 1:2 mapping between

¹⁰While PARSE_{AFFIX} and *FLOAT incur identical violations for the candidates not ruled out by higher ranked constraints in (117), both are needed because *FLOAT will not apply to the fixed segment allomorph and *FLOAT is needed to avoid an instance where a mora is parsed into a syllable or foot without associating with segmental content.

the input and output. Fission of a segment violates INTEGRITY (McCarthy & Prince 1999), defined in (118). INTEGRITY also limits the size of the reduplicant: a candidate which fills the affixed prosodic unit with the fewest violations of INTEGRITY will always be preferred (cf. Kirchner 2010).

- (118) INTEGRITY: An element of the input does not have multiple correspondents in the output.

Each instance of fission incurs a violation mark under INTEGRITY; C_1 reduplication copies a single consonant and incurs one violation. The fact that *FLOAT and MAX-FLOAT outrank faithfulness constraints, including INTEGRITY, results in EVAL (the evaluation component of the grammar) selecting candidates where the affixed prosodic unit is filled. The tableau in (119) shows how ranking *FLOAT and MAX-FLOAT above INTEGRITY results in reduplication. I use subscripts to mark correspondence between consonants in the input and output. The candidate with fission to fill the floating mora in (119a) wins over a candidate which leaves the mora floating in (119b), as well as one which deletes the mora entirely (119c).

$/\mu + C_1VC_2/$		MAXFLT	*FLOAT	INTEGRITY
(119)	☞ a. $C_1VC_{1,\mu}C_2$			*
	b. μC_1VC_2		*!	
	c. C_1VC_2	*!		

*FLOAT and MAX-FLOAT are ranked highly at every stratum relevant for reduplication and will be omitted from tableaux for the remainder of the chapter unless further discussion of their ranking is warranted.

In order to correctly predict which segments will be copied (vowels, consonants, or both), INTEGRITY is divided into two constraints: INTEGRITY-V and INTEGRITY-C. These constraints are defined in (120). INTEGRITY-C is violated by fission of a consonant, and INTEGRITY-V is violated by fission of a vowel.

- (120) a. INTEGRITY-C (INT-C): A consonant in the input does not have multiple correspondents in the output.

- b. INTEGRITY-V (INT-V): A vowel in the input does not have multiple correspondents in the output.

The relative ranking of INTEGRITY-C and INTEGRITY-V determines whether the grammar favours candidates with the reduplication of a consonant or the reduplication of a vowel.

The position of the INTEGRITY constraints relative to other faithfulness constraints determines whether the grammar uses reduplication to fill an empty prosodic unit. When DEP, defined in (121), is ranked above INTEGRITY, reduplication is preferred to epenthesis.¹¹

(121) DEP: An output segment must have an input correspondent.

Parallel to the division of INTEGRITY into INTEGRITY-C and INTEGRITY-V, I divide DEP into DEP-C and DEP-V, defined in (122).

- (122) a. DEP-C: A consonant in the output must have an input correspondent.
- b. DEP-V: A vowel in the output must have an input correspondent.

The ranking of each of the DEP constraints relative to the INTEGRITY constraints allows EVAL to select candidates with reduplication of one type of segment and epenthesis of another (e.g., reduplication of consonants and epenthesis of vowels). Diminutive reduplication shows that DEP-C is ranked above INTEGRITY-C (consonants are reduplicated) and INTEGRITY-V is ranked above DEP-V (vowels are epenthesized). This is consistent with the cross-Salish patterns in Chapter 2: fission is generally preferred for consonants, while epenthesis is generally preferred for vowels.

Throughout this dissertation, I give [i] as an example of an epenthetic moraic vowel to illustrate how unattested candidates are ruled out; this choice is motivated by the analysis of other Salish languages where the [i] associated with the diminutive may be analysed as epenthetic, rather than an underlying vowel (see, e.g., analysis of Lushootseed diminutive reduplication in Mellesmoen

¹¹Epenthesis and fission are the only options to fill prosodic units in Comox-Sliammon, but lengthening is also an option in other languages. This is discussed in Section 6.6.2.2.1 with respect to vowel length in the Tsamosan languages.

& Urbanczyk, 2024). While [h] is typically the epenthetic consonant in Salish, I give [ʔ] as an example of a default consonant in tableaux in this dissertation because [ʔ]-epenthesis is seen with V₁ reduplication in Comox-Sliammon (see Section 4.1.5.2), as well as in other Salish languages. Further, Blake (2000, p.60) states that [h] is the “epenthetic (onset) consonant” in Comox-Sliammon; as I am interested in epenthesis in both onset and coda positions, I have picked [ʔ] to represent any epenthetic consonant in unattested candidates.

The tableau in (123) for example (113b) above shows how ranking both of the DEP constraints and INTEGRITY-V above INTEGRITY-C correctly predicts which segments are copied in Level 1 C₁ reduplication. The attested candidate (123a) only violates INTEGRITY-C, which means that INTEGRITY-C must be ranked lower than the other constraints. The ranking in (123) accounts for the default -C-diminutive pattern.

/μ + supayu/		DEP-C	INT-V	DEP-V	INT-C
(123)	a. sus _μ payu				*
	b. su _μ supayu		*!		*
	c. ʔu _μ supayu	*!	*!		
	d. si _μ supayu			*!	*
	e. suʔ _μ payu	*!			

The tableau in (123) does not consider candidates where the mora surfaces in different positions. The infixed position of the reduplicant can be derived by modifying the PREFIX constraints from McCarthy (2003, p.94), which are violated by any infixation (PREFIX), infixation beyond the first mora (PREFIX/μ), and infixation beyond the first syllable (PREFIX/σ). These constraints are categorical, rather than gradient, such that a prefix preceded by a single segment and one preceded by multiple segments each incur a single violation. While McCarthy (2003) uses these constraints to make reference to a specific morpheme (e.g., PREFIX(-af-)), I adapt these as general constraints in Comox-Sliammon which affect *any* prefix.

- (124) a. PREFIX: The output correspondent(s) of a prefix are not preceded by a segment.

- b. PREFIX/ μ : The output correspondent(s) of a prefix are not preceded by a mora.
- c. PREFIX/ σ : The output correspondent(s) of a prefix are not preceded by a syllable.

modified from McCarthy (2003, p.94)

The tableau in (125) shows how ranking the PREFIX constraints below INTEGRITY-V, DEP-V motivates infixation within, but not beyond, the first syllable. The diminutive mora is indicated with a subscript μ in candidates; a period is used to show syllable breaks. The position of the diminutive reduplicant is limited by a combination of constraints penalizing the fission and epenthesis of vowels, which force the reduplicant inwards (125b–c). However, the affixed mora cannot migrate further than the first syllable without violating PREFIX/ σ (125d). This limits the diminutive to the attested position.

/ μ + supayu/		INT-V	DEP-V	PREFIX/ σ	PREFIX/ μ	PREFIX
(125)	a. sus μ .pa.yu				*	*
	b. si μ .su.pa.yu		*!			
	c. su μ .su.pa.yu	*!				
	d. su.pap μ .yu			*!	*	*

The tableau in (125) also shows that anything that violates PREFIX/ σ or PREFIX/ μ also violates PREFIX. If there is a mora or a syllable preceding the output correspondent(s) of the prefix, it will be associated with segmental content given the high ranking of *FLOAT. I only include PREFIX in subsequent tableaux for brevity.

The choice between μ (-C-) and i - (Ci-) can be captured by using preferential allomorph selection (Mascaró 2007). Violation marks are assigned under a PRIORITY constraint, given in (126), when the selected allomorph is not the preferred candidate (as noted by indices showing allomorph order).

(126) PRIORITY: Respect lexical priority (ordering)

Given an input containing allomorphs $m_1, m_2, \dots, m_n$, and a candidate m_i' , where m_i' is in correspondence with m_i , PRIORITY assigns as many

violation marks as the depth of ordering between m_i and the highest dominating morph.

(Mascaró, 2007, p.726)

I treat -C- as the default realisation of the diminutive, which means that μ is the default allomorph. The Ci- form from the /i/ allomorph occurs when the unreduplicated form of the word is monosyllabic or no vowel in the input to the first stratum. Choosing /i/ as the default allomorph would not work because words with a moraic vowel may take either μ or /i/; the choice depends on whether the word is monosyllabic. If /i/ was the priority allomorph, there would be no phonological motivation to select μ and the -C- form would never occur. The preferential order of allomorphs is provided in (127).

- (127) *Comox-Sliammon C₁ Diminutive:*
 $(\mu_1 > i_2) \Leftrightarrow \langle \text{DIM} \rangle$

*COMPLEXONSET, defined in (128), and DEP-V are ranked higher than PRIORITY, which motivates the choice of the non-default allomorph when the non-reduplicated form has a [ə]. I assume that [ə] is generally epenthized at Level 2 to ensure a well-formed prosodic word (i.e., [ə] is typically absent in at the first stratum).

- (128) *COMPLEXONSET (*COMPLEX): Syllables do not have complex onsets.
 (Blake, 2000, p.137, from McCarthy & Prince 1993)

Given that /i/ is a fixed segment, as opposed to a floating prosodic unit, I include the constraint MAX, defined in (129), which penalises the deletion of input segments. MAX sits in a comparable position to MAX-FLOAT in the constraint ranking.

- (129) MAX: All segments in the input have a correspondent in the output.
 Assign a violation mark for every segment in the input that does not have a correspondent in the output.

(McCarthy & Prince, 1995, p.16)

The tableau in (130) for example (109d) above shows the ranking of these constraints that allows for the selection of the non-preferred allomorph. The subscript numbers in the candidates correspond to which allomorph was inserted. The attested candidate (130a) violates PRIORITY, but candidates with the preferred (i.e., default) affix are ruled out because they either insert a vowel (130b violates DEP-V) or create a complex onset (130c violates *COMPLEX).¹² Deletion of a segment to satisfy *COMPLEX is ruled out under MAX, as shown in (130d).

$/(\mu_1 > i_2) + \theta\check{j}apuq^w/$		MAX	DEP-V	*COMPLEX	PRIORITY	PREFIX	INT-C
(130)	a. $\theta i_2 \theta \check{j}apuq^w$				*	*	*
	b. $\theta \check{\alpha} \theta i_1 \check{j}apuq^w$		*!			*	*
	c. $\theta \check{j}ay_1 puq^w$			*!		*	*
	d. $\check{j}ay_1 puq^w$	*!				*	*

In order to ensure that a consonant is reduplicated to provide an onset for the prefixed fixed-segment affix, ONSET, defined in (131a), must be ranked higher than PREFIX and INT-C. The constraint MAX-CONTIGUITY, defined in (131b), is also needed to ensure /i/ does not migrate into clusters to ensure an onset. MAX-CONTIGUITY penalises the loss of contiguity (immediate precedence) relations, such that if there is some segment x that immediately precedes a segment y, then there exists some segment x' (corresponding to x in the input) that immediately precedes a segment y' (corresponding to y in the input) in the output.^{13,14}

- (131) a. ONSET: Syllables have onsets (Blake, 2000, p.244).
b. MAX-CONTIGUITY (MAX-CONTIG): If $x, y \in S_1$ and x immediately precedes $(\rightarrow)y$, then $\exists x', y'$ such that $x\mathfrak{N}x'$ and $y\mathfrak{N}y'$ and x' immediately precedes $(\rightarrow)y'$ (Heinz, 2005, p.204).

The tableau in (132) shows that the correct candidate wins when ONSET and MAX-CONTIGUITY are ranked above INT-C: it is better to reduplicate a consonant to

¹²Recall from Section 4.1.1 that $[\check{j}]$ alternates with $[y]$ based on syllable position: the $[y]$ in (130c) results from fission of $\check{j}/$.

¹³ $x\mathfrak{N}x'$ and $y\mathfrak{N}y'$ in (131b) mark a relationship between input and output segments.

¹⁴Heinz (2005) motivates MAX-CONTIGUITY and DEP-CONTIGUITY as alternatives to LINEARITY constraints. I adopt MAX-CONTIGUITY because LINEARITY would be violated by both the attested diminutive in (132a) and the infixing diminutive in (132b) because fission and metathesis both result in a mismatch between linear order in the input and output.

provide an onset (132a) than to infix /i/ (132b). Infixing removes two immediate precedence relations present in the input ($i \rightarrow \theta$ and $\theta \rightarrow \check{j}$), resulting in two violations of MAX-CONTIG. The onsetless candidate (132c) fatally violates ONSET.

$/(\mu_1 > i_2) + \theta \check{j}apuq^w/$		ONSET	MAX-CONTIG	PRIORITY	PREFIX	INT-C
(132)	a. $\theta i_2 \theta \check{j}apuq^w$			*	*	*
	b. $\theta i_2 \check{j}apuq^w$		*!*	*	*	
	c. $i_2 \theta \check{j}apuq^w$	*!		*		

To account for the monosyllabic pattern, in which the affix surfaces as Ci- in words such as *pipuk^w* ‘little book’, an additional markedness constraint is needed. (*COMPLEX ensures the fixed-segment allomorph would be selected where a root had no underlying vowel, but this will not apply when the root comes with a moraic vowel.) I rule out unattested candidates with the -C- reduplicant (from the μ) by adopting a constraint violated by monomoraic outputs. I follow Black (2006, p.62) by adopting a constraint on stem binarity, modified to be specific to the level of the syllable and to specify a minimum number of syllables. I further modify the constraint to be about the prosodic word, rather than a morphological or phonological stem. The definition of MIN=2 σ is given in (133).

(133) MIN=2 σ : The prosodic word should be minimally two syllables.

The tableau in (134) for example (110) above shows that MIN=2 σ is ranked above PRIORITY, which motivates the selection of the non-priority allomorph where it would provide a second syllable (134a). The subscript *x* marks epenthesis, while the numbers correspond to the inserted allomorph. The constraint PREFIX/ σ rules out a candidate where the prefixed *i*- migrates further into the root.

$/(\mu_1 > i_2) + puk^w/$		DEP-V	MIN=2 σ	PREFIX/ σ	PRIORITY
(134)	a. pi_2puk^w				*
	b. $pi_{x,1}puk^w$	*!			
	c. $pupi_2k^w$			*!	*
	d. pup_1k^w		*!		

Further, I rank DEP-V and INT-V above MIN=2 σ to allow for monosyllabic words to emerge faithfully as a single syllable (outside of reduplication). This is shown

for *puk^w* ‘book’ in (135). $\text{MIN}=2\sigma$ will only affect the outcome when an input vowel is available.

(135)	/puk ^w /	DEP-V	INT-V	MIN=2 σ
	a. puk ^w			*
	b. pə _x puk ^w	*!		
	c. pupuk ^w		*!	

Preliminary Level 1 rankings are summarised in Table 4.3.¹⁵ Note that not all pair-wise rankings are reflected in the table: the rightmost column includes constraints that are never violated and/or outrank either multiple constraints (e.g., *FLOAT). The columns show rankings that can be established between constraints. For example, while PREFIX/ μ and PREFIX must be ranked beneath DEP-V and INT-V, it is unclear where PREFIX/ μ is ranked in relation to ONSET. The affix level constraints and REALIZE MORPHEME are excluded here, but are included as part of the Level 1 summary given in Section 4.1.3.3. Though PREFIX/ σ is not crucially ranked relative to the other constraints considered, it is included in the rightmost column of Row A because it is never violated. Appendix C includes flat linear rankings for each stratum; these flat linear rankings can be used to create tableaux.

Table 4.3: Preliminary Level 1 Grammar: Diminutive

A	DEP-V, INT-V		ONSET	*FLOAT, MAX, MAX-FLOAT, PREFIX/ σ , DEP-C, *COMPLEXONSET, MAX-CONTIG
B	MIN=2 σ	PREFIX/ μ	PREFIX	
C	PRIORITY			
D	INT-C			

¹⁵This constraint ranking will be updated in Section 4.1.3.2.2 to account for epenthesis in property-denoting C_1C_2 reduplication, which shows that DEP-V is ranked lower than INT-V.

4.1.3.1.3 Plural C₁ Reduplication

Plural C₁ reduplication in Comox-Sliammon can have either a -C- or Ca- shape. Examples with the Ca- allomorph of the plural are given in (136). These have the same distribution as the Ci- allomorph of the diminutive: they occur with words that would otherwise have [ə] as a first vowel and monosyllabic inputs.

- (136) a. **pa**~p_xit (cf. *pəxit* ‘torn’)
 C₁.PL~rip-STAT
 ‘all torn in pieces’ (BW)
- b. **ḡa**~ḡ_xit (cf. *ḡəḡit* ‘it is broken’)
 C₁.PL~break-STAT
 ‘They are broken.’ (BW)
- c. **ṭa**~ṭl[i]k-awus (root: *ṭlk*-)
 C₁.PL~make.hole-eye
 ‘piercing eyes’ (EP)
- d. **p̣a**~p̣uq̣^w (cf. *p̣uq̣^w* ‘grey’)
 C₁.PL~grey
 ‘grey things’ (EP/FL)

Examples of the -C- allomorph of the plural, which occurs with multisyllabic inputs with an underlying vowel, are given in (137). When plural reduplication has the -C- shape, it is homophonous with the *μ* allomorph of diminutive C₁ reduplication because both are evaluated under the same constraints and ranking (the Level 1 grammar).

- (137) a. **ṭu**<**ṭ**>**ṭ**-it (cf. *ṭuṭit* ‘something is on top’)
 on.top<C₁.PL>-STAT
 ‘A bunch of things are up top.’ (FL/KG)
- b. **ʔa**<**ʔ**>mut (cf. *ʔamut* ‘be home’)
 be.home<C₁.PL>
 ‘Everybody is home.’ (PD)
- c. **ḡi**<**ḡ**>m-us (cf. *ḡimus* ‘scratched face’)
 scratch<C₁.PL>-face
 ‘scratched on face (more than one scratch)’ (Watanabe, 2003, p.376)

4.1.3.1.4 Analysis of Plural C₁ Reduplication

The -C- allomorph of the C₁ plural occurs with multisyllabic words that have a moraic vowel as their first vowel. The Ca- allomorph occurs with monosyllabic inputs or words with [ə] as their first vowel. The lexical entry for C₁ reduplication and the preferential order of allomorphs is provided in (138).¹⁶ Subscript numbers refer to the order of listed allomorphs.

(138) *Comox-Sliammon C₁ Plural*:

$$(\mu_1 > a_2) \Leftrightarrow \langle C_1.PL \rangle$$

The same constraint ranking established for the diminutive also predicts the C₁ plural: the only difference is the vowel quality. The ranking that allows for the diminutive /i/ to be selected also allows /a/ to win, as shown in (139) for example (136a) above. The subscript numbers refer to which allomorph was selected.

/(\mu ₁ > a ₂) + p _x + it/		DEP-V	ONSET	PRIORITY	INT-C
(139)	a. pa ₂ p.xit			*	*
	b. pəp ₁ .xit	*!			*
	c. a ₂ p.xit		*!	*	

The mora is selected under the same conditions as was shown for the diminutive, which means no additional constraints or changes to the ranking are needed.

4.1.3.1.5 Unclear C₁ Reduplication

There are cases of C₁ reduplication which cannot be clearly classified on the basis of meaning: examples are given in (140). However, on the basis of form, these cases must involve C₁ reduplication at Level 1: if (140b) had Level 2 C₁ reduplication, the expected form would be *ʔiʔimaʃas (see description of progressive reduplication in Section 4.1.4.1.1).

- (140) a. hihiw=čx^w x^wa **qi**~qk^w-am-ax^w (cf. qək^wəmə 'stop')
 really=1SG.SUBJ NEG C₁~stop-MID-2SG.SUBJ
 'Sit down, you never keep still.' (EP/FL)

¹⁶I set aside the lexicalised subset of plural C₁ forms with a Ci- shape that were given at the outset of Section 4.1.3.1 in examples (107) and (108). However, this pattern can be derived if an /i/ replaces the /a/ as an input where lexically specified.

- b. $x^w a \text{ } \text{?i} < \text{?} > m\text{-aš-as}$ (cf. *?imaš* ‘walk’)
 NEG walk <C₁>-INTR-3SUBJ
 ‘He never walks.’ (EP/FL)

The two productive types of Level 1 C₁ reduplication mark either diminutivity or plurality, but neither meaning directly fits with the examples in (140). C₁ plural reduplication marks plurality with states, which is distributed across participants or space (see examples in Section 4.1.3.1.3); though negated events are states, it is unclear why these would be marked for plurality.

This type of C₁ reduplication may represent an additional function associated with Level 1 C₁ reduplication, with a very limited distribution: it occurs under negation (i.e., with $x^w a$). The translation of these sentences into English usually includes the word “never”.

As the form of this unclear C₁ reduplication is consistent with the Level 1 grammar, I conclude that it does not pose a challenge for my analysis of Comox-Sliammon reduplication and leave an exploration of its semantics to future work.

4.1.3.2 Level 1 C₁C₂ Reduplication

There is one type of C₁C₂ reduplication at Level 1 in Comox-Sliammon.¹⁷ A summary of Level 1 reduplication (repeated from earlier in this section) is provided in Table 4.4.

Table 4.4: Level 1 Reduplicative Affixes

	Underlying Form	Surface Position
Diminutive C ₁	μ, i	Prefix or Infix
Plural C ₁	μ, a	Prefix or Infix
Property-Denoting C ₁ C ₂	$\sigma_{N+\mu}$	Prefix

¹⁷On the basis of form, there are two types of C₁C₂ reduplication in Comox-Sliammon: one type of reduplication is at Level 1 and the other is at Level 2. See Section 4.1.4.2.1 for more on Level 2 C₁C₂ reduplication. However, in Chapter 5 (Section 5.5.10.2), I argue on the basis of meaning and multiple reduplication that there are actually three types of C₁C₂ reduplication. Plural C₁C₂ reduplication (with nouns) and property-denoting reduplication must happen at Level 1; pluractional C₁C₂ reduplication (with verbs) happens at Level 2.

4.1.3.2.1 Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication copies the first two consonants of the root with an intervening vowel, which is usually [ə] rather than a faithful copy of the root vowel. The examples in (141) show examples of the CəC form of C₁C₂ reduplication.

- (141) a. **təw̃**~tiwš (*tiwšam* ‘learn’)
 PDEN~smart
 ‘fast learner, smart’ (Watanabe, 2003, p.398)
- b. **šəṁ**~šam (root: *šam*- ‘square’)
 PDEN~square
 ‘square (shaped basket, object)’ (Watanabe, 2003, p.398)
- c. **təw̃**~tuw (cf. *tugut* ‘recognise them’)
 PDEN~remember
 ‘s.o. who remembers well’ (Watanabe, 2003, p.398)
- d. **gəỹ**~gay-ał-tən (cf. *gayat* ‘ask them’)
 PDEN~ask-CONNECT-instrument
 ‘curious’ (Watanabe, 2003, p.398)

Watanabe (2003, p.397) states that property-denoting C₁C₂ reduplication is restricted to roots with a sonorant segment as their second consonant, which allows him to establish a phonological diagnostic that differentiates property-denoting reduplication from C₁C₂ plural reduplication (described in Section 4.1.4.2.1); glottalisation falls on the first of the matching sonorants in C₂ position in property-denoting forms (142a), and on the second in plural denoting forms (142b).

- (142) a. **šəṁ**~šam (cf. *šamat* ‘dry it’)
 PDEN~dry
 ‘dry’ (Watanabe, 2003, p.398)
- b. **təṁ**~təṁx^w (*təṁx^w* ‘gooseberry’)
 C₁C₂.PL~gooseberry
 ‘gooseberries’ (Watanabe, 2003, p.372)

However, glottalisation of sonorant consonants alone is not a reliable clue for identifying property-denoting reduplication. Sonorant glottalisation is notoriously variable in Salish: it may be realised with a range of different acoustic correlates (e.g., creaky voicing or changes in pitch), even within the same language (Bird, Caldecott, Campbell, Gick, & Shaw, 2008). This diagnostic for identifying property-denoting reduplication is further complicated by the fact that glottalisation on sonorant segments is falling out of use in modern Comox-Sliammon (Watanabe, 1994a; Caldecott & Mellesmoen, 2018). Further, there are examples of C_1C_2 reduplication with an obstruent in C_2 position that are property-denoting rather than plural, which therefore suggests that property-denoting C_1C_2 reduplication is not actually restricted to roots with a sonorant segment as their second consonant. Examples are provided in (143).

- (143) a. $\text{ʔəx} \sim \text{ʔəx-tən}$ (cf. ʔəx ‘bad’)
 PDEN~bad-instrument
 ‘mischievous, misbehaved’ (WJF/MV)
- b. $\text{ʔəx} \sim \text{ʔəx-jim}$ (cf. ʔəx ‘bad’)
 PDEN~bad-susceptible
 ‘clumsy’ (WJF/MV)
- c. $\text{gə}^{\text{h}} \sim \text{gə}^{\text{h}}$ (root: gə^{h} ‘tease’)
 PDEN~tease
 ‘someone always teasing’ (FL/KG)

Where glottalisation is not present, property-denoting and plural C_1C_2 reduplication are distinguished on the basis of meaning, context, and distribution. Property-denoting reduplication occurs with adjectives and stative verbs, while plural/pluractional reduplication occurs with nouns and eventive verbs. None of these clues constitute sufficient evidence alone: for example, Huijsmans (2023b, p.184) provides $\text{gə}^{\text{h}}\text{gə}^{\text{h}}$ as an example where the meaning aligns with either a noun (‘a tease’) or an adjective (property of someone always teasing), but shows that its syntactic distribution is more consistent with that of a verb because unlike bare adjectives and nouns, it can occur as the predicate of a postnominal relative clause. Distribution alone is therefore not enough to differentiate plural and property-denoting reduplication. Further evidence beyond glottalisation for

analysing these types of reduplication at different strata is given as part of the analysis of multiple reduplication in Chapter 5 (Section 5.5).

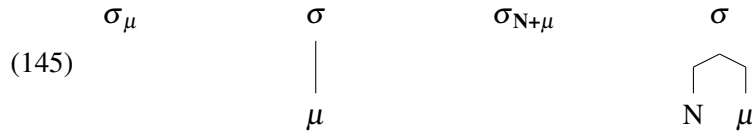
4.1.3.2.2 Analysis of Property-Denoting C_1C_2 Reduplication

Property-denoting C_1C_2 reduplication involves the affixation of a monomoraic syllable at Level 1. The lexical entry for the property-denoting morpheme contains a monomoraic syllable with a nucleus and a non-nuclear mora ($\sigma_{N+\mu}$). The lexical entry for the property-denoting morpheme is provided in (144).

(144) *Comox-Sliammon Property-Denoting Reduplication:*

$$\sigma_{N+\mu} \Leftrightarrow \langle \text{PDEN} \rangle$$

The distinction between a σ_μ and $\sigma_{N+\mu}$ was introduced in the previous chapter (Section 3.2.3) to differentiate between an open CV and a closed CəC monomoraic syllable. The two configurations are given again in (145). The crucial difference is that σ_μ only has one (a mora), while $\sigma_{N+\mu}$ has two dependants (a nucleus and non-nuclear mora). C_1C_2 reduplication requires an affixed $\sigma_{N+\mu}$.



The analysis of property-denoting reduplication builds on the Level 1 ranking established in the previous sections, given again in Table 4.5. However, this ranking will be revised below because C_1C_2 reduplication motivates updating the constraint ranking to account for all three Level 1 patterns.

Table 4.5: Preliminary Level 1 Ranking from Section 4.1.3.1.2

A	DEP-V, INT-V		ONSET	FLOAT, MAX, MAX-FLOAT, PREFIX/ σ , DEP-C, *COMPLEXONSET, MAX-CONTIG
B	MIN=2 σ	PREFIX/ μ	PREFIX	
C	PRIORITY			
D	INT-C			

I omit *FLOAT and MAX from the tableaux in this section for reasons of space. MIN=2 σ is not included because, unlike diminutive or plural C₁ reduplication, C₁C₂ reduplication always adds a syllable to monosyllabic inputs. *COMPLEXONSET also does not need to be included in the tableaux, because only a single consonant will be reduplicated; a complex onset incurs additional (and fatal) violations of DEP-C. PRIORITY is also omitted because the input for C₁C₂ reduplication does not have multiple allomorphs.

The tableau in (146) for example (141c) above shows that ranking both ONSET and PREFIX/ σ over INT-C ensures that the winning candidate in (146a) has the affixed syllable on the left edge; fission of the C₁ consonant provides an onset. The onsetless candidate in (146b) fatally violates ONSET. A candidate with the affixed syllable as the rightmost fatally violates PREFIX/ σ . The bracketed syllables in the candidates mark where the affixed syllable is positioned.

(146)	/ $\sigma_{N+\mu}$ +tuw/		PREFIX/ σ	ONSET	INT-C
		a. (təw) $\sigma_{N+\mu}$.tuw			**
		b. (əw) $\sigma_{N+\mu}$.tuw		*!	*
		c. tu.(wəw) $\sigma_{N+\mu}$	*!		*

As shown in (147), INT-V must be ranked above DEP-V to motivate epenthesis (147a) over fission (147b) of a vowel. A candidate with epenthesis of a consonant

in (147c) is ruled out because it fatally violates DEP-C.¹⁸

$/\sigma_{N+\mu} + \text{tuw}/$		DEP-C	INT-V	DEP-V	INT-C
(147)	a. $(\text{t}\dot{\text{ə}}\dot{\text{w}})_{\sigma_{N+\mu}}.\text{tuw}$			*	**
	b. $(\text{tu}\dot{\text{w}})_{\sigma_{N+\mu}}.\text{tuw}$		*!		**
	c. $(\text{?}\dot{\text{ə}}\dot{\text{w}})_{\sigma_{N+\mu}}.\text{tuw}$	*!		*	*

A candidate with vowel epenthesis filling both the nucleus and the mora $(*(\text{ti}_{N,\mu}).\text{tuw})$ was not considered in (147) because this would not be compatible with the structure of the affixed prosodic unit, which has both a nucleus and a non-nuclear mora as separate dependants that must both be filled. This requirement arises from MAX-FLOAT in combination with the constraint DEP-LINK- $\mu(\text{N})$, given in (148), which is modified from MAXLINK- $\mu(\text{C})$ in Prilop (2013). DEP-LINK- $\mu(\text{N})$ is violated whenever an association line is added between a nucleus and a mora that was not in present in the input.

(148) DEP-LINK- $\mu(\text{N})$ (DEPL- $\mu(\text{N})$) : When a nucleus is linked to a mora in the output, it is also linked to a mora in the input (modified from Prilop 2013, p.22).

The tableau in (149) shows that candidates (149b) and (149c) incur fewer violations under INT-C and DEP-V, respectively, when compared to the winning candidate because they associate both the floating nucleus and mora with the same segment; these candidates are therefore ruled out under DEP-LINK- $\mu(\text{N})$. I consider the deletion of any dependent of the floating syllable to incur a violation of MAX-FLOAT, as demonstrated by candidate (149d). The structure of $\sigma_{N+\mu}$ must therefore always be reflected in the output.¹⁹

$/\sigma_{N+\mu} + \text{tuw}/$		MAXFLT	DEPL- $\mu(\text{N})$	INT-V	DEP-V	INT-C
(149)	a. $(\text{t}\dot{\text{ə}}_{\text{N}}\dot{\text{w}}_{\mu})_{\sigma}.\text{tuw}$				*	**
	b. $(\text{ti}_{\text{N},\mu})_{\sigma}.\text{tuw}$		*!		*	*
	c. $(\text{tu}_{\text{N},\mu})_{\sigma}.\text{tuw}$		*!	*!		*
	d. $(\text{t}\dot{\text{ə}}_{\text{N}})_{\sigma}.\text{tuw}$	*!			*	*

¹⁸There are some exceptional forms where the vowel matches that of the root. These must be treated separately as lexicalised or lexically-specified reduplication; see explanation of these terms in Chapter 1 (Section 1.4.3.2).

¹⁹Except in $/\text{əy}/$ and $/\text{əw}/$ sequences, which are realised as $[\text{i}]$ and $[\text{u}]$, as discussed in Section 4.1.1.

An additional constraint is needed for the analysis of property-denoting reduplication to ensure that the epenthesized vowel is [ə]. As I treat all instances of [ə] as epenthetic, these are omitted from the underlying forms. I include output instances of [ə] in the candidates for clarity, though I assume these are inserted at subsequent strata, aside from where required to fill the nucleus of an affixed $\sigma_{N+\mu}$.

The tableau in (150) for example (143b) above shows two candidates that differ in whether the empty nucleus is filled by a [ə] (candidate 150a) or a moraic vowel (candidate 150b); the former is the attested form.²⁰ As both candidates have epenthesis, they incur the same number of violations under DEP-V. The use of ?? indicates that the two options cannot be distinguished.

(150)	$/\sigma_{N+\mu} + \text{tə} + \check{\text{jim}}/$	DEP-V	INT-C
	?? a. $(\text{tə})_{\sigma_{N+\mu}} . \text{tə} . \check{\text{jim}}$	*	**
	?? b. $(\text{tə})_{\sigma_{N+\mu}} . \text{tə} . \check{\text{jim}}$	*	**

I adopt the constraint DEP- V_μ in (151) to distinguish moraic vowels from non-moraic vowels.

- (151) DEP- V_μ : Every moraic vowel in the output must have a corresponding moraic vowel in the input.

I assume, following Blake (2000), that schwa does not have any place features of its own, while moraic vowels do. The constraint given in DEP- V_μ should therefore be more formally understood as DEP-V-Place, which is violated by the epenthesis of a V-Place node, in the sense of Clements and Hume (1995, p.276).²¹

Having both DEP- V_μ and DEP-V in the grammar results in [ə] being the least marked epenthetic vowel, which is consistent with cross-Salish phonology. The

²⁰I include the ‘susceptible’ suffix $\check{\text{jim}}$ as a Level 1 affix in (150). However, this should not be taken as an analysis of $\check{\text{jim}}$: further work would be required to determine if $\check{\text{jim}}$ is a Level 1 or Level 2 suffix.

²¹The surface quality of schwa is conditioned by adjacent consonants, which I assume entails sharing of features, though I remain agnostic as to whether a feature geometry that differentiates V-Place from C-Place is desirable for Salish. I refer to a V-Place node here as a catch-all for the relevant features pertaining to vowel place, but a general DEP-[Place] constraint could be used to penalise epenthesis of *any* place features or individual constraints targeting specific features (e.g., those under DORSAL).

revised tableau in (152) shows the inclusion of this constraint, which now correctly predicts the attested candidate (152a).

$/\sigma_{N+\mu} + \text{ɬ} \check{x} + \text{jim}/$		DEP- V_μ	DEP-V	INT-C
(152)	☞ a. $(\text{ɬ} \check{\text{ɔ}} \check{x})_{\sigma_{N+\mu}} . \text{ɬ} \check{\text{ɔ}} \check{x} . \text{jim}$		*	**
	b. $(\text{ɬ} \check{i} \check{x})_{\sigma_{N+\mu}} . \text{ɬ} \check{\text{ɔ}} \check{x} . \text{jim}$	*!	*	**

The only changes needed from the preliminary Level 1 ranking established for C_1 reduplication include the addition of DEP- V_μ and the addition of a crucial ranking between INT-V and DEP-V.

4.1.3.3 Summary of Level 1 Reduplication

Diminutive C_1 , plural C_1 , and property-denoting C_1C_2 reduplication are affixed at Level 1 in Comox-Sliammon.

The Level 1 grammar in Comox-Sliammon allows for the affix to be positioned as an infix or a prefix. In order to fill the affixed units, consonants are reduplicated and [ə] may be epenthesized, if necessary. A mora (μ) or a fixed segment (/i/ or /a/) is affixed for the diminutive C_1 and plural C_1 reduplication, while a monomoraic syllable ($\sigma_{N+\mu}$) is affixed for property-denoting reduplication. The constraint rankings needed to derive Level 1 reduplication patterns are summarised in Table 4.6. Constraints in the rightmost column are not violated at this stratum and may be crucially ranked above one or more constraints on Rows B–E (e.g., ONSET is ranked above PREFIX and INT-C).

Table 4.6: Comox-Sliammon Level 1 Grammar

A	INT-V		MAXFLT, *FLOAT, MAX, PREFIX/ σ , DEP-C, *COMPLEXONSET, DEP-V $_{\mu}$, DEPL- μ (N), ONSET
B	DEP-V		
C	MIN=2 σ	PREFIX/ μ , PREFIX	
D	PRIORITY, MAX-CONTIG		
E	INT-C		

Constraints against parsing affixes marked with a L2 (Level 2) or L3 (Level 3) diacritic are also relevant at Level 1, as argued in Chapter 3 (Section 3.5.2.3): AFFIX_{L2} and AFFIX_{L3} are ranked higher than PARSE_{AFFIX}, as shown in Table 4.7.

Table 4.7: Affix Constraints at Level 1

A	AFFIX _{L2} , AFFIX _{L3}
B	PARSE _{AFFIX}

In sum, diminutive and plural C₁ reduplication and property-denoting C₁C₂ reduplication can all be analysed as Level 1 affixes in Comox-Sliammon, which means that they are derived with the same constraint ranking.

4.1.4 Level 2

Level 2 reduplication in Comox-Sliammon includes progressive C₁ and plural C₁C₂ reduplication. Of all the reduplicative processes in Comox-Sliammon, the two Level 2 ones are the most productive. C₁ progressive reduplication is

particularly frequent in spontaneous speech because it marks a common aspectual function.

Progressive C_1 reduplication involves the affixation of a mora; the reduplicated segments (a consonant and, if applicable, the following moraic vowel) are positioned on the left-edge of the word as a prefix. Plural C_1C_2 reduplication involves the affixation of a monomoraic syllable with a nucleus node. The first two consonants of the input are copied and a [ə] is epenthesized. Plural C_1C_2 reduplication is homophonous with property-denoting C_1C_2 reduplication, except when a sonorant sound is in C_2 position (see Section 4.1.3.2.1).

The prosodic units and position of Level 2 reduplicative processes are summarised in Table 4.8. Level 2 reduplication has the following properties: (1) the reduplicated segments are positioned as prefixes (on the left edge of the prosodic word), and (2) consonants are copied and moraic vowels may also be copied (unlike at Level 1).

Table 4.8: Level 2 Reduplicative Affixes

	Underlying Form	Surface Position
Progressive C_1	μ	Prefixed or Infix
Plural C_1C_2	$\sigma_{N+\mu}$	Prefixed

4.1.4.1 Level 2 C_1 Reduplication

4.1.4.1.1 *Progressive C_1 Reduplication*

In progressive C_1 reduplication, the first consonant of the word is copied.²² If the vowel following the first consonant is moraic, then the vowel is copied as well, as shown in (153). If the vowel following the first consonant in the non-reduplicated word is non-moraic (i.e., [ə]), then only the first consonant is copied, as shown in (154). Though the reduplicative morpheme is analysed as a prefix in the input (see Section 4.1.4.1.2), the segments associated with the reduplicative morpheme

²²This type of reduplication is glossed as imperfective in other sources (e.g., Watanabe, 2003); however, H. Davis et al. (2020, p.120) argue that aspectual C_1 reduplication in Comox-Sliammon marks progressive aspect, rather than general imperfectivity.

surface as a prefix in (153) and an infix in (154).

- (153) a. **ʔa**~ʔaʔ-at (cf. *ʔaʔat* ‘chase them (sg.)’)
 PROG~chase-CTR
 ‘be chasing them (sg.)’
- b. **gu**~guh-um (cf. *guh* ‘bark’)
 PROG~bark-MID
 ‘be barking’
- c. **θa**~θap-iš (cf. *θapiš* ‘bathe them (sg.)’)
 PROG~bathe-TR
 ‘be bathing them (sg.)’
- d. **ǰi**~ǰix-im (cf. *ǰixim* ‘fall apart’)
 PROG~fall.apart-MID
 ‘be falling apart’
- e. **ya**~yaʔ-at (cf. *yaʔat* ‘call them (sg.)’)
 PROG~call-CTR
 ‘be calling them (sg.)’ (Watanabe, 2003, p.392)
- (154) a. $\check{c}\check{a}<\check{c}>\check{t}$ (cf. *čət* ‘rain’)
 rain<PROG>
 ‘raining’
- b. $k^w\check{a}<k^w>q-aʔam$ (cf. *kʷəqʔəm* ‘split’)
 split<PROG>-A.INTR
 ‘splitting’
- c. $n\check{a}<n>m-aš$ (cf. *nəmaš* ‘throw’)
 throw<PROG>-TR
 ‘throwing it’
- d. $p\check{a}<p>n-aʔam$ (cf. *pənʔəm* ‘plant’)
 plant<PROG>-A.INTR
 ‘be planting’
- e. $p\check{a}<p>\lambda š$ (cf. *pəλš* ‘surface’)
 surface<PROG>
 ‘surfacing’ (Watanabe, 2003, pp.392–393)

When the progressive is added to a CəCC- base where one of the first two consonants is a sonorant, a second [ə] follows the consonant associated with the reduplicative morpheme, as shown in (155). As [ə] is epenthetic and epenthesis is predicted to break up consonant clusters, I do not consider this vowel to be a copy; the reduplication in (155) and in (154) can receive the same analysis.

- (155) a. qə<**q**>əms-at (cf. *qəmsat* ‘put it away’)
 put.away<PROG>-CTR
 ‘be putting it away’
 b. nə<**n**>əlx^w-at (cf. *nəlx^wat* ‘stir it’)
 stir<PROG>-CTR
 ‘be stirring it’
 c. sə<**s**>əyt-aʔam (cf. *səytaʔam* ‘throw’)
 throw<PROG>-A.INTR
 ‘be throwing (something)’ (Watanabe, 2003, p.393)
 d. mə<**m**>ətk-at=č (cf. *mətkatčx^w* ‘Squash it.’)
 squash<PROG>-CTR=1SG.SUBJ
 ‘I am squashing it.’ (FL)

4.1.4.1.2 Analysis of Progressive C₁ Reduplication

The lexical entry for the progressive in Comox-Sliammon consists of a mora, as shown in (156). The mora bears an L2 diacritic to mark it as a Level 2 affix.

- (156) *Comox-Sliammon Progressive:*
 μ_{L2} ⇔ <PROG>

The constraints *FLOAT, REALIZEMORPHEME, and MAX-FLOAT are ranked highly in the Level 2 grammar, as at Level 1.^{23,24} This is shown in (157) for example (153e) above where the attested candidate violates INTEGRITY-C and

²³Constraint definitions can be found in Appendix D, which includes a key to where they are first introduced.

²⁴As proposed in Chapter 3 (Section 3.5.2.3) and discussed in Section 4.1.3.1.2, the constraint PARSE_{AFFIX} outranks AFFIX_{L2} in the Level 2 grammar, which permits affixes with an L2 diacritic to be parsed into the phonological word.

INTEGRITY-V, but none of the higher ranked constraints (*FLOAT, MAX-FLOAT, and REALIZEMORPHEME). Candidates with a floating mora (157b), no exponent of the progressive morpheme (157c), or the progressive mora associated with a vowel, which would be associated with a mora regardless, all fatally violate at least one of these constraints. *FLOAT, *MAX-FLOAT, and REALIZEMORPHEME are omitted for the remainder of the Level 2 analysis.

/μ+yaɫ+at/		*FLOAT	MAXFLT	RM	INT-V	INT-C
(157)	☞ a. ya _μ .ya.ɫat				*	*
	b. μ ya.ɫat	*!				
	c. ya.ɫat		*!			
	d. ya _μ .ɫat			*!		

The first major difference between the Level 1 and Level 2 grammar is that INTEGRITY-V is not ranked as highly at Level 2 as at Level 1. As shown in (157), the attested candidate (157a) violates INTEGRITY-V and therefore this constraint must be ranked low enough to predict the correct form. As no vowels are copied at Level 1, INTEGRITY-V is a higher ranked constraint there than at Level 2.

The second difference between Level 1 and Level 2 is the surface position of an affixed mora. At Level 1, the mora marking diminutivity or plurality was positioned as an infix following the first mora in the word (see description and analysis in Section 4.1.3.1); at Level 2, the mora marking progressive aspect is the first mora in the word. For example, the progressive form of *yaɫat* ‘call someone’ is *yayaɫat*, rather than **yayɫat*, which reflects a difference in how an affixed mora is realised across strata. The difference between Level 1 and Level 2 comes from the ranking of PREFIX/μ, which rules out infixation of prefixed material beyond the first mora. PREFIX/μ, restated again in (158), will be violated if the progressive mora is not the leftmost mora in the word. This means that the progressive must be associated with either the first nucleus in a word (if it is a moraic vowel) or the first coda consonant following a non-moraic nucleus.

(158) PREFIX/μ: A prefix is not preceded by a mora.

As shown in (159), PREFIX/μ is ranked highly in the Level 2 grammar, which rules out candidates where the mora is positioned as an infix (159b-c). Note that

the candidate in (159c) would be predicted at Level 1 and *yaylat* would be the expected form for a C₁ plural or diminutive.

/μ ₁ +ya _{μ2} .‡a _{μ3} t _{μ4} /		PREFIX/μ	INT-V	INT-C
(159)	☞ a. ya _{μ1} .ya _{μ2} .‡a _{μ3} t _{μ4}		*	*
	b. ya _{μ2} .ya _{μ1} .‡a _{μ3} t _{μ4}	*!	*	*
	c. ya _{μ2} y _{μ1} .‡a _{μ3} t _{μ4}	*!		*

The third major difference between Level 1 and Level 2 is the fact that both consonants and vowels can undergo fission, as in the attested candidate in (160a), which means that both INTEGRITY constraints must be ranked below DEP-C and DEP-V_μ at Level 2 to rule out candidates with an epenthetic consonant (160b) or moraic vowel (160c). Candidate (160d), with no onset, is ruled out by ONSET. Candidate (160e), which has C₁C₂ reduplication, motivates ranking INT-C above INT-V.

/μ ₁ +ya _{μ2} .‡a _{μ3} t _{μ4} /		DEP-C	ONSET	DEP-V _μ	INT-C	INT-V	DEP-V
(160)	☞ a. ya _{μ1} ya _{μ2} .‡a _{μ3} t _{μ4}				*	*	
	b. ?a _{μ1} ya _{μ2} .‡a _{μ3} t _{μ4}	*!				*	
	c. yi _{μ1} ya _{μ2} .‡a _{μ3} t _{μ4}			*!	*		*
	d. a _{μ1} ya _{μ2} .‡a _{μ3} t _{μ4}		*!			*	
	e. yə‡ _{μ1} ya _{μ2} .‡a _{μ3} t _{μ4}				**!		*

Next, I consider the roots without an underlying vowel, which surface with [ə]. I assume that [ə] is epenthized as part of the Level 2 derivation: this is shown for the (non-reduplicated) perfective form *nəmaš* ‘throw’ in (161).²⁵ *COMPLEXONSET rules out the faithful candidate (161b), and the attested candidate (161a) violates only DEP-V. This example further shows that INT-V is ranked above DEP-V: it is better to epenthesize /ə/ than reduplicate. The subscript numbers in (161c) show that /a/ has undergone fission.

/nm+aš/		*COMPLEX	INT-V	DEP-V
(161)	☞ a. nə.maš			*
	b. nmaš	*!		
	c. na ₁ .ma ₁ š		*!	

²⁵Other constraints deriving syllabification and the assignment of stress are omitted here but assumed to be active in the grammar.

The derivation of a progressive form built on a root with a non-moraic vowel is shown in (162) for *nəmaš* ‘throw it’ from (154c) above. The attested candidate (162a) violates DEP-V and INTEGRITY-C. Candidate (162b), which inserts a full vowel, fatally violates DEP-V_μ; candidate (162c), which resembles a C₁C₂ reduplicated word, receives a second (and fatal) violation under INT-C. Candidate (162d), which reduplicates further into the word than the other candidates, fatally violates PREFIX/μ. A potential candidate [nənəmaš] with CV reduplication is not considered in this tableau; [ə] is non-moraic and therefore there would be no segmental content to associate with the progressive mora. Subscript numbers are used to mark the position of input moras in the output.

/μ ₁ +nm+a _{μ2} š/	PREFIX/μ	DEP-V _μ	INT-C	DEP-V
(162) a. nən _{μ1} .ma _{μ2} š _{μx}			*	*
b. ni _{μ1} ma _{μ2} š _{μx}		*!		*
c. nəm _{μ1} nəma _{μ2} š _{μx}			**!	*
d. nəma _{μ2} m _{μ1} š _{μx}	*!		*	*

Progressive forms built on a CəCC root with a sonorant as one of the first two consonants do not appear to have an additional mora on the surface, contrary to the lexical entry in (156). However, progressive forms such as *qəqəmsat* ‘be putting it away’ are not counterexamples to the analysis given thus far and in fact provide further evidence for a sequential (stratum-based) model of phonology. This is because all the roots which take the “mora-less” Cə progressive have a sonorant as their second consonant. Clusters of the shape CCC are rare in Comox-Sliammon; clusters with a sonorant as a middle segment (Watanabe, 2003, p.393), since they violate the sonority hierarchy (Parker, 2011). If the second [ə] was not present in the word *qəqəmsat*, this would create an illicit cluster **qms* in the expected progressive form **qəqmsat*, which could not surface as it would violate constraints at a subsequent point in the derivation (i.e., Level 3) and trigger epenthesis of another [ə] at a later stratum than Level 2.

Thus, though on the surface, the number of moras in *qəmsat* and *qəqəmsat* is the same: as shown in (163) for example (155) above, they each have three moras despite the progressive affix itself being a mora (i.e., adding an additional mora). However, the progressive mora is affixed and filled at Level 2, while subsequent

epenthesis and resyllabification at a later stratum motivated by violations of the sonority hierarchy results in the loss of a mora. The situation is summarised in Table 4.9.²⁶

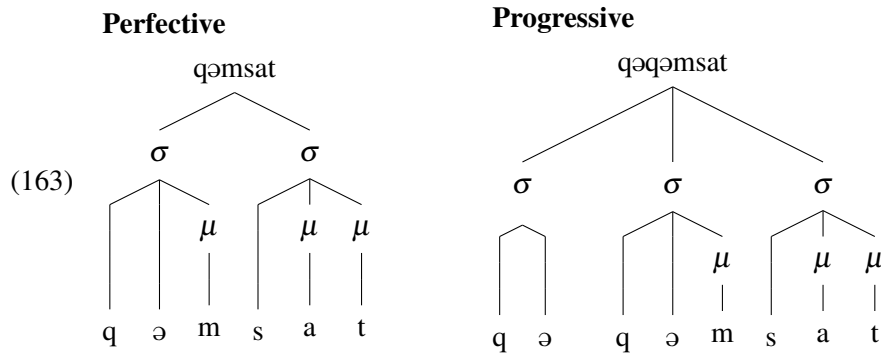


Table 4.9: Number of Underlying and Surface Moras (CəSC Base)

	Form	Number of Moras
Input _{Level 1}	qmsat	N/A
Output _{Level 1}	qmsa _μ t _μ	2
Input _{Level 2}	μ + qmsa _μ t _μ	3
Output _{Level 2} and Input _{Level 3}	qəq _μ m _μ sa _μ t _μ	4
Output _{Level 3}	qəqə _μ m _μ sa _μ t _μ	3

The reduplication of CəCC roots provides an argument against a parallel OT derivation: an additional stage of derivation following the affixation of the progressive mora is needed to explain why the progressive and perfective forms have the same number of moras, despite the fact that reduplication occurs to fill an additional (affixed) mora in the progressive forms.

Table 4.10 summarises the constraint ranking needed to derive progressive reduplication in Comox-Sliammon. As this ranking proves sufficient for the

²⁶I do not include a mora associated with the vowel in the input in Table 4.9, though one may be there underlyingly. Given that any vowel aside from /ə/ will be moraic and only moraic vowels are present in underlying forms in Comox-Sliammon, having moras either in the underlying forms or assigned by the phonological grammar at the point of syllabification will yield similar results. In the bigger picture, it is not clear if the underlying form of vowels should contain a mora. This is a question I intend to pursue in for future work as it has broader implications for the analysis of vowels in Salish, particularly in languages that reduce moraic vowels to [ə].

analysis of the other Level 2 reduplication process (discussed in the subsequent section), this ranking will not need to be revised.

Table 4.10: Comox-Sliammon Level 2 Grammar

A	PREFIX/ μ , DEP-V μ , DEP-C, ONSET	REALIZEMORPHEME, *FLOAT, *COMPLEX, MAX-FLOAT
B	INT-C	
C	INT-V	
D	DEP-V	

4.1.4.2 Level 2 C₁C₂ Reduplication

4.1.4.2.1 Plural C₁C₂ Reduplication

In plural C₁C₂ reduplication, the first two consonants of the root are copied with [ə] positioned between the first instance of C₁ and C₂. Examples are provided in (164). Descriptively, plural C₁C₂ reduplication can be characterised as a prefixed CəC- reduplicant.

- (164) a. $k^wəx^w \sim k^wax^wa$ (cf. k^wax^wa ‘box’)
 C₁C₂.PL $\sim$ box
 ‘boxes’ (JF)
- b. $jəʃ \sim jəʃən$ (cf. $jəʃən$ ‘foot’)
 C₁C₂.PL $\sim$ foot
 ‘feet’ (JF)
- c. $ʔəs \sim ʔasx^w$ (cf. $ʔasx^w$ ‘seal’)
 C₁C₂.PL $\sim$ seal
 ‘seals’ (JF)

Plural C₁C₂ reduplication is productive, as evident by its application to loanwords, such as in (165). The example in (165a) comes from *rum*, with the addition of the lexical suffix *-aya* ‘container’; (165b) comes from the English word *cherries*.

- (165) a. **ləm**~lam-aya (cf. *lamaya* ‘bottle’)
 C₁C₂.PL~bottle
 ‘bottles’ (JF)
- b. **čəl**~čəlas (cf. *čəlas* ‘cherry’)
 C₁C₂.PL~cherry
 ‘cherries’ (JF)

According to Watanabe (2003), the only formal difference between Level 1 property-denoting C₁C₂ reduplication and Level 2 plural C₁C₂ reduplication is in the glottalisation of sonorants. When the second consonant of the root is a glottalised sonorant, only one of the matching sonorants will be glottalised in the output. Glottalisation in property-denoting reduplication falls on the first of two matching sonorants (as in *təw̥tiw̥ʂ* ‘smart’), while glottalisation in the plural forms falls on the second (166).

- (166) a. **təm**~təmx^w (cf. *təmx^w* ‘gooseberry’)
 C₁C₂.PL~gooseberry
 ‘gooseberries’ (Watanabe, 2003, p.373)
- b. **čəy**~čuý (cf. *čuý* ‘child’)
 C₁C₂.PL~child
 ‘children’ (EP)

However, glottalisation is only relevant for sonorants, and as mentioned earlier, it is unreliable in modern Comox-Sliammon as a cue due to its variability.

4.1.4.2.2 Analysis of Plural C₁C₂ Reduplication

Plural C₁C₂ reduplication happens at Level 2 to fill a syllable with a nucleus and a non-nuclear mora, as shown in (167). The lexical entry has a L2 diacritic to mark it as a Level 2 affix. The strata where different types of C₁C₂ reduplication happen are revisited in the analysis of multiple reduplication in Chapter 5. See Section 5.5 for a revised classification that distinguishes between plural diminutive verbs (at Level 2) and nouns (at Level 1).

- (167) *Comox-Sliammon Plural Reduplication (Word Level):*
 ($\sigma_{N+\mu}$)_{L2} ⇔ ⟨PL⟩

The constraint ranking established for the progressive also predicts the attested form for plural C_1C_2 reduplication, as shown in the tableau in (168) for example (164c) above. The parentheses in the input to Level 2 correspond to syllables built at Level 1. The attested candidate (168a) violates DEP-V and INT-C. Candidate (168b), which copies a vowel, fatally violates INT-V; candidate (168c), which infixes the affixed syllable, violates PREFIX/ σ . A candidate with an epenthetic full vowel fatally violates DEP- V_μ (168d).

	$/\sigma_{N+\mu 1} + (? a_{\mu 2} s_{\mu 3} x_{\mu 4}^w)/$	PREFIX/ σ	DEP- V_μ	INT-C	INT-V	DEP-V
(168)	a. $?a_{\mu 1} s_{\mu 1} . ?a_{\mu 2} s_{\mu 3} x_{\mu 4}^w$			**		*
	b. $?a_{\mu x} s_{\mu 1} . ?a_{\mu 2} s_{\mu 3} x_{\mu 4}^w$			**	*!	
	c. $?a_{\mu 2} s_{\mu 3} . ?a_{\mu 1} x_{\mu 4}^w$	*!		**		*
	d. $?i_{\mu x} s_{\mu 1} . ?a_{\mu 2} s_{\mu 3} x_{\mu 4}^w$		*!	**		*

* $?a_{N,\mu} ?asx^w$ is not considered as a candidate in the tableau above because it would not satisfy the affixed prosodic unit: there must be at least one moraic segment associated with the syllable that is not the nucleus. The relevant constraint DEP-LINK- $\mu(N)$ is given in (169), restated from Section 4.1.3.2.2.²⁷ It is violated whenever an association line is added between a nucleus and a mora that was not present in the input.

(169) DEP-LINK- $\mu(N)$: When a nucleus is linked to a mora in the output, it is also linked to a mora in the input.

The tableau in (170) shows that a high ranking of either DEP-LINK- $\mu(N)$ or INT-V rules out candidate (170a), which re-associates the mora with the nucleus to allow for an open syllable to fill the affixed monomoraic syllable. Candidate (170a), which respects the structure of the input syllable, does not violate DEP-LINK- $\mu(N)$ nor INT-V; the attested candidate wins.²⁸

²⁷This candidate would also be ruled for independent reasons: it fatally violates INT-V.

²⁸DEP-LINK- $\mu(N)$ is not strictly necessary for Comox-Sliammon at Level 2, given that any candidates which associate the nucleus and the mora with the same (moraiic) vowel would also violate DEP- V_μ or INT-V; it is not possible to supply a moraic vowel through epenthesis or fission. I include DEP-LINK- $\mu(N)$ here to pre-empt questions about whether a similar analysis can be extended to other languages that allow vowels to be copied in C_1C_2 reduplication, such as Lushootseed (see Section 6.5.3.1). Specifically, this constraint is needed to ensure that an affixed $\sigma_{\mu+N}$ will yield $CV_\mu C_\mu$ rather than $CV_{\mu,N}$ because a moraic vowel undergoes fission in Lushootseed C_1C_2 reduplication. If there is a possible source of a moraic vowel, then constraints like DEP- V_μ or INT-V alone are not sufficient to rule out candidates that associate N and μ with the same vowel.

(170)	$/\sigma_{N+\mu 1} + (\text{? } a_{\mu 2} s_{\mu 3} x_{\mu 4}^w)/$	DEP-LINK- μ (N)	INT-C	INT-V	DEP-V
	a. $\text{? } a_{N, \mu 1} . \text{? } a_{\mu 2} s_{\mu 3} x_{\mu 4}^w$	*!	*	*	
	b. $\text{? } s_{\mu 1} . \text{? } a_{\mu 2} s_{\mu 3} x_{\mu 4}^w$		**		*

The form and position of plural C_1C_2 reduplication can therefore be derived with the same grammar as progressive C_1 reduplication, provided PREFIX/ σ and DEP-LINK- μ (N) are included as high-ranked constraints.

4.1.4.3 Summary of Level 2 Grammar

In the last two sections, we have seen that plural C_1C_2 and progressive C_1 reduplication occur at Level 2 in Comox-Sliammon. Reduplication at Level 2 is phonologically regular and productive. The constraint rankings at Level 1 and Level 2 are summarised in (171).

(171) a. *Level 1 Ranking:*

A	INT-V		MAXFLT, *FLOAT, MAX, PREFIX/ σ , DEP-C, *COMPLEXONSET, DEP- V_μ , DEPL- μ (N), ONSET
B	DEP-V		
C	MIN=2 σ	PREFIX/ μ , PREFIX	
D	PRIORITY, MAX-CONTIG		
E	INT-C		

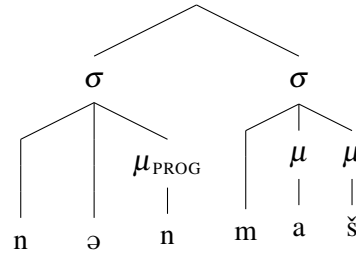
b. *Level 2 Ranking:*

A	PREFIX/ μ , DEP-V μ , DEP-C, ONSET, PREFIX/ σ	REALIZEMORPHEME, *FLOAT, *COMPLEX, MAX-FLOAT DEP-LINK- μ (N)
B	INT-C	
C	INT-V	
D	DEP-V	

Every type of Level 2 reduplication in Comox-Sliammon involves fission of consonants, but only progressive C₁ reduplication allows a vowel to be copied as well. The prosodic units for reduplication are prefixed in the input, though the segment associated with the prosodic affix may surface in an infix position, as shown in (172) for *nənmaš* ‘throwing it’ from (154c) above. However, the affixed mora remains the leftmost mora in the output due to the high ranking of PREFIX/ μ . In other words, the affixed mora remains on the left edge in the output on the prosodic tier (i.e., the first mora), but it will not be the leftmost consonant in the phonological string because onset consonants are not moraic.

(172) a. *Input to Level 2:* μ_{PROG} + nm + aš

b. *Output of Level 2:*



The constraints pertaining to diacritics on the affixes, introduced and motivated in Chapter 3 (Section 3.5.2.3), are summarised for both strata in Table 4.11. The difference between the two rankings is that AFFIX_{L2}, which is violated when an affix with an L2 diacritic is parsed into prosodic structure, is now ranked lower than PARSE_{AFFIX}, which is violated by unparsed affixes. This motivates the integration of Level 2 affixes into the prosodic word during the Level 2 derivation.

Table 4.11: Affix Constraints at Level 1 vs. Level 2

	Level 1	Level 2
A	AFFIX _{L2} , AFFIX _{L3}	AFFIX _{L3}
B	PARSE _{AFFIX}	PARSE _{AFFIX}
C		AFFIX _{L2}

Level 1 and Level 2 affixes include five of the seven reduplicative processes found in Comox-Sliammon. The last two reduplicative morphemes are discussed in the next section.

4.1.5 Level 3

All reduplication at the third stratum in Comox-Sliammon copies a vowel. The segments associated with the affixed prosodic unit associated with the reduplicative morpheme surface in an infix position.

The two types of reduplication that occur at Level 3 are change-of-state C₂ and plural V₁ reduplication. Change-of-state C₂ reduplication involves the affixation of a syllable (without any pre-associated mora or other structure); plural V₁ reduplication involves the affixation of a mora and a floating constricted glottis feature ([c.g.]). Properties of reduplication at Level 3 include (1) reduplication of both moraic and non-moraic vowels and (2) infixation of segments associated with reduplication, despite the morphemes being prefixes at the point of morphological linearisation.

Table 4.12: Level 3 Reduplication

	Underlying Form	Surface Position
Plural V ₁	μ [c.g.]	Infix
Change-of-State C ₂	σ	Infix

4.1.5.1 Level 3 C₂ Reduplication

4.1.5.1.1 Change-of-State C₂ Reduplication

A change of state is marked by the doubling of the second consonant of the root and the preceding vowel, with the copy positioned immediately following the first consonant of the root. The copied vowel may be moraic (as in 173a–e), or non-moraic vowel (as in 173f).

- (173) a. $\dot{p}<\mathbf{u}\dot{\mathbf{q}}^w>\mathbf{u}\dot{\mathbf{q}}^w$ (root: $\dot{p}\mathbf{u}\dot{\mathbf{q}}^w$ - ‘grey’)
grey<COS>
‘It is fading.’ (EP)
- b. $\dot{k}^w<\mathbf{as}>\mathbf{as}=\check{\mathbf{c}}$ (root: $\dot{k}^w\mathbf{as}$ - ‘hot’)
hot<COS>=1SG.SUBJ
‘I am getting warm.’ (MV/WJF)
- c. $\dot{k}^w<\mathbf{as}>\mathbf{as}-a\dot{\mathbf{a}}\mathbf{q}$ (root: $\dot{k}^w\mathbf{as}$ - ‘hot’)
hot<COS>-wind
‘It is getting warm.’ (Watanabe, 2003, p.349)
- d. $q^w\dot{\mathbf{a}}\dot{\mathbf{l}}\dot{\mathbf{x}}<\mathbf{i}\dot{\mathbf{q}}>\mathbf{i}\dot{\mathbf{q}}\mathbf{i}w$ (root: $\dot{\mathbf{x}}\mathbf{i}\dot{\mathbf{q}}\mathbf{i}w$ ‘dark’)
come dark<COS>
‘It is starting to get dark.’ (FL)
- e. $q^w\dot{\mathbf{a}}\dot{\mathbf{l}}\dot{\mathbf{x}}<\mathbf{a}\dot{\mathbf{x}}>\mathbf{a}\dot{\mathbf{x}}$ (root: $\dot{\mathbf{x}}\mathbf{a}\dot{\mathbf{x}}$ - ‘old’)
come old<COS>
‘He’s getting to be grown/older.’ (FL)
- f. $\dot{\mathbf{x}}<\mathbf{\dot{a}}\dot{\mathbf{q}}>\mathbf{\dot{a}}\dot{\mathbf{q}}$ (root: $\dot{\mathbf{x}}\mathbf{\dot{q}}$ - ‘rot’)
rot<COS>
‘It has gotten rotten.’ (EP/FL)

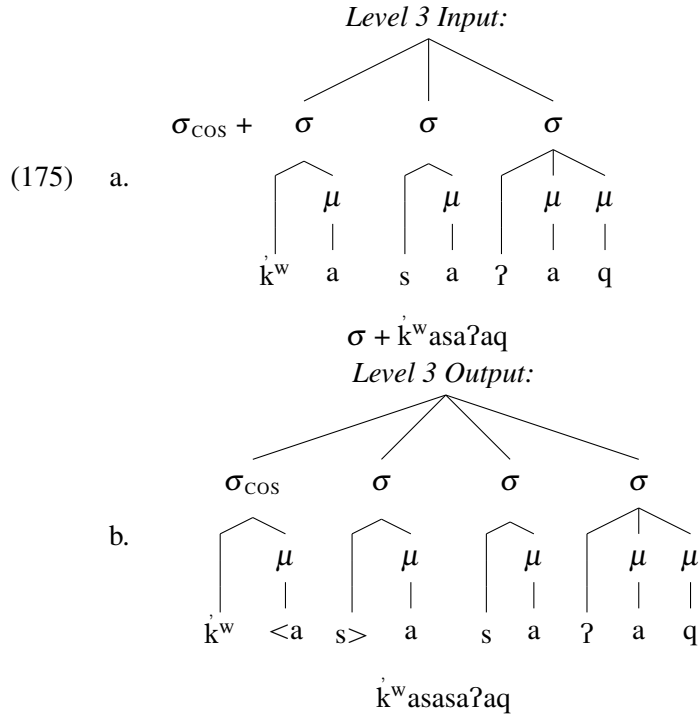
4.1.5.1.2 Analysis of Change-of-State C₂ Reduplication

Changes of state in Comox-Sliammon are formed via the affixation of a syllable with no segmental content or pre-associated structure. The lexical entry is given in (174), which bears a diacritic specifying it as a Level 3 affix.

(174) *Comox-Sliammon Change-of-State Reduplication*:

$$\sigma_{L3} \Leftrightarrow \langle \text{COS} \rangle$$

While reduplicated segments in examples such as $\dot{\lambda}i\dot{q}i\dot{q}iw$ ‘it got dark’ in (173d) above and $\dot{k}^w asasa?aq$ ‘it is getting warm’ in (173c) are positioned on the surface as infixes, I analyse the affixed syllable as a prefix, like all reduplicative morphemes in Comox-Sliammon. The proposed structure for $\dot{k}^w asasa?aq$ is given in (175). The onset of the word remains in the initial position, but becomes the onset for the affixed syllable, regardless of word length.



The constraint *PREFIX/σ* is crucial at Level 3 to ensure the affixed syllable surfaces as the leftmost syllable in the output, despite the fact that the associated segments appear as infixes (violating *PREFIX*). *PREFIX/σ*, first given in Section 4.1.3.1.2, is restated in (176).

(176) *PREFIX/σ*: A prefix is not preceded by a syllable.

The other constraint needed to derive *C*₂ reduplication is one that prevents fission of the first segment in a word. I adopt the insights of Casali (1997) with

respect to the position-sensitive faithfulness and special status of word-initial segments. Casali proposes a family of MAX constraints that make reference to different positions, and I extend this proposal to the set of INTEGRITY constraints. INTEGRITY_{WORD-INITIAL} is defined in (177).²⁹

- (177) INTEGRITY_{WORD-INITIAL} (INT-WI): No word-initial element of the input has multiple correspondents in the output.

The tableau in (178) gives the derivation of *ḵəqəq* ‘it has gotten rotten’ from (173f) above.³⁰ The subscript syllable shows the output position of the syllable that corresponds to the change-of-state morpheme. Both PREFIX/σ and INTEGRITY-WI must be highly ranked to rule out candidates including one that infixes the syllable (178a), and one that copies the initial segment (178b). Candidates (178c-d) have epenthesis and fatally violate the DEP constraints. Finally, candidate (178e), which avoids reduplicating the initial segment by creating an onsetless syllable, is ruled out by ONSET. The attested candidate (178f) only violates INT-V and INT-C, which means these are crucially ranked beneath the other constraints.

(178)

/σ+ḵəq/	INT-WI	PREFIX/σ	ONSET	DEP-V	DEP-C	INT-V	INT-C
a. ḵə.qəq _σ		*!				*	*
b. ḵə _σ .ḵəq	*!					*	*
c. ḵi _σ .qəq				*!			*
d. ʔə _{σ1} .ḵəq					*!	*	
e. ə _σ .ḵəq			*!			*	
f. ḵə _σ .qəq						*	*

The tableau in (179) for *ḵiqiqiw* ‘got dark’ in example (173d) above shows that the constraints and ranking given thus far also predict fission of the correct consonant in multi-syllabic words. ONSET, DEP-V, and DEP-C are omitted from this tableau because they are not violated by any of the candidates considered.

²⁹While INTEGRITY_{WORD-INITIAL} could be split into INTEGRITY-V_{WORD-INITIAL} and INTEGRITY-C_{WORD-INITIAL} as I have done for INTEGRITY (motivated in Section 4.1.3.1.2), this does not impact the analysis because ONSET is ranked high enough that the first segment of a word will always be a consonant. INTEGRITY_{WORD-INITIAL} will only affect consonants.

³⁰*FLOAT, MAX-FLOAT, and REALIZE MORPHEME are undominated across all strata are therefore not included in this tableau or the others for Level 3.

(179)	/σ+ ʔi.qiw/	INT-WI	PREFIX/σ	INT-V	INT-C
	a. ʔi _σ .q _i .qiw			*	*
	b. ʔi.qi _σ .qiw		*!	*	*
	c. ʔi _σ .ʔi.qiw	*!		*	*
	d. ʔi.qi.wiw _σ		*!	*	*

The preliminary rankings needed at Level 3 to predict the correct position and shape of the change-of-state C₂ reduplication are summarised in Table 4.13.

Table 4.13: Comox-Sliammon Level 3 Grammar (Preliminary)

A	DEP-V	DEP-C, ONSET	REALIZEMORPHEME, *FLOAT, MAX-FLOAT PREFIX/σ, INT-WI
B	INT-V	INT-C	

4.1.5.2 Level 3 V₁ Reduplication

4.1.5.2.1 Plural V₁ Reduplication

Plural V₁ reduplication copies the initial vowel of the word. A glottal stop separates the two matching vowels (180).

- (180) a. ʔ<uʔ>u<ʔ>ʔ-ut (root: ʔut- ‘sip’)
sip<V₁.PL><DIM>-CTR
‘sipping’ (EP/FL)
- b. ʔ<uʔ>u<y>-θut (root: ʔuθ- ‘push’)
push<V₁.PL><DIM>-CTR+REFL
‘pushing over and over’ (JF)
- c. ʔ<iʔ>i<č>ʔ-əm (root: ʔiʔ- ‘dance’)
dance<V₁.PL><DIM>-MID
‘dancing here and there’ (EP)

- d. $\acute{t}<\mathbf{u?}>\mathbf{u}<\acute{t}>\acute{t}a\acute{t}$ (cf. *tútaṭ* ‘bed’)
 bed<V₁.PL><DIM>
 ‘little beds’ (EP)
- e. $s<\mathbf{u?}>\mathbf{u}<s>payu$ (cf. *supayu* ‘axe’)
 axe<V₁.PL><DIM>
 ‘little axes’ (PD)

Plural V₁ is limited in its distribution and is found only in contexts with multiple reduplication.³¹ Even though V₁ reduplication does not occur outside of multiple reduplication, I include it in this chapter because it merits a separate lexical entry, for two reasons. First, it occurs with more than one other type of reduplication: in particular, it co-occurs with both diminutive C₁, as shown above in (180), and with plural C₁C₂ reduplication as shown in (181) below.³²

- (181) a. $n<\mathbf{a?}>a\acute{x}\sim n\acute{o}x$ -at-as (root: *nṣ*- ‘cut a line’)
 C₁C₂.PL<V₁.PL>~cut.line-CTR-3ERG
 ‘They (sg.) are cutting lines (intermittent/right now).’ (BW)
- b. $\acute{t}<\mathbf{a?}>a\acute{k}^w\sim\acute{t}u\acute{k}^w$ (root: *tuk^w*- ‘fly’)
 C₁C₂.PL<V₁.PL>~fly
 ‘flying around’ (BW)
- c. $\acute{j}<\mathbf{a?}>a\acute{\lambda}\sim\acute{j}\acute{o}\acute{\lambda}$ $t\acute{o}=\acute{c}\acute{o}y\sim\acute{c}u\acute{y}$ (root: *jṣ*- ‘run’)
 C₁C₂.PL<V₁.PL>~run DET=C₁C₂.PL~child
 ‘The children are running around.’ (BW)

Second, words with V₁ and another prosodic affix can be given a compositional semantic analysis where each morpheme contributes its own meaning. V₁ reduplication typically marks temporal pluractionality with verbs (Mellesmoen et al., 2020), as shown in (180a–d) and (181) above, though it also denotes plural entities when it occurs on diminutive nouns, as shown in (180e).

³¹My analysis does not fully explain this distributional fact. This is a topic of broader relevance to Salish that deserves further study; for example, the diminutive in Halkomelem always co-occurs with imperfective aspect, which is also a reduplicative morpheme (see Section 5.3.1.2).

³²The examples in (181) are built on C₁C₂ reduplicated forms that have an additional change in vowel quality, which is discussed in Chapter 5 (Section 5.5.6.1). The input at Level 3 for *jaʔaṣjəṣ* in (181c) is $\mu + ja\acute{\lambda}j\acute{o}\acute{\lambda}$ (not $\mu + j\acute{o}\acute{\lambda}j\acute{o}\acute{\lambda}$).

V₁ reduplication is productive within its restricted domain (multiple reduplication). Loanwords are also attested with V₁, which supports the inclusion of V₁ reduplication as a part of the synchronic grammar. For example, the diminutive plural of the loanword *čikins* ‘chicken’ is formed with V₁ reduplication, as shown in (182).³³

- (182) č<ī?>i<č>kin[i]s-uł (cf. *čikins* ‘chicken’)
 chicken<V₁.PL><DIM>-young
 ‘small chickens’ (Watanabe, 2003, p.401)

Treatments of V₁ reduplication differ across the literature: for example, (Watanabe, 2003, pp.401–404) treats these forms as single reduplication (e.g., a CV?V process), rather than multiple reduplication. However, the fact that V₁ reduplication occurs with more than one type of reduplication and has an identifiable semantic contribution requires an independent lexical entry.

While the focus of this chapter is on individual reduplication patterns, the fact that V₁ reduplication must combine with a word that already has Level 1 C₁ reduplication or Level 2 C₁C₂ reduplication provides evidence for it being a Level 3 affix, because it must apply after Level 2 reduplication.

4.1.5.2.2 Analysis of Plural V₁ Reduplication

Plural V₁ reduplication involves affixation of a μ at Level 3 together with a floating constricted glottis feature ([c.g.]), which is realised as a glottal stop. The lexical entry for V₁ reduplication is provided in (183). The lexical entry bears an L3 diacritic, requiring the affix to be parsed during the Level 3 derivation.

- (183) *Comox-Sliammon V₁ Reduplication*:
 (μ [c.g.])_{L3} ⇔ <V₁.PL>

I assume the least marked consonants in Comox-Sliammon are the glottal consonants (/h/ and /ʔ/); [h] is the default in intervocalic environments (Blake, 2000, p.45). I adopt the feature specifications for place in Table 4.14 from Blake

³³The [i] in square brackets is epenthetic.

(2000, pp.56–59). Both epenthetic consonants have a place node with a dependent pharyngeal feature ([PHAR]) in her analysis.^{34,35}

Table 4.14: Place Features

	LAB	COR	DOR	PHAR
Labial	✓			
Coronal		✓		
Alveopalatal			✓	
Velar (Plain)			✓	
Velar (Round)	✓		✓	
Uvular (Plain)			✓	✓
Uvular (Round)	✓		✓	✓
Glottal				✓

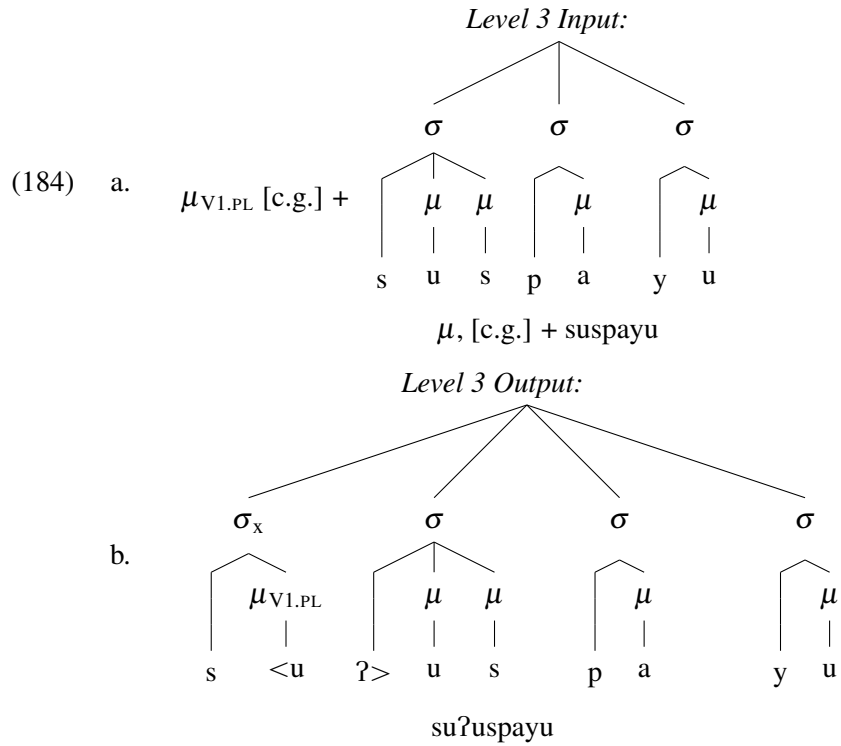
The difference between /ʔ/ and /h/ is that /ʔ/ has a laryngeal node with a constricted glottis feature ([c.g.]); this is why [ʔ] is epenthesized instead of [h] in V₁ reduplication, despite [h] being the default epenthetic consonant in an intervocalic environment.

Turning to the position of V₁ reduplication, an example of the prosodic structure is provided in (184) for example (180e) above. The affixed prosodic unit occurs on the left edge (in a prefixed position), and the fissioned segments are infixes. The epenthetic glottal stop is positioned between the fissioned vowels.

³⁴These are phonological features: [ʔ] and [h] are not pronounced as pharyngeal consonants.

³⁵In order to correctly predict that [PHAR] is the least marked place feature, I assume the constraint ranking in (i).

(i) DEP[LAB], DEP[COR], DEP[DOR] >> DEP[PHAR]



The tableau in (185) shows how the V_1 forms fare under the constraint ranking established for the derivation of C_2 reduplication.³⁶ I assume that PREFIX/μ and PREFIX/σ are both ranked high at Level 3, but I omit PREFIX/σ in this section for space. Epenthesis of the expected default consonant [h] fatally violates MAX-FLOAT (185b), as the [c.g.] feature is absent (i.e., [h] is has no [c.g.] feature). A candidate where the affixed mora is not the leftmost one fatally violates PREFIX/μ (185c); a candidate with vowel hiatus is eliminated by ONSET (185d). The glottalisation of a sonorant later in the word is therefore blocked by the need for an onset.

³⁶Though the flat linear ranking for the Level 3 grammar has $\text{DEP-C} \gg \text{INT-V/INT-C}$, a pairwise ranking was not established between DEP-C and INT-V . This is reflected in the tableau with the dotted line.

(185)	/μ ₁ [c.g.] + sus.pa.yu/	PREFIX/μ	MAXFLT	ONSET	DEP-C	INT-V
	☞ a. su _{μ1} .ʔus.pa.yu				*	*
	b. su _{μ1} .hus.pa.yu		*!		*	*
	c. su.ʔu _{μ1} s.pa.yu	*!			*	*
	d. su _{μ1} .us.pa.ʔu			*!	*	*

With the current ranking, there is indeterminacy between candidates with a glottal stop as the onset of the first syllable (unattested) and the second syllable (attested). This can be resolved with a constraint from the ANCHORING family of faithfulness constraints (e.g., McCarthy & Prince, 1994, p.37). The constraint, given in (186), is violated when the first segment of the output word does not have a corresponding segment in the same position in the input.

(186) ANCHOR-L(Word, Segment):

Any segment at the left periphery of a word in the output has a correspondent at the left periphery of an input word.

The tableau in (187) shows how ANCHOR-L rules out a candidate with a word-initial [ʔ].

(187)	/μ ₁ [c.g.] + suspayu/	ANCHOR-L	DEP-C	INT-V
	☞ a. su _{μ1} ʔuspayu		*	*
	b. ʔu _{μ1} suspayu	*!	*	*

The C₂ and V₁ reduplication patterns appear to pose a ranking paradox: DEP-C needs to be below INT-C for epenthesis to occur for V₁ reduplication, while C₂ reduplication needs the reverse ranking for fission to occur. The tableau in (188) shows that the ranking needed for C₂ reduplication (DEP-C >> INT-C) predicts the wrong candidate if [c.g.] can target a different segment in the word *in addition* to fission of a consonant.

(188)	/μ ₁ [c.g.] + sus.pa.yu/	MAXFLT	DEP-C	INT-V	INT-C
	⊖ a. su _{μ1} .ʔus.pa.yu		*!	*	
	b. su _{μ1} .sus.pa.yu	*!			
	☞ c. su _{μ1} .sus.pa.ʔu			*	*

The ranking paradox can be solved by ensuring the faithfulness constraint that penalises changes to the [c.g.] specification of a segment is ranked above DEP-C. IDENT-IO[C.G.] is defined in (189).³⁷

- (189) IDENT-IO[C.G.] (IDENT[C.G.]): The specification (or lack of specification) for the feature [c.g.] of an input segment must be preserved in its output correspondent.

When IDENT-IO[C.G.] is ranked above DEP-C, candidate (190a) – with fission providing an onset and glottalisation of a sonorant – can be ruled out. This allows for the attested candidate (190b) to win. This solution does not affect C₂ reduplication because DEP-C remains above INT-C, which permits fission to occur in the absence of [c.g.].

(190)	/μ [c.g.] + sus.pa.yu/	MAX[C.G]	IDENT[C.G.]	DEP-C	INT-V	INT-C
	a. su _μ .sus.pa.yu		*!		*	*
	 b. su _μ lʔuspayu			*	*	
	c. su _μ .sus.pa.yu	*!				

The revised constraint ranking for the Level 3 grammar is given in Table 4.15.

Table 4.15: Comox-Sliammon Level 3 Grammar (Updated)

A	ONSET, IDENT[C.G.]	DEP-V	REALIZE MORPHEME, *FLOAT, ANCHOR-L, INTEGRITY-WI, PREFIX/σ, PREFIX/μ, MAX-FLOAT
B	DEP-C	INT-V	
C	INT-C		

³⁷IDENT-IO[C.G.] would be lower ranked at the previous two strata: when a glottalised sonorant undergoes fission in C₁C₂ reduplication, only one of the sonorants in the output is glottalised (see Section 4.1.4.2.1).

4.1.5.3 Summary of Level 3 Grammar

Change-of-state C_2 and plural V_1 reduplication are Level 3 affixes. The C_2 process involves the affixation of an empty syllable, while the V_1 process involves the affixation of an empty mora and a constricted glottis feature. Similar to Level 2 reduplication, the Level 3 reduplicative processes are phonologically regular, though V_1 reduplication is highly restricted in distribution, because it is restricted to contexts of multiple reduplication.

At Level 3, vowels undergo fission to fill affixed prosodic units, which may be accompanied by fission of a consonant (C_2 reduplication), or the epenthesis of a consonant (V_1 reduplication). Segments reduplicated at this stratum are positioned as infixes, though the affixed mora or syllable remains leftmost in the output. The constraint ranking across all three strata is summarised in (191).

(191) a. *Level 1 Ranking:*

A	INT-V		MAXFLT, *FLOAT, MAX, PREFIX/ σ , DEP-C, *COMPLEXONSET, DEP- V_μ , DEPL- $\mu(N)$, ONSET
B	DEP-V		
C	MIN=2 σ	PREFIX/ μ PREFIX	
D	PRIORITY, MAX-CONTIG		
E	INT-C		

b. *Level 2 Ranking:*

A	PREFIX/ μ , DEP-V μ , DEP-C, ONSET, PREFIX/ σ	REALIZEMORPHEME, *FLOAT, *COMPLEX, MAX-FLOAT DEP-LINK- μ (N)
B	INT-C	
C	INT-V	
D	DEP-V	

c. *Level 3 Ranking:*

A	ONSET, IDENT[C.G.]	DEP-V	REALIZEMORPHEME, *FLOAT, ANCHOR-L, INTEGRITY-WI, PREFIX/ σ , PREFIX/ μ , MAX-FLOAT
B	DEP-C	INT-V	
C	INT-C		

Level 3 reduplicative processes differ from Level 1 and Level 2 processes because they not only *allow* for the reduplication of vowels (in contrast to what occurs at Level 1), but they also *always* have a reduplicated vowel (in contrast to what occurs at Level 2). The vowel copied at Level 3 may be [ə], which does not happen at earlier strata.

The constraints pertaining to affix-parsing diacritics, introduced and motivated in Chapter 3 (Section 3.5.2.3), are summarised in Table 4.16 for all three strata. In the Level 3 grammar, $\text{PARSE}_{\text{AFFIX}}$ is ranked higher than AFFIX_{L2} and AFFIX_{L3} , which ensures that all affixes in the input are parsed into the prosodic word; in other words, if a given AFFIX constraint is ranked above $\text{PARSE}_{\text{AFFIX}}$, then any affix marked for the level denoted by the AFFIX constraint is left unparsed in the output of that stratum.

Table 4.16: Comox-Sliammon Affix Constraints

	Level 1	Level 2	Level 3
A	AFFIX _{L2} , AFFIX _{L3}	AFFIX _{L3}	PARSE _{AFFIX}
B	PARSE _{AFFIX}	PARSE _{AFFIX}	AFFIX _{L2} , AFFIX _{L3}
C		AFFIX _{L2}	

4.1.6 Summary of Reduplication in Comox-Sliammon

In this section, I consider the full set of constraints introduced throughout the discussion of all three strata. These will be presented in batches that allow for comparison; batches represent partial rankings and constraints in the top row may be dominated by other constraints not given in a particular table. I focus on strata that are relevant for reduplication; the total number of strata in the language is not limited to three.

In Table 4.17, the constraints relevant to parsing affixes are considered alongside the constraints that motivate reduplication. In the Level 1 grammar, both AFFIX_{L2} and AFFIX_{L3} must be above *FLOAT and PARSE_{AFFIX}, while only AFFIX_{L3} is above them in the Level 2 grammar. The re-ranking allows for the Level 2 affixes to be parsed into the prosodic word and for their associated prosodic units to be filled at Level 2. At Level 3, both AFFIX_{L2} and AFFIX_{L3} are ranked below the other constraints, which allows for all affixes to be parsed into the prosodic word. MAX_{FLOAT}, MAX, and REALIZE_{MORPHEME} are ranked high across all strata as they are not violated in the reduplicated examples considered.

Table 4.17: Constraints Motivating Parsing and Filling

	Level 1	Level 2	Level 3
A	REALIZEMORPHEME, MAX, MAXFLT AFFIX _{L2} , AFFIX _{L3}	REALIZEMORPHEME, MAX, MAXFLT, AFFIX _{L3}	REALIZEMORPHEME, MAX, MAXFLT, PARSE _{AFFIX} , *FLOAT
B	*FLOAT, PARSE _{AFFIX}	*FLOAT, PARSE _{AFFIX}	AFFIX _{L2} , AFFIX _{L3}
C		AFFIX _{L2}	

Next, I compare constraints motivating fission and epenthesis in Table 4.18. Across all three strata, *FLOAT, *COMPLEX, and DEP- V_μ are highly ranked relative to the other constraints.³⁸ All three strata tolerate violations of INT-C, which means consonants may be reduplicated. A difference between the strata is that INT-V outranks DEP-V in the Level 1 grammar, but not in the Level 2 or Level 3 grammars. Vowels are only reduplicated at the later two strata. Finally, only the Level 3 grammar permits epenthesis of a consonant, which is reflected in the lower ranking of DEP-C at Level 3, relative to the other strata.

³⁸DEP- V_μ does not crucially outrank DEP-V at Level 1 because anything that violates DEP- V_μ also violates DEP-V; DEP- V_μ is in Row A because it is never violated, while DEP-V may be.

Table 4.18: Constraints Motivating Fission and Epenthesis

	Level 1	Level 2	Level 3
A	*FLOAT, *COMPLEX, DEP-V _μ , ONSET, DEP-C, DEPL-μ(N), INT-V	*FLOAT, *COMPLEX, DEP-V _μ , ONSET, DEP-C, DEPL-μ(N)	*FLOAT, *COMPLEX, DEP-V _μ , ONSET, DEP-V
B	DEP-V	INT-C	DEP-C, INT-V
C	INT-C	INT-V	INT-C
D		DEP-V	
Unclear			DEPL-μ(N)

Constraints that deal with consonant epenthesis and a floating [c.g.] feature are given in Table 4.19. As there are only two rows needed to display these rankings, I use the labels "high" and "low" in this table. MAX-FLOAT is ranked high across all three strata: an input [c.g.] feature must be realised.³⁹ However, when a glottalised sonorant undergoes fission at Level 1 and Level 2, only one of the resulting sonorants is glottalised, meaning IDENT[C.G.] is not ranked as highly as it is at Level 3. Further, there is no epenthesis of consonants at Level 1 and Level 2, so DEP-C is ranked high. In contrast, DEP-C may be violated at Level 3, allowing for glottal stop epenthesis.

Table 4.19: Constraints Related to Consonant Features

	Level 1	Level 2	Level 3
High	MAXFLT, DEP-C,	MAXFLT, DEP-C,	MAXFLT, IDENT[C.G.],
Low	IDENT[C.G.]	IDENT[C.G.]	DEP-C

³⁹MAX-FLOAT can be violated at later strata, however. A floating [c.g.] feature has been proposed as part of the underlying representation of the past tense marker -ʔ; a [ʔ] or glottalised sonorant is often absent in past tense forms (Blake, 2000, p.43).

The constraints that relate to alignment and the position of the reduplicative morpheme are summarised in Table 4.20. Across all strata, PREFIX/ σ is never violated; reduplicative morphemes always occur as part of the first syllable of the word. Further, PREFIX/ μ is not violated at Level 2 or Level 3; it is only ranked low enough to allow infixation beyond the first mora at Level 1. INTEGRITY-WI is only ranked high in the Level 3 grammar; the first consonant of a word can undergo fission at Level 1 or Level 2, but not at Level 3. Finally, ANCHOR-L is necessarily ranked high at Level 3; it can also be ranked high at Level 1 or Level 2 because it is not violated by any attested type of reduplication.

Table 4.20: Constraints Related to Alignment and Position

	Level 1	Level 2	Level 3
High	PREFIX/ σ ANCHOR-L	PREFIX/ σ , PREFIX/ μ ANCHOR-L	PREFIX/ σ , PREFIX/ μ , INTEGRITY-WI, ANCHOR-L
Low	PREFIX/ μ , PREFIX INTEGRITY-WI	PREFIX INTEGRITY-WI	PREFIX

Finally, the constraints MIN=2 σ and PRIORITY are considered alongside DEP-V and INT-V. The only stratum where a ranking between all four constraints can be established is Level 1. There are no Level 2 or Level 3 reduplicative morphemes with listed allomorphs, which means PRIORITY cannot be ranked. Given that monosyllabic progressive forms are possible at Level 2 (e.g., *čăčŭ* ‘raining’), MIN=2 σ cannot be ranked above DEP-V and INT-V at either Level 1 or 2. MIN=2 σ cannot be evaluated at Level 3: both reduplicative processes at this stratum add a syllable. However, MIN=2 σ is likely ranked low in the Level 3 grammar as well. I assume all words pass through all strata; if MIN=2 σ were ranked higher than DEP-V, then epenthesis should be triggered to avoid monosyllabic words.

Table 4.21: Allomorph Selection and Vowel/Syllable Constraints

	Level 1	Level 2	Level 3
A	INT-V	INT-V	DEP-V
B	DEP-V	DEP-V	INT-V
C	MIN=2 σ	MIN=2 σ	
D	PRIORITY		
Unclear		PRIORITY	PRIORITY, MIN=2 σ

I do not compare DEP-LINK- μ (N) across strata as I assume it is highly ranked across strata, though other constraints ensure the same outcome at Level 1 and Level 2, which have reduplicative morphemes with $\sigma_{N+\mu}$ as an input. DEP-LINK- μ (N) is not relevant for Level 3, which only has μ and σ as input units.

Overall, the differences between reduplicative morphemes in Comox-Sliammon arise from different constraint rankings at each stratum, in addition to a range of different inputs (lexical entries), as summarised in Table 4.22.

Table 4.22: Lexical Entries in Comox-Sliammon

	Level 1	Level 2	Level 3
μ -type	$\mu_1 > i_2$ $\mu_1 > a_2$	μ_{L2}	$(\mu \text{ [c.g.]})_{L3}$
σ -type	$\sigma_{N+\mu}$	$(\sigma_{N+\mu})_{L2}$	σ_{L3}

4.2 Case Study II: Lillooet

The second case study in this chapter concerns reduplication in Lillooet. In Section 4.2.1, I first present some relevant facts about the general phonological grammar of the language, followed by a preview of the grammar of reduplication in Section 4.2.2. Description, data, and analysis are then presented for reduplication at Level 1 (Section 4.2.3) and Level 2 (Section 4.2.4). A triplication pattern found at a later stratum is described in Section 4.2.5.

4.2.1 Lillooet Phonology

Key aspects of Lillooet phonology that pertain to reduplication include (i) stress assignment, and (ii) the behaviour of schwa (in comparison to the full vowels /a/, /i/, and /u/).

Feet in Lillooet are optimally binary (Van Eijk 1981; Roberts 1993; Revithiadou 1999). Stress assignment interacts with constraints on syllable structure, which determine the distribution of [ə]. For example, [ə] is usually not stressed if a moraic vowel is available.⁴⁰ This is a common cross-Salish pattern (see, e.g., Dyck's 2004 analysis of stress in Squamish).

Matthewson (1994) argues that the distribution of [ə] is predictable and therefore does not need to be included in underlying representations. Her arguments are similar to the ones that Blake (2000) later makes for Comox-Sliammon; however, the phonological grammar of each language yields differences in the distribution of [ə] and stress assignment. In contrast to fixed-initial stress in Comox-Sliammon, where the position of stress and edge of the prosodic word invariably coincide (see Section 4.1.1), Lillooet has a more complicated stress system. Not only is there stress assignment and shift within different domains, but the grammar that governs these domains is also different (i.e., the position of stress can be different at Level 2 than it was at Level 1). Stress is discussed alongside Level 2 reduplication in Section 4.2.4.

4.2.2 Preview of Lillooet Reduplication

Lillooet has three types of C_1 reduplication (initial, plural, and diminutive), one type of C_1C_2 reduplication (plural/property-denoting), and one type of C_2 reduplication (change-of-state). There is also a pattern of C_2^2 reduplication (i.e., triplication). The types of reduplication considered in Lillooet are previewed in Table 4.23.

⁴⁰A stressed schwa surfaces regardless of other vowels with diminutive reduplication, as discussed in Section 4.2.4.2.3, however.

Table 4.23: Preview of Lillooet Reduplicative Affixes

Stratum	Type	Affix	Section
Level 1	Initial C ₁	σ	4.2.3.1
	Plural/Property-Denoting C ₁ C ₂	$\sigma_{N+\mu}$	4.2.3.2.1
Level 2	Change-of-State C ₂	σ	4.2.4.1.1
	Pluractional C ₁	μ [c.g.]	4.2.4.2.1
	Diminutive C ₁	$\mu^{[\text{stress}]}$ [c.g.]	4.2.4.2.3
Level 2+n	Triplcation C ₂ ²	Ft	4.2.5

4.2.3 Level 1

There are two types of Level 1 reduplication in Lillooet: (i) initial C₁ reduplication, and (ii) plural/property-denoting C₁C₂ reduplication. Initial C₁ reduplication fills an affixed syllable (unspecified for structure), and plural/property-denoting C₁C₂ reduplication fills an affixed monomoraic syllable (with an empty nucleus). Level 1 reduplication in Lillooet is summarised in Table 4.24.

Table 4.24: Level 1 Reduplicative Affixes

	Underlying Form	Surface Position
Initial C ₁	σ	Prefix
Plural/Property-Denoting C ₁ C ₂	$\sigma_{N+\mu}$	Prefix

4.2.3.1 Level 1 C₁ Reduplication

4.2.3.1.1 Initial C₁ Reduplication

Initial C₁ reduplication copies the first consonant of the root. There is a [ə] epenthesized between the two matching consonants, as shown in (192).

- (192) a. $k\bar{ə} \sim kl\acute{a}x^w$ (cf. $klax^w$ ‘muskrat’)
 C₁.INIT ~ muskrat
 ‘muskrat(s)’ (Van Eijk, 1997, p.56)

- b. n-**xə**~**xzúm**-ana? (root: *xzum* ‘big’)
LOC-C₁.INIT~big-ear
‘big ears’ (Van Eijk, 1997, p.57)
- c. s-**q^wə**~q^wlíp (cf. *sq^wlip* ‘black tree moss’)
NMLZ-C₁.INIT~black.tree.moss
‘lots of black tree moss’ (Van Eijk, 1997, p.57)
- d. **q^wə**~q^wlít-az (cf. *q^wlítaž* ‘jackpine’)
C₁.INIT~jackpine-tree
‘jackpines’ (Van Eijk, 1997, p.58)
- e. **cə**~cítx^w (cf. *cítx^w* ‘house’)
C₁.INIT~house
‘houses (collective)’ (Van Eijk, 1997, p.58)

There is variability in the meaning of initial C₁ reduplication: it shows a higher degree of lexicalization in both form and function than the other C₁ reduplication processes in Lillooet (see description of diminutive and plural C₁ reduplication in Section 4.2.4.2). Initial C₁ reduplication is most often associated with plurality, as shown in the examples in (192) above; in contrast, the examples in (193) mark continuative aspect. The label “initial” refers to form (prefixed to the left edge) of C₁ reduplication rather than function.

- (193) a. **x^wí**~x^witən (cf. *x^witən* ‘whistle’)
C₁.INIT~whistle
‘keep whistling’ (H. Davis & Mellesmoen, 2023, p.27)
- b. **ʔəs-tá**~tał-ləx (cf. *ʔəstáłləx* ‘standing’)
STAT-C₁.INIT~stand-AUT
‘keep standing’ (Van Eijk, 1997, p.56)

Initial C₁ reduplication is restricted to certain lexical items (H. Davis & Mellesmoen, 2023, p.27): however, the examples in (193) above exemplify a further lexicalised pattern where the first vowel of the root is a copy (instead of an epenthetic [ə]). I assume that vowels are not copied in any productive Lillooet reduplication process. I treat these CV forms as lexically-specified reduplication

patterns at Level 1.⁴¹ The following analysis focuses on [ə] forms.

While initial reduplication has a restricted distribution, it is phonologically regular (see discussion of phonologically regular and active reduplication in Section 1.4.3.2); the [ə] forms can be derived with the same set of constraints as Level 1 C₁C₂ reduplication (see Section 4.2.3.2.2). In Chapter 5, I further show that multiple reduplication patterns with initial C₁ reduplication are also consistent with reduplication being present as an active Level 1 phonological process (see Section 5.4).

4.2.3.1.2 Analysis of Initial C₁ Reduplication

Initial C₁ reduplication in Lillooet is an affixed σ that receives segmental content at Level 1. The lexical entry is given in (194). As an affix parsed at Level 1, there is no diacritic needed on (194).

(194) *Lillooet Initial Reduplication:*

$$\sigma \Leftrightarrow \langle \text{INIT} \rangle$$

The affix parsing constraints and their ranking across strata are given in Chapter 3 (Section 3.5.2.3). As the rankings of $\text{PARSE}_{\text{AFFIX}}$ and AFFIX_{L2} are determined by the stratum, there is no need to discuss them for any individual pattern; I will discuss them as part of the summary of Level 1 (Section 4.2.3.3), Level 2 (Section 4.2.4.4), and the overall grammar (Section 4.2.6).

As in the analysis of Comox-Sliammon in Section 4.1, REALIZEMORPHEME , MAX-FLOAT , and *FLOAT are relevant for ensuring that floating prosodic units or features in the input are associated with segmental content in the output. The constraints, first defined in Section 4.1.3.1.2, are restated in (195).

⁴¹The function and form of the CV realisation of Initial C₁ reduplication resemble the Central Salish C₁ aspectual reduplication, discussed in Chapter 2 (Section 2.5.3.2); these may be borrowings from Squamish, while the [ə] forms likely have an old plural marker (H. Davis, p.c.). It occurs with fewer than five roots in Lillooet.

- (195) a. REALIZEMORPHEME (RM): Let α be a morphological form, β be a morphosyntactic category, and $F(\alpha)$ be the phonological form from which $F(\alpha+\beta)$ is derived to express a morphosyntactic category β . Then RM is satisfied with respect to β iff $F(\alpha+\beta) \neq F(\alpha)$ phonologically.

(Kurusu, 2001, p.39)

- b. MAX-FLOAT (MAXFLT): All prosodic units or features that are floating in the input have output correspondents.

(modified from Wolf, 2005, p.370)

- c. *FLOAT: $\forall p \in O$, where p is a prosodic unit or phonological feature:
 $\exists s$, where s is a segment, and p dominates or is associated with s .

(modified from Saba Kirchner 2013, p.232)

The constraints in (195) will only be shown in the first tableau for Lillooet; they will only be mentioned afterwards in summaries of the constraint rankings or where further discussion is needed (e.g., in relation to MAX-FLOAT in the context of glottalisation in Section 4.2.4.3). REALIZEMORPHEME and MAX-FLOAT are undominated in the Level 1 grammar; *FLOAT is dominated by constraints against parsing affixes with diacritics (e.g., *AFFIX_{L2}). As in Comox-Sliammon, the choice of segments to fill the affixed prosodic unit is determined by the relative ranking of faithfulness constraints. The relevant constraints, first defined in Section 4.1.3.1.2, are restated in (196). DEP is violated by epenthesis; INTEGRITY is violated by fission. DEP and INTEGRITY are each separated into two constraints: one assesses consonants and the other vowels.

- (196) a. DEP Constraints:
- i. DEP-V: A vowel in the output must have an input correspondent.
 - ii. DEP-C: A consonant in the output must have an input correspondent.
- b. INTEGRITY Constraints:
- i. INTEGRITY-V: A vowel in the input must not have multiple correspondents in the output.

- ii. INTEGRITY-C: A consonant in the input must not have multiple correspondents in the output.

While [ə] is the default epenthetic vowel, I use [i] to represent an epenthetic moraic vowel when necessary to rule out unattested candidates. I use [ʔ] to represent epenthetic consonant in unattested forms in tableaux for consistency across languages (see explanation in Section 4.1.3.1.2); however, note that the default epenthetic consonant in Lillooet is /h/.

The tableau in (197) for example (192) above shows that the attested candidate (197a), with fission of a consonant and epenthesis of a vowel, violates DEP-V and INT-C. DEP-C is above INT-C to rule out epenthesis (e.g., 197b); INT-V is ranked above DEP-V to rule out vowel reduplication (e.g., 197c). Further, DEP-V and INT-C must be ranked lower than *FLOAT and MAX-FLOAT to ensure that the affixed prosodic unit is not floating or deleted, as in candidates (197d) and (197e), respectively. Candidate (197f) fatally violates REALIZEMORPHEME because it cannot be distinguished from the output of the same derivation without the reduplicative morpheme (i.e., the root segments will be parsed into a syllable regardless of whether one was present in the input). While a candidate with C₁C₂ reduplication fills the affixed syllable (candidate 197g), it incurs an additional violation under INT-C. The attested candidate with fission of a single consonant only incurs a single violation; the candidate which fills the affixed prosodic unit with the fewest violations of the relevant faithfulness constraints will always win. Periods mark the syllable boundaries; I use parentheses and the σ in candidates to show where the floating σ surfaces in the output.

(197)

/σ ₁ + citx ^w /	MAXFLT	*FLOAT	RM	DEP-C	INT-V	DEP-V	INT-C
a. (cə) _σ .citx ^w						*	*
b. (ʔə) _σ .citx ^w				*!		*	
c. (ci) _σ .citx ^w					*!		*
d. σ citx ^w		*!					
e. citx ^w			*!				
f. (citx ^w) _σ	*!						
g. (cət) _σ .citx ^w						*	**!

In order to ensure the affixed syllable remains as the leftmost syllable, I use the constraint PREFIX/σ, first defined in Section 4.1.3.1.2, which is restated in (198).

(198) PREFIX/ σ : A prefix is not preceded by a syllable.

(modified from McCarthy 2003, p.94)

The tableau in (199) shows how PREFIX/ σ rules out any candidate where the affixed syllable is not the first. Candidate (199a) wins over candidates (199b–c) because it does not incur a fatal violation under PREFIX/ σ .

/ σ + citx ^w /		PREFIX/ σ	INT-C	DEP-V
(199)	a. (cə) _{σ} .citx ^w		*	*
	b. cit(x ^w əx ^w) _{σ}	*!	*	*
	c. cit(x ^w ə) _{σ}	*!		*

A preliminary constraint ranking for Level 1 is summarised in Table 4.25.

Table 4.25: Lillooet Level 1 Grammar (Preliminary)

A	DEP-C	INT-V	REALIZEMORPHEME *FLOAT PREFIX/ σ MAX-FLOAT
B	INT-C	DEP-V	

4.2.3.2 Level 1 C₁C₂ Reduplication

4.2.3.2.1 Plural/Property-Denoting C₁C₂ Reduplication

C₁C₂ reduplicated forms in Lillooet denote plurality (including pluractionality) or a property.⁴² While these two meanings are distinct, both types of C₁C₂ reduplication occur at Level 1.

The first two consonants are doubled in C₁C₂ reduplication; [ə] is inserted between the first instance of C₁ and the first instance of C₂. Examples with C₁C₂ reduplication marking plurality are given in (200), and examples with C₁C₂ reduplicated forms denoting a property are given in (201).

⁴²See H. Davis (2011) and the discussion in Chapter 2 (Section 2.9.4) for the argument that property-denoting C₁C₂ reduplication is semantically vacuous and serves the syntactic function of creating an adjective.

- (200) a. s-**ɣəp**~ɣáp (cf. *sɣap* ‘tree’)
 NMLZ-C₁C₂.PL~tree
 ‘trees’ (Van Eijk, 1997, p.56)
- b. s-**nək^w**~núk^waʔ (cf. *snúk^waʔ* ‘friend/relative’)
 NMLZ-C₁C₂.PL~friend/relative
 ‘friends/relatives’ (Van Eijk, 1997, p.56)
- c. s-**q^wəm**~q^wəm (cf. *sq^wəm* ‘mountain’)
 NMLZ-C₁C₂.PL~mountain
 ‘mountain range’ (Van Eijk, 1997, p.56)
- (201) a. **ləx̣**~ləx̣ (cf. *ləxp* ‘get smart’)
 PDEN~clear
 ‘clever’ (CA)
- b. **səx̣**~səx̣ (cf. *səxp* ‘get numb’)
 PDEN~silly
 ‘silly’ (Van Eijk, 1997, p.20)
- c. **cəq^w**~cíq^w (cf. *ciʔq^w* ‘turn red’)
 PDEN~red
 ‘red’ (Van Eijk, 1997, p.82)

A secondary -CəC- pattern is found when property-denoting reduplication co-occurs with the affix -t, as shown in (202). This type of reduplication surfaces as -CC- if [ə] is omitted (202b–c); this combination occurs with roots that have a moraic vowel.

- (202) a. kín~**kən**-t (root: *kín*- ‘dangerous’)
 dangerous~PDEN-IMMED
 ‘dangerous’
- b. kík~**kx**-ət (root: *kx*- ‘lively, active’)
 lively/active~PDEN-IMMED
 ‘lively, active’

- c. $q^wám \sim q^wm-ət$ (root: q^wam- ‘funny’)
 funny $\sim$ PDEN-IMMED
 ‘funny’

(H. Davis, in prep, Ch.49)

With roots without an underlying vowel that would take $-t$, $-m$ is used instead of $-t$ in combination with C_1C_2 reduplication.⁴³ As shown in (203), a word with $-m$ and C_1C_2 reduplication does not have [ə] between the consonants of the second C_1C_2 sequence.

- (203) a. $čəx \sim čx-əm$ (root: $čx-$ ‘clean’)
 PDEN $\sim$ clean-CHAR
 ‘clean’
 b. $χəp \sim χp-əm$ (root: $χp-$ ‘deep, dark’)
 PDEN $\sim$ deep-CHAR
 ‘deep, dark’
 c. $k^wəz \sim k^wz-əm$ (root: k^wz- ‘smooth’)
 PDEN $\sim$ smooth-CHAR
 ‘smooth’
 d. $χəɣ^w \sim χɣ^w-əm$ (root: $χɣ^w-$ ‘hard, tough’)
 PDEN $\sim$ hard/tough-CHAR
 ‘hard, tough’
 e. $cət \sim ct-əm$ (root: $ct-$ ‘bright, sober’)
 PDEN $\sim$ bright/sober-CHAR
 ‘bright/sober’

⁴³Van Eijk (1997, p.72) states that these affixes have an “elusive function” and may refer to a continued state. While it is possible that these affixes could use the same gloss (e.g., CHAR), I follow H. Davis (2011) and gloss $-t$ as intermediate IMMED and $-m$ as CHAR; he notes that the combination of the immediate suffix $-t$ and C_1C_2 reduplication appears to have a “distinct quasi-causative meaning” (p.37), which is not shared by forms with the characteristic suffix $-m$ and C_1C_2 reduplication.

- f. $\dot{\lambda}\dot{\sigma}\dot{k} \sim \dot{\lambda}k\text{-}\dot{\sigma}m$ (root: $\dot{\lambda}k\text{-}$ ‘silent’)
 PDEN $\sim$ silent-CHAR
 ‘silent’

(H. Davis, in prep, Ch.44)

Reduplication in words with *-m* can be analysed as prefixed (CəC-), which is the same as in the plurals and the other property-denoting forms, such as those given in (201) above; only *-t* forms correspond to a non-initial syllable (-CəC-).⁴⁴ I therefore assume that the infix pattern is motivated by the presence of the *-t* affix.

4.2.3.2.2 Analysis of Plural/Property-Denoting C_1C_2 Reduplication

The plural and property-denoting C_1C_2 reduplication processes in Lillooet result from parsing $\sigma_{N+\mu}$ at Level 1. This a light syllable (i.e., a syllable associated with an empty nucleus and a non-nuclear mora) in the input. For forms such as $k\acute{t}nk\acute{\sigma}nt$ ‘dangerous’, I assume that the input consists of a property-denoting morpheme and *-t*, where a fixed segment /t/ is already associated with the property-denoting syllable; this combination is linearised as a suffix in the morphology.⁴⁵ The lexical entries are provided in (204).⁴⁶

(204) *Lillooet Level 1 C_1C_2 Reduplication:*

- a. $\sigma_{N+\mu} \Leftrightarrow \langle PL \rangle$
- b. $\sigma_{N+\mu} \Leftrightarrow \langle PDEN \rangle$
- c. $\sigma_{N+\mu+t} \Leftrightarrow \langle PDEN+IMMED \rangle$

⁴⁴The word $\acute{\sigma}z\acute{\sigma}z\acute{k}\acute{\sigma}t$ ‘scary’ in H. Davis (2011, p.37) is an exception to the generalisation that reduplication surfaces as a prefix with *-t*, just like the words with *-əm*. It is built on a root without an underlying vowel ($\acute{\sigma}zk$). It is also the only example I have been able to find with a CVCC root.

⁴⁵While allomorphy is handled with listed allomorphs in Comox-Sliammon in Section 4.1.3.1, this represents a different type of allomorphy: $\sigma_{N+\mu+t}$ is selected prior to the phonology.

⁴⁶I based this analysis around the assumption that *-m* and *-t* were separate morphemes and assumed the form in (204c) could be analysed as a portmanteau; however, H. Davis (p.c.) points out that these are better analysed as part of PDEN with respect to the syntax and semantics. Though this has implications for lexical insertion, it does not necessarily affect the analysis in this section (aside from potentially requiring an analysis of *-m* in future), I leave the specifics of these combination as an area for future investigation.

C₁C₂ reduplication can be derived using the same constraint ranking that was used for initial C₁ reduplication with the addition of DEP-V_μ, which is restated in (205). DEP-V_μ is violated when any vowel aside from [ə] is epenthesized.

- (205) DEP-V_μ: Every moraic vowel in the output must have a corresponding moraic vowel in the input.

The tableau in (207) for *cəq^wciq^w* ‘red’ from example (201c) above shows that DEP-V_μ rules out candidate (207a), which epenthesizes a full vowel to provide segmental content to the affixed monomoraic syllable. Candidate (207b), which positions the affixed syllable as an infix, fatally violates PREFIX/σ. The attested candidate in (207c) only violates DEP-V and INT-C. Candidate (207d), which reduplicates the vowel, fatally violates INT-V. Candidate (207d) is also bad because it creates a new link between the mora and the nucleus. The relevant constraint from Section 4.1.4.2.1 is restated in (206). I assume this is ranked high in Lillooet at Level 1 to protect the reassociation of an input mora with a nucleus that was also present in the input.

- (206) DEP-LINK-μ(N): When a nucleus is linked to a mora in the output, it is also linked to a mora in the input. (modified from Prilop 2013, p.22)

	/σ _{N+μ} + ciq ^w /	PREFIX/σ	INT-V	DEP-V	INT-C	DEP-V _μ
(207)	a. (căq ^w) _σ .ciq ^w			*	**	*!
	b. ciq ^w .(căq ^w) _σ	*!		*	**	
	☞ c. (căq ^w) _σ .ciq ^w			*	**	
	d. (ci) _σ .ciq ^w		*!		*	

In order to handle examples such as *čəχčəxm* ‘clean’ in example (203) above, which surfaces without [ə] in what would have been the second syllable, I follow Matthewson (1994) in assuming that the distribution of [ə] follows from sonority-related constraints. I use the label SONORITY in the simplified tableau in (208) as a stand-in for the constraints that influence where a [ə] must be inserted. The best candidate is the attested one, given in (208a), which inserts the fewest vowels required to ensure that the requirements for sonority sequencing and prosody are met.

(208)

$/\sigma_{N+\mu} + \dot{c}\dot{x} + m/$	SONORITY	DEP-V
☞ a. $\dot{c}\dot{o}\dot{x}\dot{c}\dot{x}\dot{o}m$		**
b. $\dot{c}\dot{o}\dot{x}\dot{c}\dot{o}\dot{x}\dot{o}m$		***!
c. $\dot{c}\dot{o}\dot{x}\dot{c}\dot{x}m$	*!	*

For reduplicated adjectives without the *-m* suffix, such as *lǎxlǎx* ‘clever’ in example (201) above, [ə] is needed in both the first and second C_1C_2 sequence, following the general phonological patterns of the language. The ranking of SONORITY above DEP-V again allows for the attested candidate to win, as shown in (209). The position of the inserted [ə] differs from in (208) because the number and type of consonants in the input is different, which determines which distribution of [ə] will satisfy SONORITY.

(209)

$/\sigma_{N+\mu} + l\dot{x}/$	SONORITY	DEP-V
☞ a. $l\dot{o}x\dot{l}\dot{o}x$		**
b. $l\dot{o}x\dot{l}\dot{x}$	*!	*

For adjectives with the suffix *-t*, such as *kínkənt* ‘dangerous’ in example (202) above, I assume the input contains $\sigma_{N+\mu+t}$, with the structure in (210).



The *-CəC-t* sequence is motivated by general constraints on prosodic structure. Constraints pertaining to stress in the language, given in (211), predict the default position of an affixed syllable. *STRUC-FT motivates building as few feet as possible; at least one foot will be built to ensure the output meets the requirement for a prosodic word.⁴⁷ Building feet is also motivated by PARSE-SYLL, which is violated by syllables that are not parsed into a foot. Feet in Lillooet are trochaic, which means the leftmost syllable bears stress; iambic feet violate TROCHEE.

⁴⁷See Section 4.2.4.1.2 for more on stress-related constraints and reduplication in Lillooet.

- (211) a. *STRUC-FT: No feet. Assign a violation mark for every foot.
(modified from Zoll 1997)
- b. PARSE-SYLL: Syllables are parsed by feet.
(Kager, 1999, p.153)
- c. TROCHEE: Feet are L-dominant. One violation per R-dominant foot.
(Prince & Smolensky, 2004, as cited in Bosworth 2017)

The tableau in (212) shows the derivation of *kínkənt* ‘dangerous’. Square brackets mark foot boundaries, while rounded ones mark the output correspondent of the affixed syllable in the input. Given that the affixed syllable is not a prefix, basic constraints on prosodic structure are sufficient to predict its position. The attested candidate, given as (212a), builds a trochaic foot and therefore only violates *STRUC-FT. Candidate (212b) leaves a syllable unparsed, which fatally violates PARSE-SYLL. Candidate (212c), with the affixed syllable as the leftmost, fatally violates TROCHEE. Candidate (212d) builds two feet, which results in a fatal violation of *STRUC-FT. Candidate (212e), which satisfies TROCHEE and *STRUC-FT by building no feet, fatally violates PARSE-SYLL. PARSE-SYLL must be ranked above *STRUC-FT to motivate building feet.⁴⁸

(212)	/kín + $\sigma_{N+\mu+t}$ /	PARSE-SYLL	TROCHEE	*STRUC-FT
	☞ a. [kín(kənt) _{σ}]			*
	b. [kín](kənt _{σ})	*!		*
	c. [(kənt) _{σ} kín]		*!	*
	d. [kín][kənt _{σ}]			**!
	e. kín(kənt) _{σ}	**!		

The Level 1 constraint ranking is provided in Table 4.26.

⁴⁸I do not include [kín(kənt) _{σ}]_{FT} as a candidate in (212), but MAX- μ would rule out the association of /t/ with the empty mora as it would result in the loss of the mora associated with the /t/ underlyingly.

Table 4.26: Lillooet Level 1 Grammar (Updated)

A	DEP-C	INT-V, SONORITY	TROCHEE, PARSE-SYLL	*FLOAT, REALIZE MORPH, MAX-FLOAT, PREFIX/ σ , DEP- V_μ
B	INT-C	DEP-V	*STRUC-FT	

4.2.3.3 Summary of Level 1 Grammar

I analyse initial C_1 reduplication and plural/property-denoting C_1C_2 reduplication as Level 1 morphemes in Lillooet. The lexical entries for Level 1 reduplication are summarised in (213): the difference between C_1 and C_1C_2 reduplication reflects the difference between affixing a syllable (σ) and a monomoraic syllable with an empty nucleus ($\sigma_{N+\mu}$).

(213) *Lillooet Level 1 Reduplication:*

- a. $\sigma \Leftrightarrow \langle \text{INIT} \rangle$
- b. $\sigma_{N+\mu} \Leftrightarrow \langle \text{PL} \rangle$
- c. $\sigma_{N+\mu} \Leftrightarrow \langle \text{PDEN} \rangle$

Shared properties of Level 1 reduplication in Lillooet emerging from the ranking given in Table 4.26 above include the following: the reduplicant remains on the left edge (except when C_1C_2 reduplication is part of a portmanteau with the suffix *-t*), consonants are readily copied, and [ə] is epenthesized where a vowel is required.

Constraints against parsing affixes marked with L2 (Level 2) are also relevant at Level 1 as argued in Chapter 3 (Section 3.5.2.3): AFFIX_{L2} is ranked higher than $\text{PARSE}_{\text{AFFIX}}$, as shown in Table 4.27. AFFIX_{L2} must also be ranked higher than PARSE-SYLL to allow Level 2 affixes to remain unparsed at Level 1.

Table 4.27: Lillooet Affix Constraints

A	AFFIX_{L2}
B	$\text{PARSE}_{\text{AFFIX}}$

4.2.4 Level 2

Lillooet has three types of Level 2 reduplication: (1) change-of-state C₂ reduplication (Section 4.2.4.1.1), (2) pluractional C₁ reduplication (Section 4.2.4.2.1), and (3) diminutive C₁ reduplication (Section 4.2.4.2.3). Change-of-state reduplication occurs to fill an affixed syllable, while pluractional and diminutive reduplication occur to provide segmental content for an affixed mora. Both types of C₁ reduplication have a floating [c.g.] feature, which is discussed in Section 4.2.4.3.

Pluractional and diminutive C₁ reduplication are differentiated because the diminutive mora carries a [stress] feature, while the pluractional mora does not. Lexical entries for Level 2 reduplication and the surface position of the copied segments are previewed in Table 4.28.

Table 4.28: Level 2 Reduplicative Affixes

	Underlying Form	Surface Position
Change-of-State C ₂	σ	Infix
Pluractional C ₁	μ [c.g]	Infix
Diminutive C ₁	μ ^[stress] [c.g.]	Infix

4.2.4.1 Level 2 C₂ Reduplication

4.2.4.1.1 Change-of-State C₂ Reduplication

C₂ reduplication marks a change of state in Lillooet. The consonant following the stressed vowel is doubled with [ə] between the two matching consonants, as shown in (214).

- (214) a. ʰí<ɬə>l (root: ʰíl- ‘sprinkle’)
 sprinkle<COS>
 ‘to sprinkle, be sprinkling’ (Van Eijk, 1997, p.56)
- b. pú<ɬə>ɬ (root: puɬ-)
 boil<COS>
 ‘boiling’ (Van Eijk, 1997, p.56)

- c. $\dot{p}l\acute{i}<\dot{x}^w\dot{\alpha}>\dot{x}^w$ (root: $\dot{p}li\dot{x}^w$ -)
 flow.over<COS>
 ‘flowing over’ (Van Eijk, 1997, p.56)
- d. $q\acute{a}<x^w\dot{\alpha}>x^w$ (root: qax^w -)
 break<COS>
 ‘to break’ (Van Eijk, 1997, p.59)
- e. $z\acute{e}w\acute{a}<t\dot{\alpha}>t-s$ (root: $zwat$)
 be.known<COS>-CAUS
 ‘have learned how to do something’ (H. Davis et al., in prep)
- f. $w\grave{\alpha}l\acute{i}<\dot{k}\dot{\alpha}>\dot{k}$ (root: $w\dot{l}k$ -)
 croak<COS>
 ‘to croak (frogs)’ (Van Eijk, 2013, p.349)

The pairs in (215) and (216) show that the presence of [ə] between the matching consonants is sensitive to the general composition of the word: with the same roots, [ə] is between the doubled consonants when the change-of-state morpheme is added to a bare root, as in (215a) and (216a), but there is no [ə] in this position in other forms, as in (215b) and (216b). The [ə] between the matching consonants is required to provide a nucleus for an affixed syllable in change-of-state reduplication, but it may be omitted at a subsequent stratum if the grammar does not require a vowel in that position.

- (215) a. $\dot{p}\acute{a}<\dot{w}\dot{\alpha}>\dot{w}$ (root: $\dot{p}aw$ -)
 echo<COS>
 ‘to echo’
 b. $\dot{p}\acute{a}<\dot{w}>\dot{w}-\acute{\alpha}c$ ⁴⁹
 echo<COS>-mouth
 ‘echoing voice’ (Van Eijk, 2013, p.34)
- (216) a. $k\dot{\alpha}<\dot{n}\dot{\alpha}>\dot{n}$ (root: kn -)
 bump<COS>
 ‘got bumped’

⁴⁹The second /w/ is glottalised by the glottalising lexical suffix -c.

- b. kə<**ṇ**>**ṇ**-álq^w
 bump<COS>-hulk/mass
 ‘to bump against something hard/to crash’ (Van Eijk, 1997, p.90)

The position of the C₂ reduplicant is sensitive to the position of stress assigned at the first (Level 1) stratum. When additional affixes are added to the word as in (217b), primary stress shifts to the right while the C₂ reduplicant remains aligned to the earlier position of stress, which is shown in (217a).

- (217) a. zəwá<**tə**>t-s (root: *zwat*-)
 be.known<COS>-CAUS
 ‘have learned how to do something’ (H. Davis et al., in prep)
- b. waʔ zəwa<**tə**>t-s-twítas
 IPFV be.known<COS>-CAUS-3PL.ERG
 ‘They learned how to do it.’ (S. Mitchell, 2022, p.163)

When the change-of-state morpheme is added to a word with lexical suffixes, which are “substantival suffixes that have semantic content analogous to freestanding nominals of the same meanings” (Hinkson, 1999, p.1). The copied segment is typically positioned relative to where stress would fall if there were no lexical suffix (218). For example, compare *pútəʔ* in (214b), above, with (218a). In other words, C₂ reduplication is sensitive to the prosodic structure, but does not consider lexical suffixes.

- (218) a. pu<**t**>**t**-átq^waʔ (root: *put*-)
 boil<COS>-water
 ‘water boils’ (Van Eijk, 2013, p.11)
- b. ci<**x^w**>**x^w**-ánwas (root: *cix^w*-)
 stream.down<COS>-inside
 ‘to feel sad (as if one is crying inside)’ (Van Eijk, 2013, p.76)
- c. cù<**ḷ**>**ḷ**-átməx (root: *cul*-)
 sour<COS>-belly
 ‘to have a sour stomach from eating too many berries’ (Van Eijk, 2013, p.88)

- d. qa<**x^w**>x^w-áltəx^w (root: qax^w-)
break<COS>-rib
'to break one's rib' (Van Eijk, 2013, p.432)
- e. qa<**x^w**>x^w-áwł (root: qax^w-)
break<COS>-conveyance
'break down (car)' (Van Eijk, 2013, p.263)
- f. n-qi<**p̣**>p̣-ánk-xən (root: qip̣-)
LOC-stuck<COS>-belly-foot
'mud or snow cakes on shoes or boots, or on wagon's wheels, or on horse's hoofs.' (Van Eijk, 2013, p.270)

There are some exceptions to the generalization that C₂ reduplication overlooks the stress shift related to the addition of lexical suffixes: there are some forms with stress on a lexical suffix in which C₂ reduplication may target content from the lexical suffix and be positioned within the lexical suffix (219). These examples all have CC-shaped roots. However, it is possible that the examples in (219) are lexicalised with the lexical suffix as part of the root: the combination of root and lexical suffix is not straightforwardly compositional (H. Davis, p.c.).

- (219)
- a. ʔq^w-á<**ẉ**>ẉəs (root: ʔq^w-)
bang-together<COS>
'get together' (Van Eijk, 1997, p.59)
 - b. məč-u<**s**>s-əm (root: mč-)
deceive.cheat-face<COS>-MID
'cheat' (H. Davis et al., in prep)
 - c. təq-á<**pə**>p (root: tq-)
touch-back.part<COS>
'get dammed, jammed, or blocked by logs, ice, or a slide' (H. Davis et al., in prep)

A final set of exceptions is given in (220), which shows variation with the same lexical suffix with respect to the position of C₂ reduplication. A consonant in the suffix -ana? 'ear' in (220) is reduplicated in some words (a–b), but not in others

(c–e). Negative judgments come from fieldwork conducted by Henry Davis (see discussion of fieldwork sources in Section 1.4.1).

- (220) a. $\dot{q}\text{əmp-}\acute{a}\langle n \rangle\text{na?}$ (root: $\dot{q}\text{mp-}$)
 get.plugged-ear<COS>
 ‘ears get plugged’ (Van Eijk, 2013, p.271)
- b. $\text{cək}^w\text{-}\acute{a}\langle n \rangle\text{na?}=\text{ɬkan}$ ($*\text{cək}^w\text{ək}^w\text{ána?ɬkan}$) (root: $\text{ck}^w\text{-}$)
 pull-ear<COS>=1 SG.SUBJ
 ‘My ear got pulled.’ (CA)
- c. $n\text{-lə}\langle \dot{p}(\text{ə}) \rangle\dot{p}\text{-ána?}$ ($*\text{ləpána?}$) (root: $\text{l}\dot{p}\text{-}$)
 LOC-bury<COS>-ear
 ‘get completely buried’ (CA)
- d. $\text{ɣ}^w\text{ə}\langle \dot{l}\text{ə} \rangle\dot{l}\text{-ána?}$ ($*\text{ɣ}^w\text{əlána?}$) (root: $\text{ɣ}^w\text{l-}$)
 bubble<COS>-ear
 ‘get bubbles in your ears’ (CA)

It is not possible to predict the form of words with C_2 reduplication with certain lexical suffixes: for example, the words in (220b-d) all have a CC root. The analysis proposed in the following section will account for the phonologically predictable examples in (214-218), while the forms in (219-220) are set aside.

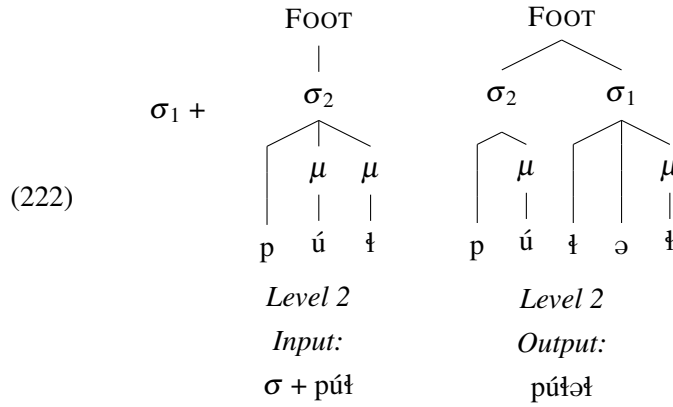
4.2.4.1.2 Analysis of Change-of-State C_2 Reduplication

Change-of-state reduplication involves the affixation of an empty syllable, as shown in (221). The syllable is marked as a Level 2 affix.

(221) *Lillooet Change-of-State Reduplication:*

$$\sigma_{L2} \Leftrightarrow \langle \text{COS} \rangle$$

When the change-of-state σ is parsed into a word, the number of syllables increases by one, while the number of moras remains unchanged. The difference between the input and output is shown in (222) for example (214b) above.



As discussed above, C_2 reduplication is sensitive to prosodic structure and the position of stress (in the root). I assume, following H. Davis (in prep) and Van Eijk (1997), that the position of stress changes over the course of the derivation.⁵⁰ A simplified overview of stress assignment and shift across strata is given in Table 4.29. The change-of-state C_2 and diminutive C_1 reduplication are filled at Level 2, and they are combined with the output of Level 1.

⁵⁰Regular stress shift can result in C_2 reduplication appearing to be aligned to the word edge, rather than stress. For example, a word-final sequence of VCC at Level 2 will cause stress to shift away from the position of C_2 reduplication (e.g., $-awt$ in $qax^w x^w áwt$ ‘break down (car)’). The constraint $*UNSTRESSED\sigma_\mu\mu\mu$ in (i) motivates this shift in combination with $*CLASH$, which bans adjacent stressed syllables.

- (i) a. $*UNSTRESSED\sigma_\mu\mu\mu$: Unstressed syllables are maximally bimoraic. Assign a violation mark for any unstressed syllable associated with three (or more) moras.
 b. $*CLASH$: rhythm is alternating; no two adjacent stressed syllables. (Alber, 2005, p.504)

Table 4.29: Stress Assignment

Stratum	Morphology	Stress	Reduplication
Level 1	[root]	First V_μ (first [ə] if no V_μ)	Plural C_1C_2 Property-Denoting C_1C_2 Initial C_1
Level 2	[[Level 1 _{output}] + lexical suffixes] (and other affixes not at Level 3)	Shift to V_μ if in $\sigma_{\mu\mu\mu}$	Change-of-State C_2 Diminutive C_1 Pluractional C_1
Level 3	[[Level 2 _{output}] + transitivizers/pronouns]	Shift to V_μ if 2+ unstressed σ from right edge	
Level 4	[[Level 3 _{output}] + clitics]	Shift to V_μ if 3+ unstressed σ from right edge	

Sample derivations are provided in Table 4.30 and Table 4.31 for examples (218e) and (220c) above. These show that stress is assigned at Level 1 (to the bare root) and then lexical suffixes and the change-of-state affix are added at Level 2. Descriptively, the position of the segments associated with reduplication depends on the position of stress in the input to Level 2. My analysis is that the affixed syllable is integrated into the existing foot. The epenthetic [ə] is deleted at a later stratum if required by the phonological grammar.⁵¹

Table 4.30: Stress and Reduplication: $qax^w x^w áw\ddot{t}$ ‘break down’

Stratum	Input	Output
Level 1	qax^w	$qáx^w$
Level 2	$\sigma + qáx^w + aw\ddot{t}$	$qax^w əx^w áw\ddot{t}$
Level 3	-	$qax^w əx^w áw\ddot{t}$
Level 4	$qax^w əx^w áw\ddot{t}$	$qax^w x^w áw\ddot{t}$

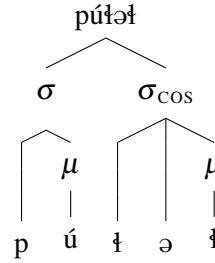
⁵¹There are also cases where [ə] may be omitted in fast speech (H. Davis, p.c.), which suggests that the vowel may also be lost due to phonetic reduction (see Section 6.7.1 for further discussion).

Table 4.31: Stress and Reduplication: *nləpəpánaʔ* ‘get completely buried’

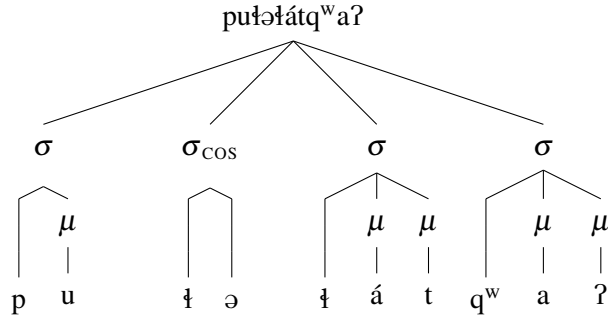
Stratum	Input	Output
Level 1	l ^ə p	l ^ə p
Level 2	σ + n + l ^ə p + anaʔ	nl ^ə pəpánaʔ
Level 3	nl ^ə pəpánaʔ	nl ^ə pəpánaʔ
Level 4	-	nl ^ə p(ə)pánaʔ

The figures in (223) show a comparison of the structure of *púʔəʔ* ‘boiling’ and *puʔəʔátq^waʔ* ‘water boils’ for examples (214b) and (218a) above.

(223) a.



b.



While a full analysis of Lillooet stress is beyond the scope of this dissertation, the constraints in (224) capture the basics of stress assignment as it pertains to the position of reduplication.⁵² The constraint PRWD=GRWD requires at least one foot in the output, while *STRUC-FT incurs a violation for every foot in the output.⁵³ The constraint PARSE-SYLL incurs a violation for every syllable that is not parsed into a foot.

⁵²See Roberts (1993) or Caldecott (2009) for analyses of Lillooet stress.

⁵³PRWD=GRWD was not discussed at earlier in Section 4.2.3 because it does not affect the form of reduplication at Level 1 because Level 1 reduplication is edge-aligned, not prosody-aligned.

- (224) a. PRWD=GRWD: A grammatical word must be a prosodic word.
(Kager, 1999, p.152)
- b. *STRUC-FT: No feet. Assign a violation mark for every foot.
(modified from Zoll 1997)
- c. PARSE-SYLL: Syllables are parsed by feet.
(Kager, 1999, p.153)

In the Level 1 grammar, PRWD=GRWD is ranked above *STRUC-FT, which means building at least one foot is required. This is shown in (225), where the candidate that builds a foot wins over the one that leaves the root unfooted. Square brackets are used to mark edges of a foot.

/puł/		PRWD=GRWD	*STRUC-FT	PARSE-SYLL
(225)	☞ a. [púł]		*	
	b. puol	*!		*

The output from Level 1 is the input to Level 2 in (226). The combination of *STRUC-FT and PARSE-SYLL motivate parsing a syllable into an existing foot, if possible. The attested candidate in (226a) only violates *STRUC-FT once; it wins against candidates with one syllable left unparsed (226b-c), and the candidate with two feet (226d).⁵⁴

/σ + [púł]/		*STRUC-FT	PARSE-SYLL
(226)	☞ a. [púłəł]	*	
	b. [púł]əł	*	*!
	c. pu[łəł]	*	*!
	d. [pú][łəł]	**!	

The position of feet in the input is protected by highly ranked ANCHOR constraints (McCarthy & Prince, 1995). I use the constraint ANCHOR_{FOOT} in (227) as a stand-in to represent the two anchor constraints used by Özçelik (2014, p.238) to ensure faithfulness at the edges of a foot.⁵⁵ ANCHOR_{FOOT} is violated when a segment on

⁵⁴There are other prosodic constraints that are not given here, but that play a role in selecting a candidate (see, e.g., Caldecott 2009).

⁵⁵The ranking of the two constraints is not relevant at present and therefore they are combined to save space.

the left/right edge in the input does not have the same position (relative to a foot) in the output.

(227) $\text{ANCHOR}_{\text{FOOT}}$

- a. ANCHOR-R : If α is at the right edge of a foot and $\alpha\Re\beta$, then β is at the right edge of a foot.
- b. ANCHOR-L : If α is at the left edge of a foot and $\alpha\Re\beta$, then β is at the left edge of a foot.

(McCarthy, 1995, p.23)

$\text{ANCHOR}_{\text{FOOT}}$ is therefore violated if affixation causes stress shift. In cases where an input segment on the edge of a foot corresponds to multiple segments in the output, $\text{ANCHOR}_{\text{FOOT}}$ merely requires that *one* of the output segments is in the same position relative to the edge of a foot.

The tableau in (228) shows that building an additional foot violates $\ast\text{STRUC-FT}$ twice (228a), while leaving three syllables unparsed incurs three violations of PARSE-SYLL (228b). The winning candidate retains a single foot which begins and starts with the same segments as in the input (228c). This shows PARSE-SYLL must be ranked lower than $\ast\text{STRUC-FT}$. Candidates (228a) and (228c) differ with respect to where the right boundary of the first foot is placed: I show Candidate (228a) with resyllabification of /t/ into an onset of the second foot. However, even if the /t/ remained in a coda and satisfied $\text{ANCHOR}_{\text{FOOT}}$, building a second foot would still result in a fatal violation of $\ast\text{STRUC-FT}$.

	/σ + [pút] + atq ^w a?/	$\ast\text{STRUC-FT}$	$\text{ANCHOR}_{\text{FOOT}}$	PARSE-SYLL
(228)	a. [pùtə _σ][tátq ^w a?]	**!	*	
	b. [pút]ə _σ tátq ^w a?	*		***!
	c. [pútə _σ]atq ^w a?	*		**

Unparsed syllables in the output at Level 2 may be parsed into feet at a later stratum. For example, if there are two or more unparsed syllables on the right edge in the input to Level 3, they will be parsed into feet; primary stress will then shift to this new foot. A tableau showing a Level 3 derivation is given in (229). The attested candidate builds a second foot in (229a). The candidate that does not build an additional foot fatally violates PARSE-SYLL (229b).

	/[pùtə́]atq ^w aʔ/	PARSE-SYLL	*STRUC-Ft	ANCHOR _{FOOT}
(229)	a. [pùtə́][tátq ^w aʔ]		**	*
	b. [pùtə́]atq ^w aʔ	*!*	*	

The candidate [pùtə́][tátq^waʔ] in (229a) is the input to Level 4, and optional deletion of the unstressed [ə] may occur. This allows for the derivation of both attested forms (pùtə́tátq^waʔ and pùtátq^waʔ). The tableaux above in (225-229) show the relationship between stress, lexical suffixes, and the position of C₂ reduplication. C₂ reduplication occurs at Level 2, as was shown in (228).

In order to predict fission of a consonant and epenthesis of a vowel at Level 2, the relative rankings of INTEGRITY and DEP must be similar to their ranking at Level 1, but PREFIX/σ must be ranked low to allow for infixation. In order to capture the position of the reduplication relative to stress (such that the affixed syllable follows the stressed syllable), the constraints TROCHEE and *P/ə are needed. TROCHEE is violated by any foot where stress does not fall on the leftmost element (mora or syllable); *P/ə is violated when stress falls on a schwa.

- (230) a. TROCHEE: Feet are L-dominant. One violation per R-dominant foot.
 (Prince & Smolensky, 2004, as cited in Bosworth 2017)
- b. *P/ə: Do not associate schwa with a prosodic head.
 (definition paraphrased from Prince & Smolensky, 2004)

The constraint *P/ə is part of a hierarchy of segment sonority with universal rankings of constraints pertaining to the best prosodic peak. The hierarchy for syllables adapted from Prince and Smolensky (2002) is given in (231a), and the version for metrical feet from Kenstowicz (1994) that Dyck (2004) uses for Squamish is given in (231b). The first constraint in the hierarchy corresponds to the most marked peak and will be universally ranked above the others (e.g., a language may allow [ə] to bear stress, but it will always more marked than another vowel). The other constraints on prosody in the language mediate when and where [ə] will be stressed, such as in the absence of other vowels (DEP-V_μ >> *P/ə in Lillooet) or when stress has a fixed position (TROCHEE, ALLFEETLEFT >> *P/ə in Comox-Sliammon).

(231) a. **Peak Hierarchy:**

*P/t >> *P/d >> ... >> *P/ə >> *P/i >> *P/a

(adapted from Prince & Smolensky, 2004, p.45)

b. **Peak Prominance for Metrical Feet:**

*P/ə >> *P/i,u >> *P/e,o >> *P/a

(Kenstowicz 1994, as cited in Dyck 2004, p.15)

The tableau in (232) shows the ranking needed to predict the attested form of C₂ reduplication in the word *ʔləl* ‘to sprinkle, be sprinkling’ from (214a) above. A candidate with fission of a vowel (232b) fatally violates INT-V, while candidates that maintain the input order fatally violate either *P/ə (232c) or TROCHEE (232d), depending on whether stress falls on the [ə].

(232)	/σ + [ʔl] /	TROCHEE	*P/ə	INT-V	DEP-V	INT-C	PREFIX/σ
	☞ a. [ʔləl _σ]				*	*	*
	b. [ʔlil _σ]			*!		*	*
	c. [ʔ _σ ʔlil]		*!		*	*	
	d. [ʔ _σ ʔlil]	*!			*	*	

There is a final consideration in the analysis of C₂ reduplication in Lillooet; the Generalised Nonlinear Affixation (GNLA) approach does not provide a clear way to specify which consonants are copied (direction of fission/copying). This must be determined by other constraints in the grammar, as demonstrated by the ambiguity in (233). Though the given constraints motivate fission and account for the position of the affixed syllable, candidate (233b) which copies the first consonant incurs the same violations as the attested candidate in (233a).

(233)	/σ ₁ + [ʔl] /	TROCHEE	*P/ə	INT-V	DEP-V	INT-C	PREFIX/σ
	?? a. [ʔləl _{σ1}]				*	*	*
	?? b. [ʔlʔl _{σ1}]				*	*	*

The direction of fission and which segment is copied in GNLA is a problem that has not received much attention in other analyses of reduplication. There are a number of ways for the grammar to categorically rule out copying in one direction. For example, the constraint INTEGRITY-WI in the analysis of Comox-Sliammon

C₂ reduplication in Section 4.1.5.1.2 penalises fission of a word-initial segment, resulting in copying the second consonant of the root. It is not possible to adopt the same strategy for Lillooet because C₂ reduplication is stress-aligned and therefore it is not sufficient to reference to a fixed position in the word.

The directionality problem illustrated by the ranking ambiguity in (233) has theoretical implications beyond the present analysis. For the present analysis, I use the constraint INT·EDGE-(σ), defined in (234).

(234) INT·EDGE-(σ): If two output segments correspond to the same input segment, then they are in the same σ .

The constraint INT·EDGE-(σ) is modelled after constraints in the CC·EDGE family of constraints, which come from the literature on surface correspondence and harmony (Bennett, 2013). A CC·Edge constraint identifies a domain; any output consonants in correspondence with each other must be within the same domain.

The tableau in (235) shows how the inclusion of INT·EDGE-(σ) correctly predicts the attested candidate.⁵⁶ Syllable boundaries are marked with periods.

/ σ + [ɬɪ] /		INT·EDGE-(σ)	DEP-V	INT-C
(235)	a. [ɬɪ .lə _x l]		*	*
	b. [ɬɪ.ɬ ə _x l]	*!	*	*

The ranking needed to derive C₂ reduplication in Lillooet is summarised in Table 4.32. This ranking will be updated following the analysis of Level 2 C₁ reduplication in the next section.

⁵⁶An additional constraint will be needed to handle directionality of fission for C₁ reduplication (see Section 4.2.4.2).

Table 4.32: Lillooet Level 2 Grammar (Preliminary)

A	PRWDGRWD	INT-V, DEP-C	*P/ə, TROCHEE	REALIZEMORPHEME, *FLOAT, MAX-FLOAT, INT·EDGE-(σ), ANCHOR _{FOOT}
B	*STRUC-FT,	DEP-V, INT-C	PREFIX- σ	
C	PARSE-SYLL			

4.2.4.2 Level 2 C₁ Reduplication

There are two types of Level 2 C₁ reduplication in Lillooet: (i) a pluractional C₁ reduplication process, and (ii) a diminutive C₁ reduplication process.

The key difference between the two reduplicative processes is in the treatment of the stressed vowel. In the diminutive form in (236a), the stressed vowel reduces to a [ə]. The form in (236b), which has the pluractional type of Level 2 C₁ reduplication, does not reduce to [ə].

- (236) a. k^ws-á<s>li'ca? (root: k^ws-)
singe-outer.cover<C₁.PL>
‘burn hair or feathers off an animal or a bird’ (H. Davis & Mellesmoen, 2023, p.21)
- b. tó<t>p-ən=łkan (root: tup-)
punch<DIM>-TR=1SG.SUBJ
‘I hit them (sg.) lightly.’ (H. Davis & Mellesmoen, 2023, p.17)

I follow H. Davis and Mellesmoen (2023) in distinguishing these two C₁ reduplicative processes; previous studies treated them as a single, variable C₁ reduplication process (e.g., Van Eijk, 1997).

4.2.4.2.1 Pluractional C₁ Reduplication

Pluractional C₁ reduplication copies the consonant before the stressed vowel, resulting in identical consonants on either side of it. There are no changes in

vowel quality between the input and output. Examples are given in (237). This type of reduplication is only with verbs. Fissioned consonants may cross into certain suffixes, including the autonomous marker and lexical suffixes, as shown in (237e–f) and (237g–h), respectively.⁵⁷ Glottalisation of a sonorant accompanies pluractional reduplication and is discussed further and analysed in Section 4.2.4.3.

- (237) a. n-ká<**k**>əṁ-xal (root: *kam*-)
 LOC-pick.up<C₁.PL>-A.INTR
 ‘pick things up here and there’
- b. n-ká<**k**>əl-xal (root: *kal*-)
 LOC-follow<C₁.PL>-A.INTR
 ‘follow s.o. around’
- c. ḵá<**ḵ**>ḵ-əṁ (root: *ḵal*-)
 bite<C₁.PL>-TR
 ‘gnaw something’
- d. cá<**c**>q-əṁ (root: *caq*-)
 pull.out<C₁.PL>-TR
 ‘pluck something (a bird)’
- e. x^wəlp-í<**p**>ləx (root: *x^wlp*-)
 turn<C₁.PL>-AUT
 ‘turn around and around’
- f. ʔəŋ^w-í<**ŋ^w**>ləx (root: *ʔŋ^w*-)
 bounce<C₁.PL>-AUT
 ‘jump up and down’
- g. cṵ-ú<**ṵ**>saʔ (root: *cṵ*-)
 kick-round.thing<C₁.PL>
 ‘play soccer’

⁵⁷The affix *-ilx/-ləx* has been classified as an intransitivizer by H. Davis (in prep), and a lexical suffix (meaning ‘body’, often with a reflexive function, by Van Eijk 1997, p.105). Steiner (in prep) reconciles these two positions in Thompson River Salish by treating the autonomous marker as a somatic lexical suffix with a zero middle marker. I assume that the autonomous marker in Lillooet has the phonology (w.r.t. stress) of a lexical suffix. Crucially, I assume that *-ilx/-ləx* is parsed at Level 2, and therefore that it may be targeted by diminutive reduplication, which other in/transitivizers cannot.

- h. $k^ws\text{-}\acute{a}\langle s \rangle li\acute{c}a?$ (root: $k^ws\text{-}$)
 singe-outer.cover<C₁.PL>
 ‘burn hair or feathers off an animal or a bird’

(H. Davis & Mellesmoen, 2023, pp.20–21)

4.2.4.2.2 Analysis of Pluractional C₁ Reduplication

Pluractional C₁ reduplication in Lillooet involves the affixation of a mora; its lexical entry is given in (238). The lexical entry bears the diacritic L2 (Level 2). The constricted glottis feature [c.g.] is set aside until Section 4.2.4.3 where I analyse glottalisation associated with both pluractional and diminutive C₁ reduplication.

- (238) *Pluractional C₁ Reduplication:*
 $(\mu [c.g.])_{L2} \Leftrightarrow \langle PL.C_1 \rangle$

The derivation of $\acute{h}ú\acute{t}t\acute{a}n$ ‘to squish it all up’ is shown in (239).⁵⁸ Stress is assigned to the root in the Level 1 derivation in (239a), followed by the addition (and filling) of the pluractional C₁ mora at Level 2 in (239b). The transitivizer is parsed at Level 3 (239c).

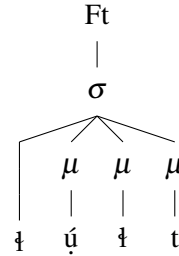
- (239) *Derivation of $\acute{h}ú\acute{t}t\acute{a}n$ ‘to squish it all up’*
 (cf. $\acute{h}út\acute{u}n$ ‘to squish something soft’)
 a. Level 1
 i. *Input:* $\acute{h}út$
 ii. *Output:* $\acute{h}út$

⁵⁸There is retraction, marked with a dot beneath the segment, on the root vowel in (239). See Bessell (1993b, pp.291–294) for a discussion of retraction in Lillooet.

b. Level 2

i. *Input*: $\mu + \text{tút}$

ii. *Output*: tút



c. Level 3

i. *Input*: $\text{tút} + \text{ən}$

ii. *Output*: tútən

The tableau in (240) shows that the constraints established for C_2 reduplication in Section 4.2.4.1.2 can also account for the fission of a consonant to fill a mora (240a). A candidate with fission of a vowel and a consonant fatally violates INT-V (240b), while epenthesis of a vowel or a consonant to fill a mora fatally violate DEP-V or DEP-C, respectively (240c,d).

$/\mu + [\text{tút}]/$		INT-V	DEP-C	DEP-V	INT-C
(240)	a. $[\text{tút}_\mu \text{t}]$				*
	b. $[\text{tút}_\mu \text{t}]$	*!			*
	c. $[\text{tút}_\mu \text{t}]$			*!	*
	d. $[\text{tút}_\mu \text{t}]$		*!		

An example of pluractional C_1 reduplication with the autonomous suffix is given in (241), using the word $\text{təŋ}^w \text{təŋ}^w \text{ləx}$ ‘jump up and down’ from (237f) above. Though the [ə] is stressed at Level 1, it is left unfooted at Level 2 when $-\text{ilx}$ is integrated.

(241) a. Level 1

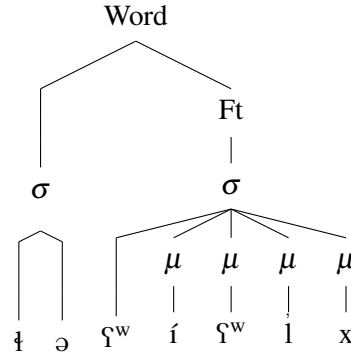
i. *Input*: təŋ^w

ii. *Output*: $[\text{təŋ}^w]$

b. Level 2

i. *Input*: $\mu + [\text{t}\text{ə}\text{ɪ}^w] + \text{ilx}$

ii. *Output*: $\text{tə}[\text{ɪ}^w\text{ɪ}\text{ɪ}^w\text{l}\text{x}]$



c. Level 3

i. *Input*: $\text{tə}[\text{ɪ}^w\text{ɪ}\text{ɪ}^w\text{l}\text{x}]$

ii. *Output*: $\text{tə}[\text{ɪ}^w\text{ɪ}\text{ɪ}^w\text{l}\text{ə}\text{x}]$

The fact that stress shifts from [ə] when *-ilx* is added means that the constraint HEAD-MATCH may be violated. The definition of HEAD-MATCH is given in (242): the prosodic head in the input should correspond to the prosodic head in the output. HEAD-MAX, with the same definition as the one given for HEAD-MATCH in (242), was used by Dyck (2004) for an analysis of Squamish stress, integrating the insight of Alderete (1996:6): “the prosodic category relevant to the meaning of HEAD-MAX is the moraic head [and] therefore, HEAD-MAX asserts that, for every underlying segment dominated by a stressed mora which has a correspondent in the output, the related output segment will also be dominated by a stressed mora”.

(242) HEAD-MATCH:

If α is the prosodic head of the word and $\alpha \mathfrak{R} \beta$, then β is the prosodic head of the word.

(McCarthy, 1995, p.23)

The tableau in (243) shows how ranking $*P/\text{ə}$, which is violated by any stressed [ə], above HEAD-MATCH correctly predicts that stress will shift to the full vowel in *-ilx*.⁵⁹ Candidate (243a), which builds a trochaic foot with the [i] as the head

⁵⁹In the absence of other vowels, [ə] will remain stressed due to the high-ranked constraint PRWD=GRWD which requires a word to have a stressed syllable.

of the stressed syllable, wins against candidates which either retain stress on [ə] (243b), or keep the foot aligned with the left edge of the word while assigning stress to [i] (243c).

	$/\mu + [\text{t}\acute{\text{ə}}\text{f}\text{w}] + \text{ilx}/$	TROCHEE	*P/ə	HEAD-MATCH
(243)	a. $\text{t}\acute{\text{ə}}[\text{f}\text{w} \text{f}\text{w}]\text{lx}$			*
	b. $[\text{t}\acute{\text{ə}}\text{f}\text{w} \text{f}\text{w}]\text{lx}$		*!	
	c. $[\text{t}\acute{\text{ə}}\text{f}\text{w} \text{f}\text{w}]\text{lx}$	*!		*

In order to ensure that the copied segment appears in the correct position (immediately after the stressed vowel), two additional constraints are needed: $\text{PARSE-}\mu$ and $*-\triangle\text{FT} \succ \triangle\text{FT}$. These two constraints, defined in (244), ensure the affixed pluractional mora will be integrated into the head syllable of a foot. $\text{PARSE-}\mu$ is violated by every mora that is not parsed into a foot, and $*-\triangle\text{FT} \succ \triangle\text{FT}$ is violated by bisyllabic feet where the head syllable is lighter than the non-head syllable.⁶⁰ Together, these constraints favour candidates where the affixed mora is integrated into the head syllable of a foot.

(244) a. $\text{PARSE-}\mu$:

Moras are parsed by feet.

(modified from Kager 1999, p.153)

b. $*-\triangle\text{FT} \succ \triangle\text{FT}$:

Assign a violation for every non-head syllable in a foot with a greater weight than the corresponding head syllable.

$\text{PARSE-}\mu$ ensures that the pluractional mora is parsed into a foot, as shown in (245). The winning candidate integrates the mora into the the stressed syllable (245a). Candidates which reduplicate the initial segment fatally violate $*\text{STRUC-FT}$ (or $*\text{P}/\text{ə}$) by building a second foot, or $\text{PARSE-}\mu$ by leaving the pluractional mora unparsed (245b vs. 245c).

⁶⁰The notation used in this constraint name draws inspiration from the constraints used by Caldecott (2009) to compare the sonority of a non-head with the sonority of a segment (e.g., $*-\triangle\text{FT} \succ \text{SCHWA}_{\text{SUFFIX}}$). $\triangle$ is used to refer to a head.

(245)	$/\mu_1 + [\text{t}\text{ə}\text{t}^w] + \text{ilx} /$	*STRUC-FT	*P/ə	ANCHOR _{FOOT}	PARSE- μ	HEAD-MATCH
	☞ a. $[\text{t}\text{ə}[\text{t}^w\text{ilx}]]$	*		*		*
	b. $[\text{t}\text{ə}[\text{t}][\text{t}^w\text{ilx}]]$	**!	*!	*		
	c. $[\text{t}\text{ə}[\text{t}^w\text{ilx}]]$	*		*	*!	*

An affixed mora increases the weight of a syllable. With bisyllabic roots with two full vowels, such as *qítíl* ‘run’, *- Δ FT $\succ$ Δ FT ensures that the pluractional mora will be parsed into the stressed syllable (246a), rather than an unstressed one (246b).

(246)	$/\mu_1 + [\text{qítíl}]/$	*- Δ FT $\succ$ Δ FT	*STRUC-FT	PARSE- μ	HEAD-MATCH
	☞ a. $[\text{qítíl}]]$		*	*	
	b. $[\text{qí.tíl}]]$	*!	*	*	

Finally, the issue outlined in Section 4.2.4.1.2 concerning directionality of fission in C_2 reduplication is also relevant for pluractional reduplication: there is nothing thus far in the analysis that decides between fission of C_1 or C_2 to fill an affixed mora. An additional constraint is needed to select between a potential candidate * $[\text{t}\text{ətt}]$ and the right form $[\text{t}\text{ət}]]$, for example, for the word *tútəñ* ‘to squish it all up’, as in (240) above.⁶¹

In order to rule out fission of a C_2 consonant to fill the pluractional mora, a constraint banning two identical adjacent consonants is needed. This is possible with reference to the Obligatory Contour Principle, which has roots in the analysis of suprasegmental phonology and gemination (e.g., Leben, 1973; McCarthy, 1986). The Obligatory Contour Principle forbids adjacent identical elements. This can be integrated into Optimality Theory as a constraint named OCP, which is given in (247).

(247) OCP: Adjacent identical elements are prohibited. Assign a violation for each sequence of identical elements.

(Myers, 1997)

The inclusion of OCP allows for the grammar to select the attested candidate (248a)

⁶¹The constraint INT-EDGE-(σ), used for C_2 reduplication in 4.2.4.1.2, is not sufficient where the fissioned consonants will be in the same syllable, which is the case when the output is monosyllabic (e.g., $[\text{t}\text{ət}]]$).

over the unattested candidate (248b), despite each incurring the same violations under INT-C and INT·EDGE-(σ).⁶²

$/\mu_1 + [\text{t}\acute{\text{u}}\text{t}]/$		INT-C	OCP	INT·EDGE-(σ)
(248)	a. $[\text{t}\acute{\text{u}}\text{t}\text{t}]$	*		
	b. $[\text{t}\acute{\text{u}}\text{tt}]$	*	*!	

An updated Level 2 constraint ranking, which now accounts for pluractional C_1 reduplication, is given in Table 4.33. This will be updated again in the following sections covering diminutive reduplication (Section 4.2.4.2.4) and glottalisation (Section 4.2.4.3).

Table 4.33: Lillooet Level 2 Grammar (Preliminary - Second Version)

A	PRWDGrWD	INT-V, DEP-C	*P/ə, TROCHEE	REALIZEMORPHEME, *FLOAT, MAX-FLOAT, INT·EDGE-(σ), PARSE- μ , * $-\Delta\text{FT} > \Delta\text{FT}$ ANCHOR _{FOOT} , OCP
B	*STRUC-FT,	DEP-V, INT-C	PREFIX/ σ HEADMATCH	
C	PARSE-SYLL			

⁶²MAX, defined in (i), penalises the deletion of input segments. MAX is ranked above OCP to protect identical consonants in the input; in other words, identical consonants may surface in the output outside of a reduplicative context. The ranking of OCP relative to MAX will vary across strata. H. Davis (p.c.) points out that OCP is violated with lexical suffixes, including *-tən* or *-mən* (e.g., *n-ŋ^wuýt-tən* ‘bed’ and *n-qóm-mən* ‘kettle’), but not with morphemes at later levels (e.g., the [t] in *naq^w-c-xi[t]-twal* ‘steal food from each other’ and *cix^w-xi[t]-táni=łkan* ‘I took it to them’ does not surface).

- (i) MAX: All segments in the input have a correspondent in the output. Assign a violation mark for every segment in the input that does not have a correspondent in the output.

(McCarthy & Prince, 1995, p.16)

4.2.4.2.3 Diminutive C₁ Reduplication

Diminutivity in Lillooet is marked with C₁ reduplication. The consonant immediately before a stressed vowel is doubled, resulting in the stressed vowel being surrounded by the fissioned consonants. The stressed vowel is usually by [ə], as shown in (249), which is a marked pattern because the language typically avoids stressed schwa due to *P/ə being highly ranked (see discussion in Section 4.2.4.2.2). Glottalisation, as shown in (249a–e), accompanies diminutive reduplication (as well as pluractional reduplication discussed in the previous section) and is discussed further and analysed in Section 4.2.4.3.

- (249) a. syə<ŷ>p (cf. *syap* ‘tree’)
 NMLZ-tree<DIM>
 ‘little tree, sapling’ (H. Davis & Mellesmoen, 2023, p.13)
- b. xwə<w̥>t̚ (cf. *xwat̚* ‘road, trail’)
 road<DIM>
 ‘little road, trail’ (H. Davis & Mellesmoen, 2023, p.13)
- c. t̚ə<t̚>pə̌ (cf. *t̚pə̌* ‘table, desk’)
 table<DIM>
 ‘little table, desk’ (H. Davis & Mellesmoen, 2023, p.13)
- d. lap̚lə̌<l̚>s (cf. *lap̚lə̌s* ‘plank, board’)
 plank<DIM>
 ‘little plank or board’ (H. Davis & Mellesmoen, 2023, p.15)
- e. kl̚ə̌<l̚>si (cf. *kl̚isi* ‘crazy’)
 crazy<DIM>
 ‘a little crazy’ (H. Davis & Mellesmoen, 2023, p.15)
- f. z̚ə̌<z̚>xt-an (cf. *z̚ə̌xtan* ‘lengthen s.t.’)
 long<DIM>-TR
 ‘lengthen s.t. a bit’ (H. Davis & Mellesmoen, 2023, p.13)
- g. c̚ə̌<c̚>s-əp (cf. *c̚ə̌sp* ‘got stretched’)
 stretch<DIM>-INCH
 ‘stretched a bit’ (H. Davis & Mellesmoen, 2023, p.15)

The examples in (250) show that the vowel is not replaced with a [ə] when the

doubled consonant is /ʔ/ (250a–d) or /h/ (250e).

- (250) a. ʔí<ʔ>mac (cf. ʔímac ‘grandchild’)
grandchild<DIM>
‘little grandchild’ (Van Eijk, 1997, p.60)
- b. ʔí<ʔ>ʔən (cf. ʔíʔən ‘eat’)
eat<DIM>
‘eat a bit’ (H. Davis & Mellesmoen, 2023, p.13)
- c. ʔú<ʔ>q^waʔ (cf. ʔúq^waʔ ‘drink’)
drink<DIM>
‘drink a bit’ (H. Davis & Mellesmoen, 2023, p.13)
- d. ʔí<ʔ>q̣-əṁ (cf. ʔíq̣əṁ ‘scrape’)
scrape<DIM>-MID
‘scrape hair off a bit’ (H. Davis & Mellesmoen, 2023, p.17)
- e. ʔəmh-á<h>əṁ (cf. ʔəmhám ‘be good at s.t.’)
good-MID<DIM>
‘to improve, get better, to get a bit better’ (H. Davis et al., in prep)

As with other types of reduplication in Lillooet, a reduplicated word may surface with an additional [ə] after the fissioned consonant, as shown in (251a–b). While reduplication creates the sequences that will require epenthesis at a later stratum for sonority purposes, this is independent of reduplication. For example, the unstressed [ə] in (251c) occurs in a position consistent with the general phonology of the language (i.e., codas should have falling sonority).

- (251) a. ʔxə́<ʔ>əṁ (cf. ʔxum ‘big’)
big<DIM>
‘a bit bigger’
- b. mə<m>əẉ (cf. maw ‘cat’)
cat<DIM>
‘kitten’

- c. zə<**ẓ**>xət (cf. *zaxt* ‘long’)
 long<DIM>
 ‘a bit longer’

(H. Davis & Mellesmoen, 2023, p.13)

The examples in (252) show that diminutive reduplication may cross into a stressed lexical suffix.

- (252) a. qɪ-ə<**ɪ̣**>kaʔ (cf. *qlákaʔ* ‘clumsy’)
 bad-hand<DIM>
 ‘a bit clumsy’
 b. s-pəq-ɪnə<**ɪ̣**>x (cf. *spəqɪnɪx* ‘swan’)
 NMLZ-white-excess<DIM>
 ‘little swan’ (U)
 c. s-kəɪ-ə<**ɪ̣**>caʔ (cf. *kəɪlɪcaʔ* ‘item of buckskin clothing’)
 NMLZ-buckskin-outer.cover<DIM>
 ‘small (item of) buckskin clothing’

(H. Davis & Mellesmoen, 2023, p.15)

4.2.4.2.4 Analysis of Diminutive C₁ Reduplication

There are two related aspects of the diminutive that require additional explanation: (1) what determines when a stressed vowel is replaced by [ə]?; and (2) why does the diminutive create/allow such a marked structure (i.e., stressed [ə])? The answer to the first question lies in the phonological grammar, where [əʔ] and [əh] are not permissible sequences within a syllable; the second lies in the lexical entry for the diminutive.

Diminutive C₁ reduplication must differ from pluractional C₁ reduplication in its lexical entry because by hypothesis both are subject to the same constraint ranking (the Level 2 grammar), but the stressed vowel matches the corresponding vowel in the input for pluractional C₁ reduplication, and becomes [ə] in diminutive C₁ reduplication. As we saw in Section 4.2.4.2.2, pluractional C₁ reduplication

happens following the affixation of a mora. Here I propose that diminutive C_1 reduplication happens following the affixation of a mora marked for lexical stress.

The lexical entry for diminutive reduplication is given in (253). The lexical entry bears the diacritic L2 (Level 2). The constricted glottis feature [c.g.] is discussed in Section 4.2.4.3 below.

(253) *Diminutive C_1 Reduplication:*

$$(\mu^{[\text{stress}]} [\text{c.g.}])_{\text{L2}} \Leftrightarrow \langle \text{DIM} \rangle$$

I adopt the privative feature [stress] to mark lexical stress in the input (de Lacy, 2020). Any segment or prosodic unit which has [stress] in the input will correspond to or be associated with a designated terminal element (DTE) in the output of that same stratum, unless blocked by other competing pressures (e.g., two elements marked for lexical stress).⁶³ More specifically, a unit “that bears [stress] in the output must also be [associated with] the ‘designated terminal element’ (DTE) of a prosodic word” de Lacy (2020, p.1). In other words, the diminutive mora in (253) must be the head mora of the head syllable of the head foot of a prosodic word.

The diminutive is represented as $\mu^{[\text{stress}]}$ in the input, while a superscript + is used to indicate the head mora in the output.⁶⁴ I diverge from de Lacy (2020) in proposing that the feature [stress] is not present in the output, which only encodes the prosodic structure. This allows for a distinction between prosodic structure built at an earlier stratum than the given one and lexical stress: an input [stress] feature will not be visible in the output, which means later strata cannot know that a morpheme was lexically stressed.

A prediction of this approach is that lexical stress must be assigned at the stratum where the lexical item is parsed into the word; later strata will not be able to distinguish between lexical stress and phonologically regular stress as both will have identical prosodic structure in the output. The head mora associated with the diminutive cannot be distinguished from any other head mora after it has been parsed into the phonological word.

The input and output of Level 1 and Level 2 are shown in (254) for $sp\acute{a}p\acute{l}\acute{a}x^w$

⁶³DTE refers to the lowest element in prosodic structure which is dominated exclusively by heads.

⁶⁴As mentioned in Chapter 3 (Section 3.2.2), the leftmost mora in a stressed syllable is the head mora.

‘to stick out a bit’ to demonstrate how [stress] and prosodic structure are related. The lexical stress feature is marked with the label [stress], while head moras are marked with a superscript +. The input to Level 1 has no prosodic structure; the output of the Level 1 grammar has prosodic structure because there must be a well-formed foot in the output. This is fed into Level 2, where the diminutive affix is added. The lexical entry for the diminutive is a mora with an associated [stress] feature. Crucially, the Level 1 output has no [stress] feature, just prosodic structure. Similarly, the [stress] feature is not present in the output of the Level 2 derivation.

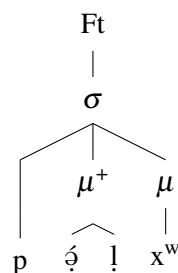
(254) *Derivation of spəp̤l̤əx^w ‘to stick out a bit’*

(cf. spə̤l̤x^w ‘to stick out’) (Van Eijk, 2013, p.14)

a. Level 1

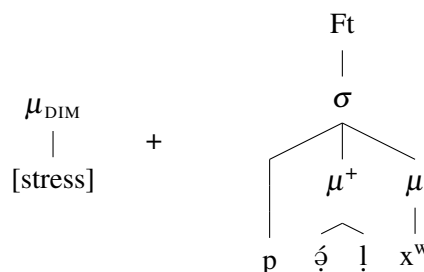
i. *Input:* pə̤l̤x^w

ii. *Output:* [pə̤̌l̤x^w]

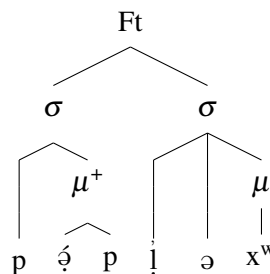


b. Level 2

i. *Input:* μ^[stress] + [pə̤̌l̤x^w]



ii. *Output:* [pə̤̌p̤̌.̌l̤ə̤x^w]



The mapping between lexical stress in the input and the appropriate prosodic structure in the output is checked by the constraint STRESSSTRESS, defined in (255a), which ensures a head in the output corresponds to a unit marked with the feature [stress] in the input. The constraint MAX- μ , defined in (255b), is also needed for diminutive reduplication. MAX- μ is violated when a mora present in the input is not present in the output.⁶⁵

- (255) a. STRESSSTRESS (STRESS²): Assign a violation mark for any prosodic unit with a [stress] feature in the input that does not correspond to a head in the output.
- b. MAX- μ : All moras in the input have a correspondent in the output. Assign a violation mark for every mora in the input that does not have a correspondent in the output.

The constraint CULMINATIVITY- μ^+ , defined in (256), is also needed to account for the loss of an input head mora, defined here as the head mora of a head foot. CULMINATIVITY- μ^+ is violated whenever a candidate has more than one head mora. CULMINATIVITY- μ^+ rules out words with two primary stresses (which would each be part of a head foot with a head mora), as well as a stressed bimoraic syllables where both moras are equally accented (two head moras).⁶⁶

- (256) CULMINATIVITY- μ^+ (CULMIN):
Every word has exactly one head foot with one head mora.

(modified from Alderete, 2001 and Hayes, 1995)

The tableau for *xwəwʔ* ‘little road, trail’ in (257) from (249) above shows that STRESSSTRESS and CULMIN must be ranked above MAX- μ : the head mora in the input is absent in the attested candidate (257a). The subscript numbers track

⁶⁵MAX- μ is needed in addition to MAX-FLOAT because MAX-FLOAT is not violated when a mora associated with segmental content in the input is deleted.

⁶⁶There are larger questions for future work concerning the relationship between head moras in the phonology and phonetic correlates of stress. Caldecott (2009, p.128) finds that acoustic correlates of primary and secondary stress in Lillooet were only significantly different for most speakers for /u/, not /a/ and /i/. As CULMINATIVITY- μ^+ requires a single mora to be the head of a prosodic word (which is in the head syllable of the head foot), the acoustic facts may indicate that phonetic implementation rules equally apply to head moras in non-head feet.

input moras. Candidate (257b) fatally violates CULMIN because both the input head mora and the affixed mora are stressed. Candidate (257c) retains all three input moras, but does not assign stress to the affixed mora, which fatally violates STRESSSTRESS. Subscript numbers are used to show the relationship between moras in the input and output; the + is used to mark a head mora (of a head foot).

$/\mu^{[\text{stress}]}_1 + [xw\acute{a}_{\mu+2}\dot{\mathfrak{t}}_{\mu 3}]/$		CULMIN	STRESS ²	MAX- μ	INT-C
(257)	a. $[xw\acute{\mathfrak{o}}_{\mu+1}\dot{\mathfrak{t}}_{\mu 3}]$			*	*
	b. $[xw\acute{a}_{\mu+2}\dot{\mathfrak{t}}_{\mu+1}\mathfrak{t}_{\mu 3}]$	*!			*
	c. $[xw\acute{a}_{\mu+2}\dot{\mathfrak{t}}_{\mu 1}\mathfrak{t}_{\mu 3}]$		*!		*

STRESSSTRESS and CULMIN ensure that the affixed mora will replace an existing head mora. They account for why diminutive reduplication yields a stressed syllable with a non-moraic nucleus when combined with core assumptions about stress in Lillooet.

Lillooet has trochaic feet where the leftmost syllable is the head syllable of a foot, which also means that the leftmost mora of the head syllable is the head mora. I modify the definition of the constraint TROCHEE, used first in the analysis of C_1C_2 reduplication in Section 4.2.3.2.2, to specifically refer to the mora. The revised constraint in (258) ensures that both the leftmost syllable and the leftmost mora in a word will be the most prominent. The constraint TROCHEE is violated whenever the head mora of a foot is not the leftmost. TROCHEE checks that the head is leftmost at both the level of the mora and of the syllable. If the leftmost mora is the head mora of a foot, the syllable dominating the head mora will also be the head syllable of the foot.

(258) TROCHEE (*revised*):

Feet are L-dominant at the level of the mora. Assign one violation per foot where the head is not the leftmost mora.

The tableau for $s\gamma\acute{\mathfrak{o}}\dot{\mathfrak{t}}p$ ‘little tree’ in (259) from (249a) above shows the constraint ranking needed to predict the shift from a stressed moraic vowel to a $[\mathfrak{o}]$.⁶⁷ The attested candidate (259a) replaces the stressed vowel with $[\mathfrak{o}]$, which violates *P/ $\mathfrak{o}$,

⁶⁷A number of constraints are omitted for space, including CULMIN and MAX- μ .

the constraint against a stressed non-moraic vowel. $*P/\partial$ must be ranked lower than DEP- V_μ (259b), STRESS² (259c), and TROCHEE (cf. 259d).

$/\mu^{[\text{stress}]}_1 + [\text{s}\acute{\text{y}}\acute{\text{a}}_{\mu+2}\text{p}_{\mu3}]/$		STRESS ²	TROCHEE	DEP- V_μ	$*P/\partial$	DEP-V	INT-C
(259)	a. $[\text{s}\acute{\text{y}}\acute{\text{x}}_{\mu+1}\text{p}_{\mu3}]$				*	*	*
	b. $[\text{s}\acute{\text{y}}\acute{\text{x}}_{\mu+1}\text{p}_{\mu3}]$			*!		*	
	c. $[\text{s}\acute{\text{y}}\acute{\text{a}}_{\mu+2}\acute{\text{y}}_{\mu1}\text{p}_{\mu3}]$	*!					*
	d. $[\text{s}\acute{\text{y}}\acute{\text{a}}_{\mu2}\acute{\text{y}}_{\mu+1}\text{p}_{\mu3}]$		*!				*

The combination of STRESS² and CULMIN ensures that the diminutive mora is the only stressed mora in the output, while TROCHEE ensures that this stressed mora is the leftmost in the foot. Associating the diminutive mora with the vowel whose input correspondent is already stressed is not a possibility, as reduplication would not happen and therefore the diminutive and non-diminutive words would be identical; having no clear exponent of the diminutive in the output violates REALIZE MORPHEME. Furthermore, the ranking of $*P/\partial$ between DEP- V_μ and DEP-V is crucial to ensure that a $/\partial/$ will be epenthesized, rather than a full vowel (which would be a better nucleus for a stressed syllable).

If the consonant that undergoes fission is an $/h/$ or $/\text{ʔ}/$, the stressed vowel is retained, rather than being replaced with schwa. The vowel-retaining forms show that STRESS² may be violated. A markedness constraint $*\partial C_{\text{ʔ/h}}]_\sigma$, given in (260), bans the sequence $[\partial\text{ʔ}]$ or $[\partial h]$ within the same syllable.

(260) $*\partial C_{\text{ʔ/h}}]_\sigma$ ($*\partial\text{ʔ}]_\sigma$): Sequences of $\partial\text{ʔ}$ or ∂h within a single syllable are not permitted.

The tableau in (261) for ‘little grandchild’ from (250a) above shows that STRESS² must be ranked beneath $*\partial\text{ʔ}]_\sigma$, REALIZE MORPHEME, and CULMIN to allow for the attested candidate in (261a), where the lexically stressed diminutive mora is not the head mora in the output, to win.

$/\mu^{[\text{stress}]}_1 + [\text{ʔ}\acute{\text{í}}_{\mu+2}\text{ma}_{\mu3}\text{c}_{\mu4}]/$		$*\partial\text{ʔ}]_\sigma$	CULMIN	RM	STRESS ²
(261)	a. $[\text{ʔ}\acute{\text{í}}_{\mu+2}\text{ʔ}_{\mu1}\text{ma}_{\mu3}\text{c}_{\mu4}]$				*
	b. $[\text{ʔ}\partial\text{ʔ}_{\mu+1}\text{ma}_{\mu3}\text{c}_{\mu4}]$	*!			
	c. $[\text{ʔ}\acute{\text{í}}_{\mu+2}\text{ma}_{\mu3}\text{c}_{\mu4}]$			*!	
	d. $[\text{ʔ}\acute{\text{í}}_{\mu+2}\text{ʔ}_{\mu+1}\text{ma}_{\mu3}\text{c}_{\mu4}]$		*!		

The constraint OCP, introduced in Section 4.2.4.2.2, bans identical adjacent consonants; OCP motivates the reduplication of the consonant before the stressed vowel in both types of Level 2 C₁ reduplication. Therefore, OCP must also be ranked above STRESS² to rule out a candidate which avoids violating *əʔ]_σ by reduplicating the consonant following the stressed vowel (262c).

$/\mu^{[stress]} + [\acute{\imath}mac]/$		OCP	*əʔ] _σ	STRESS ²
(262)	a. ʔíʔmac			*
	b. ʔəʔmac		*!	
	c. ʔəmmac	*!		

Table 4.34 provides a summary of the Level 2 constraint ranking needed for the derivation of all three Level 2 reduplicative morphemes in Lillooet. This will be updated one final time in Section 4.2.4.4 to include glottalisation.

Table 4.34: Lillooet Level 2 Grammar (Preliminary - Version 3)

A	*əʔ] _σ , OCP, CULMIN	PRWDGrWD	DEP-V _μ , TROCHEE	INT-V, DEP-C	REALIZE MORPHEME, *FLOAT, MAX-FLOAT, INT-EDGE-(σ), PARSE-μ, *-ΔFT > ΔFT, ANCHOR _{FOOT}
B	STRESS ²	*STRUC-FT	*P/ə	DEP-V, INT-C	
C	MAX-μ	PARSE-SYLL	PREFIX/σ, HEADMATCH		

4.2.4.3 Glottalisation and C₁ Reduplication

Both pluractional and diminutive C₁ reduplication trigger glottalisation on a sonorant. As the position of this glottalisation is the same for both processes, I analyse them together in this section. I first describe the distribution and position of glottalisation accompanying C₁ reduplication in Section 4.2.4.3.1. I revise both lexical entries (first stated in Section 4.2.4.2.2 and Section 4.2.4.2.4) to include a floating [c.g.] feature. I provide a phonological analysis that accounts for both types of C₁ reduplication in Lillooet in Section 4.2.4.3.2.

4.2.4.3.1 Distribution of Glottalisation in C_1 Reduplication

There is a process of glottalisation accompanying both Level 2 C_1 processes of reduplication: a sonorant consonant is glottalised. Examples with diminutive C_1 reduplication are provided in (263); examples with pluractional C_1 reduplication are provided in (264). The segment that may be glottalised includes any sonorant that falls between the position of stress and the right edge (at the output of Level 2), including the reduplicated consonant itself (e.g., in 263a). As shown in (263b–c) and (264b), it is only the rightmost sonorant that is glottalised when there is more than one available.⁶⁸

(263) Diminutive C_1 Reduplication

- a. s-mó<**m̥NMLZ-woman<DIM>
‘girl’ (H. Davis & Mellesmoen, 2023, p.13)**
- b. ɣzɔ<**zbig<DIM>
‘a bit bigger’ (H. Davis & Mellesmoen, 2023, p.13)**
- c. mó<**mcat<DIM>
‘kitten’ (H. Davis & Mellesmoen, 2023, p.13)**

(264) Pluractional C_1 Reduplication

- a. qí<**q̣run<PL>
‘run around’ (H. Davis & Mellesmoen, 2023, p.20)**
- b. q̣^wəɪ̃m̥~q̣^wm̥-í<**m̥wɔm̥q̣^wəɪ̃m̥ ‘get cramped in a small spot’)
PL~curl.up<PL>-AUT
‘curl up’ (H. Davis et al., in prep)**

When no sonorants are available for glottalisation, reduplication occurs without any glottalisation, as shown in (265) and (266): obstruents are not made ejective, nor does epenthesis occur. While /ʔ/ behaves as a sonorant in Lillooet, it is already

⁶⁸The example in (264c) has both C_1C_2 and C_1 reduplication marking plurality.

a glottal consonant. There is therefore no change resulting from the glottalisation associated with reduplication in (266a). A sonorant to the left of stress is also not a potential target for glottalisation: the /n/ in (265b) occurs before stress and is not glottalised.

(265) *Diminutive C₁ Reduplication*

- a. cʰó<c>s-əp (cf. cəsp ‘stretch’)
stretch<DIM>-INCH
‘stretched a bit’ (H. Davis & Mellesmoen, 2023, p.15)
- b. naɣ^wʰó<ɣ^w>t (cf. naɣ^wít ‘snake’)
snake<DIM>
‘worm’ (H. Davis & Mellesmoen, 2023, p.12)

(266) *Pluractional C₁ Reduplication*

- a. q^wut-á<t>kaʔ (cf. q^wutákaʔ ‘one hand’)
(one.)side<PL>-hand
‘one hand, use’ (H. Davis et al., in prep)
- b. ʰí<ɣ^w>q^wət (cf. ʰíq^wt ‘crackle or explode in fire’)
spark.flies<PL>
‘sparks fly up’ (H. Davis et al., in prep)

If a sonorant is already glottalised in the input, the [c.g.] feature is associated with another available sonorant between stress and the right edge, as shown in (267).

(267) *Diminutive C₁ Reduplication*

- nó<n̥>əm-p (cf. nəmp ‘get blind’)
blind<DIM>-INCH
‘get a little blind’ (H. Davis & Mellesmoen, 2023, p.15)

The domain for glottalisation matches that of reduplication: sonorants in lexical suffixes and the autonomous suffix *-ilx* may both be glottalised, as shown in (268) and (269), respectively.

- (268) a. s-kəɫ-ə-<ɪ>caʔ (cf. *skəɫicaʔ* ‘item of buckskin clothing’)
NMLZ-buckskin-outer.cover<DIM>
‘small item of buckskin clothing’ (H. Davis & Mellesmoen, 2023, p.14)
- b. qɫ-ə-<ɪ>kaʔ (*qɫəkaʔ* ‘clumsy’)
bad-hand<DIM>
‘a bit clumsy’ (H. Davis & Mellesmoen, 2023, p.14)
- c. cəx-m-á<m>lus (cf. *cəxəm* ‘sting’)
sting-MID-eye<PL>
‘to get blinded (from facing the sun or anything bright)’ (Van Eijk, 2013, p.438)
- d. x^wú<x^w>ɫ-xən (cf. *x^wúɫun* ‘to take smth. off’)
take.off<PL>-foot
‘barefooted’ (Van Eijk, 2013, p.438)
- (269) a. q^wəc-ə-<c>ləx (cf. *q^wəcɪlx* ‘move (oneself)’)
move-AUT<DIM>
‘wiggle’ (H. Davis & Mellesmoen, 2023, p.16)
- b. x^wəlp-í<p>ləx (cf. *x^wəlpɪlx* ‘turn around’)
turn-AUT<PL>
‘turn around and around’ (H. Davis & Mellesmoen, 2023, p.21)
- c. təɽ^w-í<ɽ^w>ləx (cf. *təɽ^wɪlx* ‘jump’)
bounce-AUT<PL>
‘jump up and down’ (H. Davis & Mellesmoen, 2023, p.21)

The middle suffix *-m* may also be glottalised in words with diminutive C₁ reduplication.⁶⁹ Examples are given in (270).

⁶⁹As with many processes of glottalisation, glottalisation on the middle marker can be variable. There are exceptions where the middle marker is not glottalised and places where the middle marker and a second sonorant are glottalised; an example of each is provided in (i).

- (i) a. ʔá<ʔ>əɫs-əm (cf. *ʔəɫsəm* ‘sick’)
sick<DIM>-MID
‘a little bit sick’ (Van Eijk, 1997, p.60)

- (270) a. $x^w\acute{o}<x^w>k-\acute{o}m$ (cf. $x^w\acute{ik}\acute{o}m$ ‘cut fish’)
 butcher<DIM>-MID
 ‘cut a bit of fish / cut small fish / small people cut fish’ (H. Davis & Mellesmoen, 2023, p.17)
- b. $\acute{r}i<\acute{r}>\acute{q}-\acute{o}m$ (cf. $\acute{r}i\acute{q}\acute{o}m$ ‘scrape’)
 scrape<DIM>-MID
 ‘scrape hair off a bit’ (H. Davis & Mellesmoen, 2023, p.17)

Unlike the middle marker, the active intransitivizer suffix *-xal* is outside of the domain for glottalisation: the /l/ in *-xal* is never glottalised with C₁ reduplication, as shown in (271).

- (271) a. $n-k\acute{a}<k>\acute{o}l-xal$ (cf. $nk\acute{a}lxal$ ‘follow s.o. or s.t.’)
 LOC-follow<PL>-A.INTR
 ‘follow s.o. around’
- b. $n-k\acute{a}<k>\acute{o}m-xal$ (cf. $nk\acute{a}m\acute{x}al$ ‘pick s.t. up’)
 LOC-pick.up<PL>-A.INTR
 ‘pick things up here and there’

(H. Davis & Mellesmoen, 2023, p.20)

Further, diminutive reduplication is typically avoided with *-xal*. (H. Davis & Mellesmoen, 2023, p.16) note that *-xal* can be replaced with the middle marker *-m* in these forms. The middle marker is glottalised in these forms, as shown in (272b). The middle marker must therefore be parsed in the word at the same stratum as C₁ reduplication; the active intransitivizer *-xal* is not parsed into the prosodic structure until the following stratum. This analysis is supported by the fact that these two intransitivizer suffixes can co-occur (i.e., they are not in strict complementary distribution); examples are given in (273).

- (272) a. $p\acute{a}k^w-x\acute{a}l=\acute{t}kan$ $\acute{r}i=\acute{c}\acute{x}^w\acute{u}t=a$
 pour.solid-A.INTR=1SG.SUBJ PL.DET=gravel=EXIS
 ‘I poured out the gravel.’
-
- b. $\acute{r}i<\acute{r}>\acute{c}\acute{o}m$ (cf. $\acute{r}i\acute{c}xal$ ‘to gnaw, intr.’)
 bite<PL>-MID
 ‘gnaw a hole, intr.’ (Van Eijk, 2013, p.328)

- b. pə<ᵑ>k^w-əᵑ=ɬkán ʔi=k^wik^wənʔ=a c̣x^wut
 pour.solid<DIM>-MID=1 SG.SUBJ PL.DET=few=EXIS gravel
 ‘I poured out a bit of gravel.’

(H. Davis & Mellesmoen, 2023, p.16)

- (273) mánx-əᵑ-xal (cf. *mánxəm* ‘smoke tobacco’)
 smoke-MID-A.INTR
 ‘give somebody a smoke’ (H. Davis et al., in prep)

Glottalisation with C₁ reduplication also does not cross into pronominal markers, as shown in (274).

- (274) a. ʔ^wə<ʔ^w>ʔət=kán (cf. ʔ^wuʔt ‘sleep’)
 sleep<DIM>=1 SG.SUBJ
 ‘I slept a little bit.’
 b. mə<ᵑ>caʔq=ɬkan (cf. *mícaʔq* ‘sit’)
 sit<DIM>=1 SG.SUBJ
 ‘I sat for awhile.’

(H. Davis & Mellesmoen, 2023, p.19)

Glottalisation is sometimes observed on transitivizers with C₁ reduplication, as shown in (275). Transitivizers are parsed at Level 3 and do not participate in reduplication. I analyse the glottalisation in (275) as an independent property associated with the transitivizer morpheme, rather than triggered by reduplication.⁷⁰ I follow Caldecott (2009) in assuming that the underlying form of the transitivizer is /n/, but this is deglottalised when left unfooted or in the head of a foot.⁷¹ I assume that glottalisation on the transitivizers with C₁ reduplication is a faithful realisation of the underlying form.

⁷⁰It is common to get glottalisation on the transitivizer independent of C₁ reduplication: e.g., *níkin* ‘cut it’ in Caldecott (2009, p.249).

⁷¹Caldecott (2009, p.251) also describes overapplication of deglottalisation; there is variation between lexical items and across different documentation that cannot be explained by regular phonological processes.

- (275) a. $c\acute{o}<c>k^w\text{-}\acute{a}n$ (cf. $c\acute{o}k^w\acute{a}n$ ‘pull s.t.’)
 pull<DIM>-TR
 ‘pull s.t. a bit’ (Van Eijk, 2013, p.70)
- b. $s\acute{u}<s>\acute{q}^w\text{-}\acute{a}n$ (cf. $s\acute{u}\acute{q}^w\acute{a}n$ ‘skin an animal’)
 skin<PL>-TR
 ‘skin pl. animals’ (H. Davis & Mellesmoen, 2023, p.20)

If glottalisation on the transitivizer is independent of C_1 reduplication, it should be possible to find forms with C_1 reduplication and the transitivizer where glottalisation falls on a sonorant to the left of the transitivizer. Examples are given in (276). If the transitivizer were in the domain for glottalisation, the /n/ should be glottalised in each example. Even in (276g), where /n/ is the only available sonorant that glottalisation could target, it is not glottalised.

- (276) a. $q^w\acute{o}<q^w>\acute{w}\text{-}\acute{a}n$ (cf. $q^w\acute{w}\acute{a}n$ ‘stick s.t. out’)
 stick.out<DIM>-TR
 ‘stick s.t out a bit’ (H. Davis & Mellesmoen, 2023, p.13)
- b. $z\acute{a}<\acute{z}>x\acute{t}\text{-}\acute{a}n$ (cf. $z\acute{a}x\acute{t}a\acute{n}$ ‘lengthen s.t.’)
 lengthen<DIM>-TR
 ‘lengthen s.t. a bit’ (H. Davis & Mellesmoen, 2023, p.13)
- c. $s\acute{a}w\sim s\acute{o}<s>\acute{w}\text{-}\acute{a}n$ (cf. $s\acute{a}w\acute{a}n$ ‘ask s.o. s.t.’)
 PL~ask<PL>-TR
 ‘ask s.o. a lot of questions’ (H. Davis & Mellesmoen, 2023, p.22)
- d. $k\acute{o}<k>\acute{l}\text{-}\acute{a}n$ (cf. $k\acute{o}\acute{l}\acute{a}n$ ‘to make a mark by scratching’)
 scratch-TR
 ‘to scratch it a little bit’ (Van Eijk, 2013, p.203)
- e. $y\acute{o}<y>p\text{-}\acute{a}n$ (cf. $y\acute{o}p\acute{a}n$ ‘to stand it up’)
 upright<DIM>-TR
 ‘to stand it up a little bit’ (Van Eijk, 2013, p.251)
- f. $p\acute{a}<p>x\text{-}us\acute{a}\text{?}\text{-}\acute{a}n$ (cf. $p\acute{a}x\acute{a}n$ ‘to scrape s.t.’)
 scrape<PL>-fruit-TR
 ‘to peel fruit’ (Van Eijk, 2013, p.22)

Finally, the presence of glottalisation is also somewhat variable. For example, if the rightmost sonorant is already glottalised, Caldecott (2009, p.14) states that this sonorant “will absorb the glottalisation”, rather than having the [c.g.] feature associate with a preceding sonorant. While this holds for some examples, such as in (277a), it is not observed in other forms, such as in (277b).

- (277) a. n-mí<**m**>ɫ-ən (cf. *miɫ* ‘get shared out or divided’)
 LOC-share<PL>-TR
 ‘share things out’ (H. Davis & Mellesmoen, 2023, p.20)
- b. n-čmí<**m̥**>x^w-laqin̥ (cf. *nčmíx^wlaqin̥* ‘peak on roof’)
 LOC-peak<PL>-top.of.head
 ‘peaked roof’ (Van Eijk, 2013, p.83)

The analysis proposed in the following section focuses on the places where glottalisation occurs, but I set aside a potential absorption or blocking effect of a glottalised sonorant or glottal stop at present.

4.2.4.3.2 Analysis of Glottalisation in C_1 Reduplication

The lexical entries for Level 2 C_1 reduplicative morphemes are provided in (278). Both include a floating [c.g.] feature. [c.g.] is a unary feature.

- (278) a. *Pluractional C_1 Reduplication*:
 $(\mu [c.g.])_{L2} \Leftrightarrow \langle PLU \rangle$
- b. *Diminutive C_1 Reduplication*:
 $(\mu^{[stress]} [c.g.])_{L2} \Leftrightarrow \langle DIM \rangle$

The constraint ranking established through Section 4.2.4.2 ensures that the correct consonant is reduplicated and that the vowel is replaced with schwa with diminutive reduplication. The analysis thus far accounts for how the mora is filled; the analysis requires the modification of an existing constraint and additional constraints to account for glottalisation associated with the [c.g.] feature.

*FLOAT makes reference to phonological features, as well as prosodic units, which means it plays a role in determining how a floating [c.g.] feature is realised. Additional constraints needed to account for the association of [c.g.] with a segment include $MAX_{[c.g.]}$ and $DEP-LINK_{[c.g.]}$, defined in (279). $MAX_{[c.g.]}$ is

violated if an input [c.g.] feature is not also present in the output, regardless of whether the feature is floating or associated with a segment.⁷² *DEP-LINK_[c.g.] is violated by any association lines between a feature and a segment that were not present in the input. This type of DEP-LINK constraint comes from Morén (1999, p.40), who adapted DEP-LINK-MORA[SEG] from NOSPREAD (see, e.g., McCarthy, 2000).

- (279) a. MAX_[c.g.]: Any [c.g.] feature must have an output correspondent.
 b. DEP-LINK_[c.g.] (DEPL_[c.g.]): Any association line between a segment and a phonological feature in the output must have an input correspondent.

The tableau in (280) for *q̣ṭq̣ṭiḷ* ‘run around’ in (264a) above shows that *DEP-LINK_[c.g.] must be ranked lower than *FLOAT and MAX_[c.g.] in order for the attested candidate (280a), with glottalisation of the final sonorant, to win. Candidates (280b) and (280c), where the [c.g.] is either absent or left floating, fatally violate MAX_[c.g.] and *FLOAT, respectively.

/μ [c.g.] + [q̣ṭq̣ṭiḷ]/		*FLOAT	MAX _[c.g.]	DEP-LINK _[c.g.]
(280)	a. [q̣ṭq̣ṭiḷ]			*
	b. [q̣ṭq̣ṭiḷ]		*!	
	c. [c.g] [q̣ṭq̣ṭiḷ]	*!		

Where more than one consonant is available, there are several factors that determine which segment is glottalised. First, only sonorant segments are glottalised: plain obstruents are never made into ejectives. Second, glottalisation always falls on a sonorant on the right side of the stressed vowel. Third, if there is more than one sonorant in the appropriate range, glottalisation will fall on the rightmost one.

I use the constraint DEP-LINK_{OBS[c.g.]} to select between candidates with the glottalisation of obstruents and sonorants.⁷³ This constraint, defined in (281), is

⁷²This is different from MAX-FLOAT, which only pertains to floating features or prosodic units. The difference between MAX-FLOAT and MAX_[c.g.] will be discussed in Section 4.2.4.3.4.

⁷³I assume that [c.g.] is associated with both glottalised sonorants and obstruents. However, there are differences in their behaviour which may reflect different feature specifications: e.g., Caldecott (2003) uses [creak] for glottalised sonorants, rather than [c.g.].

violated by the addition of an association line between a [c.g.] feature and an obstruent.

- (281) DEP-LINK_{OBS[C.G.]}(DEP_{LO[C.G.]}): Any association line between an obstruent and a phonological feature in the output must have an input correspondent.

The position of glottalisation can be predicted with reference to two additional constraints: *SON_[C.G.]/-△FT and INT·AGREE-[C.G.].⁷⁴ *SON_[C.G.]/-△FT, defined in (282), is violated by any glottalised sonorant outside of a head foot. The definition of INT·AGREE-[C.G.] is modeled off of INT·EDGE-(σ), which was first introduced in Section 4.2.4.1.2. Instead of defining a domain via the EDGE-(domain) format, I use the template AGREE-[F] for constraints satisfied when fissioned segments share a featural specification with respect to [c.g.]. INT·AGREE-[C.G.] is violated when two output segments corresponding to the same input segment do not share the same featural specification as each other.

- (282) a. *SON_[C.G.]/-△FT: Glottalised sonorants are only permitted in head feet.
 b. INT·AGREE-[C.G.] (INT·AGR-[C.G.]): If two output segments correspond to the same input segment, then either both or neither are specified for [c.g.].

Having DEP-LINK_{OBS[C.G.]} and *SON_[C.G.]/-△FT ranked above MAX_[C.G.] correctly predicts that glottalisation will not occur if there are no sonorants in the head foot. The tableau in (283) for *naχ^wéχ^wt* ‘worm’ (from *naχ^wít* ‘snake’) shows that the attested form with no glottalisation in (283a) wins over candidates which glottalise a sonorant earlier in the word (283b), glottalise an obstruent (283c), or leave the [c.g.] feature floating in the output (283d).⁷⁵

⁷⁴*SON_[C.G.]/-△FT and its ranking relative to other Level 2 constraints in this section makes a prediction about the domain for C₁ glottalisation that diverges from previous descriptions (e.g., H. Davis & Mellesmoen, 2023): glottalisation will fall on the rightmost sonorant in a foot, not the rightmost sonorant in a stem or word. See Section 4.2.4.3.3 for an evaluation of this prediction.

⁷⁵There is no /h/ in the phonemic inventory of Lillooet. However, the identity of a specific stop is not important here: DEP_{LO[C.G.]} rules out the glottalisation of *any* obstruent, regardless of whether it yields a phoneme in the language.

(283)	/μ [c.g.] + na[$\dot{x}^w \acute{x}^w t$]/	*FLOAT	SON _[c.g.] /-ΔFT	DEPL _{O[c.g.]}	MAX _[c.g.]	DEPL _[c.g.]
	a. na[$\dot{x}^w \acute{x}^w t$]				*	
	b. na[$\dot{x}^w \acute{x}^w t$]		*!			*
	c. na[$\dot{x}^w \acute{x}^w t$]			*!		*
	d. [c.g.] na[$\dot{x}^w \acute{x}^w t$]	*!				

INT·AGREE-[C.G.] determines which consonant is glottalised when there is more than one available sonorant. This is shown in (284) with the word *nċəxmámłus* ‘blinded’ from (268) above, which has pluractional reduplication: the ranking motivates selecting a sonorant further into the word than the fissioned sonorant.⁷⁶ The attested candidate in (284a) wins over a candidate which associates the [c.g.] feature with one of the fissioned segments (284b). A candidate which associates a [c.g.] feature with both of the fissioned sonorants receives a second and fatal violation of DEPL_[c.g.].⁷⁷

(284)	/μ [c.g.] + [ċəxm] + alus/	INT·AGREE-[C.G.]	MAX _[c.g.]	DEPL _[c.g.]
	a. [ċəx̣][mámłus]			*
	b. [ċəx̣][mámłus]	*!		*
	c. [ċəx̣][mámłus]			**!

The constraint INT·AGREE-[C.G.] must be ranked below DEP-LINK_{OBS[c.g.]} and MAX_[c.g.] to ensure that the fissioned consonant is associated with the [c.g.] feature when there are no other options. For example, in the diminutive word *nċəñəñp* ‘get a little blind’, the only other sonorant is underlyingly glottalised. The constraint *ONSET_{σ[c.g.]} is also required to rule out an additional candidate which satisfies INT·AGREE-[C.G.] by glottalising both of the fissioned consonants. This constraint is violated by any candidate with a [c.g.] feature associated with an onset consonant in a stressed syllable.

(285) *ONSET_{σ[c.g.]}: A segment in the onset of a stressed syllable is not associated with a [c.g.] feature.

The tableau in (286) for example (267) above shows that ranking *ONSET_{σ[c.g.]} above INT·AGREE-[C.G.] correctly rules out candidate (286a), which glottalises

⁷⁶The prefix *n-* is not included in the input because it is assumed to be affixed later in the derivation.

⁷⁷*FLOAT, SON_[c.g.]/-ΔFT, and DEPL_{O[c.g.]} are omitted for space, but are ranked above MAX_[c.g.].

two sonorants. Candidate (286b), which has associated [c.g.] with an obstruent, incurs a fatal violation under DEP-LINK_{OBS}[C.G.]. Candidate (286c) deletes the floating feature, which fatally violates MAX_[C.G.]. Candidate (286d), the anticipated form, wins.⁷⁸

(286)

/μ [c.g.] + [nə́mp]/	*ONS _σ [C.G.]	DEPL _O [C.G.]	MAX _[C.G.]	INT·AGR-[C.G.]	DEPL _[C.G.]
a. [nə́nmp]	*!				**
b. [nə́nmp]			*!		
c. [nə́nmp]		*!			*
d. [nə́nmp]				*	*

In order to protect sonorants glottalised in the underlying form, I assume that both *SON_[C.G.]/−ΔFT and *ONSET_σ[C.G.] are ranked beneath IDENT-[C.G.], a constraint violated when a glottalised segment in the input is not also glottalised in the output.

The constraints and ranking established in this section account for sonorant glottalisation with both types of C₁ reduplication.

4.2.4.3.3 *Is the domain of [c.g.] the foot or word?*

The constraint-based analysis given in Section 4.2.4.3.2 makes an empirical prediction: the relevant domain for C₁ glottalisation is the foot, not the word. In Section 4.2.4.3.1, I stated that sonorants available to host the floating [c.g.] fall between the stressed syllable and the right edge of the word (the output of Level 2). This is achieved in the analysis by ranking *SON_[C.G.]/−ΔFT high enough to limit [c.g.] to sonorants in the head foot, which rules out glottalisation to the left of the stressed syllable. However, *SON_[C.G.]/−ΔFT also defines a limit on the right edge: the furthest [c.g.] can go is the last segment in the syllable following the stressed one, provided that the head foot is a bisyllabic foot where the second syllable has a sonorant in its coda.

Caldecott (2009, p.14) proposes that glottalisation will generally surface on the rightmost sonorant, defining the prosodic word as its domain. H. Davis (in prep) proposes that the domain for C₁ glottalisation is the morphological stem; glottalisation will fall on the rightmost sonorant in the stem, which includes the

⁷⁸The attested form *nə́nə́mp* ‘get a little blind’ will be generated at a later stratum following the sonority-motivated epenthesis of [ə].

root and any aspectual, lexical, abstract, and in/transitive suffixes.⁷⁹ In H. Davis and Mellesmoen (2023, p.14), the relevant domain for glottalisation is narrowed to exclude in/transitive suffixes. The three previous descriptions and my analysis in Section 4.2.4.3.2 converge on the fact that glottalisation is rightmost in *some* domain, though the size of this domain is unclear. The four options are summarised in (287).

- (287) a. Rightmost sonorant in word (Caldecott, 2009)
 b. Rightmost sonorant in stem (H. Davis, in prep)
 c. Rightmost sonorant in stem, excluding intransitive and transitive suffixes (H. Davis & Mellesmoen, 2023)
 d. Rightmost in the foot containing the head mora of the word (this dissertation)

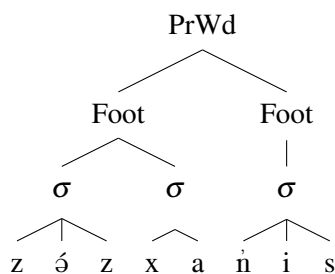
The fact that C₁ glottalisation does not cross into pronominal affixes rules out the full prosodic word as the domain for glottalisation (e.g., 288a), while the fact that C₁ glottalisation only affects certain (in)transitivity markers (i.e., the middle *-m*) and not others (i.e. the active intransitivizer *-xal* or transitivizer *-n*) rules out the morphological stem 288b–c). The only viable explanation is phonological: glottalisation targets the rightmost sonorant of the head foot in the output of Level 2.

- (288) a. $\text{ʔ}^w\text{ə} < \text{ʔ}^w > \text{yət} = \text{kán}$ (cf. $\text{ʔ}^w\text{u} \text{y} \text{t}$ ‘sleep’)
 sleep <DIM> = 1SG.SUBJ
 ‘I slept a little bit.’ (H. Davis & Mellesmoen, 2023, p.19)
 b. $\text{q}^w\text{ə} < \text{q}^w > \text{w-ən}$ (cf. $\text{q}^w \text{í} \text{win}$ ‘stick s.t. out’)
 stick.out <DIM> -TR
 ‘stick s.t out a bit’ (H. Davis & Mellesmoen, 2023, p.13)
 c. $\text{n-ká} < \text{k} > \text{əl-xal}$ (cf. nkálxal) ‘follow s.o. or s.t.’
 LOC-follow <PL> -A.INTR
 ‘follow s.o. around’ (H. Davis & Mellesmoen, 2023, p.20)

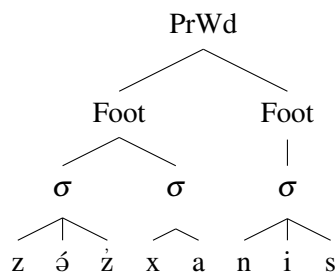
⁷⁹There are 11 abstract suffixes that are “somewhat aspectual in character” and “express notions such as ‘real’, ‘the very one’, ‘about to do smt.’, ‘willing to do smt.’, [and] ‘for fun, playfully’” (Van Eijk, 1997, p.102).

Lillooet has trochaic feet and stress typically falls on the penultimate syllable, which means that the right boundary of the head foot and the right-edge of the word will often align. This tendency is even stronger when the relevant domain is limited to the root (most often a single syllable) and lexical/abstract suffixes. In fact, previous documentation is consistent with either a morpheme edge-based domain, as proposed by H. Davis and Mellesmoen (2023), or a (foot) prosody-based domain, as I predict. The crucial test case is a word with a syllable containing a sonorant that falls outside the head foot. Specifically, the relevant non-reduplicated words must have at least three syllables with stress not falling on the last two syllables. As an example, if H. Davis and Mellesmoen (2023) are correct, the diminutive form of the word *záxanis* ‘long board’ will be *zézxanís*. If the foot-based approach is correct, the diminutive will be *zéẓxanis*. The two possibilities are summarised in (289): [n̥] is outside of the head foot; [ẓ] is within it.⁸⁰

(289) a. Edge-Based:



b. Prosody-Based:



The following examples, provided in (290), were elicited by Henry Davis to differentiate between a morpheme-based and foot-based domain for glottalisation. Glottalisation does not occur on the rightmost sonorant if the sonorant is not part of the stressed syllable or the syllable immediately following. Note that stress

⁸⁰I exclude moras in these diagrams for simplicity.

shifts in (290b–c) with the addition of the EXIS =a, which is affixed at a later stratum.

(290) a. *záxanis* ‘long board’

xá^h-mín=łkan k^wu=zó<^hz>x-anis laplás
want-RLT=1 SG.SUBJ DET=long<DIM>-board board

‘I want a slightly longer board.’ (CA, sf/repeated)

b. *łux^walmix^w* ‘indigenous person’

pzán=łkan ta=łu<ł>x^walmíx^w=a
meet+TR=1 SG.SUBJ DET=person<DIM>=EXIS

‘I met a little person.’ (CA, vf)

c. *x^wúk^wawíł* ‘haul-vehicle’

təq-álk-ən=łkan ta=x^wə<x^w>k^w-awíł=a
touch-string-TR=1 SG.SUBJ DET=pull-vehicle=EXIS

‘I drove a little skidder.’ (CA, vf)

The elicited data in (290) shows that the domain for C₁ glottalisation is the foot, as predicted by the high ranking of *SON_[c.g.]/–ΔFT established in Section 4.2.4.3.2.

4.2.4.3.4 Unifying C₁ and Glottalisation Analyses

The analysis of glottalisation proposed in Section 4.2.4.3.2 focused solely on the [c.g.] component of the lexical entry, abstracting away from how this might interact with the analysis of C₁ reduplication in Section 4.2.4.2, which focused on how a μ is filled. I conclude the analysis of Level 2 C₁ reduplication in Lillooet by demonstrating that the glottalisation analysis is compatible with the analysis of reduplication proposed in Section 4.2.4.2.

One potential way to fill the mora and to associate segmental content with the [c.g.] feature would be epenthesis of a glottal stop into coda position, which could satisfy both parts of the input for the diminutive or pluractional affix with a single segment. The fact that this does not occur means that DEP-C must be ranked high enough to rule out epenthesis of a consonant.

The tableau in (291) for example (264) above shows that the attested candidate *qíqíłil* ‘run around’ in (291a), which has both reduplication and glottalisation, is

selected over one with glottal stop epenthesis (291b), as well as over those where either the mora or feature is not present in the output (291c–e). The constraint *FLOAT is omitted from this tableau for space, but rules out candidates which leave either μ or [c.g.] floating in the output.

(291)	/μ [c.g.] + [qíʔíł]/	MAX _μ	DEP-C	MAX _[c.g.]	DEP-LINK _[c.g.]	INT-C
	a. [qíqíł]				*	*
	b. [qíʔíł]		*!			
	c. [qíqíł]			*!		*
	d. [qíł]	*!			*	
	e. [qíł]	*!		*		

The [c.g.] feature is not present in the output when there is no available sonorant, which means DEP-LINK_{O[c.g.]} must be ranked above MAX_[c.g.] and MAX-FLOAT. This ranking ensures that [c.g.] will not associate with an obstruent. Having both DEP-C and DEP-LINK_{O[c.g.]} above MAX_[c.g.] correctly yields the attested candidate $\lambda\acute{i}\lambda q^w\acute{o}t$ ‘sparks fly up’, shown in (292a) for example (266) above.

(292)	/μ [c.g.] + (λíq ^w t)/	DEP _L _{O[c.g.]}	MAX _μ	DEP-C	MAX _{FLT}	MAX _[c.g.]	DEP _L _[c.g.]	INT-C
	a. λíλq ^w ót				*	*		*
	b. λíλq ^w ót	*!					*	*
	c. λíʔq ^w ót			*!				

The proposed constraint ranking can account for both the floating μ and the [c.g.] associated with C₁ diminutive and pluractional reduplication at Level 2.

4.2.4.4 Summary of Level 2 Grammar

Level 2 reduplication in Lillooet includes C₂ change-of-state reduplication, C₁ pluractional reduplication, and C₁ diminutive reduplication. Level 2 reduplication differs from Level 1 reduplication in how constraints are ranked. First, there are more constraints relevant for predicting the attested form and position of reduplicative morphemes at Level 2 than at Level 1 (compare 293a and 293b).

(293) a. *Lillooet Level 1 Grammar:*

A	DEP-C	INT-V, SONORITY	TROCHEE, PARSE-SYLL	*FLOAT, REALIZEMORPH, MAX-FLOAT, PREFIX/ σ , DEP-V $_{\mu}$
B	INT-C	DEP-V	*STRUC-FT	

b. *Lillooet Level 2 Grammar:*

A	* σ ?] σ , OCP, CULMIN	PRWDGRWD	DEP-V $_{\mu}$, TROCHEE	INT-V, DEP-C	IDENT[C.G.]	REALIZEM, ANCHOR _{FOOT} INT-EDGE-(σ), PARSE- μ , * Δ FT > Δ FT, *FLOAT, DEPL _O [C.G.]
B	STRESS ²	*STRUC-FT	*P/ σ	DEP-V, INT-C	*SON _[C.G.] /- Δ FT, *ONSET $_{\sigma}$ [C.G.]	
C	MAX- μ	PARSE- σ	PREFIX/ σ , HDMATCH			
D	MAX _[C.G.] , MAX-FLOAT					
E	INT-AGR-[C.G.]					
F	DEPL _[C.G.]					

Second, Level 2 reduplication processes in Lillooet are sensitive to prosodic structure. This pattern emerges because constraints like HEADMATCH and ANCHORL are ranked higher in the Level 2 grammar than they would be in the Level 1 grammar to ensure that prosodic structure built at Level 1 is retained where possible; in other words, it is better to integrate a floating prosodic unit into an existing foot than build a new one on the edge of the word.

While Level 2 reduplication in Lillooet cannot be analysed without also considering stress assignment, Level 1 processes do not require reference to prosodic structure. A key difference in the constraints at Level 1 and Level 2 is that PREFIX/ σ is ranked much lower at Level 2, allowing for C₂ reduplication to appear on the surface as an infix, rather than a prefix. With a high ranking of PREFIX constraints at Level 1, reduplication defaults to edge-aligned, since the position of stress does not matter. When PREFIX constraints have a low ranking

relative to other constraints, other properties, such as sensitivity to stress, may emerge.

At Level 2, the affixed mora (diminutive or pluractional) or syllable (change-of-state) are incorporated into existing syllables or feet to avoid incurring additional violations of *STRUC-FT. The position of C₁ reduplication is also motivated by * $-\Delta FT \succ \Delta FT$, which prefers adding an additional mora to the head of a foot over adding weight to a foot non-head. Level 2 reduplication processes are thus stress-adjacent, unlike Level 1 processes, which are adjacent to the left edge.

As argued in Chapter 3 (Section 3.5.2.3) and shown in Table 4.35, PARSE_{AFFIX} must be ranked higher than AFFIX_{L2} at Level 2. PARSE_{AFFIX} motivates parsing affixes into the prosodic word; AFFIX_{L2} is violated when a Level 2 affix is parsed into the word. When PARSE_{AFFIX} is ranked above AFFIX_{L2}, all affixes marked with an L2 diacritic will be parsed into the word; the inverse order of the constraints is required for the Level 1 grammar because Level 2 affixes are left unparsed.

Table 4.35: Level 2 Affix Constraints

	Level 1	Level 2
A	AFFIX _{L2}	PARSE _{AFFIX}
B	PARSE _{AFFIX}	AFFIX _{L2}

Level 2 reduplicative morphemes also partially differ in underlying (input) representations from those at Level 1. The lexical entry for change-of-state C₂ reduplication is an affixed syllable, identical to initial C₁ reduplication at Level 1, but the lexical entries for the diminutive and pluractional reduplicative morphemes consist of a mora and a [c.g] feature. The difference between the latter two is that the diminutive mora carries a [stress] feature, while the pluractional mora does not. The inventory of Lillooet reduplicative morphemes is given in (294).

(294) a. *Lillooet Level 1 Reduplication:*

- i. $\sigma \Leftrightarrow \langle \text{INIT.C}_1 \rangle$
- ii. $\sigma_{N+\mu} \Leftrightarrow \langle \text{C}_1 \text{C}_2.\text{PL} \rangle$
- iii. $\sigma_{N+\mu} \Leftrightarrow \langle \text{PDEN} \rangle$

b. *Lillooet Level 2 Reduplication*:

- i. $\sigma_{L2} \Leftrightarrow \langle \text{COS} \rangle$
- ii. $(\mu^{\text{[stress]}} [\text{c.g.}])_{L2} \Leftrightarrow \langle \text{DIM} \rangle$
- iii. $(\mu [\text{c.g.}])_{L2} \Leftrightarrow \langle \text{C}_1.\text{PL} \rangle$

4.2.5 C_2^2 Reduplication (Triplication)

In addition to Level 1 and Level 2 reduplication, there is another type of reduplication that occurs late in the derivation (Level 2+n): C_2^2 reduplication (triplication). While this pattern was first introduced in Chapter 2 (Section 2.6.4) as an apparent case of two instances of C_2 reduplication, I reclassify it as a distinct reduplicative morpheme in this section.⁸¹

I first provide examples and generalisations in Section 4.2.5.1. A full formal analysis is not provided for this type of reduplication due to high amounts of variation related to the vowel source and quality, but the necessary components are explored, including its lexical entry (Section 4.2.5.2) and level (Section 4.2.5.3).

4.2.5.1 Form of C_2^2 Reduplication (Triplication)

Words with C_2^2 reduplication are associated with semelfactive events (i.e., events with a short duration that are often repeated), often those that involve sensory information about vision (flashing lights) or sound (noises). There are two copies of the consonant following the stressed vowel in C_2^2 reduplication, which resembles two instances of C_2 reduplication. While each copy is preceded by a vowel, it may be a vowel of the same quality as the stressed one, as shown in (295a-b), which is unlike C_2 reduplication. Words with C_2^2 reduplication may also surface with [ə] as the first vowel and [a] as the second, as shown in (295c-d). The [ə] is epenthetic, while the source of the [a] is unclear.⁸²

- (295) a. $\text{p}l\langle \text{ix}^w \text{ix}^w \rangle \text{ix}^w$ (cf. $\text{kap}l\acute{\text{ix}}^w a$ ‘to boil over suddenly’)
 overflow <TRIP>
 ‘to keep overflowing’ (Van Eijk, 2013, p.32)

⁸¹See Chapter 2 (Section 2.2.3.2.2) for a discussion of forms that appear to have two instances of C_1C_2 reduplication.

⁸²It is possible that [a] comes from a process of ablaut.

- b. q<əḵ^wəḵ^w>əḵ^w-c (cf. qəḵ^wcánəḵ 'mouth is shivering')
get.slammed<TRIP>-mouth
'rattle one's teeth' (Van Eijk, 2013, p.263)
- c. ʔ<ələl>íl (cf. ʔlín 'to sprinkle s.t.')
sprinkle<TRIP>
'to keep sprinkling' (Van Eijk, 1997, p.56)
- d. p<əṭəṭ>əṭ (root: p(ə)t- 'bubble')
bubble<TRIP>
'to make a bubbling, gurgling noise' (Van Eijk, 2013, p.4)

I have not been able to identify any phonological factors that condition or distinguish between the two vowel patterns. The vowel patterns are further complicated by variation. The forms in (296) show variation in whether the [a] is present (296a-b), as well as in which position(s) it is recorded.

- (296)
- a. ʔ<ənən>ən ~ ʔ<ənən>án (cf. ʔənp 'sound of things vibrating')
vibrate<TRIP>
'sound of things vibrating (e.g., a logging truck passes by)' (H. Davis et al., in prep)
 - b. ʔ^w<əməm>ám ~ ʔ^w<əməm>əm ~ ʔ^w<əmam>əm
quick<TRIP>
'hurry' (cf. ʔ^wəm 'quick') (H. Davis et al., in prep)
 - c. ʔ^w<ələl>əl ~ ʔ^w<ələl>ál (cf. kaʔ^wála 'get lit')
light<TRIP>
'be flickering or blinking' (H. Davis et al., in prep)

Further elicitation is needed to explore the nature of this variation and whether there are any categorical patterns that predict when and where [a] occurs.

Words with C₂² reduplication are also notable for their prosody: they show invariable word-final stress, which is not sensitive to vowel weight. Additionally, vowels are not copied in any of the other reduplicative patterns, while here they are. However, given the variability in vowel identity and position in the C₂² reduplication, it is not possible to present a full analysis of this type of reduplication

at present. Instead, I will provide a brief sketch of what the lexical entry and grammar must include in order for C_2^2 reduplication to be possible.

4.2.5.2 Input for C_2^2 Reduplication (Triplication)

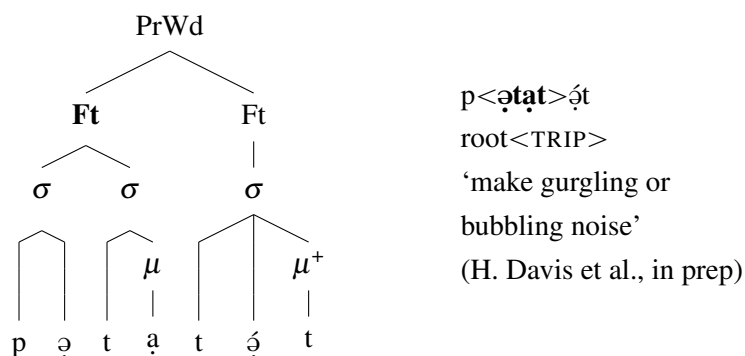
I propose that the input for C_2^2 reduplication is an affixed foot, as shown in (297). At present, I leave the question open as to whether this foot has further specification (e.g., two syllables or a fixed /a/). I use the diacritic L2+n for Level 2+n as a placeholder: further work on the phonology of post-Level 2 affixes is needed to determine the exact level of triplication.

(297) *Triplicating (C_2^2) Reduplication:*

$$Ft_{L2+n} \Leftrightarrow \langle \text{TRIP} \rangle$$

The foot introduced by the triplicating morpheme is bolded in (298).

(298) C_2^2 Reduplication



While the presence of [a] in the C_2^2 triplicated forms is variable, it does mean that the foot has moraic weight. When both of the vowels associated with the foot were [ə], the foot has no moraic weight.

The alternative analyses to positing a single reduplicative morpheme are that C_2^2 reduplication involves two instances of C_2 reduplication (multiple reduplication), or that the reduplicated forms are lexicalised.

Multiple reduplication can be ruled out because if these forms involved the addition of the same reduplicative morpheme twice, this should be reflected in the meaning. For example, when applying C_2 reduplication to a form already

marked with C_2 reduplication, compositional semantics would predict change-of-state semantics applied to a change of state. This is not the case: C_2^2 reduplication is associated with repetition of a semelfactive event, not a change of state. There is also at least one documented form, provided in (299), which has C_2 reduplication in addition to the C_2^2 pattern. The copied consonant in the regular C_2 reduplication is different from the consonant tripled as part of the C_2^2 process.

- (299) $qəl<\dot{l}><\dot{\alpha}x^w\dot{\alpha}x^w>\acute{\alpha}x^w$ (root: $q\dot{l}x^w$ - ‘hurry’)
 hurry<COS><TRIP>
 ‘rushing to get s.t. done, rushing to get ready’ (H. Davis et al., in prep)

The example in (299) suggests that either the input or the grammar, or both, differ between C_2 reduplication and triplication. In other words, C_2^2 cannot be analysed as C_2 reduplication added to words that already have C_2 reduplication.

A second possibility is that these examples are fully lexicalised. However, it should be noted that while these forms lack phonological regularity in their vowels, they have a consistent meaning. A better interpretation is that these forms are limited in occurrence because they are restricted to a narrow semantic domain, rather than due to lack of productivity.⁸³

C_2^2 reduplication is best analysed as a standalone reduplicative morpheme which affixes an empty foot.

4.2.5.3 C_2^2 Reduplication (Triplication) at a Later Stratum

Words with C_2^2 reduplication have a number of phonological properties that suggest that the affixed foot is not filled at Level 1 or Level 2.

First, the grammars proposed for Level 1 and Level 2 do not permit copying vowels. However, with a full vowel root, the root vowel is sometimes tripled, as shown in (300).

- (300) $\dot{p}l<\dot{i}x^w\dot{i}x^w>\acute{i}x^w$ (cf. $k\dot{a}p\dot{l}\acute{i}x^w$ ‘to boil over suddenly’)
 overflow<TRIP>
 ‘to keep overflowing’ (Van Eijk, 2013, p.32)

⁸³Henry Davis (p.c.) suspects that new words with this type of reduplication could be elicited provided that the lexical item was appropriate.

If triplication occurred as a Level 1 or Level 2 process, [ə] would always be predicted by the Level 2 grammar proposed in Section 4.2.4. In addition, the invariable presence of a vowel in these forms marks a departure from reduplication at the early strata, which may lose an epenthetic [ə] later in the derivation. For example, the [ə] in C₂ reduplication is dropped in (301). In comparison, the triplicated forms do not lose their vowel. This suggests that triplication occurs considerably later in the derivation, such that [ə] is retained.

- (301) qa<x^w>x^w-áwł (cf. qáx^wəx^w ‘break’)
 break<COS>-conveyance
 ‘break down (car)’ (Van Eijk, 2013, p.263)

The stress pattern associated with triplication is also notable. Stress remains on the rightmost of the three syllables in (302), even when there is a better candidate for stress (i.e., moraic [a]). This suggests that stress assigned earlier in the derivation is retained and that the affixation of the foot does not cause stress to shift, as might be expected if the foot were parsed into the word earlier in the derivation.

- (302) ʔ^w<ə́ləl>ə́l (cf. ʔ^wə́ləm ‘a bubble’)
 bubble<TRIP>
 ‘bubbles come up when someone dives in’ (Van Eijk, 2013, p.336)

While I do not offer a constraint-based analysis of the patterns, I have provided a sketch of the components of a solution that would best fit the available data and that is consistent with the larger analysis of Lillooet reduplication considered in this chapter. In Chapter 5 (Section 5.4.6), I will compare this type of reduplication with multiple reduplication (i.e., forms exhibiting more than one type of reduplication).

4.2.6 Summary of Reduplication in Lillooet

Reduplication in Lillooet occurs at three different strata, which each has a unique constraint ranking. I have proposed constraint rankings to account for Level 1 and Level 2 reduplication in Sections 4.2.3 and 4.2.4, respectively. In this section, I directly compare the Level 1 and Level 2 rankings.

The constraints relevant to parsing affixes and the constraints motivating reduplication and glottalisation are compared in Table 4.36.

Table 4.36: Constraints Motivating Parsing and Filling

	Level 1	Level 2
High	AFFIX _{L2} , REALIZEMORPHEME	PARSE _{AFFIX} , REALIZEMORPHEME, *FLOAT
Low	PARSE _{AFFIX} , *FLOAT	AFFIX _{L2}

The crucial rankings of the INTEGRITY and DEP constraints are the same across Level 1 and Level 2 in Lillooet: consonants are copied and vowels are epenthesized.

Table 4.37: Fission and Epenthesis Constraints

	Level 1	Level 2
High	DEP-C, INT-V	DEP-C, INT-V
Low	DEP-V, INT-C	DEP-V, INT-C

The position of copied consonants relates to how highly PREFIX/ σ (as well as PREFIX and PREFIX/ μ) and INT·EDGE-(σ) are ranked. At Level 1, all reduplication occurs on the left edge, which means the PREFIX constraints have a high ranking. At Level 2, reduplication is infix, which means the PREFIX constraints are ranked low. At Level 1, the fissioned consonants are in separate syllables, which means INT·EDGE-(σ) is low; at Level 2, the copied consonants are always in the same syllable, which means INT·EDGE-(σ) is high.

Table 4.38: Position of Fissioned Consonants

	Level 1	Level 2
High	PREFIX/ σ , PREFIX/ μ , PREFIX	INT·EDGE-(σ)
Low	INT·EDGE-(σ)	PREFIX/ σ , PREFIX/ μ , PREFIX

The constraint SONORITY was used in the analysis of Level 1 to abstract away from prosodic constraints that motivated building a foot but were not otherwise significant to the analysis. Table 4.39 shows how the constraints used in Level 2 must be ranked at Level 1 to select candidates with a single foot. The main difference between Level 1 and Level 2 is that the constraints dealing with prosodic structure in the input (HEADMATCH and ANCHOR_{FOOT}) are not crucially ranked at Level 1 because no segments will have been parsed into feet yet. The rankings of *- Δ FT > Δ FT and PARSE- μ relative to other constraints are also unclear at Level 1.

Table 4.39: Building Prosodic Structure

	Level 1	Level 2
A	GRWDPRWD, TROCHEE, DEP- V_μ , PARSESYLL	GRWDPRWD, TROCHEE, DEP- V_μ , * $-\Delta FT > \Delta FT$ ANCHOR _{FOOT} , PARSE- μ
B	*P/ ∂ , *STRUC-FT	*P/ ∂ , *STRUC-FT
C		PARSESYLL, HEADMATCH
Unclear	HEADMATCH, ANCHOR _{FOOT} , * $-\Delta FT > \Delta FT$, PARSE- μ	

Table 4.40 compares the constraints motivating the assignment of lexical stress and those dealing with general phonotactics. Lexical stress is observed in morphologically simple nouns, which suggests that STRESS² is undominated at Level 1. There are also no lexical items with a tautosyllabic $\partial?$ (or ∂h) and primary stress is limited to one per word, which supports a high ranking of $*\partial?$ _{σ} and CULMIN at Level 1, also necessary at Level 2. The rankings of OCP and MAX- μ are unclear at Level 1.

Table 4.40: Constraints Dealing with Lexical Stress and Phonotactics

	Level 1	Level 2
A	* $\partial?$ _{σ} , CULMIN, STRESS ²	* $\partial?$ _{σ} , CULMIN, OCP
B		STRESS ²
C		MAX- μ
Unclear	OCP, MAX- μ	

The constraints pertaining to glottalisation are summarised in Table 4.41. IDENT-[C.G.] must be ranked above the two positional restrictions on [c.g.] segments (*SON_[C.G.]/-△FT and *ONSET_σ[C.G.]) to protect against deglottalisation of underlyingly glottalised sonorants. However at Level 1, the positions of the other glottalisation constraints are unclear because the reduplicative morphemes do not contain a floating [c.g.] feature, and therefore the constraints are all vacuously satisfied.

Table 4.41: Glottalisation and [c.g.] Constraints

	Level 1	Level 2
A	IDENT-[C.G.], MAX-FLOAT	IDENT-[C.G.], DEPL _O [C.G.]
B	*SON _[C.G.] /-△FT, *ONSET _σ [C.G.]	*SON _[C.G.] /-△FT, *ONSET _σ [C.G.]
C		MAX-[C.G.], MAX-FLOAT
D		INT·AGR-[C.G.]
E		DEPL _[C.G.]
Unclear	MAX-[C.G.], INT·AGR-[C.G.] DEPL _O [C.G.], DEPL _[C.G.]	

Differences between reduplicative morphemes in Lillooet arise from different constraint rankings at each stratum, in addition to a range of different inputs (lexical entries), as summarised in Table 4.42.

Table 4.42: Lexical Entries in Lillooet

	Level 1	Level 2	Level 2+n
μ -type		(μ [c.g.]) _{L2} (μ _[stress] [c.g.]) _{L2}	
σ -type	$\sigma_{N+\mu}$ σ	σ_{L2}	
Ft-type			Ft _{L2+n}

Many of the key rankings are the same across Level 1 and Level 2 in Lillooet, unlike in Comox-Sliammon; the two languages differ more strongly in the underlying form of the affixes and the stratum where each is parsed into the word.

4.3 Conclusion

A combination of assumptions from Stratal OT and Prosodic Morphology provides the framework needed for an analysis of reduplicative morphemes in languages with large inventories of reduplicative processes, such as Comox-Sliammon (seven) and Lillooet (six). Reduplicative morphemes in Comox-Sliammon and Lillooet include affixed moras and syllables (σ or $\sigma_{N+\mu}$). In addition to these prosodic units, Comox-Sliammon uses fixed segment allomorph (Section 4.1.3.1); Lillooet uses a foot (Section 4.2.5).

Certain prosodic units are accompanied by a floating [c.g.] feature, which corresponds to glottalisation in both languages. The diminutive mora in Lillooet is also marked for lexical stress. The realization of these morphemes follows from the general phonological grammar of each language, resulting in differences in the surface form (how the prosodic affix is filled). For example, the [c.g.] feature accompanies a mora in Lillooet Level 2 C_1 reduplication and is realised as glottalisation on a sonorant in combination with fission of a consonant. In Comox-Sliammon, the [c.g.] feature accompanies a mora as part of V_1 reduplication, but is realised as an epenthetic glottal stop alongside a fissioned vowel.

The levels where reduplicative morphemes are parsed into the word also differ between the two languages. While the diminutive is a Level 1 affix and the plural/pluractional C_1C_2 morpheme is a Level 2 affix in Comox-Sliammon, the order of the two is reversed in Lillooet. When triplication in Lillooet is included, both languages require positing three strata where reduplication can happen.

The analysis developed in this chapter (and in Chapter 5 on multiple reduplication) leads to a set of predictions about reduplication in Salish, which are first outlined and then evaluated against cross-Salish patterns in Chapter 6.

Chapter 5

Multiple Reduplication

5.1 Introduction

Multiple reduplication refers to the combination of at least two reduplicative morphemes in the same word. Multiple reduplication in Salish has played a key role in providing evidence for a variety of phonological theories; for example, it has been used to argue for subadjacency in morphology (Broselow, 1983), as well as to argue against theories with cyclicity in favour of parallel derivation (e.g., Urbanczyk, 1999).

Previous chapters laid out crucial theoretical assumptions about reduplication and the grammar (Chapter 3), and showed how these assumptions allow for the analysis of reduplicative morphemes in Comox-Sliammon and Lillooet (Chapter 4). This chapter presents case studies and analyses of multiple reduplication patterns in three languages: Halkomelem, Lillooet, and Comox-Sliammon.

The first case study, given in Section 5.3, focuses on multiple reduplication in Halkomelem and outlines the core proposal in this chapter: multiple reduplication is only possible if *prosodic affixes* are parsed at different strata from each other. The second case study, presented in Section 5.4, confirms this prediction through a study of combinations and gaps in Lillooet multiple reduplication. I consider multiple reduplication in Comox-Sliammon as a final case study, given in Section 5.5, to test an additional prediction that *fixed segment affixes* are exempt from the restriction on multiple reduplication.

Through the analysis of multiple reduplication in Halkomelem, Lillooet, and Comox-Sliammon, this chapter explores the patterns, restrictions, and allomorphy that arise when various reduplicative morphemes are combined. These case studies highlight potential challenges for other theoretical approaches and showcase the benefits of adopting the theoretical framework proposed in this dissertation.

5.2 Background

Phonological interest in Salish multiple reduplication started in the 1980s, with work by Broselow (1983) covering Lushootseed, Thompson River Salish, and Shuswap.

Broselow (1983) uses multiple reduplication to argue that subadjacency is relevant not just at the level of the sentence, but also the level of the word. She proposes a cyclic derivation where reduplication can only access material contained in a subjacent cycle. In multiple reduplication, this means the outermost of two reduplicative morphemes is limited to copying material associated with the innermost reduplicative morpheme, rather than the root. While Broselow (1983) takes a different theoretical approach to reduplication than I use (see Section 3.5.2 for my assumptions about stratification and the phonological grammar), both approaches converge on a need for multiple stages of derivation to account for the empirical facts in Salish.

Despite featuring prominently in arguments for particular phonological theories, the description and analysis of multiple reduplication in Salish rarely considers the full range of reduplicative morphemes and the full range of possible combinations within a single language, much less across multiple languages or the family as a whole. Combinations of diminutive C_1 and plural C_1C_2 reduplication are those best represented in the literature (see, e.g., Broselow 1983; Bjorkman 2006), though some works have included the combination of C_1C_2 and C_2 reduplication as well (e.g., Urbanczyk, 2006).

The survey of multiple reduplication in Chapter 2 (Section 2.8) established that multiple reduplication is found in 15 Salish languages. A primary goal of this chapter is to explore possible combinations and to identify circumstances where multiple reduplication is ungrammatical, absent in existing documentation, or

avoided through the use of non-reduplicative allomorphs.

5.3 Case Study I: Halkomelem (Allomorphy)

Imperfectivity in Halkomelem can be expressed with a range of non-concatenative processes, including ablaut, reduplication, metathesis, glottal stop epenthesis, sonorant aspiration, schwa deletion, and schwa insertion (see, e.g., Zimmermann 2013; Mellesmoen & Urbanczyk, 2020).

Aside from imperfective C_1 reduplication, the types of reduplication (and their allomorphs) included in this case study are diminutive C_1 reduplication and plural C_1C_2 reduplication. An example of each is shown in (303).¹

- (303) a. *Imperfective C_1 Reduplication*
 $\text{tá} \sim \text{tə́k}^w$ (cf. tak^w ‘fly’)
 IPFV $\sim$ fly
 ‘flying’ (Hukari, 1978, p.163)
- b. *Diminutive C_1 Reduplication*
 $\text{k}^w\text{i}ʔ \sim \text{k}^w \langle \text{é} \rangle \text{t}$ (cf. $\text{k}^w\text{e}t$ ‘pouring it’)
 DIM $\sim$ pour <IPFV>-CTR
 ‘pouring it (dim.)’ (Hukari, 1978, p.194)
- c. *Plural C_1C_2 Reduplication*
 $\text{θə́x} \sim \text{θə́x-ə́t}$ (cf. θə́t ‘push it’)
 PL $\sim$ push-CTR
 ‘push them’ (Hukari, 1978, p.189)

Mellesmoen and Urbanczyk (2021a) propose that Halkomelem exhibits *variable* multiple reduplication. Variable multiple reduplication means that a language allows multiple reduplication in some instances, but blocks it in others. Halkomelem allows for combinations of reduplicative morphemes, as shown by the co-occurrence of the imperfective and diminutive in (304); note that the diminutive is marked both with C_1 reduplication and the infixation of a glottal stop in these forms.

¹I leave ‘(dim)’ in translations following the source because diminutive verbs may have a range of different meanings, including small participants or events; it is not clear what the translation would be. See Chapter 1 (Section 1.4.4.3) for discussion of glossing.

- (304) a. $\dot{\text{t}}\text{ə} \sim \dot{\text{t}}\text{á} < \text{ʔ} > \sim \dot{\text{t}}\text{ə} \text{k}^{\text{w}}$ (cf. $\dot{\text{t}}\text{ak}^{\text{w}}$ ‘fly’)
 DIM ~ IPFV < DIM > ~ fly
 ‘flying (dim.)’
 b. $\dot{\text{t}}\text{ə} \sim \dot{\text{t}}\text{í} < \text{ʔ} > \sim \dot{\text{t}}\text{əl} \sim \text{əm}$ (cf. $\dot{\text{t}}\text{iləm}$ ‘to sing’)
 DIM ~ IPFV < DIM > ~ sing-MID
 ‘singing (dim.)’

(Hukari, 1978, p.196)

Not all combinations of reduplicative morphemes are attested in Halkomelem. In the examples in (305) which include diminutive, plural, and imperfective morphology, an infix *-l-* is used instead of C₁C₂ reduplication to mark plurality.

- (305) a. $\dot{\text{t}} < \text{ə} \text{l} > \text{i} \text{ʔ} \sim \dot{\text{t}}\text{ə} \sim \dot{\text{t}}\text{əl} \text{n} \sim \text{əm}$ (cf. $\dot{\text{t}}\text{əl} \text{n} \text{əm}$ ‘weaving’)
 DIM < PL > ~ IPFV ~ weave-MID
 ‘weaving (dim. pl.)’
 b. $\text{p} < \text{ə} \text{l} > \text{i} \text{ʔ} \sim \text{p} < \text{a} > \text{q}^{\text{w}} \text{-t}$ (cf. $\text{paq}^{\text{w}} \text{t}$ ‘breaking it’)
 DIM < PL > ~ break < IPFV > -CTR
 ‘breaking them (dim.)’

(Hukari, 1978, p.196)

Given that Halkomelem is a language with variable multiple reduplication, an analysis of reduplication in Halkomelem must predict attested combinations while ruling out unattested types of multiple reduplication.

Unlike the case studies of Lillooet and Comox-Sliammon that follow, this case study relies exclusively on secondary sources. Data in this section comes from the Island and Downriver dialects of Halkomelem.²

5.3.1 Single Reduplication in Halkomelem

In this section, I provide a brief summary of imperfective (Section 5.3.1.1), diminutive (Section 5.3.1.2), and plural (Section 5.3.1.3) reduplication patterns in Halkomelem.

²The Upriver dialect has similar reduplication patterns, though see Section 6.7.1.2 for discussion of compensatory lengthening not found in Island or Downriver Halkomelem.

5.3.1.1 Level 1: Imperfective

5.3.1.1.1 Allomorphs of the Imperfective

Mellesmoen and Urbanczyk (2020) note that phonological, morphological, and lexical factors condition the surface form of the imperfective in Halkomelem. I focus on the phonologically conditioned allomorphs here.³ The surface form of the imperfective is determined by the shape of the root (and sometimes a following affix), in particular whether it starts with a consonant cluster, has a moraic vowel or [ə], as well as which consonants are in the root.

The first set of imperfective forms, given in (306), are built on words with a CVC root with a moraic vowel, excluding those with a glottal stop or /h/ as the second consonant. The initial consonant and vowel are copied; there is subsequent reduction of a moraic unstressed vowel to [ə] at a later stratum.

- (306) a. $\mathbf{\acute{h}i} \sim \mathbf{\acute{h}ə\acute{c}}\text{-}\acute{ə}t$ (cf. $\mathbf{\acute{h}i\acute{c}\acute{ə}t}$ ‘cut it’)
IPFV $\sim$ cut-CTR
‘cutting it’
- b. $\mathbf{\acute{h}i} \sim \mathbf{\acute{h}ə\acute{m}}\text{-}\acute{ə}t$ (cf. $\mathbf{\acute{h}i\acute{m}\acute{ə}t}$ ‘lick it’)
IPFV $\sim$ lick-CTR
‘licking it’
- c. $\mathbf{y\acute{e}} \sim \mathbf{y\acute{ə}q}$ (cf. $\mathbf{yeq}$ ‘topple down’)
IPFV $\sim$ topple
‘toppling down’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2020, p.240)

Glottalisation of sonorant consonants is also common in imperfective forms, such as in (306b–c) above. As the goal of this chapter is to discuss the possible combinations of reduplicative morphemes, I will not provide further discussion of glottalisation with respect to the imperfective in Halkomelem.⁴

³I assume that the other forms represent morphologically conditioned allomorphy or irregular forms, which are beyond the scope of the present analysis.

⁴There is likely a [c.g.] feature, as was proposed for Level 3 V₁ reduplication in Comox-Sliammon in 4.1.5.2.2, as well as for both types of Level 2 C₁ reduplication in Lillooet in Section 4.2.4.3.

When the root starts with a glottal stop or an /h/, the imperfective is marked by an infix *ʔ*, as shown in (307).

- (307) a. *ʔí<ʔ>m-əš* (cf. *ʔíməš* ‘walk’)
 walk<IPFV>-INTR
 ‘walking’
- b. *ʔá<ʔ>k^w-ət* (cf. *ʔák^wət* ‘hook it’)
 hook<IPFV>-CTR
 ‘hooking it’
- c. *hé<ʔ>s-əm* (cf. *hésəm* ‘sneeze’)
 sneeze<IPFV>-MID
 ‘sneezing’
- d. *há<ʔ>k^w-əš* (cf. *hákwəš* ‘use/wear it’)
 use.wear<IPFV>-INTR
 ‘using it, wearing it’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2020, p.240)

When the word starts with a CəCC- sequence, the imperfective is formed by ablaut. As shown in (308), a moraic vowel (/a/ or /e/) surfaces instead of [ə].

- (308) a. *ʔ<e>pt^θ-t* (cf. *ʔəpt^θt* ‘slurp it’)
 slurp<IPFV>-CTR
 ‘slurping it’
- b. *s<e>wq* (cf. *səwq* ‘look for’)
 look<IPFV>
 ‘looking for’
- c. *č<a>tq^w-t* (cf. *čətq^wt* ‘grind it’)
 grind<IPFV>-CTR
 ‘grinding it’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2020, p.241)

Metathesis, shown in (309), marks the imperfective form of words which begin with a consonant cluster comprised of obstruents. The vowel and the second consonant are switched in the imperfective form.

- (309) a. **paq^w**-t (cf. *pq^wat* ‘break it’)
 break+IPFV-CTR
 ‘breaking it’
 b. **taq^w**-t (cf. *tq^wat* ‘break it’)
 break+IPFV-CTR
 ‘breaking it’
 c. **seq**-t (cf. *sqet* ‘tear/split it’)
 tear.split+IPFV-CTR
 ‘tearing/splitting it’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2020, p.241)

If the word starts with a sonorant (excluding /ʔ/) followed by a /ə/, then the imperfective word has an infix copy of that consonant; the initial consonant of the word surfaces as [h].⁵

- (310) a. **hó<ḷ>m-ət** (cf. *lómət* ‘fold it’)
 fold<IPFV>-CTR
 ‘folding it’
 b. **hó<ṁ>q-ət** (cf. *móqət* ‘swallow it’)
 swallow<IPFV>-CTR
 ‘swallowing it’
 c. **hó<ṇ>qəm** (cf. *nóqəm* ‘dive’)
 dive<IPFV>-MID
 ‘diving’
 d. **hó<ṽ>k^w-ət** (cf. *yók^wət* ‘scrub/rub together’)
 scrub.rub<IPFV>-CTR
 ‘scrubbing, rubbing together’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2020, p.241)

When a word starts with an obstruent-schwa-sonorant-schwa (OəSə) sequence, its imperfective form has a copy of the shape Cə- and stress falls on the second [ə], as shown in (311).

⁵This process can be analysed as sonorant “aspiration”, devoicing, debuccalisation, spirantisation, or as some combination of these processes.

(311) a. $tə \sim tǝm-ə$ (cf. $tǝmət$ ‘pound on it’)

IPFV~drum-CTR

‘pounding on it, beating a drum’

b. $tə \sim tǝl-ə$ (cf. $tǝlət$ ‘bail it out’)

IPFV~bail-CTR

‘bailing it out’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2020, p.243)

Table 5.1 summarizes the phonologically conditioned allomorphs of the imperfective in Halkomelem; I use “C_{Obs}” to refer to an obstruent, while “C_{Son}” refers to a sonorant consonant.

Table 5.1: Phonologically Conditioned Allomorphs of the Imperfective (Halkomelem)

Type	Shape	First Segments of Non-Reduplicated Word
Reduplication	CV- Cə- hǝ-	CV _μ C... , e.g., (306) C _{Obs} əC _{Son} ə..., e.g., (311) C _{Son} ə..., e.g., (310)
Infix	-ʔ-	C _{ʔ/h} VC, e.g., (307)
Ablaut	-a- or -e-	C _{Obs} əCC..., e.g., (308)
Metathesis	Metathesis	C _{Obs} C _{Obs} VC _{Obs} ..., e.g., (309)

5.3.1.1.2 Analysis of the Imperfective

The lexical entry for the imperfective in Halkomelem is a mora, as shown in (312). The mora is unmarked for stratum, given that it is parsed at Level 1 (following the assumptions laid out in Section 3.5.2.3). This analysis is modified from Mellesmoen and Urbanczyk (2020) to match the phonological framework presented in Chapter 3 and the analyses of reduplication given in Chapter 4.

(312) *Halkomelem Imperfective:*

$\mu \Leftrightarrow \langle \text{IPFV} \rangle$

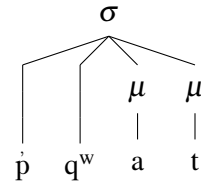
Each surface form of the imperfective can be derived through the addition of a mora. In this section, I will show where the mora is integrated into prosodic

structure for each surface pattern to show that they all can be derived from the same input μ .⁶

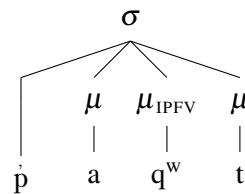
Metathesis can fill a mora, as shown in (313). When a mora is affixed to a CCVC input, as shown in (313a), switching an onset consonant with a vowel puts the consonant into a coda position, which then can be associated with a mora, as shown in (313b). I assume no prosodic structure is associated with the root in the input to Level 1 (aside from the fact that underlying vowels are moraic); the prosodic structure given beside the input in the following examples is strictly for comparison and shows the structure that would be built if the imperfective were not present. The imperfective form must necessarily be different than this comparison form to satisfy REALIZEMORPHEME, as defined first in Section 4.1.3.1.2. I label the mora as μ_{IPFV} to show where the imperfective mora is integrated into prosodic structure, though this is for illustrative purposes: the phonological grammar does not have access to information about the function associated with a particular morpheme.

(313) Derivation of paq^wt ‘breaking it’

a. *Level 1 Input*: $\mu_{\text{IPFV}} + pq^wa + t$



b. *Level 1 Output*: paq^wt

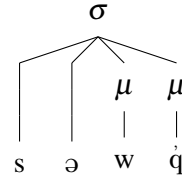


Ablaut (moraic [e] or [a] surfacing where non-moraic [ə] would otherwise occur) is another way the grammar provides segmental content for an affixed mora, as shown in (314).

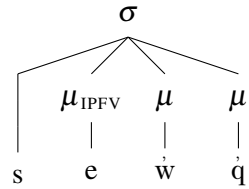
⁶A full constraint ranking is not provided because it is beyond the scope of this chapter and would require a deeper dive into the phonology of Halkomelem, including a consideration of feature geometry and the structure of sonorants and glottal segments.

(314) Derivation of *sewq̣* ‘looking for’

a. *Level 1 Input*: $\mu_{\text{IPFV}} + \text{səwq̣}$



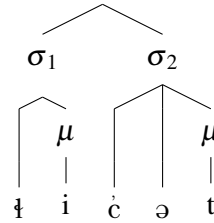
b. *Level 1 Output*: *sewq̣*



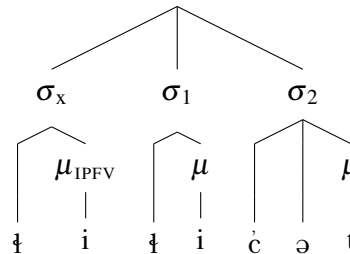
CV reduplication is fission of the first consonant and vowel of the root, where the fissioned vowel fills the mora, as shown (315). The Level 1 output is shown in (315b). The intermediate stage represented by the Level 1 output has two identical moraic vowels; however, the unstressed vowel is reduced at a later stratum, resulting in surface [ə], as shown in (315c). The reduction of an unstressed to [ə] follows a general pattern in the language: root vowels typically reduce to [ə] or ∅ if they do not bear primary stress (see, e.g., Jones, 1976, p.27–31; Suttles, 2004, p.11).

(315) Derivation of *ɬɬəcət* ‘cutting it’

a. *Level 1 Input*: $\mu_{\text{IPFV}} + \text{ɬicət}$



b. *Level 1 Output*:

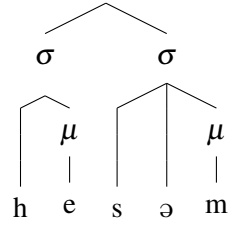


c. *Output of a Later Stratum*: ʰɛʔsəm

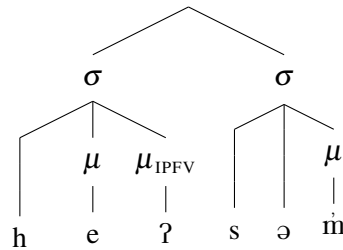
When the perfective form starts with a glottal consonant (/ʔ/ or /h/), glottal stop epenthesis fills the mora, as shown in (316).

(316) Derivation of *hɛʔsəm* ‘sneezing’

a. *Level 1 Input*: μ_{IPFV} + *hɛsəm*



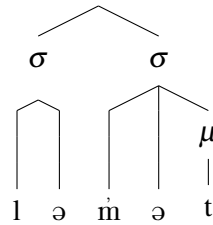
b. *Level 1 Output*: *hɛʔsəm*



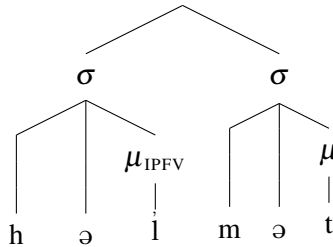
When the perfective form of the verb begins with a sonorant-schwa sequence, the imperfective form has reduplication with concurrent change of the first sonorant to [h]. The second sonorant is in a coda position and hosts the affixed mora, as shown in (317).

(317) Derivation of *həlmət* ‘folding it’

a. *Level 1 Input*: μ_{IPFV} + *ləmət*



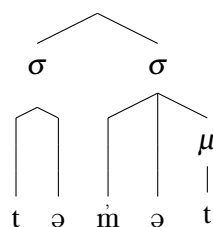
b. *Level 1 Output*: *həlmət*



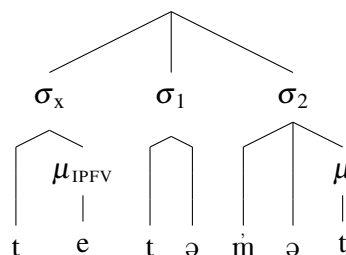
Finally, stems starting with the sequence OəSə- (obstruent-schwa-sonorant-schwa), surface with Cə- reduplication. I analyse this allomorph as reduplication and ablaut with subsequent reduction of an unstressed vowel (i.e., Ce → Cə), as shown in (318). Reduction happens at a later stratum (318c), as in (315) above.

(318) Derivation of *tətə́mət* ‘pounding on it’

a. *Level 1 Input*: $\mu_{\text{IPFV}} + \text{tə́mət}$



b. *Level 1 Output*: *tətə́mət*



c. *Output of a Later Stratum*: *tətə́mət*

In summary, the imperfective in Halkomelem across all forms can be analysed as the affixation of a mora.⁷

5.3.1.2 Level 2: Diminutive

5.3.1.2.1 Allomorphs of the Diminutive

The diminutive in Halkomelem is marked by C₁ reduplication. Diminutive verbs in Halkomelem are always build on bases with imperfective aspect (Suttles, 2004, p.178).⁸

⁷Though not discussed here, this is also true of morphologically and lexically conditioned allomorphy (see Mellesmoen & Urbanczyk, 2020).

⁸Though there is no obvious semantic reason for diminutive verbs to always be imperfective, suggesting a possible phonological motivation, diminutive nouns do not require additional morphology, as shown in the examples from the Downriver (hənqə́minən) dialect of Halkomelem in (i).

Some diminutives have [ə] between the matching consonants, as shown in (319a-b), and other diminutives have an epenthetic [i], as shown in (319c-e). This alternation is phonologically conditioned (Hukari, 1978, p.195): the [ə] form occurs when the unreduplicated word starts with a CV_μC... sequence. There is also an infixated glottal stop, which either follows the stressed vowel (in [ə] forms) or the epenthetic vowel (in [i] forms).

- (319) a. $\text{fə} \sim \text{fá} < \text{?} > \sim \text{fə} \text{k}^w$ (cf. fak^w ‘fly’)
 DIM ~ IPFV < DIM > ~ fly
 ‘flying (dim.)’
- b. $\text{fə} \sim \text{fé} < \text{?} > \sim \text{fə} \text{q} \text{ə} \text{m}$ (cf. $\text{fəqə} \text{m}$ ‘whisper’)
 DIM ~ IPFV < DIM > ~ whisper- MID
 ‘whispering (dim.)’
- c. $\text{k}^w \text{i} \text{?} \sim \text{k}^w < \text{é} > \text{f} \text{-t}$ (cf. $\text{k}^w \text{ə} \text{f}$ ‘spill’)
 DIM ~ pour < IPFV > -CTR
 ‘pouring it (dim.)’
- d. $\text{x} \text{i} \text{?} \sim \text{x} < \text{é} > \text{p} \text{x} \text{-t}$ (cf. $\text{xəp} \text{x} \text{t}$ ‘scatter it’)
 DIM ~ sprinkle < IPFV > -CTR
 ‘sprinkling it (dim.)’
- e. $\text{hi} \text{?} \sim \text{hó} < \text{l} > \text{c} \text{-t}$ (cf. ləc ‘full’)
 DIM ~ fill < IPFV > -CTR
 ‘filling it (dim.)’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.172)

5.3.1.2.2 Analysis of the Diminutive

The lexical entry for the diminutive in Halkomelem has two components: an empty

-
- (i) a. sput
 ‘little boat’
- b. mémən
 ‘little father’

(Suttles, 2004, p.212, Downriver Halkomelem)

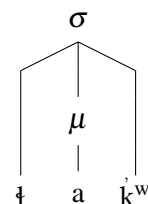
syllable and a constricted glottis ([c.g.]) feature, as shown in (320). The morpheme bears a Level 2 diacritic.

- (320) *Halkomelem Diminutive*:
 $(\sigma \text{ [c.g.]})_{L2} \Leftrightarrow \langle \text{DIM} \rangle$

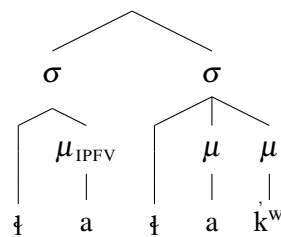
The empty syllable is parsed into the word at Level 2; the input to Level 2 already contains the imperfective parsed at Level 1. Fission occurs to fill the empty syllable and the glottal stop is positioned as an infix following the first moraic (non-schwa) vowel of the word. This is shown in (321).

- (321) Derivation of ʔəʔáʔə́k^w ‘flying (dim.)’

a. *Input to Level 1*: $\mu_{\text{IPFV}} + \text{ʔak}^w$

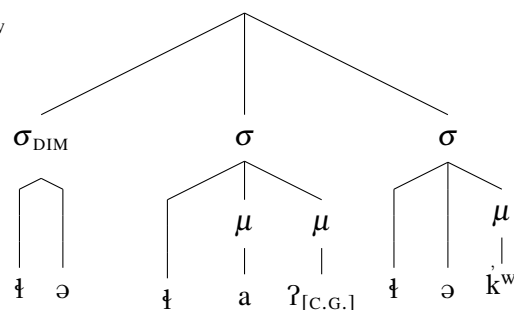


b. *Output of Level 1*: ʔáʔak^w



c. *Input to Level 2*: $\sigma_{\text{DIM}} \text{ [c.g.]} + \text{ʔáʔak}^w$

d. *Output of Level 2*: ʔəʔáʔə́k^w



The difference between the [ə] and [i] forms of the diminutive corresponds to whether a moraic vowel is copied (and then reduced to [ə]) or provided through epenthesis ([i]).

5.3.1.3 Level 2: Plural

5.3.1.3.1 Allomorphs of the Plural

Plurality can be marked on verbs with one of two allomorphs: C₁C₂ reduplication, shown in (322), or an infix -l-, shown in (323). Hukari (1978, p.175) states that he is “not aware of phonological criteria distinguishing forms” with C₁C₂ reduplication from those with an infix -l-.

- (322) a. **təl**~tɪl-əm (cf. *tɪləm* ‘sing’)
PL~sing-MID
‘they sing’
- b. **ləm**~ləm-ət (cf. *ləmət* ‘look at it’)
PL~look-CTR
‘look at them’
- c. **ʔəm**~ʔim-əʃ (cf. *ʔiməʃ* ‘walk’)
PL~walk-INTR
‘walk’ (pl)
- d. **məq̣**~məq̣-ət (cf. *məq̣ət* ‘swallow it’)
PL~swallow-CTR
‘swallowing them’
- e. **səṭ**~səṭq̣-t (cf. *səṭq̣t* ‘disturb’)
PL~disturb-CTR
‘disturb (repeatedly)’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.173)

- (323) a. **né<lə>m̄** (cf. *nem̄* ‘go’)
go<PL>
‘they go’
- b. **k^wé<lə>s** (cf. *k^wes* ‘hot’)
hot<PL>
‘hot, warm weather’

- c. $té<\overset{\cdot}{l}ə>cəl$ (cf. *técəl* ‘arrive’)
 arrive<PL>
 ‘arrive (pl.)’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.173)

Hukari (1976, p.180) notes that *-l-* is “productive outside the domain of actions and processes, where $[C_1C_2]$ reduplication predominates”. Given that I am considering verbal reduplication, I treat C_1C_2 reduplication as the default allomorph of the plural.

5.3.1.3.2 Analysis of the Plural

The lexical entry for the plural morpheme in Halkomelem is given in (324). There are two allomorphs: a monomoraic syllable with a non-moraic nucleus and an infix *-l-*. These allomorphs are ordered: $\sigma_{N+\mu}$ form is the prioritised allomorph.

(324) *Halkomelem Plural:*

$$(\sigma_{N+\mu 1} > l_2)_{L2} \Leftrightarrow \langle PL \rangle$$

The ordered allomorphs in (324) are subject to the constraint PRIORITY, first used in the analysis of C_1 plural and diminutive reduplication in Comox-Sliammon in Chapter 4 (Section 4.1.3.1.2); PRIORITY is restated in (325a). The ranking of PRIORITY relative to INTEGRITY-C, with its definition restated in (325b), determines if the preferred allomorph is selected.

(325) a. PRIORITY: Respect lexical priority (ordering)

Given an input containing allomorphs $m_1, m_2, \dots, m_n$, and a candidate m_i' , where m_i' is in correspondence with m_i , PRIORITY assigns as many violation marks as the depth of ordering between m_i and the highest dominating morph.

(Mascaró, 2007, p.726)

b. INTEGRITY-C (INT-C): A consonant in the input does not have multiple correspondents in the output.

Ranking PRIORITY over INTEGRITY-C motivates the insertion of the monomoraic syllable, as shown in (326). The subscript numbers correspond to the selected

allomorph. It is better to select the preferred allomorph and fill it via fission (violating INTEGRITY-C) than to pick the fixed segment infix.

	$/(\sigma_{N+\mu 1} > l_2) + m\acute{o}q\acute{o}t/$	PRIORITY	INTEGRITY-C
(326)	a. $m\acute{o}q\acute{o}m\acute{o}q\acute{o}t_1$		**
	b. $m\acute{o}l\acute{o}q\acute{o}t_2$	*!	

Though *-l-* is not the preferred allomorph, it is the one that will be shown to occur in multiple reduplication when two types of Level 2 reduplication would co-occur (diminutive and plural), as will be discussed further in Section 5.3.2.3.

5.3.2 Multiple Reduplication

5.3.2.1 Attested: Diminutive Imperfective

Combinations involving the diminutive plus imperfective morpheme were already discussed in Section 5.3.1.2. Given that diminutive verbs are always imperfective, if the imperfective form of a verb surfaces with reduplication, then the diminutive imperfective form will necessarily be expressed with multiple reduplication, as shown in (327).

- (327) a. $\acute{t}\acute{o} \sim \acute{t}\acute{a} < ? > \sim \acute{t}\acute{o}k^w$ (cf. $\acute{t}\acute{a}k^w$ ‘fly’)
 DIM~IPFV<DIM>~fly
 ‘flying (dim.)’
 b. $\acute{t}\acute{o} \sim \acute{t}\acute{u} < ? > \sim \acute{t}\acute{o}l\text{-}\acute{o}m$ (cf. $\acute{t}\acute{u}l\acute{o}m$ ‘to sing’)
 DIM~IPFV<DIM>~sing-MID
 ‘singing (dim.)’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.169)

Imperfective reduplication is parsed at Level 1, while diminutive reduplication is parsed at Level 2. Diminutive imperfective forms therefore involve single reduplication at each of two successive strata.

5.3.2.2 Attested: Plural Imperfective

The imperfective and plural morphemes can result in multiple reduplication. For words where the imperfective is formed by reduplication, there are two main imperfective plural patterns. The first of these, shown in (328–329), has imperfective C₁ reduplication (with the sonorant > [h] change described in Section 5.3.1.1) with subsequent plural C₁C₂ reduplication.

(328) Unreduplicated: *méʔx* ‘remove it’

a. Imperfective:

hó<ṁ>x

remove<IPFV>

b. Plural Imperfective:

həṁ~hó<ṁ>əx

PL~remove<IPFV>

(Suttles, 2004, p.169, Downriver Halkomelem)

(329) Unreduplicated: *wəqʷ* ‘drift away’

a. Imperfective:

hó<ẉ>qʷ

drift.away<IPFV>

b. Plural Imperfective:

həw~hó<ẉ>qʷ

PL~drift.away<IPFV>

(Suttles, 2004, p.171, Downriver Halkomelem)

Note that the multiple reduplication pattern shown in (328) provides further evidence for the imperfective being a Level 1 affix, because imperfective must apply before the plural in order for the [h] (a debuccalised copy of the root-initial /w/) to be copied in plural C₁C₂ reduplication.

With other non-reduplicated allomorphs of the imperfective, C₁C₂ reduplication is added to the imperfective form as predicted. An example with an imperfective word formed by metathesis is shown in (330).

- (330) **pəq^w~páq^w-t** (cf. *páq^wt* ‘breaking it’)
 PL~break+IPFV-CTR
 ‘breaking them’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.175)

The second pattern observed with imperfective plus plural reduplication is shown in (331). This pattern appears to have only one type of reduplication, with C₁C₂ reduplication marking both plurality and imperfectivity, as shown in (331d). Crucially, however, the perfective plural (331b) and the plural imperfective (331c) differ in the position of vowels: the plural perfective has a stressed [i] as the first vowel in the word, while the plural imperfective has a stressed [i] as the second vowel in the word.

- (331) a. **tí~təl-əm** (cf. *tíləm* ‘sing’)
 IPFV~sing-MID
 ‘singing’
 b. **təl~tíləm**
 PL~sing-MID
 ‘they sing’
 c. **tíl~təl-əm**
 IPFV+PL~sing-MID
 ‘they are singing’

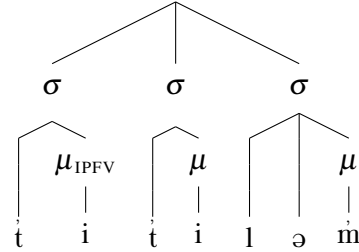
(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.175)

I propose, following Mellesmoen and Urbanczyk (2021a), that the plural imperfective involves the regular affixation of a mora marking imperfective aspect at Level 1, with subsequent affixation of a monomoraic syllable marking plurality at Level 2. However, the Level 2 affix is realised differently than usual: instead of adding another syllable, the plural syllable coalesces with the first syllable of the word. The association of segments under this analysis is shown in (332). Imperfective reduplication fills a mora at Level 1; reduplication of the consonant in C₂ position of the root ([l]) provides a coda to fill the mora associated with a monomoraic syllable.

(332) Derivation of $\acute{t}\acute{ɪ}l\acute{t}\acute{ə}l\acute{ə}m$ ‘They are singing.’

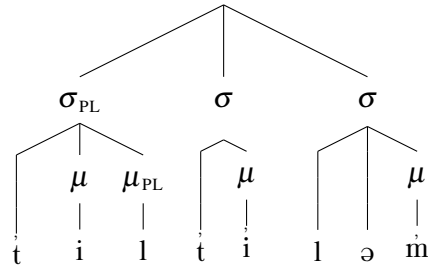
a. *Input to Level 1:* $\mu_{IPFV} + \acute{t}\acute{ɪ}l\acute{ə}m$

b. *Output of Level 1:* $\acute{t}\acute{ɪ}l\acute{ə}m$



c. *Input to Level 2:* $\sigma_{PL} + \acute{t}\acute{ɪ}l\acute{ə}m$

d. *Output of Level 2:* $\acute{t}\acute{ɪ}l\acute{t}\acute{ə}l\acute{ə}m$



e. *Output (after vowel reduction):* $\acute{t}\acute{ɪ}l\acute{t}\acute{ə}l\acute{ə}m$

The affixed syllable coalesces with existing prosodic structure at Level 2, which allows for multiple reduplication to show up as a truncated pattern (see Zimmermann 2021 for discussion of faithful vs. truncating multiple reduplication patterns). I leave questions related to the implementation of this analysis via constraint ranking for future work; we can, however, conclude that plural imperfectives involve reduplication at two strata: Level 1 (imperfective) and Level 2 (plural).

5.3.2.3 Unattested: Diminutive Plural

Recall that the diminutive only occurs with imperfective aspect, as mentioned in Section 5.3.1.2. The combination of diminutive plus plural marking in Halkomelem is realized as C_1 reduplication plus the *-l-* allomorph of the plural, as shown in (333) and (334); forms with both diminutive and plural reduplication such as $*\acute{t}\acute{ə}?\acute{t}\acute{ɪ}?\acute{t}\acute{ə}?\acute{t}\acute{ə}n\acute{ə}m$ (or $*\acute{t}\acute{ɪ}?\acute{t}\acute{ə}?\acute{t}\acute{ə}?\acute{t}\acute{ə}n\acute{ə}m$) and $*p\acute{ə}?\acute{p}\acute{ɪ}?\acute{p}\acute{ə}q^w\acute{t}$ (or $*p\acute{ɪ}?\acute{p}\acute{ə}q^w\acute{p}\acute{ə}q^w\acute{t}$) are not attested.

- (333) a. **ḥiʔ~ḥə~ḥón-əḥ** (cf. *ḥónəm* ‘weave’)

DIM~IPFV~weave-MID

‘weaving (dim.)’

- b. **ḥ<əl>ḥiʔ~ḥə~ḥón-əḥ**

DIM<PL>~IPFV~weave-MID

‘weaving (dim. pl.)’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.176)

- (334) a. **piʔ~p<á>q^w-t** (cf. *pq^wat* ‘break it’)

DIM~break<IPFV>-CTR

‘breaking it (dim.)’

- b. **p<əl>piʔ~p<á>q^w-t**

DIM<PL>~break<IPFV>-CTR

‘breaking them (dim.)’

(Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.176)

In plural-diminutive-imperfective words, only the diminutive and imperfective can be marked by reduplication: the plural is always marked with *-l-*, as in (333b) and (334b). While imperfective reduplication involves a Level 1 affix, the lexical entries for the diminutive and plural are marked as Level 2. The unattested multiple reduplication pattern would therefore require two types of reduplication to occur at the same stratum (Level 2).

5.3.3 Restrictions on Multiple Reduplication

Multiple reduplication entails the existence of at least two reduplicative morphemes in a language; however, having two reduplicative morphemes does not entail multiple reduplication (Zimmermann, 2021). Halkomelem is a language with variable multiple reduplication: multiple reduplication is attested in the language through plural imperfectives and diminutive imperfective forms, though a combination of two types of reduplication (plural and diminutive) at the same stratum is systematically avoided. It is not possible to provide a syntactic or semantic motivation for the missing combination, because the plural and

diminutive morphemes *can* be combined if a fixed segment allomorph (-l-) rather than reduplication marks plurality.

I propose in Section 5.3.3.1 that multiple reduplication is limited by the morphological affiliation of input segments to prosodic units. If any two prosodic affixes would be filled at the same stratum in the derivation, I predict that they will not co-occur. In Section 5.3.3.2, I compare my analysis to two alternative approaches that restrict multiple reduplication by limiting *how many* or *how* segments undergo fission, which I label as *Segment-Based Fission* and *Segment-Based Binarity* approaches, respectively.

5.3.3.1 Multiple Reduplication is Limited by Morphological Affiliation

The lexical entries for the diminutive and plural morphemes bear a Level 2 (L2) diacritic, as shown in (335). I omit the [c.g.] when discussing the diminutive for the remainder of this case study.

(335) a. *Halkomelem Imperfective*:

$\mu \Leftrightarrow \langle \text{IPFV} \rangle$

b. *Halkomelem Diminutive*:

$(\sigma [\text{c.g.}])_{\text{L2}} \Leftrightarrow \langle \text{DIM} \rangle$

c. *Halkomelem Plural*:

$(\sigma_{\text{N}+\mu 1} > l_2)_{\text{L2}} \Leftrightarrow \langle \text{PL} \rangle$

The diminutive-plural gap in Halkomelem follows from limiting reduplication to one prosodic affix per stratum. The plural and diminutive morphemes are parsed into the prosodic word at the same stratum (Level 2); multiple reduplication is therefore not expected with plural diminutive forms. However, the imperfective is parsed into the word at the first stratum (Level 1). On the basis of phonology, the imperfective is compatible with either the diminutive or the plural morpheme because they are integrated into the word at Level 2. In other words, limiting multiple reduplication with respect to the number of prosodic affixes per stratum is able to account for variable multiple reduplication languages, which only permit certain combinations of reduplicative morphemes.

I implement my analysis with the constraint MORPHEME-AFFILIATION,

defined in (336), which is violated when segments associated with a given morpheme in the input are associated with prosodic units in the output that have a different morphological affiliation (i.e., occur at a different stratum). Specifically, the grammar will tolerate segments associated with prosodic units that have one (and only one) morphological affiliation that differs from the segment's own.

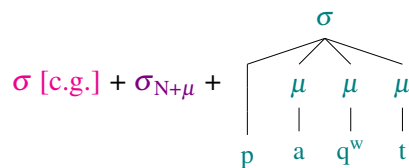
- (336) MORPHEME-AFFILIATION (MAFFIL): For any three output segments S (S_1, S_2, S_3) that belong to the same morpheme M_a , if S_1 is associated with a prosodic unit belonging to M_a and S_2 is associated with a prosodic unit belonging to some other morpheme M_b , then S_3 must be associated with a prosodic unit belonging to either M_a or M_b . Assign a violation mark for any three segments where all three segments belong to the same morpheme and each segment is associated with a prosodic unit belonging to a different morpheme ($*S_1-M_a, S_2-M_b, S_3-M_c$).

In the following diagrams, I use colour in lieu of indices to reflect morphological affiliation, though I do not adopt the assumptions of so-called coloured containment approaches (e.g., Van Oostendorp, 2006). MORPHEME-AFFILIATION looks at association lines directly linking segments and prosodic units in the output.

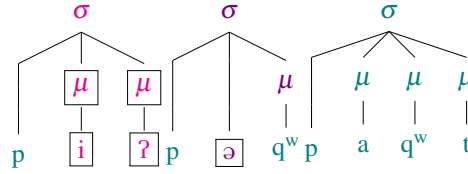
A potential plural diminutive form $*pi?pəq^wpaq^wt$ based on paq^wt 'breaking it' would violate MORPHEME-AFFILIATION. As shown in (337), there are prosodic units with three different morphological affiliations in the input, as marked by three different colours. In the output, segmental content with a single colour (teal) is associated with prosodic units of all three colours (teal, pink, and purple). The boxes mark epenthetic content, which takes on the morphological affiliation of the unit that immediately dominates it (i.e., colour spreads downward to epenthetic content).

- (337) Unattested Diminutive Plural: $*pi?pəq^wpaq^wt$

a. *Level 2 Input:*



b. *Level 2 Output (Unattested):*

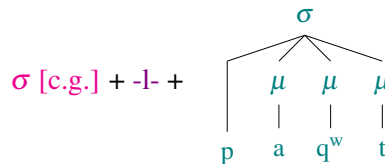


I assume that structure built at the preceding stratum is collectively treated as a single group in the input, distinct from unparsed affixes. For example, the imperfective μ , root, and transitivity affix $-t$ are all treated as a single (teal) group in the input to Level 2 in (337a) because they were combined into a single prosodic word at Level 1 (not shown here). The unattested form in (337) violates MORPHEME-AFFILIATION because there are teal segments (affiliated with the prosodic word in the input) associated with prosodic units of three different colours in the output (pink, purple, and teal).

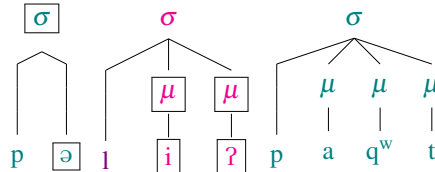
Compare (337) above, which shows an unattested form, with the attested one in (338): the leftmost syllable in (338b), which was not present in the input, receives the colour associated with the prosodic word as a whole in the input (teal). All teal segments are associated with teal prosodic units; the only mismatch between segment and prosodic unit colour is a purple (plural) segment linked to a pink (diminutive) syllable. MORPHEME-AFFILIATION is not violated by the form in (338): the attested form does not have any single segment colour associated with prosodic units of more than two colours.

(338) Attested Diminutive Plural: $pəli?paq^wt$ ‘breaking them (dim.)’

a. *Level 2 Input:*



b. *Level 2 Output:*



The limitation on multiple reduplication at the same stratum emerges when MORPHEME-AFFILIATION is ranked higher than constraints that pertain to allomorph selection and word-building. The relevant constraints $PARSE_{AFFIX}$ and $PARSE_{L2}$ from Chapter 3 (Section 3.5.2.3), as well as PRIORITY from Chapter 4 (Section 4.1.3.1.2) are restated in (339).

- (339) a. $PARSE_{AFFIX}$ ($PARSE_{AFF}$): Assign a violation mark for any affix that is left unparsed.
- b. $AFFIX_{L2}$ (AFF_{L2}): Assign a violation mark for every affix with a L2 diacritic that is parsed into the phonological domain defined by the stratum.
- c. PRIORITY: Respect lexical priority (ordering)
- Given an input containing allomorphs $m_1, m_2, \dots, m_n$, and a candidate m_i' , where m_i' is in correspondence with m_i , PRIORITY assigns as many violation marks as the depth of ordering between m_i and the highest dominating morph (Mascaró, 2007)

The tableau for *pəlipaq^{wt}* ‘breaking them (dim.)’ in (340) demonstrates that MORPHEME-AFFILIATION and $PARSE_{AFFIX}$ are crucially ranked above PRIORITY and $AFFIX_{L2}$ at Level 2. The attested candidate (340a), which corresponds to (338b) above, avoids a violation of MORPHEME-AFFILIATION by selecting the second of the listed allomorphs. Candidate (340b), which corresponds to (337b), satisfies PRIORITY by picking the preferred allomorph, but incurs a fatal violation of MORPHEME-AFFILIATION. Candidate (340c) vacuously satisfies PRIORITY by not parsing the affix (and therefore not choosing an allomorph), which incurs a fatal violation of $PARSE_{AFF}$.⁹ Square brackets contain the listed allomorphs; parentheses show which affixes have been integrated into the word.

/[$\sigma_{N+\mu 1} > l_2$] _{L2} + σ_{L2} + (paq ^{wt})/		MAFFIL	$PARSE_{AFF}$	PRIORITY	AFF_{L2}
(340)	a. (pəlipaq ^{wt})			*	**
	b. (piʔpəq ^w paq ^{wt})	*!			**
	c. [$\sigma_{N+\mu 1} > l_2$] _{L2} (piʔpaq ^{wt})		*!		*

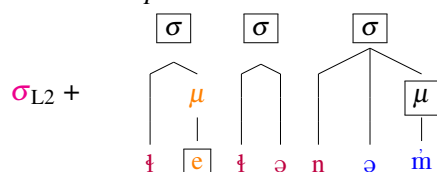
⁹MAX, REALIZEMORPHEME, and MAX-FLOAT are not shown here, but they would rule out a potential candidate that avoids violating any of the constraints in (340) by deleting the plural morpheme entirely.

MORPHEME-AFFILIATION is violated when multiple reduplication happens at the same stratum. The example in (341) shows how the diminutive imperfective form *ʔiʔəʔənəm* ‘weaving (dim.)’ is derived, with imperfective reduplication occurring at Level 1 and diminutive reduplication at Level 2. Though I assume schwa is epenthetic in the input, I show schwa in the correct output positions in the Level 1 input for the root (*ʔən-*) and the middle marker (*-əm*) for simplicity. The imperfective morpheme is parsed into the prosodic structure built at the first stratum, which is fed into Level 2 with a single colour (teal). The initial consonant of the word may undergo reduplication again at Level 2, as segments with a teal colour are associated with prosodic units of only one additional colour (pink). The unstressed moraic vowel is reduced at a later stratum.

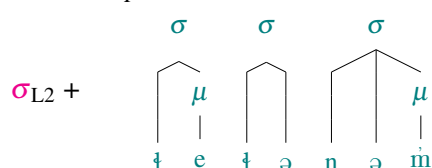
(341) Diminutive Imperfective: *ʔiʔəʔənəm* ‘weaving (dim.)’

a. *Level 1 Input*: $\sigma_{L2} + \mu + ʔən + əm$

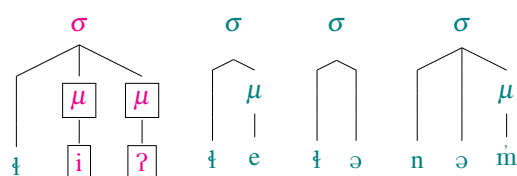
b. *Level 1 Output*:



c. *Level 2 Input*:



d. *Level 2 Output*:



Restricting reduplication to one prosodic affix per stratum via the inclusion of MORPHEME-AFFILIATION predicts both the attested forms and the gap in Halkomelem multiple reduplication.

5.3.3.2 Alternate Approach: Multiple Reduplication is Limited by Segments

I have proposed a morpheme-based limit on multiple reduplication in Section 5.3.3.1, which means that segments from one morpheme are maximally associated with prosodic units from one other morpheme. An alternative analysis would be to place restrictions on the process of reduplication itself (fission of segments) with reference to the segment. Segment-based approaches place general limits on fission that will equally apply regardless of the number of reduplicative morphemes in the input. A single reduplicative morpheme will not be distinguished from two reduplicative morphemes in the input under a segment-based approach.

Analyses of multiple reduplication that treat reduplication as prosodic affixation and fission have accounted for avoidance of multiple reduplication by using INTEGRITY to limit the overall number of copied segments (Zimmermann, 2021b), or by proposing a restriction on GEN that only allows for 1:2 mappings between input and output (Mellesmoen & Urbanczyk, 2021b). I refer to these approaches as *Segment-Based Fission* and *Segment-Based Binariness* approaches, respectively.

5.3.3.2.1 Segment-Based Fission – Zimmermann (2021)

Zimmermann (2021a) proposes a segment-based analysis restricting the number of times fission occurs to account for the typology of multiple reduplication across languages; she uses the INTEGRITY constraint INT_S , which is defined in (342).

(342) INT_S : Assign * to every pair of output segments that correspond to the same input segment. (Zimmermann, 2021a, p.14)

INT_S evaluates every pair of co-occurring output segments and assigns a violation mark if they correspond to the same input segment. Table 5.2 shows how pairs of segments incur violations under INT_S for the plural form *məq̣məq̣ət* ‘swallowing them’, with one type of reduplication (= 2 instances of a 1-to-2 input-output mapping). The numbers in the left column count up the number of INT_S violations: different types of dot are used to distinguish violations involving fission of different input segments (e.g., pairs involving *m* are ○ and those involving *q̣* are ●).

Table 5.2: INT_S Violations for Plural

	m ₁	ə	q̇ ₂	m ₁	ə	q̇ ₂	ə	t
1	○			○				
2			●			●		

Violations of INT_S are shown in Table 5.3 for **piʔpəq^wpaq^wt*, which is an unattested plural diminutive (imperfective) form; the 1:3 mapping (○) for one of the consonants incurs three violations, while the 1:2 mapping (●) incurs a single violation.

Table 5.3: INT_S Violations for Diminutive Plural Imperfective (Unattested)

	p ₁	i	ʔ	p ₁	ə	q ^w ₂	p ₁	a	q ^w ₂	t
1	○			○						
2	○						○			
3				○			○			
4						●			●	

Under a segment-based approach that limits fission by number, the plural diminutive gap in Halkomelem might be taken as evidence that the grammar will tolerate a maximum of two violations of INT_S at any given stratum: otherwise the alternate allomorph will surface. This approach would predicts that the *-l-* allomorph is employed to avoid “too many” INT_S violations.

Note that a fission-based approach is able to account for Halkomelem, only if it is integrated into a stratum-based model. If cast in a mono-stratal framework, as in Zimmermann (2021b), INT_S is not sufficient because plural imperfective forms actually allow a larger number of violations than unattested combinations do. For example, the word *wəq̇^w* ‘drift away’ has the plural imperfective form *həwhəẇq̇^w*, which in a mono-stratal analysis involves a 1:4 mapping between a single input segment /w/ and four output segments (2 * [h], 1 * [w], 2 * [ẇ]). Table 5.4 summarises the six violations that a plural imperfective would incur under INT_S; this is a greater number of violation marks than the unattested form in Table 5.3 above.

Table 5.4: INT_S Violations for Plural Imperfective

	h_1	$\emptyset$	w_1	h_1	$\emptyset$	$\dot{w}_1$	$\dot{q}^w$
1	○		○				
2	○			○			
3	○					○	
4			○	○			
5			○			○	
6				○		○	

The segment-based approach using INT_S is therefore only able to account for Halkomelem reduplication if the other components of my analysis, including Stratal OT and the proposed mappings between morphemes and strata, are adopted.

5.3.3.2.2 Segment-Based Binarity – Mellesmoen and Urbanczyk (2021)

Mellesmoen and Urbanczyk (2021b) assume the definition of INTEGRITY in (343); S₁ is the input and S₂ is the output.

(343) INTEGRITY: No element of S₁ has multiple correspondents in S₂.
(McCarthy & Prince, 1999, p.296)

They argue that counting INTEGRITY violations does not predict that the grammar will distinguish between single and multiple reduplication, nor between 1:2 or 1:3 input-output mappings if the number of INTEGRITY violations are the same.¹⁰ The grammar cannot select between a form with single reduplication and one with multiple reduplication in cases where the violations are equal.

They further contend that true 1:3 mappings are rare cross-linguistically. Where 1:3 mappings do appear on the surface, they explain that these can usually be accounted for in multiple steps of derivation, such as having fission occur at different strata. They propose that the limitation on multiple reduplication is better characterised as a limit on how fission may occur. Their Binarity restriction, given in (344), predicts only 1:2 mappings are possible; 1:3 mappings will never arise at a single stratum.

¹⁰Mellesmoen and Urbanczyk (2021b) adopt a different definition of INTEGRITY than the pairwise definition Zimmermann (2021) provides for INT_S (in 342 above).

This is a restriction on GEN and therefore no candidates with 1:3 mappings will be generated.

- (344) Binariness restriction on GEN: an element in the input can correspond to a maximum of two output elements.

(Mellesmoen & Urbanczyk, 2021b, p.176)

The Binariness restriction in (344) limits GEN to generating candidates with a maximally binary mapping between the input and the output (e.g., excluding 1:3 mappings/triplication at the same stratum). This means multiple reduplication is permitted when the reduplication occurs at different strata (e.g., Level 1 imperfective and Level 2 diminutive), but GEN is not able to generate forms with multiple reduplication at the same stratum (e.g., Level 2 diminutive and plural). The form in (345a) with a 1-to-3 mapping is not a possible candidate, and therefore will not be considered under EVAL.

- (345) a. *Impossible Candidate* (PL-DIM)

$$\sigma_{N+\mu_{PL}} + \sigma_{DIM} + p_1 a q^w_2 t$$

$$*p_1 i ? p_1 \partial q^w_2 p_1 a q^w_2 t$$

- b. *Attested Form* (PL-DIM)

$$-l_{-PL} + \sigma_{DIM} + p_1 a q^w t$$

$$p_1 \partial l i p_1 a q^w t$$

The segment-based binariness approach proposed in Mellesmoen and Urbanczyk (2021b) crucially assumes Stratal OT to derive the variable patterns, which is a departure from the mono-stratal framework used by Zimmermann (2021b); the segment-based binariness approach can therefore account for attested combinations and gaps in Halkomelem without additional modifications. However, both segment-based approaches will be shown to be insufficient for Salish reduplication based on patterns in Lillooet (Section 5.4).

5.3.3.3 Summary of Approaches

In this case study, I have presented an analysis of multiple reduplication in Halkomelem that correctly predicts the co-occurrence possibilities. Specifically, I

showed that multiple reduplication is impossible when the plural and diminutive morphemes co-occur; a non-reduplicative allomorph of the plural is used instead.

Overall, the case study of Halkomelem reduplication alone cannot be used to select between a morpheme-based or segment-based approach: all three approaches are sufficient for Halkomelem if they are considered within a stratum-based model of phonology.

5.4 Case Study II: Lillooet (Multiple Reduplication Gaps)

Lillooet provides an ideal test case for exploring attested and avoided multiple reduplication. First, multiple reduplication is well-documented (Van Eijk, 1997, 2013; H. Davis & Mellesmoen, 2023; H. Davis et al., in prep.). Second, the analysis of single reduplication in Lillooet that I provide in Chapter 4 (Section 4.2) can be used to make concrete predictions concerning possible combinations with respect to each stratum. Third, investigation is not limited to previous documentation: empirical predictions may be tested explicitly in elicitation, which furthers documentation and provides crucial negative evidence: fieldwork data confirms that gaps are systematic (ungrammatical), rather than accidental.

The data presented in this case study come from a combination of published sources and elicitation conducted by Henry Davis; examples from unpublished fieldwork are given with speaker initials.

5.4.1 Predictions for Multiple Reduplication in Lillooet

Assuming the morpheme-based approach to multiple reduplication proposed in Section 5.3.3.1, I predict that combinations involving two reduplicative morphemes at the same stratum will not be possible because MORPHEME-AFFILIATION has a high ranking; attested multiple reduplication patterns must involve reduplicative morphemes at different strata.

Reduplication in Lillooet is summarized in Table 5.5 (adapted from Section 4.2.2). Reduplicative affixes may receive segmental content at three strata: Level 1, Level 2, and Level 2+n. At Level 1, there are two types of affixed prosodic unit: a monomoraic syllable and a plain syllable. An affixed monomoraic syllable

yields property-denoting and plural C_1C_2 reduplication, while a plain syllable corresponds to initial C_1 reduplication. At Level 2, there are three affixed prosodic units: a mora with a floating [c.g.] feature (C_1 pluractional), a mora with a [stress] feature and a floating [c.g.] (C_1 diminutive), and a plain syllable (C_2 change-of-state). Triplication is at a later stratum (Level 2+n) and will be discussed separately in Section 5.4.6.

Table 5.5: Inventory of Lillooet Reduplication

Reduplication	Underlying Form	Stratum
Property-Denoting/Plural	$\sigma_{N+\mu}$	Level 1
Initial	σ	Level 1
Pluractional	$(\mu \text{ [c.g.]})_{L2}$	Level 2
Diminutive	$(\mu^{[\text{stress}]} \text{ [c.g.]})_{L2}$	Level 2
Change-of-State	σ_{L2}	Level 2
Triplication	Ft_{L2+n}	Level 2+n

I predict that no two Level 1 reduplicative morphemes (i.e., C_1 initial and C_1C_2 property-denoting reduplication) or two Level 2 reduplicative morphemes (i.e., C_1 diminutive/pluractional and C_2 change-of-state reduplication) will be combined in the same word. My predictions based on the analysis in Chapter 4 for possible multiple reduplication in Lillooet are summarized in Table 5.6. Shading is used to exclude redundant cells (e.g., C_2-C_1 C_2 and C_1 C_2-C_2 are the same combination).

Table 5.6: Predictions for Possible Lillooet Multiple Reduplication

		Level 1		Level 2	
		Property/Plural C_1C_2	Initial C_1	Dim/Pluractional C_1	COS C_2
Level 1	Property/Plural		*	✓	✓
	Initial			✓	✓
Level 2	Dim/Pluractional				*
	Change-of-State				

In this case study, I group C_1 diminutive and C_1 pluractional reduplication

together because they are identical aside from stressed vowel quality and are in complementary distribution (H. Davis & Mellesmoen, 2023).

5.4.2 Multiple Reduplication in Lillooet

Van Eijk (1997) describes diminutive plural and diminutive property-denoting forms (C_1C_2 and C_1 reduplication), as well as plural change-of-state forms (C_1C_2 and C_2 reduplication). However, this list does not include all possible combinations, which raises the question of whether their absence is a gap in description/documentation, or if these combinations are actually ungrammatical.

In the following sections (Sections 5.4.2.1–5.4.2.6), I describe multiple reduplication in Lillooet, providing either positive evidence (attested forms) or negative evidence (grammaticality judgments) to determine which combinations are possible. The positive evidence comes from a combination of published sources and elicitation conducted by Henry Davis; the negative evidence comes exclusively from fieldwork by Henry Davis. Where negative evidence is provided, I conclude that a combination is ungrammatical.

5.4.2.1 Attested: Level 1 C_1C_2 and Level 2 C_1 Reduplication

Plural C_1C_2 and diminutive C_1 reduplication occur together, as shown in (346).

- (346) a. s-**qəw**~qə<q>wəc (cf. *sqawc* ‘potato’)
 NMLZ-PL~potato<DIM>
 ‘little potatoes’ (Van Eijk, 1981a, p.57)
- b. s-**qəx**~qə<q>xəʔ (cf. *sqəxəʔ* ‘dog’)
 NMLZ-PL~dog<DIM>
 ‘pups’ (Van Eijk, 1997, p.59)
- c. **ʔək**~ʔə<ʔ>k=kaʔ (cf. *ʔək* ‘go along’)
 PL~go.along<DIM>=1 PL.SUBJ
 ‘We went along a little ways’ (CA)
- d. **təw**~tə<t>wən (cf. *təwən* ‘visit’, from English *town*)
 PL~visit<DIM>
 ‘go visiting’ (CA)

- e. $q^w\acute{a}l\sim q^wal\acute{o}<\acute{l}>t$ (cf. $q^wal\acute{u}t$ ‘speak’)
 PL $\sim$ speak<DIM>
 ‘have a conversation’ (CA)

Property-denoting C_1C_2 reduplication and C_1 diminutive reduplication can also be combined, as shown in (347).

- (347) a. $s-\acute{x}\acute{a}q^w\sim\acute{x}\acute{o}<\acute{x}>q^w$ (cf. $s\acute{x}\acute{a}q^w$ ‘to have a spot on one’s body’)
 STAT-PDEN $\sim$ spot<DIM>
 ‘to have little spots, variegated’ (Van Eijk, 1997, p.150)
- b. $k\acute{o}<k\acute{o}>\acute{a}n\sim k\acute{n}-\acute{a}t$ (cf. $k\acute{i}nk\acute{a}nt$ ‘dangerous’)
 PDEN<DIM> $\sim$ dangerous-CHAR
 ‘a little bit dangerous’ (CA)
- c. $z\acute{o}<z\acute{o}>\acute{a}w\sim z\acute{w}-\acute{a}t$ (cf. $z\acute{a}wz\acute{w}at$ ‘annoying’)
 PDEN<DIM> $\sim$ annoy-CHAR
 ‘a little bit annoying’ (CA, vf)
- d. $q^w\acute{o}<q^w\acute{o}>\acute{a}l\sim q^w\acute{a}l\acute{t}$ (cf. $q^wal\acute{q}^w\acute{a}l\acute{t}$ ‘hurting’)
 PDEN<DIM> $\sim$ hurt-CHAR
 ‘hurting a little bit’ (CA, vf)
- e. $k\acute{o}<k\acute{o}>\acute{a}w\sim k\acute{w}-\acute{a}t$ (cf. $k\acute{a}wk\acute{w}at$ ‘gross, filthy’)
 PDEN<DIM> $\sim$ dirty-CHAR
 ‘a little bit gross/filthy’ (CA, vf)

Level 2 C_1 reduplication must follow Level 1 C_1C_2 reduplication. Table 5.7 shows two ordering possibilities for multiple reduplication: C_1C_2 reduplication (plural: $\sigma_{N+\mu}$) before C_1 reduplication (diminutive: $\mu^{[stress]}$ [c.g.]) in the attested column, and C_1 reduplication before C_1C_2 reduplication in the ungrammatical column. If the order were reversed, an ungrammatical form like $*sq\acute{a}qq\acute{a}xa?$ would be expected, where the diminutive is copied by subsequent plural reduplication.

Table 5.7: Multiple Reduplication with C₁C₂ (Level 1) and C₁ (Level 2)

	PL-DIM	
	Attested	Ungrammatical
Input	sqáxaʔ	sqáxaʔ
Morpheme 1	σ _{N+μ} + sqáxaʔ	μ ^[stress] [c.g.] + sqáxaʔ
Morpheme 2	μ ^[stress] [c.g.] + sqəxqáxaʔ	σ _{N+μ} + sqəqxaʔ
Output:	sqəxqəqxaʔ	*sqəqəqəqxaʔ

The diminutive plural and diminutive property-denoting forms can be derived using the analysis in Chapter 4. Further, the multiple reduplication patterns provide evidence for Level 1 reduplication (property-denoting/plural) applying before Level 2 (diminutive) reduplication. If the order of application had been reversed, the C₁C₂ reduplication process would result in a copy of the diminutive C₁, as shown in the “ungrammatical” column in Table 5.7.

5.4.2.2 Attested: Level 1 C₁C₂ and Level 2 C₂ Reduplication

Plural C₁C₂ reduplication and change-of-state C₂ reduplication can be combined in Lillooet. The examples in (348) show plural change-of-state forms, marked with two types of reduplication.

- (348) a. **pə́t**~pú<**tə**>t (cf. *pú́tə́t* ‘to be boiling’)
 PL~boil<COS>
 ‘rapids’ (Van Eijk, 1997, p.57)
- b. **cəq**~cí<**qə**>q (cf. *cíqín* ‘to stab s.o.’)
 PL~stab<COS>
 ‘to get stabbed all over’ (Van Eijk, 1997, p.57)
- c. **cək^w**~ck^w-á<**n**>naʔ=łkan (cf. *cək^wən* ‘pull something’)
 PL~pull-ear<COS>=1 SG.SUBJ
 ‘Both my ears got pulled.’ (CA)
- d. **ḵəp**~ḵú<**pə**>p (root: *ḵup*- ‘twist, wrap’)
 PL~twist<COS>
 ‘twist or wrap (pl.)’ (H. Davis et al., in prep.)

- e. $\dot{\text{h}}\acute{\text{a}}\text{ŋ}^{\text{w}} \sim \text{t} < \text{ŋ}^{\text{w}} \acute{\text{a}} > \text{ŋ}^{\text{w}}$ (root: hŋ^{w} - ‘jump’)
 PL $\sim$ bounce/hop <COS>
 ‘bound, hop or jump, of animal (pl.)’ (H. Davis et al., in prep.)

The combination of Level 1 C_1C_2 and Level 2 C_2 reduplication is predicted because the reduplication occurs at different strata. The form of this type of multiple reduplication is also consistent with the proposed order of application (Level 1 before Level 2): C_1C_2 reduplication is applied first, followed by the C_2 reduplication, which is positioned relative to stress.

5.4.2.3 Attested: Level 1 C_1 and Level 2 C_1 Reduplication

Double C_1 reduplication involving Level 1 (initial) and Level 2 (diminutive/plural) reduplication is attested in Lillooet, as shown in (349). This combination has received little previous description; examples with diminutive C_1 reduplication are given under “remaining cases” in Van Eijk (1997, p.57).

- (349) a. $\dot{\text{x}}\text{ak}$ $\text{k}^{\text{w}}\text{u}=\text{k}\acute{\text{a}} \sim \text{k}\acute{\text{a}} < \text{k} > \acute{\text{a}}\text{w}$ (cf. *nkáwlax* ‘go far from shore’)
 go.along DET=INIT. $\text{C}_1 \sim$ far <DIM>
 ‘Go a little further!’ (CA)
- b. $\text{x}\acute{\text{a}} \sim \text{x}\text{z}\acute{\text{a}} < \text{z} > \acute{\text{a}}\text{m}$ ʔizá (cf. *xzum* ‘big’)
 INIT. $\text{C}_1 \sim$ big <DIM> DEM
 ‘These ones are a little bigger.’ (CA)
- c. $\text{x}^{\text{w}}\text{ʔit}$ $\text{ʔi}=\text{s}-\text{q}^{\text{w}}\acute{\text{a}} \sim \text{q}^{\text{w}}\text{y}\acute{\text{a}} < \text{y} > \text{c}=\text{a}$
 many PL.DET=NMLZ-INIT. $\text{C}_1 \sim$ rabbit <DIM>=EXIS
 $\dot{\text{q}}^{\text{w}}\text{u}\text{ʔ}-\text{xít}-\text{an}$
 trap-IND-1 SG.ERG
 ‘I trapped a lot of little rabbits.’ (cf. *sqʷ[ə]yíc* ‘rabbit’) (CA)

The combination of initial C_1 and diminutive/pluractional C_1 reduplication is predicted to be possible as it involves reduplication at two strata. The two types of C_1 reduplication are Level 1 (initial) C_1 reduplication and Level 2 (diminutive/pluractional) C_1 reduplication.

5.4.2.4 Attested: Level 1 C₁ and Level 2 C₂ Reduplication

There are only a few cases that can be elicited with a combination of Level 1 (initial) C₁ and Level 2 (change-of-state) C₂ reduplication (H. Davis, p.c.), such as the one in (350c). This pattern has not been described in previous work.

(350) a. *Change-of-State C₂ Reduplication* (cf. *ɣzum* ‘big’)

ɣzú<mə>m=tu? ʔi=sít-k-s=a ta=s-tłáyn=a
big<COS>=REM PL.DET=mesh-3POSS=EXIS DET=NMLZ-net=EXIS
‘The meshes of the net got bigger.’ (CA)

b. *Initial C₁ Reduplication*

ɣə~ɣzú ʔi=sít-k-s=a ta=s-tłáyn=a
INIT.C₁~big PL.DET=mesh-3POSS=EXIS DET=NMLZ-net=EXIS
‘The meshes of the net were big.’ (CA)

c. *Change-of-State C₂ and Initial C₁ Reduplication*

ɣə~ɣzú<mə>m ʔi=sít-k-s=a
INIT.C₁~big<COS> PL.DET=mesh-3POSS=EXIS
ta=s-tłáyn=a
DET=NMLZ-net=EXIS
‘The meshes of the net got bigger.’ (CA)

The acceptability of (350c) suggests that the two types of reduplication are phonologically compatible. The scarcity of examples is due to the very limited distribution of initial C₁ reduplication in Lillooet, which restricts the number of possible forms with which C₂ reduplication could combine (H. Davis & Mellesmoen, 2023).

5.4.2.5 Ungrammatical: Level 1 C₁C₂ and Level 1 C₁ Reduplication

The combination of C₁C₂ plus initial C₁ reduplication has not been described in previous documentation. Elicitation confirms that this combination is judged as ungrammatical by consultants, as shown in (351) and (352).

- (351) a. *Initial C₁* (cf. *q^walút* ‘speak’)
- papt wa? **q^wə~q^walút** ti? k^wu=s-qáyx^w
 always IPFV INIT.C₁~speak that.VIS DET=NMLZ-man
 ‘That man is always talking.’ (CA)
- b. *Plural C₁C₂*
- papt wa7 **q^wəl~q^walút** ?iz k^wu=s-qáyx^w
 always IPFV PL~talk those.VIS DET=NMLZ-man
 ‘Those men are always talking.’ (CA)
- c. *Rejected: Initial C₁ + Plural C₁C₂*
- *papt wa? **q^wə~q^wəl~q^walút** ?iz k^wu=s-qáy~qəyx^w
 always IPFV INIT.C₁~PL~talk those.VIS DET=NMLZ-PL~man
 Intended: ‘Those men are always talking.’ (CA)
- d. *Rejected: Plural C₁C₂ + Initial C₁*
- *papt wa? **q^wəl~q^wə~q^walút** ?iz k^wu=s-qáy~qəyx^w
 always IPFV PL~INIT.C₁~talk those.VIS DET=NMLZ-PL~man
 Intended: ‘Those men are always talking.’ (CA)
- (352) a. *Initial C₁* (root: *tał-* ‘stand’)
- wa? s-**tał**~tał-ləx ?i=twə<w>wət=a
 IPFV STAT-INIT.C₁~stand-AUT PL.DET=young.person<DIM>=EXIS
 ‘The boys are all standing.’ (CA)
- b. *Plural C₁C₂*
- wa? s-**təł**~táł-ləx ?i=twíwt=a
 IPFV STAT-PL~stand-AUT PL.DET=young.person=EXIS
 ‘The young people are all standing.’ (CA)
- c. *Rejected: Initial C₁ + Plural C₁C₂*
- *wa? s-**ta**~**təł**~táł-ləx ?i=twíwt=a
 IPFV STAT-INIT.C₁~PL~stand-AUT PL.DET=young.person=EXIS
 Intended: ‘The young people are all standing around.’ (CA)

d. *Rejected: Plural C₁C₂ + Initial C₁*

*waʔ s-**təɫ**~**tá**~taɫ-ləx ʔi=twíwt=a
 IPFV STAT-PL~INIT.C₁~stand-AUT PL.DET=young.person=EXIS
 Intended: ‘The young people are standing around.’ (CA)
 Corrected to: s-tá~taɫ-ləx
 STAT-INIT.C₁~stand-AUT (CA)

The combination of initial C₁ and C₁C₂ reduplication is rejected, or corrected to a single type of reduplication, as in (352d). The combination of initial C₁ plus C₁C₂ reduplication is predicted to be impossible, given that it involves two reduplicative morphemes at the same stratum (Level 1).

5.4.2.6 Ungrammatical: Level 2 C₁ and Level 2 C₂ Reduplication

The combination of Level 2 C₁ and C₂ reduplication has not been previously described; elicitation confirms a systematic gap because forms with both diminutive C₁ reduplication and change-of-state C₂ reduplication are rejected (H. Davis & Mellesmoen, 2023), as shown in (353).

- (353) a. *má<**m**><**lə**>ɫ (cf. *múləl* ‘It got flooded.’)
 flood<DIM><COS>
 Intended: ‘It got a little bit flooded.’ (CA)
- b. *xzá<**z**><**mə**>ɪ (cf. *xzúməm* ‘It got bigger.’)
 big<DIM><COS>
 Intended: ‘It got a little bit bigger.’ (CA)
- c. *ká<**k**><**ɲə**>ɲ (cf. *káɲəɲ* ‘It got bumped.’)
 bump<DIM><COS>
 Intended: ‘It got a bit bumped.’ (CA)

There is no reason to suspect that Level 2 C₁ reduplication would be semantically incompatible with change-of-state C₂ reduplication. In fact, in a context where the semantics associated with both diminutive and change-of-state reduplication are appropriate, the consultant’s paraphrase includes C₂ reduplication and the word *k^wik^ws* ‘small’, as shown in (354).

- (354) a. i. *nó<**ḥ**><**sə**>s (cf. *núsəs* ‘It got damp.’)
 damp<DIM><COS>
 Intended: ‘It got a little bit damp.’ (CA)
- ii. **k^wik^ws** k^w=s=nus=c
 small DET=NMLZ=damp=3POSS
 ‘It got a little bit damp.’ (CA)
- b. i. ***ḥ**ó<**ḥ**><**pə**>p (cf. *ḥúpəp* ‘It twisted.’)
 twist<DIM><COS>
 Intended: ‘It twisted a little bit.’
 Corrected to: **k^wik^ws** k^w=s=**ḥ**ú<**pə**>p=s
 small DET=NMLZ=twist<COS>=3POSS
 ‘It twisted a little bit.’ (CA, vf)

There is also another way to mark a change of state, which has a near-identical semantic contribution to C₂ reduplication (H. Davis, in prep): the inchoative marker, which either takes the form of an infix -ʔ- (with roots containing a moraic vowel) or a suffixed -p (with roots without an underlying vowel). The example in (355) shows diminutive C₁ reduplication plus the inchoative affix -p.

- (355) pḥ<**p**>**m-əp** tiʔ k^wu=káḥ (cf. *pəmp* ‘fast’)
 fast<DIM>-INCH DEM DET=car
 ‘That car is a little faster.’ (CA, vg)

Where the inchoative marker -p is used instead of C₂ reduplication, there is nothing blocking the addition of diminutive reduplication. This shows the compatibility of diminutive and change-of-state semantics, and supports a phonological source for the restriction on multiple reduplication.

My analysis predicts the ungrammaticality of pluractional/diminutive plus change-of-state reduplication: both are Level 2 reduplicative morphemes and therefore cannot surface at the same time.

5.4.3 Summary: Multiple Reduplication in Lillooet

Table 5.8 shows a summary of multiple reduplication in Lillooet; a ✓ is used to mark possible combinations and × is used to mark incompatible pairings; the results match the predictions given in Table 5.6 above.

Table 5.8: Lillooet Multiple Reduplication Inventory

		Level 1		Level 2	
		C ₁ C ₂	C ₁	C ₁	C ₂
Level 1	C ₁ C ₂		×	✓	✓
	C ₁			✓	✓
Level 2	C ₁				×
	C ₂				

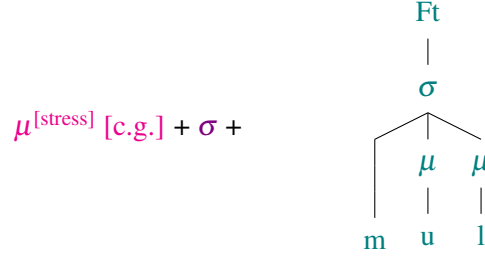
The possible multiple reduplication patterns in Lillooet all involve the combination of a Level 1 and Level 2 morpheme. Rejected combinations involve two reduplicative morphemes that would receive segmental content at the same stratum.

5.4.4 Analysis: Multiple Reduplication in Lillooet

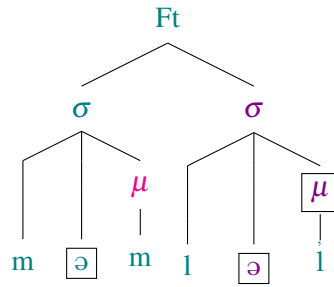
The morpheme-based approach proposed in Section 5.3.3.1 can account for both permitted and impossible patterns in Lillooet. All permitted combinations of reduplicative morphemes in Lillooet involve reduplication at different strata. Where two types of reduplication would occur at the same stratum, multiple reduplication is not possible.

Both diminutive C₁ and change-of-state C₂ reduplication are Level 2 affixes, which makes them incompatible. The diagram in (356) shows the structure of the ungrammatical form **mámləl*, which was intended to mean ‘it got a little bit flooded’: see (353a) above. The unattested output in (356b) will violate MORPHEME-AFFILIATION because segments are associated with prosodic units associated with two other morphemes; colours show that teal segments are associated with pink and purple prosodic units, in addition to teal ones.

(356) a. *Level 2 Input:*



b. (*Ungrammatical*) *Level 2 Output:*



While Halkomelem uses a non-reduplicative allomorph to avoid multiple reduplication (see Section 5.3.2.3), Lillooet lacks an alternative “escape hatch” in the form of a non-reduplicative allomorph; there is a paradigmatic gap where combining the two morphemes is simply ungrammatical. Speakers must use a different morpheme (e.g., inchoative $-\text{?}/-p$) or independent word (e.g., $k^w ik^w s$ ‘small’). It is beyond the scope of the present case study to analyse the choice of alternative forms; I instead provide an analysis that accounts for the fact that forms with multiple reduplication at the same stratum are avoided by using a “null” or “gap” candidate, represented by $\odot$ (McCarthy & Wolf, 2007, p.1). The ranking of a constraint MPARSE, defined in (357), sets a “harmony threshold” (Legendre, Smolensky, & Wilson, 1998, p.8). Any constraint ranked above MPARSE will never be violated: $\odot$ will always be preferable. Paradigmatic gaps thus occur when GEN can produce no possible candidate that will fare better than the null parse $\odot$.

(357) MPARSE: Given a candidate $(o, \langle i \rangle, \langle o \rangle, \mathfrak{A})$, if $\mathfrak{A}$ is not a total bijective function from $\langle i \rangle$ to $\langle o \rangle$, assign a violation mark. (McCarthy & Wolf, 2007, p.14)

The definition of MPARSE in (357) considers ordered 4-tuples consisting of the candidate (o), the concatenative decomposition of the input (<i>), the concatenative decomposition of the output (<o>), and a relation between the input and the output ($\mathfrak{R}$). Crucially, the correspondence $\mathfrak{R}$ between the input and output is over strings, rather than segments; a monosegmental string S in the input can correspond to a null string # in the output. While MPARSE requires a bijective (1-to-1) mapping between input and output, this can be satisfied even when processes such as deletion (<i> = S, <o> = #), epenthesis (<i> = #, <o> = S), fission (<i> = S, <o> = SS), and coalescence (<i> = SS, <o> = S) have occurred. The null candidate $\odot$ has nothing in <o>, which means there is no relation $\mathfrak{R}$ between <i> and <o>: $\odot$ will always violate MPARSE.

MORPHEME-AFFILIATION must be ranked above MPARSE to predict the null candidate will be selected over one with multiple reduplication (at a given stratum). This ranking is shown in (358).

$/\mu + \sigma + C_1VC_2/$		MAFFIL	MPARSE
(358)	a. $\odot$		*
	b. $C_1VC_1C_2VC_2$	*!	

Given that the input for the diminutive and pluractional C_1 reduplication has a floating [c.g.] feature in addition to a μ , MAX_μ must also be ranked above MPARSE to rule out a candidate where the diminutive is realised just as glottalisation on a sonorant (see Section 4.2.4.3). It is better to pick the gap candidate in (359a) over the one that deletes the mora, realising the diminutive by just the [c.g.] feature in (359b). Candidate (359c) fatally violates MORPHEME-AFFILIATION because it has the structure shown for the ungrammatical form **məmləl* in (356b) above.

$/\mu^{[stress]} [c.g.] + \sigma + mul/$		MAFFIL	MAX_μ	MPARSE	$MAX_{[c.g.]}$
(359)	a. $\odot$			*	
	b. <i>muləl</i>		*!		
	c. <i>məmləl</i>	*!			

The morpheme-based approach is therefore sufficient for the analysis of multiple reduplication in Lillooet. MORPHEME-AFFILIATION has a high ranking

in Lillooet, as in Halkomelem. While Halkomelem is able to avoid multiple reduplication through selecting another allomorph, Lillooet cannot generate any output that is more harmonic than a null candidate $\odot$. In other words, violations of MPARSE are tolerated if the alternative would be to generate a candidate with multiple reduplication at the same stratum.

5.4.5 Evaluating Alternative Approaches: Multiple Reduplication

Restricting multiple reduplication by morpheme affiliation predicts that multiple reduplication will only be possible if the reduplication happens at different strata; this is compatible with the data. A stratum-based phonology is needed to account for the patterns. I therefore evaluate the segment-based alternatives assuming the morpheme-to-stratum mappings proposed in this chapter.

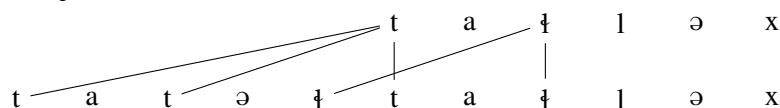
While the Binariness restriction proposed by Mellesmoen and Urbanczyk (2021b), restated in (360), was viable for the analysis of Halkomelem, it is not sufficient for multiple reduplication in Lillooet.

- (360) Binariness restriction on GEN: an element in the input can correspond to a maximum of two output elements.

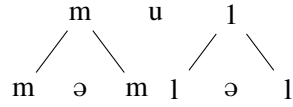
(Mellesmoen & Urbanczyk, 2021b, p.176)

The combination of C_1 and C_1C_2 reduplication at Level 1 is correctly ruled out by the Binariness restriction because it yields a 1:3 (ternary) mapping, as shown in (361a); however, the Binariness restriction offers no explanation for why Level 2 C_1 and C_2 reduplication patterns are also not possible, despite targeting different segments (i.e., two separate 1:2 mappings). The unattested form in (361b) is permitted under the Binariness restriction, which means this approach is not sufficient for Lillooet.

- (361) a. *Ungrammatical Combination: Level 1 C_1 and Level 1 C_1C_2 Reduplication*



- b. *Ungrammatical Combination: Level 2 C₁ and Level 2 C₂ Reduplication*



The segment-based fission approach proposed by Zimmermann (2021b) does not perform any better than the segment-based binarity approach: plural C₁C₂ reduplication receives two violations of INT_S on its own, as shown in (362a). The ungrammatical combination with Level 2 morphemes also incurs two violations, as shown in (362b).

- (362) a. INT_S Violations for Single Reduplication (Plural)

	q ^w ₁	ə	l ₂	q ^w ₁	a	l ₂	u	t
1	○			○				
2			●			●		

- b. INT_S Violations for Ungrammatical Level 2 Combination

	m ₁	ə	m ₁	l ₂	ə	l ₂
1	○		○			
2				●		●

A segment-based fission approach is unable to explain why a single instance of C₁C₂ reduplication, which copies two consonants, is tolerated, while a combination of diminutive/plural C₁ plus change-of-state C₂ reduplication is rejected. Both combinations would receive the same number of violations under INT_S.

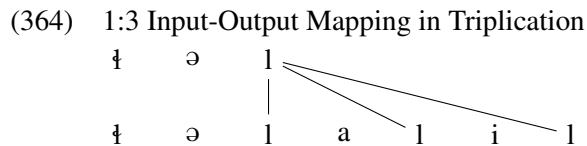
Neither the fission or binarity approaches to multiple reduplication are sufficient for Lillooet; only the morpheme-based approach is able to account for both attested and ungrammatical multiple reduplication patterns in Lillooet.

5.4.6 Evaluating Alternative Approaches: Triplication

The Binariness restriction proposed by Mellesmoen and Urbanczyk (2021b) is not only unable to rule out one of the two ungrammatical combinations (C_1 and C_2): it also predicts that triplication should be impossible because it yields a 1:3 input-output mapping. Examples of triplication, which were discussed in Chapter 4 (Section 4.2.5), are restated in (363).

- (363) a. $\text{ʔ} < \text{ə} \text{la} \text{l} > \text{ɪ} \text{l}$ (cf. *ʔilɪn* ‘to sprinkle s.t.’)
 sprinkle <TRIP>
 ‘to keep sprinkling’ (Van Eijk, 1997, p.56)
- b. $\text{p} < \text{ə} \text{t} \text{a} \text{t} > \text{ə} \text{t}$ (root: $\text{p}(\text{ə})\text{t-}$)
 bubble <TRIP>
 ‘to make a bubbling, gurgling noise’ (Van Eijk, 2013, p.4)

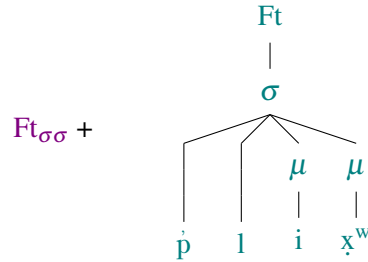
The diagram in (364) shows how a single input segment is associated with three output segments in *ʔəlaɪlɪl* ‘to keep sprinkling’.



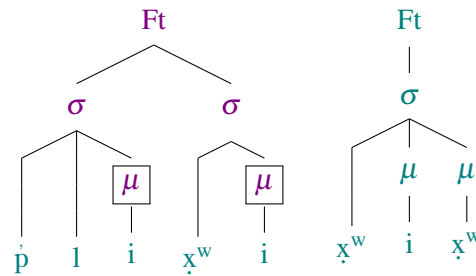
In Chapter 4 (Section 4.2.5), I argue that the input for triplication is a foot, which means that it is an instance of single reduplication, which is filled at a single stratum. The Binariness restriction is not able to differentiate between single reduplication resulting in a 1:3 mapping, as seen in triplication, and multiple reduplication resulting in a 1:3 mapping. The segment-based binarity approach therefore predicts that GEN should be unable to create candidates with triplication, which is contrary to observed patterns.

A morpheme-based approach using MORPHEME-AFFILIATION allows for triplication. While the triplicated word *plix^wix^wix^w* ‘keep overflowing’ in (365) has three consonants that correspond to the same input segment, all segments are associated with prosodic units that share their morpheme affiliation (denoted by teal) or that of one other morpheme (denoted by purple). Triplication does not violate MORPHEME-AFFILIATION.

(365) a. *Input:*



b. *Output:*



Only the morpheme-based restriction on multiple reduplication proposed in this dissertation can distinguish between attested triplication and ungrammatical multiple reduplication patterns that copy the same number of segments; the Binarity restriction cannot account for Lillooet reduplication.

5.4.7 Summary: Morpheme vs. Segment Restrictions

Lillooet allows multiple reduplication when the two reduplicative morphemes are realised at different points in the derivation. Differences in the number of fissioned segments or the type of input-output mapping (i.e., binarity) cannot differentiate attested multiple reduplication patterns from ungrammatical ones. While both segment-based approaches were sufficient for Halkomelem when considered within Stratal OT, neither is compatible with Lillooet.

Possible and impossible combinations of reduplicative morphemes in Lillooet are only predicted by an analysis that restricts multiple reduplication with reference to morphemes. Alternative approaches that propose limitations based on permissible types or amount of fission are at odds with attested patterns.

Having established the inadequacy of alternative approaches, I now turn to Comox-Sliammon to evaluate whether the morpheme-based approach makes the

correct predictions for a language with all four core patterns (C_1C_2 , C_1 , C_2 , and V_1).

5.5 Case Study III: Comox-Sliammon

If a morpheme-based approach to multiple reduplication is to be extended to the rest of Salish, it must be compatible with languages with the full range of core patterns (C_1C_2 , C_1 , C_2 , and V_1). The third case study therefore examines multiple reduplication in Comox-Sliammon, a language with a large inventory of reduplicative morphemes, which are parsed into the word at three different levels (see the case study of single reduplication in Section 4.1). Data presented in this case study come from my own elicitation unless otherwise noted; examples from fieldwork are given with speaker initials.

5.5.1 Predictions for Multiple Reduplication in Comox-Sliammon

If MORPHEME-AFFILIATION has a high ranking in Comox-Sliammon, as in Halkomelem and Lillooet, then combinations of two prosodic affixes at the same stratum are predicted to be absent.

Reduplication in Comox-Sliammon is summarised in Table 5.9, adapted from Table 4.1 given in Chapter 4 (Section 4.1.2). Reduplicative morphemes are parsed into the prosodic word at one of three strata: Level 1 (diminutive/plural C_1 and property-denoting C_1C_2 reduplication), Level 2 (plural C_1C_2 and progressive C_1 reduplication), and Level 3 (plural V_1 and change-of-state C_2 reduplication).

Table 5.9: Inventory of Comox-Sliammon Reduplication

Reduplicative Morpheme	Underlying Form	Stratum
Diminutive C_1	$\mu_1 > i_2$	Level 1
Plural C_1	$\mu_1 > a_2$	Level 1
Property-Denoting C_1C_2	$\sigma_{N+\mu}$	Level 1
Plural C_1C_2	$(\sigma_{N+\mu})_{L2}$	Level 2
Progressive C_1	μ_{L2}	Level 2
Plural V_1	$(\mu [c.g.])_{L3}$	Level 3
Change-of-State C_2	σ_{L3}	Level 3

Diminutive and plural C₁ reduplication in Comox-Sliammon, which are parsed at Level 1, have two allomorphs: a mora or a fixed segment. MORPHEME-AFFILIATION is violated when multiple prosodic affixes are parsed into the prosodic word at the same stratum; a morpheme-based approach using this constraint therefore makes different predictions than fission-based approaches (discussed in 5.3.3.2 and 5.4.5) with respect to reduplication associated with an underlying fixed segment. Using MORPHEME-AFFILIATION to restrict multiple reduplication predicts that two reduplicative morphemes at Level 1 could occur in the same word if at least one selects the fixed segment allomorph. In contrast, fission associated with underlying prosodic units or fixed segments will be evaluated identically by INT_S (Zimmermann, 2021b) or the Binariness restriction on GEN (Mellesmoen & Urbanczyk, 2021b).

Multiple reduplication at Level 2 and Level 3 is predicted to be impossible because all types of reduplication at these strata involve underlying prosodic affixes. These predictions are summarised in Table 5.10; I use ✓ to indicate combinations that are predicted to be possible, * to indicate combinations that are predicted to be impossible, and % to indicate combinations that are predicted to be possible *only* if a fixed segment allomorph is selected. Shading is used to mark redundant cells (i.e., the information in Level 1-Level 2 and Level 2-Level 1 cells is the same).

Table 5.10: Predictions for Comox-Sliammon Multiple Reduplication

	Level 1	Level 2	Level 3
Level 1	%	✓	✓
Level 2		*	✓
Level 3			*

As reduplicative morphemes are parsed into the word across three strata in Comox-Sliammon, an analysis using MORPHEME-AFFILIATION to restrict multiple reduplication also predicts that combinations of more than two reduplicative morphemes in the same word are phonologically possible provided each is parsed at a different stratum.

5.5.2 Combinations Excluded

MORPHEME-AFFILIATION predicts that multiple reduplication that involves two prosodic affixes at the same stratum should be impossible. In this section, I focus on (i) attested patterns, and (ii) combinations predicted to be impossible under MORPHEME-AFFILIATION (i.e., combinations at the same stratum). I set aside unattested or marginal/unclear combinations that include reduplicative morphemes at two different strata, as they are not crucial for testing the predictions of the morpheme-based restriction on multiple reduplication: these combinations are marked with – in Table 5.11. Cells in gray are redundant (i.e., they involve two instances of same type of reduplication or contain the same information as another cell).

Table 5.11: Excluded Pairs

		Level 1			Level 2		Level 3	
		PDEN C ₁ C ₂	DIM C ₁	PL C ₁	PL C ₁ C ₂	PROG C ₁	COS C ₂	PL V ₁
Level 1	PDEN				–	–	–	–
	DIM				–		–	
	PL				–	–	–	–
Level 2	PL						–	
	PROG							
Level 3	COS							
	PL							

Even though I set aside 11/21 possible combinations, I consider every same-stratum combination, which are the crucial cases (i.e. the ones that are relevant for MORPHEME-AFFILIATION).¹¹ I also consider at least one potential combination of reduplicative morphemes for each unique combination of strata (i.e., L1-L1, L1-L2, L1-L3, L2-L2, L2-L3, L3-L3, L1-L2-L3).

¹¹Unattested/ungrammatical combinations that involve reduplication at different strata are not evidence for or against morpheme-based limitations on multiple reduplication.

5.5.3 L1-L1 Combinations

5.5.3.1 Attested (Allomorphy): Level 1 C₁ and Level 1 C₁C₂ Reduplication

Examples in (366) show that diminutive C₁ reduplication co-occurs with property-denoting C₁C₂ reduplication; these forms are formed with the *i*- allomorph of the diminutive, which consists of only a fixed segment, instead of the prosodic affix (μ).

- (366) a. $\dot{x}i \sim \dot{x}\dot{\epsilon}t \sim \dot{x}\dot{\epsilon}t$ (cf. $\dot{x}\dot{\epsilon}t$ ‘get angry’)
 DIM~PDEN~angry
 ‘a little angry’ (FL, sf)
- b. $\dot{s}i \sim \dot{s}\dot{\epsilon}m \sim \dot{s}\dot{\epsilon}m$ (cf. $k^w u \dot{s}\dot{\epsilon}m$ ‘It got dry.’)
 DIM~PDEN~dry
 ‘a little dry’ (FL, vf)
- c. $\dot{x}i \sim \dot{x}\dot{\epsilon}m \sim \dot{x}\dot{\epsilon}m$ $t\dot{\epsilon}=?apust\dot{\epsilon}n$ (cf. $\dot{x}\dot{\epsilon}m$ ‘get wet’)
 DIM~PDEN~wet DET=towel
 ‘The towel is a little wet/damp.’ (BW, vf)
- d. $\dot{t}^{\theta}i \sim \dot{t}^{\theta}\dot{\epsilon}y \sim \dot{t}^{\theta}ay$ (cf. $\dot{t}^{\theta}\dot{\epsilon}y\dot{t}^{\theta}ay$ ‘loud’)
 DIM~PDEN~loud
 ‘a little loud’ (FL, sf/vt)
- e. $\dot{x}i \sim \dot{x}\dot{\epsilon}s \sim \dot{x}\dot{\epsilon}s\dot{\epsilon}m$ (cf. $\dot{x}\dot{\epsilon}s\dot{\epsilon}m$ ‘strong’)
 DIM~PDEN~strong
 ‘kinda strong’ (FL, sf/vt)

As MORPHEME-AFFILIATION is only violated when fission provides segmental content to two prosodic affixes at the same stratum, the combination of two Level 1 morphemes is predicted to be possible, if the fixed segment allomorph is selected (see Section 5.5.1); the attested forms in (366) are consistent with this prediction.

Plural C₁ reduplication and property-denoting C₁C₂ reduplication can also be combined. Both types of reduplication are parsed at Level 1, but multiple reduplication is formed with the *a*- allomorph of the plural, rather than μ . Examples are shown in (367).¹²

¹²Thank you to Marianne Huijsmans for providing these examples.

- (367) a. hihiw **ḵa~ḵəs~ḵasəm** tə=təq~tiqiw (cf. *ḵasəm* ‘strong’)
 really PL~PDEN~strong DET=PL~horse
 ‘Those horses are really strong.’ (EP)
- b. hihiw **t^θa~t^θəy~t^θay-mut** (cf. *t^θəyt^θay* ‘loud’)
 really PL~PDEN~noisy-INTS
 ‘They’re really noisy.’ (EP)
- c. hihiw **ḵa~ḵəṁ~ḵəm-mut** tə=təm~tumiš (cf. *ḵəṁ* ‘get wet’)
 really PL~PDEN~wet-INTS DET=PL~man
 ‘Those men are really wet.’ (EP)

The fixed segment affix *a-* is seen in plural properties in (367), parallel to *i-* in diminutive properties in (366); these combinations of two Level 1 morphemes are possible, since they involve only one prosodic affix (i.e., property-denoting $\sigma_{N+\mu}$). Multiple reduplication and allomorph selection at Level 1 are analysed further in Section 5.5.10.

5.5.3.2 Attested (Allomorphy): Level 1 C₁ and Level 1 C₁ Reduplication

Plural C₁ and diminutive C₁ reduplication can be combined, as shown in (368).¹³ The fixed segment allomorph is selected for both morphemes. The plural and diminutive may occur in either order for some words, as shown in (368a–b), though not in others, as shown in (368c).

- (368) a. **pi~pa~pḵ-it** sayayaḵan-s qəsnaɣ-s (✓ **pa~pi~pḵ-it**)
 DIM~PL~rip-STAT elbow-3POSS shirt-3POSS (PL~DIM~rip-STAT)
 ‘The sleeves of the shirt are ripped a little.’ (root: *pḵ-* ‘rip’) (FL, vf)
- b. **ki~ka~kp-it** (✓ **ka~ki~kp-it**)
 DIM~PL~cut-STAT (PL~DIM~cut-STAT)
 ‘has small little cuts’ (root: *kp-* ‘cut’) (FL, sf)
- c. **ḵa~ḵi~ḵm-it** (* **ḵi~ḵa~ḵm-it**)
 PL~DIM~wet-STAT (DIM~PL~wet-STAT)
 ‘They are a little wet.’ (root: *ḵm-* ‘wet’) (FL, sf/vt)

I have not elicited any examples of plural C₁ and diminutive C₁ reduplication with a root with a moraic vowel; each type of reduplication would select a prosodic

¹³Thank you to Marianne Huijsmans for eliciting these examples for me.

affix (μ) in this context (see Section 4.1.3.1).¹⁴ However, an example of a word with a moraic vowel that takes lexicalised C_1 (diminutive) reduplication with the μ allomorph may further take the a - allomorph of plural C_1 reduplication is shown in (369).

- (369) **?a~?u<?>mat** (cf. *x^wa?ut ?umatas* ‘She’s not lazy.’)
 PL~lazy< C_1 >
 ‘They are really lazy.’ (EP, vf)

While further elicitation is needed to investigate the combination of two types of Level 1 C_1 reduplication with roots that have a moraic vowel, it is notable that the form in (369) above involves both a prosodic affix and a fixed segment; this allomorphy follows the anticipated pattern because it avoids parsing two prosodic affixes at the same stratum because a fixed segment allomorph is used for one of the morphemes.

5.5.4 L1-L2 Combinations

5.5.4.1 Attested: Level 1 C_1 and Level 2 C_1C_2 Reduplication

Multiple reduplication involving Level 1 diminutive C_1 plus Level 2 plural C_1C_2 reduplication is predicted to be possible, given that the morphemes are not parsed at the same stratum; examples showing this combination are given in (370).¹⁵

- (370) a. **ti~tə~tułat** (cf. *tułat* ‘bed’)
 DIM~PL~bed
 ‘a bunch of little beds’ (EP/FL, sf)
 b. **si~səp~supayu** (cf. *supayu* ‘axe’)
 DIM~PL~axe
 ‘lots of little axes’ (EP/FL, sf)

¹⁴This is due to time constraints; I only noticed this gap while revising this chapter.

¹⁵This combination of diminutive C_1 and plural C_1C_2 reduplication is only found with nouns; diminutive plural verbs are formed with V_1 reduplication instead of C_1C_2 reduplication, as discussed in Section 5.5.5.1.

- c. **q̣i~q̣as~q̣asnaŋ** (cf. *q̣asnaŋ* ‘shirt/dress’)
 DIM~PL~shirt/dress
 ‘a bunch of little shirts or dresses’ (EP/FL, sf)
- d. **pi~pəč~pəču** (cf. *pəču* ‘basket’)
 DIM~PL~basket
 ‘small baskets’ (EP/FL, sf)

The combination of diminutive C_1 and C_1C_2 plural reduplication is predicted to be possible, given that C_1 and C_1C_2 reduplication were assigned to different strata in Chapter 4 (Section 4.1). However, the surface order of morphemes in (370) is unexpected: if diminutive reduplication happens at Level 1 and plural reduplication happens at Level 2, then the plural ($\sigma_{N+\mu}$) should be parsed *after* the diminutive (μ or i -). In other words, the plural should be leftmost because it is prefixed at a later stratum than the diminutive.

Allomorph selection presents an additional complication: the i - allomorph of the diminutive is always selected, even when the μ allomorph is chosen for a diminutive form with single reduplication. For example, the diminutive form of *tut̩aɬ* ‘bed’ in (370a) is *tut̩taɬ* ‘little bed’. The plural is clearly not added to the Level 1 form because the anticipated form would be **tətt̩tut̩taɬ*, which suggests that the order of reduplicative morphemes proposed in Chapter 4 needs to be revised. This will be addressed in Section 5.5.10.

5.5.5 L1-L3 Combinations

5.5.5.1 Attested: Level 1 C_1 and Level 3 V_1 Reduplication

The combination of Level 1 diminutive C_1 and Level 3 plural V_1 reduplication is attested. This type of multiple reduplication is used to form diminutive plural nouns, as shown in (371), or to mark intermittent events where each event typically has a short duration, as shown in (372).¹⁶

¹⁶The diminutive plural nouns in (371) have the same meaning as those given in Section 5.5.4.1. This was confirmed explicitly by speakers EP and FL for pairs like *tu?ut̩taɬ* in (371b) and *tut̩tut̩taɬ* in (370a), which both can be used to mean ‘little beds’.

- (371) a. s<**u?**>u<**s**>payu (cf. *supayu* ‘axe’)
 axe<PL><DIM>
 ‘little axes’ (JF, sf)
- b. ^ˈt<**u?**>u<^ˈt>ɬaɬ (cf. *tɬaɬ* ‘bed’)
 DIM<PL>~bed
 ‘little beds’ (EP, vf)
- c. m<**i?**>i<**m**>xat (cf. *mixat* ‘black bear’)
 DIM<PL>~bear
 ‘baby black bears’ (FL, sf)
- d. č<**i?**>i~čuy (cf. *čuy* ‘child’)
 DIM<PL>~child
 ‘little children’ (FL, vf)
- (372) a. **p**<**i?**>i~pčəm Bailey
 DIM<PL>~wake Bailey
 ‘Bailey wakes up now and then.’ (BW, sf/vt)
- b. č<**i?**>i~čɬəm Freddie
 DIM<PL>~dance-MID Freddie
 ‘Freddie does a little jig now and then.’ (BW, vf)
- c. namaɬ=ʔut **q**<**i?**>i~qk^w-əm
 little.bit=EXCL DIM<PL>~stop-MID
 ‘It kept stopping and starting for a little while.’ (BW, sf/vt)
- d. ^ˈk^w<**i?**>i~^ˈk^wən-[t]=čan Daniel
 DIM<PL>~see-[CTR]=1SG.SUBJ Daniel
 ‘I’m looking at Daniel (several times to see what he’s doing).’ (BW, vf)

Both the prosodic affix (μ) and fixed segment (/i/) allomorphs of C₁ reduplication are observed: compare (371a–c) with (371d) and (372). As plural V₁ reduplication is always a prosodic affix (μ [c.g.]), two prosodic affixes are present in (371a–c); this is predicted to be possible because each reduplicative morpheme is parsed at a different stratum (Level 1 vs. Level 3).

These forms provide independent evidence for diminutive happening at an early stratum: allomorph selection for the diminutive matches the phonological

environments described in Chapter 4 for the diminutive on its own (Section 4.1.3.1.2), which means the choice between listed allomorphs must happen before plural V₁ reduplication. For example, if the order were reversed, plural V₁ reduplication would apply to *čuy* ‘child’ in (371d), resulting in an intermediate form **čuyuy*, which would no longer be monosyllabic and therefore pick the preferred allomorph *μ*, which would yield **čuyčuy*. These forms therefore provide further support for the morpheme to stratum mappings proposed in Chapter 4.

5.5.6 L2-L2 Combinations

5.5.6.1 Ungrammatical: Level 2 C₁ and Level 2 C₁C₂ Reduplication

The combination of Level 2 progressive C₁ plus Level 2 plural C₁C₂ reduplication is not predicted to be possible because both require prosodic affixes – *μ* and *σ_{N+μ}*, respectively – to be parsed into the word at Level 2; if this combination were possible, it would violate MORPHEME-AFFILIATION.

The examples in (373) show that forms with this combination are rejected. As there are no attested examples of this combination to guide elicitation with respect to the form and position of reduplicated segments, I tested at least two forms for each.

- (373) a. **x^wə<x^w>[ə]y~x^way-əṁ/*x^wəy~x^wa~x^way-əṁ*
 PL<PROG>~dive-MID/PL~PROG~dive-MID
 Intended: ‘They are diving here and there.’ (cf. *x^wayəṁ* ‘dive’) (FL)
- b. **ʔi~ʔəṁ~ʔim-aš/*ʔəṁ~ʔi~ʔim-aš* Marianne
 PROG~PL~walk-INTR/PL~PROG~walk-INTR Marianne
 Intended: ‘Marianne is walking around.’ (cf. *ʔimaš* ‘walk’) (FL/BW)
- c. **ʔa~ʔaḳ~ʔaḳ-at-as/*ʔaḳ~ʔa~ʔaḳ-at-as* tə=čanu
 PROG~PL~chase-CTR-3ERG/PL~PROG~chase-CTR-3ERG DET=dog
 Intended: ‘They are chasing the dog around.’ (cf. *ʔaḳatas* ‘He chases it.’) (FL/BW)

The combination of two reduplicative morphemes at Level 2 is ungrammatical, as predicted. Where this combination is avoided, *-a-* replaces the [ə] in forms with plural C₁C₂ reduplication. The pairs in (374–376) compare the meaning

associated with the CəC and CaC forms.¹⁷ When speakers were asked to translate an imperative, they volunteered the CəC form; they volunteered the CaC when describing something in-progress.¹⁸

- (374) a. $\dot{k}^w <a> n \sim \dot{k}^w \text{ən-t-as}$
 PL<ASP>~look-CTR-3ERG
 tə=pipa (cf. $\dot{k}^w \text{ət gi Betty!}$ ‘Look at Betty!’)
 DET=paper
 ‘He’s looking over the paper.’ (Huijsmans & Mellesmoen, 2021, p.353)
- b. $\dot{k}^w \text{ən} \sim \dot{k}^w \text{ən-t=ga ti?i}$
 PL~look-CTR=PRT DEM
 ‘You look it over.’ (Huijsmans & Mellesmoen, 2021, p.353)
- (375) a. $h <a> k^w \sim hək^w \text{-t}$ (cf. $hu \text{ ga } hək^w t$ ‘Go hang it.’)
 PL<ASP>~hang-CTR
 ‘They (blankets) are hanging.’ (EP, vf)
- b. $hək^w \sim hək^w \text{-t-čx}^w$
 PL~hang-CTR=2SG.SUBJ
 ‘You hang them up. (EP, sf)
- (376) a. $\dot{t}^\theta <a> k^w \sim \dot{t}^\theta uk^w \text{-t=čan}$ $tə=\theta iwu\theta itən ?i$
 PL<ASP>~wipe-CTR=1SG.SUBJ DET=table CONJ
 $hu=\dot{s}təm$ $\dot{j}u?$ $t^\theta=huy$ (cf. $\dot{t}^\theta \text{ək}^w tčan$ ‘I wiped it.’)
 $go=1PL.SUBJ+FUT \text{ home } 1SG.POSS=finish$
 ‘I’m wiping down the tables and then we’ll head home.’ (EP, sf)
- b. $\dot{t}^\theta \text{ək}^w \sim \dot{t}^\theta \text{ək}^w \text{-t=ga}$ $tə=\theta iwu\theta itən$
 PL~wipe-CTR=PRT DET=table
 ‘Wipe the tables down.’ (FL, vf)

Given that the distribution of *-a-* and the meaning of forms with CaC are consistent with the progressive, a preliminary hypothesis is that CaC forms mark a plural progressive as a strategy to avoid multiple reduplication; however, further semantic

¹⁷There are also cases where no contrast or difference in meaning can be elicited and the presence of [a] may represent a lexicalized exception (Huijsmans & Mellesmoen, 2021, p.353).

¹⁸Elicitation was conducted as part of joint work with Marianne Huijsmans.

exploration is needed to determine if *-a-* is an allomorph of the progressive or if it is another aspectual morpheme with a similar contribution.¹⁹

5.5.7 L2-L3 Combinations

5.5.7.1 Attested: Level 2 C₁C₂ and Level 3 V₁ Reduplication

Level 2 plural C₁C₂ reduplication plus Level 3 plural V₁ reduplication can occur in the same word. Examples are provided in (377).

- (377) a. hiɬ Bailey ʔə=t^θ<aʔ>ak^w~t^θək^wt tə θək^w~θuk^wnačtən
 CLF Bailey CLF=PL<PL>~wipe-CTR DET PL~chair
 ‘It was Bailey who wiped down the chairs.’ (cf. t^θək^wtčan ‘I wiped it.’)
 (EP, vf)
- b. t<aʔ>ak^w~t^θək^w tə=hiyum (cf. t^θək^w Fred ‘Fred flew.’)
 PL<PL>~fly DET=seagull
 ‘The seagull flies around (go a little distance, stop, and repeat).’ (BW, sf/vt)
- c. j<aʔ>ač~jəč tə=čanu (cf. jəč ‘run’)
 PL<PL>~run DET=dog
 ‘The dog is running around the yard (starting and stopping, sniffing at things).’ (BW, vf)

The combination of Level 2 plural C₁C₂ reduplication and Level 3 plural V₁ reduplication is predicted to be phonologically possible: the underlying form of both morphemes includes a prosodic affix, but each is parsed a different stratum and therefore they do not violate MORPHEME-AFFILIATION.

¹⁹Previous descriptions highlight the need for further work in this area. J. Davis (1970b, p.88) notes that sometimes C₁C₂ reduplication marks imperfective (= progressive) aspect, without mentioning *-a-*, while Watanabe (2003, p.399) suggests that *-a-* with C₁C₂ reduplication has an unclear function but may be related to a process of plural ablaut. Huijsmans and Mellesmoen (2021, p.352) mention that *-a-* with CəC reduplication is “volunteered in a situation where the action described was happening at the time of utterance, whereas the forms without ... were used for contexts such as completed actions and imperative[s]”, suggesting an aspectual function.

5.5.7.2 Attested: Level 2 C₁ and Level 3 C₂ Reduplication

Level 2 progressive C₁ and Level 3 change-of-state reduplication C₂ reduplication can be combined. Examples are provided in (378). This type of multiple reduplication has been described in discussions of aspect in previous work (Kroeber, 1988; Watanabe, 2003).

- (378) a. $t^{\theta}u \sim t^{\theta} < uk^w > uk^w$ (cf. $\lambda_{\partial}qt \ t^{\theta}uk^w$ ‘The day is long.’)
 PROG~day<COS>
 ‘becoming daylight’ (EP, vf)
- b. $q^w\partial l \ ?a \sim ? < ah > ah$ $t^{\theta} = \check{c}ayis$ (cf. $?ah$ ‘sore’)
 come PROG~sore<COS> 1SG.POSS=hand
 ‘My hand is starting to get sore.’ (BW/FL, vf)
- c. $hu = \check{c}an$ $k^wa \sim k^w < as > as$ (cf. k^was ‘warm/hot’)
 go=1SG.SUBJ PROG~warm/hot<COS>
 ‘I am getting warm.’ (EP, vf)
- d. $q\partial \sim q < x\partial > x$ (cf. $q\partial x$ ‘more’)
 PROG~more<COS>
 ‘increasing, getting to be more’ (EP/FL, sf/vt)
- e. $p\partial \sim p < s\partial > s$ $t^{\theta} = \check{c}ayiš$ (cf. $p\partial s$ ‘numb’)
 PROG~more<COS> 1SG.POSS=hand
 ‘My hand is getting numb.’ (BW, sf)

Given that the forms in (378) involve prosodic affixes at two different strata, this type of multiple reduplication does not violate MORPHEME-AFFILIATION and therefore this data is consistent with the predictions of the morpheme-based approach.

5.5.7.3 Attested: Level 2 C₁ and Level 3 V₁ Reduplication

The combinations of progressive C₁ reduplication and plural V₁ reduplication is attested in the literature: examples are given in (379).²⁰

²⁰I keep the vowels in these examples as transcribed in Blake (2000, p.285) because I am unsure if it corresponds to an /i/ or /a/.

- (379) a. $\dot{t}^{\theta} < \epsilon ? > \epsilon \sim \dot{t}^{\theta} \dot{q} ? \dot{a} m$ (cf. $\dot{t}^{\theta} \dot{a} \dot{q} ? \dot{a} m$ ‘throw (s.t.)’)
 PROG<PL>~throw-A.INTR
 ‘throwing lots’ (Blake, 2000, p.285)
- b. $?ut = \check{c}x^w = k^w a$ $\dot{t} < a ? > a \sim \dot{t}aw - \check{s} \dots$
 if=2SG.SUBJ=QUOT PROG<PL>~leave-TR
 ‘If you keep leaving them (sg.) alone...’ (cf. $\dot{t}awax^w as$ ‘Someone left them (sg.).’) (Watanabe, 2021, p.101)

The combination in (379) involves prosodic affixes at two strata, which means that MORPHEME-AFFILIATION will not be violated; these forms are correctly predicted to be phonologically possible.

5.5.8 L3-L3 Combinations

5.5.8.1 Ungrammatical: Level 3 C₂ and Level 3 V₁ Reduplication

There are two types of reduplication that occur at Level 3: change-of-state C₂ and plural V₁ reduplication. If the combination of these two Level 3 morphemes were attested, it would mean two prosodic affixes were parsed at the same stratum (Level 3). The prediction that these will not co-occur is borne out, as shown by the negative data in (380).

- (380) a. $*\lambda < i ? > < i q > i q i w$ (cf. $\lambda i q i w$ ‘dark’)
 dark<PL><COS>
 Intended: ‘got dark again and again’
 Intended: ‘got dark here and there’ (FL)
- b. $*\dot{t} < u ? > < u s > u s$ (cf. $\dot{t} u s \dot{a} m$ ‘quiet’)
 quiet<PL><COS>
 Intended: ‘They kept getting quiet.’ (FL)
- c. $*p < a ? > < a s > \dot{a} s$ (cf. $p \dot{a} s$ ‘numb’)
 numb<PL><COS>
 Intended: ‘I got numb (throughout the day).’ (FL)

- d. * k^w < $a?$ >< as > as (cf. k^was ‘warm/hot’)
 warm/hot<PL><COS>
 Intended: ‘I got hot again and again.’
 Intended: ‘I got hot throughout the day.’ (FL)

5.5.9 L1-L2-L3 Combinations

5.5.9.1 Attested: Level 1 C₁, Level 2 C₁, and Level 3 V₁ Reduplication

Given that Comox-Sliammon has productive reduplication at three strata, three types of reduplication are predicted to be phonologically compatible in the same word. The examples in (381) show that Level 1 C₁ (diminutive), Level 2 C₁ (progressive), and Level 3 V₁ reduplication may occur together.

- (381) a. $\check{c}$ < $i?$ > i ~ $\check{c}i$ ~ $\check{c}\partial$ γasq (cf. $\check{c}\partial$ ‘rain’)
 PROG<PL>~DIM~rain outside
 ‘It is raining a little (throughout the day).’ (BW, vf)
 b. $\check{j}$ < $i?$ > i ~ $\check{j}i$ ~ $\check{j}\partial\lambda$ (cf. $\check{j}\partial\lambda$ ‘run’)
 PROG<PL>~DIM~run
 ‘running around’ (Watanabe, 1994a, p.70)

While the survey in Chapter 2 reveals that multiple reduplication in Salish typically only involves two types of reduplication in a single word (Section 2.8), the morpheme-based limit on multiple reduplication does not rule out the possibility of more than reduplicative two morphemes in a single word. Comox-Sliammon has three strata, which mean forms with three reduplicative morphemes such as those in (381) are expected, despite the fact that this type of reduplication is very limited in occurrence.²¹

²¹Another example of three reduplicative morphemes is provided in (i), though this involves two Level 1 morphemes (assuming the reanalysis of nominal plurals as Level 1 in Section 5.5.10.3). This involves C₁ diminutive (Level 1), C₁C₂ plural (Level 1), and V₁ plural (Level 3) reduplication; the *i*-allomorph of the diminutive is selected.

- (i) s < $i?$ > i ~ $s\partial p$ ~ $saplin$ (cf. $saplin$ ‘bread’)
 DIM<PL>~PL~bread
 ‘lots of little bits of bread’ (Blake, 2000, p.285)

5.5.10 Analysis

In Section 5.5.1, I showed that the Morpheme Affiliation constraint predicts that multiple reduplication at the same stratum will be possible, provided that only one type of reduplication involves a prosodic affix; this prediction follows from the fact that MORPHEME-AFFILIATION is only violated if fission supplies segmental content to more than one prosodic affix at a given stratum.

The only combinations involving two types of reduplication at the same stratum in Comox-Sliammon happen at Level 1: (i) property-denoting C_1C_2 plus diminutive C_1 reduplication, (ii) property-denoting C_1C_2 plus plural C_1 reduplication, and (iii) diminutive C_1 plus plural C_1 reduplication. As multiple reduplication occurs at the same stratum and yields a 1:3 input-output mapping, this combination would be ruled out under the Binariness restriction proposed by Mellesmoen and Urbanczyk (2021b); in this section, I show that the morpheme-based approach developed in this chapter does not face the same limitations.

I first demonstrate in Sections 5.5.10.1 and 5.5.10.2 that multiple reduplication at Level 1 does not violate MORPHEME-AFFILIATION in Comox-Sliammon; in fact, including MORPHEME-AFFILIATION provides an explanation for allomorph choice in multiple reduplication. The unexpected form of diminutive C_1 plus plural C_1C_2 reduplication described in Section 5.5.4.1 above is addressed in Section 5.5.10.3; only a minor adjustment to the analysis proposed in Chapter 4 is required.

5.5.10.1 Diminutive and Plural Properties

Diminutive C_1 , plural C_1 , and property-denoting C_1C_2 reduplication are all Level 1 affixes; their lexical entries are given again in (382) repeated from Chapter 4 (Section 4.1.3.1.2). As argued there, the lexical entry for the diminutive and the plural contains two listed allomorphs: an affixed mora, and a fixed vowel (/i/ for diminutive and /a/ for plural). Of the two allomorphs, μ is the default. The property-denoting morpheme is always an affixed monomoraic syllable.

(382) *Lexical Entries for Diminutive, Plural, and Property-Denoting Reduplication:*

- a. $(\mu_1 > i_2) \Leftrightarrow \langle \text{DIM} \rangle$
- b. $(\mu_1 > a_2) \Leftrightarrow \langle \text{PL} \rangle$

c. $\sigma_{N+\mu} \Leftrightarrow \langle \text{PDEN} \rangle$

Diminutive properties and plural properties are identical aside from vowel quality: [i] marks a diminutive, while [a] marks a plural. A pair is given in (383).²²

(383) a. *Diminutive Property*

$\lambda i \sim \lambda \dot{\alpha} m \sim \lambda \dot{\alpha} m$ $t\dot{\alpha} = ?apust\dot{\alpha}n$ (cf. $\lambda \dot{\alpha} m$ ‘get wet’)
 DIM $\sim$ PDEN $\sim$ wet DET=towel
 ‘The towel is a little wet/damp.’ (BW, vf)

b. *Plural Property*

hihiw $\lambda a \sim \lambda \dot{\alpha} m \sim \lambda \dot{\alpha} m$ -mut $t\dot{\alpha} = t\dot{\alpha} m \sim tumi\check{s}$
 really PL $\sim$ PDEN $\sim$ wet-INTS DET=PL $\sim$ man
 ‘Those men are really wet.’ (EP)

For the remainder of this section, I will focus on diminutive properties; everything discussed here applies to plural properties as well, if [a] is substituted for [i].

The tree structures in (384b) and (384c) compare two potential forms of the diminutive property $\lambda i \lambda \dot{\alpha} t \lambda \dot{\alpha} t$ ‘a little angry’. In (384b), the preferred allomorph (μ) is inserted, resulting in the unattested form $\lambda \dot{\alpha} \lambda \dot{\alpha} t \lambda \dot{\alpha} t$. This results in segmental content from a single morpheme (marked with teal) being associated with prosodic units from both the diminutive and property-denoting morpheme (marked with pink and purple), which violates MORPHEME-AFFILIATION. In comparison, the attested form $\lambda i \lambda \dot{\alpha} t \lambda \dot{\alpha} t$ is shown in (384c); segments marked in teal are only associated with prosodic content belonging to the property-denoting morpheme (marked in purple). Selecting the fixed segment allomorph therefore avoids a violation of MORPHEME-AFFILIATION.

²²Example (383b) was elicited by Marianne Huijsmans.

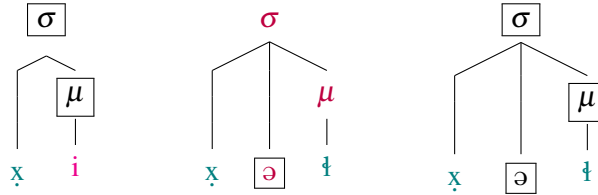
(384) a. *Input:*

$(\mu > i) + \sigma_{N+\mu} + \dot{x}\dot{t}$

b. *Unattested: μ Allomorph*



c. *Attested: /i/ Allomorph*



A tableau showing the ranking needed to derive (384c) is given in (385) below; MORPHEME-AFFILIATION must be ranked above PRIORITY, which is the constraint violated when the preferred allomorph is not selected (see Section 4.1.3.1.2).

	$/(\mu_1 > i_2) + \sigma_{N+\mu} + \dot{x}\dot{t}/$	MAFFIL	PRIORITY
(385)	a. $\dot{x}\dot{\epsilon}\dot{x}_1\dot{t}\dot{x}\dot{\epsilon}\dot{t}$	*!	
	b. $\dot{x}\dot{i}_2\dot{x}\dot{\epsilon}\dot{t}\dot{x}\dot{\epsilon}\dot{t}$		*

Ranking MORPHEME-AFFILIATION above PRIORITY predicts that the fixed segment affix will be selected over the preferred one to avoid fissioned segments being associated with more than one prosodic affix at the same stratum.

In these cases, MORPHEME-AFFILIATION correctly allows more than one type of reduplication at the same stratum while at the same time it provides an explanation for allomorph choice. In contrast, the attested form is not something GEN can produce under Mellesmoen and Urbanczyk's (2021) Binarity restriction, because the attested form has a 1:3 mapping. The morpheme-based approach therefore yields a more successful analysis of multiple reduplication in Comox-Sliammon.

5.5.10.2 Double C₁ Diminutive Plurals at Level 1

The combination of plural C₁ plus diminutive C₁ reduplication at Level 1 is also attested in Comox-Sliammon, as described in Section 5.5.3.2. The combination of two types of reduplication at the same stratum is expected, provided that the fixed segment allomorph is selected for at least one of the morphemes. The relevant lexical entries are given in (386).

(386) *Comox-Sliammon Level 1 Diminutive and Plural:*

a. $(\mu_1 > i_2) \Leftrightarrow \langle \text{Dim} \rangle$

b. $(\mu_1 > a_2) \Leftrightarrow \langle \text{Pl} \rangle$

Recall from Section 5.5.3.2 that fixed segment affixes may be selected for both morphemes in double C₁ diminutive plurals. For example, both *kikakpit* and *kakikpit* are translated by the speaker as ‘small little cuts’. The tableau in (387) shows that a potential candidate in (387a) with both preferred allomorphs is ruled out by MAFFIL.²³ Candidates (387b–c), which select one fixed segment allomorph, violate PRIORITY once; candidates (387d–e), which select two fixed segment allomorphs, receive an additional (and fatal) violation under PRIORITY. EVAL would therefore incorrectly select either (387b) or (387c), rather than the attested forms (387d–e).

(387)	$/(\mu_1 > i_2) + (\mu_1 > a_2) + \text{kp} + \text{it}/$	MAFFIL	PRIORITY
	a. $\text{k}\acute{\text{e}}\text{k}_1\text{k}_1\text{pit}$	*!	
	☞ b. $\text{k}\text{a}_2\text{k}_1\text{pit}$		*
	☞ c. $\text{k}\text{i}_2\text{k}_1\text{pit}$		*
	☹ d. $\text{k}\text{i}_2\text{k}\text{a}_2\text{k}\text{pit}$		**!
	☹ e. $\text{k}\text{a}_2\text{k}\text{i}_2\text{k}\text{pit}$		**!

Candidates (387b) or (387c) must be possible outputs of the grammar when a single reduplicative morpheme is present, as evident from the pair in (388) which show diminutive and plural forms. These forms have the same shape as (387b) and (387c) in the tableau above. The fact that these forms are identical words with single reduplication means that these combinations violate REALIZEMORPHEME,

²³This particular form would also be ruled out by DEP-V independently: see Section 4.1.3.1.

first defined in Chapter 4 (Section 4.1.3.1.2) and restated in (389): if DIM+PL is identical to DIM (or PL), then there is no surface expression of PL (or DIM).

(388) a. *Diminutive C₁ Reduplication*

$\dot{\text{t}}\text{i} \sim \text{tl}[\text{i}]\text{k}$ (cf. $\dot{\text{t}}\text{ə} \text{lk}$ ‘hole’)

DIM~hole

‘small hole’ (BW)

b. *Plural C₁ Reduplication*

$\dot{\text{t}}\text{a} \sim \text{tl}[\text{i}]\text{k}$ (cf. $\dot{\text{t}}\text{ə} \text{lk}$ ‘hole’)

C₁.PL~hole

‘bunch of holes’ (KG)

(389) REALIZEMORPHEME (RM): Let α be a morphological form, β be a morphosyntactic category, and $F(\alpha)$ be the phonological form from which $F(\alpha+\beta)$ is derived to express a morphosyntactic category β . Then RM is satisfied with respect to β iff $F(\alpha + \beta) \neq F(\alpha)$ phonologically.

(Kurusu, 2001, p.39)

The tableau in (390) shows that ranking REALIZEMORPHEME above PRIORITY ensures that one of the attested forms is chosen.

$/(\mu_1 > i_2) + (\mu_1 > a_2) + \text{kp} + \text{it}/$		MAFFIL	RM	PRIORITY
(390)	a. $\text{kə} \text{k}_1 \text{k}_1 \text{pit}$	*!		
	b. $\text{ka}_2 \text{k}_1 \text{pit}$		*!	*
	c. $\text{ki}_2 \text{k}_1 \text{pit}$		*!	*
	 d. $\text{ki}_2 \text{ka}_2 \text{kpit}$			**
	 e. $\text{ka}_2 \text{ki}_2 \text{kpit}$			**

Fixed segment allomorphs are selected for both morphemes in double C₁ diminutive plural reduplication because selecting two prosodic affixes would violate MORPHEME-AFFILIATION and selecting one prosodic affix would result in an output for multiple reduplication that is not sufficiently different from the form with single reduplication. I leave the variation in the order of diminutive C₁ and plural C₁ reduplication as an area for future investigation.

5.5.10.3 Diminutive Plurals

We have seen that there are no attested multiple reduplication patterns in Comox-Sliammon that are predicted to be impossible, which means that MORPHEME-AFFILIATION is sufficient to explain limitations on multiple reduplication. However, there is one pattern that poses a potential problem for unifying the analysis of individual reduplication patterns with their combinations. Recall from Section 5.5.4.1 that the form and position of the diminutive in plural diminutive nouns formed with diminutive C_1 plus Level 2 plural C_1C_2 reduplication is not compatible with diminutive reduplication occurring before plural reduplication.

If diminutive C_1 reduplication happens at Level 1 and plural C_1C_2 reduplication happens at Level 2, then the choice between prosodic affix and fixed segment allomorphs must happen prior to plural reduplication. The diminutive plural form of *supayu* ‘axe’ predicted by this analysis is **səssuspayu*, as shown in Table 5.12: μ is selected because *supayu* has more than one syllable and has a moraic vowel, which means the segment associated with the diminutive mora is in a C_2 position in the input to Level 2.

Table 5.12: Box - Predicted Derivation (Ungrammatical Output)

Input:	$\{\mu, i\} + \text{supayu}$
Output _{Level 1}	suspayu
Input _{Level 2}	$\sigma_\mu + \text{suspayu}$
Output:	<i>*səssuspayu</i>

In addition to the wrong choice of allomorph, the attested form *sisəpsupayu* also has an unanticipated order; even if the fixed segment was picked at Level 1, Level 2 plural reduplication would yield **səssisupayu*.

The evidence from multiple reduplication suggests that the analysis in Chapter 4 must be revised to be compatible with the patterns described in this chapter. The diminutive plural forms resemble the diminutive properties analysed in Section 5.5.10.1, as shown with the pair in (391). Given that the correct form of C_1C_2 reduplication can be derived at either Level 1 or Level 2, this suggests that plural C_1C_2 reduplication might be better analysed as a Level 1 affix.

- (391) a. $\dot{\text{c}}\dot{\text{i}}\sim\dot{\text{c}}\dot{\text{a}}\dot{\text{m}}\sim\dot{\text{c}}\dot{\text{a}}\dot{\text{m}}$ root: $\dot{\text{c}}\dot{\text{m}}$ - ‘cold’)
 DIM~PDEN~cold
 ‘a little cold’ (EP)
- b. $\dot{\text{t}}\dot{\text{i}}\sim\dot{\text{t}}\dot{\text{a}}\dot{\text{t}}\sim\dot{\text{t}}\dot{\text{a}}\dot{\text{t}}$ (cf. $\dot{\text{t}}\dot{\text{a}}\dot{\text{t}}$ ‘bed’)
 DIM~PL~bed
 ‘a bunch of little beds’ (EP)

In Section 5.5.6.1, I showed that the combination of C₁ progressive reduplication and C₁C₂ plural reduplication is systematically rejected; if plural C₁C₂ reduplication is at Level 1, as indicated by the nouns, and progressive C₁ at Level 2, there is no explanation for their incompatibility. However, the problematic diminutive plural forms are all nouns while the potential plural progressive forms are necessarily verbs (because progressive aspect only applies to verbs).²⁴ I therefore revise the original classification of plural reduplication to differentiate between Level 1 C₁C₂ nominal plurals and Level 2 C₁C₂ verbal plurals. Given that plural nouns refer to plural entities while plural verbs refer to spatio-temporally distributed events (Huijsmans & Mellesmoen, 2021), there are independent semantic differences between the two which further support treating them as separate processes.

Distinguishing Level 1 and Level 2 C₁C₂ reduplication means there are two strata where plural C₁C₂ reduplication can occur. The order of diminutive relative to plural reduplication for diminutive plural nouns ceases to be an issue if both are affixed at Level 1.

5.5.11 Summary

Table 5.13 shows the revised classification of Comox-Sliammon reduplication, based on the case study in Chapter 4 and revised with evidence from multiple reduplication.

²⁴Diminutive plural verbs can only be formed with C₁ plus V₁ reduplication, as described in Section 5.5.5.1.

Table 5.13: Comox-Sliammon System: Revised

Reduplication	Underlying Form	Level
Plural (Entities)	$\sigma_{N+\mu}$	Level 1
Diminutive	$(\mu_1 > i_2)$	Level 1
Plural (Spatial/States)	$(\mu_1 > a_2)$	Level 1
Property-Denoting	$\sigma_{N+\mu} + ?$	Level 1
Plural (Events)	$\sigma_{N+\mu}$	Level 2
Progressive	μ	Level 2
Change-of-State	σ	Level 3
Plural	μ [c.g.]	Level 3

Table 5.14 summarises the different combinations of reduplicative morphemes. C_1C_2 reduplication is separated into N.PL for plural entities (Level 1) and V.PL for plural events (Level 2). I use ✓ to indicate combinations that are attested, * to indicate combinations that are ungrammatical, and % to indicate combinations that are possible only where a fixed segment allomorph is selected. Cells with – have not yet been tested (see Section 5.5.2).

Table 5.14: Comox-Sliammon Multiple Reduplication

		Level 1			Level 2		Level 3	
		PDEN/N.PL	DIM	PL	V.PL	PROG	COS	PL
		C_1C_2	C_1	C_1	C_1C_2	C_1	C_2	V_1
Level 1	PDEN/N.PL		%	%	–	–	–	–
	DIM			%	✓	–	–	✓
	PL				–	–	–	–
Level 2	V.PL					*	–	✓
	PROG						✓	✓
Level 3	COS							*
	PL							

The analysis of individual reduplication patterns in Chapter 4 and the analysis of multiple reduplication patterns in this chapter complement each other and together provide a firm empirical foundation for both the overall model of grammar

proposed in this dissertation and the specific constraints and constraint rankings which characterize each of the the three case study languages.

5.6 Conclusion

More than one reduplicative morpheme may occur in a single word in Halkomelem, Lillooet, and Comox-Sliammon. However, not all potential types of multiple reduplication are possible; if two types of reduplication co-occur at the same stratum, then only one type of reduplication can fill a prosodic affix. Where two reduplicative morphemes are affixed at the same point in the derivation, a fixed segment allomorph may be inserted if one is available, as in Halkomelem and Comox-Sliammon. If there is no fixed segment allomorph, then there is simply a gap and the intended meaning requires a periphrastic construction, as in Lillooet. Systematic gaps in the distribution of multiple reduplication can therefore be explained in a morpheme-based approach using MORPHEME-AFFILIATION, a constraint limiting segments to associating with prosodic units affiliated with only one other morpheme.

Chapter 6

A Model of Reduplication: Cross-Salish Variation and Implications for Theory

In this chapter, I further elaborate my theory of reduplication, showing how the model proposed in Chapter 3 and further developed through the case studies in Chapters 4 and 5 predicts the variation across Salish, as detailed in Chapter 2. An overview of the architecture of the grammar is provided in Section 6.1. Sections 6.2–6.7 cover each component of the model, starting from syntactic structure and concluding with a discussion of post-reduplication phonological processes. Section 6.8 concludes the chapter with a discussion of theoretical implications of the model.

One of the main claims of this dissertation, summarised in this chapter, is that phonological words are built across ordered strata, which each have their own constraint ranking; these strata define phonological domains where different affixes are parsed into prosodic structure. However, the smallest *morphosyntactic* domain relevant for reduplicative morphemes in Salish is the morphological word (w). Salish reduplication shows that phonological and morphosyntactic domains beneath the level of the word are not isomorphic. Further, in the theory of reduplication presented here, concatenative and non-concatenative morphemes are not inherently different; I provide further evidence in this chapter that these

morphemes are treated the same with respect to linearisation and allomorphy prior to being parsed into the phonological word. Differences between them arise when the phonological content associated with a given lexical entry cannot surface faithfully, which in turn motivates infixation or other non-concatenative processes, including reduplication.

6.1 Architecture of the Grammar: Overview

I assume a standard Minimalist approach to the syntax (Chomsky, 2014), paired with a view of the lexicon and morphology broadly in the tradition of Distributed Morphology (Halle & Marantz, 1993). The model I adopt for the spell-out component of the grammar is shown in Figure 6.1.

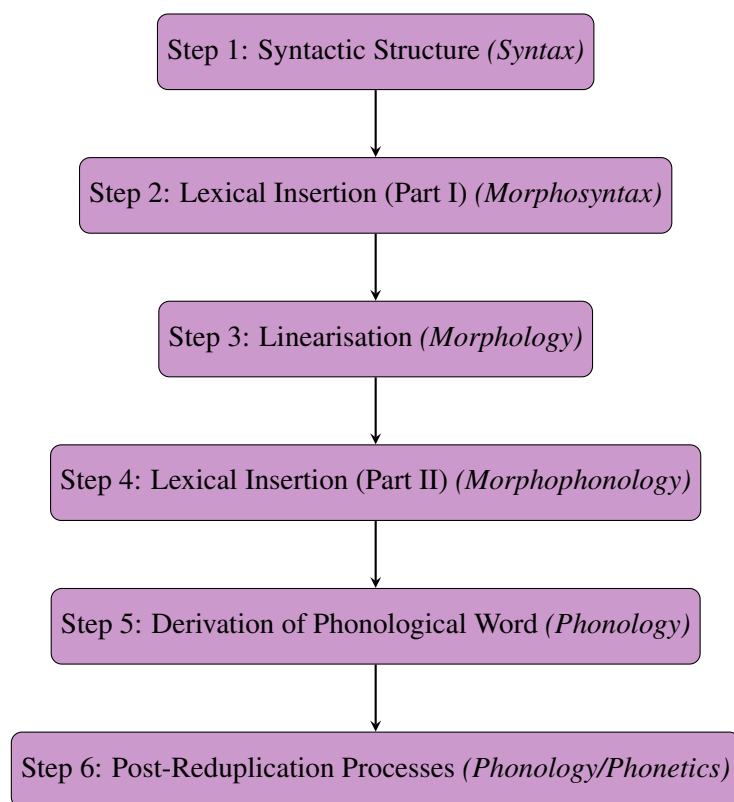


Figure 6.1: Architecture of the PF Component of the Grammar

The path from the syntax to surface realisation is broken down into six steps. The output of one component of the grammar is the input to the next: any earlier steps in the derivation cannot be “seen” (e.g., the lexical entry is not accessible at the point of phonetic implementation). In phonological terms, this is a *modular feedforward architecture* (see, e.g., Pierrehumbert, 2002; Bermúdez-Otero, 2007), which is relatively straightforward to integrate into a derivational model of the syntax.

Hierarchical relationships between morphemes are established by the syntax, as I discuss in Section 6.2. The domain where reduplication applies is the morphological word; the word corresponds to a *span* of syntactic heads up to and including one marked with a [w] feature, as proposed by Svenonius (2016). Section 6.3 explains the first stage of lexical insertion (Step 2): syntactic heads are matched to bundles of morphosyntactic features. Linearisation happens at Step 3, as summarised in Section 6.4: the morphology works up from the lowest head in the w-span and linearises morphemes following instructions provided by the linearisation features ([prefix] and [suffix]).

The phonological component of the model combines key elements from Prosodic Morphology (e.g., McCarthy & Prince, 1986), moraic theory (e.g., Hayes, 1989), Generalised Nonlinear Affixation (e.g., Svenonius & Bye, 2012), and Stratal Optimality Theory (e.g., Kiparsky, 2015). Section 6.5 discusses Step 4, the second stage of lexical insertion, where phonological content is inserted. Reduplication is fission, which occurs during the phonological derivation in Step 5, as described in Section 6.6. A phonological word is built over a series of strata, which then are further affected at subsequent strata by phonological processes and phonetic implementation, which are discussed in Section 6.7.

Cross-linguistic variation in reduplication patterns is predicted to reflect differences in lexical entries (linearisation features, phonological content, and stratum designation) and phonological grammars (constraint rankings). Throughout this chapter, I show how differences among the reduplication patterns surveyed in Chapter 2 correspond either to differences in specific lexical entries or to aspects of the phonological grammars of the languages in question. I also consider the broader implications of this analysis for phonological theory, particularly where data from Salish languages pose challenges to existing

approaches to non-concatenative morphology.

6.2 Step 1: Structure Defines Span (Syntax)

The domain for reduplication is the morphosyntactic word; this is constrained by syntactic constituency, because a feature [w] marks the highest syntactic head that will be part of the word. The feature [w] comes from the span-based approach to spell-out proposed by Svenonius (2016, 2022). A span is a sequence of heads which stand in a head-complement relationship. The feature [w] is an instruction for spell-out: it marks the top boundary of a *w-span*, which includes the [w]-marked syntactic head and all heads lower in the span.

The syntactic component of the grammar specifies hierarchical relationships between different morphemes. The only order between morphemes in the syntax is order imposed by phrase structure. For example, the tree in Figure 6.2 defines a span of heads in a complement relationship including A, which bears the [w] feature, B, and C. The syntactic structure thus defines a relationship between these heads: B must be linearised relative to C and A must be linearised relative to the combination of B and C (headed by B).

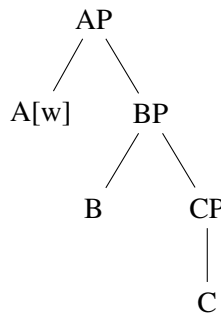


Figure 6.2: Example Tree

The orders BAC or CAB are ruled out by the structure in Figure 6.2 because B and C must be linearised adjacent to each other. However, the syntax alone is not enough to completely determine linear order: $[[BC]A]$, $[A[BC]]$, $[A[CB]]$, and $[[CB]A]$ are all equally compatible with the structure in Figure 6.2. The choice between the four possible constituent orders cannot be read off the syntax

and instead is determined in Step 3, by linearisation in the morphology, which is discussed in Section 6.4.

I assume a set of hierarchically ordered functional projections above the VP (see e.g., H. Davis 2016 on Lillooet). There is no reason to suspect substantial variation in the hierarchy of functional projections across Salish (though see Huijsmans 2023a on variation in the clitic string across the family). However, languages may certainly differ in which head hosts the [w] feature. Variation in the size of the w-span was discussed in a comparison of Comox-Sliammon and Lillooet in Chapter 3 (Section 3.5.1.1): viewpoint aspect is part of the w-span for Comox-Sliammon (progressive is realised as C₁ reduplication), but not for Lillooet (imperfective is realised as *wa?*, a separate word). The model is therefore able to account for variation related to the size and composition of w-spans across Salish.

6.3 Step 2: Lexical Insertion, Step I (Morphosyntax)

I assume that lexical insertion takes place in two stages, following Svenonius and Bye (2012) and Svenonius (2012). The first stage is called *L-Match*. L-Match links syntactic heads with morphosyntactic feature bundles; there is no phonological content inserted until the second step of lexical insertion, *Insert*.

Following work by H. Davis and Huijsmans (2024) and Huijsmans (2023a) I assume that feature bundles contain *linearisation features*. These features provide instructions for linearisation in the morphological component of the grammar, discussed further in the subsequent section.

While the lexical entries and their associated linearisation features may vary cross-linguistically, I assume that lexical insertion operates in the same way for all languages.

6.4 Step 3: Linearisation (Morphology)

6.4.1 Feature-Based Linearisation

Linearisation takes place in the morphological component of the grammar; it follows the instructions provided by features on each affix. The linear order of

adjacent heads in a span is therefore lexically specified, which means the prefixal or suffixal status of a particular affix cannot be read off the syntax prior to the insertion of morphosyntactic features at the first stage of lexical insertion.

Linearisation starts at the lowest head in the w-span and determines whether the head immediately above should be linearised on its left (prefix) or right (suffix). Figure 6.3 provides an example where B has a [suffix] feature and A has a [prefix] feature. B is first linearised to the right of C, which yields CB; A is linearised to the left of CB, which yields ACB.

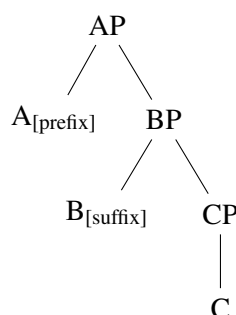


Figure 6.3: Example Tree

Assuming that the direction of affixation is lexically specified is a departure from work that assumes a more general algorithm for linearisation, such as Mirror Theory (e.g., Brody, 2000; Svenonius, 2016). Alternative theories of linearisation are considered in Section 6.8.2.1.

6.4.2 Cross-Salish Variation: [prefix] vs. [suffix]

Most reduplicative morphemes in Salish bear the feature [prefix], which means they are linearised as prefixes. Hyman, Inkelas, and Sibanda (2008, pp.316–317) state the following cross-linguistic generalisations: (i) reduplication tends to be prefixal, and (ii) reduplication tends to favour the edge opposite from typical affixation. As in Salish almost all affixation is suffixal, its reduplication patterns are consistent with the larger typological picture. However, there is nothing in the feature-based approach to linearisation that requires all reduplicative morphemes in any given language or family would need to have the same feature.

Variation in reduplication patterns across Salish provides support for a feature-based approach to linearisation. In Chapter 4 (Section 4.2.3.2.2), I propose that C₁C₂ reduplication in Lillooet is a suffix when it co-occurs with *-t*, as shown in (392). This shows that there is no “default” linearisation for reduplicative morphemes, each lexical entry may be specified its own [prefix] or [suffix] feature.

- (392) *kín~kənt* (root: *kin-* ‘dangerous’)
 dangerous~PDEN+IMMED
 ‘dangerous’ (H. Davis, in prep, Ch.49, Lillooet)

Fixed segments associated with reduplicative morphemes provide good evidence for both [prefix] and [suffix] features. Nonlinear representations, in the sense of Sagey (1986), contain units on different tiers (e.g., moras vs. segments). Precedence (linear order) cannot be established between a *floating* prosodic unit and segment in the phonology. Units on different tiers must be linked via association lines during the phonological derivation, which establishes that units on different tiers overlap in time. As many reduplicative morphemes consist only of a mora or syllable, they have no precedence relations with phonological content on the segmental tier. In contrast, precedence can be established between an affix with fixed segmental content and other morphemes in the input. When precedence can be determined in the input, then the grammar can evaluate whether the same relationships are present in the output. Affixes with fixed segments are more constrained with respect to position than prosodic affixes because any process of metathesis, infixation, or moving an affix will incur violations under faithfulness constraints pertaining to linear order.

The diminutive morpheme in Bella Coola is another example of a reduplicative morpheme that carries a [suffix] feature; this is clear from the position of the fixed segmental content associated with the morpheme. Examples of diminutive reduplication with *-i* are given in (393). The fixed segment is suffixed while reduplication occurs earlier in the word.

- (393) a. *ni<ni><ɪ>q^w-i* (cf. *niz^w* ‘fire’)
 fire<DIM><DIM>-DIM
 ‘fire (dim.)’ (Bagemihl, 1991, p.599, Bella Coola)

- b. qa<qa>yt-i (cf. *qayt* ‘hat’)
 hat<DIM>-DIM
 ‘toadstool (dim.)’ (Bagemihl, 1991, p.598, Bella Coola)
- c. s-tq^wl~q^wl-us-i (cf. *stq^wlus* ‘black bear snare’)
 NMLZ-weave~DIM-face-DIM
 ‘black bear snare (dim.)’ (Bagemihl, 1991, p.615, Bella Coola)

An example of the input to the phonological grammar is shown in (394). Prosodic units are on one tier, while the segments are on another. The underlying form of the Bella Coola diminutive contains μ and /i/, which means that the mora is the only prosodic unit in the input.¹ While the fixed segment /i/ stands in a linear precedence relation with other segments, a floating prosodic unit does not. The linear position of the fixed segment across the examples in (393) therefore shows that the the diminutive affix (as a whole) is linearised as a suffix. The floating mora must have the same linearisation feature as the fixed segment because they are part of the same lexical entry, even though they are not linked by an association line in the underlying form.²

- (394) a. CVC + μ i (Prefix)
 Prosodic Tier: μ
- Segmental Tier: i C V C
- b. CVC + μ i (Suffix)
 Prosodic Tier: μ
- Segmental Tier: C V C i

Fixed segments resist reordering due to pressure to remain faithful between input and output. Metathesis, which involves changes to linear order of segments between input and output, violates LINEARITY, defined in (395). S_1 refers to the phonological string in the input; S_2 refers to the phonological string in the output.

¹ See Section 6.7.1.2 for discussion of a [c.g.] feature that is also part of the lexical entry for the diminutive in Bella Coola.

² Linearisation happens prior to the insertion of lexical content, which raises questions about the representation and linearisation of discontinuous fixed segment morphemes with prefixed and suffixed components. If they are inserted as a single morpheme, they must be linearised as either a prefix or suffix prior to the phonology.

- (395) LINEARITY: S_1 is consistent with the precedence structure of S_2 , and vice versa. (McCarthy & Prince, 1995, p.123)

Any changes to linear order after morphological linearisation must be entirely predictable and attributable to the general phonology of the language. For example, metathesis is only permitted if LINEARITY is ranked low enough to tolerate violations. Even if LINEARITY is ranked low, it will always favour a faithful candidate over an unfaithful one (i.e., metathesis will only be tolerated required to satisfy a markedness constraint ranked above LINEARITY). However, there is no systematic phonological process that would motivate such a constraint. Fixed segments associated with reduplicative morphemes therefore provide evidence for a [suffix] linearisation feature. The fact that *-i* surfaces on the right edge of the word in Bella Coola therefore means the diminutive must have a [suffix] feature.

In contrast to the suffixed diminutive in Bella Coola, the diminutive in Comox-Sliammon, which is discussed in Chapter 4 (Section 4.1.3.1), must bear a [prefix] feature because *i-* is infixed relative to the left edge of the word; examples are given in (396).

- (396) a. $\dot{\mathbf{t}}\mathbf{i} \sim \dot{\mathbf{t}}\mathbf{a}\dot{\mathbf{q}}\mathbf{t}$ (cf. $\dot{\mathbf{t}}\mathbf{a}\dot{\mathbf{q}}\mathbf{t}$ ‘mountain’)
 DIM~mountain
 ‘small mountain’ (JF, Comox-Sliammon)
- b. $\mathbf{t}\mathbf{i} \sim \mathbf{t}\mathbf{\acute{a}}\mathbf{\acute{w}}$ (cf. $\mathbf{t}\mathbf{\acute{a}}\mathbf{\acute{w}}$ ‘ice’)
 DIM~ice
 ‘little bit of ice’ (JF, Comox-Sliammon)
- c. $\mathbf{p}\mathbf{i} \sim \mathbf{puk}^w$ (cf. $\mathbf{puk}^w$ ‘book’)
 DIM~book
 ‘little book’ (JF, Comox-Sliammon)

An approach that assumes a default direction for linearisation, such as Mirror Theory (see discussion of alternative approaches to linearisation in Section 6.8.2.1) would require the phonology to do long-range “metathesis” when the lexical entry for the reduplicative morpheme contains fixed segments for either Bella Coola (if default = prefix) or Comox-Sliammon (if default = suffix); this requires a

phonological grammar that produces patterns that are otherwise unattested and impossible.

6.5 Step 4: Lexical Insertion, Step II (Morphophonology)

The second stage of lexical insertion takes place after linearisation; this is when phonological content is inserted. This is called *Insert* in Svenonius and Bye (2012) and Svenonius (2012). I discuss allomorphy and lexical insertion in Section 6.5.1; underlying forms of reduplicative morphemes are covered in Section 6.5.2.

6.5.1 Allomorphy and Lexical Insertion

The phonological representations inserted at Stage 2 may contain either a single allomorph or a set of listed allomorphs, which may be partially or totally ordered. This allows for allomorph selection either at the point of lexical insertion or at a later point where the phonological grammar selects from the set of allomorphs. The choice between listed allomorphs does not occur until the morpheme is parsed into the word, which means that the choice can only be sensitive to information present in the phonological representation at that point in the derivation.

If allomorphy references morphosyntactic features, it must happen at (the second stage of) lexical insertion (Stage 4 of the model in Figure 6.1); later allomorphy in the phonological grammar cannot see the features associated with the morpheme (Stage 5 of the model).

A distinction between allomorph selection at the point of lexical insertion and at a later point in the phonological derivation is motivated by the data. For example, as argued in Section 5.3.3.1, the choice between the σ and *-l-* in Halkomelem for marking plurality depends on whether another floating prosodic unit is parsed into the prosodic word at the same stratum (Level 2). In order for the grammar to select the correct allomorph, both σ and *-l-* must be present in the input to the phonological derivation. If allomorphy is conditioned by factors related to a particular stratum, then lexical insertion must supply a set of allomorphs; the phonological grammar then selects between the allomorphs at the stratum when the morpheme is parsed into the prosodic word.

6.5.2 Lexical Entries

Reduplicative morphemes, and other non-concatenative morphemes, are not distinguished from concatenative morphemes in the approach taken here; underlying forms for reduplicative morphemes are only constrained by what is possible for any other morpheme in the language. Lexical entries can therefore include segmental and/or suprasegmental content. In other words, reduplication can be triggered both by linear and non-linear representations. I summarise the prosodic units, fixed segments, and features found across the case study languages in Sections 6.5.2.1–6.5.2.3.

Lexical entries also carry *level diacritics*, which allow the morpheme to be parsed into prosodic structure at the right stratum. See Chapter 3 (Section 3.5.2.3) for further discussion of how the diacritics work; morpheme-level and grammar-level mappings are discussed as part of Step 5 of the model in Section 6.6.

6.5.2.1 Suprasegmental: Prosodic Units

The inventory of possible prosodic affixes is constrained by the prosodic structure of the language, as per Prosodic Morphology (McCarthy & Prince, 1986). I adopt moraic theory, which proposes that syllabification is based on the mora (see, e.g., Hyman, 1985; Hayes, 1989).

I assume the prosodic structure given in (397), adapted from Zec (1988); I add a nucleus as a possible constituent of a syllable. Prosodic units beneath the level of the prosodic word (PrWd) include the foot (Ft), syllable (σ), mora (μ) and nucleus (N).³



³While both N and μ may be direct dependents of a syllable, I leave the issue of the constituency of syllables with moraic nuclei for future work.

The hierarchy in (397) identifies the set of prosodic units that can be present in underlying representations, either on their own or in combination with other segmental or suprasegmental content. A prosodic affix requiring just an onset or rime is predicted to be impossible because there is no corresponding prosodic unit. This prediction is supported by the cross-Salish survey: only a mora (Section 6.5.2.1.1), syllable (Section 6.5.2.1.2), monomoraic syllable with a nucleus (Section 6.5.2.1.3), and foot (Section 6.5.2.1.4) are needed to account for Salish reduplication patterns.⁴

6.5.2.1.1 Mora

Seven types of reduplication analysed in the case studies from Chapters 4 and 5 are derived from an underlying mora, as summarised in Table 6.1. With respect to the core patterns defined in Chapter 2 (Section 2.3), an input mora corresponds to both C₁ and V₁ reduplication.

Table 6.1: Mora in Case Studies

Reduplication Type	Input	Form
Comox-Sliammon (n=4)		
Diminutive C ₁ Reduplication	{ μ , i}	-C- from μ (§4.1.3.1.2)
Plural C ₁ Reduplication	{ μ , a}	-C- from μ (§4.1.3.1.3)
Progressive C ₁ Reduplication	μ	-C- or CV- (§4.1.4.1.2)
Plural V ₁ Reduplication	μ [c.g.]	-V?- (§4.1.5.2.2)
Lillooet (n=2)		
Diminutive C ₁ Reduplication	$\mu^{[stress]}$ [c.g.]	-C- (§4.2.4.2.4)
Plural C ₁ Reduplication	μ [c.g.]	-C- (§4.2.4.2.2)
Halkomelem (n=1)		
Imperfective C ₁ Reduplication	μ	C reduplication or other realisation (§5.3.1.1.1)

The case studies show that an affixed mora can be realised with substantial surface variation. This is exemplified by the progressive in Comox-Sliammon (Section 4.1.4.1.2), which surfaces as either -C- or CV-, as shown in (398); the affixed mora is associated with an infixed coda in the former and a prefixed vowel in the latter.

⁴See Section 6.5.2.3 for discussion of fixed segment inputs, which can trigger reduplication to provide an onset. Other apparent cases of “onset” reduplication are considered in Section 6.5.3.3.

The analysis of imperfective reduplication in Halkomelem in Chapter 5 (Section 5.3.1.1.1) also shows that an input mora has a number of non-concatenative realisations, including C_1 reduplication and ablaut, as shown in (399).⁵ In other words, considerable variation in surface position and realisation are associated with affixed moras, even for a single lexical entry.

(398) *Comox-Sliammon Progressive C_1 Reduplication*

- a. $\dot{h}i \sim \dot{h}ik^w$ -it-as (cf. $\dot{h}ik^wit$ 'it is sewn')
 PROG~sew-CTR-3ERG
 'She is sewing.' (BW)
- b. $\dot{c}\acute{a} < \dot{c} > \dot{t}$ (cf. $\dot{c}\acute{a}t$ 'rain')
 rain<PROG>
 'It's raining.' (BW)

(399) *Halkomelem Imperfective C_1 Reduplication and Ablaut*

- a. $\dot{h}i \sim \dot{h}\acute{a}c\acute{-}t$ (cf. $\dot{h}\acute{c}at$ 'cut it')
 IPFV~cut-CTR
 'cutting it'
- b. $p < a > q^w$ -t (from pq^wat)
 break<IPFV>-CTR
 'breaking it'

(Hukari & Peter, 1995 in Mellesmoen & Urbanczyk, 2020,
 pp.240–241)

Mora affixation is a topic of debate in previous work. For example, while mora affixation was proposed in work on prosodic circumscription in Choctaw and Alabama (Lombardi & McCarthy, 1991; Matthewson, 1992), it was also rejected in other analyses of the same languages (Hammond, 1993). My analysis of Salish reduplication motivates the inclusion of moras as prosodic affixes; I use μ to unify disparate realisations of reduplicative morphemes within and across Salish languages.

⁵Other non-concatenative realisations associated with reduplicative morphemes, including length, are discussed further in Section 6.6.

Salish reduplication provides support for moraic theory (Hyman, 1985). However, the mora as a standalone unit was not included in the original set of prosodic templates given by McCarthy and Prince (1986), which included core syllables (σ_c), light syllables (σ_μ), syllables (σ), and heavy syllables ($\sigma_{\mu\mu}$). My approach addresses a concern raised by Hendricks (1999, p.14), who states that patterns such as the Lillooet diminutive are incompatible with prosodic approaches to reduplication because a single consonant does not correspond to a prosodic unit. As shown in Chapter 4 (Section 4.2.4.2.4), an affixed mora may be satisfied by a coda consonant.

6.5.2.1.2 Syllable

Four types of reduplication in the case studies are derived from an affixed syllable (i.e., a plain syllable without any additional structure), as shown in Table 6.2. With respect to the core patterns defined in Chapter 2 (Section 2.3), a σ affix corresponds to both C_1 and C_2 reduplication.

Table 6.2: Syllable in Case Studies

Reduplication Type	Input	Form
Comox-Sliammon (n=1)		
Change-of-State C_2 Reduplication	σ	-VC- (§4.1.5.1.2)
Lillooet (n=2)		
Change-of-State C_2 Reduplication	σ	-əC- (§4.2.4.1.2)
Initial C_1 Reduplication	σ	Cə- (§4.2.3.1.2)
Halkomelem (n=1)		
Diminutive C_1 Reduplication	σ [c.g.]	Cə-...(-ʔ-) or Ci(ʔ)- (§5.3.1.2.2)

An affixed syllable can account for a range of surface patterns. For example, an affixed syllable yields both C_1 and C_2 reduplication in Lillooet (Section 4.2.3.1.2 and Section 4.2.4.1.2), which each copy a different consonant and surface in a different position (prefix vs. infix); examples are given in (400).

- (400) a. *Lillooet Initial C₁ Reduplication*
 n-**x̣ə**~x̣zúm-ana? (cf. x̣zum ‘big’)
 LOC-C₁.INIT~big-ear
 ‘big ears’ (Van Eijk, 1997, p.57)
- b. *Lillooet Change-of-State C₂ Reduplication*
 qá<**x̣^wə**>x̣^w (root: qax̣^w-)
 break<COS>
 ‘to break’ (Van Eijk, 1997, p.59)

An affixed syllable minimally requires a nucleus, though this may be either moraic or non-moraic (i.e., a schwa). For example, a faithful copy of the root vowel surfaces in change-of-state C₂ reduplication in Comox-Sliammon (Section 4.1.5.1.1), even if the vowel is [ə]. Unlike the affixed mora discussed in the previous section, an affixed syllable is agnostic with respect to weight.

- (401) *Comox-Sliammon Change-of-State C₂ Reduplication*
- a. q^wəl **ḵ**<**iḵ**>iḵiw (root: ḵiḵiw ‘dark’)
 come dark<COS>
 ‘It is starting to get dark.’ (FL)
- b. q^wəl **ḵ**<**aḵ**>aḵ (root: ḵaḵ- ‘old’)
 come old<COS>
 ‘He’s getting to be grown/older.’ (FL)
- c. **ḵ**<**əḵ**>əḵ (root: ḵḵ- ‘rot’)
 rot<COS>
 ‘It has gotten rotten.’ (EP/FL)

6.5.2.1.3 Syllable with Nucleus and Mora

C₁C₂ reduplication in all case study languages is derived from $\sigma_{N+\mu}$, which is a syllable with a nucleus and a mora as dependents; N and μ are crucially independent of each other. As shown in Table 6.3, out of the core patterns given Chapter 2 (Section 2.3), only C₁C₂ reduplication is associated with a syllable containing a non-moraic nucleus and an additional mora. There are fewer surface

configurations associated with $\sigma_{N+\mu}$ than μ or σ because $\sigma_{N+\mu}$ always requires a coda consonant to fill the mora.⁶

Table 6.3: Syllable with Nucleus and Mora in Case Studies

Reduplication Type	Input	Form
Comox-Sliammon (n=2)		
Plural/Property-Denoting C ₁ C ₂ Reduplication	$\sigma_{N+\mu}$	CəC- (§4.1.4.2.2/4.1.3.2.2)
Lillooet (n=2)		
Plural/Property-Denoting C ₁ C ₂ Reduplication	$\sigma_{N+\mu}$	CəC- (§4.2.3.2.2)
Halkomelem (n=1)		
Plural/Property-Denoting C ₁ C ₂ Reduplication	$\sigma_{N+\mu}$	CəC- (§5.3.1.3.2)

A monomoraic syllable σ_μ is a well-accepted unit (e.g., McCarthy & Prince, 1986); Salish provides evidence that $\sigma_{N+\mu}$ is also needed. A nucleus node is required because while σ_μ maps to open syllables and $\sigma_{\mu\mu}$ to closed syllables cross-linguistically, monomoraic closed syllables are frequent in Salish due to schwa being nuclear and non-moraic (see, e.g., Shaw, 1993; Shaw et al., 1999; Blake, 2000).

An empty nucleus plays a crucial role in distinguishing different types of reduplication that yield a monomoraic syllable in the output. For example, progressive C₁ reduplication in Comox-Sliammon can yield an open monomoraic syllable (Section 4.1.4.1.2), while plural C₁C₂ reduplication yields a closed monomoraic syllable (Section 4.1.4.2.2). The tableau in (402) shows two candidates which each correspond to an attested surface form of a different reduplicative morpheme. Candidate (402a) is a progressive verb (*?i?imaš* ‘walking’); candidate (402b) is a plural verb (*?əm?imaš* ‘walking around’). I use ??? in the input because the violation marks would be the same for an input σ_μ or μ , which means the grammar will always favour *either* an open or closed

⁶ $\sigma_{N+\mu}$ can surface as [CV] across Salish if reduplication yields /Cəw/ or /Cəy/ followed by vocalisation of the glide, resulting in [Cu] or [Ci], respectively. This process occurs independently of reduplication. See discussion of glide vocalisation in Comox-Sliammon in Blake (2000, pp.35–38).

monomoraic syllable, irrespective of whether the input is a syllable or a mora. If DEP-V is ranked above INT-C (or INT-V), then the grammar will favour C_1 reduplication for either input; if INT-C (or INT-V) is ranked above DEP-V, then the grammar will favour C_1C_2 reduplication, regardless of whether the input is σ_μ or μ .⁷

(402)

/??? + ʔimaš/	INT-V	INT-C	DEP-V
a. ʔiʔimaš	*	*	
b. ʔəmʔimaš		**	*

The attested C_1 and C_1C_2 reduplication patterns in (402) above are both Level 2 processes (see Section 4.1.4), which means that both are subject to the same phonological grammar (constraint ranking). In other words, the grammar alone cannot derive both closed and open monomoraic syllables if the input options are restricted to σ_μ and μ (or σ , for that matter). An empty nucleus is therefore needed to distinguish between monomoraic syllables with a moraic nucleus (CV) and those with a non-moraic nucleus but a moraic coda (CəC). With the ranking INT-C >> DEP-V, candidate (402a) is selected for μ or σ_μ , while candidate (402b) is selected for $\sigma_{N+\mu}$ to ensure all floating units are associated with segmental content. Deriving the two patterns with the same grammar is only possible with an empty nucleus.

Jimmie (1994, p.115) proposes an alternate analysis of C_1C_2 reduplication using prosodic morphology: the lexical entry for Thompson River Salish plural reduplication is $\mu\mu$. This analysis requires two further assumptions to predict the correct surface forms: (i) schwa is moraic, and (ii) (at least some) schwas are underlying. Given that these assumptions are not compatible with those required for the general phonology of other Salish languages and reduplicative morphemes, $\mu\mu$ (or $\sigma_{\mu\mu}$) cannot be extended to all 23 Salish languages as an alternative to $\sigma_{N+\mu}$.

⁷See Section 6.6.1.3 for discussion of how the relative ranking of INTEGRITY and DEP determine the shape of reduplicative morphemes with respect to fission (INTEGRITY violation) and epenthesis (DEP violation).

6.5.2.1.4 Foot

Lillooet triplication affixes a foot, as discussed in Chapter 4 (Section 4.2.5). This type of reduplication is also attested in Shuswap and Okanagan, as discussed in Chapter 2 (Section 2.6.4).

Table 6.4: Summary: Foot

Branch	Number	Ft = C ₂ ²
Central Salish (n=10)	0	
Interior Salish (n=7)	3	Lillooet Shuswap Okanagan
Tsamosan (n=4)	0	
Bella Coola (n=1)	0	
Tillamook (n=1)	0	
Total (n=23)	3	

Though there is evidence for an affixed foot in these three languages, there is limited documentation of the pattern. In Lillooet, the limited frequency of C₂² forms follows from its semantics: C₂² reduplication marks a type of iterativity that does not seem to be compatible with many roots (H. Davis, p.c.).

Given the small number of forms recorded for languages with C₂² reduplication, it is possible that other Salish languages beyond these three may have, or may have once had, a reduplicative morpheme consisting of a foot.⁸

6.5.2.2 Suprasegmental: Features

Underlying floating features have been used in previous Generalized Nonlinear Affixation analyses (e.g., Paschen, 2018). Each of the case studies in Chapters 4 and 5 requires [c.g.] in the lexical entry for at least one morpheme: pluractional V₁ reduplication in Comox-Sliammon (Section 4.1.3.1), pluractional and diminutive C₁ reduplication in Lillooet (Section 4.2.4.3), and diminutive reduplication in Halkomelem (Section 5.3.1.2). A lexical stress feature ([stress]) was also needed

⁸Some of the additional/peripheral patterns discussed in Chapter 2 (Section 2.2.3.2) may also involve an affixed foot, such as the CVCVC-shaped pattern in Lower Chehalis in Section 2.2.3.2.1 and some of the repetition patterns in Section 2.2.3.2.2, such as the C₁C₂² pattern.

for diminutive reduplication in Lillooet (Section 4.2.4.2.4).

The features [c.g.] and [stress] differ in their relationship to other phonological content in the lexical entry: [c.g.] is a floating feature and [stress] is associated with a prosodic unit. The lexical entry for the diminutive in Lillooet, first given in Section 4.2.4.3.2 and restated in (403), contains both features: [stress] is directly associated with μ in the input, but [c.g.] is not.

(403) *Diminutive C₁ Reduplication:*

$$(\mu^{[\text{stress}]} [\text{c.g.}])_{L2} \Leftrightarrow \langle \text{DIM} \rangle$$

Proposing that prosodic units can bear [stress] is a novel contribution, but is supported by analyses of lexical stress that use diacritics or features marking stress on particular morphemes (e.g., Dyck 2004; de Lacy 2020). Furthermore, proposing that certain features may associate with a mora, rather than a segment, has been proposed for Danish stød by Iosad (2016), as well as for tone features. Including a [stress] feature associated with a mora is motivated by diminutive reduplication in Lillooet, which could not otherwise be distinguished from μ -triggered pluractional C₁ reduplication (see Section 4.2.4.2).⁹

Despite the fact that the theory predicts that any feature used in the language may be present in the lexical entry for a reduplicative morpheme, [c.g.] and [stress] are sufficient to account for Salish reduplication. A floating [c.g.] feature is not uncommon in analyses of Salish reduplication (see, e.g., Shaw 2005, pp.162,177); only Lillooet needs [stress].

6.5.2.3 Segmental: Fixed Segments

The realisation of underlying fixed segments can also be accompanied by reduplication; my analysis does not distinguish reduplication triggered by a prosodic affix from reduplication triggered by a segmental affix.

Examples of reduplication triggered by fixed segments were given in the case study of Comox-Sliammon in Chapter 4 (Section 4.1.3.1): C₁ diminutive and pluractional reduplication can either correspond to a fixed segment input (/i/

⁹I leave larger questions of how a [stress]-bearing μ relates to a head mora for future work. For examples of analyses including head or strong moras, see Prilop (2013) and work by Björn Köhnlein (e.g., Köhnlein, 2018).

or /a/) or a μ . Examples of the fixed segment allomorph are shown in (404). Reduplication provides an onset.

- (404) a. $\dot{k}^w i \sim \dot{k}^w a s$ (cf. $\dot{k}^w a s$ ‘hot’)
 DIM~hot
 ‘It is warm.’ (BW)
- b. $\dot{p} a \sim \dot{p} u \dot{q}^w$ (cf. $\dot{p} u \dot{q}^w$ ‘grey’)
 C₁.PL~grey
 ‘grey things’ (EP/FL)

Treating reduplicative morphemes as “special” and distinct from other morphemes would mean that certain types of C₁ reduplication, as well as V₁ reduplication (see discussion of Southern Interior patterns in Section 6.5.3.4), would be considered different from “true” reduplicative morphemes. However, there is no surface difference between using fission to provide an onset, as in (404), and using fission to fill a mora. Distinguishing between concatenative morphemes (i.e., segments) and non-concatenative morphemes (i.e., prosodic units) morphemes is not necessary in my approach because both cases can require fission to provide segmental content to repair prosodic structure.

6.5.3 Cross-Salish Variation: Underlying Forms

Underlying forms of reduplicative morphemes across Salish make use of the same set of suprasegmental and segmental representations, though specific configurations and combinations of these units may differ (e.g., with respect to the presence or absence of [c.g.] with μ). Table 6.5 summarises the suprasegmental and segmental units present in lexical entries for reduplicative morphemes across Comox-Sliammon, Lillooet, and Halkomelem.

Table 6.5: Underlying Forms in Comox-Sliammon, Lillooet, and Halkomelem

	Comox-Sliammon	Lillooet	Halkomelem
C_1C_2	$\sigma_{N+\mu}$	$\sigma_{N+\mu}$	$\{\sigma_{N+\mu}, l\}$
C_1	μ $\{\mu, i\}$ $\{\mu, a\}$	μ [c.g.] $\mu^{[stress]}$ [c.g.] σ	μ σ [c.g.]
C_2	σ	σ	n/a
V_1	μ [c.g.]	n/a	n/a
C_2^2	n/a	Ft	n/a

The phonological material needed for Comox-Sliammon, Lillooet, and Halkomelem can account for the rest of the family with minimal modification.

All languages with C_2 reduplication use σ ; all languages with C_2^2 reduplication use a foot. C_1 reduplication may surface from an underlying μ , σ , or fixed segment, while V_1 reduplication may involve an underlying μ or fixed segment. All Salish languages have C_1C_2 reduplication, which can be analysed as $\sigma_{N+\mu}$ across the family. In Section 6.5.3.1, I discuss a pattern where $\sigma_{N+\mu}$ can be realised as a surface bimoraic syllable ($\sigma_{\mu\mu}$): moraic vowels are copied in Lushootseed C_1C_2 reduplication.

Additional patterns beyond those covered by the underlying forms given above in Table 6.5 include reduplicative affixes consisting *only* of segmental content (Section 6.5.3.2 and Section 6.5.3.4), “onset” reduplication that does not appear to correspond to an underlying prosodic unit (Section 6.5.3.3), and fixed segmental affixes consisting of a consonant (Section 6.5.3.4). Though the patterns in Sections 6.5.3.2–6.5.3.4 warrant additional discussion, they are all consistent with the predictions of the model.

6.5.3.1 Surface Bimoraic Syllable: Lushootseed, Tsamosan, Bella Coola

Moraic root vowels are copied in C_1C_2 reduplication in Lushootseed and the Tsamosan languages, and sometimes in Bella Coola, which results in a bimoraic syllable on the surface. Examples from Lushootseed are given in (405). The vowel

associated with the plural morpheme matches the one in the root instead of always being [ə].

- (405) a. **sáq^w** ~ saq^w (cf. *?usáq^w* ‘It flew.’)
 PL ~ fly
 ‘fly here and there’ (Urbanczyk, 1996a, p.78, from Bates et al. 1994, Lushootseed)
- b. **ǰəs** ~ ǰəsəd (cf. *ǰəsəd* ‘foot’)
 PL ~ foot
 ‘feet’ (Urbanczyk, 2006, p.180, from Bates et al. 1994, Lushootseed)

Reduplication of a moraic vowel in (405) above yields a bimoraic syllable ($\sigma_{\mu\mu}$). Though proposing an underlying $\sigma_{\mu\mu}$ is compatible with my approach, it is not necessary because a bimoraic syllable is also a possible realisation of an underlying $\sigma_{N+\mu}$. Specifically, there is no reason why the empty nucleus of an affixed $\sigma_{N+\mu}$ needs to be associated with a non-moraic vowel. If the grammar prefers copying over epenthesis, then reduplicating a moraic vowel to fill the nucleus is predicted. Furthermore, proposing a bimoraic syllable to account for forms like those in (405a) above would mean two different allomorphs are needed for the same type of reduplication because the form with [ə] in (405b) is not bimoraic. All C_1C_2 patterns in Salish can therefore be attributed to an underlying $\sigma_{N+\mu}$.

6.5.3.2 Single Fixed Segmental Allomorphs: Montana Salish-Spokane-Kalispel and Moses Columbia Repetitive

Montana Salish-Spokane-Kalispel and Moses Columbia use C_1 reduplication to mark repetition; this type of reduplication diverges from the C_1 patterns described in the case study of Comox-Sliammon in Chapter 4 (Section 4.1.3.1) because the lexical entry consists only of a fixed segment. Furthermore, the repetitive morpheme in Montana Salish-Spokane-Kalispel and Moses Columbia does not always trigger reduplication: *-e-* or *-a-* may be infixes without reduplication (fission) of any consonant, as shown in (406b) and (407b).

- (406) a. $\dot{\text{č}}\text{e}\sim\dot{\text{č}}\text{ip}\text{-}\dot{\text{s}}$ (cf. $\dot{\text{č}}\text{ip}\dot{\text{p}}$ ‘It accidentally got pinched.’)
 REP~pinch-PL
 ‘scissors, shears’ (Bates & Carlson, 1998, p.104, Spokane)
- b. $\text{s-m}\langle\text{e}\rangle\dot{\text{l}}\text{k}^{\text{w}}\text{-}\dot{\text{é}}\text{ye}?$ (cf. $\text{mlk}^{\text{w}}\dot{\text{é}}\text{lx}^{\text{w}}$ ‘a fur rug’)
 NMLZ-solid<REP>-seem
 ‘shadow’ (Bates & Carlson, 1998, p.105, Spokane)
- (407) a. $\dot{\text{w}}\text{a}\sim\dot{\text{w}}\text{ál}$ (root: $\dot{\text{w}}\text{ál}$ - ‘tilt’)
 REP~tilt
 ‘rock (tilt back and forth)’ (Willett, 2003, p.207, Moses Columbia)
- b. $\dot{\text{x}}\langle\text{a}\rangle\text{x}^{\text{w}}\text{up}$ (cf. $\text{sx}^{\text{w}}\text{s}\dot{\text{x}}\text{x}^{\text{w}}\text{up}$ ‘winner’)
 win<REP>
 ‘keep winning’ (Willett, 2003, p.208, Moses Columbia)

The alternation between *-e-* and *Ce-* in Montana Salish-Spokane-Kalispel and the one between *-a-* and *Ca-* in Moses Columbia can be analysed with a single fixed segment in their lexical entries, as shown in (408). The phonological grammar determines if the prefixed segment should remain on the left edge or be positioned as an infix. The choice to reduplicate a consonant with the repetitive morpheme is therefore motivated by the prosodic structure: if the fixed segment remains as a prefix, fission provides an onset.

- (408) a. *Montana Salish-Spokane-Kalispel Repetitive Reduplication:*
 $\text{e-} \Leftrightarrow \langle \text{REP} \rangle$
- b. *Moses Columbia Repetitive Reduplication:*
 $\text{a-} \Leftrightarrow \langle \text{REP} \rangle$

It would be challenging to classify the repetitive morpheme in Montana Salish-Spokane-Kalispel and Moses Columbia in theories that treat reduplicative morphemes as inherently different from other morphemes. In both languages, there would need to be two allomorphs in Base-Reduplicant Correspondence Theory (McCarthy & Prince, 1995), such that one allomorph would be a regular concatenative *e-* or *a-* and the other would be a reduplicative morpheme (RED) with a fixed segment /e/ or /a/. In contrast, my analysis only requires a single allomorph that consists of a fixed segment.

6.5.3.3 “Onset” Reduplication

Reduplication may provide an onset for a prefixed vowel (see previous section for discussion of this pattern). However, no prosodic unit corresponds to an onset, which means that reduplication of a single consonant into an onset position cannot be triggered by an underlying prosodic affix. There are two patterns that appear to be exceptions: (i) diminutive reduplication in Southern Interior Salish (Section 6.5.3.3.1) and (ii) “wrong-side” reduplication in Tillamook and Twana (Section 6.5.3.3.2). While these patterns appear to be examples of “onset” reduplication on the surface, they can be derived with σ and $\sigma_{N+\mu}$, respectively.

6.5.3.3.1 “Onset” Reduplication: Syllable

The first consonant of the root is copied in diminutive C_1 reduplication in three Southern Interior Salish languages (Moses Columbia, Montana Salish-Spokane-Kalispel, and Coeur d’Alene), which results in a complex onset, as shown in (409).

- (409) a. $t \sim twit$ (cf. $twit$ ‘boy’)
 DIM~boy
 ‘little boy’ (Willett, 2003, p.278, Moses Columbia)
- b. $\dot{x} \sim \dot{x}\dot{x}acín$ (cf. $\dot{x}\dot{x}acín$ ‘horse’)
 DIM~horse
 ‘It’s a dog.’ (B. Carlson, 1972a, p.115, Spokane)
- c. $c \sim cetx^w$ (cf. $cetx^w$ ‘house’)
 DIM~house
 ‘house, cottage’ (Doak, 1997, p.27, Coeur d’Alene)

Since there is no fixed segment in the output nor any possible prosodic unit that would reduplicate a consonant to yield a complex onset, it is unclear how my analysis of reduplication in Salish would extend to the examples in (409). However, the examples in (410) show that Southern Interior Salish diminutive reduplication does not always yield a complex onset. A $[ə]$ (or $[a]$ in 410c) is present when the first consonant of the root is a sonorant.

- (410) a. s-**mə**~məʔím (cf. *smíyem* ‘woman’)
 NMLZ-DIM~woman
 ‘little woman’ (Reichard, 1938, p.647, Coeur d’Alene)
- b. ní<**nə**>č-əmən (cf. *níčəmən* ‘(It’s a) knife.’)
 cut<DIM>-instrument
 ‘(It’s a) little knife.’ (B. Carlson, 1972a, p.115, Spokane)
- c. s-**ʔa**~ʔiswəl-ált (cf. *ʔíswal* ‘loon’)
 NMLZ-DIM-loon-child
 ‘baby loon chicks’ (Czaykowska-Higgins, 1993b, p.258, Moses Columbia)

The difference between the diminutive forms in (409) and (410) follows from the general phonology of Salish languages: certain clusters are not permissible and reduction of unstressed vowels is blocked in environments where it would give rise to such clusters. More specifically, two sonorants are not permitted in an onset. This suggests that the diminutive in Southern Interior Salish is an underlying syllable; an unstressed [ə] may be deleted later in the derivation if it would not create an illicit sequence. A lexical entry is given in (411).

- (411) *Southern Interior Salish Diminutive Reduplication:*
 $\sigma \Leftrightarrow \langle \text{DIM} \rangle$

An alternative analysis of diminutive reduplication has been proposed for Moses Columbia by Willett and Czaykowska-Higgins (1995) and Willett (2003), who argue that the diminutive is a mora. The two analyses are compared in (412).¹⁰

- (412) a. μ Analysis (Willett, 2003; Willett & Czaykowska-Higgins, 1995)
- i. *Input:*
 $\mu + \text{p}^{\text{w}}\text{əq}^{\text{w}}$
 DIM + spill
- ii. *Output:*
 $\text{p}^{\text{w}}_{\mu}\text{p}^{\text{w}}\text{əq}^{\text{w}}$
 ‘spill a little bit’

¹⁰I include schwa in the inputs in (412) for clarity; they are not present in the underlying form.

b. σ Analysis

i. *Input:*

$\sigma + p\acute{\sigma}q^w$

DIM + spill

ii. *(Intermediate) Output:*

$(p\acute{\sigma})\sigma p\acute{\sigma}q^w$

iii. *Surface Form:*

$p\acute{\sigma}p\acute{\sigma}q^w$

‘spill a little bit’

Adopting a μ analysis, as in (412a) above, requires an assumption that onsets may be moraic (cf. Topintzi, 2010), or that obstruents are intrinsically moraic (or are moraic outside of a coda position). Neither assumption is compatible with my assumptions about moraicity. In contrast, the proposed σ analysis in (412b) draws inspiration from the fact that the grammar of Moses Columbia allows for extensive vowel deletion. Czaykowska-Higgins (1993a, p.250) states that “vowel deletion deletes all unstressed vowels situated to the right of surface stress ... and many but not all unstressed vowels situated to the left of surface stress”. It is therefore reasonable to suggest that a syllable with [ə] as a nucleus may lose this vowel where not strictly necessary for sonority sequencing, yielding a complex onset.¹¹

6.5.3.3.2 “Wrong Side” Reduplication

Another putative example of “onset” reduplication is “wrong side” reduplication, as found in Tillamook and Twana: examples are given in (413) and (414), respectively. The second consonant of the root is copied, as in C_2 reduplication, but the consonant is positioned on the left edge of the word, yielding a complex onset.

- (413) a. $de \quad \check{s}-q^w \sim t\acute{\sigma}q^w-\acute{\sigma}n$ (root: tq^w - ‘to break’)
 DET DESID-PL $\sim$ break-3
 ‘They tried to break it.’ (Edel, 1939, p.15, Tillamook)

¹¹See Section 6.7 for further discussion of post-reduplication phonological and phonetic processes of deletion and reduction.

- b. s-**q**~**ʔ**éq-il (cf. *steqísi* 'I sit down beside them (sg.).')
 NMLZ-PL~sit-DVL
 'Someone is sitting.' (Egesdal & Thompson, 1996, p.11, Tillamook)
- (414) a. **q̣**^w~sóq̣^way (cf. *sóq̣^way* 'elder sister')
 PL~elder.sister
 'elder sisters' (Drachman, 1969, p.32, Twana)
- b. s-**q**~téqaw (cf. *steqéw* 'horse')¹²
 NMLZ-PL~horse
 'horses' (Drachman, 1969, p.86, Twana)
- c. ʔəs-**č**~**ʔ**áç (cf. *ʔasʔáç* 'measure')
 STAT-PL~measure
 'measure (pl.)' (Drachman, 1969, p.55, Twana)

On the surface, “wrong side” reduplication appears to copy a segment into a complex onset. While there is no prosodic unit that could directly yield the surface forms in (413) and (414) above, these patterns can be made compatible with a stratum-based view of phonology if the unexpected surface form can be derived by phonological processes applying after the reduplicative morpheme has been parsed into the word.

I propose that the underlying form for “wrong side” reduplication is $\sigma_{N+\mu}$. This is supported by its meaning: “wrong side” reduplication marks the same kind of plurality otherwise associated with $\sigma_{N+\mu}$ in both languages. Examples of plural C_1C_2 reduplication, with the usual form, are given in (415).¹³ Note that /b/ and /d/ behave as sonorants in Twana, which means that both examples in (415) have a sonorant in C_2 position; “wrong-side” reduplication would yield a poor onset consisting of two sonorants.

- (415) a. **dən**~**t**əni (cf. *təni* 'ear')
 PL~ear
 'ears' (Egesdal & Thompson, 1996, p.3, Tillamook)

¹²I am unsure why the [e] in the final syllable becomes [a] in this reduplicated form.

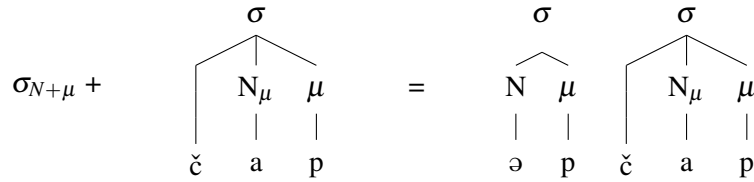
¹³Reduplicated consonants in onsets in Tillamook are deglottalised and voiced, which is why [d] surfaces in (415a). See (L. C. Thompson & Thompson, 1985, pp.140–142).

- b. **bəd**~bədəh (cf. *bədəh* ‘child’)
 PL~child
 ‘children’ (Drachman, 1969, p.81, Twana)

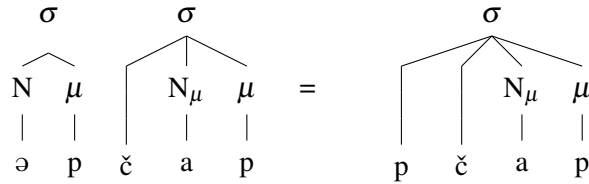
I analyse plural C₁C₂ reduplication and “wrong side” reduplication as realisations of the same morpheme and therefore propose that they share the same underlying form: $\sigma_{N+\mu}$. Given there is less documentation for Tillamook reduplication, I focus on Twana for the discussion of “wrong side” reduplication in the remainder of this section.

As shown in (416a) and (416b), the grammar fills the nucleus and mora, but does not provide an onset at the relevant stratum; this follows from the ranking INTEGRITY, DEP-C >> ONSET. The intermediate representation for both examples is əC, which will trigger a repair at a later stratum to avoid a syllable without an onset; the phonological grammar of a later stratum may delete [ə] or reduplicate another consonant to provide an onset, as shown in (416a) and (416b), respectively.¹⁴

- (416) a. *Stratum with Reduplication:*

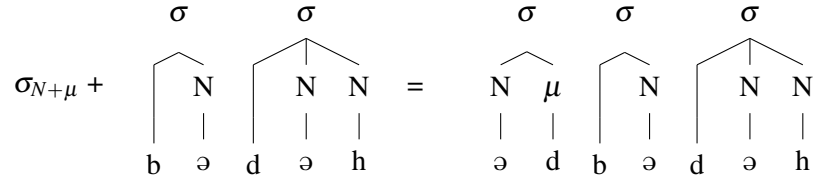


- b. *Subsequent Stratum with Onset-Related Repair (Deletion):*

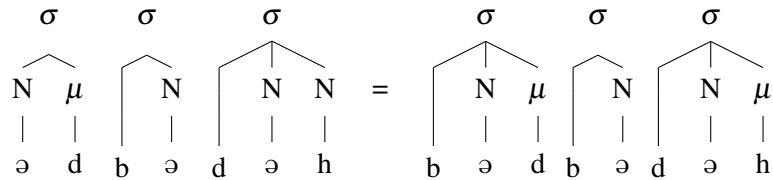


¹⁴The choice in strategies depends on sonority and the prosodic structure of the word: see Drachman (1969) and Kim (2017b) for further discussion of these patterns.

(417) a. *Stratum with Reduplication:*



b. *Subsequent Stratum with Onset-Related Repair (Fission):*



The “wrong side” reduplication pattern shown in (416) above is therefore not an example of onset reduplication; a complex onset only results from [ə] deletion at a later stratum than reduplication.

“Wrong side” reduplication patterns in Salish are described in Kim and Gardiner (2016), Kim (2017b), and Kim (2017a). These previous analyses assume rule ordering, such that reduplication involves a faithful CVC copy and subsequent phonological processes motivate the loss of a copied vowel; the loss of a vowel yields a complex consonant cluster, which then meets the criteria for rules related to cluster reduction to apply. A stratum-based approach is able to capture “wrong side” reduplication patterns in a similar manner because it allows for intermediate representations which can undergo additional phonological processes following reduplication.¹⁵

6.5.3.4 Segmental (Consonant-Only): Inchoative in Southern Interior Salish

As discussed in Chapter 2 (Section 2.7.2), three of the Southern Interior Salish languages (Montana Salish-Kalispel-Spokane, Coeur d’Alene, and Okanagan) have V₁ reduplication as an allomorph of the inchoative morpheme -ʔ-; the inchoative can surface as [ʔ] or [Vʔ], where V is a copy of the vowel following the

¹⁵Parallel OT is not sufficient to handle this type of reduplication because it requires multiple steps.

glottal stop.¹⁶ Examples of the Vʔ allomorph of the inchoative are given in (418).

- (418) a. q^w<uʔ>úʔ (cf. q^wúʔ ‘dusty’)
dusty<INCH>
‘it gets dusty’ (Vogt, 1940, p.64, Kalispel)
- b. hən-tá<ʔa>š-álp (cf. táš ‘be inherently sweet’)
LOC-sweet<INCH>-feeling
‘I felt sweetness’ (Reichard, 1938, p.612, Coeur d’Alene)
- c. q^w<aʔ>ác (cf. q^waʔcínaʔ ‘warm weather’)
warm<INCH>
‘It is getting warm.’ (Watkins, 1970, p.100, Okanagan)

The lexical entry for the inchoative morpheme illustrated in (418) is given in (419). The prefixed glottal stop is positioned as an infix, which then may trigger reduplication to provide a nucleus.¹⁷

- (419) *Lexical Entry for Montana Salish-Spokane-Kalispel, Coeur d’Alene, and Okanagan Inchoative:*
ʔ- ⇔ <INCH>

While the other types of reduplication triggered by a fixed segment described thus far all involve an affixed vowel, the theory also predicts that affixed consonants could trigger reduplication if they create an illicit sequence of consonants and the constraint ranking favours vowel fission (reduplication) over epenthesis (see Section 6.6.1.2).

6.5.3.5 Summary: Lexical Entries for Reduplicative Morphemes

Lexical entries for reduplicative morphemes in Salish include affixed prosodic units, features, and fixed segments. While the approach taken here predicts that any combination of licit prosodic structure, features, and segments in a given language could comprise the underlying form of a reduplicative morpheme, the

¹⁶There is also a V₁ plural in Montana Salish-Kalispel-Spokane, which was discussed in Chapter 2 (Section 2.7.2).

¹⁷For further discussion of these patterns, including alternate analyses and related phenomena, such as translaryngeal harmony, see Mellesmoen (2024a).

actual inventory of reduplicative morphemes across Salish is much more limited. In fact, the core patterns and C_2^2 triplication in Salish can be captured with four different prosodic units and two different types of fixed segment (consonant or vowel), as shown in Table 6.6.¹⁸

Table 6.6: Prosodic and Segmental Content in Lexical Entries

	C_1C_2	C_1	C_2	V_1	C_2^2
$\sigma_{N+\mu}$	✓				
μ		✓		✓	
σ		✓	✓		
Ft					✓
V		✓			
C				✓	

6.6 Step 5: Phonological Derivation (Phonology)

The phonological grammar is comprised of a series of strata, which each have their own constraint ranking. I discuss stratification of the grammar in Section 6.6.1, laying out two key assumptions: (i) strata are phonologically defined and (ii) each stratum has its own constraint ranking. The rankings of markedness, faithfulness, and alignment constraints across strata determine the surface forms of reduplication in Salish. Cross-Salish patterns are discussed in Section 6.6.2.

6.6.1 Stratified Word-Level Grammar

I adopt Stratal Optimality Theory (Stratal OT) (Kiparsky, 2015). Stratal OT assumes a series of discrete levels, or strata, between lexical insertion and phonetic implementation; this is unlike the single global evaluation assumed in classic (Parallel) Optimality Theory (e.g., Prince & Smolensky, 2008).

Each stratum in Stratal OT has its own constraint ranking, allowing for a series of parallel (in the sense of classic Parallel OT) evaluations. The output of the first stratum is fed into the next, and this repeats until the final stratum.

¹⁸C and V in the leftmost column are used to stand in for specific segments present in the underlying form; they are not templates for reduplication.

6.6.1.1 Phonologically-Defined Levels

My analysis diverges from traditional Stratal OT by proposing that strata are phonologically, not morphologically, defined. All phonological content associated with the word, which corresponds to the w-span in Section 6.2, is present in the input to the first stratum. The strata in my approach are steps in the derivation of the phonological word; affixes are parsed into phonological structure at different strata.

Each morpheme either bears a diacritic specifying the level at which it is parsed into the phonological word or is unmarked for phonological level, as described in Chapter 3 (Section 3.5.2.3). Unmarked affixes are parsed at Level 1. Each level has a different constraint ranking, which determines how a reduplicative affix is realised. The level also determines how much existing prosodic structure is present in the input and which other morphemes have already been parsed into the phonological word.

6.6.1.2 Reduplication is Fission

I use Generalised Nonlinear Affixation (GNLA) (e.g., Bermúdez-Otero, 2012), which proposes that non-concatenative morphemes involve nonlinear representations that trigger phonological repairs to ensure a well-formed output (Zimmermann, 2017).

High rankings of MAXFLOAT and *FLOAT motivate phonological repairs to fill affixed prosodic units (see Section 4.1.3.1.2 for constraint definitions). The grammar of a language determines which phonological processes ensure necessary segmental content is present in the output. Reduplication is simply the phonological process of fission: one input segment corresponds to more than one output segment. The output of fission is two segments with equal status: there is no distinction between a “base” and “reduplicant”, as in Base-Reduplicant Correspondence Theory (e.g., McCarthy & Prince, 1995).

6.6.1.3 Different Constraint Rankings at Each Strata

Each stratum has its own constraint ranking. Fission is possible at any stratum where INTEGRITY is ranked lower than other faithfulness constraints. However, if

DEP is ranked lower than INTEGRITY, epenthesis is the preferred phonological repair. The preferred repair may differ between consonants and vowels. See Chapter 4 (Section 4.1.3.1.2) for definitions of these constraints and examples of how they work.

Table 6.7: Epenthesis vs. Fission: Constraint Rankings

Ranking	Result
DEP >> INTEGRITY	epenthesis preferred
INTEGRITY >> DEP	fission preferred
DEP-C >> INTEGRITY-C	fission of consonants preferred
INTEGRITY-C >> DEP-C	epenthesis of consonants preferred
DEP-V >> INTEGRITY-V	fission of vowels preferred
INTEGRITY-V >> DEP-V	epenthesis of vowels preferred

Reduplication and epenthesis are the two most common processes observed with non-concatenative morphemes in Salish. For languages like Lillooet and Comox-Sliammon, where segmental content is only provided by either fission or epenthesis, any other faithfulness constraints that concern alternative strategies are ranked high enough that only fission and epenthesis are used for phonological repairs.

6.6.1.4 Number of Segments Associated with Reduplicative Morphemes

The number of segments that result from either fission or epenthesis is minimal. While violations of faithfulness constraints are tolerated to ensure floating prosodic units are associated with segmental content and all syllables have onsets, candidates with fewer violations will always be preferred over candidates with more violations of the same constraint. This means that reduplicative morphemes are only as big as required to satisfy the requirements of the affixed prosodic unit and other pressures from the grammar, which limits the surface size of a reduplicative morpheme (cf. Kirchner, 2010).

Little variation is anticipated in the maximum size of a reduplicant with the same underlying form (e.g., σ) because the relative ranking of faithfulness constraints at any given stratum will not influence the maximum number of

segments in the output. For example, a mora can be filled by a (coda) consonant or a vowel; when filled by a vowel, a consonant may be needed to provide an onset. C_1 reduplication with an underlying μ therefore always requires minimally one segment and maximally two segments.

The predicted minimal nature of reduplication is supported by cross-Salish patterns: despite languages like Bella Coola allowing syllables with 5+ segments in complex onsets or codas, the maximum number of segments fissioned and epenthesized to fill a floating syllable with a nucleus and mora is three (input: $\sigma_{N+\mu}$), while two segments is the maximum for a plain syllable (input: σ).

6.6.1.5 Linearisation in the Phonological Derivation: Prefix Constraints

The position of segments associated with a reduplicative morpheme is mediated by the ranking of PREFIX constraints, which were defined in Chapter 4 (Section 4.1.3.1.2). The PREFIX constraints are violated by any infixation (PREFIX), infixation past the first mora (PREFIX/ μ), or infixation past the first syllable (PREFIX/ σ).

Undominated PREFIX rules out infixation completely, while undominated PREFIX/ μ and PREFIX/ σ allows infixation but not beyond the first mora or syllable, respectively. The PREFIX constraints place pressure on prefixes to remain as prefixes. Edge-aligned reduplication patterns therefore require PREFIX constraints to be ranked high relative to other constraints on prosodic structure.

Infixation will arise only if one or more of the the PREFIX constraints are ranked lower than markedness constraints related to prosodic structure, such as *STRUC-FT, which penalises any feet in the output (see, e.g., Section 4.2.4.1.2).

Table 6.8 describes the surface position associated with different rankings of PREFIX constraints.

Table 6.8: Ranking of Prefix Constraints

Ranking	Result
PREFIX >> Other Constraints	first segment associated with affix (prefix) (e.g., INIT C ₁ in Lillooet - §4.2.3.1.2)
PREFIX/ μ >> Other Constraints >> PREFIX	segments may be infixed but not beyond the first mora of word (e.g., PROG C ₁ in Comox-Sliammon - §4.1.4.1.2)
PREFIX/ σ >> Other Constraints >> PREFIX	segments may be infixed but only within the first syllable of word (e.g., DIM C ₁ in Comox-Sliammon - §4.1.3.1.2)
Other Constraints >> PREFIX/ σ	segments may be infixed beyond the first syllable (e.g., DIM C ₁ in Lillooet - §4.2.4.2.4)

Reduplication in Salish is either aligned to the left edge (within the syllable) or it falls in the head foot. The patterns discussed in the case studies of Comox-Sliammon and Lillooet in Chapter 4 are therefore representative of cross-Salish patterns. PREFIX constraints are sufficient to account for the position of reduplicative morphemes in the output when considered alongside other constraints related to prosodic structure (e.g., *STRUC constraints and ONSET).

6.6.1.6 Restrictions on Multiple Reduplication

There are limits on multiple reduplication in Salish, which is the focus of Chapter 5. Only one prosodic affix may be filled per stratum. If there are two reduplicative morphemes at the same stratum, multiple reduplication is only possible if one of the reduplicative morphemes is a fixed segment. For example, property-denoting and diminutive reduplication are both at Level 1 in Comox-Sliammon and may co-occur if the fixed segment allomorph of the diminutive is selected (see Section 5.5.3.1).

Languages have different strategies for avoiding multiple reduplication. Halkomelem, like Comox-Sliammon, may use a different allomorph, as discussed

in Section 5.3.2.3. In contrast, Lillooet needs to use a separate inchoative morpheme instead of change-of-state reduplication, or a syntactic paraphrase with the adjective *k^wik^ws* ‘small’ instead of diminutive reduplication, as discussed in Section 5.4.2.6.

MORPHEME-AFFILIATION, a constraint proposed in Chapter 5 (Section 5.3.3.1), requires that segments belonging to one morpheme to be associated with prosodic units that belong to at most one other morpheme; the constraint is repeated in (420). The phonological grammar knows which phonological material belongs to a single unparsed morpheme in the input: in other words, while it does not know that the diminutive and plural have these labels, it does know that content is associated with two separate strings (morphemes).

- (420) MORPHEME-AFFILIATION (MAFFIL): For any three output segments S (S_1, S_2, S_3) that belong to the same morpheme M_a , if S_1 is associated with a prosodic unit belonging to M_a and S_2 is associated with a prosodic unit belonging to some other morpheme M_b , then S_3 must be associated with a prosodic unit belonging to either M_a or M_b . Assign a violation mark for any three segments where all three segments belong to the same morpheme and each segment is associated with a prosodic unit belonging to a different morpheme (* S_1 - M_a, S_2 - M_b, S_3 - M_c).

There is no evidence that MORPHEME-AFFILIATION is violated in Salish, which thus provides a diagnostic for level-hood: any language with reduplication has at least one stratum where prosodic affixes are parsed into the word, whereas any language with multiple reduplication has at least two strata where prosodic affixes are parsed into the word, because otherwise multiple reduplication would not be possible, due to MORPHEME-AFFILIATION.

6.6.2 Cross-Salish Variation - Word-Level Grammar

Morpheme-to-level and grammar-to-level relationships vary across the family. I first summarise the case studies of Comox-Sliammon, Lillooet, and Halkomelem in Section 6.6.2.1. In Section 6.6.2.2, I discuss patterns that differ from the case studies where additional constraints are relevant. Variation related to the number of levels where reduplication occurs is discussed in Section 6.6.2.3.

6.6.2.1 Cross-Salish Variation: Summary of Case Studies

Different constraint rankings and morpheme-level mappings are demonstrated through the case studies in Chapters 4 and 5. I discuss morpheme-level mappings in Section 6.6.2.1.1 and grammar-level mappings in Section 6.6.2.1.2.

6.6.2.1.1 Morpheme-Level Mappings: Case Studies

Table 6.9 compares lexical entries for Comox-Sliammon, Lillooet, and Halkomelem, with levels indicated.

Table 6.9: Reduplicative Morphemes in Comox-Sliammon, Lillooet, and Halkomelem

	Comox-Sliammon	Lillooet	Halkomelem
Plural C_1C_2 Property-Denoting C_1C_2	$(\sigma_{N+\mu})_{L1/L2}$	$(\sigma_{N+\mu})_{L1}$	$(\sigma_{N+\mu})_{L2}$
Aspectual C_1	μ_{L2}	n/a	μ_{L1}
Diminutive C_1	$\{\mu, i\}_{L1}$	$(\mu^{[stress]} [c.g.])_{L2}$	$(\sigma [c.g.])_{L2}$
Plural C_1	$\{\mu, a\}_{L1}$	$(\mu [c.g.])_{L2}$	n/a
Initial C_1	n/a	σ_{L1}	n/a
Change of State C_2	σ_{L3}	σ_{L2}	n/a
Plural V_1	$(\mu [c.g.])_{L3}$	n/a	n/a
Triplification C_2^2	n/a	Ft_{L2+n}	n/a

There is no consistent pattern with respect to which morphemes are associated with which level in Table 6.9 above: for example in Comox-Sliammon, the diminutive is at Level 1 and the progressive is at Level 2, while in Halkomelem the imperfective is at Level 1 and the diminutive is at Level 2. This type of variation in morpheme-level mapping across Salish provides further motivation for phonologically-defined rather than morphologically-defined strata. If levels were morphosyntactically defined, more consistency would be expected across Salish; motivation for phonologically-defined domains will be discussed further in Section 6.8.3.

6.6.2.1.2 Grammar-Level Mappings: Case Studies

Each level has its own grammar, which is defined as a constraint ranking. The choice between fission or epenthesis to provide segmental content is mediated by

the ranking of INTEGRITY and DEP relative to other constraints.

Table 6.10 summarises whether INTEGRITY and DEP constraints are violated when reduplication occurs at each stratum in Comox-Sliammon, Lillooet, and Halkomelem. A ↓ tells us that these constraints may be violated, which indicates that they must be ranked lower than other relevant constraints. Only Comox-Sliammon appears in the row for Level 3 because all core reduplication patterns are at Level 1 and Level 2 in Halkomelem and Lillooet.

Table 6.10: Levels in Comox-Sliammon, Lillooet, and Halkomelem

	INT-C	INT-V	DEP-C	DEP-V _μ
L1	↓C _x SI, Li, Hk	↑C _x SI, Li ↓Hk	↑C _x SI, Li ↓Hk	↑C _x SI, Li ↓Hk
L2	↓C _x SI, Li, Hk	↑Li ↓C _x SI, Hk	↑C _x SI, Li ↓Hk	↑C _x SI, Li ↓Hk
L3	↓C _x SI	↓C _x SI	↓C _x SI	↑C _x SI

In all three languages, INTEGRITY-C is violated at any level where reduplication occurs (i.e., consonants may be copied at every stratum). In Comox-Sliammon, fission of a vowel (violation of *Integrity-V*) can only occur at Levels 2 and 3, while epenthesis of a consonant (violation of DEP-C) can only occur at Level 3. In Lillooet, Levels 1 and 2 are identical in terms of the availability of consonant fission and the unavailability of vowel fission: INTEGRITY-C may be violated while INTEGRITY-V may not. All four constraints may be violated in Halkomelem reduplication, though INT-V is only violated at Level 1. Aside from INTEGRITY-C, there is no generalisation related to the relative ranking of DEP and INTEGRITY constraints that holds across all three languages.

Table 6.11 shows which of the PREFIX constraints are violated at each stratum for the three languages. At Level 2, Halkomelem is marked with * because the reduplicated segments themselves are always on the left edge, violating none of the PREFIX constraints; however, part of the diminutive morpheme (/ʔ/ from [c.g.]) is aligned relative to stress, which can result in a discontinuous expression of the diminutive morpheme (σ...ʔ: see Section 5.3.1.2).

Table 6.11: Levels in Comox-Sliammon, Lillooet, and Halkomelem

	PREFIX	PREFIX/ μ	PREFIX/ σ
L1	$\downarrow_{\text{CxSl,Hk}} \uparrow_{\text{Li}}$	$\downarrow_{\text{CxSl,Hk}} \uparrow_{\text{Li}}$	$\uparrow_{\text{CxSl,Li,Hk}}$
L2	$\downarrow_{\text{CxSl,Li,Hk*}}$	$\uparrow_{\text{CxSl}} \downarrow_{\text{Li,Hk*}}$	$\uparrow_{\text{CxSl}} \downarrow_{\text{Li,Hk*}}$
L3	$\downarrow_{\text{CxSl}}$	$\uparrow_{\text{CxSl}}$	$\uparrow_{\text{CxSl}}$

PREFIX/ σ is not violated at Level 1 for any of the languages, which means that all prefixed material parsed at that level (or “stratum”) is limited to the first syllable of the word. Lillooet requires all prefixed material at Level 1 to remain on the left edge, while prefixed material may surface in an infixed position at Level 2. All PREFIX constraints may therefore be violated at Level 2 in Lillooet, which allows for stress-aligned reduplication. Halkomelem shows the same pattern as Lillooet with respect to the position of [?]: it is positioned adjacent to the stressed syllable, which may be further inside the word than the first syllable. The fact that Halkomelem Level 2 (diminutive and plural) reduplication is otherwise aligned with the left edge suggests that while PREFIX constraints are ranked low, LINEARITY- σ may be ranked high enough to keep the affixed syllable to the left of syllables built at Level 1.

Unlike with INTEGRITY and DEP, it is possible to make a generalisation about the constraint ranking to level mappings for PREFIX constraints in the three case study languages: PREFIX/ σ is not violated at Level 1, but it may be violated at Level 2. When PREFIX/ σ is violated, prefixed material ends up positioned relative to stress. Given that the model developed here assumes prosodic structure is built up gradually across levels, reduplication patterns that show alignment to word-level stress are not predicted at Level 1; attested patterns are consistent with this prediction.

6.6.2.1.3 Morphemes and Number of Relevant Levels: Case Studies

I assume that the derivation of a phonological word in Salish takes place over multiple strata; however, reduplicative morphemes are not necessarily parsed into the word at every stratum. Further, for fission to occur, INTEGRITY must be ranked low enough relative to other faithfulness constraints to tolerate fission. I

define the number of relevant levels in this dissertation as the number of levels at which reduplicative morphemes are parsed into the phonological word and where INTEGRITY is ranked low enough to allow fission.¹⁹

Constraint rankings and interactions with non-reduplicative morphemes provide insight into the number of strata where reduplicative morphemes are parsed into the phonological word. In Chapter 4 (Section 4.2), it was shown that the reduplicative inventory of Lillooet motivates three strata; this is necessary to account for the fact that patterns are edge-aligned (e.g., plural C_1C_2 reduplication), prosody-aligned (e.g., change-of-state C_2 reduplication), or receive marked word-final stress (e.g., triplication).²⁰ Each type of reduplication shown in (421) operates within a different stress domain and requires a different grammar.

- (421) a. *Plural C_2C_2 Reduplication*
 s-**ɣəp**~ɣáp (cf. *sɣap* ‘tree’)
 NMLZ- C_1C_2 .PL~tree
 ‘trees’ (Van Eijk, 1997, p.56, Lillooet)
- b. *Change-of-State C_2 Reduplication*
 zəwá<**tə**>t-s (root: *zwat*)
 be.known<COS>-CAUS
 ‘have learned how to do something’ (H. Davis et al., in prep, Lillooet)
- c. *C_2^2 Triplication*
 ɬ<**əlal**>ɬl (cf. *ɬlin* ‘to sprinkle s.t.’)
 sprinkle<TRIP>
 ‘to keep sprinkling’ (Van Eijk, 1997, p.56, Lillooet)

A consideration of the grammar also indicates that three strata are needed to account for the full inventory of reduplicative morphemes in Comox-Sliammon

¹⁹This definition excludes processes of fission and gemination that happen at a later stratum than the one where the affix in question was parsed into the word, such as later fission to provide an onset in “wrong side” reduplication following the original reduplication of the C_2 consonant, which is discussed in Section 6.6.2.2.3. These cases are excluded because a reduplicative affix parsed into the prosodic word at a previous stratum can not be distinguished as a separate morpheme at subsequent strata; a hypothetical monomorphemic input of the same shape must trigger reduplication at the same stratum.

²⁰The only other process in Lillooet that shows consistent final stress is compounding; stress falls on the second root in the compound (Van Eijk, 1997, p.54).

(see summary in Section 4.1): the same affixed prosodic unit (μ) yields three different surface forms, as shown in (422). Three different constraint rankings are required to derive the different patterns, which means that three strata are needed.

- (422) a. *Diminutive C₁ Reduplication*
 sa<**ssaplin ‘bread’)
 bread<DIM>
 ‘small piece of bread’ (JF, vf, Comox-Sliammon)**
- b. *Progressive C₁ Reduplication*
gu~guh-um (cf. *guh* ‘bark’)
 PROG~bark-MID
 ‘be barking’ (Watanabe, 2003, p.392, Comox-Sliammon)
- c. *Plural V₁ Reduplication*
 p<**aʔpw=ʔut (cf. *paʔa* ‘one’)
 one<V₁.PL><DIM>-CAUS=EXCL
 ‘She did them one by one.’ (MV, vf, Comox-Sliammon)**

Recall from Section 6.6.1.6 that only one prosodic affix can be parsed into the word per stratum due to MORPHEME-AFFILIATION. This means that multiple reduplication can be used to determine which reduplicative patterns occur at different strata: the existence of any attested multiple reduplication pattern involving two prosodic affixes (as opposed to a prosodic affix and a fixed-segment affix or two fixed-segment affixes) is evidence that reduplicative morphemes are parsed into the phonological word at a minimum of two strata.

While predictions based on multiple reduplication are strongest where negative data is available or can be elicited to check for same-level combinations, only a small number of attested forms with multiple reduplication are required to make predictions about the minimum number of reduplication-relevant strata in a language.

Across the three case study languages, the number of strata where reduplicative morphemes are parsed into the word does not vary much: there are two strata in Halkomelem and three in Lillooet and Comox-Sliammon. The three languages only share two types of reduplication, diminutive C₁ and plural C₁C₂, which means

this is the only specific combination that can be compared directly across the languages.

Diminutive C_1 and plural C_1C_2 reduplication can co-occur in Lillooet and Comox-Sliammon, but not in Halkomelem. There is no semantic reason for this cross-linguistic difference: diminutive plurals are perfectly fine in Halkomelem if the *-l-* allomorph of the plural is used (see Section 5.3.2.3). This cross-Salish variation is accommodated in my model because strata are phonologically defined; each morpheme is lexically specified for a particular stratum. If domains were defined by the morphosyntax, there would be no reason to expect that diminutive C_1 and plural C_1C_2 reduplication would be mutually incompatible, in Halkomelem but not in the other two languages.²¹

6.6.2.2 Cross-Salish Variation: Extending the Analysis

The majority of Salish reduplication patterns can be captured by one of the rankings already used for the case study languages. There are three additional patterns that warrant discussion: vowel length in Upper Chehalis and Cowlitz (Section 6.6.2.2.1); non-moraic codas in Straits (Klallam/Northern Straits) and Bella Coola (Section 6.6.2.2.2); and “wrong-side” reduplication in Twana and Tillamook (Section 6.6.2.2.3).

6.6.2.2.1 Length: Upper Chehalis and Cowlitz

Alternations between short and long segments are predicted to be possible under the theoretical assumptions I adopt. For example, Hayes (1989, p.281) argues for moraic theory on the basis of compensatory lengthening, which frequently involves lengthening a vowel following the deletion of a coda consonant. Furthermore, lengthening a vowel is a common realisation of an affixed prosodic unit (see, e.g., the typology in Zimmermann 2017). Length alternations triggered by mora affixation are found in two Tsamosan languages (Upper Chehalis and Cowlitz).

Diminutivity is marked in Upper Chehalis and Cowlitz by length, as well as changes in vowel quality and epenthesis of a glottal stop (see Mellesmoen, 2022, for discussion of this allomorphy). Examples of the *V:* form of the diminutive in

²¹See Section 6.8.3 for further discussion of phonological vs. morphosyntactic domains.

both languages are given in (423).

- (423) a. $latá<:>m$ (cf. *latám* ‘table’)
 table<DIM>
 ‘little table’ (Upper Chehalis)
 b. $ʔé<:>m̥k^wu$ (cf. *ʔəm̥k^wu* ‘cedar basket’)
 cedar.basket<DIM>
 ‘little cedar-basket’ (Cowlitz)

(Kinkade, 1993, p.15)

Upper Chehalis and Cowlitz provide evidence that the ranking of WT-IDENT-IO relative to INTEGRITY is relevant for an analysis of Salish reduplication. WT-IDENT-IO is a faithfulness constraint penalising changes in segment weight between input and output; it is defined in (424).

- (424) WT-IDENT-IO:
 If $\alpha \in \text{Domain}(f)$,
 if α is non-moraic, then $f(\alpha)$ is non-moraic.
 if α is monomoraic, then $f(\alpha)$ is monomoraic.
 if α is bimoraic, then $f(\alpha)$ is bimoraic.

(Mellesmoen, 2022, p.73)

Lengthening a vowel to fill a mora is only possible if INTEGRITY is ranked higher than WT-IDENT-IO, such that lengthening a vowel is preferable to fission. The tableau in (425) shows a “lengthening” grammar with the ranking INTEGRITY >> WT-IDENT-IO. The ranking in (425) is needed to derive the diminutive in Upper Chehalis and Cowlitz.

(425)	$/\mu + C_1VC/$	*FLOAT	INTEGRITY	WT-IDENT-IO
	a. CVC μ	*!		
	 b. CV:C			*
	c. C_1VC_1C		*!	

If reduplication (i.e., fission) is attested, rather than vowel length, INTEGRITY must be ranked lower than WT-IDENT-IO, as shown by the “reduplicating” grammar in

(426). This ranking is needed for reduplicative morphemes in other languages; it is also needed for the stratum in Upper Chehalis and Cowlitz where reduplication does occur (see Section 6.6.2.3.2).

	$/\mu + C_1VC/$	*FLOAT	WT-IDENT-IO	INTEGRITY
(426)	a. CVC μ	*!		
	b. CV:C		*!	
	 c. C_1VC_1C			*

The two rankings are summarised in Table 6.12. Note that a highly ranked markedness constraint against long vowels would suffice for Salish languages without long vowels; WT-IDENT-IO is necessary for languages like Lushootseed which have phonemic vowel length but do not permit changes in vowel length with reduplicative morphemes.

Table 6.12: Lengthening vs. Reduplicating Grammars

Ranking	Result
INTEGRITY >> WT-IDENT-IO	vowel lengthening preferred
WT-IDENT-IO >> INTEGRITY	fission preferred

6.6.2.2.2 Variation in Moraic Codas: Northern Straits, Klallam, Bella Coola

I assume that coda consonants are moraic in Salish, as discussed in Chapter 3 (Section 3.2.1); however, C_1C_2 reduplication in Northern Straits, Klallam, and Bella Coola shows that certain consonants cannot fill a mora at the stratum where reduplication occurs. The ranking of markedness constraints pertaining to which segments may be associated with a mora relative to other constraints is therefore relevant to understanding cross-Salish variation.

A uvular or velar stop in C_2 position in Bella Coola undergoes spirantisation to [x] in C_1C_2 reduplication (Saunders & Davis, 1972, p.9), as shown in (427); this is not seen in other Salish languages, which retain a faithful output in fission.²²

²²In addition to spirantisation, C_1C_2 reduplication may involve concurrent uvular > velar change, as seen in (427b–c), and unrounding, as seen in (427a–c).

- (427) a. **six**~sik^w (cf. sik^w - ‘pull’)
 PL~pull
 ‘pull (cont.)’ (Bagemihl, 1991, p.601, Bella Coola)
- b. **fix**~fiq^w (root: fq^w ‘to fade’)
 PL~fade
 ‘It is fading’ (Nater, 1990, p.63, Bella Coola)
- c. **fix**~fiq^w (root: fq^w - ‘to sob’)
 PL~sob
 ‘to be sobbing’ (Nater, 1990, p.60, Bella Coola)

A bilabial or alveolar stop in Bella Coola is replaced with [n] in C₁C₂ reduplication, as shown in (428).²³

- (428) a. **sun**~supt-c (cf. *suptc* ‘I whistle.’)
 PL~whistle-1SG.SUBJ
 ‘I whistle (iter. act.).’ (Saunders & Davis, 1972, p.9, Bella Coola)
- b. **cin**~citumix-c (cf. *citumixc* ‘I take a nap.’)
 PL~nap-1SG.SUBJ
 ‘I take a nap (iter. act.).’ (Saunders & Davis, 1972, p.9, Bella Coola)
- c. **?un**~?utak-c (cf. *?utak* ‘I vomit.’)
 PL~vomit-1SG.SUBJ
 ‘I vomit (iter. act.).’ (Saunders & Davis, 1972, p.9, Bella Coola)

²³Saunders and Davis (1972, p.2) report that [n] with C₁C₂ reduplication can mark an “iterative potential”, which indicates that an “action is not occurring at the time of the utterance”. They show a contrast between “iterative potential” and “iterative actual” forms, the latter without [n]. A pair of examples is provided in (i).

- (i) a. **li<n>**~lis-ic (cf. *lisic* ‘I push it.’)
 PL<n>~push-1SG.SUBJ
 ‘I push it again and again, but not now.’ (Saunders & Davis, 1972, p.2, Bella Coola)
- b. **lis**~lis-ic (cf. *lisic* ‘I push it.’)
 PL~push-1SG.SUBJ
 ‘I push it again and again.’ (Saunders & Davis, 1972, p.1, Bella Coola)

The contrast in (i) is neutralised with roots with a bilabial or alveolar stop, which have [n] in both the “iterative potential” and “iterative actual” forms (Saunders & Davis, 1972, pp.9–10). I focus here on the forms that have [n], regardless of function.

- d. $\dot{\lambda}an \sim \dot{\lambda}ap$ - (cf. $\dot{\lambda}p$ ‘to go, start’)
 CONT $\sim$ go-
 ‘go (cont.)’ (Bagemihl, 1991, p.601, Bella Coola)

The alternations with [x] and [n] in (427) and (428) above suggest that stops cannot fill the μ of $\sigma_{N+\mu}$.²⁴

Varieties of Straits (Northern Straits and Klallam) also restrict which consonants may be associated with the moraic coda of an affixed $\sigma_{N+\mu}$; though this is only seen when marking plural entities in Straits. In Klallam, plural C_1C_2 reduplication is only found with nouns that have a sonorant in C_2 position: otherwise an infixed $-\dot{a}y/-a\dot{y}-$ is used.²⁵ Plural nouns in Northern Straits are typically marked with a fixed segment infix $-l-$ (see, e.g., plural forms in Montler’s 2018 dictionary); the ones formed with C_1C_2 reduplication also involve sonorants in C_2 position. Both $-\dot{a}y/-a\dot{y}-$ in Klallam and $-l-$ in Northern Straits are cognates of the $-l-$ plural in Halkomelem, discussed in Chapter 5 (Section 5.3.1.3). Examples from the Saanich dialect of Northern Straits are given in (429).

- (429) a. $\dot{s}-\dot{t}^{\theta}\dot{a}m \sim \dot{t}^{\theta}\dot{a}m-\dot{a}čən$ (cf. $\dot{s}\dot{t}^{\theta}\dot{a}m\dot{a}čən$ ‘bracelet’)
 for-PL $\sim$ bone-centre
 ‘bracelets’ (Montler, 2018, p.1431, Saanich - Northern Straits)
 b. $\dot{x}\dot{a}l \sim \dot{x}\dot{e}l\dot{a}w$ (cf. $\dot{x}el\dot{a}w$ ‘spoon’)
 PL $\sim$ spoon
 ‘spoons’ (Montler, 2018, p.1464, Saanich - Northern Straits)
 c. $s-\dot{x}\dot{a}n \sim \dot{x}\dot{a}n\dot{a}?$ (cf. $\dot{x}\dot{a}n\dot{a}?$ ‘foot’)
 NMLZ-PL $\sim$ foot
 ‘feet’ (Montler, 2018, p.1466, Saanich - Northern Straits)
 d. $\dot{x}<\dot{a}l>\dot{a}ws-\dot{a}l\dot{a}k^w\dot{a}?$ (cf. $\dot{x}\dot{a}ws\dot{a}l\dot{a}k^w\dot{a}?$ ‘new dancer’)
 new<PL>-bodyside
 ‘new dancers’ (Montler, 2018, p.1468, Saanich - Northern Straits)

²⁴I have argued elsewhere that stops in Bella Coola are not moraic (Mellesmoen, 2021b); while I remain agnostic here about their status in the language as a whole, the evidence presented here does show that new associations between a stop and a mora do not occur at the same stratum as fission.

²⁵ C_1C_2 reduplication is found with 10% of plural words in Klallam (Montler, 2015, p.175–180). There is also a rare C_1 reduplication pattern that occurs with approximately 20 words (Montler, 2015, p.181).

- e. $s-\dot{x}^w <\dot{a}l> \acute{a}q^w -\acute{e}n$ (cf. $s\dot{x}^w \acute{a}q^w \acute{e}n$ ‘canoe pole’)
 NMLZ-prop<PL>-instrument
 ‘canoe poles’ (Montler, 2018, p.1477, Saanich - Northern Straits)
- f. $\dot{x}^w <\dot{a}l> \acute{e}n$ (cf. $\dot{x}^w \acute{e}n$ ‘fast’)
 fast<PL>
 ‘swift water’ (Montler, 2018, p.1480, Saanich - Northern Straits)

I use the *MORA family of constraints from Morén (1997) to account for the patterns in Bella Coola and varieties of Straits discussed here. These constraints draw on the sonority-based typology of Zec (1988). While Morén (1997, p.6) uses *MORA[segment], where specific segments are listed, I use the format *MORA[feature]. The constraint *MORA[-sonorant] in (430a) is needed to account for the patterns in Klallam and Northern Straits; *MORA[-continuant] in (430b) is needed for Bella Coola. These constraints are violated by moraic obstruents and stops, respectively.

- (430) a. *MORA[-sonorant]: Assign a violation mark for every obstruent associated with a mora.
 b. *MORA[-continuant]: Assign a violation mark for every [-continuant] segment associated with a mora.

Assuming that Straits has listed allomorphs for the plural morpheme and that $\sigma_{N+\mu}$ is the preferred allomorph, as in Halkomelem (see Section 5.3), ranking *MORA[-sonorant] above PRIORITY in Northern Straits predicts that the *-l-* allomorph will be inserted to avoid an obstruent coda.²⁶ The ranking must be reversed in Halkomelem (see Section 5.3): *-l-* is selected to avoid multiple reduplication, but obstruent codas are permitted. In Bella Coola, the constraint *MORA[-continuant]

²⁶The constraint *MORA[-continuant] is not ranked as high at other strata in Saanich: when C_1C_2 reduplication marks plural events (or a property), rather than participants, an obstruent is copied. C_1C_2 reduplication marking plural events must be at different strata than reduplication marking plural entities in Saanich, as was proposed for Comox-Sliammon in Chapter 5 (Section 5.5.10). This suggests that plural reduplication is not cross-categorial, which has been claimed in previous work (e.g., Wiltschko, 2008).

- (i) a. $m\acute{o}k^w \sim m\acute{o}k^w -\acute{e}n$
 REP~claim-MID
 ‘picking up (repeatedly)’ (Montler, 2018, p.1290, Saanich - Northern Straits)

outranks *MORA[-sonorant], which allows for fricatives, but not stops, to be moraic. The constraints *MORA[-sonorant] and *MORA[-continuant] are ranked lower in other Salish languages than in Straits and Bella Coola, respectively: obstruent codas are tolerated in C₁C₂ reduplication.

Table 6.13: Mora Constraints

Ranking at Stratum where $\sigma_{N+\mu}$ Parsed	Languages
*MORA[-sonorant] >> PRIORITY	Northern Straits and Klallam
PRIORITY >> *MORA[-sonorant]	Halkomelem
*FLOAT, MAX- μ , *MORA[-continuant] >> *MORA[-sonorant]	Bella Coola
*FLOAT, MAX- μ >> *MORA[-sonorant], *MORA[-continuant]	all other Salish languages

6.6.2.2.3 “Wrong Side” Reduplication: Twana and Tillamook

Affixed $\sigma_{N+\mu}$ is filled by əC in Twana and Tillamook, as discussed earlier in this chapter (Section 6.5.3.3.2); əC is an intermediate representation present in the output of the stratum where $\sigma_{N+\mu}$ is parsed into the word.

None of the rankings used in Chapters 4 or 5 predict əC as the output corresponding to an input $\sigma_{N+\mu}$. Twana and Tillamook require INTEGRITY and DEP to be ranked above ONSET, which is violated by any syllable without an onset (see Section 4.1.3.1.2), such that the grammar will only do the bare minimum (i.e., provide segment content to *just* the N and μ nodes associated with $\sigma_{N+\mu}$). The fact that ONSET is ranked lower than INTEGRITY and DEP means that a candidate with a syllable without an onset will be preferred over one that incurs an additional violation of INTEGRITY or DEP to provide an onset.

The ranking of STROLE, which comes from McCarthy and Prince (1994, p.34) and is defined in (431), is also important in predicting the correct form

-
- b. mə<I>~mǽk^w-əŋ
 PL<PL>~claim-MID
 ‘pick up (pl.)’ (Montler, 2018, p.1290, Saanich - Northern Straits)

of C_1C_2 reduplication. STROLE is violated when an output segment does not have the same role in the syllable as its input correspondent. The "St" in STROLE is not directly defined in McCarthy and Prince (1994); I assume STROLE stands for structural role (of a syllable).

- (431) STROLE: A segment in the input and its correspondent in the output must have identical syllabic roles.

STROLE ensures that the second consonant of the root, which is a coda, undergoes fission to fill a coda, instead of the first consonant, which is an onset.

The tableau in (432) shows the ranking needed to derive the correct intermediate form for *pčap* 'aunts' in Twana, which is given as candidate (432a). Candidate (432b), with fission providing an onset, incurs two violations under INTEGRITY-C, with the second being fatal. Candidate (432c), with fission of the first consonant, fatally violates STROLE because one of the fissioned consonants is an onset and the other is a coda.

/ $\sigma_{N+\mu}$ + čap/		STROLE	DEP-V	INTEGRITY-C	ONSET
(432)	a. əpčap		*	*	*
	b. čəpčap		*	**!	
	c. əččap	*!	*	*	*

The ranking shown in (432) yields the correct intermediate form of plural C_1C_2 reduplication in Twana. The [ə] will be deleted at subsequent strata to yield *pčap* 'aunts'. In cases where deletion of a [ə] would yield an illicit cluster with sonorants, fission occurs at a later stratum to provide an onset; the intermediate form *ədbədəh* for example (415) above yields *bədbədəh* 'children', not **dbədəh*.

6.6.2.3 Cross-Salish Variation: Morpheme to Level Mappings and Number of Relevant Levels

Evidence for the number of levels relevant to reduplication was discussed in Section 6.6.2.1.3 for the case study languages. There are three types of evidence that can be used to determine morpheme to level mappings: (i) multiple reduplication, (ii) grammar (constraint rankings), and (iii) interaction with other morphemes and phonological processes.

Multiple reduplication is the most straightforward diagnostic because it only requires a small number of attested forms to determine the minimum number of strata that relevant for reduplication: if two types of reduplication occur in the same word (and each reflects an affixed prosodic unit, rather than a fixed-segment affix), then they must be parsed into the word at different levels. This follows from the constraint MORPHEME-AFFILIATION, discussed in Section 6.6.1.6 above, which is violated if segmental content from one morpheme is associated with prosodic units affiliated with more than one other morpheme.

Multiple reduplication patterns indicate a minimum number of reduplication-relevant strata; a more concrete number is only possible if an analysis of individual reduplication patterns and the general phonological grammar of the language is undertaken.

6.6.2.3.1 Overview of Cross-Salish Variation in Number of Relevant Strata

Reduplicative morphemes are parsed into the phonological word at one to four strata across Salish languages. The number of strata per language relevant to reduplication as evidenced by available documentation are summarised in Table 6.14.

Table 6.14: Number of Reduplication Relevant Strata

Levels	Languages
4	Thompson River Salish
3	Moses Columbia, Lillooet, Montana Salish-Spokane-Kalispel, Coeur d’Alene Comox-Sliammon, Sechelt, Shuswap, Okanagan
2	Lushootseed, Halkomelem, Tillamook, Twana, Straits
1	Upper Chehalis, Lower Chehalis, Cowlitz, Bella Coola, Nooksack, Squamish
Insufficient Evidence	Pentlatch, Quinault

The number of reduplication-relevant strata in Table 6.14 come from multiple

reduplication patterns, though these have been refined where different constraint rankings are motivated by individual reduplication patterns. For example, Okanagan and Shuswap have a C_2^2 triplication pattern similar to the one in Lillooet; triplication happens at a later stratum than the other types of reduplication. For languages where documentation of multiple reduplication patterns is sparse, the number of relevant strata given is the minimum number motivated by the available evidence. For example, the existence of multiple reduplication in Tillamook minimally requires two strata; further analysis may motivate additional strata. Finally, the data for Quinault and Pentlatch are too sparse to allow for any conclusion to be drawn from the absence of attested multiple reduplication.

Instances of multiple reduplication in Salish tend to involve only two reduplicative morphemes at a time, as noted in Chapter 2 (Section 2.8); this restriction is only predicted for two-stratum languages, since a single reduplicative morpheme can be parsed into the word at each stratum. For languages with more than two strata, there is nothing to stop multiple reduplication with three or more reduplicative morphemes. And, in fact, though there are few attested examples, the co-occurrence of three reduplicative morphemes in a single word is documented in a handful of Salish languages, including Montana Salish-Spokane-Kalispel, Thompson River Salish, and Coeur d'Alene. Examples are given in (433).²⁷ Each of these languages needs at least three strata on the basis of pairwise combinations of reduplicative morphemes. These forms, albeit rare, show that there are no phonological restrictions against three reduplicative morphemes in the same word. Having three types of reduplication can be taken as evidence for reduplication at three different stratum, provided that each reduplicative morpheme is a prosodic affix.

- (433) a. $\acute{x} \sim \acute{x}\acute{a}l \sim \acute{x} \langle e \rangle \langle \acute{l} \rangle \acute{l}$ (cf. $\acute{x}lip$ 'It stopped.')
- DIM ~ PL ~ still <REP> <COS>
- 'They (sg.) keep having little (uncontrolled) seizures (as in epilepsy).'
- (B. Carlson, 1989, p.211, Spokane)

²⁷I have added the [ə] in $C_1 C_2$ reduplication (433a) and (433c), though they are not present in the source material.

- b. $\text{ʔe s-t-paʔ} \sim \text{pá} < \text{p} > \text{ʔ} \sim \text{paʔ-t}$
 INT STAT-QUAL-PL $\sim$ fade $\sim$ PDEN-CHAR
 ‘little grey faded ones’ (cf. *ʔe stpaʔpáʔpaʔ* ‘several are grey, faded’)
 (L. C. Thompson & Thompson, 1996, p.227, Thompson River Salish)
- c. $\text{n-ləm} \sim \text{lé} < \text{l} > \text{m} \sim \text{əlm}$ (cf. *nlélməm* ‘to carry unannounced’)
 LOC-PL $\sim$ enter $< \text{DIM} > \sim$ COS
 ‘Small things are carried unnoticed.’ (Jimmie, 1994, p.90, Thompson River Salish)
- d. $\text{cə} \sim \text{cəq} \sim \text{cəq} \sim \text{əq-l-íp-m}$
 DIM $\sim$ C₁ C₂ $\sim$ place.vessel $\sim$ COS-CONN-back-MID
 ‘The little one fell on his back.’ (cf. *čicáqʔc* ‘They (sg.) set themselves before them.’) (Doak, 2004, p.121, Coeur d’Alene)

Returning to the question of cross-Salish variation in the number of reduplication-relevant strata, recall that the case-study languages have either two or three strata, which means that only the one and four stratum languages warrant additional discussion. This is provided in Sections 6.6.2.3.2 and 6.6.2.3.3, respectively; the one-stratum languages are of particular interest because they show that even languages with reduplication at only one stratum may vary with respect to whether reduplication occurs at Level 1 or at a later stratum.

6.6.2.3.2 One Reduplication-Relevant Stratum (*Bella Coola, Tsamosan, Nooksack, Squamish*)

Languages with one reduplication-relevant stratum include Bella Coola, all four Tsamosan languages (Upper Chehalis, Cowlitz, Lower Chehalis, and Quinault), Nooksack, and Squamish. None of these languages uses multiple reduplication. I focus on Upper Chehalis as an example of this type of language.

Upper Chehalis has plural C₁C₂ reduplication and change-of-state C₂ reduplication, which are shown in (434) and (435), respectively; it makes use of vowel length to mark the diminutive, rather than C₁ reduplication, as discussed in Section 6.6.2.2.1. All three processes are triggered by a prosodic affix, which means by hypothesis they cannot co-occur at the same stratum.

- (434) a. ʔit $\text{wák}^w \sim \text{wak}^w\text{s}$ (cf. wak^ws ‘go’)
 DEF.DET PL $\sim$ go
 ‘they (sg.) went again and again’ (Kinkade, 1991, p.150, Upper Chehalis)
- b. $\text{ʔac-yám} < \text{yám} > \text{x}^w\text{-t}$ (cf. $\text{yám}\text{x}^w\text{-t}$ ‘become round’)
 STAT-round<PL>-INTR
 ‘bumps’ (Kinkade, 1991, p.174, Upper Chehalis)
- (435) a. ʔit $\text{yáx} \sim \text{ax}$ (cf. yaxt ‘go home’)
 DEF.DET go.home $\sim$ COS
 ‘They (sg.) went home slowly.’ (Kinkade, 1991, p.171, Upper Chehalis)
- b. $\text{sák}^w \sim \text{k}^w\text{-mit-n}$ (cf. sak^wm ‘whisper’)
 whisper $\sim$ COS-MID-3SUBJ(IPFV)
 ‘They (sg.) are whispering while going.’ (Kinkade, 1991, p.123, Upper Chehalis)

While C_1C_2 reduplication and change-of-state C_2 reduplication do not co-occur, both may combine with diminutive vowel lengthening, as shown in (436a) and (436b), respectively. The combination of plural and change-of-state is not semantically impossible, as it is attested in many other Salish languages, such as Lillooet (Section 5.4.2.2).

- (436) a. $\text{yó} < \text{ʔ} > \text{t} \sim \text{yut-n}$ (cf. yút-n ‘stir it’)
 stir<DIM> $\sim$ PL-TR
 ‘stir it a little (freq.)’ (Kinkade, 1991, p.176, Upper Chehalis)
- b. $\text{s-ʔé} < \text{:} > \text{t} < \text{it} > \text{n-n}$ (cf. ʔitn ‘eat’)
 eat<DIM><COS>-3SUBJ(IPFV)
 ‘eat slowly away’ (Kinkade, 1991, p.12, Upper Chehalis)

The combination of diminutive vowel lengthening with reduplication in (436), together with the absence of multiple reduplication, suggests that reduplication happens at a single stratum, while diminutive is parsed at a different stratum. It is possible to derive both C_1C_2 and C_2 reduplication with the same grammar: both are edge-aligned and may copy a vowel. The diminutive morpheme requires a

different ranking because it is aligned to stress and realised as length rather than reduplication (see discussion of reduplicating vs. lengthening grammars in Section 6.6.2.2.1). The fact that the diminutive is aligned to prosodic structure, which is not present in the input to Level 1, further means that it must be parsed at Level 2 (or later).

The absence of multiple reduplication in Upper Chehalis is explained on the assumption that only Level 1 has a constraint ranking allowing reduplication (i.e., WT-IDENT-IO >> INTEGRITY). If Level 2 had the same ranking rather than the reverse (i.e., INTEGRITY >> WT-IDENT-IO), diminutivity would be marked by reduplication; multiple reduplication would be possible in this case because reduplication would occur at two different strata.

The lexical entries for change of state and plural reduplication, together with diminutive vowel lengthening, are given in (437a–c), respectively. The lexical entries for change-of-state and plural morphemes are left unmarked for stratum and therefore are parsed at Level 1, while the diminutive is marked as a Level 2 affix.

- (437) a. $\sigma \Leftrightarrow \langle \text{COS} \rangle$
 b. $\sigma_{N+\mu} \Leftrightarrow \langle \text{PL} \rangle$
 c. $\mu_{L2} \Leftrightarrow \langle \text{DIM} \rangle$

Not all languages with one relevant stratum have reduplication at the first stratum, as in Upper Chehalis. Reduplication in Bella Coola is positioned to the first moraic segment (see discussion of C_1C_2 reduplication in Section 2.4.1), which means that some prosodic structure must have been built prior to reduplication. Reduplication in Bella Coola must therefore happen at Level 2 (or later). Taken together, one-stratum languages show that reduplication is not restricted to the same stratum across Salish (e.g., Level 1), even in languages with only one reduplication-relevant stratum.

6.6.2.3.3 *Four Reduplication-Relevant Strata (Thompson River Salish)*

There are four strata relevant to reduplication in Thompson River Salish. Examples of multiple reduplication with three morphemes were given in (433b) and (433c)

above: further cases are given below in (438); each type of reduplication in each example must happen at a different stratum.²⁸

- (438) a. **cəm**~cm<e?>~**m-é**<**ṃ**>kst (root: *cm-* ‘small’)
 PL~small<REP+SPZG>~COS-hand<DIM>
 ‘have small hands’ (L. C. Thompson & Thompson, 1996, p.35, Thompson River Salish)
- b. **nəx**~né<**ṇ**>x~**əx** (cf. *néxk^wu* ‘have waves’)
 PL~wave<DIM>~COS
 ‘small, tiny waves, ripples’ (L. C. Thompson & Thompson, 1996, p.217, Thompson River Salish)
- c. **təp**~tə<**ttəp**-t (cf. *təptép* ‘between black and grey’)
 PL~black<DIM>~PDEN-CHAR
 ‘a very little black one’ (L. C. Thompson & Thompson, 1996, p.351, Thompson River Salish)

Reduplicative morphemes, excluding the affective/initial C₁ pattern (discussed in Section 2.5.3.3), are summarised in Table 6.15.

Table 6.15: Reduplication in Thompson River Salish

Morpheme	Position	Combinations
PDEN	Edge-Aligned (C ₁ C ₂ - or -C ₁ C ₂ -)	PL, DIM
PL	Edge-Aligned (C ₁ C ₂ -)	COS, DIM, PDEN
COS	Stress-Aligned (-C ₂ -)	PL, DIM
DIM	Stress-Aligned (-C ₁ -)	PL, PDEN, COS

On the basis of multiple reduplication, at least three levels are relevant; the grammar motivates an additional level to distinguish change-of-state and property-denoting reduplication, which require different constraint rankings to predict stress-aligned and edge-aligned patterns, respectively. A morpheme-level mapping is provided in Table 6.16.

²⁸See Section B.1.7 in the appendix for examples with two types of reduplication.

Table 6.16: Levels in Thompson River Salish

Level	Morpheme
1	PDEN (and AFF)
2	PL
3	COS
4	DIM

The order of levels in Table 6.16 is motivated by interactions between morphemes. PL (and DIM) reduplication can copy the inchoative infix *-ʔ-* (L. C. Thompson & Thompson, 1992, p.97), which means both must occur later than Level 1, where that infix is parsed into the word. DIM and COS are stress-aligned and may reduplicate suffixed content, which means they must be later than Level 1; DIM interacts with morphemes beyond the ones COS interacts with, which means it may be at a later stratum. DIM must be later than PL because otherwise $yəq^w \sim yú < y > q^w -n-s$ ‘splash something/someone a little repeatedly’ (Jimmie, 1994, p.31) would be $*yəy^w \sim yú < y > q^w -n-s$. As the evidence includes a number of non-reduplicative morphemes, a full account with examples is beyond the scope of this section.

6.6.2.3.4 Summary

MORPHEME-AFFILIATION and the fact that morphemes are lexically specified for stratum means that the attested combinations of reduplicative morphemes in one language will not necessarily match those in another. For this reason, the inventory of multiple reduplication can vary across the family and even within branches of the family. This is supported by attested variation in the number of strata relevant to reduplication and which morphemes are parsed into the word at each stratum.

6.7 Step 6: Post-Reduplication Phonological Processes and Phonetic Implementation (Phonology/Phonetics)

The output of any stratum in the word-level derivation is the input to the next one, as described in Section 6.6.1. The output of the final stratum is part of the input to

the postlexical phonology, where phonetic implementation rules apply (see, e.g., Bermúdez-Otero 2007).

In this section, I discuss how constraint rankings at later strata can affect the form of reduplicative morphemes that are parsed at an earlier stratum (Section 6.7.1). Phonetic implementation rules are briefly discussed in Section 6.7.2.

6.7.1 Post-Reduplication Phonology

The morphosyntactic word, which corresponds to a w-span (see Step 1 in Section 6.2), is the domain where reduplication occurs in Salish. Constraint rankings at strata late in the word-level derivation or in the post-lexical component of the grammar can impact the surface form of a reduplicated word. I use the term “post-reduplication phonology” to refer to phonological processes that result from the grammar at any stratum following the one where a given reduplicative morpheme is parsed into the word.

Post-reduplication phonology accounts for cases where the surface form of a reduplicative morpheme does not correspond to the prosodic unit or fixed segment in its underlying form, such as the “onset” reduplication in Southern Interior Salish discussed in Section 6.5.3.3.1: a complex onset results from an affixed syllable with later vowel deletion. Common post-reduplication phonological processes in Salish include glide vocalisation (see, e.g., discussion of general phonological processes in Comox Sliammon in Section 4.1.1), stress shift (discussed for word-level reduplication in Lillooet in Section 4.2.4), vowel reduction (see, e.g., Halkomelem C₁ reduplication in Section 5.3.1.1), schwa epenthesis (see, e.g., Comox-Sliammon C₁ reduplication in Section 4.1.4.1.1), and schwa deletion (see, e.g., Lillooet C₂ reduplication in Section 4.2.4.1.2).

Post-reduplication phonology is independent of reduplication and variation is fully dependent on the grammar of individual languages. I provide an example of a variable post-lexical reduction process in Section 6.7.1.1, followed by an example of a word-level post-reduplication process in Section 6.7.1.2.

6.7.1.1 Post-Lexical Phonology: Schwa Deletion in Lillooet

Post-lexical phonology differs from word-level phonology because processes may be optional or show variable application across different speech rates (see, e.g., Kaisse & Shaw, 1985).

Schwa deletion in Lillooet is an example of a post-lexical phonological process. The [ə] needed to fill an input $\sigma_{N+\mu}$ for C_1C_2 reduplication may be omitted in the surface form (see Section 4.2.3.2.1); forms with either [ə] or $\emptyset$ are acceptable, as shown in (439).²⁹

- (439) a. **k^wʔz~k^wz-əm -or- k^wʔz~k^wəz-əm** (root: *k^wz-* ‘smooth’)
 PDEN~smooth-CHAR
 ‘smooth’ (H. Davis et al., in prep, Lillooet)
- b. **cəɬ~cɬ-əm -or- cəɬ~cəɬ-əm** (root: *cɬ-* ‘bright, sober’)
 PDEN~bright/sober-CHAR
 ‘bright/sober’ (H. Davis et al., in prep, Lillooet)
- c. **kín~kən-t -or- kín~kn-ət** (root: *kín-*)
 dangerous~PDEN-CHAR
 ‘dangerous’ (H. Davis et al., in prep, Lillooet)

Deletion of [ə] is not limited to reduplicative contexts; it is common in fast speech (H. Davis, p.c.). Variation in the surface form of C_1C_2 reduplication between [ə] or $\emptyset$ is therefore consistent with the general post-lexical patterns in the language (i.e., any unstressed schwa may be optionally deleted unless its deletion leads to a sonority hierarchy violation).

6.7.1.2 Word-Level Post-Reduplication Phonology: Compensatory Lengthening in Bella Coola and Upriver Halkomelem

Compensatory lengthening in Bella Coola and Upriver Halkomelem is a word-level phonological process that follows reduplication. I propose that vowel length in these languages is derived from an intermediate /ʔ/, which itself results from an

²⁹ An epenthetic [ə] appears later in the word for the $\emptyset$ form in (439c).

underlying μ (Upriver Halkomelem) or a [c.g.] feature (Bella Coola).³⁰

Vowel length marks the imperfective form of roots that begin with /ʔ/ or /h/ in Upriver Halkomelem, as shown in (440). These forms correspond to those with an infix glottal stop in the other two dialects (see data and description in Section 5.3.1.1.1). The glottal stop infix is one of the realisations of the imperfective μ in Halkomelem, which was discussed in Chapter 5 (Section 5.3.1.1.2).³¹

- (440) a. ʔí<: >ʔtəl (cf. ʔéʔtəl ‘eat a meal’)
eat<IPFV>
‘eating a meal’ (Galloway, 2009, p.10, Upriver Halkomelem)
- b. ʔí<: >əs (cf. ʔiwəst ‘advise/teach someone’)
advise/teach<IPFV>
‘directing, training, teaching, guiding’ (Galloway, 2009, p.128, Upriver Halkomelem)
- c. hé<: >ᵏʷ-ələs (cf. héᵏʷələs ‘remember someone’)
remember<IPFV>-GOAL
‘remembering someone’ (Galloway, 2009, p.97, Upriver Halkomelem)
- d. há<: >qʷ-ət (cf. háqʷət ‘smell something’)
smell<IPFV>-TR
‘smelling something’ (Galloway, 2009, p.106, Upriver Halkomelem)

In Bella Coola, the diminutive is expressed with some combination of reduplication, vowel length, and a suffix (-i). Examples with all three components are given in (441). I propose that the lexical entry for the diminutive in Bella Coola contains three components: a prosodic unit (μ), a floating feature ([c.g.]), and a fixed segment (-i).

³⁰This analysis of Bella Coola is similar to one proposed by Bagemihl (1991, p.641), who argues that compensatory lengthening occurs as a general process in Bella Coola following the loss of a glottal stop in coda position. However, he does not connect this analysis to vowel length in diminutive forms, nor does he propose an underlying glottal component associated with the diminutive morpheme.

³¹Zimmermann (2017, p.220) analyses the imperfective as an affixed foot and proposes that lengthening the vowel is needed to create a heavy syllable ($\sigma_{\mu\mu}$) which can be parsed as a binary foot. In her analysis, a possible reduplicated candidate *ʔiʔəmæx fares worse than the attested form ʔimæx ‘walking’ due to a constraint *ʔə, which is violated by any syllable consisting of [ʔə].

- (441) a. **xi**~xna<: >s-**i** (cf. *xnas* ‘woman’)
 DIM~woman<DIM>-DIM
 ‘woman (dim.)’ (Bagemihl, 1991, p.601, Bella Coola)
- b. qa<**qa**>yt-**i** (cf. *qayt* ‘hat’)
 hat<DIM>-DIM
 ‘toadstool (dim.)’ (Bagemihl, 1991, p.598, Bella Coola)
- c. x^wŋ<**x^wn**>a<: >t-**i** (cf. *x^wnat* ‘spring of water’)
 spring.of.water<DIM><DIM>-DIM
 ‘spring of water (dim.)’ (Bagemihl, 1991, p.615, Bella Coola)

The glottal stop is epenthetic in Upriver Halkomelem, as in other dialects (see analysis in Section 5.3.1.1.2). The glottal stop in Bella Coola is the realisation of [c.g.], which is realised as a glottal stop following the stressed vowel, similar to the pattern described for the diminutive in Halkomelem in Chapter 5 (Section 5.3.1.2). Both languages have a highly ranked markedness constraint that is violated by [ʔ] in a coda, defined in (442).³² This constraint is ranked high enough at a stratum following reduplication to result in [ʔ] deletion, which leaves its mora to trigger compensatory lengthening of the adjacent vowel.

- (442) *ʔ]_σ: Assign a violation mark for any [ʔ] in a coda.

This constraint is independently motivated by the general phonology of each language because glottal stops do not occur as codas in the Upriver dialect of Halkomelem (Elmendorf & Suttles, 1960, p.8), nor in Bella Coola (Nater, 1984, p.20).³³

Table 6.17 shows the position of *ʔ]_σ relative to other constraints for both languages. At the stratum where the reduplicative morpheme is parsed into the prosodic word, MAX outranks *ʔ]_σ (and DEP), which allows for epenthesis of

³²Given that I assume all codas are moraic in Halkomelem, I could use a *MORA constraint that more generally penalises moraic glottal consonants, as in Section 6.6.2.2.2. However, given the fact that codas were argued to be non-moraic in Bella Coola, this approach would incorrectly predict that the glottal stop could lose its mora but remain a coda in Bella Coola.

³³The Upriver Halkomelem pattern has largely been discussed from a diachronic perspective, as in Brown and Thompson (2005, p.49). However, the imperfective pattern is a synchronic process that yields a coda [ʔ] at an intermediate level of the derivation. The ban on coda [ʔ] must still be active in the grammar.

[ʔ] into a coda position to provide segmental content to a floating μ in Upriver Halkomelem and a floating [c.g.] feature in Bella Coola. At a subsequent stratum, $*ʔ]_{\sigma}$ and $\text{MAX-}\mu$ are ranked above $\text{MAX-}[\text{C.G.}]$ and WT-IDENT-IO (see Section 6.6.2.2.1 for discussion of this constraint), which results in [ʔ] deletion and compensatory lengthening.

Table 6.17: Post-Reduplication Compensatory Lengthening

Stratum	Bella Coola	Upriver Halkomelem
Reduplication	$\text{Max-}[\text{c.g.}] >> *ʔ]_{\sigma}$	$\text{Max-}\mu >> *ʔ]_{\sigma}$
Post-Reduplication	$\text{Max-}\mu, *ʔ]_{\sigma} >> \text{Max-}[\text{c.g.}], \text{WT-IDENT-IO}$	

Compensatory lengthening must occur after the insertion of a glottal stop to fill the imperfective μ or after diminutive reduplication has occurred. Deletion of [ʔ] is a categorical process, in contrast to the variable [ə] deletion in Lillooet discussed in the previous section, meaning that that glottal stops never surface in a coda position in Upriver Halkomelem or Bella Coola.

6.7.2 Phonetic Implementation

The difference between phonological and phonetic processes can be viewed as a contrast between discrete and continuous values, such that representations in the lexical and phonological components of the grammar have discrete values (Bermúdez-Otero, 2007, p.502). Discrete phonological representations crucially lack the gradient phonetic detail associated with surface forms. For example, place features may be shared between two segments in the phonology, but the options are limited to [+feature], [–feature], or being under-specified for the feature because phonological representations are always categorical.

In contrast to categorical features present in phonological representations, phonetic realisations are gradient and continuous. For example, a vowel adjacent to a uvular consonant may undergo some amount of lowering that is strongest near the boundary between the two segments (see, e.g., Bessell 1993; Nolan 2017; Mellesmoen & Huijsmans, 2019a). This allows considerable variation in the surface realisation of vowels as a result of phonetic implementation.

Differences between Salish languages in the surface form of reduplication, particularly in vowel quality, may sometimes reflect differences in language-specific patterns of phonetic implementation (i.e., how discrete phonological representations are mapped to continuous acoustic space). Phonetic implementation is an area that deserves further examination where acoustic data is available.

6.8 Implications of the Model

The model proposed in this chapter is motivated by reduplication patterns found across Salish; it also has larger implications for the overall architecture of the phonological side of the grammar and the relationship between its different components. In the remainder of this chapter, I discuss modularity and the interface between morphology and phonology (Section 6.8.1), the status of non-concatenative morphology with respect to linearisation and allomorph selection (Section 6.8.2), and the mapping of morphosyntax to phonological domains (Section 6.8.3).

6.8.1 Modularity and the Interface between Morphology and Phonology

The phonological grammar must have access, directly or indirectly, to certain types of information about morphology. First, in order to derive the correct surface order of the phonological exponents of morphemes, the phonology must know whether a morpheme was linearised as a prefix or a suffix (see discussion of PREFIX in Section 6.6.1.5). Second, in order to predict gaps in multiple reduplication, the phonology must know, for any pair of phonological elements, whether they belong to the same morpheme (see discussion of MORPHEME-AFFILIATION in Section 6.6.1.6).³⁴ Both of these requirements are met by the model of the grammar adopted here.

Only information that can be read off the ordered string of phonological content following Stage II of lexical insertion (described in Section 6.5) is available to the phonology. All phonological representations associated with the heads that

³⁴The phonology is also able to see a diacritic specifying phonological content for a particular level (see Section 6.6.1.1 and the following section), though this is marked explicitly on the lexical entry and therefore not discussed in this section.

comprise a w-span (defined in Section 6.2) are present in the input to the first stratum, though they remain as unparsed groups of phonological material until the stratum where they are parsed into the phonological word. Information related to a prefix or suffix position and which phonological content belongs to different morphemes is therefore present in the input to Level 1 and remains accessible to the phonology until the morpheme is parsed into the phonological word.

While my approach requires the phonology to be aware of which phonological material belongs to the same morpheme, the major assumptions about modularity in Stratal Phonology remain the same (see, e.g., Kiparsky 2015; Bermúdez-Otero 2017): all phonological content parsed into a phonological word is treated as a single unit in the input to the next stratum. In other words, if the string *xxx + yyy + zzz* is present in input to Level 1 and the output is *(xxxyyy)_{PrWd} + zzz*, the phonology “sees” two separate units in the input to Level 2: *(xxxyyy)_{PrWd}* and *zzz*. The phonology cannot see, for example, that *xxx* was a prefix or distinct from *yyy*, which was a root. These assumptions explain why segments associated with reduplicative morphemes may be deleted by later phonological processes (see Step 6 in Section 6.7): the grammar does not see reduplicated segments as any more meaningful than other segments in the word. The phonology has no access to earlier stages of the derivation, to the specific identity of a morpheme (e.g., which morphosyntactic feature bundle it corresponds to), or its syntactic position.

6.8.2 Non-Concatenative Morphology

Reduplicative morphemes are not “special” under the analysis presented here; non-concatenative morphemes behave like concatenative morphemes and are subject to the same mechanisms of linearisation and allomorphy.

The fact that Salish reduplication can be triggered by both fixed segments and prosodic units motivates a theory that can unify linear and non-linear representations. Generalised Nonlinear Affixation is “a line of research that seeks to reduce the role of morphology in all instances of apparently nonconcatenative exponence to the insertion of pieces of nonlinear phonological representation whose existence is independently motivated” (Bermúdez-Otero, 2012, p.53). I follow this approach by unifying non-concatenative processes such as reduplication

with vowel length in Section 6.6.2.2.1, but further extend the analysis to include instances where reduplication is triggered by linear representations, which include underlying fixed segments (see Section 6.5.2.3).

In a unified analysis of non-concatenative and concatenative morphology, morphemes with prosodic (non-linear) or segmental (non-linear) representations are subject to linearisation and allomorphy at the same points in the derivation. Linearisation is determined at three points in the grammar (syntax, morphology, and phonology), while allomorphy is determined at two (morphophonology and phonology); these are discussed in Sections 6.8.2.1 and 6.8.2.2, respectively.

6.8.2.1 Linearisation

6.8.2.1.1 Linearisation: Syntax, Morphology, Phonology

In the model I adopt, linearisation is determined across three components: syntax, morphology, and phonology. The syntax partially determines linear order via basic conditions on phrase structure, in particular the No-Tangling Condition, which prevents crossing branches. However, the No-Tangling Condition does not determine the direction of linearisation between sisters in a phrase structure tree. In these cases, the morphology and the phonology jointly determine the surface position of fissioned segments.

The surface order of morphemes is determined first by linearisation in the morphology and then can be affected by the phonology. In the morphology, linear order is determined based on the lexical entry of a morpheme and whether a [prefix] or [suffix] feature is present, as described in Section 6.4. The features determine linear order between morphemes that are sisters in the syntactic structure. The relevant part of the lexical entry is therefore its morphological features, not its phonological content.

Linearisation in the phonology is affected by constraint rankings at relevant strata. For example, the faithfulness constraint LINEARITY is violated if the order of segments in the output does not match the input; metathesis and infixation violate LINEARITY. Infixation and metathesis are tolerated if certain markedness constraints outrank LINEARITY, as discussed in Section 6.4.2.

6.8.2.1.2 *Mirror Theory and Mirror Principle-Based Approaches*

I assume that the direction of affixation is lexically specified, which is a departure from approaches that assume Mirror Theory, as originally developed by Brody (2000). Mirror Theory includes the axiomatic assumption that higher material in a span must follow lower material when linearised (Svenonius, 2016, p.13). A Mirror Theory analysis therefore entails that all reduplicative affixes in Salish are suffixes in the morphosyntax, despite typically behaving as prefixes in the phonology.

Linearisation in a Mirror Theory approach requires additional stipulation to account for prefixes. For example, Svenonius and Bye (2012, p.467) propose an antitropal feature, which marks segments within an affix as “not aligned to the edge [they were] introduced in the syntax”. A suffix is moved to the left edge to satisfy IDENT[ANTITROPAL], a phonological constraint violated if [antitropal] segments remain on the right edge of the word (Svenonius & Bye, 2012, p.470). The purpose of the antitropal feature is therefore to resolve problems that arise from linearisation based on Mirror Theory; [antitropal] is not necessary in the present approach because morphemes are linearised as prefixes or suffixes prior to the phonology. The phonological grammar therefore is only responsible for infixation, not turning suffixes into prefixes.

An alternative, purely phonology-based approach to linearization is advocated by Zukoff (2023). He proposes that morphosyntactic representations are mapped directly to phonological representations by the Mirror Alignment Principle (MAP), which is effectively a phonological implementation of Baker’s (1985) Mirror Principle (MP). On this approach, there is no linearisation until the phonology; phonological alignment constraints are used to derive mirror effects directly in the phonology. The ranking of alignment constraints depends on the morphemes present: MAP provides rankings based on the morphosyntactic structure, and language-specific “Default Ranking Statements” are used when MAP rankings alone are not sufficient (Zukoff, 2023, p.8). The stated goal of the MAP approach is to provide an account of “the formal means by which MP-compliance is enacted in the grammar” (Zukoff, 2023, p.400).

The crucial difference between the [antitropal] (Mirror Theory) and MAP (Mirror Principle) approaches is whether morphemes are linearised prior to the phonology: Svenonius and Bye (2012) assume that the input is linearised in the

reverse order to its syntactic representation (i.e., all affixes are suffixes), while Zukoff (2023) assumes morphemes are unordered in the input (i.e., mirror order is a result of constraint ranking in the phonological grammar).

The two approaches also differ in their assumptions about how much information the phonology needs to have about the morphosyntax. Svenonius and Bye (2012) posit an [antitropical] feature which is present in the phonological representation of a lexical item; no additional stipulations beyond what I have proposed in this dissertation need to be made about the phonological grammar to implement this solution. In contrast, a MAP approach requires different assumptions about the phonological grammar and where it fits in the overall architecture. Unlike the approach in Svenonius and Bye (2012) or the one in this dissertation, Zukoff (2023) assumes that constraints can reference information beyond what is present in the phonological representation of a morpheme. If morphemes are unordered in the input, as assumed in MAP, the phonology must be able to access not only morphological information, but also information about individual lexical items; it must also be able to access the syntactic structure following lexical insertion to determine the appropriate set of constraints and their ranking for a given word. The MAP approach requires the phonological component of the grammar to be more powerful than I have proposed in this dissertation; I assume a modular framework where the phonology can only see and act upon phonological representations.

Both Mirror-based approaches can handle the surface position of fixed segment allomorphs, such as the repetitive morpheme in Montana Salish-Spokane-Kalispel, discussed in Section 6.5.3.2; neither is sufficient, however, for morphemes containing non-linear representations (i.e., floating prosodic units). The diminutive in Comox-Sliammon can be used to show why neither approach predicts the correct surface form when the input consists of both linear and non-linear phonological content.

Recall from Chapter 4 (Section 4.1.3.1.2) that the diminutive has both a fixed segment and a prosodic affix allomorph, as shown in (443a) and (443b), respectively.

- (443) a. *Comox-Sliammon Diminutive: /i/ Allomorph*
 $\dot{\mathbf{t}}\mathbf{i} \sim \dot{\mathbf{t}}\mathbf{a}\dot{\mathbf{q}}\mathbf{t}$ (cf. $\dot{\mathbf{t}}\mathbf{a}\dot{\mathbf{q}}\mathbf{t}$ ‘mountain’)
 DIM~mountain
 ‘small mountain’ (JF)
- b. *Comox-Sliammon Diminutive: μ Allomorph*
 $\dot{\mathbf{r}}\mathbf{i} < \dot{\mathbf{r}} > \mathbf{m}\mathbf{\acute{a}}\mathbf{n}$ (cf. $\dot{\mathbf{r}}\mathbf{i}\mathbf{m}\mathbf{\acute{a}}\mathbf{n}$ ‘road, door’)
 road/door<DIM>
 ‘narrow walkway (small trail)’ (JF)

The tableau in (444) shows violation marks for IDENT[ANTITROPAL] as conceived by Svenonius and Bye (2012); subscript numbers mark the choice of diminutive allomorph. The only candidate that violates IDENT[ANTITROPAL] is candidate (444a), which leaves the fixed segment on the right edge. None of the other candidates violate this constraint; (444b–c) satisfy it because the /i/ is not on the right edge; candidates (444d–e) vacuously satisfy it because [antitropical] is a feature of segments. A prosodic unit cannot be antitropical because it has no linear order with respect to segments. In other words, the [antitropical] approach only accounts for segmental affixes.

	/CVC + $\{\mu_1, i_2\}_{[\text{antitropical}]}$ /	IDENT[ANTITROPAL]
(444) a.	CVCi ₂	*
b.	Ci ₂ CVC	
c.	CVC ₁ C	
d.	CVCC ₁	

The MAP approach also faces challenges with morphemes that include non-linear phonological representations that must be linearised relative to segments, such as the Comox-Sliammon diminutive. The tableau in (445) shows the alignment constraints indicated by a MAP analysis of a word consisting of a root and a diminutive morpheme in Comox-Sliammon; letters mark whether the segments belong to the root (a) or the diminutive morpheme (b). The constraints ALIGN-R-ROOT and ALIGN-R-DIM require segments associated with the root and diminutive respectively to be aligned with the right edge of the word; however, the μ allomorph of the diminutive always vacuously satisfies ALIGN-R-DIM,

regardless of where copied material is positioned in the word.

(445)	$/\{\mu_1, i_2\}_b + CVC_a/$	ALIGN-R-DIM	ALIGN-R-ROOT
	a. $C_a i_{b,2} C_a V_a C_a$	***	
	b. $i_{b,2} C_a V_a C_a$	***	
	c. $C_a V_a C_a i_{b,2}$		*
	d. $C_a V_a C_{a,1} C_a$		
	e. $C_a V_a C_a C_{a,1}$		

If a floating mora cannot be directly linearised relative to segments, for reasons laid out in Section 6.4.2, and the segments associated with the mora in (445c–d) are evaluated as root segments in the phonology, then there is no way to ensure that both prosodic unit and fixed segment allomorphs will be aligned to the same edge.

A feature-based approach to linearisation does not face the same challenges as the Mirror Theory and Mirror Principle-based approaches because the morphology linearises all morphemes, including prefixes, prior to the insertion of phonological content. Morphological linearisation involves material of the same kind (feature bundles) and therefore linear order can be established across morphemes, including both fixed segments and prosodic units, which explains their parallel behaviour with respect to edge-aligned infixation. Following the insertion of phonological content, both types of input enter the derivation as prefixes. If prefixation and infixation are both motivated by an [antitropical] feature, then there is nothing aligning a prosodic affix with either edge. If linearisation does not happen until the phonology, as in the MAP approach, prosodic affixes cannot be ordered relative to segments, which are on a different tier. Neither Mirror-based approach can account for the fact that fixed segment and prosodic affixes may both be aligned relative to the same edge as each other.

6.8.2.2 Allomorph Selection

In the framework developed in this dissertation, allomorph selection can happen at the second step of lexical insertion (see Section 6.5.1), or during the phonological derivation (see Section 6.6.1.3).

Allomorph selection at the point of lexical insertion happens when there are multiple underlying forms for a morpheme and one is inserted. Allomorph

selection at this stage does not necessarily have phonological motivation; it can be determined by morphosyntactic features/identity of an adjacent morpheme or include lexically-specified allomorphy where a root determines the choice of allomorph.

Allomorph selection during the phonological derivation involves listed allomorphs. When a set of listed allomorphs is inserted at the point of lexical insertion, as described in Section 6.5.1, the choice between these possible allomorphs is left to the phonological grammar. Listed allomorphs allow the grammar to wait until the morpheme is parsed into the phonological word before selecting from available allomorphs; I refer to this as *late allomorph selection*.

Late allomorph selection is necessary for listed allomorphs in Salish languages. For example, Section 5.3.1.3 describes how the plural in Halkomelem may be realised either as reduplication with prosodic unit input or *-l-* with a fixed segment input. The choice between the two allomorphs provides a way to avoid multiple reduplication at the same stratum, because M-AFFILIATION outranks PRIORITY, the constraint favouring the first of the listed allomorphs. Selecting between the two allomorphs is impossible prior to the relevant stratum (Level 2).

For both listed allomorphs and repairs, the choice between phonologically conditioned allomorphs depends on prosodic structure, which the grammar does not start building until the first stratum.

6.8.3 Phonological, not Morphological Domains

Work in Stratal Phonology typically assumes a tight relationship between morphological and phonological domains. In fact, Kiparsky (2015, p.6) states that “the reason stem, word, and phrase are likely to be universal is that these are the three universal cross-categorical morphosyntactic units”. However, it is a major finding of this dissertation that the strata motivated by Salish reduplication do not correspond to morphosyntactic domains: they must instead be phonological domains.

If the strata were defined with respect to morphosyntactic domains, closely related languages would be expected to show less variation with respect to which types of reduplication occur where in the derivation. Specifically, differences in the

level at which reduplicative morphemes are parsed into the prosodic word would directly reflect differences in their morphosyntax, which in turn reflects differences in semantic composition; this is not the case.

For example, the imperfective form of a verb is the base for diminutive reduplication in Halkomelem (Hukari, 1978; Suttles, 2004; Galloway, 2009), which means that the imperfective is parsed into the word at an earlier stratum than diminutive reduplication (see Section 5.3.1.2). If the strata were defined by morphosyntactic domains, this would mean that the imperfective must be lower in the syntactic tree than the diminutive. However, in Lillooet, the imperfective is an independent word *waʔ* (see Section 3.5.1.1), which means it is in a separate w-span from diminutive reduplication. Since the diminutive is part of the same w-span as the verb in Halkomelem, it must therefore be lower than the imperfective in Lillooet. The imperfective in Halkomelem is not compositionally different from the imperfective in Lillooet, however; the differences are purely phonological.

Considering microvariation across related languages shows that variation in morpheme-level mappings does not clearly map to variation in morphosyntax or semantics. There are also ordering paradoxes within a single language that are not predicted by morphosyntactically defined phonological domains. For example, change-of-state reduplication happens at Level 3 in Comox-Sliammon (analysis in Section 4.1.5.1.2), but progressive reduplication happens at Level 2 (Section 4.1.4.1.2). Given that progressive is a type of viewpoint (outer) aspect, whereas change-of-state is a type of lexical (inner) aspect, we would expect the opposite order of derivation were levels to reflect morphosyntactic domains.

Defining strata by morphosyntactic domains also predicts more consistency in permissible and unattested multiple reduplication patterns between languages than is the case. For example, a comparison of Interior Salish languages in Appendix B (Section B.1) shows that diminutive C₁ reduplication can combine with change-of-state C₂ reduplication in Thompson River Salish, Moses Columbia, Montana Salish-Kalispel-Spokane, and Coeur d'Alene, but not in the closely related languages Lillooet, Shuswap, and Okanagan. There is no semantic reason why the diminutive and change-of-state are compatible in the first group but not in the second. If the strata were defined by morphosyntactic domain, it would mean the syntax and hence the semantics (order of composition) would have to differ

across Interior Salish languages to account for incompatibility in one case and not the other.

Attested gaps in multiple reduplication are better explained by the phonology (see discussion of restrictions on multiple reduplication in Section 6.6.1.6), which is only possible if strata are not defined by the morphosyntax.

A further implication of my analysis is that there is no place for a morphological stem in the grammar of reduplication; the only morphosyntactic domain that matters is the word. While a discussion of stem phenomena in non-Salish languages is beyond the scope of my dissertation, I believe that stems listed “nonanalytically”, in the sense of Bermúdez-Otero (2017), can be handled in my model at the point of lexical insertion. For example, I proposed a portmanteau of property-denoting reduplication and the characteristic suffix *-t* in Lillooet in Chapter 4 (Section 4.2.3.2.2) with the lexical entry in (446b); “stored stems” could instead involve lexical insertion of a combined root and affix.

(446) *Lillooet Property-Denoting C_1C_2 Reduplication*:

- a. $\sigma_{N+\mu} \Leftrightarrow \langle \text{PDEN} \rangle$
- b. $\sigma_{N+\mu+t} \Leftrightarrow \langle \text{PDEN+IMMED} \rangle$

6.9 Conclusion

The model of the grammar and reduplication described in this chapter and throughout this dissertation predicts variation arising from two key areas: lexical entries and grammar. Reduplication in Salish varies in both areas; the approach taken here can account for the attested patterns, while providing an explanation for unattested ones. With respect to lexical entries, reduplicative morphemes vary in their underlying representations (i.e., whether they contain segmental or suprasegmental units), linearisation features (i.e., whether they are prefixes or suffixes), and level diacritics (i.e., when they are parsed into the phonological word). With respect to the grammar, languages differ in morpheme-level and grammar-level mappings, which correspond to differences in how reduplicative morphemes are realised.

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Appendix A

Survey of Reduplication Patterns

A.1 Survey Methods, Sources, and Transcription

This survey is not a complete overview of everything that has been published on Salish reduplication: it largely draws on phonological analyses and descriptive grammars. Other work, such as syntactic analyses, are considered if they explicitly consider reduplication. Those interested in learning more about the syntax or semantics of reduplication will find that searching for “function” labels, such as “plural” or “pluractional”, will bring up additional sources beyond this literature review. Further, the bibliography includes additional sources.

The amount of detail and description given for each language and reduplicative process varies based on available documentation. The available sources for each language are discussed in Chapter 1. The appendix is sorted by branch: Interior Salish (Section A.2), Central Salish (Section A.3), Tsamosan (Section A.4), Tillamook (Section A.5), and Bella Coola (Section A.6). The languages are organised alphabetically within branch. Different types of reduplication are given in subsections under each language; C_1C_2 processes are always given first, followed by C_1 , C_2 , and V_1 processes, as applicable. Additional patterns with unknown status are given in a final subsection for each language.

The first line of each example is a phonemic transcription. A morphemic transcription is also provided when it provides additional information: for example, the identity of the morphemes is obscured by phonological processes in (447).

I use // as brackets for morphemic transcriptions. Bold text marks the relevant reduplicative morpheme in each example.

(447) *kʷekʷúliʔs* (cf. *kʷúləntxʷ* ‘You did it.’)

//**kʷe**~kʷú-l-n-t-es//

REP~make.do-CTR-TR-3ERG

‘They (sg.) kept making things to see if they worked (like a puzzle).’

(B. Carlson, 1978, p.89, Spokane)

Transcription lines are followed by glossing and translation lines. Glossing labels vary between languages for some affixes, for example, CTR (“control transitive”), TR (“transitive”), and DRV/DIR (“directive transitive”) are all used to refer to the *-n* and *-t* transitive markers found across the family. I have followed the original sources for glossing for in many places. I provide a non-reduplicated form or a root for comparison with the reduplicated form. The non-reduplicated form is given as part of the last line if there is not space in the first line.

I distinguish between [ə] and any other vowel when describing patterns of reduplication. I use the term “moraiic vowel” to refer to any non-[ə] vowel.¹ There are examples where I follow the original sources and do not write [ə] even though it is likely present in the surface form; I only add [ə] where it is associated with a reduplicative morpheme, supported by other documentation, or informed by phonological descriptions.

The position of morpheme breaks and which segments to gloss as the reduplicative morpheme is informed by:

- assumptions necessary to derive the correct surface forms in the phonological analyses given in the case studies in this dissertation (Comox-Sliammon, Lillooet, Halkomelem)
- absence of stress and/or presence of [ə], rather than an underlying vowel (Coeur d’Alene, Moses Columbia, Okanagan, Montana Salish-Spokane-

¹Moraiic vowels are known as “full vowels” elsewhere in the literature on Salish languages. I do not use the term “full vowel” because it is often used outside of Salish linguistics to discuss specific alternations between stressed (full) and unstressed (reduced) vowels. Schwa may be stressed in Salish languages; it is more often epenthized than the product of vowel reduction.

Kalispel, Shuswap, and Thompson River Salish, Northern Straits and Klallam, Sechelt, Cowlitz, Quinault, Upper Chehalis, Bella Coola)

- comparison with closely related languages and/or consideration of previous descriptions (Lushootseed, Nooksack, Pentlatch, Squamish, Twana, Lower Chehalis, Tillamook)

Other transcription and glossing conventions that apply to the whole dissertation, including the appendices, can be found in Section 1.4.4 and Section 2.2.1.

Decisions related to standardising or formatting examples for specific languages are given at the start of the relevant section; I also note other sources used to identify unreduplicated forms, which are provided for each example. I only note changes in glossing and transcription where they mark divergences in analysis (e.g., splitting C₁ reduplication in Lillooet documented in Van Eijk, 1997 into DIM and C1.PL following the reanalysis in H. Davis & Mellesmoen, 2023).

A.2 Interior Salish Reduplication

A.2.1 Coeur d’Alene Reduplication

Attested reduplication in Coeur d’Alene is summarised in Table A.1.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
COS			✓	
INCH				✓
Other	“Intensive” C ₁ Reduplication			

Table A.1: Reduplication in Coeur d’Alene

A.2.1.1 Coeur d’Alene Sources and Transcription

Examples come from Reichard (1938), Doak (1997), and Doak (2023). Some unreduplicated forms provided for comparison come from Barthmaier (1996), drawing on file slips from Reichard (1938), or Lyon (2010), drawing on file cards from Nicodemus (1975). I also refer to the glossary provided on Ivy Doak’s website (Doak, 2023, <http://ivydoak.com/Coeurd’Alene>).

Sources for Coeur d’Alene typically omit schwa; I only write schwas where a vowel is present in one of the source materials, because it is unclear when syllables are best analysed as having schwa-sonorant sequences or sonorant consonants as nuclei (e.g., [tʃeɪnɪ] or [tʃeɪnəm] for ‘take hold of’ from Doak 2023). Some instances of unstressed *e* in the following sections may correspond to schwas. I transcribe Coeur d’Alene examples from Reichard (1938) in this section. Decisions about vowels are based on Doak (1997), as well as texts available on her website.

In several examples, such as (454f), schwa is lowered to [a] because the vowel is next to [ʃ] or [r]. Moraic vowels, rather than [ə], are also seen in some examples before [ʔ] to avoid a [əʔ] sequence.

A.2.1.2 Coeur d'Alene C₁C₂ Plural/Pluractional Reduplication

In plural/pluractional C₁C₂ reduplication, the first two consonants of the root are copied, as shown in (448); the surface form of this type of reduplication is -CəC-.

- (448) a. hən-čém~čəm-enč (cf. *hnčémenč* 'wall')
 LOC-surface~PL-interior
 'walls' (Doak, 2023, Coeur d'Alene)
- b. ʔec-wíš~wəš (cf. *ʔecwíš* 'house')
 CUST-build~PL
 'houses' (Doak, 2023, Coeur d'Alene)
- c. ʔi[c]-čél~čəl (cf. *čéliš* 's/he stood up')
 CONT-upright~PL
 'trees' (Doak, 2023, Coeur d'Alene)
- d. yár~yər-p (cf. *ʔuyár* 'circle')
 round~PL-INCH
 'They roll.' (Reichard, 1938, p.639, Coeur d'Alene)
- e. t-čén~čən-əm (cf. *tčénm* 'take hold of')
 LOC-hold~PL-MID
 'They held on with both hands.' (Reichard, 1938, p.635, Coeur d'Alene)
- f. dés~dəs-t (cf. *tusdəst* 'went to camp')
 camp-PL-STAT
 'They camped.' (Doak, 2023, Coeur d'Alene)

In addition to infixed -CəC- forms, such as those in (448) above, Reichard (1938) also describes prefixed CəC- forms. Infixed and prefixed forms are distinguished by stress: stress falls on the first syllable in (448) above, but on the second syllable in (449). Doak (1997, p.29) notes that the prefixed pattern is uncommon.

- (449) a. t-čəq^w~céq^w-eʔst (cf. *céq^wilš* 'tamarack')
 CISL-PL~light.red-round.obj
 'They're red hot on the surface.' (Doak, 1997, p.29, Coeur d'Alene)

- b. ʔu-čəp~číp (cf. čip ‘soften’)
 INHER-PL~soft
 ‘Each is soft.’ / ‘It is soft here and there.’ (Reichard, 1938, p.636, Coeur d’Alene)
- c. ʔu-šət~šét (cf. ʔu:šét ‘It is level.’)
 INHER-PL~be.level
 ‘They are each level, it is level here and there.’ (Reichard, 1938, p.636, Coeur d’Alene)

C₁C₂ reduplication in Coeur d’Alene can mark plural entities with nouns, as shown in (448a–c) above, as well as plural participants and events with verbs, as shown in (448d–f, 450) and (451), respectively. C₁C₂ reduplication can mark either plural participants or spatial distribution with adjectives, as in (449b–c) above.

- (450) a. qét~qət-p (cf. qetp ‘climb’)
 climb~PL-INCH
 ‘They each climbed.’ (Reichard, 1938, p.635, Coeur d’Alene)
- b. sél~səl-p (root: sel- ‘turn rapidly’)
 turn.rapidly~PL-INCH
 ‘They each spin about.’ (Reichard, 1938, p.635, Coeur d’Alene)
- c. č-séx^w<səx^w>q (root: sex^wq- ‘splash’)
 LOC-splash<PL>
 ‘Each of them splashed.’ (Reichard, 1938, p.636, Coeur d’Alene)
- d. sét<sət>č-ən-c (cf. sétčənc ‘They (sg.) twisted it.’)
 twist<PL>-TR-CTR+3ERG
 ‘They (sg.) twisted them one by one.’ (Reichard, 1938, p.562, Coeur d’Alene)
- (451) a. ʔáw~ʔaw-p (cf. ʔawp ‘liquid dropped’)
 drop~PL-INCH
 ‘It dropped here and there.’ (Reichard, 1938, p.635, Coeur d’Alene)
- b. g^wél~g^wəl-p (cf. sg^welp ‘fire’)
 burn~PL-INCH
 ‘It blazed here and there.’ (Reichard, 1938, p.635, Coeur d’Alene)

- c. $\acute{c}ál < \acute{c}əl > \acute{x}-ən-c$ (cf. $\acute{c}álx$ ‘claw’)
 claw <PL>-TR-CTR+3ERG
 ‘They (sg.) clawed it here and there.’ (Reichard, 1938, p.636, Coeur d’Alene)

A.2.1.3 Coeur d’Alene C₁C₂ Property-Denoting Reduplication

Property-denoting C₁C₂ reduplication in Coeur d’Alene has the same form as plural/pluractional reduplication discussed in the preceding section, as shown in (452). It often co-occurs with the adjectival suffix *-t*, as shown in (452c–g). I treat this type of reduplication as an infix because stress falls on the first vowel; this is further supported by the fact that [ə] typically surfaces in the second syllable.²

- (452) a. $pəs \sim pəs-úl$ (cf. $pəsáčstmən$ ‘I play a trick on you.’)
 astonish ~ PDEN-HAB
 ‘It has astonishing effect.’ (Reichard, 1938, p.578, Coeur d’Alene)
- b. $?ec-pu? < pu? > s-ínč$ (cf. $hi:cpúʔs$ ‘my heart’)
 CUST-heart <PDEN>-belly
 ‘sorrowful’ (Doak, 2023, Coeur d’Alene)
- c. $\acute{c}ín \sim \acute{c}ən-t$ (root: $\acute{c}ín-$ ‘dangerous’)
 dangerous ~ PDEN-CHAR
 ‘It is dangerous.’ (Doak, 2023, Coeur d’Alene)
- d. $tík^w \sim tək^w-t$ (cf. $sniʔtík^wt$ ‘oldest one’)
 old ~ PDEN-CHAR
 ‘elder’ (Doak, 2023, Coeur d’Alene)
- e. $ýc-x^wél \sim x^wəl-t$ (cf. x^wel ‘be alive’)
 CONT-live ~ PDEN-CHAR
 ‘be living’ (Doak, 2023, Coeur d’Alene)
- f. $lúp \sim ləp-t$ (cf. lup ‘be dry’)
 dry ~ PDEN-CHAR
 ‘It (shed) keeps dry.’ (Reichard, 1938, p.578, Coeur d’Alene)

²The example in (452) is exceptional because it has /ʔ/ in C₂ position; moraic vowels are often recorded before /ʔ/, rather than schwa.

- g. šár~šar-t (cf. *lut ?ecšár* ‘They’re not lazy.’)
 be.difficult~PDEN-CHAR
 ‘It is too difficult to attempt.’ (Reichard, 1938, p.635, Coeur d’Alene)

While property-denoting C₁C₂ reduplication applies to roots across different lexical categories, the resulting form is an adjective. Property-denoting reduplication marks a state or property.

A.2.1.4 Coeur d’Alene Diminutive C₁ Reduplication

The initial consonant of a root is copied in diminutive C₁ reduplication, as shown in (453); [ə] may be epenthesized when either the first or second consonant is a sonorant, as shown in (453f–h).³ Diminutive reduplication is also accompanied by the glottalisation of sonorants in the word.

- (453) a. c~cetx^w (cf. *cetx^w* ‘house’)
 DIM~house
 ‘house, cottage’ (Doak, 1997, p.27, Coeur d’Alene)
- b. q̣^w~q̣^wuc-t (cf. *sq̣^wuct* ‘fat’)
 DIM~fat-CHAR
 ‘little fat child’ (Doak, 1997, p.27, Coeur d’Alene)
- c. k^w~k^wul-íg^wt-us (root: *k^wul-*)
 DIM~red-far-fire
 ‘small red fire in the distance’ (Doak, 2023, Coeur d’Alene)
- d. hən-x^w~x^wát-p-alqs (cf. *hnx^wátpalqs* ‘the end of the road’)
 LOC-DIM~end-NVOL-end
 ‘He finished telling a little story.’ (Doak, 2023, Coeur d’Alene)
- e. q̣ə~q̣ad-m-úl (root: *q̣ed-*)
 DIM~balk-MID-HAB
 ‘little stubborn one’ (Doak, 2023, Coeur d’Alene)
- f. s-mə~məʔím (cf. *smíyem* ‘woman’)
 NMLZ-DIM~woman
 ‘little woman’ (Reichard, 1938, p.647, Coeur d’Alene)

³/d/ behaves phonologically as a sonorant in Salish.

- g. $\text{ʔe} \sim \text{ʔex}^w \text{ús-} \dot{\text{n}}\text{-c}$ (cf. $\text{ʔex}^w \text{úsn}$ ‘I look for it.’)
 DIM $\sim$ look.for-CTR-TR+3ERG
 ‘They (sg.) looked for the little thing.’ (Doak, 2023, Coeur d’Alene)

The examples in (453) show that diminutive C₁ reduplication marks a small entity with a noun (e.g., 453f), a small participant with a verb (453g), and that a small entity that possesses that quality or state denoted by an adjective (e.g., 453b). I have not found clear examples of diminutive reduplication marking a small amount or degree with an adjective or verb.

A.2.1.5 Coeur d’Alene Change-of-State C₂ Reduplication

Change-of-state C₂ reduplication doubles the second consonant of the root; an epenthetic schwa is inserted between the identical consonants, as shown in (454).

- (454) a. $\text{pén} \sim \text{əñ}$ (cf. ʔecpén ‘It is bent.’)
 bend $\sim$ COS
 ‘It has come to be bent.’ (Reichard, 1938, p.579, Coeur d’Alene)
- b. $\text{ʔem} \sim \text{əṁ}$ (cf. ʔem ‘one sits’)
 sit $\sim$ COS
 ‘It lighted.’ (Reichard, 1938, p.579, Coeur d’Alene)
- c. $\text{táx}^w \sim \text{əx}^w$ (cf. táx^w ‘quiet’)
 stop $\sim$ COS
 ‘die’ (Doak, 2023, Coeur d’Alene)
- d. $\text{ʔen-piy} \sim \text{əy-} \dot{\text{ílg}}^w \text{es}$ (root: piy- ‘joy’)
 CUST+LOC-joy $\sim$ COS-heart
 ‘become happy’ (Doak, 2023, Coeur d’Alene)
- e. $\text{cél} \sim \text{əl}$ (root: cél- ‘arrive’)
 arrive $\sim$ COS
 ‘arrive’ (Doak, 2023, Coeur d’Alene)
- f. $\text{yaŋ} \sim \text{aŋ}$ (cf. yaŋ ‘all’)
 assemble $\sim$ COS
 ‘They assembled.’ (Reichard, 1938, p.836, Coeur d’Alene)

- g. $\text{ʔe:} \sim \text{t}^{\text{w}} \text{é} \text{k}^{\text{w}} \sim \text{ə} \text{k}^{\text{w}}$ (cf. $\text{t}^{\text{w}} \text{é} \text{k}^{\text{w}} \text{ncut}$ ‘He lay down.’)
 CUST-fall~COS
 ‘They would fall.’ (Doak, 2023, Coeur d’Alene)

C₂ reduplication marks a change of state; it is found with verbs.⁴

A.2.1.6 Coeur d’Alene V₁ Reduplication

V₁ reduplication doubles the root vowel; the two identical vowels are separated by a glottal stop, as shown in (455). This is a surface form of the inchoative marker -ʔ-. As stress falls on the first of the two matching vowels, I gloss the second one as the reduplicative infix.

- (455) a. $\text{h} \text{ə} \text{n} \sim \text{t}^{\text{a}} < \text{ʔ} \text{a} > \text{š} \sim \text{á} \text{lp}$ (cf. $\text{t}^{\text{a}} \text{š}$ ‘be inherently sweet’)
 LOC-sweet<INCH>-feeling
 ‘I felt sweetness’ (Reichard, 1938, p.612, Coeur d’Alene)
- b. $\text{h} \text{ə} \text{n} \sim \text{t}^{\text{a}} < \text{ʔ} \text{a} > \text{x} \sim \text{í} \text{lg}^{\text{w}} \text{es}$ (cf. $\text{t}^{\text{a}} \text{x}$ ‘be bitter’)
 LOC-bitter<INCH>-heart
 ‘It became bitter in heart.’ (Reichard, 1938, p.637, Coeur d’Alene)
- c. $\text{l} \text{ú} < \text{ʔ} \text{u} > \text{p}$ (cf. $\text{ʔe:} \text{lup}$ ‘It has been dried.’)
 be.dry<INCH>
 ‘it became dry’ (Reichard, 1938, p.637, Coeur d’Alene)
- d. $\text{n} \text{á} < \text{ʔ} \text{a} > \text{s}$ (cf. nas ‘wet’)
 be.wet<INCH>
 ‘It became wet.’ (Reichard, 1938, p.637, Coeur d’Alene)

⁴Reichard (1938, p.607) states that C₂ reduplication marks “possibility” when combined with the stative suffix -ut; examples are provided in (i). However, given that the “possibility” reading is reported in association with an additional suffix, it is unlikely that the modal interpretation is comes from reduplication itself.

- (i) a. $\text{q} \text{é} \text{y} \sim \text{ə} \text{y} \sim \text{út} \sim \text{əm}$ (cf. $\text{s} \text{q} \text{é} \text{y} \text{əm}$ ‘writing’)
 write-COS-STAT-INDEF.OBJ
 ‘It is possible to write.’ (Reichard, 1938, p.607, Coeur d’Alene)
- b. $\text{n} \text{í} \text{č} \sim \text{ə} \text{č} \sim \text{út} \sim \text{əm}$ (cf. $\text{c} \text{i} \text{n} \text{í} \text{č} \text{əm}$ ‘S/he cut some.’)
 cut-COS-STAT-INDEF.OBJ
 ‘It is cut-able’ (Reichard, 1938, p.607, Coeur d’Alene)

Reichard (1938) states that V₁ reduplication marks a progression or that something is “gradually becoming X”. V₁ reduplication marks a verb; it is attested with adjectival roots.

A.2.1.7 Other Reduplication in Coeur d’Alene

Doak (1997, pp.28–29) describes an additional type of C₁ reduplication in Coeur d’Alene, which copies the initial consonant and vowel of the root (456); she labels this C₁ reduplication process as “intensive”.⁵ I gloss this as a reduplicative prefix to highlight its similarity to repetitive reduplication in Moses Columbia and Montana Salish-Spokane-Kalispel, initial C₁ reduplication in Lillooet, and “affective” reduplication in Thompson River Salish.

- (456) a. **šá**~šag^w-t (root: šag^w-)
 INTS~sharp-STAT
 ‘It is sharp.’ (Doak, 1997, p.28, Coeur d’Alene)
- b. **še**~šet-út (cf. šétut ‘rock’)
 INTS~rock-place
 ‘cliff’ (Doak, 1997, p.28, Coeur d’Alene)
- c. n-**čí**~cip-s-m (cf. ncípsm ‘They (sg.) napped.’)
 LOC-INTS~close-eye-LOC
 ‘They (sg.) closed their eyes.’ (Doak, 1997, p.29, Coeur d’Alene)

There is no clear pattern with respect to the distribution of intensive reduplication, it is found with adjectives (456a), nouns (456b), and verbs (456c). This type of reduplication may be lexicalised.

⁵Doak (1997, p.28) notes that this type of C₁ reduplication is rarer than the diminutive.

A.2.2 Moses Columbia Reduplication

Attested reduplication in Moses Columbia is summarised in Table A.2.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
REP		✓		
COS			✓	
Other	Onomatopoeia/Repetition			

Table A.2: Reduplication in Moses Columbia

A.2.2.1 Moses Columbia Sources and Transcription

Examples come from Kinkade (1982), Czaykowska-Higgins (1993a), Czaykowska-Higgins (1993b), Willett and Czaykowska-Higgins (1995), Czaykowska-Higgins and Willett (1997), Van Eijk (1998a), Willett (2003), and Lyon and Czaykowska-Higgins (2020). Unreduplicated forms come from the original sources. I use Willett (2003) to guide decisions about transcription.

Sources for Moses Columbia typically omit schwa; I only regularly add [ə] to reduplicative morphemes.

A.2.2.2 Moses Columbia Plural/Pluractional C₁C₂ Reduplication

C₁C₂ reduplication copies the initial two consonants of the root with concurrent epenthesis of [ə]. As stress falls on the second syllable, I treat this as a reduplicative prefix, such that stress falls on the original root vowel. Examples are given in (457).

- (457) a. **čəl~čəlχ-nt-n** (cf. *č(ə)lxusnc* ‘It scratched my face.’)
 PL~scratch-TR-1SG.ERG
 ‘I scratched it (with several/all fingers).’ (Willett, 2003, p.275, Moses Columbia)

- b. $\dot{t}\dot{a}m \sim \dot{t}\dot{a}m\text{-nt-n}$ (cf. $\dot{t}\dot{a}mnc\dot{u}t$ ‘They (sg.) cut themselves.’)
 PL~cut-TR-1SG.ERG
 ‘I cut it up.’ (Willett, 2003, p.275, Moses Columbia)
- c. $n\dot{a}w \sim n\dot{a}w\dot{l}x$ (cf. $n\dot{a}w\dot{l}x$ ‘run’)
 PL~run
 ‘run around’ (Willett, 2003, p.275, Moses Columbia)
- d. $s\text{-}\dot{l}\dot{a}x \sim \dot{l}\dot{a}x\dot{s}x\dot{n}$ (cf. $\dot{s}\dot{l}\dot{a}x\dot{s}x\dot{n}$ ‘one toe’)
 NMLZ-PL~toe
 ‘all toes, more than one toe’ (Willett, 2003, p.271, Moses Columbia)

C₁C₂ reduplication is used to mark pluractionality on verbs with distribution across participants (457a) and events (457b–c), as well as plurality on nouns (457d).

A.2.2.3 Moses Columbia Property-Denoting C₁C₂ Reduplication

The first two consonants are copied and stress falls on the first syllable in property-denoting C₁C₂ reduplication (if not on a stress-attracting suffix), as shown in (458); the root vowel is also copied. I treat the reduplicative as an infix in these words due to the position of stress. These forms typically have a *-t* suffix labeled here as a CHAR (characteristic) but also labeled as “immediate” in L. C. Thompson and Thompson (1992) and “state” in Kinkade (1996). The *-t* suffix is absent with a lexical suffix, as shown in (458d–e).

- (458) a. $\dot{c}\dot{a}l \sim \dot{c}\dot{a}l\text{-t}$ (cf. $\dot{c}\dot{a}l\dot{l}$ ‘get shaded’)
 shade~PDEN-CHAR
 ‘shady’ (Willett, 2003, p.243, Moses Columbia)
- b. $q^w\acute{a}y \sim q^w\acute{a}y\text{-t}$ (cf. $q^w\acute{a}y$ ‘plenty’)
 rich~PDEN-CHAR
 ‘rich’ (Willett, 2003, p.243, Moses Columbia)
- c. $x^w\acute{a}\dot{x} \sim x^w\acute{a}\dot{x}\text{-t}$ (cf. $x^w\acute{a}\dot{x}$ ‘get well’)
 alive~PDEN-CHAR
 ‘lively’ (Willett, 2003, p.243, Moses Columbia)

- d. $\dot{x}^w \acute{u}s \sim \dot{x}^w \text{us}$ -cn (cf. $x^w us$ ‘hurry’)
 hurry~PDEN-mouth
 ‘talk fast’ (Czaykowska-Higgins, 1993b, p.70, Moses Columbia)
- e. $miy \sim \text{miy}$ -áws (root: miy -)
 ?~PDEN-middle
 ‘middle, half-way’ (Czaykowska-Higgins, 1993b, p.61, Moses Columbia)

This type of reduplication is found with adjectives.

A.2.2.4 Moses Columbia Diminutive C₁ Reduplication

Diminutive C₁ reduplication doubles the initial consonant of the root, as shown in (459).

- (459) a. $t \sim twit$ (cf. $twit$ ‘boy’)
 DIM~boy
 ‘little boy’ (Willett, 2003, p.278, Moses Columbia)
- b. $\dot{x} \sim \dot{x}x^w up$ (cf. $\dot{x}x^w up$ ‘win’)
 DIM~win
 ‘win a small bit’ (Willett, 2003, p.279, Moses Columbia)
- c. $s\text{-c-p} \sim pu?s$ (cf. $scpu?s$ ‘heart’)
 NMLZ-STAT-DIM~heart
 ‘small heart’ (Willett, 2003, p.45, Moses Columbia)
- d. $s\text{-}\dot{q} \sim \dot{q}\acute{a}xn$ (cf. $s\dot{q}a?xn$ ‘shoe’)
 NMLZ-DIM~shoe
 ‘little shoe’ (Willett, 2003, p.278, Moses Columbia)
- e. $c\text{-}\dot{p} \sim \dot{p}\acute{e}\dot{q}^w\text{-stú-nn}$ (cf. $swat \dot{p}\acute{e}\dot{q}^w s$ ‘Who spilled it?’)
 IM-DIM~spill(st.dry)-TR-1 SG.SUBJ
 ‘I’m spilling a little bit.’ (Willett, 2003, p.55, Moses Columbia)
- f. $kn \quad kat\text{-}s \sim s\acute{e}k^w\text{-p-cín}$ (cf. $s\acute{e}k^w p$ ‘It’s swollen.’)
 1 SG.SUBJ on/over-DIM-swell.up-INCH-mouth
 ‘I have a little bump on my lip.’ (Willett, 2003, p.278, Moses Columbia)

When necessary to avoid a violation of the sonority sequencing principle (in particular, when a sonorant is reduplicated), [ə] is added, as shown in (460).⁶ The fact that stress falls on the second vowel in (460) supports treating this type of reduplication as a prefix.

- (460) a. s-ʔa~ʔiswəl-ált (cf. ʔíswal ‘loon’)
 NMLZ-DIM-loon-child
 ‘baby loon chicks’ (Czaykowska-Higgins, 1993b, p.258, Moses Columbia)
- b. ʔa~ʔálk^wmn (cf. ʔálk^wmn ‘slave’)
 DIM~slave
 ‘little slave’ (Willett, 2003, p.278, Moses Columbia)

C₁ reduplication marks small size when attached to a noun, as in (459a,c,d) and (460); it may mark either smallness of an argument or diminished action with verbs as shown in (459f) and (459b,c), respectively.

A.2.2.5 Moses Columbia Repetitive C₁ Reduplication

Repetitive C₁ reduplication is associated with a fixed segment /a/, which surfaces between the matching consonants, as shown in (461). The repetitive morpheme is realised without reduplication with cluster-initial roots (e.g., CCVC), as shown in (462); -a- is infix between the first two consonants of the root in these forms. I analyse the forms with C₁ reduplication as a prefix, such that reduplication is motivated to provide a word-initial onset.⁷

- (461) a. ʔa~wál (root: wál- ‘tilt’)
 REP~tilt
 ‘rock (tilt back and forth)’ (Willett, 2003, p.207, Moses Columbia)

⁶The vowel is realised as [a] in these examples because it is adjacent to [ʔ].

⁷If this were an infix, regular resolution of vowel hiatus may predict deletion of a vowel or epenthesis of a laryngeal consonant rather than reduplication if it yields C₁-a-VC (infixation of /a/ following first consonant); it is not clear that the grammar of the language would predict the attested form C₁a-C₁-VC (reduplication to break up vowel hiatus).

- b. **ka**~kíc-stu-m-s (cf. *kicc* ‘someone came’)
 REP~arrive-TR-1 SG.OBJ-3ERG
 ‘They came here (to see me) several times.’ (Willett, 2003, p.209, Moses Columbia)
- c. **c-ʔa**~ʔácḵ-stu-n (cf. *ʔacḵncútn* ‘mirror’)
 IM-REP~look.at-TR-1 SG.ERG
 ‘I am watching them (sg.).’ (Willett, 2003, p.209, Moses Columbia)

- (462) **ḵ<a>x^wup** (cf. *sx^wsḵx^wup* ‘winner’)
 win<REP>
 ‘keep winning’ (Willett, 2003, p.208, Moses Columbia)

The repetitive marks the repetition of an event; it is found with verbs.

A.2.2.6 Moses Columbia Change-of-State C₂ Reduplication

C₂ reduplication in Moses Columbia copies the second consonant of the root. [ə] is found between the matching consonants in C₂ reduplication, as shown in (463).

- (463) a. **cók~ək** (cf. *cək-n* ‘I [throw and] hit it.’)
 hit~COS
 ‘get hit’ (Czaykowska-Higgins & Willett, 1997, p.395, Moses Columbia)
- b. **kíp~əp** (cf. *kípən* ‘I pinch it.’)
 pinch~COS
 ‘get pinched’ (Czaykowska-Higgins & Willett, 1997, p.395, Moses Columbia)

Kinkade (1982, p.66) notes that C₂ reduplication is often realized without [ə], as in (464); the [ə] is recorded more often in careful or slow speech.

- (464) a. **čq̣^w<q̣^w>ún-ləx^w** (cf. *čq̣^wun-m* ‘say, pronounce, name’)
 name<COS>-land
 ‘land gets named’ (Willett & Czaykowska-Higgins, 1995, p.7, Moses Columbia)

- b. sac-kás~s-əx^w (cf. *kássn* ‘I argue with them (sg.).’)
 IPFV-fight~COS-NONPFV
 ‘quarreling, fighting’ (Kinkade, 1982, p.67, Moses Columbia)
- c. šá=k^wp k-pəq^w~q^w-ána?
 Q=2PL.INTR DRV-spill~COS-ear
 ‘Did it spill on you folks?’ (cf. *niʔpəq^wn* ‘I spilled it.’) (Lyon & Czakowska-Higgins, 2020, p.205, Moses Columbia)

C₂ reduplication is found with verbs. The most common translation associated with C₂ reduplication is “got X-ed”, which is associated with something happening to an object outside of an agent’s control, or with translations that entail a change of state (Kinkade, 1982, p.66). C₂ reduplication can also be used where an action took considerable effort, as shown in (465).

- (465) a. kʔkəmn̄nús (cf. *kʔkəṁ* ‘win’)
 //kʔkəṁ~ṁ-nún-nt-s//
 win~COS-LC-TR-3ERG
 ‘They (sg.) managed to win it.’ (Willett, 2003, p.47, Moses Columbia)
- b. t-xətt̄nús (root: *xṭ-* ‘take care of’)
 //t-xəṭ~ṭ-nún-nt-s//
 LOC-take.care.of~COS-LC-TR-3ERG
 ‘They (sg.) finally began to take care of it.’ (Willett, 2003, p.47, Moses Columbia)

A.2.2.7 Repetition in Moses Columbia

An onomatopoeic repetition of a root is shown in (466). The fact that primary stress falls on each repetition and that they may surface with no vowels suggests that these forms represent a different phenomenon than reduplication; I classify these forms as having repetition rather than an independently meaningful process of reduplication.

- (466) a. kəx^w kəx^w kəx^w
 ‘knock, knock, knock’ (Van Eijk, 1998a, p.465, Moses Columbia)

b. ɬx^w ɬx^w ɬx^w ɬx^w

‘fluttering or shaking noise, as when a bird shakes its feathers’ (Van Eijk, 1998a, p.465, Moses Columbia)

A.2.3 Okanagan Reduplication

Attested reduplication in Okanagan is summarised in Table A.3.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
COS			✓	
INCH				✓
Other	C ₂ ² Triplication; Onomatopoeia/Repetition			

Table A.3: Reduplication in Okanagan

A.2.3.1 Okanagan Sources and Transcription

Examples come from Watkins (1970), A. Mattina (1973), A. Mattina (1987), N. Mattina (1996), A. Mattina and Peterson (1997), and Thomason (1999). Unreduplicated forms come from the original sources, as well as A. Mattina (1987).

I have retranscribed a few obstruents. While A. Mattina (1973, p.24) states that “no reduplicated root initial consonant remains [an ejective], except /*χ*/”, I have confirmed with John Lyon (p.c.) that reduplicated ejectives remain ejectives.

There are a number of roots given in A. Mattina (1987) that alternate between having a moraic vowel and no vowel (or [ə]). Additionally, a root vowel and an underlying C₂ consonant may undergo metathesis: for example, the root *χ**l*- ‘still’ surfaces as CVC in *χ*<ə*l*>*ál* ‘they (sg.) are dead’ but CCV in *χ**lá*-*p* ‘stop’. As schwas are never stressed, unexplained alternations between [ə] and moraic vowels across forms with the same root follow from the general phonology of the language (see, e.g., A. Mattina, 1973, p.25).

A.2.3.2 Okanagan C₁C₂ Plural/Pluractional Reduplication

Plural/pluractional C₁C₂ reduplication doubles the first two consonants of the root and [ə] is inserted, as shown in (467). I treat this type of reduplication as a prefix because stress typically falls on the second syllable.

- (467) a. **kəl**~kílʰx (cf. *kílʰx* ‘hand’)
 PL~hand
 ‘They are hands.’ (A. Mattina, 1973, p.61, Okanagan)
- b. **xəm**~xəmʰáʰt (cf. *xəmʰáʰt* ‘fly’)
 PL~fly
 ‘They are flies.’ (A. Mattina, 1973, p.62, Okanagan)
- c. **kəs**~ks-əlqʷ (root: *kʰs-* ‘bad’)
 PL~bad-cylindrical.object
 ‘bad twigs, sticks’ (A. Mattina, 1987, p.49, Okanagan)

C₁C₂ reduplication with a noun marks plurality, as shown in (467) above; C₁C₂ reduplication with a verb marks pluractionality, as shown in (468). It can mark plural participants (468a), distribution in space (468b), and distribution in time (468c).

- (468) a. **kəw**~káw-əlʰx (cf. *káwəlʰx* ‘They are gone.’)
 PL~gone-3PL
 ‘They are all gone.’ (A. Mattina, 1973, p.61, Okanagan)
- b. **təxʷ**~təxʷ-lílʰx (cf. *təxʷlwlʰs* ‘airplane’)
 PL~fly-DEV
 ‘fly around’ (A. Mattina, 1987, p.219, Okanagan)
- c. **təqʷ**~təqʷ-ənt-ís (cf. *təqʷam* ‘slap’)
 PL~slap-TR-3ERG
 ‘He kept slapping it.’ (A. Mattina, 1987, p.219, Okanagan)

A.2.3.3 Okanagan C₁C₂ Property-Denoting Reduplication

Property-denoting C₁C₂ reduplication copies the first two consonants of the root, as shown in (469); epenthetic [ə] is optionally omitted where not required to obey the sonority sequencing principle, as shown in (469a). This type of reduplication is typically found in combination with the suffix *-t*, which N. Mattina (1996) as a “stative” suffix.⁸

⁸This is the same suffix found on adjectives in other Salish languages and so I gloss it as a CHAR for consistency in languages other than Lillooet, where I follow H. Davis (2011) and gloss it as IMMED to distinguish it from the characteristic suffix *-m*.

- (469) a. $k\acute{a}n$ $t\acute{h}\sim ta\acute{h}\text{-}t$ (cf. $ta\acute{h}$ ‘straight, certain’)
 1SG.SUBJ PDEN~straight-CHAR
 ‘I am honest.’ (A. Mattina, 1987, p.200, Okanagan)
- b. $k^w\acute{a}c\sim k^w\acute{a}c\text{-}t$ (root: $k^wc\text{-}$ ‘strong, stiff’)
 PL~strong-CHAR
 ‘strong’ (A. Mattina, 1987, p.60, Okanagan)
- c. $c\acute{i}\acute{x}\sim c\acute{x}\text{-}t$ (root: $c\acute{x}$ ‘hot’)
 PDEN~hot-CHAR
 ‘very hot’ (A. Mattina, 1987, p.12, Okanagan)
- d. $p\acute{x}\sim p\acute{x}\text{-}t$ (cf. $p\acute{x}am$ ‘advise’)
 PDEN~advise-CHAR
 ‘smart’ (N. Mattina, 1996, p.110, Okanagan)

Property-denoting C_1C_2 reduplication occurs with adjectives.

A.2.3.4 Okanagan Diminutive C_1 Reduplication

Diminutive C_1 reduplication doubles the first consonant of the root, as shown in (470). The reduplicated forms may have an epenthetic [ə], as shown in (470), or no additional vowel, as shown in (471). When there is no additional vowel, the reduplicated consonant is infixal following the root vowel. The choice to treat diminutive reduplication as both a prefix and an infix on the surface is motivated by the position of the C_1 consonants relative to the stressed vowel. Glottalisation of a sonorant consonant accompanies diminutive reduplication.

- (470) a. $p\acute{a}\sim p\acute{u}k^wla?$ (cf. $p\acute{u}k^wla?$ ‘ball’)
 DIM~ball
 ‘(little) ball’ (A. Mattina & Peterson, 1997, p.318, Okanagan)
- b. $s\text{-}t\acute{a}\sim t\acute{a}t\acute{a}m$ (cf. $st\acute{a}t\acute{a}m$ ‘boat’)
 NMLZ-DIM~boat
 ‘(little) boat’ (A. Mattina & Peterson, 1997, p.318, Okanagan)
- c. $t\acute{a}\sim t\acute{?}iwl\acute{t}k$ (cf. $t\acute{?}iwl\acute{t}k$ ‘on shore’)
 DIM~on.shore
 ‘a little ways from the water’ (A. Mattina, 1987, p.206, Okanagan)

- (471) a. kí<k>íx (cf. *kíl*x ‘hand’)
hand<DIM>
‘little hand’ (A. Mattina & Peterson, 1997, p.321, Okanagan)
- b. q^wá<q^w>c-qə́n (cf. *q^wacqə́n* ‘hat’)
warm<DIM>-head
‘little hat’ (A. Mattina & Peterson, 1997, p.321, Okanagan)
- c. s-n-íí<ʔ>íñ-tə́n (cf. *snííə́ntə́* ‘restaurant’)
NMLZ-LOC-eat<DIM>-instrument
‘little restaurant’ (A. Mattina & Peterson, 1997, p.321, Okanagan)

An infix [aʔ] is also found in some forms with diminutive C₁ reduplication. The infix is positioned following the consonant that directly follows the stressed vowel, as shown in (472).⁹

- (472) a. cə́~cí<aʔ>x^w (cf. *cít*x^w ‘house’)
DIM~house<DIM>
‘(out)house’ (A. Mattina & Peterson, 1997, p.320, Okanagan)
- b. tí<t><aʔ>k^wt (cf. *tík^wt* ‘lake’)
lake<DIM><DIM>
‘little lake’ (A. Mattina & Peterson, 1997, p.322, Okanagan)

Diminutive C₁ reduplication in Okanagan marks smallness with nouns, as shown in (470b–d), (471), and (472). It marks a small degree or amount of an adjective, as shown in (473a). Diminutive verbs may refer to small participants or events, as shown in (473b–c) and (473d), respectively.

- (473) a. íə́~ííá́t (cf. *cííat* ‘be/get wet’)
DIM~wet
‘a little wet’ (A. Mattina, 1987, p.88, Okanagan)
- b. n-čə́~čáx^w (cf. *nčáx^w* ‘something poured’)
LOC-DIM~pour
‘something small poured’ (A. Mattina & Peterson, 1997, p.318, Okanagan)

⁹I analyse this as a floating [c.g.] feature realised as [ʔ] with an epenthetic [a]; I assume the vowel is not [ə] because that would result in a [əʔ] sequence within a single syllable.

- c. $x^w i < x^w > \acute{c}$ -xt (cf. $x^w i \acute{c}$ -nt ‘give something to somebody’)
 give<DIM>-BEN
 ‘give a little something to somebody’ (A. Mattina, 1987, p.242, Okanagan)
- d. $\acute{t} \sim \acute{t} k^w$ -əncút (cf. $\acute{t} k^w$ am ‘put down’)
 DIM~put.down-REFL
 ‘He lay down a bit.’ (A. Mattina, 1987, p.10, Okanagan)

A.2.3.5 Okanagan Change-of-State C₂ Reduplication

C₂ change-of-state reduplication copies the second consonant of the root, as shown in (474). I analyse this reduplicative morpheme as an infix that follows the first consonant of the root with an intervening [ə] where sonorants are present, as in (474b): this decision is informed by the position of stress and comparison of these CCVC forms in Okanagan with the CVCC forms that arise from C₂ reduplication in other Interior Salish languages, such as Moses Columbia.

- (474) a. $c < k^w > u k^w$ (cf. $c k^w$ am ‘haul’)
 haul<COS>
 ‘arrive’ (A. Mattina, 1973, p.4, Okanagan)
- b. $\acute{x} < \acute{a} l > \acute{a} l$ (cf. $\acute{x} l$ ap ‘stop’)
 still<COS>
 ‘They (sg.) are dead.’ (A. Mattina, 1973, p.64, Okanagan)
- c. $c < \acute{q} > \acute{a} \acute{q}$ (cf. $c \acute{q}$ am ‘hit something by throwing’)
 hit.by.throwing<COS>
 ‘They (sg.) hit the target.’ (A. Mattina, 1973, p.64, Okanagan)
- d. $x < \acute{x} > \acute{a} \acute{x}$ (cf. $x \acute{x}$ ap ‘complete’)
 grow<COS>
 ‘They (sg.) grew.’ (A. Mattina, 1973, p.64, Okanagan)

Change-of-state C₂ reduplication applies to verb roots; it is not clear if this type of reduplication applies to adjectives as well. A. Mattina (1973, p.65) calls this a “resultive” suffix.

A.2.3.6 Okanagan V₁ Reduplication

Inchoative V₁ reduplication copies the root vowel, as shown in (475). V₁ reduplication is one surface realisation of the inchoative infix -ʔ-.¹⁰ The copied vowel comes before the inchoative infix; this decision is motivated by the position of stress on the second of the matching vowels.

- (475) a. q^w<aʔ>ác (cf. q^waʔcínaʔ ‘warm weather’)
 warm<INCH>
 ‘It is getting warm.’ (Watkins, 1970, p.100, Okanagan)
- b. kən ʔ<aʔ>át (cf. cʰfat ‘be/get wet’)
 1SG.SUBJ wet<INCH>
 ‘I got wet.’ (Watkins, 1970, p.107, Okanagan)
- c. q̣^w<uʔ>úc (cf. q̣^wuct ‘fat’)
 fat<INCH>
 ‘They (sg.) are getting fat’ (Watkins, 1970, p.107, Okanagan)

V₁ reduplication is attested with adjective roots; it yields an inchoative verb.

A.2.3.7 Other Reduplication in Okanagan

A. Mattina (1993) describes a rare C₂² triplication pattern in Okanagan, shown in (476). The final consonant of the root is tripled.¹¹

- (476) a. qiʔ<x^wəx^w>áx^w (cf. q̣yax^w ‘something stinks’)
 stink<TRIP>
 ‘stink’ (A. Mattina, 1987, p.153, Okanagan)
- b. waỵ sic ixiʔ s-təɾ<q̣q>áq-səlx (cf. ṭiq-ncut ‘be kicking s.t.’)
 yes before DEM CONT-kick<TRIP>-3PL.POSS
 ‘Then they started jumping up and down.’ (A. Mattina, 1987, p.206, Okanagan)

¹⁰The description of this realisation, and the decision to treat it as phonological instead of phonetic, is variable across descriptions: it is described in Watkins (1970), but not elsewhere. See Mellesmoen (2024a) for discussion of the innovation and different statuses of V₁ reduplication patterns across Salish.

¹¹This has cognates in Lillooet and Shuswap: see discussion of triplication.

c. $t\langle\text{əq}^w\text{q}^w\rangle\acute{\text{íq}}^w$ (cf. $\acute{\text{tiq}}^wt$ ‘burst, explode’)

burst<TRIP>

‘He made a banging noise.’ (A. Mattina, 1987, p.219, Okanagan)

There is also pattern where a root is repeated to form an onomatopoeia, as shown in (477). I classify this process as a type of repetition, rather than reduplication.

(477) $\acute{\text{pəs}}\acute{\text{pəs}}\acute{\text{pəs}}\acute{\text{pəs}}$

‘[Coyote’s call to his helpers]’ (Thomason, 1999, p.255, Okanagan)

A.2.4 Montana Salish-Spokane-Kalispel Reduplication

Attested reduplication in Montana Salish-Spokane-Kalispel is summarised in Table A.4.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
DIM		✓		
REP				✓
COS			✓	
INCH				✓
PL				✓
Other	Onomatopoeia/Repetition			

Table A.4: Reduplication in Montana Salish-Spokane-Kalispel

A.2.4.1 Montana Salish-Spokane-Kalispel Sources and Transcription

Examples come from Vogt (1940), B. Carlson (1972a), Speck (1977), B. Carlson (1978), B. Carlson and Thompson (1981), B. Carlson and Flett (1989), B. Carlson (1990), Black (1996), Thomason (1997), Bates and Carlson (1998), Thomason (1999), and Pete (2010). Unreduplicated forms come from either the source materials or B. Carlson and Flett (1989). Each example is marked for dialect.

Many potential instances of [ə] are omitted in the source material for Montana Salish-Spokane-Kalispel. I include [ə] where it can be confirmed from at least one source or where it is present by hypothesis (e.g., reduplication adds a syllable that requires a vocalic nucleus); schwa in other affixes or non-reduplicated forms may be missing if they were excluded in the original material.¹²

Montana Salish-Spokane-Kalispel has root metathesis where a vowel and C₂ consonant may be switched (B. Carlson, 1972a, p.26–27); this accounts for alternations in vowel position between some unreduplicated and reduplicated forms.

¹²B. Carlson (1972a, p.13) states that schwa-sonorant sequences and syllabic sonorants are in free variation.

A.2.4.2 Montana Salish-Spokane-Kalispel Plural/Pluractional C₁C₂ Reduplication

Plural/pluractional C₁C₂ reduplication doubles the first two consonants of the root; there is an epenthetic [ə] between the copied consonants, as shown in as shown in (478), unless both consonants are obstruents, as shown in (479). I mark this type of reduplication as a prefix because stress falls on the vowel in the second syllable (when it does not fall on a suffix), rather than the first.

- (478) a. s-**təm**~tém (cf. *stem* ‘thing’)
 NMLZ-PL~what
 ‘(They’re) things.’ (B. Carlson, 1972a, p.111, Spokane)
- b. s-**ɣəl**~ɣál-t (cf. *hi ɣal* ‘It’s light.’)
 NMLZ-PL~light-CHAR
 ‘(They’re) days.’ (B. Carlson, 1972a, p.111, Spokane)
- c. **pəl**~pələc-mí-m (cf. *pəlčúsəm* ‘turn back’)
 PL~turn-DER.TRANS-TRANS.CONT
 ‘They (sg.) are turning it [the meat] over and over.’ (Thomason, 1996, p.313, Montana Salish)
- (479) a. **ɣs**~ɣés-t (cf. *ɣést* ‘good’)
 PL~good-CHAR
 ‘(They’re) good things.’ (B. Carlson, 1972a, p.111, Spokane)
- b. **tx^w**~tíx^wcč (cf. *tíx^wcč* ‘tongue’)
 PL~tongue
 ‘(They’re) tongues.’ (B. Carlson, 1972a, p.111, Spokane)
- c. **tq**~tqə-n-té-n (cf. *tqəncín* ‘I hit you.’)
 PL~hit-CTR-TR-1SG.SUBJ
 ‘I hit them.’ (B. Carlson, 1972a, p.112, Spokane)
- d. **k^wt**~k^wtún-šən (cf. *k^wtúnt* ‘It’s big.’)
 PL~big-foot
 ‘(They’re) big feet.’ (B. Carlson, 1972a, p.112, Spokane)

In places where C₁C₂ reduplication would result in a /əy/ or /əw/ sequence, these sequences are realised as [i:] or [u:], respectively; examples are given in (480).

- (480) a. **q̣i:**~q̣éy (cf. q̣éy ‘They (sg.) lived.’)
 PL~live
 ‘They were camped.’ (B. Carlson, 1972a, p.114, Spokane)
- b. **su:**~séwłk^w (cf. sewłk^w ‘water’)
 PL~water
 ‘(They’re) rivers.’ (B. Carlson, 1972a, p.111, Spokane)

C₁C₂ reduplication marks plural entities with nouns, as shown in (478a–b), (479b), and (480b); it marks pluractionality with verbs, which may involve plural arguments, as shown in (479c) and (480a), or plural events, as shown in (478c). C₁C₂ reduplication also marks plural arguments with adjectival roots, as shown in (479a,d).

A.2.4.3 Montana Salish-Spokane-Kalispel Diminutive C₁ Reduplication

Diminutive C₁ reduplication doubles the initial consonant of a root, as shown in (481). When a sonorant participates in diminutive reduplication, the matching consonants are both glottalised, as shown in (481a–b). I mark the diminutive morpheme as an infix based on the fact that stress falls on the first vowel when it does not fall later in the word (e.g., 481c).

- (481) a. ní<**n̥ə**>č-əmən (cf. níčəmən ‘knife’)
 cut<DIM>-instrument
 ‘(It’s a) little knife.’ (B. Carlson, 1972a, p.115, Spokane)
- b. lú<**l̥ə**>k^w (cf. luk^w ‘stick of wood’)
 stick.of.wood<DIM>
 ‘It’s a little stick of wood’ (B. Carlson, 1972a, p.115, Spokane)
- c. stə<**t̥ə**>ŋ^wáne? (cf. stəŋ^wáne? ‘flint’)
 NMLZ-flint<DIM>
 ‘(It’s a) little flint.’ (B. Carlson, 1972a, p.115, Spokane)
- d. mú<**m̥ə**>s (cf. mus ‘four’)
 four<DIM>
 ‘four little (things)’ (B. Carlson, 1972a, p.116, Spokane)

- e. cí<œ>l (cf. *cil* ‘five’)
 five<DIM>
 ‘five little (things)’ (B. Carlson, 1972a, p.116, Spokane)

C₁ reduplication with obstruent-only roots or roots that start with an obstruent cluster yield a complex onset, as shown in (482a) and (482b), respectively. I mark these as prefixes for simplicity (i.e., C~CCV...), but an infix analysis may be preferable for consistency with the examples in (481) above (i.e., (i.e., C<C>CV...).

- (482) a. x̣~x̣l̥œcín (cf. x̣l̥œcín ‘horse’)
 DIM~horse
 ‘It’s a dog.’ (B. Carlson, 1972a, p.115, Spokane)
 b. p~ptáx-m̥ (cf. *sptax^w* ‘They (sg.) spit.’)
 DIM~spit
 ‘spit (diminutive)’ (Bates & Carlson, 1998, p.119, Spokane)

This type of reduplication marks a small entity with a noun, as shown in (481a–c) and (482) above. It marks small referents with a number (i.e., “X small ones”), as shown in (481d–e). With verbs, it can mark a small participant in an event, as in (483a–b), or a small event, as in (483c). Diminutive reduplication can also refer to a small amount of a state or that a state holds for a small referent, as in (483d) and (483e), respectively.

- (483) a. k̥^wu<k̥^wœ>l̥ (cf. *k̥^wúl̥œntx^w* ‘You did it.’)
 make/do<DIM>
 ‘Something small is created/made.’ (Bates & Carlson, 1998, p.118, Spokane)
 b. š~šil̥ (cf. *səl̥ínt* ‘chop it!’)
 chop<DIM>
 ‘A small thing is chopped.’ (Bates & Carlson, 1998, p.118, Spokane)
 c. čin ʔes-k̥^wu<k̥^wœ>l̥-i (cf. *k̥^wúl̥œntx^w* ‘You did it.’)
 1SG.SUBJ STAT-make/do<DIM>-CONT
 ‘I work a little.’ (Speck, 1977, p.59, Kalispel)

- d. $\dot{c}\dot{a}\sim\dot{c}ur$ (cf. $qe? \dot{c}ur$ ‘(it’s) our salt’)
 DIM~sour/salty
 ‘a little thing is sour/salty’ (Bates & Carlson, 1998, p.118, Spokane)
- e. $p\sim pin$ (cf. $?espin$ ‘it’s bent’)
 bent<DIM>
 ‘a little bent’ (Bates & Carlson, 1998, p.118, Spokane)

A.2.4.4 Montana Salish-Spokane-Kalispel Repetitive C₁ Reduplication

Repetitive C₁ reduplication affixes a fixed segment /e/; the first consonant of the root is doubled to provide an onset if the root has a simple onset (Bates & Carlson, 1998; Yu, 2005), as shown in (484). I mark the repetitive morpheme as a prefix in these examples, for consistency between these forms and the cognate morpheme in Moses Columbia (discussed in Section A.2.2.5).

- (484) a. $\dot{c}e\sim\dot{c}ip\dot{s}$ (cf. $\dot{c}ip\dot{p}$ ‘It accidentally got pinched.’)
 REP~pinch-PL
 ‘scissors, shears’ (Bates & Carlson, 1998, p.104, Spokane)
- b. $k^we\sim k^w\dot{il}\dot{e}?$ (cf. k^wilx^w ‘red cloth’)
 REP~red-thing
 ‘pet name for any red domesticated animal’ (Bates & Carlson, 1998, p.104, Spokane)
- c. $k^wek^w\dot{ul}i?s$ (cf. $k^w\dot{ul}\dot{ont}x^w$ ‘You did it.’)
 // $k^we\sim k^w\dot{ul}\dot{n}\dot{t}\dot{e}s$ //
 REP~make.do-CTR-TR-3ERG
 ‘They (sg.) kept making things to see if they worked (like a puzzle).’
 (B. Carlson, 1978, p.89, Spokane)

If the root starts with a consonant cluster, /e/ surfaces as an infix to break up a complex onset, as shown in (485); C₁ reduplication does not occur.

- (485) a. $s\dot{m}\langle e\rangle\dot{l}k^w\dot{e}ye?$ (cf. $mlk^w\dot{e}lx^w$ ‘a fur rug’)
 NMLZ-solid<REP>-seem
 ‘shadow’ (Bates & Carlson, 1998, p.105, Spokane)

- b. $\text{ɬ} < \mathbf{e} > \mathbf{k}^w$ -p-min (cf. $\text{ɬk}^w \text{pim}$ ‘They (sg.) speared it.’)
 pierce/spear<REP>-INCH-instrument
 ‘pitchfork’ (Bates & Carlson, 1998, p.105, Spokane)

The /e/ in the repetitive morpheme may be realised as [a] when added to a root with uvular or pharyngeal consonants, as shown in (486). Bates and Carlson (1998, p.104) describe [a] as a lowered variant of /e/.¹³

- (486) a. $\text{n-}\dot{\text{la}} \sim \dot{\text{lo}} \dot{\text{ɬ}}^w\text{-e?}$ (cf. *hesčloɬ* ‘It’s hooked up.’)
 LOC-REP~fit.together-thing
 ‘wore a loose dress without a belt’ (Bates & Carlson, 1998, p.104, Spokane)
- b. $\dot{\text{na}} \sim \dot{\text{na}} \dot{\text{q}} \sim \dot{\text{sn}}$ (cf. *hi naq* ‘It stinks.’)
 REP~smell-feet
 ‘smelly feet’ (Bates & Carlson, 1998, p.105, Spokane)
- c. $\dot{\text{la}} \dot{\text{la}} \dot{\text{q}} \dot{\text{ə}} \dot{\text{n}}$ (cf. *laq-n* ‘I buried it.’)
 // $\dot{\text{la}} \sim \dot{\text{la}} \dot{\text{q}} \text{-n-t-əñ//}$
 REP~bury-CTR-TR-1SG.ERG
 ‘I just covered things as I went along.’ (B. Carlson, 1978, p.89, Spokane)
- d. $\text{s-n-p} < \mathbf{a} > \text{táx}^w \text{-mñ}$ (cf. *sptáx^w* ‘They (sg.) spit.’)
 NMLZ-LOC-spit<REP>-instrument
 ‘spittoon’ (Bates & Carlson, 1998, p.105, Spokane)

Sonorants in C₁ position are glottalised in repetitive forms, as seen in (485a) and (486a–c) above.

Repetitive C₁ reduplication is found with verbs and adjectives, as well as nouns derived from verbs. It can refer to an iterative or repeated event, as in (484c) and 486c) above; the function reduplication is not compositional in all nouns (see, e.g., 485a).

¹³This is best described as retraction harmony, rather than feature sharing between adjacent segments. It does not require segments to be adjacent (see Bessell, 1993b and Schillo, 2024 for discussions of retraction in Interior Salish).

A.2.4.5 Montana Salish-Spokane-Kalispel Change-of-State C₂ Reduplication

Change-of-state C₂ reduplication copies the second consonant of the root. The consonant associated with the reduplicative morpheme is positioned before the root vowel, as shown in (487), unless the root starts with a sonorant-vowel-obstruent sequence; the examples in (488) show that change-of-state reduplication follows the root vowel in these forms.^{14,15}

- (487) a. m<ə́l>íł (cf. *mlip* ‘It overflowed.’)
 flow<COS>
 ‘It bled’ (B. Carlson & Thompson, 1981, p.113, Spokane)
- b. ł<ᵑ>íp (cf. *łip* ‘marked’)
 mark<COS>
 ‘It got marked unexpectedly.’ (B. Carlson & Thompson, 1981, p.112, Spokane)
- c. x^w<ᵑ^{ww} (cf. *x^wuk^w* ‘He is clean.’)
 clean<COS>
 ‘He got clean.’ (B. Carlson, 1972a, p.117, Spokane)
- d. c<q>áq (cf. *hescáq* ‘It’s placed.’)
 place<COS>
 ‘It stopped.’ (B. Carlson & Thompson, 1981, p.112, Spokane)
- e. q<č>íc (cf. *qic* ‘braided, woven’)
 braid/weave<COS>
 ‘It got tangled up (as a thread might do as you are sewing).’ (B. Carlson & Thompson, 1981, p.112, Spokane)
- (488) a. ní<čə>č (cf. *níčəntx^w* ‘You cut him/them.’)
 cut<COS>
 ‘It got cut.’ (B. Carlson, 1972a, p.117, Spokane)

¹⁴[h] patterns as a sonorant in (488c).

¹⁵I mark the reduplication in (488) in an infix position with CVC roots (i.e., CV-C₂-C₂) rather than suffixed (i.e., CVC₂-C₂) following my analysis of C₂ reduplication in Lillooet.

- b. mí<[˙]λ>[˙]λ (cf. *miλ* ‘It is smeared on.’)
 smear<COS>
 ‘It got smeared by accident.’ (B. Carlson & Thompson, 1981, p.112, Spokane)
- c. hé<^wk>^wk (cf. *hechék^w* ‘It’s open a crack.’)
 peel<COS>
 ‘It came open a crack without my knowing it.’ (B. Carlson & Thompson, 1981, p.112, Spokane)
- d. lá<[˙]q>[˙]q (cf. *heláq* ‘It’s buried.’)
 bury<COS>
 ‘It got buried.’ (B. Carlson & Thompson, 1981, p.113, Spokane)

C₂ reduplication marks a change of state, which may happen by accident, spontaneously, as a result of natural phenomena, without control, or with great effect or patience (B. Carlson & Thompson, 1981). This type of reduplication is typically found with verbs: it is most frequently found with stative roots, but is also found with some eventive ones (Kroeber, 1988).

A.2.4.6 Montana Salish-Spokane-Kalispel V₁ Reduplication

Two types of V₁ reduplication have been described in Montana Salish-Spokane-Kalispel. One type marks inchoativity; the other marks plurality.

Inchoative V₁ reduplication is associated with the inchoative suffix -ʔ-, which is inserted before the root vowel. [ʔ] is preceded by a copy of the root vowel in forms with V₁ reduplication, as shown in (489). I mark the first vowel as part of the reduplicative morpheme because stress falls on the second vowel. This type of reduplication is recorded with adjectives; it yields an inchoative verb.

- (489) a. n<^aʔ>ás (cf. *hi nas* ‘It is wet.’)
 wet<INCH>
 ‘They (sg.) get wet.’ (Vogt, 1940, p.64, Kalispel)
- b. ^qq̣<^uʔ>úc (cf. *q̣^wuct* ‘They (sg.) get fat.’)
 fat<INCH>
 ‘They (sg.) get fat.’ (Vogt, 1940, p.64, Kalispel)

- c. q^w<uʔ>úʔ (cf. *hi q^wuʔ* ‘It is dusty.’)
dusty<INCH>
‘It gets dusty.’ (Vogt, 1940, p.64, Kalispel)
- d. t̥<iʔ>ím (cf. *tím* ‘damp’)
damp<INCH>
‘It got damp.’ (Pete, 2010, p.153, Montana Salish)
- e. čʰ-p̥<eʔ>éč-sšn (cf. *p̥ečis* ‘They (sg.) put s.t. into the heat.’)
surface-heat<INCH>-round.object
‘get a hot forehead’ (Pete, 2010, p.302, Montana Salish)
- f. ʰ<aʔ>áq̣ (cf. *ʰa q̣* ‘hot’)
hot<INCH>
‘It got hot.’ (Pete, 2010, p.317, Montana Salish)

The other type of V₁ reduplication marks plurality of an argument; it is found with verbs. A glottal stop and copy of the stressed vowel are positioned following the stressed vowel, as shown in (490). I mark the second vowel as part of the reduplicative morpheme because stress falls on the second vowel.

- (490)
- a. čʰí<ʔi>p (cf. *čʰip* ‘They (sg.) hunted something.’)
hunt.something<PL>
‘They hunted something.’ (Thomason, 1997, p.354, Montana Salish)
 - b. n-pí<ʔi>lš (cf. *qe npilš* ‘We went in.’)
LOC-enter<PL>
‘They went in (all at once).’ (Thomason, 1997, p.354, Montana Salish)
 - c. cí<ʔi>tx^w-s (cf. *citx^w* ‘house’)
house<PL>-3POSS
‘their house’ (Thomason, 1997, p.354, Montana Salish)
 - d. mí<ʔi>ʰ-is (cf. *miʰis* ‘It’s smeared.’)
smear<PL>-3ERG
‘They smeared it.’ (Pete, 2010, p.568, Montana Salish)
 - e. tm-p-í<ʔi>lš (cf. *tím* ‘damp’)
damp-INCH-DEV<PL>
‘They gradually get damp.’ (Pete, 2010, p.153, Montana Salish)

- f. $\acute{p}\acute{e}<\textcolor{red}{?e}>\acute{c}$ -is (cf. $\acute{p}\acute{e}\acute{c}is$ ‘S/he/it put s.t. into the heat.’)
 heat<PL>-3ERG
 ‘They put s.t. into the heat.’ (Pete, 2010, p.302, Montana Salish)
- g. $nunx^w\acute{e}<\textcolor{red}{?e}>n\textcolor{red}{?}$ -em (cf. $nunx^w\acute{e}ne\textcolor{red}{?}$ ‘believe’)
 believe<PL>-PASS
 ‘They believed.’ (Pete, 2010, p.51, Montana Salish)

The presence of V_1 reduplication varies across dialects. Both types are present in Kalispel (Vogt, 1940, Camp, 2007), as well as Montana Salish (Pete, 2010). Neither type have been included in descriptions of Spokane.¹⁶

A.2.4.7 Repetition in Montana Salish-Spokane-Kalispel

A pattern of repetition is also found in some sound symbolic/onomatopoeic words Thomason (1999). One pattern involves repeating a root, where stress is marked on each of the reduplicated vowels, as shown in (491). The second involves the repetition of a final consonant, as shown in (492). I treat these as patterns of repetition, rather than reduplication.

- (491) a. $\acute{c}\acute{a} \acute{c}\acute{a} \acute{c}\acute{a} \acute{c}\acute{a}$
 ‘the hollering noise one someone makes when getting beaten up’
 (Thomason, 1999, p.255, Montana Salish)
- b. $l\acute{i}w l\acute{i}w l\acute{i}w l\acute{i}w$
 ‘sound of continuous ringing bells’ (Thomason, 1999, p.255, Spokane)
- c. $\acute{t}\acute{a}p \acute{t}\acute{a}p \acute{t}\acute{a}p \acute{t}\acute{a}p$
 ‘sound of feet slapping on the floor’ (Thomason, 1999, p.255, Montana Salish)
- (492) a. $\acute{c}\acute{u}ssss$
 ‘rustling noise when you walk in dry brush or pushing through brush’
 (Thomason, 1999, p.253, Montana Salish)

¹⁶See Mellesmoen (2024a) for the potential stages leading to the development of V_1 reduplication. Spokane may be at an earlier stage where the inchoative triggers epenthesis of [ə], rather than reduplication.

b. i cáxxx

‘the sound of frying food’ (Thomason, 1999, p.253, Montana Salish)

c. ma^{''''}lll

‘sound of a gurgling stomach or brook’ (B. Carlson & Flett, 1989, p.50, Spokane)

A.2.5 Lillooet Reduplication

Attested reduplication in Lillooet is summarised in Table A.5.

	C_1C_2	C_1	C_2	V_1
PL	✓			
PDEN	✓			
DIM		✓		
PL		✓		
INIT		✓		
COS			✓	
Other	C_2^2 Triplication			

Table A.5: Reduplication in Lillooet

A.2.5.1 Lillooet Sources and Transcription

Examples come from Van Eijk (1997), H. Davis and Matthewson (2003), H. Davis (2004), Matthewson (2005), Matthewson, Rullman, and Davis (2005), H. Davis (2011), Van Eijk (2013), H. Davis and Matthewson (2022), H. Davis and Mellesmoen (2023), and from fieldwork conducted by Henry Davis. Unreduplicated forms come from the original sources, Van Eijk (2013), or H. Davis et al. (in prep).

I have updated the glosses associated with C_1 reduplication from work prior to the first description of pluractional C_1 reduplication in H. Davis and Mellesmoen (2023). There are numerous other examples of Lillooet reduplication in this dissertation (see the case studies in Section 4.2 and Section 5.4 for more in-depth description, further examples, and phonological analysis).

Decisions about which segments are associated with a reduplicative morpheme, such as whether a reduplicative morpheme surfaces as a prefix or infix, are guided by the phonological analysis of Lillooet reduplication provided in Section 4.2.

A.2.5.2 Lillooet Plural/Pluractional C_1C_2 Reduplication

Plural/pluractional C_1C_2 reduplication copies the first two consonants of the root with the addition of an epenthetic [ə]; the reduplicative morpheme is realised as

CəC, as shown in (493).

- (493) a. s-**ɣəp**~**ɣáp** (cf. *sɣap* ‘tree’)
 NMLZ-PL~tree
 ‘trees’ (Van Eijk, 1997, p.56, Lillooet)
- b. s-**nək^w**~**núk^wa?** (cf. *snuk^wa?* ‘friend’)
 NMLZ-PL~friend
 ‘friends’ (Van Eijk, 1997, p.56, Lillooet)
- c. s-**q^wə́m**~**q^wə́m** (cf. *sq^wə́m* ‘mountain’)
 NMLZ-PL~mountain
 ‘mountain range’ (Van Eijk, 1997, p.56, Lillooet)

C₁C₂ reduplication in Lillooet can be used to describe plural entities with nouns, as in (493) above, or pluractionality with verbs, as in (494). Pluractionality includes events distributed across participants (494a–b,e), space (494a–c,e), and time (494a–d).

- (494) a. **ɬas** **plan** **wa?** **ɣi<?>p**
 COMP+IPFV+3CONJ already IPFV grow<INCH>
ʔi=píns=a **wa?** **cək^w~cək^w-ən-ə́m**
 DET.PL=beans=EXIS IPFV PL~pull-DIR-1 PL.ERG
ɬəl=ti=tmíx^w=a...
 from=DET=land=EXIS
 ‘When the beans were grown we pulled them from the ground...’ (cf. *cək^wən* ‘to pull s.t.’) (Matthewson, 2005, p.100, Lillooet)
- b. *Context: For the used cars, you have to lock the doors one by one.*
lək~**lák-c-añ=ɬkan** **tákə́m ʔi=sóps=a**
 PL~lock-mouth-DIR=1 SG.SUBJ all PL.DET=door=EXIS
ʔi=plán=a **wa?** **q^wə́z kəh**
 PL.DET=already=EXIS IPFV used car
 ‘I locked all the doors of the used cars.’ (cf. *nłákcəñ* ‘lock something’)
 (CA, Lillooet)

- c. níł=ka k^w=s-Henry wa? pəŋ^w~pəŋ^w-c-ám
 COP=EVID DET=NMLZ-Henry IPFV PL~knock-mouth-MID
 ‘That’ll be Henry knocking.’ (cf. pəŋ^w cá^ám ‘knock’) (Matthewson et al., 2005, p.3, Lillooet)
- d. *Context: One light is broken: it keeps going off and on again.*
 papt ka-ŋ^wól~ŋ^wəl-a tí? k^wu=qəl=wíl^x sčák
 always CIRC-PL~burn-CIRC that.VIS DET=bad=become light
 ‘That broken light kept going on and off.’ (cf. ŋ^wəl ‘get burned/lit’) (CA, Lillooet)
- e. *Context: There’s a power cut, and all the lights go off. Then the power is restored, and they all come on at once:*
 łapt tákəm ʔi=sčák^w=a l=ta=táwn=a, níł
 get.extinguished all PL.DET=light=EXIS at=DET=town=EXIS COP
 s=lóp=s=łu? ka-ŋ^wól~ŋ^wəl-a múta?
 NMLZ=suddenly=3POSS=EXCL CIRC-PL~burn-CIRC again
 ‘The lights went off in the town, and then all at once they came on again.’ (cf. ŋ^wəl ‘get burned/lit’) (CA, Lillooet)

A.2.5.3 Lillooet Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication is shown in (495); the form is identical to forms that receive a plural or pluractional interpretation in the previous section (Section A.2.5.2), except where the suffix *-t* is also present (see Section 4.2.3.2 for examples and discussion of this combination).

- (495) a. cəq^w~cíq^w (root: cíq^w- ‘red’)
 PDEN~red
 ‘red’ (Van Eijk, 1997, p.82, Lillooet)
- b. ʔəl~ʔəl=łkák^w łəl=sʔón^c
 PDEN~strong=2SG.SUBJ from=1SG.INDEP
 ‘You’re stronger than me.’ (cf. ʔəlíl^x ‘exert oneself’) (H. Davis, 2011, p.41, Lillooet)

- c. plan waʔ n-təq^w~tíq^w ti=q^wúʔ=a
 already IPFV LOC-PDEN~dirty DET=water=EXIS
 †=án=a=tuʔ n-qay-lx-áz
 COMP=IPFV+1SG.SJV=EXIS=REM LOC-jump-AUT-for.fun
 ‘The water where I used to swim is already dirty.’ (cf. *ntiʔq^w* ‘water gets dirty’) (H. Davis, 2004, p.97, Lillooet)

Property-denoting reduplication is only found with adjectives.

A.2.5.4 Lillooet Diminutive C₁ Reduplication

Diminutive C₁ reduplication copies the consonant before the stressed vowel; the diminutive morpheme surfaces as an infix following the stressed vowel, as shown in (496). The stressed moraic vowel is replaced with [ə], unless a glottal consonant is copied, as shown in (496b), in which case the stressed vowel remains unchanged. The last sonorant of a foot is glottalised, as shown in (496c–f).¹⁷

- (496) a. kapó<ɸ>uh (cf. *kapúh* ‘coat’)
 coat<DIM>
 ‘small coat’ (Van Eijk, 1997, p.257, Lillooet)
- b. ʔí<ʔ>mac (cf. *ʔímac* ‘grandchild’)
 grandchild<DIM>
 ‘little grandchild’ (Van Eijk, 1997, p.60, Lillooet)
- c. tó<t>pə́l̥ (cf. *típał* ‘table’)
 table<DIM>
 ‘little table, desk’ (H. Davis & Mellesmoen, 2023, p. 13)
- d. x^wó<x^w>ktən̥ (cf. *x^wíktən* ‘knife’)
 knife<DIM>
 ‘small knife’ (Van Eijk, 1997, p.254, Lillooet)
- e. ʔzə<z>ə́m̥ (cf. *ʔzum* ‘big’)
 big<DIM>
 ‘a little bit bigger’ (Van Eijk, 1997, p.56, Lillooet)

¹⁷See Section 4.2.4.3 for an analysis of the process of sonorant glottalisation that accompanies C₁ diminutive and pluractional reduplication in Lillooet..

- f. zó<ẏ>xət (cf. *zaxt* ‘long’)
long<DIM>
‘a bit longer’ (H. Davis & Mellesmoen, 2023, p.13, Lillooet)
- g. s-pəq-mó<ṁ>x (cf. *spəqmíx* ‘swan’)
NMLZ-white-excess<DIM>
‘little swan’ (H. Davis & Mellesmoen, 2023, p.15, Lillooet)
- h. s-kəl-ə<ḷ>ca? (cf. *kəlíca?* ‘item of buckskin clothing’)
NMLZ-buckskin-outer.cover<DIM>
‘small (item of) buckskin clothing’ (H. Davis & Mellesmoen, 2023, p.15, Lillooet)

C₁ reduplication refers to the small size of a noun, as shown in (496a–d,g–h). With an adjective, it can mark a small amount of a quality, as shown in (497), or a small degree of difference in a comparison, as shown in (496e–f) above. Diminutive reduplication with a verb can mark diminished action, short duration, or partial completion, as shown in (498a–d); it can also mark a small participant, as shown in (498e–f).

- (497) a. k^wó<k^wə>m-p-əqs ta=wətkál-mən=a
dull<DIM>-INCH-point DET=pry-instrument=EXIS
‘The pry bar has gotten a bit blunt.’ (cf. *k^wəmp* ‘get blunt or dull’)
(H. Davis et al., in prep, Lillooet)
- b. ɬəmḵ^w-ə<ḵ^{ww}u=s-k^wú<k<sup>wdirty-face<DIM> that.VIS DET=NMLZ-child<DIM>
‘That child’s face is a little dirty.’ (cf. *ɬəmḵús* ‘have a dirty face’)
(H. Davis & Mellesmoen, 2023, p.27, Lillooet)</sup>
- (498) a. təp~tə<t>p-aḵan=ɬkán k^w=s-Henry
PL~punch<DIM>-arm-1 SG.SUBJ DET=NMLZ-Henry
‘I punched Henry in the arm lightly a bunch of times.’ (cf. *túpun* ‘punch something’) (CA, Lillooet)
- b. pə<p>ɣ^w-əṁ=ɬkán l=ta=səps=a
knock<DIM>-MID=1 SG.SUBJ on=DET=door=EXIS
‘I knocked a little bit on the door.’ (cf. *pəɣ^wcám* ‘knock’) (H. Davis & Mellesmoen, 2023, p.17, Lillooet)

- c. mʰ<ṁ>caʔq=ʔkan ʔəʔ xʷúʔ=ʔkan ʔayʔ ʔalkst
sit<DIM>=1 SG.SUBJ and.then PROS=1 SG.SUBJ then work
'I sat down for a little while before going back to work.' (cf. *míca(ʔ)q*
'sit down or up') (H. Davis & Mellesmoen, 2023, p.19, Lillooet)
- d. xʷə<xʷ>k-əṁ=ʔkán ta=sćúqʷaʔ=a, ʕuʔ kʷəṁ-p
cut<DIM>-MID=1 SG.SUBJ DET=fish=EXIS but dull-INCH
ta=n-xʷík-tən=a, níʔ s=xʷʔay=s
DET=1 SG.POSS-cut-instrument=EXIS COP NMLZ=NEG=3POSS
kʷ=n-s-ka-cukʷ-s-a
DET=1 SG.POSS=NMLZ=CIRC-finish-CAUS-CIRC
'I cut the fish a little bit, but my knife got dull, so I couldn't finish it.'
(cf. *xʷíkəṁ* 'butcher fish') (CA, Lillooet)
- e. waʔ xʷə<xʷ>k-əṁ ta=s-kʷú<kʷ>miʔt=a
IPFV cut.fish<DIM>-MID DET=NMLZ-child<DIM>=EXIS
'The child was cutting fish.' (cf. *xʷíkəṁ* 'butcher fish') (CA, Lillooet)
- f. mʰ<ṁ>caʔq ta=s-kʷú<kʷ>miʔt=a níʔ=ʕuʔ
sit<DIM> DET=NMLZ-child<DIM>=EXIS COP=EXCL
s=zapʷun[-n]-án l=ta=qílq-s=a.
NMLZ=belt[-DIR]-1 SG.ERG in=DET=seat-3POSS=EXIS
'The child sat down and I buckled them in their seat.' (cf. *míca(ʔ)q* 'sit
down or up') (H. Davis & Mellesmoen, 2023, p.19, Lillooet)

A.2.5.5 Lillooet Pluractional C₁ Reduplication

Pluractional C₁ reduplication copies the consonant before the stressed vowel; it occurs in the same position as diminutive C₁ reduplication, as discussed in the previous section. The difference between the two stress-aligned C₁ reduplication processes is that the vowel is not replaced by [ə] in the pluractional forms, as shown in (499).

- (499) a. kʷs-á<s>lićaʔ (root: *kʷas-*)
singe-outer.cover<PL>
'burn hair or feathers off an animal or a bird' (H. Davis & Mellesmoen, 2023, p.21, Lillooet)

- b. $x^w\text{əlp-í}<\text{p}>\text{ləx}$ (cf. $x^w\text{əlpilx}$ ‘turn around’)
 turn<PL>-AUT
 ‘turn around and around’ (H. Davis & Mellesmoen, 2023, p.21, Lillooet)
- c. $zwát-\text{ən}=\text{íkan}$ $sónaʔ$ $k^w=s=x^w\text{uʔ}$ $k^w\text{is}$...
 know-DIR=1 SG.SUBJ CNTR DET=NMLZ=PROSP rain
 $məs=kán=\text{ḵ}$ cix^w $má<\text{m}>\text{təq}$
 but=1 SG.SUBJ=EXCL get.there walk<PL>
 ‘I knew it was going to rain ... but I went for a walk anyways.’
 (cf. *matq* ‘to walk, go on foot’) (H. Davis & Matthewson, 2022, p.1343, Lillooet)

Pluractional C_1 is only found with verbs and shows a tendency toward event-internal readings, such as those in (500a–b). Event-internal readings involve “a multiplicity of actions which form part of the same event” (H. Davis & Mellesmoen, 2023, p.24). However, there are also examples of event-external readings, such as one with multiple participants in (500c).

- (500) a. $\text{ḵá}<\text{ḵ}>\text{í-ən}$ (cf. ḵáqlan ‘bite something’)
 bite<PL>-DIR
 ‘gnaw something’ (H. Davis & Mellesmoen, 2023, p.20, Lillooet)
- b. $\text{čá}<\text{č}>\text{q-ən}$ (cf. čáqan ‘pull something out’)
 pull.out<PL>-DIR
 ‘pluck something (a bird)’ (H. Davis & Mellesmoen, 2023, p.24, Lillooet)
- c. $sú<s>\text{q}^w\text{-ən}$ (cf. $súq^w\text{ən}$ ‘skin an animal’)
 skin<PL>-DIR
 ‘skin pl. animals’ (H. Davis & Mellesmoen, 2023, p.20, Lillooet)

A.2.5.6 Lillooet Plural (Initial) C_1 Reduplication

Initial C_1 reduplication copies the first consonant of the root and adds either a [ə] (501). It has a limited distribution (H. Davis, p.c.).

- (501) a. **kə~kláx^w** (cf. *klax^w* ‘muskrat’)
 C₁.INIT~muskrat
 ‘muskrat(s)’ (Van Eijk, 1997, p.56, Lillooet)
- b. **n-xə~xzúm-ana?** (cf. *xzum* ‘big’)
 LOC-C₁.INIT~big-ear
 ‘big ears’ (Van Eijk, 1997, p.57, Lillooet)
- c. **s-q^wə~q^wlíp** (cf. *sq^wlíp* ‘black tree moss’)
 NMLZ-C₁.INIT~black.tree.moss
 ‘lots of black tree moss’ (Van Eijk, 1997, p.57, Lillooet)
- d. **q^wə~q^wítaz** (cf. *q^wlítaz* ‘jackpine’)
 C₁.INIT~jackpine
 ‘jackpines’ (Van Eijk, 1997, p.58)
- e. **cə~cítx^w** (cf. *cítx^w* ‘house’)
 C₁.INIT~house
 ‘houses (= village, settlement)’ (Van Eijk, 1997, p.58, Lillooet)

Initial C₁ reduplication is found with nominal and adjectival roots; it marks plural entities.

A second type of initial C₁ reduplication marks continuative aspect with three verbs (H. Davis, p.c.). The first consonant and vowel are copied, as shown in (502).

- (502) a. **x^wí~x^witən** (cf. *x^witən* ‘to whistle’)
 C₁.INIT~whistle
 ‘to keep whistling’ (Van Eijk, 1997, p.56, Lillooet)
- b. **s-tá~tał-ləx** (*stáłləx* ‘be standing’)
 STAT-C₁.INIT~upright-AUT
 ‘to keep standing up’ (Van Eijk, 1997, p.56, Lillooet)

A.2.5.7 Lillooet Change-of-State C₂ Reduplication

Change-of-state C₂ reduplication copies the consonant after the stressed vowel; [ə] is epenthesized between the matching consonants, as shown in (503).¹⁸

¹⁸I segment these as CV<Cə>C rather than CVC~əC following the analysis in Chapter 4 (Section 4.2.4.1).

- (503) a. $\acute{h}i < \mathbf{l}\mathbf{\acute{o}} > l$ (root: $h\acute{i}l$ - ‘sprinkle’)
 sprinkle<COS>
 ‘to sprinkle, be sprinkling’ (Van Eijk, 1997, p.56, Lillooet)
- b. $p\acute{u} < \mathbf{l}\mathbf{\acute{o}} > \acute{t} = tu?$ $ta = q^w \acute{u} ? = a$ (root: $pu\acute{t}$ - ‘to boil’)
 boil<COS>=REM DET=water=EXIS
 ‘The water was boiling (but hadn’t finished boiling yet).’ (H. Davis & Matthewson, 2003, p.54, Lillooet)
- c. $\acute{p}l\acute{i} < \mathbf{x}^w \mathbf{\acute{o}} > \mathbf{x}^w$ (root: $\acute{p}li\mathbf{x}^w$ -)
 flow.over<COS>
 ‘flowing over’ (Van Eijk, 1997, p.56, Lillooet)
- d. $c\acute{i} < \mathbf{x}^w \mathbf{\acute{o}} > \mathbf{x}^w = kan$ $s\acute{o}na?$, $\acute{\lambda}u?$ $? \acute{a}y = \acute{\lambda}u?$
 get.there<COS>=1SG.SUBJ CNTR but NEG=EXCL
 $k^w as$ $wa?$ $k = Laura$
 DET+NMLZ+IPFV+3POSS IPFV DET=Laura
 ‘I got there, but Laura wasn’t there.’ (cf. cix^w ‘to arrive (over there)’)
 (H. Davis & Matthewson, 2022, p.1351, Lillooet)
- e. $nas = kan = k\acute{a} = tu?$ $z\acute{a}w - \acute{o}m$ $\acute{\lambda}u?$ $q\acute{a} < \mathbf{x}^w \mathbf{\acute{o}} > \mathbf{x}^w$
 go-1SG.SUBJ=IRR=REM dipnet-INTR but break<COS>
 $ta = k\acute{a}h - s = a$ $ta = s - mu\acute{t} \acute{a}c = a$
 DET=car-3POSS=EXIS DET=NMLZ-female=EXIS
 $n - s - k^w uza?$
 1SG.POSS-NMLZ-offspring
 ‘I would have gone fishing, but my daughter’s car broke down.’ (cf. $sqa\acute{x}^w$ ‘broken’) (Matthewson et al., 2005, p.7, Lillooet)
- f. $nka?$ $\acute{t} = k^w \acute{a} < \mathbf{n}\mathbf{\acute{o}} > n - s = ax^w$ $ta = xz\acute{u}m = a$
 where COMP=get.caught<COS>-CAUS=2SG.ERG DET=big=EXIS
 $\mathbf{x}^w u? \acute{\lambda}$
 sturgeon
 ‘Where did you catch the big sturgeon?’ (cf. k^wan ‘to take, catch, grab’) (H. Davis, 2004, p.90, Lillooet)

If [ə] is not obligatory for phonotactic reasons, it may be omitted, as shown in (504); however, note that an additional [ə] appears after the reduplicant in these forms.

- (504) a. ʔá<ḡ>ḡ-ət (cf. ʔawt ‘to be late’)
 behind<COS>-CHAR
 ‘too late, to miss the train or bus’ (Van Eijk, 1997, p.59, Lillooet)
- b. ká<p>pəḡ (cf. kəpḡxál ‘to knock’)
 knock<COS>
 ‘to knock constantly’ (Van Eijk, 1997, p.59, Lillooet)

C₂ reduplication marks a change-of-state in Lillooet; it is found with verbs.

A.2.5.8 Other Reduplication in Lillooet

Triplication, which appears to have two instances of C₂ reduplication, is shown in (505). This pattern is discussed in Section 4.2.5.

- (505) a. pl<ix^wix^w>íx^w (cf. kaplíx^wa ‘to boil over suddenly’)
 overflow<TRIP>
 ‘to keep overflowing’ (Van Eijk, 2013, p.32, Lillooet)
- b. q<əḡ^wəḡ^w>əḡ^w-c (cf. qəḡ^wcánəm ‘mouth is shivering’)
 get.slammed<TRIP>-mouth
 ‘rattle one’s teeth’ (Van Eijk, 2013, p.263, Lillooet)

A.2.6 Shuswap Reduplication

Attested reduplication in Shuswap is summarised in Table A.6.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
1 SG		✓		
COS			✓	
Other	C ₂ ² Triplication			

Table A.6: Reduplication in Shuswap

A.2.6.1 Shuswap Sources and Transcription

Examples come from J. Gibson (1973), Kuipers (1974a), Kuipers (1989), Lai (1998), and my fieldwork. Unreduplicated forms come from the original sources, Kuipers (1974a), Kuipers (1989), or my fieldwork. I have added [ə] in many places where they are omitted in source material.

Note that only one obstruent and one sonorant may be glottalised in a single word in Shuswap (see, e.g., L. C. Thompson & Thompson, 1985), which means that deglottalisation must accompany fission (i.e., ...C'V... > ...C'VC'...). While this affects reduplicated consonants, it is a general phonological process in the language and is not explicitly discussed in the following sections.¹⁹

A.2.6.2 Shuswap Plural/Pluractional C₁C₂ Reduplication

Plural/pluractional C₁C₂ reduplication involves a prefixed copy of the initial two consonants of the root with an epenthetic [ə], as shown in (506). I mark the reduplicative morpheme as a prefix because stress falls on the second syllable.

- (506) a. **pəs**~pəsətk^we (cf. *pəsətk^we* 'lake')
 PL~lake
 'lates' (Kuipers, 1974a, p.138, Shuswap)

¹⁹Ejective lateral affricates are reduplicated as /t/.

- b. x^wʔit ɣə=s-**pəq**~péq (root: *peq*-)
many DET=NMLZ-PL~berry
'There are many berries.' / 'The berries are many.' (Lai, 1998, p.9, Shuswap)
- c. s-**pəl**~púl-tən (cf. *pultən* 'bed, lair')
NMLZ-PL~lie-instrument
'lairs' (Kuipers, 1974a, p.39, Shuswap)
- d. s-**təm**~témtət-tən (cf. *stəmet* 'friend')
NMLZ-PL~friend-PL
'in-laws' (Kuipers, 1974a, p.152, Shuswap)
- e. **kəc**~kícx (cf. *kicx* 'arrive')
PL~arrive
'arrive (pl.)' (Kuipers, 1974a, p.39, Shuswap)
- f. (c-)**ʔəł**~ʔúłx^w (cf. (c)*ʔułx^w* 'enter')
(STAT-)PL~enter
'enter (pl.)' (Kuipers, 1974a, p.277, Shuswap)
- g. **wət**~wétkx (cf. *wetkx* 'vomit')
PL~vomit
'vomit (pl.)' (Kuipers, 1974a, p.262, Shuswap)
- h. **nək**~ník-əm (cf. *nikəm* 'to cut')
PL~cut-MID
'cut, saw (pl.)' (Kuipers, 1974a, p.191, Shuswap)

C₁C₂ reduplication marks plural entities with nouns, as shown in (506a–d) above, in addition to marking pluractionality on verbs, as shown in (506e–h). With verbs, C₁C₂ reduplication frequently marks plural participants, as shown in (507a–c). There are also examples that show a spatial or temporal distribution, such as those in (507d) and (507e), respectively.

- (507) a. c-**yəm**~yémt (cf. *cyem* 'to sit, be camped')
STAT-PL~sit/camp
'several groups are at home/camped' (Kuipers, 1974a, p.39, Shuswap)

- b. (w)ʔex ɣə=yəw~yéw-m-əs (cf. *yewəm* ‘fish with scoop net’)
IPFV DET=PL~fish-MID-3SBJV
‘They’re fishing.’ (Kuipers, 1974a, p.80, Shuswap)
- c. ɣ̣^wəɣ̣^wéyt ɣə=cék^wstən **ceʔ**~cé<ʔ>k^w-t-əs (cf. *cʔek^w* ‘got bright’)
all DET=light PL~bright<INCH>-TR-3ERG
‘All of the lights were lit up (at the same time).’ (GD, Shuswap)
- d. ɣə=pus **k^wəp**~k^wup-ən-s ɣə=nəkúʔ tə=s-ɣ̣cyétəm^w ɣ̣^wəɣ̣^wéyt
DET=cat PL~push-CTR-3ERG DET=one DET=NMLZ-box all
ʔ(ə)héʔən
to.where
‘The cat kept pushing one box all over the place.’ (cf. *k^wupəm* ‘to push’) (GD, Shuswap)
- e. ɣə=səməyúỵə **təq^w**~ʔq^wésxn-əm neʔ tqeltk x̣^wúʔcəm ʔeʔ
DET=small.fish PL~swim-MID DEM top/above again CONJ
x̣^wúʔcəm
again
‘The little fish swam to the top (again and again).’ (cf. *ʔəq^w-ésxn-m* ‘to tread water/swim’) (GD, Shuswap)

A.2.6.3 Shuswap Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication has the same form as plural/pluractional C₁C₂ reduplication. The first two consonants of the root are copied and positioned as a prefix with epenthetic [ə], as shown in (508). Property-denoting reduplication is often found in combination with the characteristic suffix *-t*.

- (508) a. **qəp**~qép-t (cf. *qépəm* ‘to soften hides’)
PDEN~soft-CHAR
‘soft (as flannel)’ (Kuipers, 1989, p.204, Shuswap)
- b. **tək^w**~túk^w-t (cf. *ʔəstuk^w* ‘fall silent’)
PDEN~silent-CHAR
‘silent’ (Kuipers, 1974a, p.39, Shuswap)

- c. $\gamma\dot{a}l \sim \gamma\dot{a}l-t$ (root: $\gamma\dot{a}l-$ ‘strong’)
 PDEN~strong-CHAR
 (Kuipers, 1974a, p.55, Shuswap)
- d. $t\dot{a}x^w \sim t\dot{u}x^w-t$ (root: $t\dot{u}x^w-$ ‘straight, right’)
 PDEN~straight-CHAR
 ‘straight’ (GD, Shuswap)

Property-denoting C_1C_2 reduplication is found with adjectives.

A.2.6.4 Shuswap Diminutive C_1 Reduplication

Diminutive C_1 reduplication copies the consonant before the stressed vowel in the word. The reduplicated consonant is positioned directly after the stressed vowel, as shown in (509).

- (509) a. $cq\dot{e} < \dot{q} > \dot{t}p$ (cf. $c\dot{q}e\dot{t}p$ ‘tree’)
 tree<DIM>
 ‘small tree’ (Kuipers, 1974a, p.39, Shuswap)
- b. $\dot{t}ke < k > p$ (cf. $\dot{t}kep$ ‘pot’)
 pot<DIM>
 ‘little pot’ (J. Gibson, 1973, p.47, Shuswap)
- c. $tiq\dot{u}qst\dot{a}n$ (cf. $ti\dot{q}u\dot{s}t\dot{a}n$ ‘butter’)
 //t-yq-ú< $\dot{q}$ >s-t\dot{a}n//
 LOC-mash-face<DIM>-instrument
 ‘a little butter’ (GD, Shuswap)
- d. Eve $t\dot{a}x^w-\dot{a}n-t-\dot{e} < t > s$ $\gamma\dot{a}=s-\dot{\lambda}x\dot{u}m\dot{a}n$
 Eve straight-CTR-TR-3ERG<DIM> DET=NMLZ-spoon
 ‘Eve straightened the spoon a little.’ (cf. $t\dot{a}x^wt\dot{u}x^wt$ ‘straight’) (GD, Shuswap)
- e. $\dot{p}?\dot{e} < ? > x^w$ (cf. $\dot{p}?\dot{e}x^w$ ‘more’)
 more<DIM>
 ‘a little more’ (Kuipers, 1989, p.153, Shuswap)

- f. fəqt-ʔú<ʔ>əy ʔə=lətép (cf. fəqtʔúy ‘It’s too wide.’)
 wide-real<DIM> DET=table
 ‘The table is a little too wide.’ (GD, Shuswap)

Diminutive reduplication occurs with nouns, verbs, and adjectives. It marks a small entity or amount with nouns, as shown in (509a–c) above. With adjectives, such as those in (509d–f), diminutive reduplication marks either a small degree (i.e., “a little bit X”) or a small difference between two degrees (i.e., “a little bit more X”). With verbs, diminutive reduplication refers to partial completion, as shown in (510a); it cannot be used to describe a short duration or small participant, as shown in (510b) and (510c) respectively.

- (510) a. ʔə=s-ʔʷəs~ʔʷés-t $\text{m<ʔ>ixʷ-ənwé<ʔʷ>ən-s}$
 DET=NMLZ-PDEN~sunny-CHAR melt<INCH>-NTR<DIM>-3ERG
 ʔə=s-wuxʷt
 DET=NMLZ=snow
 ‘The sunshine was able to melt the snow a little bit.’ (cf. míxʷən ‘They (sg.) melt it.’) (GD, Shuswap)
- b. $\text{\#ʔə=s-kʷékʷʔəs qʷé<qʷ>c-ən-s}$ ʔə=ʔúqʷləxʷ
 DET=NMLZ-sun warm<DIM>-CTR-3ERG DET=ground/soil
 Intended: ‘The sun warmed the ground for a little while.’ (cf. qʷécəm ‘to warm up’) (GD, Shuswap)
- c. *ʔə=nú<n>xʷən-xʷ pən-mí<m>ən-s
 DET=woman<DIM> find-RLT<DIM>-3ERG
 ʔə=x-ʔúqʷ-mən
 DET=LOC-scoop-instrument
 Intended: ‘The little girl found the cup.’ (cf. pənminx ‘You will find it.’) (GD, Shuswap)

A.2.6.5 Shuswap First Person C₁ Reduplication

First person C₁ reduplication, which has the same form as the diminutive described in the previous section, marks first person subjects or objects. Like the diminutive, it is sensitive to stress. Examples are provided in (511).

- (511) a. $\gamma=n-ci< c >tx^w$ (cf. *citx^w* ‘house’)
 DET=1SG.POSS-house<1SG>
 ‘my house’ (Kuipers, 1974a, p.39, Shuswap)
- b. $kí< k >cx-kən$ (cf. *tkicx* ‘arrived there’)
 arrive<1SG>-1SG.SUBJ
 ‘I arrive.’ (Kuipers, 1974a, p.39, Shuswap)
- c. $x-kəł-cí< c >ən$ $məł$ $\gammaə=x-kəmcənéłx^w$
 LOC-open-mouth<1SG> already DET=LOC-door
 ‘I already opened the door.’ (cf. *xkəłcínš* ‘to open’) (GD, Shuswap)
- d. $m-wí< w >k-t-səm-s$ $rə=Mary$
 COMPL-see<1SG>-TR-1SG.OBJ-3SUBJ DET=Mary
 ‘Mary saw me.’ (cf. *swéty kwikts* ‘Whom did they (sg.) see?’)
 (Gardiner, 1996, p.175, Shuswap)

First person C_1 reduplication is also used in the nominal domain to mark first person singular possessives, as in (511a) above. With verbs, it marks a first person subject or object, as shown in (511b–c) and (511d), respectively.

There is variation between dialects and speakers with respect to the use of first person C_1 reduplication (Kuipers, 1974a); Lyon and Ignace (2021) also observe that C_1 reduplication is not found in quoted speech in the Kuipers (1974a) narratives.

A.2.6.6 Shuswap Change-of-State C_2 Reduplication

Change-of-state C_2 reduplication doubles the consonant following the root vowel and places it before the root vowel. A [ə] is also epenthesized, as shown in (512). I mark the reduplicant as an infix based on the fact that based on the position of stress.²⁰

- (512) a. $p<ə k^w>úk^w$ (cf. *pk^wum* ‘to scatter’)
 spill<COS>
 ‘be spilled’ (J. Gibson, 1973, p.49, Shuswap)

²⁰Interior Salish languages differ in whether C_2 reduplication comes before or after the stressed or root vowel: Okanagan and Shuswap both have a $C<əC_2>\acute{V}C_2$ pattern, while the other Interior Salish languages have a $C\acute{V}<C_2ə>C_2$ one.

- b. $\dot{p}<\acute{a}t>\acute{a}t$ (cf. *cpať* ‘flat’)
flat<COS>
‘flattened out’ (Kuipers, 1989, p.153, Shuswap)
- c. $\dot{\lambda}<\acute{a}x^w>\acute{u}x^w$ (cf. *łux^w* ‘numb’)
numb<COS>
‘numb’ (Kuipers, 1974a, p.39, Shuswap)
- d. $\dot{\lambda}<\acute{a}s>\acute{a}s$ (cf. *łas* ‘make a splashing sound’)
splash<COS>
‘soggy’ (Kuipers, 1989, p.163, Shuswap)

Change-of-state C_2 reduplication applies to adjective and verb roots; it does not occur with nouns. Kuipers (1989, p.22) states that forms with C_2 reduplication involve either an “ongoing process of change” or “a quality (and not to a process leading to it)”.

A.2.6.7 Other Reduplication in Shuswap

Kuipers (1989) describes a C_2^2 triplication pattern in Shuswap, which is shown in (513). The final consonant of the root is tripled and two instances of [ə] are epenthesized. This appears to have an iterative meaning, much like the cognate triplication pattern found in Lillooet.

- (513)
- a. $p<\acute{a}t\acute{a}t>\acute{é}t$ (cf. *snptóle łx^w* ‘dumplings’)
bubble/boil<TRIP>
bubble, boil (Kuipers, 1989, p.22, Shuswap)
 - b. $t<\acute{a}k^w\acute{a}k^w>\acute{í}k^w$ (cf. *łik^w* ‘fire’)
fire<TRIP>
‘throw off sparks’ (Kuipers, 1989, p.22, Shuswap)

A.2.7 Thompson River Salish Reduplication

Attested reduplication in Thompson River Salish is summarised in Table A.7.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
AFF		✓		
COS			✓	

Table A.7: Reduplication in Thompson River Salish

A.2.7.1 Thompson River Salish Sources and Transcription

Examples come from B. Carlson and Thompson (1981), L. C. Thompson and Thompson (1992), Jimmie (1994), and L. C. Thompson and Thompson (1996). I draw on L. C. Thompson and Thompson (1996) for most unreduplicated forms. Sources for Thompson River Salish typically omit schwa; I only regularly add [ə] to reduplicative morphemes.

See Chapter 6 (Section 6.6.2.3.3) for further discussion of reduplication in Thompson River Salish.

A.2.7.2 Thompson River Salish Plural/Pluractional C₁C₂ Reduplication

Plural/pluractional C₁C₂ reduplication copies the initial two consonants of the root; [ə] is epenthesized between the two reduplicated consonants, as shown in (514).

- (514) a. kéykix (cf. *keyx* ‘hand’)
 //kéy<**kəy**>x//
 hand<PL>
 ‘hands, arms’ (L. C. Thompson & Thompson, 1992, p.84, Thompson River Salish)
- b. s-k^wóz~k^wəz (cf. *sk^woz* ‘aunt’)
 NMLZ-aunt~PL

‘aunts’ (L. C. Thompson & Thompson, 1992, p.84, Thompson River Salish)

- c. **ᵗᵗᵗᵗ**~ᵗᵗᵗᵗ (cf. ᵗᵗᵗᵗ ‘ear’)

PL~ear

‘ears’ (L. C. Thompson & Thompson, 1992, p.82, Thompson River Salish)

- d. s-**ᵗᵗᵗᵗ**~ᵗᵗᵗᵗ-ᵗᵗᵗᵗ (cf. sᵗᵗᵗᵗᵗᵗᵗ ‘niece’)

NMLZ-PL~niece

‘nieces’ (L. C. Thompson & Thompson, 1992, p.82, Thompson River Salish)

C₁C₂ reduplication marks plural entities with nouns, as shown in (514) above; with verbs or adjectives, it can refer to plural participants, as shown in (515a–c). C₁C₂ reduplication on verbs can also mark events distributed in time or space, as shown in (515d–e).

- (515) a. **ᵗᵗᵗᵗ**~ᵗᵗᵗᵗ (cf. sᵗᵗᵗᵗᵗᵗ ‘be cold’)

PL~feel.cold

‘They are cold.’ (L. C. Thompson & Thompson, 1992, p.82, Thompson River Salish)

- b. ᵗᵗᵗᵗᵗᵗ (cf. ᵗᵗᵗᵗ ‘good’)

//ᵗᵗᵗᵗ~ᵗᵗᵗᵗ//

PL~good

‘plural persons are good’ (L. C. Thompson & Thompson, 1996, p.450, Thompson River Salish)

- c. **wᵗᵗᵗ**~wᵗᵗᵗ-t-x^w (cf. ᵗᵗᵗᵗᵗᵗᵗ ‘seen, observed’)

PL~see-CTR-2SG.ERG

‘You see them.’ (L. C. Thompson & Thompson, 1992, p.82, Thompson River Salish)

- d. **sᵗᵗᵗ**~sᵗᵗᵗ-t-ᵗᵗᵗ (cf. sᵗᵗᵗᵗ ‘breeze, draft’)

PL~aspirate-CTR-3ERG

‘blows repeatedly on’ (L. C. Thompson & Thompson, 1992, p.82, Thompson River Salish)

- e. $\text{p}^{\text{h}}\text{a}\text{ŋ}^{\text{h}}\text{p}^{\text{h}}\text{a}\text{ŋ}^{\text{h}}\text{t}$ (cf. $\text{p}^{\text{h}}\text{a}\text{ŋ}^{\text{h}}\text{t}$ ‘burned, singed’)
 //p^həŋ~p^həŋ-t//
 PL~burn.flesh-CHAR
 ‘burned in several places’ (L. C. Thompson & Thompson, 1992, p.82,
 Thompson River Salish)

A.2.7.3 Thompson River Salish Property-Denoting C₁C₂ Reduplication

In words with property-denoting C₁C₂ reduplication, stress falls on the initial syllable, as shown in (516). An epenthetic [ə] appears in the second syllable. Property-denoting C₁C₂ reduplication often co-occurs with a -t suffix, which is cognate to the adjectival or stative -t suffixes in other Salish languages, though L. C. Thompson and Thompson (1992) label it as “intermediate”.

- (516) a. $\text{ʔ}^{\text{h}}\text{a}\text{q}^{\text{h}}\sim\text{ʔ}^{\text{h}}\text{ə}\text{q}^{\text{h}}\text{-t}$ (cf. $\text{ʔ}^{\text{h}}\text{a}\text{q}^{\text{h}}\text{t}$ ‘pricked, stuck’)
 perforate~PDEN-CHAR
 ‘full of thorns, thorny’ (L. C. Thompson & Thompson, 1992, p.89,
 Thompson River Salish)
- b. $\text{m}^{\text{h}}\text{í}\text{ʔ}^{\text{h}}\text{m}^{\text{h}}\text{í}\text{ʔ}^{\text{h}}\text{t}$ (cf. $\text{m}^{\text{h}}\text{í}\text{ʔ}^{\text{h}}\text{t}$ ‘They (sg.) are infected.’)
 //m^həy~m^həy-t//
 distribute~PDEN-CHAR
 ‘It’s very contagious.’ (L. C. Thompson & Thompson, 1992, p.89,
 Thompson River Salish)
- c. n-wén~wən (cf. *nwením* ‘arise early’)
 LOC-early~PDEN
 ‘It’s early’ (L. C. Thompson & Thompson, 1992, p.87, Thompson
 River Salish)
- d. $\text{w}^{\text{h}}\text{í}\text{ʔ}^{\text{h}}\sim\text{w}^{\text{h}}\text{í}\text{ʔ}^{\text{h}}\text{-t}$ (cf. *wíʔes* ‘disable someone’)
 disable~PDEN-CHAR
 ‘hard, difficult’ (L. C. Thompson & Thompson, 1996, p.379,
 Thompson River Salish)

Property-denoting C₁C₂ reduplication occurs as an infix following the first syllable. I mark property-denoting C₁C₂ reduplication as an infix based the fact

that the first syllable is stressed; this is unlike the plural/pluractional forms in the previous section where stress typically falls on the second syllable, suggesting that plural/pluractional reduplication is best marked as a prefix.²¹

Forms with property-denoting C₁C₂ reduplication appear to be adjectives, though they typically are formed from verb roots.

A.2.7.4 Thompson River Salish Diminutive C₁ Reduplication

Diminutive C₁ reduplication in Thompson River Salish copies the consonant before the stressed vowel; the copied consonant is placed after the stressed vowel. Examples are given in (517). Sonorants may also be glottalised, as shown in (517b,d); see Luntzlara (in prep) for discussion of glottalisation.

- (517) a. *č̣ye~ȳ* (cf. *č̣ye* ‘basket’)
 basket~DIM
 ‘little basket’ (Jimmie, 1994, p.75, Thompson River Salish)
- b. *qemú<ṁ>t* (cf. *qemút* ‘hat’)
 hat<DIM>
 ‘small hat’ (Jimmie, 1994, p.73, Thompson River Salish)
- c. *ti(y-e)-t-é<t>s* (cf. *ti(ye)tés* ‘praise s.o.’)
 praise-(DRV)-CTR-3ERG<DIM>
 ‘flatter s.o.’ (L. C. Thompson & Thompson, 1996, p.335, Thompson River Salish)
- d. *s-nú<ṁ>ye?* (cf. *snúye?* ‘money’)
 NMLZ-money<DIM>
 ‘a small amount of money’ (Jimmie, 1994, p.73, Thompson River Salish)

If the consonant following the stressed vowel is /ʔ/ or /h/, a vowel is also copied, as shown in (518).

²¹While stress generally does not fall on the initial syllable in C₁C₂ reduplicated words, there are some plural/pluractional forms (e.g., 514a-b) that resemble property-denoting C₁C₂ reduplicated forms because they have stress on the first syllable (L. C. Thompson & Thompson, 1992, p.84).

- (518) a. k^wesú<ṣu>ʔ (cf. k^waʃo(h) ‘pig’)²²
 pig<DIM>
 ‘little pig’ (Jimmie, 1994, p.76, Thompson River Salish)
- b. ká<ka>h (cf. ka(h) ‘car’)
 car<DIM>
 ‘little car’ (Jimmie, 1994, p.76, Thompson River Salish)
- c. xé<xē>ʔ (cf. xeʔ ‘high’)
 high<DIM>
 ‘a little high’ (Jimmie, 1994, p.75, Thompson River Salish)

C₁ reduplication in Thompson River Salish marks a small or young noun, as shown in (517a–b) and 518a–b) above; it can also refer to a small amount with a noun, as shown in (517d). With adjectives, diminutive reduplication can refer to a reduced or small amount or a difference between two measures (i.e., a diminutive comparative adjective), as shown in (519) (and 518c above). It can also mark a small argument of a verb, as shown in (520) or a diminutive event, as shown in (521). With stative marking, it can mark a partial state or that the state holds of a small argument or area, as shown in (522).

- (519) a. c<ʔ>é<ʔ>k (cf. cʔek ‘get cool’)
 cool<INCH><DIM>
 ‘get a little cooler’ (Jimmie, 1994, p.76, Thompson River Salish)
- b. ʔes-lə<ḷ>ṣ ~ ʔes-lə<ḷ>s (cf. ʔeslás ~ ʔeslṣ ‘low’)
 STAT-low<DIM>
 ‘a little lower’ (L. C. Thompson & Thompson, 1996, p.139, Thompson River Salish)
- c. x^wy-ekst-é<t>ṁ (cf. xyekst ‘do s.t. fast’)
 fast-hand-MID<DIM>
 ‘do s.t. a little faster. (L. C. Thompson & Thompson, 1992, p.105, Thompson River Salish)

²²This is a retracting root with variable effect (i.e., [u] ~ [o] - see (L. C. Thompson & Thompson, 1996, p.119).

- (520) a. ʔaʔx-á<ʔ>ns (cf. ʔaʔxáns ‘(a person) eats’)
eat-teeth<DIM>
‘a little person eats’ (Jimmie, 1994, p.76, Thompson River Salish)
- b. sú<ʂ>qʷ-m (cf. sóqʷ-m ‘he butchers (deer)’)
butcher<DIM>-MID
‘(little child helps) butcher (deer)’ (L. C. Thompson & Thompson, 1992, p.92, Thompson River Salish)
- (521) a. te<ʈ>kʈ (cf. stekʈ ‘rain’)
rain<DIM>
‘sprinkle, rain lightly’ (L. C. Thompson & Thompson, 1996, p.344, Thompson River Salish)
- (522) a. ʔes-wlǎ<ǎ>qʷ (cf. ʔəswlóqʷ ‘cleared (of brush)’)
STAT-clear.land<DIM>
‘somewhat cleared (of brush)’ (L. C. Thompson & Thompson, 1996, p.701, Thompson River Salish)
- b. ʔes-ʔu<ʈ>qʷ (cf. ʔesʈóqʷe ‘(undergrowth) is clear now’)
STAT-clear<DIM>
‘a small clear, open spot [in brush]’ (L. C. Thompson & Thompson, 1996, p.701, Thompson River Salish)

A.2.7.5 Thompson River Salish Affective C₁ Reduplication

“Affective” C₁ reduplication copies the first consonant of the root, with a [ə] between the matching consonants. Examples are given in (523).

- (523) a. tə~təz-eʔ (root: tʂ-)
AFF~arrowhead-FMV
‘arrow head’ (Jimmie, 1994, p.139, Thompson River Salish)
- b. cə~cp-ʔkʰ (cf. nɕpəp ‘come to fit in’)
AFF~fit.in-back
‘sun goes down over mountain (sunset)’ (Jimmie, 1994, p.84, Thompson River Salish)

- c. $x\dot{\alpha} \sim x\dot{w}\acute{e}l$ (cf. $x\dot{w}e\dot{l}$ ‘road’)
AFF~road
‘trail’ (Jimmie, 1994, p.84, Thompson River Salish)
- d. $n\text{-}\dot{\alpha} \sim c\acute{e}w\dot{s}$ (root: $ce\dot{w}s$)
LOC-AFF~fatigued
‘exhausted, tired out’ (Jimmie, 1994, p.84, Thompson River Salish)

While the label “affective” comes from L. C. Thompson and Thompson (1992), the meaning associated with this type of reduplication is unclear. They suggest that the affective C_1 reduplication may have developed from diminutive reduplication; however, unlike diminutive C_1 reduplication, affective C_1 reduplication is not sensitive to stress.²³ Affective C_1 reduplication does not appear to be restricted to a specific lexical category.

A.2.7.6 Thompson River Salish Change-of-State C_2 Reduplication

The consonant following the stressed vowel is doubled in change-of-state C_2 reduplication and [ə] intervenes. Examples are given in (524).²⁴

- (524) a. $\dot{x}^w\acute{u}s \sim \dot{\alpha}s$ (cf. $\dot{x}ust$ ‘frothy, soapy’)
foam~COS
‘It foams on its own.’ (Jimmie, 1994, p.79, Thompson River Salish)
- b. $\dot{c}\acute{e}l \sim \dot{\alpha}l$ (cf. $n\dot{c}\acute{\alpha}l$ ‘cold water’)
cold~COS
‘cooled off’ (Jimmie, 1994, p.79, Thompson River Salish)
- c. $t\acute{u}p \sim \dot{\alpha}p\text{-s-ne}$ (cf. $\dot{?}est\acute{u}p$ ‘broken into pieces’)
smash~COS-CAUS-1SG.ERG
‘I smash it up accidentally.’ (Jimmie, 1994, p.79, Thompson River Salish)

²³Affective C_1 reduplication resembles initial C_1 reduplication in Lillooet, which marks plurality (see Section A.2.5.6).

²⁴The vowel in (524h-i) in the source is [u], but this is /ə/ with subsequent rounding (due to its position between two rounded consonants).

- d. $\acute{c}\acute{?}\acute{í}k<\acute{\text{ə}}k>\acute{n}$ (cf. $\acute{?}es(n)\acute{c}\acute{?}íkn$ ‘wrapped (up)’)
 lay-back<COS>
 ‘be or get wrapped (with difficulty or accidentally, or by outside agency’ (L. C. Thompson & Thompson, 1996, p.47, Thompson River Salish)
- e. $n\acute{é}k\sim\acute{\text{ə}}k$ (cf. $n\acute{é}kes$ ‘change/transform s.t.’)
 change~COS
 ‘It changes.’ (B. Carlson & Thompson, 1981, p.113, Thompson River Salish)
- f. $sux^w\sim\acute{\text{ə}}x^w$ (cf. sux^wtx^w ‘You recognize them (sg.).’)
 recognize~COS
 ‘They (sg.) were accidentally recognized.’ (B. Carlson & Thompson, 1981, p.112, Thompson River Salish)
- g. $pik^w\sim\acute{\text{ə}}k^w$ (cf. $\acute{?}espik^w$ ‘sectioned, sliced’)
 slice~COS
 ‘It comes to pieces.’ (B. Carlson & Thompson, 1981, p.112, Thompson River Salish)

C₂ reduplication in Thompson River Salish is attested with verbs; Kroeber (1988) states that C₂ reduplication may combine with stative and eventive roots. Change-of-state reduplication marks something that is “out of control” (L. C. Thompson & Thompson, 1992), or under “limited control” (Jimmie, 1994).

A.3 Central Salish Reduplication

A.3.1 Comox-Sliammon Reduplication

Attested reduplication in Comox-Sliammon is summarised in Table A.8.

	C_1C_2	C_1	C_2	V_1
PL	✓			
PDEN	✓			
DIM		✓		
PL		✓		
PROG		✓		
COS			✓	
PL				✓

Table A.8: Reduplication in Comox-Sliammon

A.3.1.1 Comox-Sliammon Sources and Transcription

Examples come from Blake (2000), Watanabe (2003), J. Davis (2020), and my fieldwork. Unreduplicated forms come from the original sources or my fieldwork. I do not mark stress in Comox-Sliammon because it is predictable (i.e., fixed-initial stress). There are numerous other examples of reduplication in Comox-Sliammon in the case studies in Section 4.1 and Section 5.5.

Decisions about which segments are associated with a reduplicative morpheme, such as whether a reduplicative morpheme surfaces as a prefix or infix, are guided by the phonological analysis of Comox-Sliammon reduplication provided in Section 4.1.

A.3.1.2 Comox-Sliammon Plural/Pluractional C_1C_2 Reduplication

Plural/pluractional C_1C_2 reduplication copies the initial two consonants of the root with an intervening (epenthetic) [ə], as shown in (525).

- (525) a. **k^wəs~k^wusən** (cf. *k^wusən* ‘star’)
 PL~star
 ‘stars’ (Watanabe, 2003, p.372, Comox-Sliammon)
- b. **čəy~čuý** (cf. *čuý* ‘child’)
 PL~child
 ‘children’ (Watanabe, 2003, p.327, Comox-Sliammon)
- c. **ṭən~ṭin** (cf. *ṭin* ‘barbecued fish’)
 PL~barbecued.fish
 ‘lots of barbecued fish’ (Watanabe, 2003, p.372, Comox-Sliammon)
- d. **həs~has-am** (cf. *hasam* ‘sneeze’)
 PL~sneeze-MID
 ‘sneeze all the time’ (Watanabe, 2003, p.373, Comox-Sliammon)
- e. **ʔət~ʔutq^wu** (cf. *ʔutq^wu* ‘dig clams’)
 PL~dig.clams
 ‘always be digging clams’ (Watanabe, 2003, p.373, Comox-Sliammon)
- f. **təq~təq** (cf. *təqtčx^w* ‘close it’)
 PL~close
 ‘all of them closed’ (Watanabe, 2003, p.373, Comox-Sliammon)
- g. hu=čat **ʔəm~ʔim-aš-uł** (cf. *ʔiməš* ‘walk’)
 go=2PL.SUBJ PL~walk-INTR-PST
 ‘We went walking (around).’ (JF, Comox-Sliammon)
- h. **t^θək^w~t^θək^w-t-an** tiʔi (cf. *t^θək^wtčan* ‘I wiped it.’)
 PL~wipe-CTR-1SG.ERG DEM
 ‘I’m wiping it all over.’ (PD, Comox-Sliammon)

C₁C₂ reduplication in Comox-Sliammon marks plural entities with nouns, as shown in (525a–c) above. With verbs, such as those in (525d–h), C₁C₂ reduplication marks a spatiotemporal distribution with (Huijsmans & Mellesmoen, 2021). Pluractional C₁C₂ reduplication requires distribution in both time and space: it is not sufficient to distribute events over multiple times or or spaces/participants, as shown in in (526) and (527), respectively.²⁵

²⁵The [a] in (527) may involve a process of ablaut (see Section 5.5.6.1 and Huijsmans & Mellesmoen, 2021, p.351).

- (526) a. **təq**~təq-t-an məmkiyustən (cf. *təqtčx^w* ‘close it’)
 PL~close-CTR-1 SG.ERG window
 ‘I closed the windows (around the house).’ (PD, Comox-Sliammon)
- b. # **təq**~təq-t=č tə=məmkiyustən
 PL~close-CTR=1 SG.SUBJ DET=window
 Intended: ‘I repeatedly closed the (one) window.’ (EP, Comox-Sliammon)
- (527) a. *Context: A girl is jumping from rock to rock to cross a river.*
k^wat^h~k^wit^h-im (cf. *k^wit^him* ‘jump’)
 PL~jump-MID
 ‘She is jumping across.’ (EP/FL, Comox-Sliammon)
- b. *Context: A bunch of students jumping for a graduation photo. Everyone jumps at the same time, just once.*
 # **k^wat^h**~k^wit^h-im
 PL~jump-MID
 Intended: ‘They jumped (at the same time).’ (Huijsmans & Mellesmoen, 2021, p.351, Comox-Sliammon)

C₁C₂ reduplication can also mark plurality with adjectives, as shown in (528).²⁶

- (528) a. ^huk^w **łəq**~łəqt yəw~yawup (cf. *łəqt* ‘long’)
 all PL~long PL~fabric
 ‘The (two) pieces of fabric are both long.’ (JF, Comox-Sliammon)
- b. hihiw **tih**~tih tə=təm~tumiš (cf. *tih* ‘big’)
 really PL~big DET=PL~man
 ‘Those men are really big.’ (BW, Comox-Sliammon)

A.3.1.3 Comox-Sliammon Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication copies the first two consonants of the root are copied. The forms given in (529a–c) forms would distinguishable from the plural/pluractional forms when the root has a sonorant in C₂ position, because only

²⁶Example (528b) has [i] in the reduplicant: this is due to restrictions related to coda /h/ and schwa before a coda glottal consonant.

the first rather than the second sonorant is glottalized (cf. 525b in the previous section).

- (529) a. **təw̃**~tiwš (*tiwšam* ‘learn’)
 PDEN~smart
 ‘fast learner, smart’ (Watanabe, 2003, p.398, Comox-Sliammon)
- b. **šəṁ**~šam (cf. *šamat* ‘dry it’)
 PDEN~dry
 ‘dry’ (Watanabe, 2003, p.398, Comox-Sliammon)
- c. **q^wəỵ**~q^way (cf. *q^way* ‘talk’)
 PDEN~talk
 ‘talkative’ (Watanabe, 2003, p.398, Comox-Sliammon)
- d. **łəx̣**~łəx̣-tən (cf. *łəx̣* ‘bad’)
 PDEN~bad-instrument
 ‘mischievous, misbehaved’ (WJF/MV, Comox-Sliammon)
- e. **łəx̣**~łəx̣-ṵim (cf. *łəx̣* ‘bad’)
 PDEN~bad-susceptible
 ‘clumsy’ (WJF/MV, Comox-Sliammon)
- f. **gəṭ^θ**~gəṭ^θ (root: *gəṭ^θ*- ‘tease’)
 PDEN~tease
 ‘someone always teasing’ (FL/KG, Comox-Sliammon)

Property-denoting reduplication is recorded with verb roots, such as *q^way*- ‘talk’, and adjective roots, such as *łəx̣*- ‘bad’. Huijsmans (2023b, p.184) states that “words with [property-denoting] reduplication are [...] difficult to categorize” with respect to lexical category. Though they typically are translated as adjectives, she shows that that *gəṭ^θgəṭ^θ* ‘someone always teasing’ “patterns as a verb, since it is able to precede and follow a noun it modifies” (Huijsmans, 2023b, p.184). Further work is needed to determine the distribution of property-denoting reduplication.

A.3.1.4 Comox-Sliammon Diminutive C₁ Reduplication

Diminutive C₁ reduplication copies the first consonant of the root, as shown in (530a–d). A fixed segment /i/ is found between the two matching consonants

with roots that surface with schwa as the first vowel when they occur without reduplication, as shown in (530e–f). The fixed segment /i/ is also used with monosyllabic roots with a moraic vowel, as shown in (531).

- (530) a. su<s>payu (cf. *supayu* ‘axe’)
 axe<DIM>
 ‘small ax’ (Watanabe, 2003, p.385, Comox-Sliammon)
- b. ʔa<ʔ>yaʔ (cf. *ʔayaʔ* ‘house’)
 house<DIM>
 ‘small house’ (Watanabe, 2003, p.385, Comox-Sliammon)
- c. ʔan-aθ ga šu<š>k^wa (cf. *šuk^wa* ‘sugar’)
 give-CTR+1SG.OBJ IMP sugar<DIM>
 ‘Give me a little sugar.’ (FL, Comox-Sliammon)
- d. ʔiʔtan-uʔ=č la<l>sup (cf. *lasup* ‘soup’)
 eat-PST=1SG.SUBJ soup<DIM>
 ‘I ate a little bit of soup.’ (RT/FL, Comox-Sliammon)
- e. **mi**~mθ[i]k^w (cf. *məθk^w* ‘blackcap berry’)
 DIM~blackcap
 ‘small blackcap berry’ (Watanabe, 2003, p.386, Comox-Sliammon)
- f. **qi**~q̣snaʔ (cf. *q̣snaʔ* ‘shirt’)
 DIM~shirt
 ‘small shirt, dress’ (Watanabe, 2003, p.386, Comox-Sliammon)
- (531) a. **ʔi**~ʔax^w (cf. *ʔax^w* ‘snow’)
 DIM~snow
 ‘It is snowing a little bit.’ (Watanabe, 2003, p.386, Comox-Sliammon)
- b. **pi**~puk^w (cf. *puk^w* ‘book’)
 DIM~book
 ‘small book’ (Watanabe, 2003, p.386, Comox-Sliammon)

Diminutive C₁ reduplication is found with nouns, verbs, and adjectives. With nouns, C₁ diminutive reduplication marks a small size with count nouns, as shown in (530a–b,e–f) and (531b) above; with a mass noun, it marks small amount, as shown in (530c–d). With adjectives, diminutive reduplication can mark a small

amount of a property or that there is a small difference between two degrees (i.e., a diminutive comparison), or that the property exceeds a threshold by a small amount. Examples of the three diminutive readings available for adjectives are given in (532).²⁷

- (532) a. *hihiw qi~qaʔtəm tə=k^wan~k^win-at-an* (cf. *qaʔtəm* ‘heavy’)
 really DIM~heavy DET=PL~carry-TR-1 SG.ERG
 ‘The ones (bags) that I’m carrying are a little heavy.’ (BW, Comox-Sliammon)
- b. *qi~qaʔtəm tiʔi θu tita*
 DIM~heavy DEM go DEM
 ‘This one (bag) is heavier than that one.’ (BW, Comox-Sliammon)
- c. *qi~qaʔtəm tiʔi. hi-sx^w-səm Daniel qəh-t*
 DIM~heavy DEM it’s-CAUS-FUT Daniel lift-CTR
 ‘This one is a little too heavy (for me). Let Daniel lift it.’ (BW, Comox-Sliammon)

With verbs, diminutive reduplication indicates that an event is diminished or reduced in a particular way, as shown in (531a) and (533a), or that it involves a small degree of change or progress toward completion, as shown in (533b–c).

- (533) a. *ṣa<x>s-əm* (root: *ṣas-*)
 rustle<DIM>-MID
 ‘It is rustling a little bit.’ (FL, Comox-Sliammon)
- b. *ti<t>h-at=čx^w* (cf. *tih* ‘big’)
 big<DIM>-CTR=2SG.SBJ
 ‘Make it a little bigger.’ (FL, Comox-Sliammon)
- c. *θa<θ>p-at-as mimaw-s* (cf. *θapiš* ‘bathe’)
 bathe<DIM>-CTR-3ERG cat-3POSS
 ‘She bathed the cat a little bit (only part of it).’ (FL, Comox-Sliammon)

²⁷The fact that the Ci- reduplicant is used for *qaʔtəm* ‘heavy’ suggests that [a] may not be the underlying vowel; this may be a root without a vowel, with [a] surfacing instead of [ə] before a coda glottal stop.

A.3.1.5 Comox-Sliammon Plural C₁ Reduplication

Plural C₁ reduplication is almost identical to diminutive C₁ reduplication in shape. The first consonant of the root is copied, as shown in (534). The vowel between the matching consonants is [a], rather than the diminutive [i], if the root is weak or the word is monosyllabic, as shown in (535).

- (534) a. ?a<ʔ>mut (cf. ?amut ‘be home’)
be.home<PL>
‘They are all home.’ (Watanabe, 2003, p.376, Comox-Sliammon)
- b. hu<h>ǰ-igis (cf. hujǰigis ‘They (sg.) are dressed up.’)
finish<C₁PL>-body
‘They were all dressed up’ (Watanabe, 2003, p.376, Comox-Sliammon)
- xi<x>m-us (cf. ximus ‘scratched face’)
scratch<PL>-face
‘be scratched on the face (with more than one scratch)’ (Watanabe, 2003, p. 376, Comox-Sliammon)
- (535) a. x^wa~x^ws-awus (root: x^ws- ‘black’)
PL~black-eye
‘black eyes’ (Watanabe, 2003, p.378, Comox-Sliammon)
- b. ča~čpχ (cf. čəpχ ‘dirty’)
PL~dirty
‘They are all dirty.’ (Watanabe, 2003, p.377, Comox-Sliammon)

Plural C₁ reduplication only applies to states (Watanabe, 2003); it combines with adjectives and stative-marked verbs. The examples in (536) show that the state may hold of plural participants or be spatially distributed (e.g., apply to different parts or spots on a single object).²⁸ A singular argument is sufficient for the felicitous use of this type of reduplication if the state is distributed in space.²⁹

²⁸Plural C₁ reduplication differs from pluractional C₁C₂ reduplication because C₁ reduplication does not require a temporal distribution (see Huijsmans & Mellesmoen, 2021).

²⁹This type of reduplication is not used for contiguous states that hold of the full object. Further elicitation is needed to determine whether overlap in space, and to what extent, is permitted with C₁ reduplication.

- (536) a. **ʔa**~ʔp<i>x^w (root: ʔp^w- ‘break’)
 PL~break<STV>
 ‘They (multiple) are broken.’ (EP/FL)
 ‘It (one) is broken up.’ (BW, Comox-Sliammon)
- b. **pa**~p^w-it (root: p^w- ‘rip’)
 PL~rip-STV
 ‘multiple are ripped’ (JF)
 ‘multiple rips (on same jacket)’ (JF, Comox-Sliammon)

A.3.1.6 Comox-Sliammon Progressive C₁ Reduplication

Progressive aspect in Comox-Sliammon is marked by C₁ reduplication. If the root has a moraic vowel, then it is copied along with the first consonant, as shown in (537).

- (537) a. **ʔa**~ʔa^w-at (cf. ʔa^wat ‘chase it/them (sg.).’)
 PROG~chase-CTR
 ‘be chasing it/them (sg.)’ (Watanabe, 2003, p.392, Comox-Sliammon)
- b. **θa**~θa^w-iš (cf. θa^wiš ‘bathe’)
 PROG~bathe-TR
 ‘be bathing’ (Watanabe, 2003, p.392, Comox-Sliammon)
- c. **ti**~ti^w-um (cf. ti^wum ‘disembark’)
 PROG~disembark-MID
 ‘be disembarking’ (Watanabe, 2003, p.392, Comox-Sliammon)
- d. **q^wu**~q^wum-ut (cf. q^wumut ‘put in mouth’)
 PROG~root-CTR
 ‘be putting it in mouth’ (Watanabe, 2003, p.392, Comox-Sliammon)

If the corresponding perfective form would have [ə] as its first vowel (i.e., it has no underlying vowel), then only a consonant is copied, as shown in (538).

- (538) a. pə<**p**>n-aʔam (cf. pənaš ‘bury/plant it’)
 plant<PROG>-A.INTR
 ‘be burying/planting’ (Watanabe, 2003, p.393, Comox-Sliammon)

- b. tə<t>m-aš (cf. *təmaš* ‘wish for it’)
wish<PROG>-TR
‘be wishing for it’ (Watanabe, 2003, p.393, Comox-Sliammon)
- c. pə<p>λš (cf. *pəλš* ‘surface’)
surface<PROG>
‘surfacing’ (Watanabe, 2003, p.393, Comox-Sliammon)

An additional [ə] is epenthesized when the root has a CəCC shape and the second consonant is a sonorant, as shown in (539).

- (539) a. nə<n>əlx^w-at (cf. *nəlx^wat* ‘stir it’)
stir<PROG>-CTR
‘be stirring it’ (Watanabe, 2003, p.393, Comox-Sliammon)
- b. pə<p>əlg-at (cf. *pəlgat* ‘roll it up’)
roll<PROG>-CTR
‘be rolling it up’ (Watanabe, 2003, p.393, Comox-Sliammon)

Progressive C₁ reduplication is only found with verbs; it marks action that are ongoing (J. Davis, 2019; Watanabe, 2003).³⁰ Further examples of progressive forms are provided in (540).

- (540) a. θi~θiq-əx^w-an tə=qawθ
PROG~dig-NTR-1SG.ERG DET=potato
‘I am digging all of the potatoes (still digging, but some are already dug up).’ (cf. *θiqəx^wan* ‘I dug it up.’) (Watanabe, 2003, p.209, Comox-Sliammon)
- b. hu~huj-əx^w-an (cf. *hujəx^wan* ‘I finished it.’)
PROG~finish-NTR-1SG.ERG
‘I am finishing it off (e.g., my work).’ (Watanabe, 2003, p.209, Comox-Sliammon)
- c. qə<q>h-əx^w-as (cf. *qəhiytut* ‘It has been lifted up.’)
lift<PROG>-NTR-3ERG
‘They (sg.) are lifting it up slowly.’ (Watanabe, 2003, p.210, Comox-Sliammon)

³⁰Watanabe (2003) uses the label “imperfective” for progressive reduplication.

A.3.1.7 Comox-Sliammon Change-of-State C₂ Reduplication

The second consonant of the root and the preceding vowel is copied in change-of-state C₂ reduplication, as shown in (541).³¹

- (541) a. $\dot{\lambda}<\mathbf{ax}>\mathbf{ax}$ (cf. $\dot{\lambda}axay$ ‘old person’)
old<COS>
‘get old’ (Watanabe, 2003, p.397, Comox-Sliammon)
- b. $p<\mathbf{əs}>\mathbf{əs}$ (cf. $pəs$ ‘numb’)
numb<COS>
‘get numb’ (Watanabe, 2003, p.397, Comox-Sliammon)
- c. $t<\mathbf{ih}>\mathbf{ih}$ (cf. tih ‘big’)
big<COS>
‘get big’ (Watanabe, 2003, p.397, Comox-Sliammon)
- d. $\dot{p}<\mathbf{uq}^w>\mathbf{uq}^w$ (cf. $\dot{p}uq^w$ ‘moldy’)
mold<COS>
‘get moldy’ (Watanabe, 2003, p.397, Comox-Sliammon)
- e. $\dot{\lambda}<\mathbf{iq}>\mathbf{iqiw}$ (cf. $\dot{\lambda}iqiw$ ‘dark’)
dark<COS>
‘get dark’ (Watanabe, 2003, p.397, Comox-Sliammon)
- f. $p<\mathbf{əł}>\mathbf{əłt}$ (cf. $pəłtsx^w$ ‘Make it thick!’)
thick<COS>
‘getting thick’ (Watanabe, 2003, p.397, Comox-Sliammon)

C₂ reduplication is used to derive a change of state from adjectives (Huijsmans, 2023b; Kroeber, 1988); it has been called an “inceptive” by Kroeber (1988) and an “inchoative” by Watanabe (2003).

A.3.1.8 Comox-Sliammon V₁ Reduplication

Plural/pluractional V₁ reduplication only occurs in combination with other types of reduplication. V₁ reduplication copies a root vowel; [ʔ] provides an onset to break up hiatus, as shown in (542). If the unreduplicated counterpart of a word with V₁

³¹I segment these as CV<Cə>C rather than CVC~əC following the analysis in Chapter 4 (Section 4.1.5.1).

reduplication has [ə] as its first vowel, then the vowel will be replaced with (and copied as) [a] (or [i] when combined with diminutive reduplication).

- (542) a. $\check{j} < \mathbf{a?} > a\check{j}a?$ (cf. $\check{j}a? \check{j}a$ ‘relative’)
 relative<PL>
 ‘relatives’ (Blake, 2000, p.279, Comox-Sliammon)
- b. $\check{j} < \mathbf{a?} > a\check{\lambda}$ (cf. $\check{j}\check{\lambda}$ ‘run’)
 run<PL>
 ‘several running together’ (J. Davis, 2020, p.145, Comox-Sliammon)

While V_1 reduplication typically occurs in combination with another type of reduplication, it has a unique semantic contribution (plurality/pluractionality) and can be combined with different reduplicative morphemes, as discussed in Section 4.1.5.2. It marks plural entities with nouns, as shown in (543a–c), and plural events with verbs, as shown in (543d–f). Plural events marked by V_1 reduplication are distributed in time (Mellesmoen, 2018a; Mellesmoen et al., 2020), which allows them to be used to mean “over and over”, “one by one”, and “on and off”.

- (543) a. $\check{c} < \mathbf{i?} > i < \check{c} > kin[i]s-u\check{\lambda}$ (cf. $\check{c}ikins$ ‘chicken’)
 chicken<PL><DIM>-young
 ‘small chickens’ (Watanabe, 2003, p.401, Comox-Sliammon)
- b. $m < \mathbf{i?} > i \sim m?in$ (cf. $mi?in$ ‘carrot’)
 DIM<PL>~carrot
 ‘small carrots’ (Watanabe, 2003, p.401, Comox-Sliammon)
- c. $n < \mathbf{a?} > a < n > x^w i\check{\lambda}$ (cf. $n\check{x}^w i\check{\lambda}$ ‘canoe’)
 canoe<PL>< C_1 >
 ‘many boats together’ (J. Davis, 2020, p.145, Comox-Sliammon)
- d. $q < \mathbf{i?} > i \sim qpa?am$ (cf. $q\check{a}p?am$ ‘touch (s.t.)’)
 PROG<PL>~touch-A.INTR
 ‘touch repeatedly’ (Watanabe, 2003, p.403, Comox-Sliammon)
- e. $y < \mathbf{i?} > iym-a?am$ (cf. $y\check{a}mt$ ‘kick it’)
 kick<PL><DIM>-A.INTR
 ‘playing soccer’ (J. Davis, 2020, p.145, Comox-Sliammon)

- f. $q^w < \mathbf{u?} > u \sim q^w u-t$ (cf. $q^w u?$ 'get water')
 PROG < PL > $\sim$ get.water-CTR
 'drinking it' (Watanabe, 2003, p.403, Comox-Sliammon)

A.3.2 Halkomelem Reduplication

Attested reduplication in Halkomelem is summarised in Table A.9.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
IPFV		✓		
DIM		✓		
Other	Unclear C ₁ Reduplication			

Table A.9: Reduplication in Halkomelem

A.3.2.1 Halkomelem Sources and Transcription

Examples come from Galloway (1977), Leslie (1979), Jones (1976), Hukari and Peter (1995), Suttles (2004), J. Thompson (2012), and Mellesmoen and Urbanczyk (2021a). Unreduplicated forms come from a range of sources, including the original sources for the example, Galloway (2009), and Suttles (2004). I use [e] where Galloway (1977) uses [ɛ] for Upriver Halkomelem.

Each example is marked for dialect. Cross-dialect differences in glottalisation with reduplication reflect general phonological differences that are unrelated to reduplication; Upriver Halkomelem differs from the other two dialects because it has lost coda glottal stops and has no glottalised sonorants (Elmendorf & Suttles, 1960; J. Thompson, 2005b).

More in-depth discussion of Halkomelem with additional data can be found in Chapter 5 (Section 5.3) and Chapter 6 (Section 6.7.1.2).

A.3.2.2 Halkomelem Plural/Pluractional C₁C₂ Reduplication

The first two consonants of the root are copied in C₁C₂ plural/pluractional reduplication in Halkomelem, as shown in (544). The copied consonants have [ə] between them and stress falls on the second syllable of the word. I mark this type of reduplication as a prefix due to the position of stress.

- (544) a. **ʔəm**~ʔíməθ (cf. *ʔíməθ* ‘grandchild’)
 PL~grandchild
 ‘grandchildren’ (Leslie, 1979, p.49, Island Halkomelem)
- b. **ʔəm**~ʔí:məθ (cf. *ʔí:məθ* ‘grandchild’)
 PL~grandchild
 ‘grandchildren’ (Galloway, 1977, p.108, Upriver Halkomelem)
- c. s-**ʔən**~ʔéniʔ (cf. *sʔéniʔ* ‘woman’)
 NMLZ-PL~woman
 ‘women’ (Leslie, 1979, p.49, Island Halkomelem)
- d. s-**ʔən**~ʔénəy (cf. *sʔénəy* ‘female, woman’)
 NMLZ-PL~woman
 ‘females, women’ (Suttles, 2004, p.209, Downriver Halkomelem)
- e. **cəl**~céləš (cf. *céləš* ‘hand’)
 PL~hand
 ‘hands’ (Leslie, 1979, p.9, Island Halkomelem)
- f. **səl**~sílə (cf. *sílə* ‘grandparent’)
 PL~grandparent
 ‘grandparents’ (Leslie, 1979, p.10, Island Halkomelem)
- g. **səl**~sí:lə (cf. *sí:lə* ‘grandparent’)
 PL~grandparent
 ‘grandparents’ (Galloway, 1977, p.108, Upriver Halkomelem)
- h. **mən**~mént (cf. *ment* ‘rock’)
 PL~rock
 ‘rocks’ (Leslie, 1979, p.9, Island Halkomelem)
- i. s-**təl**~tá:ləs (cf. *stá:ləs* ‘wife’)
 NMLZ-PL~wife
 ‘wives’ (Galloway, 1977, p.108, Upriver Halkomelem)
- j. s-**təl**~táləw (cf. *stáləw* ‘river’)
 NMLZ-PL~river
 ‘rivers’ (Suttles, 2004, p.209, Downriver Halkomelem)

- k. **θəm**~θámən (cf. *θámən* ‘eyebrow’)
 PL~eyebrow
 ‘eyebrows’ (Suttles, 2004, p.189, Downriver Halkomelem)

C₁C₂ reduplication marks plurality when applied to nouns, as shown in (544) above. C₁C₂ reduplication on verbs marks plurality of a subject (545), object (546), or that the event happens multiple times (547). Suttles (2004, pp.178–181) observes that an iterative use of C₁C₂ reduplication has a different stress pattern in Downriver Halkomelem: stress falls on the first syllable when reduplication marks multiple events, as in (547c–d), but the second syllable with plural entities or arguments. When C₁C₂ reduplication receives an iterative interpretation in Downriver Halkomelem, it has the same form as property-denoting reduplication, which discussed in the following section.

- (545) a. **təl**~tíl-əm (cf. *tíləm* ‘sing’)
 PL~sing-MID
 ‘They sing.’ (Leslie, 1979, p.49, Island Halkomelem)
- b. **nəq**~nəq-əm (cf. *nəqəm* ‘dive’)
 PL~dive-MID
 ‘several dive’ (Suttles, 2004, p.170, Downriver Halkomelem)
- c. **ləm**~lí:m (cf. *lí:m* ‘pick’)
 PL~pick
 ‘lots of people picking (fruit, leaves’ (Galloway, 1977, p.309, Upriver Halkomelem)
- (546) a. **qəp**~qəp-ət (cf. *qəpət* ‘gather it’)
 PL~gather-CTR
 ‘gather them’ (Leslie, 1979, p.49, Island Halkomelem)
- b. **θək^w**~θək^w-t (cf. *θək^wət* ‘pull it taut’)
 PL~pull.taut-CTR
 ‘pull several taut’ (Suttles, 2004, p.170, Downriver Halkomelem)

- b. q^wəl~q^wəl (cf. q^wal ‘say, speak’)
talk~PDEN
‘talkative’ (Jones, 1976, p.23, Island Halkomelem)
- c. qə́n~qə́n (cf. qen ‘steal’)
steal~PDEN
‘thief, likely to steal’ (Suttles, 2004, p.179, Downriver Halkomelem)
- d. tám~təm (cf. táməθət ‘be wary, be suspicious’)
wary~PDEN
‘cautious, wary’ (Suttles, 2004, p.179, Downriver Halkomelem)
- e. tək^w~tək^w (cf. tək^w ‘stuck, get stuck’)
stuck~PDEN
‘muddy’ (Hukari & Peter, 1995, p.158, Island Halkomelem)
- f. sə́q~sə́q (cf. səq ‘get split, get torn, get cracked’)
split~PDEN
‘easy to split, easy to tear’ (Suttles, 2004, p.179, Downriver Halkomelem)
- g. mə́t~mə́t (cf. mə́tət ‘bend it’)
bend~PDEN
‘limber’ (Suttles, 2004, p.179, Downriver Halkomelem)

Suttles (2004, p.179) labels property-denoting reduplication as “dispositional” and states “it indicates that there is an inclination or propensity toward the action of quality expressed by the root”. Property-denoting reduplication applies to verbal roots; it derives a state or property.

A.3.2.4 Halkomelem Imperfective C₁ Reduplication

Imperfective C₁ reduplication copies the initial consonant of the root, with [ə] following the second of the matching consonants. Non-initial sonorant segments in the word undergo glottalisation in Downriver Halkomelem and Island Halkomelem; this excludes sonorants in “prefixes, stem-initial positions, before a stressed vowel, and certain other positions”, according to Hukari (1978, p.206).³²

³²I have added glottalisation to the example in (549g) to be consistent with the form given in Hukari (1978).

See Section 5.3.1.1 for further discussion of this imperfective reduplication and related allomorphy in Halkomelem.³³

- (549) a. **łá~łək^w** (cf. *łak^w* ‘fly’)
 IPFV~fly
 ‘be flying’ (Suttles, 2004, p.139, Downriver Halkomelem)
- b. **łí~təq^w** (cf. *łíq^wət* ‘bump it’)
 IPFV~bump
 ‘be bumping’ (Suttles, 2004, p.139, Downriver Halkomelem)
- c. **yé~yəq** (cf. *yəq* ‘topple’)
 IPFV~topple
 ‘be toppling’ (Suttles, 2004, p.140, Downriver Halkomelem)
- d. **yí~yəq** (cf. *yíq* ‘snow’)
 IPFV~snow
 ‘be snowing’ (Suttles, 2004, p.140, Downriver Halkomelem)
- e. **łé~łəw** (cf. *łew* ‘flee’)
 IPFV~flee
 ‘be fleeing’ (Suttles, 2004, p.140, Downriver Halkomelem)
- f. **s-łé~pəł^θ** (cf. *pəł^θət* ‘sew it’)
 STAT-IPFV~sew
 ‘sewed’ (Suttles, 2004, p.182, Downriver Halkomelem)
- g. **łé~łəłəm** (cf. *łéłəm* ‘salt, salty’)
 IPFV~salt
 ‘salty, taste salty’ (Suttles, 2004, p.139, Downriver Halkomelem)
- h. *təli?* *k^wə=nə-s-mi* *k^wən-nəx^w* *k^wə=nə-šqíləs* *łá*
 from DET=my-NMLZ-come be.taken-TR DET=my-sense BE3P
nə-s-wə-k^wé~k^wəc-ət *tə=xəθí:lə* **mí~mələ**
 my-NMLZ-EST-IPFV~see-TR DET=four.person IPFV~dance.possessed
 ‘From when I first came to my senses, I used to see four dancers.’
 (cf. *k^wəcnəx^w* ‘see it’/ *míle?* ‘dance possessed’) (Suttles, 2004, p.138,

³³In Section 5.3.1.1, I analyse this reduplication as a copy of the first CV sequence of the root with subsequent reduction of an unstressed vowel to [ə]. I mark imperfective C₁ as a prefix here following this analysis.

Downriver Halkomelem)

- i. ʔəy-stəx^w=cən k^wə=nə=s ʔi
 good-CAUS+3OBJ=1SG.SUBJ COMP=1SG.POSS=NOM AUX
 ti~təl-əm
 IPFV~sing-MID
 'I like singing.' (cf. *tíləm* 'sing') (J. Thompson, 2012, p.155, Island Halkomelem)
- j. *tí*~təl-əm (cf. *tí:ləm* 'sing')
 IPFV~sing-MID
 'singing' (Galloway, 1977, p.102, Upriver Halkomelem)
- k. *tá*~təq^w-əm (cf. *tá:q^wəm* 'to cough')
 IPFV~cough-MID
 'coughing' (Galloway, 1977, p.53, Upriver Halkomelem)

If the unreduplicated form begins with a sonorant followed by [ə], the initial sonorant of the imperfective word undergoes debuccalisation to [h] following reduplication, as shown in (550). Note that there is no [ə] following the second consonant in these forms.

- (550) a. hə<*n̥*>qəm̐ (cf. *nəqəm* 'dive')
 dive<IPFV>
 'diving' (Leslie, 1979, p.46, Island Halkomelem)
- b. hə<*n̥*>q^w (cf. *nəq^w* 'fall asleep')
 fall.asleep<IPFV>
 'falling asleep' (Leslie, 1979, p.46, Island Halkomelem)
- c. hə<*m̐*>qət (cf. *məqət* 'to swallow')
 swallow<IPFV>-CTR
 'swallowing' (Galloway, 1977, p.60, Upriver Halkomelem)

C₁ reduplication in Halkomelem marks imperfective aspect (Hukari, 1984).³⁴
 C₁ reduplication is one of a number of non-concatenative processes that mark

³⁴It is an imperfective, despite being labeled "progressive" in Suttles (2004), "continuative" in Galloway (1977), and "actual" in Leslie (1979) and Jones (1976).

imperfective aspect; the others are discussed in Chapter 5 (Section 5.3.1.1.1). This type of reduplication occurs with verbs.

A.3.2.5 Halkomelem Diminutive C₁ Reduplication

Diminutive C₁ reduplication copies the initial consonant of the root. When the root has a moraic vowel as its first (or only) vowel, the form of the diminutive is -C-; the copied consonant is placed after the first vowel, as shown in (551).

- (551) a. pú<**p**>t (cf. *put* ‘boat’)
 boat<DIM>
 ‘little boat’ (Suttles, 2004, p.212, Downriver Halkomelem)
- b. k^wá<**k^w**>x^waʔ (cf. *k^wáx^waʔ* ‘box’)
 box<DIM>
 ‘little box’ (Suttles, 2004, p.213, Downriver Halkomelem)
- c. pu<**p**>s (cf. *pus* ‘cat’)
 cat<DIM>
 ‘little cat, kitten’ (Suttles, 2004, p.212, Downriver Halkomelem)
- d. xá<**x**>ce (cf. *xá:ce* ‘lake’)
 lake<DIM>
 ‘pond, small lake’ (Galloway, 1977, p.59, Upriver Halkomelem)
- e. s-ɬé<**ɬ**>niʔ (cf. *sɬéniʔ* ‘woman’)
 NMLZ-woman<DIM>
 ‘little girl’ (Leslie, 1979, p.52, Island Halkomelem)

When the first vowel in the unreduplicated word is [ə] or the word starts with a complex onset, reduplication takes the form Ci- instead of -C-; examples are given in (552).

- (552) a. p^hí~pəwəy (cf. *pəwəy* ‘flounder’)
 DIM~flounder
 ‘little flounder’ (Suttles, 2004, p.213, Downriver Halkomelem)
- b. θí~θqət (cf. *θəqét* ‘tree’)
 DIM~tree
 ‘little tree’ (Galloway, 1977, p.59, Upriver Halkomelem)

- c. s-**k^wí**~k^wəl (cf. *sk^wəl* ‘waterfall’)
NMLZ-DIM~waterfall
‘small waterfall’ (Galloway, 1977, p.111, Upriver Halkomelem)
- d. **ǵí**~ǵcəs (cf. *ǵəcəs* ‘island’)
DIM~island
‘little island’ (Galloway, 1977, p.59, Upriver Halkomelem)
- e. **ǵí**~ǵcs (cf. *ǵces* ‘island’)
DIM~island
‘islet’ (Suttles, 2004, p.532, Downriver Halkomelem)
- f. **číʔ**~čsəy (cf. *čsey* ‘Douglas fir’)
DIM~douglas.fir
‘little Douglas fir’ (Suttles, 2004, p.214, Downriver Halkomelem)

Glottalisation often co-occurs with diminutive reduplication in the Island and Downriver dialects of Halkomelem. It may be realized as a glottal stop (553a), or as glottalisation of a sonorant (553b–d).

- (553)
- a. s-**k^wíʔ**~k^wθeʔ (cf. *k^wθeʔ* ‘island’)
NMLZ-DIM~island
‘little island’ (Leslie, 1979, p.53, Island Halkomelem)
 - b. **mí**~mqsən (cf. *máqsən* ‘nose’)
DIM~nose
‘little nose’ (Suttles, 2004, p.213, Downriver Halkomelem)
 - c. s-**mí**~myəθ (cf. *smáyəθ* ‘deer’)
NMLZ-DIM~deer
‘fawn, little deer’ (Suttles, 2004, p.213, Downriver Halkomelem)
 - d. s-**ní**~nx^wəʔ (cf. *snáx^wəʔ* ‘canoe’)
NMLZ-DIM~canoe
‘little canoe’ (Leslie, 1979, p.53, Island Halkomelem)

The diminutive in Halkomelem is used with nouns to reflect smallness of a referent, as in (551–553) above, or to express endearment, as shown in (554).

- (554) a. **mí~mñe?** (cf. *mənə* ‘child’)
 DIM~child
 ‘dear child’ (Jones, 1976, p.22, Island Halkomelem)
- b. **lí~ləm** (root: *ləm-* ‘house’)
 DIM~house
 ‘small/dear house’ (Jones, 1976, p.22, Island Halkomelem)

Diminutive C₁ reduplication can also be used with verbs, as shown in (555). Diminutive verbs are always imperfective (Suttles, 2004, p.172). It is not clear why there are no perfective diminutive forms.

- (555) a. **xi?~x<é>pɣ-t** (cf. *xəpɣt* ‘scatter it’)
 DIM~sprinkle<IPFV>-CTR
 ‘sprinkling it (dim.)’ (Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.172, Island Halkomelem)
- b. **hi?~hó<l>ç-t** (cf. *ləç* ‘full’)
 DIM~fill<IPFV>-CTR
 ‘filling it (dim.)’ (Hukari & Peter, 1995, in Mellesmoen & Urbanczyk, 2021a, p.172, Island Halkomelem)
- c. **ti~tés-t** (cf. *tsət* ‘approach it’)
 DIM~approach+IPFV-CTR
 ‘be approaching it (dim.)’ (Suttles, 2004, p.173, Downriver Halkomelem)

Diminutive verbs in Halkomelem are typically associated with a small participant, as shown in (556a–c); they may also be associated with a diminished event, as shown in (556d–e).

- (556) a. **ʔi~ʔexíθ** (cf. *ʔéxəθ* ‘lie down’)
 DIM~lie.down
 ‘little baby lying down’ (Galloway, 1977, p.313, Upriver Halkomelem)
- b. **tə~tí<ʔ><tə>l-əm** (cf. *tíləm* ‘sing’)
 DIM~sing<DIM><PROG>-MID
 ‘A little one is singing.’ (Suttles, 2004, p.172, Downriver Halkomelem)

- c. **k^wi**~k^wə~k^wc-é:m (cf. *k^wécəm* ‘scream’)
 DIM~IPFV~scream-MID
 ‘scream a little, squeal’ (Galloway, 1977, p.313, Upriver Halkomelem)
- d. **hi**~hí:m (cf. *hi:m* ‘picking’)
 DIM~pick+IPFV
 ‘picking a little (bit)’ (Galloway, 1977, p.313, Upriver Halkomelem)

Suttles (2004, p.222) finds a small number of diminutive adjectives, which are used to refer to a small entity that possesses the quality denoted by the root, as shown in (557).

- (557) a. **xé**<**?xə**>ws (cf. *xəws* ‘new’)
 new<DIM>
 ‘new little (one)’ (Suttles, 2004, p.222, Downriver Halkomelem)
- b. **xi**~xəlá^w (cf. *xəlá^w* ‘round’)
 round<DIM>
 ‘round little (one)’ (Suttles, 2004, p.222, Downriver Halkomelem)

A.3.2.6 Other Reduplication in Halkomelem

An additional C₁ reduplication process marks plurality. It is marginal and limited to particular lexical items (see, e.g., Leslie, 1979, p.50); its status across the dialects is unclear. The examples in (558) show that stressed [ə] is replaced with a full vowel with this type of reduplication.

- (558) a. **qə**<**qí**>ləm (cf. *qələm* ‘eye’)
 PL~eye
 ‘eyes’ (Suttles, 2004, p.208, Downriver Halkomelem)
- b. **s-xə**<**xí**>nə (cf. *sxənə* ‘leg’)
 NMLZ-PL~leg
 ‘legs’ (Leslie, 1979, p.50, Island Halkomelem)

A.3.3 Lushootseed Reduplication

Attested reduplication in Lushootseed is summarised in Table A.10.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
DIM		✓		
PL		✓		
COS			✓	
PL				✓

Table A.10: Reduplication in Lushootseed

A.3.3.1 Lushootseed Sources and Transcription

Examples come from Tweddell (1950), Hess (1967), Kirkham (1991), Bates et al. (1994), and Beck and Hess (2014). I draw on Bates et al. (1994) for unreduplicated forms.

Stress is missing from some examples in original sources; I have chosen to leave these as given in the source material. Vowels from Tweddell (1950) have been standardised to be consistency with the other examples. Note that /b/ and /d/ pattern as sonorants in Lushootseed. The sound written [d^z] is a voiced alveolar affricate.

A.3.3.2 Lushootseed Plural/Pluractional C₁C₂ Reduplication

The first CVC sequence of a root is copied in plural/pluractional reduplication. There is a faithful copy of the vowel in the unreduplicated form, regardless of whether it is [ə] or a moraic vowel, as shown in (559) and (560), respectively.

- (559) a. **ǰəs**~ǰəsəd (cf. *ǰəsəd* ‘foot’)
 PL~foot
 ‘feet’ (Bates et al., 1994, p.117, Lushootseed)
- b. s-**čət**~čətx^wəd (cf. *sčət^wəd* ‘bear’)
 NMLZ-PL~bear
 ‘black bears all about’ (Hess, 1967, p.65, Lushootseed)

- c. **dʰəx̣**~dʰəx̣ (cf. *ʔudʰəxtəd* ‘move it’)
PL~move
‘move household’ (Bates et al., 1994, p.90, Lushootseed)
- d. **čəgʷ**~čəgʷás (cf. *čəgʷás* ‘wife’)
PL~wife
‘seeking a woman to marry’ (Bates et al., 1994, p.63, Lushootseed)
- e. huy dxʷ-**ləgʷ**~ləgʷəb (cf. *ləgʷəb* ‘youth’)
SCONJ DIST-PL~youth
‘They were youths.’ (Beck & Hess, 2014, p.35, Lushootseed)
- f. gʷəl ʔas-bəs-s-**qʷəb**~qʷəbayʔ əlgʷəʔ ʔə tə bə-saliʔ
CONJ STAT-PROP-NMLZ-PL~dog PL PREP NSPEC ADD-two
‘And they have two dogs.’ (cf. *sqʷəbáyʔ* ‘dog’) (Beck & Hess, 2014, p.35, Lushootseed)
- g. ʔəs-**təq̣**~təq̣ (cf. *təq̣əd* ‘patch it’)
STAT-PL~patch
‘patched with mass of patches’ (Tweddell, 1950, p.15, Lushootseed)
- h. **gʷəd**~gʷəd-ɪl (cf. *gʷədɪl* ‘sit down’)
PL~sit.down-INCH
‘They are all sitting about.’ (Hess, 1967, p.65, Lushootseed)
- (560) a. **pás**~pastəd (cf. *pástəd* ‘white people’)
PL~white.people
‘many white folks’ (Bates et al., 1994, p.160, Lushootseed)
- b. **sáqʷ**~saq̣ʷ (cf. *ʔusáqʷ* ‘It flew.’)
PL~fly
‘fly here and there’ (Bates et al., 1994, p.200, Lushootseed)
- c. **síd**~sidq̣-cut (cf. *sídq̣əd* ‘turn it’)
PL~turn-REFL
‘spin self around’ (Bates et al., 1994, p.203, Lushootseed)
- d. **ʔib**~ʔibəš (cf. *ʔibəš* ‘walk or journey over land’)
PL~walk
‘walk all about’ (Bates et al., 1994, p.15, Lushootseed)

- e. $\text{ʔu-túq}^w \sim \text{tuq}^w\text{-ub}$ (cf. $\text{túq}^w\text{ub}$ ‘cough’)
 PFV-PL~cough-MID
 ‘They (sg.) coughed and coughed.’ (Tweddell, 1950, p.16, Lushootseed)
- f. $\text{sáx}^w \sim \text{sax}^w\text{-əb}$ (cf. $\text{sáx}^w\text{əb}$ ‘jump, leap, run’)
 PL~run-MID
 ‘running all over’ (Hess, 1967, p.65, Lushootseed)
- g. $\text{cút} \sim \text{cut}$ (cf. cut ‘say, talk’)
 PL~say
 ‘Several are talking.’
 ‘One person talking or saying something repeatedly.’ (Bates et al., 1994, as cited in Kye, 2023, p.279, Lushootseed)

C_1C_2 reduplication marks plural entities when applied to nouns; it marks plural events with verbs. Events with C_1C_2 reduplication may be temporally distributed (e.g., 560e) or spatially distributed (e.g., 560f). Kye (2023, p.278–279) states that pluractionality may “denote actions that are carried out repeatedly (or many times) or the action is done (or distributed) across many individuals”.

A.3.3.3 Lushootseed Diminutive C_1 Reduplication

Diminutive C_1 reduplication in Lushootseed copies the initial consonant of a root; the following vowel may also be copied in roots with a full vowel, as shown in (561). There is also a set of lexical items which do not copy a vowel and surface as an infix, such as those given in (562). The choice between CV- or -C- is lexically specified (see Mellesmoen & Urbanczyk, 2024). Glottalisation, which is typically realised as a glottal stop, may accompany C_1 diminutive reduplication, as shown in (562).

- (561) a. $\text{ʔú} \sim \text{ʔuq}^w\text{-ud}$ (cf. $\text{ʔúq}^w\text{ud}$ ‘pull it out’)
 DIM~pull.out-CTR
 ‘pull it part way out’ (Bates et al., 1994, p.22, Lushootseed)

- b. **hí**~hiw-il (cf. *híwil* ‘go ahead’)
 DIM~go.ahead-INCH
 ‘go on ahead a bit’ (Bates et al., 1994, p.111, Lushootseed)
- c. **qí**~qix^w (cf. *dx^w qix^w* ‘located upstream’)
 DIM~upstream
 ‘a little upstream’ (Bates et al., 1994, p.187, Lushootseed)
- d. **x^wí**~x^wit-il (cf. *x^wítíl* ‘fall’)
 DIM~fall-INCH
 ‘small item(s) fall(s) from a height’ (Hess, 1967, p.62, Lushootseed)
- (562) a. **hú<?í>p** (cf. *hup* ‘early’)
 morning<DIM>
 ‘a little early’ (Hess, 1967, p.63, Lushootseed)
- b. **sá<?s>x^wəb** (cf. *sáx^wəb* ‘jump, leap, run’)
 run<DIM>-MID
 ‘run a few steps’ (Hess, 1967, p.62, Lushootseed)
- c. **sá<?s>q^w** (cf. *?usáq^w* ‘It flew.’)
 fly<DIM>
 ‘fly just a bit’ (Hess, 1967, p.62, Lushootseed)

Epenthetic [i] is inserted between the matching consonants when the unreduplicated word has [ə] (563), a long vowel (564), or begins with a consonant cluster (565).

- (563) a. **ǰí**~ǰəsəd (cf. *ǰəsəd* ‘foot’)
 DIM~foot
 ‘little foot’ (Bates et al., 1994, p.117, Lushootseed)
- b. **tí**~təd^z-il (cf. *təd^zíl* ‘go to bed’)
 DIM~go.to.bed-INCH
 ‘lie down for a little while’ (Bates et al., 1994, p.220, Lushootseed)
- c. **bí**~bəč (cf. *bəčád* ‘set smth down’)
 DIM~fall
 ‘drop in from time to time’ (Bates et al., 1994, p.35, Lushootseed)

- d. s-**k^w**í~k^wəbšəd (cf. *sk^wəbšəd* ‘animal hide’)
NMLZ-DIM~animal.hide
‘small hide’ (Bates et al., 1994, p.123, Lushootseed)
- e. **tí**~təlàw-il (cf. *təláwil* ‘run’)
DIM-run-INCH
‘jog’ (Hess, 1967, p.62, Lushootseed)
- f. **pí**~pí^w (cf. *ʔupəq^w* ‘drifted’)
DIM~drift
‘small objects drift’
‘object drifts slowly’ (Hess, 1967, p.63, Lushootseed)
- g. **hí**?~íq^w (cf. *íəq^w* ‘wet’)
DIM-wet
‘damp’ (Hess, 1967, p.8, Lushootseed)
- (564) a. s-**dí**~du:k^w (cf. *sdu:k^w* ‘knife’)
NMLZ-DIM~knife
‘small knife’ (Bates et al., 1994, p.85, Lushootseed)
- b. **bí**?~bu:s (cf. *buz* ‘four’)
DIM~four
‘four little items’ (Bates et al., 1994, p.42, Lushootseed)
- c. **lí**?~lu:d (cf. *lud* ‘hear’)
DIM~hear+CTR
‘hear something a little’ (Bates et al., 1994, p.139, Lushootseed)
- (565) **čí**~čá? (cf. *čá?* ‘rock’)
DIM~rock
‘little rock’ (Bates et al., 1994, p.74, Lushootseed)

C₁ reduplication in Lushootseed is attested with a range of different lexical items across lexical categories (e.g., nouns, verbs, and adjectives). With nouns, it marks a small entity, as shown in (563a,d). With verbs, it can mark the smallness of an argument, as shown in (561d) and (563f), or a diminutive event; diminutive events include those that have short duration (e.g., 563b), involve a limited degree

of change (e.g., 561a), or are reduced in intensity (e.g., 563e–f). With adjectives, it marks a small amount of a state, as shown in (563g).

A.3.3.4 Lushootseed Plural C₁ Reduplication

Plural C₁ reduplication copies the first consonant of a root, as in diminutive reduplication, but the vowel [a] occurs between the two matching consonants, as shown in (566).

- (566) a. s-**da**~du:k^w (cf. *sdu:k^w* ‘knife’)
 NMLZ-PL~knife
 ‘knives’ (Kirkham, 1991, p.1, Lushootseed)
- b. **sa**~sax^w-əb (cf. *sáx^w* ‘jump, leap, run’)
 PL~run-MID
 ‘many run away’ (Kirkham, 1991, p.1, Lushootseed)
- c. **sa**~saq^w (cf. *ʔusáq^w* ‘It flew.’)
 PL~fly
 ‘flocks fly away’ (Kirkham, 1991, p.1, Lushootseed)

C₁ plural reduplication occurs with nouns and verbs. Kirkham (1991) and Kim (1992) claim that this type of C₁ reduplication marks a collective sense of plurality; Hess (1967, p.64) describes it as a plural morpheme that is used in contexts where “more than one perform an act simultaneously”.

A.3.3.5 Lushootseed C₂ Reduplication

C₂ reduplication in Lushootseed copies of the second consonant of the root and the preceding vowel, as shown in (567); stress falls on the first of the matching vowels.

- (567) a. *sáq^w~a^w* (cf. *ʔusáq^w* ‘It flew.’)
 fly~C₂
 ‘wheeling in sky’ (Hess, 1967, p.66, Lushootseed)
- b. *yúb~ub-il* (cf. *yúbil* ‘starve’)
 starve~C₂-INCH
 ‘run down, sickish’ (Hess, 1967, p.66, Lushootseed)

- c. $dáy \sim \mathbf{áy}$ (cf. *dáy* ‘only’)
only $\sim C_2$
‘alone’ (Hess, 1967, p.66, Lushootseed)
- d. $?át \sim \mathbf{at}$ (cf. *?at* ‘fast’)
fast $\sim C_2$
‘hurry up’ (Bates et al., 1994, p.7, Lushootseed)
- e. $d^z áq \sim \mathbf{aq}$ (cf. *d^z áqad* ‘fell a tree’)
fall $\sim C_2$
‘totter, stagger’ (Bates et al., 1994, p.88, Lushootseed)
- f. $sáx^w \sim \mathbf{ax}^w - \text{əb}$ (cf. *sáx^w əb* ‘jump, leap, run’)
run $\sim C_2$ -MID
‘scurrying about trying to get things done’ (Hess, 1967, p.66, Lushootseed)
- g. $g^w əd \sim \mathbf{əd} - \text{íl}$ (cf. *g^w ədíl* ‘sit down’)
sit.down $\sim C_2$ -INCH
‘sit for awhile’ (Hess, 1967, p.66, Lushootseed)
- h. $pəq^w \sim \mathbf{əq}^w$ (cf. *?upəq^w* ‘drifted’)
drift $\sim C_2$
‘drift up and down’ (Hess, 1967, p.66, Lushootseed)

The function of C_2 reduplication is unclear in many examples. The available descriptions and translations suggest a range of possible interpretations. This type of reduplication has previously been labeled as “restrictive” by Hess (1967). Hess (1967, p.66) further states that C_2 reduplication with verbs indicates that the “act stops short of completion”, or that it is somewhat aimless or otherwise all over the place. When C_2 reduplication occurs with nouns or independent pronouns, it can mark the “exclusiveness” or “aloneness” of the referent, as shown in (568).

- (568) a. $s-łád < \mathbf{ad} > əy$ (cf. *słádəy* ‘woman’)
NMLZ-woman $< C_2 >$
‘woman (living) alone’ (Hess, 1967, p.66, Lushootseed)

- b. $d\acute{f}b<\mathbf{ib}>\acute{o}t$ (cf. $d\acute{f}b\acute{o}t$ ‘we, us’)
1PL.INDEP<C₂>
‘we alone (we, us)’ (Hess, 1967, p.66, Lushootseed)
- c. $g^w\acute{e}l<\mathbf{el}>\acute{a}pu$ (cf. $g^w\acute{e}l\acute{a}pu$ ‘you pl.’)
2PL.INDEP<C₂>
‘just you folks’ (Hess, 1967, p.66, Lushootseed)
- d. $c\acute{a}d<\mathbf{ad}>\acute{f}t$ (cf. $c\acute{a}d\acute{f}t$ ‘they (sg.)’)
3.INDEP<C₂>
‘just the one who’ (Hess, 1967, p.66, Lushootseed)

A.3.3.6 Lushootseed V₁ Reduplication

V₁ reduplication doubles the root vowel, as shown in (569). I treat this as a type of reduplication, rather than lengthening, following Hess (1967, p.64); however, it is realised as a long vowel on the surface, given that sequences transcribed as VV in Lushootseed are realised as [V:]. If the unreduplicated word would have a [ə], an [a] is inserted and doubled, as shown in (569d–e).

- (569)
- a. $\acute{c}\acute{a}<\mathbf{a}>g^w\acute{e}s$ (cf. $\acute{c}\acute{e}g^w\acute{e}s$ ‘wife’)
wife<PL>
‘wives’ (Hess, 1967, p.65, Lushootseed)
 - b. $si<\mathbf{i}>?ab$ (cf. $si?ab$ ‘noble person’)
noble.person<PL>
‘noble people’ (Kirkham, 1991, p.242, Lushootseed)
 - c. $\acute{c}\acute{a}<\mathbf{a}>\acute{c}\acute{a}s$ (cf. $\acute{c}\acute{a}\acute{c}\acute{a}s$ ‘child’)
child<PL>
‘still a child’ (Kim, 1992, p.72, Lushootseed)
 - d. $k^wa<\mathbf{a}>l\acute{e}q$ (cf. $k^w\acute{e}l\acute{e}q$ ‘other things’)
?<PL>
‘other people’ (Kirkham, 1991, p.242, Lushootseed)
 - e. $\acute{c}\acute{a}<\mathbf{a}>di\acute{f}t$ (cf. $c\acute{a}d\acute{f}t$ ‘the one who’)
3.INDEP<PL>
‘the ones who’ (Hess, 1967, p.65, Lushootseed)

V₁ reduplication marks plural entities; however, there are some forms where the contribution is unclear, such as (569c) above.

A.3.4 Nooksack Reduplication

Attested reduplication in Nooksack is summarised in Table A.11.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
Other	Unclear C ₁ Reduplication			

Table A.11: Reduplication in Nooksack

A.3.4.1 Nooksack Sources and Transcription

Examples and unreduplicated forms come from Galloway (1984b). Galloway (1984b, 1993b) treats glottalised sonorants as sequences of sonorant and glottal stop; I retranscribe these as glottalised sonorants. I also replace "æ" with /a/ where it is a moraic vowel and /ə/ otherwise; this decision follows the one made for the morphology and syntax chapters by Galloway, Adams, Burton, Davis, and Trotter (in prep). However, it should be noted that this low vowel has a more front realisation than elsewhere in the family, though it has not become /e/ as in Halkomelem and Northern Straits (Galloway, 1988, p.298).

A.3.4.2 Nooksack Plural/Pluractional C₁C₂ Reduplication

Plural/pluractional C₁C₂ reduplication copies the first two consonants of the root and [ə] is epenthesized, as shown in (570).

- (570) a. s-**łən**~łánəy̓ (cf. łénəy̓ 'woman')
 NMLZ-PL~woman
 'women' (Galloway, 1984b, p.85, Nooksack)
- b. **pəč̣**~píč̣t (cf. píč̣t 'charcoal')
 PL~charcoal
 'lots of charcoal' (Galloway, 1984b, p.85, Nooksack)

- c. **ʔəm**~ʔímac (cf. *ʔímac* ‘grandchild’)
PL~grandchild
‘grandchildren’ (Galloway, 1984b, p.85, Nooksack)
- d. **təl**~táləw (cf. *táləw* ‘arm’)
PL~arm
‘both arms’ (Galloway, 1984b, p.85, Nooksack)
- e. **kəm**~kámič (cf. *kámič* ‘cabbage’)³⁵
PL~cabbage
‘cabbages’ (Galloway, 1984b, p.85, Nooksack)

C₁C₂ reduplication marks plural entities with nouns, as shown in the examples above; with verbs, C₁C₂ reduplication can mark plural participants or repeated events, as shown in (571) and (572), respectively.³⁶ The example in (572b) suggests that both plural argument and event interpretations are available for a word with C₁C₂ reduplication.

- (571) a. **nəy**~náy-əm (cf. *náyəm* ‘One person laughing.’)
PL~laugh-MID
‘people laughing’ (Galloway, 1984b, p.85, Nooksack)
- b. **ʔəm**~ʔím-əš (cf. *ʔíməš* ‘to walk’)
PL~walk-INTR
‘They (group of people) [are] walking.’ (Galloway, 1984b, p.85, Nooksack)
- (572) a. ʔay **təp**~túp-un-əm (cf. *ʔəwáčax^w túpuntsax^w* ‘Don’t hit me.’)
IPFV PL~hit-CTR-PASS
‘being hit (rep.)’ (Galloway, 1984b, p.85, Nooksack)

³⁵The [i] transcribed by Galloway (1984b) may actually be co-articulation between /ə/ and the following palatal.

³⁶The unreduplicated form given in (572a) comes from Galloway et al. (in prep).

- b. **pən**~pún-əx^w (cf. *punəx^w* ‘see someone’)
 PL~see-NTR
 ‘see them, see lots of times’ (Galloway, 1984b, p.85, Nooksack)

A.3.4.3 Nooksack Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication copies the first two segments of the root with an epenthetic [ə], as shown in (573).

- (573) a. **k^wəḵ**~k^wḵ (cf. *sk^wḵil* ‘kind of/partly red/pink’)
 PDEN~red
 ‘red’ (Galloway, 1984b, p.84, Nooksack)
- b. **q^wəc**~q^wíc (cf. *q^wic* ‘drift downstream’)
 PDEN~drift.downstream
 ‘drowned’ (Galloway, 1984b, p.84, Nooksack)
- c. **qəx**~qíx (cf. *qíxil* ‘kinda black’)
 PDEN~black
 ‘black’ (Galloway, 1984b, p.84, Nooksack)

Property-denoting C₁C₂ reduplication applies to adjectival and verbal roots. Words with this type of reduplication refer to either properties or states.

A.3.4.4 Nooksack Diminutive C₁ Reduplication

Diminutive C₁ reduplication copies the initial consonant of the root; the root vowel is copied as well if it is moraic, as shown in (574). For roots that surface with a consonant cluster or with [ə] as the first vowel, [i] separates the matching consonants, as shown in (575).

- (574) a. **yú<yū>məč** (cf. *yúməč* ‘spring salmon’)
 spring.salmon<DIM>
 ‘small spring salmon’ (Galloway, 1984b, p.83, Nooksack)
- b. **qá<qā>məý** (cf. *qáməý* ‘adolescent girl’)
 woman<DIM>
 ‘young girl (2 to 7 years)’ (Galloway, 1984b, p.83, Nooksack)

- c. ná<na>təx^w (cf. *snat* ‘night’)
morning<DIM>
‘early in the morning’ (Galloway, 1984b, p.83, Nooksack)
 - d. tá<ta>n (cf. *tan* ‘mother’)
mother<DIM>
‘mom, mother (affectionate)’ (Galloway, 1984b, p.83, Nooksack)
- (575)
- a. čí~cəqət (cf. *c(ə)qat* ‘tree’)
DIM~tree
‘a little tree’ (Galloway, 1984b, p.86, Nooksack)
 - b. s-q^wí~q^wəmay̓ (cf. *q^wəmay̓* ‘dog’)
NMLZ-DIM~dog
‘little dog’ (Galloway, 1984b, p.86, Nooksack)
 - c. čí~čq^wáy̓ (cf. *sčəq^wəy̓* ‘salmon’)
DIM~salmon
‘trout’ (Galloway, 1984b, p.86, Nooksack)

Diminutive nouns are associated with smallness, youth, and affection. Most of the forms provided by Galloway (1984b) are built on nominal roots, with a few exceptions involving verbs, such as those given in (576). From the limited data, it appears that a “small event” reading is possible for diminutive verbs. It is not clear if a small participant reading is possible.

- (576)
- a. ʔá<ʔa>xíc (cf. *ʔáxic* ‘lie down’)
lie.down<DIM>
‘lie down awhile (a little)’ (Galloway, 1984b, p.83, Nooksack)
 - b. tí<ti>č-im (cf. *tičim* ‘swim’)
swim<DIM>-MID
‘to swim around or (possibly) to swim a little’ (Galloway, 1984b, p.86, Nooksack)

A.3.4.5 Other Reduplication in Nooksack

There are a number of examples that appear to have C₁ reduplication with verbal roots which are not clearly diminutive, such as those shown in (577). Galloway

(1984b) labels these as “resultative/dispositional” or “durative”.

- (577) a. **ši**~š^ʷk^ʷú^ʷm (cf. Island Halkomelem šəš^ʷk^ʷə^ʷm ‘swim’)³⁷
 C₁~swim-A.INTR
 ‘swim for pleasure’ (Galloway, 1984b, p.85, Nooksack)
- b. pú<**pu**>n-əx^ʷ (cf. *punəx*^ʷ ‘see someone’)
 see<C₁>-NTR
 ‘see something/someone many times’ (Galloway, 1984b, p.83, Nooksack)

The source of these forms, is unclear. It is possible that they were either originally diminutive, reflect an aspectual C₁ reduplication process that was lost, are borrowed from nearby languages, or are pluractional.

There are also a handful of forms with C₁ reduplication that include [a], as shown in (578). It is likely that these have [ə], rather than [a].

- (578) a. **ma**~mú^ʷk^ʷ (cf. *muk*/*mək*^ʷ ‘all’)
 C₁~all
 ‘all (of subgroup)’ (Galloway, 1984b, p.85, Nooksack)
- b. **na**~nəč^ʷú^ʷ? (cf. *nəču?* ‘one’)
 DIM~one
 ‘one young one’ (Galloway, 1984b, p.85, Nooksack)

³⁷The unreduplicated form comes from (Kuipers, 2002, p.119). A cognate is found in Downriver Halkomelem: *xixk^ʷám* ‘be in swimming’ from *xək^ʷám* ‘bathe’ with a combination of C₁ imperfective reduplication and the durative marker -i- (Suttles, 2004, p.176).

A.3.5 Northern Straits and Klallam Reduplication

Given similarities in the reduplicative systems of the two Straits languages (Northern Straits and Klallam), I group them together in this dissertation. The choice to group them together is motivated by the fact that there is no clean distinction between the reduplication systems of Northern Straits and Klallam.

Attested reduplication in Northern Straits and Klallam is summarised in Table A.12.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
IPFV		✓		
Other	Unclear C ₁ Reduplication			

Table A.12: Reduplication in Northern Straits and Klallam

A.3.5.1 Northern Straits and Klallam Sources and Transcription

Examples come from Efrat (1969), Raffo (1972), Demers (1974), Demers and Jelinek (1984), Montler (1986), Galloway (1990), Turner (2010), Montler (2012), Montler (2015), and Montler (2018); unreduplicated forms come from one of these sources. Glottalised sonorant segments are written as [Cʔ] or [ʔC] in Raffo (1970) (Songish), Demers (1974) (Lummi), and at certain points in Galloway (1990) (Samish); I have retranscribed these as [Cʰ].

Additional discussion of Northern Straits and Klallam can be found in a section on moraic codas in Chapter 6 (6.6.2.2.2).

A.3.5.2 Northern Straits and Klallam Plural/Pluractional C₁C₂ Reduplication

C₁C₂ reduplication marks plurality and pluractionality in Northern Straits and Klallam, but is somewhat limited in distribution: it is found with roots with a sonorant in C₂ position, as shown in (579). A non-reduplicative allomorph of

the plural is used elsewhere (Montler, 2012, p.176).³⁸ In roots with a sonorant as the second consonant, the first and second consonants are copied and [ə] is epenthesized.

- (579) a. s-**łən**~łéniʔ (cf. *słéniʔ* ‘woman’)
NMLZ-PL-woman
‘women’ (Raffo, 1972, p.22, Songish - Northern Straits)
- b. **səl**~séləs (cf. *séləs* ‘hand’)
PL-hand
‘hands’ (Raffo, 1972, p.77, Songish - Northern Straits)
- c. **ʔəl**~ʔéləx^w (cf. *ʔéləx^w* ‘old person’)
PL~old.person
‘old people’ (Raffo, 1972, p.136, Songish - Northern Straits)
- d. **ŋən**~ŋénəʔ (cf. *ŋénəʔ* ‘son or daughter’)
PL~son/daughter
‘sons and daughters’ (Galloway, 1990, p.45, Samish - Northern Straits)
- e. s-**čəŋ**~čé:nəx^w (cf. *sčé:nəx^w* ‘fish’)
NMLZ-PL~fish
‘lots of fish’ (Galloway, 1990, p.45, Samish - Northern Straits)
- f. **łəl**~łéləw (cf. *łéləw* ‘arm’)
PL~arm
‘arms’ (Galloway, 1990, p.45, Samish - Northern Straits)
- g. **łəm**~łəməx^w-t-əŋ (cf. *łəməx^wtəŋ* ‘be rained on’)
PL~rain-CTR-PASS
‘get much rain’ (Montler, 2018, p.1264, Saanich - Northern Straits)
- h. s-**łən**~łéni (cf. *słéni* ‘woman’)
NMLZ-PL~woman
‘women’ (Demers & Jelinek, 1984, p.44, Lummi - Northern Straits)
- i. **ləm**~ləmətú (cf. *ləmətú* ‘sheep’)
PL-sheep

³⁸Montler (2015, p.179) estimates that C₁C₂ reduplication applies to 10% of lexical items in Klallam, reflecting the limited distribution of the reduplicative allomorph.

‘bunch of sheep’ (Montler, 2015, p.179, Klallam)

- j. s-**łən**~łániʔ (cf. *słániʔ* ‘woman’)

NMLZ-PL~woman

‘women’ (Montler, 2015, p.179, Klallam)

- k. nəx^w-s-**k^wən**~k^wánəŋət=caʔ=st (cf. *k^wánəŋətən* ‘I run.’)

LOC-NMLZ-PL~run=FUT=1PL.SUBJ

‘We’ll race.’ (Montler, 2015, p.92, Klallam)

The C₁C₂ reduplication process shown in (579) above marks plurality with nouns and plural arguments with verbs. This reading is restricted to words with sonorants in the C₂ position. Saanich and Samish diverge from Klallam and Songish because they allow plural C₁C₂ reduplication when a root has an obstruent in C₂ position, as shown in (580) and (581). This pattern (i.e., copying an obstruent in C₂ position) marks pluractionality; it therefore only applies to verbs. When an obstruent in C₂ position is copied, the events are distributed in time (i.e., repeated).³⁹

- (580) a. **nəq**~nəq-əŋ (cf. *nəqəŋ* ‘dive’)

PL~dive-MID

‘dive repeatedly’ (Montler, 2018, p.1301, Saanich - Northern Straits)

- b. **čəq^w**~čəq^w-t (cf. *čəq^wət* ‘burn it’)

PL~burn-CTR

‘turn on repeatedly’ (Montler, 2018, p.1490, Saanich - Northern Straits)

- (581) a. **nəq**~nəq-əŋ (cf. *nəqəŋ* ‘dive’)

PL~dive-MID

‘keeps diving repeatedly’ (Galloway, 1990, p.45, Samish - Northern Straits)

³⁹See Chapter 6 (Section 6.6.2.2.2) for further discussion of phonological restrictions related to which roots may undergo C₁C₂ reduplication.

- b. $\dot{t}^{\theta}\dot{\text{ə}}\dot{\text{q}} \sim \dot{t}^{\theta}\dot{\text{ə}}\dot{\text{q}}\text{-ə}\dot{\eta}/\dot{\text{c}}\dot{\text{ə}}\dot{\text{q}} \sim \dot{\text{c}}\dot{\text{ə}}\dot{\text{q}}\text{-ə}\dot{\eta}$ (cf. $\dot{t}^{\theta}\dot{\text{q}}\dot{\text{ə}}\eta/\dot{\text{c}}\dot{\text{q}}\dot{\text{ə}}\eta$ ‘drip’)
 PL~drip-MID
 ‘It’s dripping lots.’ (Galloway, 1990, p.45, Samish - Northern Straits)

A.3.5.3 Northern Straits and Klallam Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication copies the consonants in C₁ and C₂ position of the root, as in the plural forms discussed in the previous section. However, unlike with the plural forms, [ə] is in the second syllable with the property-denoting forms. This type of reduplication is not limited to roots with a sonorant in C₂ position. Examples are given in (582).

- (582) a. $\text{m}\dot{\text{ə}}\dot{\text{t}} \sim \text{m}\dot{\text{ə}}\dot{\text{t}}$ (cf. $\text{m}\dot{\text{ə}}\dot{\text{t}}\dot{\text{ə}}\eta$ ‘be bent’)
 bend~PDEN
 ‘It’s flexible’ (Montler, 1986, p.97, Saanich - Northern Straits)
- b. $\dot{\text{q}}^{\text{w}}\dot{\text{a}}\dot{\text{m}} \sim \dot{\text{q}}^{\text{w}}\dot{\text{ə}}\dot{\text{m}}$ (cf. $\dot{\text{q}}^{\text{w}}\dot{\text{a}}\dot{\text{m}}$ ‘strong’)
 strong~PDEN
 ‘very strong’ (Raffo, 1972, p.14, Songish - Northern Straits)
- c. $\text{s}\dot{\text{e}}\text{x}^{\text{w}} \sim \text{s}\dot{\text{ə}}\text{x}^{\text{w}} = \text{s}\text{x}^{\text{w}}$ (root: $\text{s}\dot{\text{e}}\text{x}^{\text{w}}$ -)
 lazy~PDEN=2SG.SUBJ
 ‘You’re really lazy.’ (Efrat, 1969, p.80, Sooke - Northern Straits)

In Klallam, [ə] may be deleted if both consonants are obstruents, as shown in (583).

- (583) a. $\dot{\text{c}}\dot{\text{ə}}\dot{\text{q}}^{\text{w}} \sim \dot{\text{c}}\dot{\text{q}}^{\text{w}}$ (cf. $\dot{\text{c}}\dot{\text{ə}}\dot{\text{q}}^{\text{w}}$ ‘dirt’)
 dirt~PDEN
 ‘dirty’ (Montler, 2015, p.286, Klallam)
- b. $\dot{\text{x}}\dot{\text{ə}}\dot{\text{q}}^{\text{w}} \sim \dot{\text{x}}\dot{\text{q}}^{\text{w}}$ (cf. $\dot{\text{x}}\dot{\text{ə}}\dot{\text{q}}^{\text{w}}$ ‘get stuck’)
 get.stuck~PDEN
 ‘sticky’ (Montler, 2015, p.286, Klallam)

A difference between the Songish and other dialects of Northern Straits, as well as Klallam, concerns the treatment of glottalised sonorants in property-denoting C₁C₂ reduplication: glottalised sonorants are copied faithfully in Klallam, Samish,

and Saanich, but the copy surfaces as a plain sonorant in Songish. Compare the Songish word for thief in (584a) with the Klallam word in (584b).

- (584) a. $q\acute{e}n \sim q\acute{e}n$ (cf. *qen* ‘to steal’)
steal ~ PDEN
‘thief, stealing continuously’ (Raffo, 1972, p.156, Songish)
- b. $q\acute{a}n \sim q\acute{a}n$ (cf. *qan* ‘rob’)
rob ~ PDEN
‘thief’ (Montler, 2015, p.286, Klallam)
- c. $q\acute{e}n \sim q\acute{e}n$ (cf. *qen* ‘to steal’)
steal ~ PDEN
‘thief’ (Galloway, 1990, p.7, Samish - Northern Straits)
- d. $q\acute{e}n \sim q\acute{e}n$ (cf. *qen* ‘steal’)
steal ~ PDEN
‘thief, robber’ (Montler, 2018, p.1324, Saanich - Northern Straits)
- e. $q^w\acute{e}l \sim q^w\acute{e}l$ (root: *q^wel-*)
talk ~ PDEN
‘talkative’ (Demers, 1974, p.19, Lummi - Northern Straits)

Property-denoting reduplication means something is “typical of” or “characteristic of” something, according to Montler (2015, p.287); it applies to eventive verbs and denotes a state or a property. It is not clear whether a word with property-denoting reduplication patterns as a verb, noun, or adjective.

A.3.5.4 Northern Straits and Klallam Diminutive C_1 Reduplication

The initial consonant of a root is copied in C_1 diminutive reduplication in Northern Straits and Klallam, as shown in (585); a glottal stop is also infixes following the stressed vowel in some forms, such as those in (585d–e). The vowel between the matching consonants is [ə].

- (585) a. $\acute{x}\acute{e} \sim \acute{x}\acute{e}y\acute{e}q\acute{e}s$ (cf. *xyeqes* ‘box’)
DIM ~ box
‘small box’ (Raffo, 1972, p.84, Songish - Northern Straits)

- b. s-**pə**~pəčáʔ (cf. *spəčáʔ* ‘berry basket’)
NMLZ-DIM~berry.basket
‘small berry basket’ (Raffo, 1972, p.84, Songish - Northern Straits)
- c. s-**tə**~táəkən (cf. *stáəkən* ‘sock’)
NMLZ-DIM~sock
‘little sock’ (Montler, 1986, p.42, Saanich - Northern Straits)
- d. **ʔə**~ʔá<ʔ>čeʔ (cf. *ʔáčeʔ* ‘lake’)
DIM~lake<DIM>
‘small lake’ (Galloway, 1990, p.46, Samish - Northern Straits)
- e. **pə**~p<á><ʔ>x^w-ən (cf. *páx^wən* ‘sail’)
DIM~blow<RES><DIM>-instrument
‘small sail’ (Montler, 1986, p.98, Saanich - Northern Straits)
- f. **tə**~tə́m̩x^w (cf. *tə́m̩x^w* ‘raining’)
DIM~rain
‘light rain’ (Montler, 2018, p.1264, Saanich - Northern Straits)
- g. s-**q^w**ə~q^wəmóy (cf. *sq^wəmóy* ‘dog’)
NMLZ-DIM~dog
‘little dog’ (Demers & Jelinek, 1984, p.44, Lummi - Northern Straits)

In some Songish diminutive forms, there is no [ə] recorded in the diminutive forms between the identical consonants, as shown in (586).

- (586) a. **ʔ̌**~ʔ̌čés (cf. *ʔ̌čés* ‘island’)
DIM~island
‘little island’ (Raffo, 1972, p.11, Songish - Northern Straits)
- b. **č̌**~č̌éʔwiʔ (cf. *č̌éʔwiʔ* ‘dish’)
DIM~dish
‘little dish’ (Raffo, 1972, p.162, Songish - Northern Straits)

Diminutive marking is found with nouns, verbs, and adjectives. With a noun, as in the examples above, diminutive C₁ reduplication marks something small in size. With a verb, diminutive reduplication may mark a small argument (587a–b), a reduced amount of an action (587c), or a small amount of change or progress (587d).

- (587) a. $\dot{p}\dot{a}\sim\dot{p}<\dot{a}l>\dot{a}\dot{\lambda}\sim\dot{a}\eta$ $t\theta\dot{a}$ $s\dot{\lambda}\dot{a}l\dot{i}\dot{\lambda}\dot{t}q\dot{a}\dot{t}$ (cf. $\dot{p}\dot{a}l\dot{a}\dot{\lambda}\dot{a}\eta$ ‘be smoking’)
 DIM~smoke<PL>-MID DET child
 ‘The kids are smoking.’ (Montler, 2018, p.14, Saanich - Northern Straits)
- b. $k^w\dot{a}\sim k^w\dot{a}n\sim\dot{i}\sim t$ (cf. $k^w\dot{a}n\dot{e}s$ ‘hold me’)
 DIM~take-DUR-TR
 ‘to hold something (a small subject or object)’ (Montler, 2018, p.77, Saanich - Northern Straits)
- c. $\dot{t}\dot{a}\sim\dot{t}\dot{e}\dot{?}\sim s\dot{a}t$ (cf. $\dot{t}\dot{e}\dot{?}stx^w$ ‘to give someone a taste’)
 DIM~try-REFL
 ‘to be trying (some)’ (Montler, 2018, p.117, Saanich - Northern Straits)
- d. $\check{c}\dot{a}\sim\check{c}\dot{x}^w\sim\dot{a}t\sim\dot{a}\eta$ $t\theta\dot{a}$ $\dot{?}\dot{a}yskl\dot{i}m$ ($s\check{c}\dot{a}\check{c}\dot{x}^w$ to be melted)
 DIM~melt-TR-PASS DET ice.cream
 ‘The ice cream is melted a little.’ (Montler, 2018, p.34, Saanich - Northern Straits)

When combined with an adjective, diminutive reduplication indicates either a small argument or a diminished amount of the quality; examples are given in (588).

- (588) a. $k^w\dot{a}\sim k^w\dot{a}t^{\dot{\theta}}\dot{e}\sim s\dot{a}t$ (cf. $k^w\dot{a}t^{\dot{\theta}}\dot{e}s\dot{a}t$ ‘ornery, crabby, hard to get along with’)
 DIM~pry-RFLX
 ‘to get ornery, crabby, difficult, hard to get along with (a little bit or of a little one)’ (Montler, 2018, p.78, Saanich - Northern Straits)
- b. $\dot{?}\dot{a}\sim\dot{?}i\dot{y}\dot{a}s$ (cf. $\dot{?}i\dot{y}\dot{a}s$ ‘pleasant’)
 DIM~pleasant
 ‘to be little and cute’ (Montler, 2018, p.139, Saanich - Northern Straits)
- c. $q\dot{a}\sim q\dot{e}\sim q\dot{a}\dot{t}$ (cf. $q\dot{e}q\dot{a}\dot{t}$ ‘sore’)
 DIM~IPFV~sore
 ‘to be sore (of a little one or a little sore)’ (Montler, 2018, p.259, Saanich - Northern Straits)
- d. $s\sim\check{c}\dot{a}\sim\check{c}\dot{a}nt^{\dot{\theta}}$ (cf. $s\check{c}\dot{a}nt^{\dot{\theta}}$ ‘to be tight’)
 STAT-DIM~tight
 ‘to be tight (of something small)’ (Montler, 2018, p.1202, Saanich - Northern Straits)

- e. s-mə~mé<ʔ>í^θən (cf. mət^θénəŋ ‘to act proud’)
 STAT-DIM~proud<DIM>
 ‘to be kind of proud, stuck up’ (Montler, 2018, p.1289, Saanich - Northern Straits)

A.3.5.5 Northern Straits and Klallam Imperfective C₁ Reduplication

A second type of C₁ reduplication pattern marks imperfective aspect in Northern Straits and Klallam.⁴⁰ The initial consonant of the root is copied, as shown in (589). I mark these as infixes based on the position of stress.

- (589) a. qé<q>ən (cf. sgeŋ ‘It’s stolen.’)
 steal<IPFV>
 ‘They (sg.) are stealing (something).’ (Montler, 1986, p.117, Saanich - Northern Straits)
- b. s-k^wú<k^w>əl=sən (cf. sk^wul ‘school’)
 STAT-school<IPFV>=1 SG.SUBJ
 ‘I’m going to school.’ (Montler, 1986, p.42, Saanich - Northern Straits)
- c. í^θé<í^θ>əʔ (cf. st^θəʔsənəŋ ‘steps’)
 on.top<IPFV>
 ‘They (sg.) are riding a horse (Montler, 1986, p.117, Saanich - Northern Straits)
- d. k^wí íá<í>əp=sən (cf. íəpəqθínəŋ ‘slurping’)
 REAL slurp<IPFV>=1 SG.SUBJ
 ‘I’m eating (soup).’ (Montler, 1986, p.117, Saanich - Northern Straits)
- e. šə<š>t-əŋ-ás-əŋ (cf. štəŋ/šətəŋ ‘walk’)
 walk<IPFV>-face-MID
 ‘They (sg.) are walking around.’ (Galloway, 1990, p.40, Samish - Northern Straits)

⁴⁰Other labels have been used to describe imperfective C₁ reduplication. For example, Montler (1986) uses the label “actual” for the Saanich dialect of Northern Straits, which is equivalent to what Galloway (1984b) calls “continuative” in his description of the Samish dialect.

- f. $\text{ʔú} < \text{ʔ} > \text{p} = \text{cn}$ (cf. *ʔup cn* ‘I eat soup.’)
 eat.soup <IPFV> = 1SG.SUBJ
 ‘I am eating soup.’ (Montler, 2015, p.227, Klallam)
- g. $\text{qá} < \text{qə} > \text{n} = \text{cn}$ (cf. *qán cn* ‘I steal.’)
 steal <IPFV> = 1SG.SUBJ
 ‘I am stealing.’ (Montler, 2015, p.227, Klallam)
- h. $\text{s-k}^{\text{w}} \text{é} < \text{k}^{\text{w}} > \text{əl}$ (cf. *k^wélət* ‘hide it’)
 STAT-hide <IPFV>
 ‘hide’ (Montler, 2018, p.1224, Saanich - Northern Straits)

A second allomorph of aspectual C₁ reduplication is given in (590). Stress falls on the second syllable; reduplication surfaces a Cə- prefix. Turner (2006, p.26) finds that this form with roots that surface with the shape CVCəC without reduplication.⁴¹

- (590) a. $\text{tə} \sim \text{téy} \text{əq}$ (cf. *téčəq* ‘to be angry’)
 IPFV ~ angry
 ‘being angry’ (Raffo, 1972, p.45, Songish - Northern Straits)
- b. $\text{s-θə} \sim \text{θímə}?$ (cf. *θímə?* ‘freeze’)
 STAT-IPFV ~ freeze
 ‘It’s frozen.’ (Montler, 1986, p.131, Saanich - Northern Straits)

Reduplication is just one way imperfective aspect is marked in Northern Straits and Klallam: there are a range of non-concatenative allomorphs including affixed (fixed segmental) content, the C₁ reduplication process, and metathesis. A discussion of this allomorphy is beyond the scope of the present dissertation (though see Section 5.3.1.1.1 for discussion of similar allomorphy in Halkomelem): see Montler (2015) for a more detailed explanation of the allomorphy in Klallam and Svenonius and Bye (2012) for an analysis of these patterns in Saanich.

Imperfective C₁ reduplication is found with verbs.

⁴¹As I assume [ə] is never present in the underlying form, I assume that this pattern applies to CVCC roots with a sonorant as one of the last two consonants, which will motivate [ə] epenthesis.

A.3.5.6 Other Reduplication in Northern Straits and Klallam

An additional type of C₁ reduplication is documented in Klallam, which Montler (2015, p.287) labels “affective”. He finds that it has a diminutive-like function, but is more focused on a notion of cuteness than a small size. Though the two appear to largely overlap in function, minimal pairs are attested: for example, (591a) is a diminutive form and (591b) is an “affective” form.

- (591) a. **ŋaʔ**~ŋáqsən (cf. *ŋéqsən* ‘nose’)
DIM~nose
‘small nose’ (Montler, 2012, p.286, Klallam)
- b. **ŋi**~ŋéqsən (cf. *ŋéqsən* ‘nose’)
AFF~nose
‘cute little nose’ (Montler, 2012, p.286, Klallam)

An additional type of C₁ reduplication with /i/ is attested with a few lexical items in Songish, Samish, and Lummi; Raffo (1972) states that it marks a “collective” plural. Examples are given in (592).

- (592) a. čə<**čí**>nəs (cf. *čénəs* ‘tooth’)
tooth<PL>
‘bunch of teeth’ (Raffo, 1972, p.11, Songish - Northern Straits)
- b. s-qə<**qí**>məl (cf. *sqéməl* ‘canoe paddle’)
NMLZ-canoe.paddle<PL>
‘lots of canoe paddles (not just 3)’ (Galloway, 1990, p.45, Samish - Northern Straits)
- c. čə<**čí**>nəs (cf. *ŋən čénəs* ‘a lot of teeth’)
tooth<PL>
‘all one’s teeth’ (Galloway, 1990, p.45, Samish - Northern Straits)
- d. č<**čí**>nəs (cf. *čénəs* ‘tooth’)
tooth<PL>
‘teeth’ (Demers & Jelinek, 1984, p.77, Lummi - Northern Straits)

A.3.6 Pentlatch Reduplication

Little is known about reduplication in Pentlatch. From existing documentation (Boas, 1886), it can be concluded that C_1 and C_1C_2 reduplication were present, as shown in Table A.13.

C_1C_2	C_1	C_2	V_1
✓	✓		

Table A.13: Reduplication in Pentlatch

While I discuss potential meanings of the reduplication in each section, there is not enough information to ascertain how many distinct types of reduplication were present (e.g., closely related languages Sechelt and Comox-Sliammon have 2 types of C_1C_2 reduplication and 3 types of C_1 reduplication).

A.3.6.1 Pentlatch Sources and Transcription

Examples come from Boas (1886). I include unreduplicated forms from other languages: Sechelt examples come from Beaumont (2011), Northern Straits from Montler (2018), Squamish from Squamish Nation Education Department (2011), Proto-Coast-Salish from Kuipers (2002), and Comox-Sliammon from fieldwork.

A.3.6.2 Pentlatch C_1C_2 Reduplication

Examples of C_1C_2 reduplication in Pentlatch are given in (593). The first two consonants of the root are copied and there is an epenthetic [ə], as shown in (593).⁴²

- (593) a. **qəl**~qálaḵ (cf. Sechelt *qalaḵay* ‘oceanspray bush, ironwood’)
 C_1C_2 ~ironwood
 ‘harpoon points’ (Boas, 1886, Pentlatch)

⁴²Ironwood is used in the creation of various tools traditionally, including harpoons; I assume that the Pentlatch word in (593a), which Boas (1886) translates as “harpoon points”, is related to the word for “ironwood” in Sechelt. It is not clear whether the first consonant in (593b) is a ʔ or ɬ , I have gone with ɬ based on Sechelt.

- b. $\dot{\lambda}\acute{a}\acute{c} \sim \dot{\lambda}\acute{a}\acute{c}$ (cf. Sechelt $\dot{\lambda}\acute{a}\acute{c}\dot{\lambda}\acute{a}\acute{c}$ ‘steady (boat, canoe)’)
 $C_1C_2 \sim$ steady
 ‘steady (boat)’ (Boas, 1886, Pentlatch)
- c. $q\acute{o}n \sim qan$ (cf. Saanich - Northern Straits $q\acute{e}nq\acute{o}n$ ‘thief’)
 $C_1C_2 \sim$ steal
 ‘steal’ (Boas, 1886, Pentlatch)
- d. $\dot{c}\acute{a}m \sim \dot{c}\acute{a}mq^w\dot{t}\text{-}am$ (cf. Sechelt $\dot{c}\acute{a}m\dot{c}\acute{a}mq^w\acute{o}lam$ ‘foggy’)
 $C_1C_2 \sim$ cloud-MID
 ‘fog’ (Boas, 1886, Pentlatch)
- e. $\dot{t}\acute{a}n \sim \dot{t}\acute{a}nay$ (cf. Sechelt $s\dot{t}\acute{a}nay$ ‘woman’)
 $C_1C_2 \sim$ woman
 ‘all women’ (Boas, 1886, Pentlatch)
- f. $ti \quad s\text{-}\dot{m}\acute{a}n \sim \dot{m}\acute{a}nit$ (cf. Sechelt $s\dot{m}\acute{a}nit$ ‘mountain’)
 $DET \text{ NMLZ-}C_1C_2 \sim$ mountain
 ‘the mountains’ (Boas, 1886, Pentlatch)
- g. $y\acute{a}m \sim y\acute{a}m\theta ut$ (cf. Sechelt $y\acute{a}m\acute{a}liw\acute{a}t$ ‘lose o.’s appetite for s.th.’)
 $C_1C_2 \sim$ sick
 ‘be sick all over’ (Boas, 1886, Pentlatch)

C_1C_2 reduplication occurs across different syntactic categories. Pentlatch appears to have the standard plural or pluractional function found across Salish languages in both the nominal and verbal domains (e.g., 593f and 593g); it also appears to have property-denoting C_1C_2 reduplication (e.g., 593b–c).

A.3.6.3 Pentlatch C_1 Reduplication

C_1 reduplication in Pentlatch might be expected to have similar reduplication to C_1 reduplication in Comox-Sliammon and Sechelt, which each have three types of C_1 reduplication (aspectual, diminutive, and plural). In a reconstruction of Proto-Salish aspect, Kinkade (1996) includes a C_1 reduplicative process in Pentlatch as a cognate to the imperfective in Sechelt (Beaumont, 1985), and the progressive in Comox-Sliammon (Watanabe, 2003).

There are examples of C_1 reduplication that have the initial consonant and a vowel copied that appear to mark imperfective aspect, as shown in (594). There

are also examples with just the consonant copied that appear to mark plurality or diminutivity, as shown in (595); note that the addition of /a/ in (595a) and /i/ in (595b) is comparable to patterns found in plural and diminutive C₁ reduplication in Comox-Sliammon and Sechelt.

- (594) a. **łú**~łuk^w (cf. Comox-Sliammon *łuluk^w* ‘flying’)
 C₁-fly
 ‘to fly’ (Boas, 1886, Pentlatch)
- b. **łí**~łiq^w-əm (cf. Squamish *łétəyq^wm* ‘throw off sparks’)
 C₁~shine?-MID
 ‘to shine, rays’ (Boas, 1886, Pentlatch)
- c. **k^wa**~k^wáy=čì (cf. Sechelt *k^wak^way* ‘[being] very hungry’)
 C₁~hungry=1 SG.SUBJ
 ‘I am hungry’ (Boas, 1886, Pentlatch)
- d. s-**qí**~qíws=čì (cf. Comox-Sliammon *qayíws* ‘tired’)
 STAT-C₁~?=1 SG.SUBJ
 ‘I am hurt all over.’ (Boas, 1886, Pentlatch)
- e. s-**łá**~łáyáq=čì (cf. Squamish *łáyáq* ‘get angry’)
 STAT-C₁~angry=1 SG.SUBJ
 ‘I am angry.’ (Boas, 1886, Pentlatch)
- (595) a. **łá**~ləm-əm (cf. Proto-Coast Salish **ləm-* ‘thump, kick’)
 C₁~thump?-MID
 ‘lightning’ (Boas, 1886, Pentlatch)
- b. **qí**~qí (cf. Sechelt *qíqí* ‘youngest child’ from *qéti*)
 C₁~young.child
 ‘youngest child’ (Boas, 1886, Pentlatch)
- c. **ší**~šəwaθ (cf. ?)
 C₁~man
 ‘boy’ (Boas, 1886, Pentlatch)
- d. si<s>k^w-ím (cf. Sechelt *sískím* ‘keep shaking’)
 shake<C₁>MID(+STAT?)
 ‘tremble with old age’ (Boas, 1886, Pentlatch)

- e. s-ła<ł>nay (cf. Squamish *słanáy* ‘woman’)
 NMLZ-woman<C₁>
 ‘girl’ (Boas, 1886, Pentlatch)

Examples (595c) and (595e) are consistent with C₁ diminutive reduplication, as found in other Salish languages. These are attested with nouns, though it is not possible to tell if this process was productive or lexicalized.

A.3.7 Sechelt Reduplication

Attested reduplication in Sechelt is summarised in Table A.14.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
DIM		✓		
PL		✓		
IPFV		✓		
COS			✓	

Table A.14: Reduplication in Sechelt

A.3.7.1 Sechelt Sources and Transcription

Examples come from Beaumont (1985) and Beaumont (2011). All unreduplicated forms provided come from the original source.

Words are marked with multiple diacritics over vowels to mark stress (i.e., $\acute{V}$), following Beaumont (1985). Since this notation does not distinguish primary and secondary stress. I follow Beaumont's convention as it is unclear where primary stress falls. I use my analysis of Comox-Sliammon reduplication to guide decisions regarding which segments correspond to the reduplicative morpheme.

A.3.7.2 Sechelt Plural/Pluractional C₁C₂ Reduplication

C₁C₂ reduplication in Sechelt is used to mark plurality on nouns and pluractionality on verbs. The first two consonants of the root are copied and [ə] is epenthesized, as shown in (596).

- (596) a. **cəl**~cəl^á (cf. *cəla* 'lake')
 PL~lake
 'lates' (Beaumont, 2011, p.248, Sechelt)
- b. **čám**~čámq^w (cf. *ščámq^wí* 'cloudy')
 NMLZ-PL~cloud
 'clouds' (Beaumont, 2011, p.886, Sechelt)

- c. **ʃ^wǎp̥**~ʃ^wúp̥-nis-tən (cf. ʃ^wúp̥nistən ‘safety pin’)
PL~pin-tooth-instrument
‘safety pins’ (Beaumont, 2011, p.940, Sechelt)
- d. **k^wǎs**~k^wúsən (cf. k^wúsən ‘star’)
PL~star
‘stars’ (Beaumont, 2011, p.430, Sechelt)
- e. **lám**~lám-áli (cf. lámáli ‘bottle’)
PL~alcoholic.drink-container
‘bottles (general)’ (Beaumont, 2011, p.666, Sechelt)
- f. **q^wǎt**~q^wǎtíy-šən (cf. q^wǎtíyšən ‘boot’)
PL~-wash.ashore-foot
‘boots’ (Beaumont, 2011, p.54, Sechelt)

C₁C₂ reduplication marks a basic sense of plurality with nouns, as shown in (596) above; verbs with C₁C₂ reduplication have translations suggesting pluractionality, including those that indicate repetition (event plurality) and plural participants, as shown in (597).

- (597)
- a. **lǫp**~lúp-um (cf. *lúput* ‘poke s.t./s.o.’)
PL~poke-MID
‘always poking/prodding’ (Beaumont, 2011, p.331, Sechelt)
 - b. **čǫp**~čǫpχ-ús (cf. *čǫpxús* ‘have a dirty face’)
PL~dirty-face
‘have dirty faces’ (Beaumont, 2011, p.574, Sechelt)
 - c. **qǫm**~qǫms-ála (cf. *qǫmsála* ‘be(ing) buried’)
PL~bury-inside
‘be(ing) buried (more than one person)’ (Beaumont, 2011, p.630, Sechelt)
 - d. **yǫx**~yíχ-im (cf. *syíχít* ‘apart, collapsed, in pieces’)
PL~collapse-MID
‘collapse/collapsing/fall(ing) apart (everything, the whole thing)’
(Beaumont, 2011, p.954, Sechelt)

A.3.7.3 Sechelt Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication is shown in (598): the first two consonants of the root are copied and [ə] is epenthesized.

- (598) a. **q̣əl**~q̣əl (cf. *q̣əl* ‘get wet’)
PDEN~wet
‘wet’ (Beaumont, 2011, p.518, Sechelt)
- b. **ʰəq^w**~ʰəq^w (cf. *ʰəq^w* ‘hard’)
PDEN~hard
‘very hard (glass, rock, shell, wood, etc.)’ (Beaumont, 2011, p.204, Sechelt)
- c. **ləx̣**~ləx̣-át-k^wu (cf. *ləx̣* ‘get clear’)
PDEN~clear-CONN-water
‘clear water’ (Beaumont, 2011, p.668, Sechelt)
- d. s-**x̣ət**~x̣at (cf. *x̣átám* ‘stare (watch)’)
NMLZ-STAT~stare
‘curious person, s.o. who keeps staring’ (Beaumont, 2011, p.803, Sechelt)

Property-denoting C₁C₂ reduplication appears to combine with eventive and stative roots. The category of the reduplicated forms is not clear. Translations of the examples in (598) suggest that this type of reduplication may be either nouns or adjectives.

Property-denoting forms look similar to those with C₁C₂ plural/pluractional reduplication, and the translations of the two suggest that the distinction between plurality and property-denoting is unclear for some forms (or the distinction is not reflected clearly in the English translations). For example, (598c) is translated as a property of a person (‘curious person’) and as a person engaging in a repeated action (‘keep(s) staring’). This pairing of translations suggesting both pluractional and property-denoting readings is found for a range of forms; it is not clear how the two are distinguished.

Where minimal pairs can be found between property-denoting and pluractional C₁C₂ reduplication, the pluractional forms are further marked by an ablaut vowel

/a/) instead of a schwa in the reduplicant, as shown in (599–600).

- (599) a. s-**ṣ**Ṷ**Ṷ**w~ṣáw (cf. ṣáwam ‘cry’)
 STAT-PDEN~cry
 ‘someone who is always crying, cry-baby’ (Beaumont, 2011, p.803, Sechelt)
- b. **ṣ**<**á**>**w**~ṣáw-am (cf. ṣáwam ‘cry’)
 PL<PL>~cry-MID
 ‘crying incessantly’ (Beaumont, 2011, p.924, Sechelt)
- (600) a. s-**k**^wṶ^s~k^wás (cf. k^was ‘(quite) warm’)
 STAT-PDEN~warm
 ‘someone who is always warm’ (Beaumont, 2011, p.511, Sechelt)
- b. **k**^w<**á**>**s**~k^wás-aw (cf. k^was ‘(quite) warm’)
 PL<PL>~hot-PLU
 ‘burn/scald oneself in more than one place’ (Beaumont, 2011, p.616, Sechelt)

A.3.7.4 Sechelt Diminutive C₁ Reduplication

Diminutive C₁ reduplication copies the first consonant of the root. The diminutive surfaces as an infix when the root has a moraic vowel, as shown in (601).

- (601) a. s-čí<**č**>q-ít (cf. sčíqít ‘ragged’)
 STAT-shred<DIM>-STAT
 ‘a bit ragged’ (Beaumont, 2011, p.744, Sechelt)
- b. cí<**c**>č-ómin (cf. cíčómin ‘fork’)
 fork<DIM>-instrument
 ‘little fork’ (Beaumont, 2011, p.874, Sechelt)
- c. s-tí<**t**>qíw (cf. stíqíw ‘horse’)
 NMLZ-horse<DIM>
 ‘colt’ (Beaumont, 2011, p.787, Sechelt)
- d. k^wí<**k**^w>t-ús-əm (cf. k^wítúsəm ‘look out of the window’)
 look<DIM>-face-MID
 ‘look briefly out of the window’ (Beaumont, 2011, p.621, Sechelt)

- e. $x^w\acute{í}<\mathbf{x}^w>tím$ (cf. x^witim ‘jump’)
jump<DIM>
‘hop’ (Beaumont, 2011, p.221, Sechelt)
- f. $s-k^wú<\mathbf{k}^w>\acute{c}-ít$ (cf. $sk^wú\acute{c}ít$ ‘bent’)
STAT-bend<DIM>-STAT
‘bent slightly, slightly crooked’ (Beaumont, 2011, p.42, Sechelt)
- g. $\acute{x}^wú<\mathbf{x}^w>\acute{q}^w-ít$ (cf. $\acute{x}^wú\acute{q}^wít$ ‘snore’)
snore<DIM>-STAT
‘snore softly’ (Beaumont, 2011, p.945, Sechelt)

If the root has a sonorant as one of the first two consonants, an [ə] follows the second of the matching consonants, as shown in (602).

- (602)
- a. $cí<\mathbf{c}\mathbf{ə}>l-náč-tən$ (cf. $cílnáčtən$ ‘pad’)
put.on.flat.surface<DIM>-bottom-instrument
‘little pad’ (Beaumont, 2011, p.316, Sechelt)
 - b. $\acute{x}^wá<\mathbf{x}^w\mathbf{ə}>l-ám$ (cf. $\acute{x}^wálám$ ‘glow(ing) from heat/light’)
glow<DIM>-MID
‘dim/weak (light), dim glow from heat/light’ (Beaumont, 2011, p.939, Sechelt)
 - c. $ká<\mathbf{k}\mathbf{ə}>n-át-cut$ (cf. $kánátcut$ ‘lower one’s voice’)
lower.voice<DIM>-CTR-REFL
‘lower voice a bit, speak a bit more softly’ (Beaumont, 2011, p.272, Sechelt)
 - d. $mú<\mathbf{m}\mathbf{ə}>lik$ (cf. $múlik$ ‘lynx’)
lynx<DIM>
‘lynx kitten’ (Beaumont, 2011, p.273, Sechelt)
 - e. $k^wú<\mathbf{k}^w\mathbf{ə}>yuk^w-úya$ (cf. $k^wúyuk^w$ ‘hook’)
hook<DIM>-hand
‘have a little hook for a hand’ (Beaumont, 2011, p.622, Sechelt)

- f. lá<ɭ>tíx-ím (cf. *látíxím* ‘rattling’)
 rattle<DIM>-MID+STAT
 ‘rattling a bit, (making) light rattling sound’ (Beaumont, 2011, p.662, Sechelt)
- g. s-wá<wə>t-ít (cf. *swátít* ‘levered out/up’)
 STAT-lever<DIM>-STAT
 ‘levered/pried out/up a bit’ (Beaumont, 2011, p.339, Sechelt)

The diminutive has a Ci prefix if the root starts with a cluster or has no underlying vowels (i.e., the surface vowel is [ə]); this is shown in (603). However, there are exceptions: some diminutive forms of words with [ə] do not surface with /i/, as shown in (604).

- (603) a. s-čí~cqay (cf. *scéqay* ‘sockeye’)
 NMLZ-DIM~sockeye
 ‘young sockeye (salmon)’ (Beaumont, 2011, p.795, Sechelt)
- b. čí~čłq^w-óm (cf. *čólkwóm* ‘scoop(ing) (in liquid)’)
 DIM~scoop-CTR
 ‘scoop up a small amount of something’ (Beaumont, 2011, p.376, Sechelt)
- c. tí~tł-at (cf. *tól-óm* ‘taste things’)
 DIM~taste-CTR
 ‘have a little taste of something’ (Beaumont, 2011, p.459, Sechelt)
- d. tí~tx^w-álus-əm (cf. *stáx^wálús* ‘see(ing) clearly’)
 DIM~see.clearly-eye-MID
 ‘try(ing) briefly to see clearly’ (Beaumont, 2011, p.840, Sechelt)
- e. s-wí~wəq-šáwíł (cf. *wəq* ‘come open’)
 STAT-DIM~open-door/road
 ‘partly open(ed) (door, road)’ (Beaumont, 2011, p.308, Sechelt)
- f. hí~łəlq-óm (cf. *łəqát* ‘inhale something’)
 DIM~inhale-MID
 ‘inhale a bit (air, fumes, gas, smoke)’ (Beaumont, 2011, p.682, Sechelt)

- (604) a. s-má<mə>l-ít (cf. *məl* ‘sink’)
 STAT-root<DIM>-STAT
 ‘almost sunk’ (Beaumont, 2011, p.775, Sechelt)
- b. łə<ł>q̣^w (cf. *łəq̣^wt* ‘peel something’)
 peel<DIM>
 ‘peeling easily (bark)’ (Beaumont, 2011, p.322, Sechelt)

Diminutive C₁ reduplication can occur with a range of roots, including nouns, verbs, and adjectives. With a noun, diminutive reduplication marks a small or young entity, as shown in (601b–c), (602a,d), and (603a) above. With eventive verbs, diminutive reduplication marks an event or action with low intensity (601e,g, 602b,c,f, and 604b), a small amount (603b,c,f), a short duration (601d and 603d). It can also translate to ‘almost X’, as shown in (604a). With states and adjectives, diminutive reduplication marks a small degree, as shown in (601a,f), (602g), and (603e).

A.3.7.5 Sechelt Imperfective C₁Reduplication

Imperfective C₁ reduplication copies the first consonant and vowel of the root, as shown in (605). If the root has no underlying vowel (i.e., the perfective form surfaces with [ə]), there is no vowel copied, as shown in (606). If the root has a sonorant as its first segment, the vowel is a [ə] instead of a faithful copy, as shown in (607).⁴³

- (605) a. k^wí~k^win-át (cf. *k^wíntm* ‘carry/hold things in arms/hands’)
 IPFV~carry-CTR
 ‘carrying/holding someone/something in arm(s)/hand(s)’ (Beaumont, 2011, p.218, Sechelt)
- b. k^wá~k^way-át (cf. *k^wáyátcut* ‘make room’)
 IPFV~move
 ‘moving/taking something out of the way’
 ‘taking something away from somewhere’ (Beaumont, 2011, p.602, Sechelt)

⁴³/l/ behaves as a sonorant in Salish.

- c. **cí**~cíč-it (cf. *cíčámin* ‘fork’)
 IPFV~fork-CTR
 ‘forking something, spearing something’ (Beaumont, 2011, p.875, Sechelt)
- (606) **pó<p>sq^w-əw-át** (cf. *pəsq^w* ‘burst’)
 burst<IPFV>-enter.state-CTR
 ‘bursting/crushing something’ (Beaumont, 2011, p.67, Sechelt)
- (607) a. **má**~məl (cf. *məl* ‘sink, drown’)
 IPFV~sink
 ‘sinking’ (Beaumont, 2011, p.694, Sechelt)
- b. **má**~məq^w-t (cf. *məq^wt* ‘swallow s.t.’)
 IPFV~swallow-CTR
 ‘swallowing something’ (Beaumont, 2011, p.694, Sechelt)
- c. **?á**~?ayq-át (cf. *?áyqát* ‘ghost/spook s.o.’)
 IPFV~ghost/spook-CTR
 ‘ghosting/spooking someone’ (Beaumont, 2011, p.974, Sechelt)
- d. **má**~muy-út (cf. *múyám* ‘aim’)
 IPFV~root-CTR
 ‘aiming (weapon) at something’ (Beaumont, 2011, p.694, Sechelt)
- e. **?á**~?íftən (cf. *?íftən* ‘eat’)
 IPFV~eat
 ‘eating’ (Beaumont, 1985, p.18, Sechelt)
- f. **?á**~?ítut (cf. *?ítut* ‘sleep’)
 IPFV~sleep
 ‘sleeping’ (Beaumont, 1985, p.18, Sechelt)

This type of C₁ reduplication marks imperfectivity; it only occurs with verbs.

A.3.7.6 Sechelt Plural C₁ Reduplication

Plural C₁ reduplication is shown in (608). As with diminutive C₁ reduplication, plural C₁ reduplication involves a copy of the initial consonant of the root and

[ə] is inserted following the second of the matching consonants if either the first or second consonant of the root is a sonorant. [a] is inserted if the root has no underlying vowel and the first two consonants are obstruents; these are the same roots that take [i] in the diminutive.⁴⁴

- (608) a. s- $\acute{x}í$ < $\acute{x}ə$ >m< $\acute{i}$ >s (cf. $s\acute{x}ámís$ ‘built up with houses’)
 STAT-settle<PL><STAT>
 ‘built up with houses, settled (more than one area)’ (Beaumont, 2011, p.794, Sechelt)
- b. s-yá< $yə$ > $\acute{q}$ -ít (cf. $ya\acute{q}$ ‘fall from standing position’)
 STAT-fall<PL>-STAT
 ‘fallen from a standing position (more than one thing)
 ‘felled (more than one thing)’ (Beaumont, 2011, p.156, Sechelt)
- c. $\acute{c}á$ ~ $\acute{c}p\acute{x}$ -iyús (cf. $\acute{c}épxús$ ‘have a dirty face’)
 PL~dirty-round.object
 ‘dirty (lots of spherical objects)’ (Beaumont, 2011, p.366, Sechelt)
- d. s- $\acute{p}á$ ~ $\acute{p}s$ < $\acute{i}$ > $\acute{q}^w$ (cf. $s\acute{p}əsí\acute{q}^w$ ‘crushed’)
 STAT-PL-burst/crush<STAT>
 ‘dented all over’ (Beaumont, 2011, p.781, Sechelt)

Plural C₁ reduplication reduplication is attested with stative verbs and adjectives (cf. plural C₁ reduplication in Comox-Sliammon discussed in Section 4.1.3.1.3); it marks plural participants or a spatial distribution, as shown in (608a–c) and (608b,d), respectively.

A.3.7.7 Sechelt Change-of-State C₂ Reduplication

Change-of-state C₂ reduplication doubles the second consonant of the root and the preceding vowel, as shown in (609).⁴⁵ The position of the copy is clearly infixal in roots that are longer than CVC, such as in (609g).

⁴⁴The vowel qualities associated with diminutive and plural C₁ reduplication in Sechelt matches the one found in Comox-Sliammon.

⁴⁵While C₂ reduplication is not explicitly described by Hill-Tout (1904), Haeberlin (1918), or Beaumont (1985), there are examples in the dictionary (Beaumont, 2011).

- (609) a. $\acute{t}íq \sim \mathbf{i}q$ (cf. $\acute{t}íqím$ ‘black’)
 black~COS
 ‘getting black (dark, dirty, discoloured)’
 ‘tarnishing (losing brightness)’ (Beaumont, 2011, p.855, Sechelt)
- b. $táx^w \sim \mathbf{ə}x^w$ (cf. $təx^wt$ ‘take up slack’)
 take.slack~COS
 ‘get very tight’ (Beaumont, 2011, p.477, Sechelt)
- c. $púq^w \sim \mathbf{u}q^w$ (cf. $púq^w$ ‘grey’)
 grey~COS
 ‘getting grey, getting dilapidated (house, etc.)’
 ‘faded, getting old and grey.’ (Beaumont, 2011, p.150, Sechelt)
- d. $híx \sim \mathbf{i}x$ (cf. $híx$ ‘thin (underweight)’)
 thin~COS
 ‘get thin’ (Beaumont, 2011, p.684, Sechelt)
- e. $ǵáǵ \sim \mathbf{ə}ǵ$ (cf. $ǵáǵ$ ‘congeal’)
 stiff~COS
 ‘congealed’ (Beaumont, 2011, p.862, Sechelt)
- f. $k^wám \sim \mathbf{ə}m$ (cf. $k^wámáqín$ ‘(have) red hair’)
 red~COS
 ‘getting red’ (Beaumont, 2011, p.607, Sechelt)
- g. $qíl <\mathbf{i}l>t$ (cf. $qílt$ ‘bright’)
 bright<COS>
 ‘daybreak, dawn, getting light’ (Beaumont, 2011, p.645, Sechelt)

C₂ reduplication in Sechelt marks a change of state; it applies to adjective roots. (

A.3.8 Squamish Reduplication

Attested reduplication in Squamish is summarised in Table A.15.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
PDEN	✓			
PROG		✓		
DIM		(✓)		

Table A.15: Reduplication in Squamish

A.3.8.1 Squamish Sources and Transcription

Examples come from Kuipers (1967), Currie (1997), Bar-el and Watt (1998), Bar-el (1998), Bar-el and Watt (2001), Dyck (2004), Gillon (2006), Bar-el (2006), Bar-el (2008), Jacobs (2011), and Squamish Nation Education Department (2011). Unreduplicated forms come from the original sources or the dictionary (Squamish Nation Education Department, 2011).

A.3.8.2 Squamish Plural/Pluractional C₁C₂ Reduplication

C₁C₂ reduplication is used to mark plurality on nouns and pluractionality on verbs in Squamish. The first two consonants of the root are doubled and [ə] is epenthesized between the copied consonants, as shown in (610).

- (610) a. $\dot{k}^w\dot{x}^w \sim k^w\acute{a}x^w a?$ (cf. $k^w\acute{a}x^w a?$ ‘box’)
 PL~box
 ‘boxes’ (Kuipers, 1967, p.100, Squamish)
- b. $s-n\dot{x}^w \sim n\acute{x}^w i?$ (cf. $sn\acute{x}^w i?$ ‘canoe’)
 NMLZ-PL~canoe
 ‘canoes’ (Kuipers, 1967, p.100, Squamish)
- c. $\dot{t}\dot{ə}q \sim t\acute{a}q a?$ (cf. $t\acute{a}q a?$ ‘bruised’)
 PL~bruised
 ‘be bruised all over’ (Kuipers, 1967, p.101, Squamish)

- d. $\text{tə}^{\text{w}}\text{k}^{\text{w}} \sim \text{t}^{\text{w}}\text{k}^{\text{w}}\text{-}\check{\text{s}}\text{ə}^{\text{w}}\text{n}$ (cf. $\text{t}^{\text{w}}\text{k}^{\text{w}}\check{\text{s}}\text{ə}^{\text{w}}\text{n}$ ‘stumble’)
 PL~get.hooked-foot
 ‘stumble along’ (Kuipers, 1967, p.101, Squamish)
- e. $\text{lə}^{\text{w}}\text{m} \sim \text{l}^{\text{w}}\text{a}^{\text{w}}\text{m}$ (cf. $\text{l}^{\text{w}}\text{a}^{\text{w}}\text{m}$ ‘house’)
 PL~house
 ‘houses’ (Kuipers, 1967, p.100, Squamish)
- f. $\text{qə}^{\text{w}}\text{x} \text{ s-}\text{tə}^{\text{w}}\text{n} \sim \text{t}^{\text{w}}\text{a}^{\text{w}}\text{nay}$ na $\text{č}^{\text{w}}\text{icá}^{\text{w}}\text{p}$ (cf. $\text{s}^{\text{w}}\text{t}^{\text{w}}\text{a}^{\text{w}}\text{nay}$ ‘woman’)
 many NMLZ-PL~woman RL work
 ‘Many woman worked.’ (Gillon, 2006, p.29, Squamish)
- g. $\check{\text{c}}\text{ə}^{\text{w}}\text{n} \text{ k}^{\text{w}}\text{áč-nə}^{\text{w}}\text{x}^{\text{w}} \text{ ta}=\text{pə}^{\text{w}}\check{\text{s}} \sim \text{pú}^{\text{w}}\check{\text{s}}$ (cf. $\text{pú}^{\text{w}}\check{\text{s}}$ ‘cat’)
 1SG.SUBJ look-NTR DET=PL~cat
 ‘I saw (the) cats.’ (Gillon, 2006, p.17, Squamish)
- h. $\check{\text{c}}\text{ə}^{\text{w}}\text{n} \text{ k}^{\text{w}}\text{áč-nə}^{\text{w}}\text{x}^{\text{w}} \text{ ʔíʔx}^{\text{w}} \text{ ta}=\text{s-}\text{q}^{\text{w}}\text{ə}^{\text{w}}\text{m} \sim \text{q}^{\text{w}}\text{ə}^{\text{w}}\text{má}^{\text{w}}\text{y}$
 1SG.SUBJ look-CTR all DET=PL~dog
 ‘I saw all the dogs.’ (cf. $\text{s}^{\text{w}}\text{q}^{\text{w}}\text{ə}^{\text{w}}\text{má}^{\text{w}}\text{y}$ ‘dog’) (Gillon, 2006, p.29, Squamish)
- i. $\text{t}^{\text{w}}\text{iq} \text{ wa} \text{ hiyí} \text{ ta}=\text{mə}^{\text{w}}\text{x} \sim \text{mí}^{\text{w}}\text{xat}$ (cf. $\text{mí}^{\text{w}}\text{xat}$ ‘bear’)
 always PA big DET=PL~bear
 ‘These bears are always big (a group of bears in the area).’ (Bar-el, 1998, p.76, Squamish)
- j. na $\text{k}^{\text{w}}\text{ə}^{\text{w}}\text{l} \sim \text{k}^{\text{w}}\text{ə}^{\text{w}}\text{laš-t-}\emptyset\text{-as} \text{ ta}=\text{mí}^{\text{w}}\text{xat}$ (cf. $\text{k}^{\text{w}}\text{ə}^{\text{w}}\text{laš}$ ‘shoot’)
 RL PL~shoot-TR-3OBJ-3SUBJ DET=bear
 ‘He shot a number of bears (at the same time).’
 ‘He shot the bear repeatedly.’ (Jacobs, 2011, p.45, Squamish)
- k. $\check{\text{c}}\text{ə}^{\text{w}}\text{n} \text{ ʔə}^{\text{w}}\text{x} \sim \text{ʔə}^{\text{w}}\text{x}^{\text{w}}\text{ə}^{\text{w}}\text{nq}$ (cf. $\text{ʔə}^{\text{w}}\text{x}^{\text{w}}\text{ə}^{\text{w}}\text{nq}$ ‘win’)
 1SG.SUBJ PL~win
 ‘I’m winning all the time.’ (Bar-el, 2006, p.300, Squamish)

The (first) $\text{C}_1\text{əC}_2$ sequence is usually not stressed, except when reduplication is added to monosyllabic words with [ə], as shown in (611).

- (611) a. na $\text{yut} \text{ ta}=\text{s-}\text{cə}^{\text{w}}\text{q} \sim \text{cə}^{\text{w}}\text{q}$ (cf. $\text{scə}^{\text{w}}\text{q}$ ‘tree’)
 RL burn DET=NMLZ-PL~tree
 ‘The forest is burning.’ (Jacobs, 2011, p.11, Squamish)

- b. $\dot{k}^w\acute{\text{ʔ}}\dot{\text{ʔ}}\sim\dot{k}^w\acute{\text{ʔ}}\dot{\text{ʔ}}$ (cf. $\dot{k}^w\acute{\text{ʔ}}\dot{\text{ʔ}}$ ‘spill something’)
 PL $\sim$ spill
 ‘spill repeatedly’ (Bar-el, 2000, p.90, Squamish)
- c. $s\text{-}\dot{x}^w\acute{\text{ʔ}}\dot{\text{ʔ}}\sim\dot{x}^w\acute{\text{ʔ}}\dot{\text{ʔ}}$ (cf. $s\acute{\text{ʔ}}\dot{\text{ʔ}}$ ‘leg’)
 NMLZ-PL $\sim$ leg
 ‘legs’ (Dyck, 2004, p.177, Squamish)

Plural C₁C₂ reduplication occurs with nouns and verbs. It marks plural entities when applied to nouns, as shown in (610a–b,e–i) and (611a,c); it marks pluractionality with verbs. C₁C₂ reduplication can mark multiple participants (610j), temporally-distributed events (610j,k, 611b), or a spatial distribution (610c,d). Bar-el (2008) argues that both habitual and iterative readings are possible on verbs with C₁C₂ reduplication.

A.3.8.3 Squamish Property-Denoting C₁C₂ Reduplication

Property-denoting C₁C₂ reduplication copies the first two consonants of the root, as shown in (612).

- (612) a. $\text{ʔ}\acute{\text{ʔ}}\text{-}\dot{\text{c}}\acute{\text{ʔ}}\dot{\text{x}}^w\sim\dot{\text{c}}\acute{\text{ʔ}}\dot{\text{x}}^w$ (cf. $\dot{\text{c}}\acute{\text{ʔ}}\dot{\text{x}}^w$ ‘help s.o. (out of trouble)’)
 STAT-PDEN $\sim$ help
 ‘willing to help, helpful’ (Kuipers, 1967, p.101, Squamish)
- b. $\text{ʔ}\acute{\text{ʔ}}\text{-}\dot{\text{c}}\acute{\text{ʔ}}\dot{\text{q}}\sim\dot{\text{c}}\acute{\text{ʔ}}\dot{\text{q}}$ (cf. $s\dot{\text{c}}\acute{\text{ʔ}}\dot{\text{q}}$ ‘dirt’)
 STAT-PDEN $\sim$ dirt
 ‘dirty’ (Kuipers, 1967, p.113, Squamish)
- c. $\dot{\text{p}}\acute{\text{ʔ}}\dot{\text{q}}^w\sim\dot{\text{p}}\acute{\text{ʔ}}\dot{\text{q}}^w$ (cf. $\dot{\text{p}}\acute{\text{ʔ}}\dot{\text{q}}^w$ ‘become yellow’)
 PDEN $\sim$ yellow
 ‘yellow’ (Kuipers, 1967, p.101, Squamish)
- d. $\text{qal}\acute{\text{ʔ}}\text{qal}\text{ wa } \dot{\text{q}}\acute{\text{ʔ}}\dot{\text{x}}\sim\dot{\text{q}}\acute{\text{ʔ}}\dot{\text{x}}$ (cf. $\dot{\text{q}}\acute{\text{ʔ}}\dot{\text{x}}$ ‘become black’)
 crow IPFV PDEN $\sim$ black
 ‘Crows are black.’ (Gillon, 2006, p.29, Squamish)

Property-denoting reduplication is attested with adjectives and stative-marked verbs.

A.3.8.4 Squamish Progressive C₁ Reduplication

C₁ reduplication marks progressive aspect (Bar-el, 2006, 2008). The first consonant and vowel are copied and stress falls on the first vowel in forms with progressive C₁ reduplication, as shown in (613).

- (613) a. *tá~tayaq* (cf. *táyaq* ‘get angry’)
 PROG~angry
 ‘be angry’ (Dyck, 2004, p.163, Squamish)
- b. *tú~túk^w* (cf. *túk^ws* ‘take (something) home’)
 PROG~go.home
 ‘be going home’ (Dyck, 2004, p.200, Squamish)
- c. *k^wa~k^way* ta=*míxał* (cf. *k^way* ‘be hungry’)
 PROG~hungry DET=bear
 ‘The bear is hungry’ / ‘The bear is getting hungry.’ (Bar-el & Watt, 1998, p.41, Squamish)
- d. *ʔíʔx^w* ta=*staʔəx^wł* nam *ʔí~ʔimaš* naʔ ta=*sxałnat*
 all DET=children got PROG~walk RL DET=Sunday
 ‘All the children/every child went walking on Sunday.’ (cf. *ʔíməš* ‘walk’) (Currie, 1997, p.52, Squamish)
- e. na *yuł-Ø* ta=*yú~yul* (cf. *yuł* ‘burn’)
 RL burn-3SUBJ DET=PROG~fire
 ‘The fire is burning.’ (Jacobs, 2011, p.52, Squamish)
- f. na *pa~payaq-ant-as* ta=*s-nəx^wil-s* (cf. *páyayaq* ‘be fixed’)
 RL PROG~fix-TR-3ERG DET=NMLZ-canoe-3POSS
 ‘He’s in the process of fixing the canoe.’ (Bar-el, 2008, p.47, Squamish)

Progressive C₁ reduplication applies to verbs.

A.3.8.5 Squamish Diminutive C₁ Reduplication

The use of C₁ reduplication to mark a diminutive is less frequent in Squamish than in other Salish languages (Dyck, 2004, p.171). Where it does occur, it involves a copy of the first consonant of the root, and it may be accompanied by epenthetic [i], as shown in (614a).

- (614) a. $\text{li} \sim \text{lam}$ (cf. lam 'house')
 DIM ~ house
 'little house' (Squamish Nation Education Department, 2011, p.103, Squamish)
- b. $\text{k}^w \acute{a} < \text{k}^w > \text{x}^w \text{a}?$ (cf. $\text{k}^w \acute{a} \text{x}^w \text{a}?$ 'box')
 DIM ~ box
 'small box' (Kuipers, 1967, p.346, Squamish)
- c. $\text{x}^w \text{f} \sim \text{x}^w \text{ik}^w$ (root: $\text{x}^w \text{ik}^w$ - 'grey')
 DIM ~ grey
 'small blueberry' (Kuipers, 1967, p.350, Squamish)

Diminutive C₁ reduplication in Squamish is found with nouns; it marks something small in size.

A.3.9 Twana Reduplication

Attested reduplication in Twana is summarised in Table A.16.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
DIM		✓		
DIM				✓
Other	Unclear C ₂ Reduplication			

Table A.16: Reduplication in Twana

A.3.9.1 Twana Sources and Transcription

Examples and corresponding unreduplicated forms come from Drachman (1969). I leave most vowels (e.g., “ɛ” and “ɔ”) as transcribed by Drachman (1969). I re-transcribe sequences of glottal stop and a sonorant as a glottalised sonorant where appropriate. Note that /b/ and /d/ pattern as sonorants in Twana.

A variant of plural C₁C₂ labeled “wrong side” reduplication is discussed in Chapter 6 (Section 6.5.3.3.2 and Section 6.6.2.2.3).

A.3.9.2 Twana Plural C₁C₂ Reduplication

C₁C₂ reduplication in Twana copies the first two consonants of the root. If one of the first two consonants in the stem is a sonorant, there is a [ə] between the copied consonants, resulting in a CəC shape.

- (615) a. ʔəs-**x^wəl**~x^wél-čəd (cf. ʔasx^wélčəd ‘I’m lost.’)
 STAT-PL~lost-1SG.SUBJ
 ‘I’ve lost them.’ (Drachman, 1969, p.35, Twana)
- b. **təb**~təbax^w (cf. təbéx^w ‘land’)
 PL~land
 ‘lands’ (Drachman, 1969, p.68, Twana)
- c. ʔəs-**təb**~túb (cf. túb ‘heals, scars’)
 STAT-PL~heal/scar
 ‘scarred’ (aug.) (Drachman, 1969, p.54, Twana)

A sonorant in C₂ position may be glottalised in the reduplicant, as shown in (616).

- (616) a. $\dot{q}^w\dot{a}l \sim \dot{q}^w\acute{e}le(h)$ (cf. $\dot{q}^w\acute{e}le(h)$ ‘cedar tree’)
 PL~cedar.tree
 ‘cedar trees’ (Drachman, 1969, p.45, Twana)
- b. $\dot{q}^w\dot{a}l \sim \dot{q}^w\acute{o}l-ab$ (cf. $ʔas\dot{q}^w\acute{o}l-ab$ ‘sunk’)
 PL~sink-MID
 ‘all sunk’ (Drachman, 1969, p.76, Twana)

[ə] may be omitted when both of the initial two consonants are obstruents, as shown in (617).

- (617) a. $\acute{s}-\acute{c}t \sim \acute{c}úta\grave{x}$ (cf. $\acute{s}\acute{c}úta\grave{x}$ ‘halibut’)
 NMLZ-PL~halibut
 ‘halibut (collective)’ (Drachman, 1969, p.35, Twana)
- b. $ʔ\acute{o}\acute{s}-\acute{s}\acute{c} \sim \acute{s}\acute{e}\acute{c}$ (cf. $ʔa\acute{s}\acute{s}\acute{e}\acute{c}$ ‘paid’)
 STAT-PL~pay
 ‘all paid’ (Drachman, 1969, p.35, Twana)
- c. $s-p\dot{q}^w \sim p\acute{o}\dot{q}^w$ (cf. $sp\acute{o}\dot{q}^w$ ‘feather’)
 NMLZ-PL~feather
 ‘feathers’ (Drachman, 1969, p.54, Twana)
- d. $s-p\acute{c} \sim p\acute{o}\acute{c}u$ (cf. $sp\acute{o}\acute{c}u$ ‘berry-basket’)
 NMLZ-PL~berry.basket
 ‘berry-baskets’ (Drachman, 1969, p.35, Twana)

A version of plural C₁C₂ reduplication, without the first consonant copied, is shown in (618). This is discussed further in Chapter 6 (Section 6.5.3.3.2).

- (618) a. $\dot{q}^w \sim s\acute{o}\dot{q}^w\acute{a}y$ (cf. $s\acute{o}\dot{q}^w\acute{a}y$ ‘elder sister’)
 PL~elder.sister
 ‘elder sisters’ (Drachman, 1969, p.32, Twana)
- b. $s-q \sim t\acute{e}qaw$ (cf. $steq\acute{e}w$ ‘horse’)
 NMLZ-PL~horse
 ‘horses’ (Drachman, 1969, p.86, Twana)

- c. ʔəs-č̣~x̣ác (cf. ʔasx̣ác ‘measure’)
 STAT-PL~measure
 ‘measure (pl.)’ (Drachman, 1969, p.55, Twana)

Plural C₁C₂ reduplication is attested with nouns and verbs. Most examples of C₁C₂ reduplication given by Drachman (1969) are not given translations. Where translations are provided, C₁C₂ reduplication marks plural entities with nouns, as shown in (615b), (616a), (617a,b,d), and (618a–b). With verbs, C₁C₂ reduplication marks plural participants with verbs, as shown in (615a), (616b), and (617b).

A.3.9.3 Twana Diminutive C₁ Reduplication

Diminutive C₁ reduplication in Twana doubles the initial consonant and vowel of the root, as shown in (619). However, for some words that start with /ʔ/, the vowel is not copied in C₁ reduplication, as shown in (620).

- (619) a. ʔa~ʔáxac (cf. ʔax̣ác ‘bed’)
 DIM~go.to.bed
 ‘go to bed (dim.)’ (Drachman, 1969, p.149, Twana)
- b. ʔé~ʔeyəʔuč̣ád (cf. ʔeyəʔó(h)č̣ád ‘untie’)
 DIM~untie
 ‘untie (dim.)’ (Drachman, 1969, p.149, Twana)
- c. bə~bəqsád (cf. bəqsád ‘nose’)
 DIM~nose
 ‘nose (dim.)’ (Drachman, 1969, p.162, Twana)
- (620) a. s-ʔé<ʔ>lətx^w (cf. sʔélatx^w ‘timber, board’)
 NMLZ-timber<DIM>
 ‘timber (dim.)’ (Drachman, 1969, p.226, Twana)
- b. ʔé<ʔ>x̣ád (cf. ʔáxad ‘clear the throat’)
 clear.the.throat<DIM>
 ‘clear the throat (dim.)’ (Drachman, 1969, p.149, Twana)

With words that are monosyllabic (prior to reduplication), /ʔ/ is also epenthesized following the first vowel in C₁ reduplication, as shown in (621).

- (621) a. **tʰɛʔ**~ts (cf. *tes* ‘shoot’)
 DIM~shoot
 ‘shoot (dim.)’ (Drachman, 1969, p.147, Twana)
- b. **sɛʔ**~səl (cf. *sel* ‘cloth, rag’)
 DIM~cloth/rag
 ‘cloth, rag’ (dim.) (Drachman, 1969, p.147, Twana)
- c. **dɛʔ**~dəx (cf. *dəx* ‘converse, gossip’)
 DIM~converse
 ‘converse, gossip (dim.)’ (Drachman, 1969, p.147, Twana)

Diminutive C₁ reduplication applies to nouns and verbs. There are no translations given for most of the diminutive forms in Drachman (1969). It is likely that forms with C₁ reduplication is comparable to diminutive forms found in other Salish languages, such that (619c) can be assumed to indicate a nose that is small in size. As the meanings associated with diminutive verbs across the family are quite broad, nothing further can be about possible readings of diminutive verbs in Twana at present.

A.3.9.4 Twana Diminutive V₁ Reduplication

V₁ reduplication in Twana copies a root vowel across a glottal stop. Though Drachman (1969) attributes a diminutive function to V₁ reduplication (e.g., 622e), it is not possible to confirm this from the data he provides because there are no translations.

- (622) a. y<**ɛʔ**>ɛqʷ-əb (cf. *yəqʷ* ‘wash’)
 wash<DIM>-MID
 ‘wash (dim.)’ (Drachman, 1969, p.170, Twana)
- b. p<**ɛʔ**>ɛt-šəd-əb (cf. *pátšədəb* ‘set out’)
 ?<DIM>-foot-MID
 ‘set out (dim.)’ (Drachman, 1969, p.170, Twana)

The attested forms with V₁ reduplication are verbs; their meaning is unclear.

A.3.9.5 Other Reduplication in Twana

Twana has a C₂ reduplication process with an unclear function. It takes the shape -aC and follows the first CVC sequence. The vowel in the reduplicant is [a].

- (623) a. s-ʔəq^w ~ **aq^w** (root: ʔəq-)
NMLZ-snow ~ C₂
'snow' (Drachman, 1969, p.172, Twana)
- b. bix^w ~ **ax^w** (root: bix^w - 'smile')
smile ~ C₂
'smile' (Drachman, 1969, p.279, Twana)
- c. ʔil ~ **al** (root: ʔil- 'cry')
cry ~ C₂
'sing' (Drachman, 1969, p.279, Twana)

A.4 Tsamosan Reduplication

A.4.1 Cowlitz Reduplication

Attested reduplication in Cowlitz is summarised in Table A.17.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
Other	Unclear/Lexicalised C ₂ Reduplication			

Table A.17: Reduplication in Cowlitz

A.4.1.1 Cowlitz Sources and Transcription

All examples and unreduplicated forms come from Kinkade (2004). Cowlitz vowel length, which marks diminutivity, is discussed in Chapter 6 (Section 6.6.2.2.1). There are instances of [ə] missing in these examples.

A.4.1.2 Cowlitz Pluractional C₁C₂ Reduplication

Pluractionality is marked on verbs by C₁C₂ reduplication. The first CVC sequence of the root is copied; stress falls on the first syllable, excluding fixed segment prefixes, as shown in (624). Kinkade (2004) labels this type of reduplication as “repetitive”; it has a pluractional function.

- (624) a. s-cúy<**cuy**>qs-mit-n (cf. *cúyqs-m* ‘stumble’)
IPFV-stumble<PL>-MID-3SG.SUBJ
‘He’s stumbling.’ (Kinkade, 2004, p.259, Cowlitz)
- b. ?it káy~**kay**-n (root: *kay*-)
PFV tickle~PL-3SG.SUBJ
‘He tickled him.’ (Kinkade, 2004, p.259, Cowlitz)
- c. ?ac-łúm~**lum**-ł (root: *łuma*-)
STAT-wrinkle~PL-INTR
‘It’s wrinkled.’ (Kinkade, 2004, p.259, Cowlitz)

- d. s-mə́ʔx̣~**mə́ʔx̣**-cši-t-n (root: mə́ʔx̣a-)
 IPFV-take.a.chance~PL-REFL-TR-3SG.SUBJ
 ‘He’s shooting at targets.’ (Kinkade, 2004, p.259, Cowlitz)
- e. téṃ~**tem**-ṃ (root: tíṃ-)
 across~PL-MID
 ‘striped horizontally’ (Kinkade, 2004, p.259, Cowlitz)

If the root has obstruents in both C₁ and C₂ position, there is no vowel in the reduplicant, as shown in (625).

- (625) a. s-čéq~**čq**-mit-n (root: čiq- ‘whine, squeal’)
 IPFV-whine~PL-MID-3SUBJ
 ‘he’s whining’ (Kinkade, 2004, p.259, Cowlitz)
- b. ʔac-pús~**ps**-m (cf. *pust* ‘swell up’)
 STAT-swell~PL-MID
 ‘mumps’ (Kinkade, 2004, p.260, Cowlitz)

C₁C₂ reduplication applies to verbs. While the precise type of plurality marked by C₁C₂ reduplication is unclear, distribution in space or time are sufficient for its use in circumstances with without a plural arguments (e.g., 624a).

A.4.1.3 Other Reduplication in Cowlitz

C₂ reduplication is attested, but rare, in Cowlitz (Kinkade, 2004); the second consonant of the root is copied, as shown in (626). The function of this type of reduplication is unclear.

- (626) a. s-ʔə́x̣~**x̣**-t-n (cf. ʔə́xn ‘look at’)
 IPFV-look~C₂-TR-3SUBJ
 ‘He’s examining it.’ (Kinkade, 2004, p.226, Cowlitz)
- b. k^wós~**s**-w-n (cf. *k^wəsmn* ‘supper’)
 evening~C₂-INTR-3SUBJ
 ‘evening’ (Kinkade, 2004, p.226, Cowlitz)

- c. s-[́]tíc~**c**-w-n (cf. *s[́]tícšitn* ‘splashing on’)
IPFV-splash~C₂-INTR-3SUBJ
‘It’s splashing.’ (Kinkade, 2004, p.226, Cowlitz)

A.4.2 Lower Chehalis Reduplication

Attested reduplication in Lower Chehalis is summarised in Table A.18.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
Other	Unclear C ₁ Reduplication Unclear C ₂ Reduplication Unclear V ₁ Reduplication Unclear CVCVC Reduplication			

Table A.18: Reduplication in Lower Chehalis

A.4.2.1 Lower Chehalis Sources and Transcription

All examples and unreduplicated forms come from Robertson (2014). There are instances of [ə] missing in these examples.

A.4.2.2 Lower Chehalis C₁C₂ Pluractional Reduplication

Pluractional C₁C₂ reduplication involves a copy of the first and second consonants of the root, as shown in (627). It is unclear from the available data what determines the quality of the vowel in (627b), though it is possible that an epenthetic schwa could be realized as [i] adjacent to a /y/.

- (627) a. ?i-cáx^w~cəx^w-č-w-n (root: cáx- ‘drip’)
 IPFV-drip~PL-water-INTR-3.SUBJ.IPFV
 ‘drip continuously’ (Robertson, 2014, p.135, Lower Chehalis)
- b. s-yíl~yíl-i? (root: yíl- ‘walk’)
 NMLZ-walk~PL-INTNS.DISTR
 ‘walking around’ (Robertson, 2014, p.135, Lower Chehalis)

Examples of C₁C₂ reduplication are attested with verbs and translated in a manner that indicates an exclusively pluractional (rather than plural) function, as in the other Tsamosan languages, supporting the analysis of this reduplication as specifically a pluractional, rather than general plural, marker.

A.4.2.3 Other Reduplication in Lower Chehalis

The scarcity of data poses a challenge in determining which types of reduplication were productive in Lower Chehalis. Among the documented types, there is a type of C₂ reduplication that involves the doubling of the second consonant of the root with no vowel between the matching consonants, as shown in (628), or [i], as shown in (629). The function of this C₂ reduplication is unclear.

- (628) $\lambda\acute{u}k^w \sim k^w$ -a? (cf. $\lambda u k^w$ ‘high’)
high~C₂-IMPER
‘a little higher’ (Robertson, 2014, p.134, Lower Chehalis)
- (629) a. $\acute{x}i\acute{w} \sim i\acute{w}$ (root: $\acute{x}i\acute{w}$ - ‘fear’)
fear~C₂
‘fear, frighten’ (Robertson, 2014, p.135, Lower Chehalis)
- b. $s-w\acute{e}q \sim i\acute{q}$ (root: $w\acute{e}q$ - ‘frog’)
NMLZ-frog~C₂
‘frog’ (Robertson, 2014, p.135, Lower Chehalis)

While there is very little evidence for a productive C₁ reduplication process in Lower Chehalis, Robertson (2014) documents a handful of words with a copy of the first consonant of the root, as shown in (630). C₁ reduplication co-occurs with diminutive morphology in examples where it marks plurality.

- (630) a. $s-k^w\acute{e} \sim k^w\acute{a}<?>m-u?$ (root: $k^w m$ - ‘inland’)
NMLZ-PL~inland<DIM>-DIM
‘animal, insect, misbehaved (children)’ (Robertson, 2014, p.134, Lower Chehalis)
- b. $\acute{c}\acute{e} \sim \acute{c}\acute{a}t-u?$ (root: $\acute{c}a\acute{t}$ - ‘maggot’)
PL~maggot-DIM
‘maggots’ (Robertson, 2014, p.134, Lower Chehalis)

There is also a V₁ reduplication process in Lower Chehalis. The root vowel is copied and the copied vowel is positioned following the root, as shown in (631). This reduplication marks plurality; the only examples Robertson (2014) provides

involve nouns. In each of the examples, the reduplicated vowel is followed by a glottal stop.

- (631) a. cúʔ~**uʔ** (root: *cúʔ*- ‘foot’)
 foot~PL
 ‘feet’ (Robertson, 2014, p.119, Lower Chehalis)
- b. ʔáʔq~**aʔ** (cf. *ʔaʔq*- ‘child’)
 child~PL
 ‘children’ (Robertson, 2014, p.119, Lower Chehalis)

Robertson (2014) also documents a longer C₁C₂C₃ (CVCVC) reduplication pattern that has a similar semantic function to pluractional C₁C₂ reduplication (632).⁴⁶

- (632) a. čəpáʔx~čə**pəx**-ẉ-n
 lightning~PL-INTR-3.SUBJ.IPFV
 ‘It is lightening off and on.’ (Robertson, 2014, p.136, Lower Chehalis)
- b. yəx^wáx^w~yə**x^w****ax^w**-ẉ-n
 light(?)~PL-INTR-3.SUBJ.IPFV
 ‘blinking lights’ (Robertson, 2014, p.136, Lower Chehalis)
- c. ʔi-xəpəq̣~xə**pəq̣**-ẉ-n
 IPFV-strike~PL-INTR-3.SUBJ.IPFV
 ‘They (stones) always strike together.’ (Robertson, 2014, p.136, Lower Chehalis)

⁴⁶Robertson (2014) labels it as *intermittent* and notes that it “and the words involved in it, may be borrowings from Chinookan” (p.135).

A.4.3 Quinault Reduplication

Attested reduplication in Quinault is summarised in Table A.19.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL COMP	✓			✓
Other	Unclear C ₂ Reduplication (or Gemination)			

Table A.19: Reduplication in Quinault

A.4.3.1 Quinault Sources and Transcription

Examples come from Modrow (1971). Unreduplicated forms come from Cowlitz, Lower Chehalis, and Upper Chehalis come from Kinkade (2004), Robertson (2014), and Kinkade (1991), respectively; Proto-Salish forms come from Kuipers (2002). As Modrow (1971) did not mark ejectives, I have retranscribed plain stops as ejectives where appropriate.

I provide examples to show a process labelled as reduplication by Modrow (1971) under “Other”, though I consider it to be gemination rather than C₂ reduplication.

A.4.3.2 Quinault Pluractional C₁C₂ Reduplication

There is no description of C₁C₂ reduplication in either J. Gibson (1964) or Modrow (1971), but there are examples of C₁C₂ reduplication in the examples provided in Modrow (1971) and Rowicka (2006). The first two consonants of the root are copied, as shown in (633). It is not clear what determines vowel quality.

- (633) a. kəlt ǰáǰ~ǰǰǰ-əm (cf. L. Chehalis *ʔiǰáǰǰščln* ‘They (sg.) told me.’)
 1PL.PFV tell~PL-MID
 ‘We told stories.’ (Rowicka, 2006, p.463, Quinault)
- b. šám~šǰm-təm (cf. U. Chehalis *šǰmsʔ* ‘feel faint’)
 dizzy~C₁C₂-PASS
 ‘to feel dizzy’ (Modrow, 1971, p.155, Quinault)

- c. $\acute{s}úk^w \sim \acute{s}úk^w -\acute{a}m$ (cf. U. Chehalis $\acute{s}úk^w \acute{s}úk^w mitn$ 'whooping cough')
whooping.cough $\sim C_1 C_2$ -MID
'whooping cough' (Modrow, 1971, p.157, Quinault)
- d. $mí\acute{x} \sim ma\acute{x}$ (cf. Cowlitz $m\acute{a}\acute{x}n$ 'to try, taste, guess')⁴⁷
look $\sim$ PL
'look around' (Modrow, 1971, p.121, Quinault)
- e. $k^wi \quad ?ín \sim ?ín$ (cf. L. Chehalis $?ini-$ 'do what')
2SG.CONT do $\sim$ PL
'What are you doing?' (Rowicka, 2006, p.466, Quinault)
- f. $\acute{j}il \sim \acute{j}il-m-g^wal$ (cf. L. Chehalis $yíl-$ 'walk')
walk $\sim$ PL-?-RECP
'[walking or whirling] around and around' (Modrow, 1971, p.63, Quinault)

The reduplication in (633) is glossed as an “iterative” in Rowicka (2006); this translation is consistent with the patterns found across the Tsamosan languages (i.e., a pluractional types of $C_1 C_2$ reduplication restricted to verbs).

A.4.3.3 Quinault V_1 Reduplication

In V_1 reduplication, the stressed vowel is copied and a glottal stop separates the identical vowels, as shown in (634). This type of reduplication marks comparison.

- (634) a. $q^w umí < ?i > n$ (cf. $q^w umín$ 'slow')
slow <COMP>
'slower' (Modrow, 1971, p.356, Quinault)
- b. $ná < ?a > w$ (cf. $náw$ 'big')
big <COMP>
'bigger' (Modrow, 1971, p.356, Quinault)

⁴⁷I am unsure about the vowels in this form; I have left them as transcribed in Modrow (1971).

V₁ reduplication in Quinault is attested with adjectives.

A.4.3.4 Other Reduplication in Quinault

Plurals in Quinault can be formed by doubling the last consonant of the word and adding a plural suffix (-*əlma* or -*elma*), as shown in (635). It is not clear that C₂ reduplication has a semantic contribution independent of the plural suffix.

- (635) a. *cínnup~p-əlmá* (cf. Cowlitz *cənúp-* ‘bed’)
 bed~C₂-PL
 ‘beds’ (Modrow, 1971, p.23)
- b. *s-k^wúʔm~m-əlmá* (cf. Proto-Salish **mək^w(uʔ)* ‘to cover (w. blanket)’)
 NMLZ-cover~C₂-PL
 ‘blankets’ (Modrow, 1971, p.23, Quinault)

Though Modrow (1971) does not mention reduplication in the formation of diminutives, the forms in (636) suggest that the addition of the diminutive suffix also triggers this same process.

- (636) a. *músmus~s-uʔ* (cf. *músmus* ‘cow’)⁴⁸
 cow~C₂-DIM
 ‘small cow’ (Modrow, 1971, p.353, Quinault)
- b. *cáʔalaʔ~ʔ-uʔ* (cf. Cowlitz *celʔ* ‘lake’)
 lake~C₂-DIM
 ‘small lake’ (Modrow, 1971, p.353, Quinault)

Given that C₂ “reduplication” occurs in combination with both plural and diminutive non-reduplicative suffixes, this does not appear to involve a reduplicative morpheme.

⁴⁸The C₁C₂ reduplication in (636a) is lexicalized.

A.4.4 Upper Chehalis Reduplication

Attested reduplication in Upper Chehalis is summarised in Table A.20.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
COS			✓	
SUPR			✓	

Table A.20: Reduplication in Upper Chehalis

A.4.4.1 Upper Chehalis Sources and Transcription

Examples come from Kinkade (1963), Kinkade (1985), and Kinkade (1995). Unreduplicated forms come from Kinkade (1991), as well as the source materials. Many instances of [ə] are omitted in the source materials; I do not reinsert them in most places. Upper Chehalis vowel length, which marks diminutivity, is discussed in Chapter 6 (Section 6.6.2.2.1).

A.4.4.2 Upper Chehalis Pluractional C₁C₂ Reduplication

C₁C₂ reduplication is used to mark pluractionality in Upper Chehalis. The first two consonants of the root are copied, as shown in (637). If the root has a moraic vowel, the vowel is copied too, as shown in (638).

- (637) a. s-pət<**pt**>q^wx^w (cf. *spətq^wx^w* ‘tonight’)
 NMLZ-night<PL>
 ‘night by night’ (Kinkade, 1991, p.98, Upper Chehalis)
- b. s-təq̣<**tq̣**>x^w-t-n (*təq̣x^wm̥t* ‘slap’)
 IPFV-slap<PL>-TR-IPFV
 ‘They (sg.) are slapping it again and again.’ (Kinkade, 1995, p.358, Upper Chehalis)

- (638) a. s-čís<čís>-n (cf. *sčís* ‘come’)
 IPFV-come~PL-3SUBJ.IPFV
 ‘They (sg.) are coming from time to time.’ (Kinkade, 1995, p.358, Upper Chehalis)
- b. s-pán<pan>api-t-n (cf. *spánapim̓* ‘tree falls’)
 IPFV-tree.falls<PL>-TR-3SUBJ.IPFV
 ‘They (sg.) are felling trees.’ (Kinkade, 1995, p.358, Upper Chehalis)
- c. ?é:ł<?el>n (cf. *?im̓* ‘eat’)
 eat<PL>
 ‘eat slowly away’ (Kinkade, 1963, p.20, Upper Chehalis)
- d. ?ac-tup<tup>áy-m (cf. *tupáyn* ‘become decayed’)
 STAT-decayed<PL>-PFV.MID
 ‘It is decaying in spots.’ (Kinkade, 1995, p.358, Upper Chehalis)

There are some instances of C₁C₂ reduplication with roots that surface with [a] in the reduplicant, instead of [ə]. Examples are given in (639).⁴⁹

- (639) a. ?it wəq̓~waq̓-ł (cf. *wəq̓á:lač̓a* ‘runner’)
 PFV PL~run-INTR
 ‘They (sg.) ran around, they (sg.) ran to and fro, they (sg.) ran back and forth.’ (Kinkade, 1995, p.358, Upper Chehalis)
- b. łəx̑~łax̑-n (cf. *łəx̑* ‘burn’)
 burn~PL-TR/3OBJ
 ‘start fires here and there’ (Kinkade, 1995, p.358, Upper Chehalis)

C₁C₂ pluractional reduplication in Upper Chehalis is associated with a distributive function: events must be distributed in time or space (Kinkade, 1995). Example (637b) shows distribution in time, while example (638d) shows distribution in space.

⁴⁹[a] is sometimes found with these roots outside of reduplication in some words; Kinkade (1991, p.194) gives *łax̑áča* ‘burn one’s hand’, for example, which has the same root as (639b).

A.4.4.3 Upper Chehalis Change-of-State (“Slow”) C₂ Reduplication

Change-of-state C₂ reduplication in Upper Chehalis is formed by copying the stressed vowel and the following consonant, as shown in (640).

- (640) a. s-típ<**ip**>i-w-n (cf. *típł* ‘get dark’)
IPFV-dark<COS>-INCH-IPFV
‘It gets dark gradually.’ (Kinkade, 1985, p.193, Upper Chehalis)
- b. *łiś-áw*~**aw**-m (cf. *łiś* ‘cold’)
cold-INCH~COS-MID
‘It got cold slowly.’ (Kinkade, 1985, p.191, Upper Chehalis)

The vowel in C₂ reduplication may be omitted as part of regular vowel deletion rules, though Kinkade (1985, p.190) notes that the precise factors that condition this deletion are unclear. Examples with vowel deletion are given in (641).

- (641) a. *yép*~**pł** (root: *yp-*)
walk~COS-INTR
‘He walked slowly back and forth.’ (Kinkade, 1985, p.193, Upper Chehalis)
- b. s-*łéq^w*~**q^w**-w-n (root: *łq^wi-*)
IPFV-climb~COS-INCH-IPFV
‘They (sg.) climb a little slowly.’ (Kinkade, 1985, p.192, Upper Chehalis)
- c. s-pət~**t**-w-n (root: *pt-*)
IPFV-stick~COS-INCH-IPFV
‘It sticks and sticks slowly.’ (Kinkade, 1985, p.192, Upper Chehalis)

Change-of-state C₂ reduplication applies to verbs; it is frequently found in combination with the inchoative suffix *-(a)w*. It refers to a change of state that unfolds slowly or gradually (Kinkade, 1985, p.189).

A.4.4.4 Upper Chehalis Superlative C₂ Reduplication

Superlative C₂ reduplication has the same form as change-of-state C₂ reduplication: it is formed by copying the consonant and vowel following

the stressed vowel, as shown in (642).⁵⁰ The vowel may be deleted, as shown in (643), though the circumstances surrounding vowel deletion or retention are unclear.

- (642) a. $sx^w-ʔáw<aw>t$ (cf. Lillooet $sʔawt$ ‘behind’)
 SUPR-behind<SUPR>
 ‘the last one’ (Kinkade, 1985, p.193, Upper Chehalis)
- b. $sx^w-ǰúk^w \sim uk^w-t-s$ (cf. $ǰuk^w$ ‘high’)
 SUPR-high~SUPR-INTR-SUPR
 ‘highest one’ (Kinkade, 1985, p.193, Upper Chehalis)
- c. $sx^w-tó:m \sim um$ (cf. $tó:mylps$ ‘short-tailed’)
 SUPR-short~SUPR
 ‘the shortest’ (Kinkade, 1985, p.194, Upper Chehalis)
- (643) a. $sx^w-qóx \sim x-s$ (cf. $qəxt$ ‘many’)
 SUPR-many~SUPR-SUPR
 ‘most’ (Kinkade, 1963, p.21, Upper Chehalis)
- b. $sx^w-ǰúk^w \sim k^w-st$ (cf. $ǰuk^w$ ‘high’)
 SUPR-high~SUPR-SUPR
 ‘highest’ (Kinkade, 1963, p.21, Upper Chehalis)
- c. $sx^w-čap-us \sim s$ (cf. $čapálwn$ ‘strong-looking’)
 SUPR-strong-face~SUPR
 ‘strongest’ (Kinkade, 1985, p.193, Upper Chehalis)
- d. $sx^w-čəp \sim p$ (cf. $čapálwn$ ‘strong-looking’)
 SUPR-strong-face~SUPR
 ‘the strongest’ (Kinkade, 1985, p.193, Upper Chehalis)

The type of C_2 reduplication shown in (642) and (643) is found with adjectives; it is used in superlative constructions. Superlative C_2 reduplication is found in combination with the superlative marking $sx^w-...(-st)$.

⁵⁰The vowel lowers in the stressed syllable when it is lengthened in (642c): /u/ > [o:].

A.5 Tillamook Reduplication

Attested reduplication in Tillamook is summarised in Table A.21.

	C ₁ C ₂	C ₁	C ₂	V ₁
PL	✓			
DIM		✓		
PL/CONT		✓		
COS			✓	

Table A.21: Reduplication in Tillamook

A.5.0.1 Tillamook Sources and Transcription

Examples come from Edel (1939), Egedal and Thompson (1996), Kroeber (1999), and Kuipers (2002); unreduplicated forms come from the original sources. Stress is not reliably marked in the Tillamook materials; vowel quality also varies. I use Egedal and Thompson (1996) and Kroeber (1999) to help guide transcription and glossing decisions.

Reduplicated consonants in onsets in Tillamook are deglottalised and voiced, which is why /d/ surfaces in C₁C₂ reduplication in (415a) below. See L. C. Thompson and Thompson (1985, pp.140–142) for discussion of deglottalisation and voicing in Tillamook.

A.5.1 Tillamook Plural/Pluractional C₁C₂ Reduplication

The first two consonants of the root are copied in C₁C₂ reduplication in Tillamook, as shown in (644).⁵¹ Examples are given in (644a) and (644d).

- (644) a. **dən**~təni (cf. *təni* ‘ear’)
 PL~ear
 ‘ears’ (Egedal & Thompson, 1996, p.3, Tillamook)
- b. **ṭəh**~ṭəh-əys-əw (cf. *ṭəhəys* ‘to sleep’)
 PL~close-eye-MID
 ‘They were all sleeping together.’ (Edel, 1939, p.15, Tillamook)

⁵¹ Voiceless stops are voiced in reduplication (Edel, 1939, p.14)

- c. c-**yəl**~yəl-əwi(n) (cf. *cyəlháʔwin* ‘turn s.t. over’)
PL~turn.over-RLT-DRV
‘crooked’ (Kuipers, 2002, p.130, Tillamook)
- d. **gəl**~qels (root: *qels-*)
PL~cook.in.earth.pit
‘cook in pit’ (Kuipers, 2002, p.87, Tillamook)
- e. c-**ʔəh**~ʔah-əwí-n-i (cf. *ʔaha* ‘carry’)
STAT-PL~take-RLT-DRV-LS.SUBJ
‘I always take it.’ (Edel, 1939, p.14, Tillamook)

A realisation of plural/pluractional C₁C₂ reduplication, without the first consonant copied, is shown in (645). This is discussed further in Chapter 6 (Section 6.5.3.3.2).

- (645) a. de š-**q**^w~təq^w-ən (root: *tq^w*- ‘to break’)
DET DESID-PL~break-3
‘They tried to break it.’ (Edel, 1939, p.15, Tillamook)
- b. s-**q**~təq-il (cf. *səqísi* ‘I sit down beside them (sg.).’)
NMLZ-PL~sit-DVL
‘Someone is sitting.’ (Egesdal & Thompson, 1996, p.11, Tillamook)

C₁C₂ reduplication is found with nouns and verbs. With nouns, C₁C₂ reduplication marks plural entities, as shown in (644a) above. With verbs, it marks plural participants, as shown in (644b) and (645a), or events, as shown in (644e). While (644e) involves a habitual reading with distribution in time, there is not enough data to determine whether distribution in space is also required.

A.5.2 Tillamook Diminutive C₁ Reduplication

Diminutive C₁ reduplication copies the first consonant of the root, with [u] between the identical consonants, as shown in (646).

- (646) a. **du**~tiyəlhu (cf. *tiyəlhu* ‘man’)
DIM~man
‘boy’ (Edel, 1939, p.14, Tillamook)

- b. **du**~tuq^wúsu (cf. *tuq^wúsu* ‘beaver’)
 DIM~beaver
 ‘little beaver’ (Edel, 1939, p.14, Tillamook)
- c. k(ɬ)-**šu**~šenš (cf. *šenš* ‘rock’)
 PL-DIM~rock
 ‘gravel’ (Edel, 1939, p.14, Tillamook)
- d. s-**wu**-wiłac (cf. *swiłac* ‘woman’)
 NMLZ-DIM~woman
 ‘girl’ (Kroeber, 1999, p.142, Tillamook)
- e. s-**?u**~ʔəhɬ (cf. *ʔəhɬ* ‘flounder’)
 NMLZ-DIM~flounder
 ‘little flounder’ (Edel, 1939, p.14, Tillamook)

Diminutive reduplication is only attested with nouns in Edel (1939); it marks a small referent. It is unclear if diminutive reduplication is possible with adjectives and verbs in Tillamook.

A.5.3 Tillamook Pluractional and Continuative C₁ Reduplication

C₁ reduplication can also be used to mark either a pluractional or aspectual (continuative) function; the first consonant and a root vowel of the root are reduplicated, as shown in (647–648).⁵² It is not clear if pluractional and continuative C₁ reduplication are two distinct processes or the same one with different meanings when applied to different roots.

- (647) a. gəʔ **či**~čic-iɬ (cf. *čic* ‘reach’)
 FUT PL~reach-?
 ‘They’ll get to them (sg.).’ (Edel, 1939, p.15, Tillamook)
- b. t gəʔ **?i**~ʔíwuč (cf. *?iwuč* ‘go upstream’)
 DET FUT PL~go.upstream
 ‘They will paddle upstream.’ (Edel, 1939, p.15, Tillamook)

⁵²The ? in the gloss for (648a) comes from Egesdal and Thompson (1996, p.8).

- c. $\text{ɬe}=\text{wə-s-nə-niʃ}$ $\text{cik(ɬ)}=\text{sig}^{\text{w}}\text{ək}^{\text{w}}\text{əlwin}$
 PL=back-ASP-PL-arrive DEM(PL)-child
 ‘The young men returned.’ (cf. $\text{g}^{\text{w}}\text{u}ʔ \text{wa}ɬ \text{niʃ(s)}\text{tx}^{\text{w}}\text{i}$ ‘I’m coming with it.’) (Kroeber, 1999, p.388, Tillamook)
- d. $\text{nəʃ-ʃə-ʃəl-wi-t-əg}^{\text{w}}\text{əl}$ (cf. $\text{ʃəlwitəg}^{\text{w}}\text{əl}$ ‘They hate each other.’)
 LOC-PL~dislike-RLT-TR-RCP
 ‘They hate each other.’ (Egesdal & Thompson, 1996, p.20, Tillamook)
- (648) a. $\text{də c-yə} \sim \text{yələh-ačəʔ-ən}$ (cf. cyələháʔwin ‘turn s.t. over’)
 DET STAT-CONT~turn-hand-DRV
 ‘He is cheating on him.’ (Egesdal & Thompson, 1996, p.4, Tillamook)
- b. $\text{ci g}^{\text{w}}\text{ə} \sim \text{g}^{\text{w}}\text{əh-əs-wəʃ-əs}$
 DEM CONT~call-PUR-1SG.OBJ-3SUBJ
 ‘If they call me...’ (cf. $\text{ci g}^{\text{w}}\text{u g}^{\text{w}}\text{əhəsi}$ ‘I’m going to call him.’) (Egesdal & Thompson, 1996, p.6, Tillamook)
- c. $\text{c-yə} \sim \text{yeh-s-íwɬ}$ $\text{g}^{\text{w}}\text{əlg}^{\text{w}}\text{ələ}$ (cf. yehis ‘see’)
 STAT-CONT~cause-eye-2PL.OBJ EMPH.2PL
 ‘They (sg.) are looking at you folks.’ (Egesdal & Thompson, 1996, p.9, Tillamook)
- d. $\text{ne} \sim \text{né} <ʔ> \text{g-il}$ (cf. negil ‘run’)
 CONT~run<INCH>-DVL
 ‘run’ (Egesdal & Thompson, 1996, p.11, Tillamook)

A.5.4 Tillamook Change-of-State C₂ Reduplication

C₂ reduplication is attested in Tillamook and marks a change of state. The second consonant of the root is copied, as shown in (649); an epenthetic [ə] is present.

- (649) a. $\text{de c-q}^{\text{w}}\text{x}^{\text{w}} \sim \text{əx}^{\text{w}}\text{-əw-i}$ (cf. $\text{q}^{\text{w}}\text{əx}^{\text{w}}$ ‘sleep’)
 DET STAT-sleep~COS-MID-1SG.SUBJ
 ‘I fell asleep’ (Edel, 1939, p.16, Tillamook)
- b. ʔəncə $\text{g}^{\text{w}}\text{əʔ yəɬ} \sim \text{əɬ-ní-wʃ}$ (cf. yəɬ ‘to be next’)
 EMPH.1SG FUT next~COS-NTR-1SG.OBJ
 ‘He will be next to me.’ (Egesdal & Thompson, 1996, p.9, Tillamook)

- c. tən~ən-s-ə-t-i (cf. *ʔe stənəcityəʔ* ‘We burned ourselves.’)
 burn~COS-PUR-DRV-TR-1SG.SUBJ
 ‘I burned it for him.’ (Egesdal & Thompson, 1996, p.14, Tillamook)
- d. de (s-)səḡ^w~əḡ^w-əwí-n-i (cf. *səḡ^w* ‘to jump’)
 DET (STAT)-jump~COS-RLT-DRV-1SG.SUBJ
 ‘I jumped over it.’ (Egesdal & Thompson, 1996, p.14, Tillamook)
- e. də ləš~əš-əwi-c-í (root: *ləš-*)
 DET angry~COS-RLT-2SG.OBJ-1SG.SUBJ
 ‘I am angry at you.’ (Egesdal & Thompson, 1996, p.14, Tillamook)
- f. ʔe c-wəḡ^w<əḡ^w>ál-i (root: *wəḡ^w-*)
 DET STAT-live<COS>-1SG.SUBJ
 ‘I am alive.’ (Egesdal & Thompson, 1996, p.15, Tillamook)

A.6 Bella Coola Reduplication

C₁C₂ reduplication in Bella Coola is associated with a range of functions including plural, continuative, and diminutive (Nater, 1978); diminutivity is also marked with C₁ reduplication.

C ₁ C ₂	C ₁	C ₂	V ₁
✓	✓		

Table A.22: Reduplication in Bella Coola

I discuss plural C₁C₂ reduplication in Section A.6.2; I treat the continuative as a subtype of pluractional. Diminutive C₁C₂ and C₁ reduplication are discussed in Section A.6.3.

Patterns shared by plural and diminutive reduplication are discussed together in Section A.6.4. Further discussion of Bella Coola reduplication can be found in Chapter 6, including discussion of linearisation (Section 6.4.2), spirantisation (Section 6.6.2.2.2), and glottalisation-triggered lengthening (Section 6.7.1.2).

A.6.1 Bella Coola Sources and Transcription

Sources include Saunders and Davis (1972), Nater (1978), Nater (1984), Bagemihl (1991), and Nater (1990).

Stress is not marked in the examples because it is not indicated in the source materials. Many instances of [ə] are omitted in the source material. For forms with diminutive C₁C₂ reduplication with a sonorant in C₂ position, I re-transcribe the first consonant-sonorant (CS) instance (which marks the diminutive) as “CŞ”, following Bagemihl (1991), though I suspect these are realised as “CəS”. I do not add vowels not present in the source nor the unreduplicated forms.

A.6.2 Bella Coola Plural C₁C₂ Reduplication

The first CVC sequence in the word, excluding any prefixes, is copied in plural C₁C₂, as shown in (650). The reduplicative morpheme is infixes following the first CVC in the word.

- (650) a. mil<**mil**>ix^w-ɬp (cf. *milix^w* ‘bearberry’)
 bearberry<PL>-plant
 ‘bearberry/plant’ (Nater, 1978, p.3, Bella Coola)
- b. s-tan~**tan**-mc (cf. *stan* ‘mother’)
 NMLZ-mother~PL-relatives
 ‘female ancestors’ (Nater, 1978, p.2, Bella Coola)
- c. ck^w-aɬ~**k^waɬ** (cf. *ck^waɬ* ‘heavy feet’)
 heavy-feet~PL
 ‘to walk heavily’ (Bagemihl, 1991, p.609, Bella Coola)
- d. x^wul~**x^wul**-m (cf. *x^wulta* ‘drill’)
 drill~PL-MID
 ‘make fire with drill’ (Nater, 1978, p.2, Bella Coola)
- e. ya-n<**yan**>k-layx-m (cf. *yank* ‘to pole a boat adroitly’)
 good-bottom<PL>-AUT-MID
 ‘to be poling a boat’ (Nater, 1990, p.168, Bella Coola)

If a root has obstruents in both C₁ and C₂ position, the vowel in the reduplicated syllable may be omitted, as shown in (651).

- (651) a. ʔa-sɕ-aɬ~**ɕɬ** (cf. *ʔasɕaɬ* ‘waterfall’)
 IPFV-disperse-motion~PL
 ‘heavy rain’ (Nater, 1978, p.2, Bella Coola)
- b. ʔaɬ-q^wəɬ~**q^wɬ**-anɬ (cf. *ʔaɬq^wəɬ* ‘printed’)
 IPFV-design/write~PL-cloth
 ‘variegated dress’ (Nater, 1978, p.2, Bella Coola)

With words with a bilabial or alveolar stop in C₂ position, [n] replaces the stop in the reduplicant, as shown in (652a–b); velar and uvular stops in C₂ position undergo spirantisation, as shown in (652c). I analyse this as a prefix.

- (652) a. **sun**~supt-c (cf. *suptc* ‘I whistle.’)
 PL~whistle-1 SG.SUBJ
 ‘I whistle (iterative).’ (Saunders & Davis, 1972, p.9, Bella Coola)

- b. **cin**~citumix-c (cf. *citumixc* ‘I take a nap.’)
 PL~nap-1 SG.SUBJ
 ‘I take a nap (iterative).’ (Saunders & Davis, 1972, p.9, Bella Coola)
- c. **?un**~?utak-c (cf. *?utak* ‘I vomit.’)
 PL~vomit-1 SG.SUBJ
 ‘I vomit (iterative).’ (Saunders & Davis, 1972, p.9, Bella Coola)
- d. **six**~sik^w (cf. *sik^w* ‘pull’)
 CONT~pull
 ‘pull (cont.)’ (Bagemihl, 1991, p.601, Bella Coola)

With obstruent-only words (i.e., those without vowels), [i] is epenthesized and copied, as shown in (653b–c).

- (653) a. **ṭix**~ṭixt (cf. *ṭxt* ‘stone’)
 PL~stone
 ‘large stones’ (Bagemihl, 1991, p.608, Bella Coola)
- b. **ḥix**~ḥiq^w (cf. *ḥq^w* ‘sob’)
 PL~sob
 ‘to be sobbing’ (Nater, 1990, p.60, Bella Coola)

C₁C₂ reduplication in Bella Coola includes plural and pluractional uses. It marks plural entities with nouns (e.g., 650b) and plural events with verbs. The pluractional uses are labelled as “continuative” by Nater (1984); they include events that are distributed in time, such as the iterative forms in (653a–c). It is not clear if a plural participant reading is possible with verbs in Bella Coola. There are a number of words where the contribution of plural C₁C₂ reduplication is unclear or where it may be lexicalised, such as in (650a).

A.6.3 Bella Coola Diminutive C₁C₂ and C₁ Reduplication

Diminutivity in Bella Coola is marked by a combination of the suffix *-i* (realised as *-y* or *-yi* following a vowel), reduplication (either C₁ or C₁C₂ type), and vowel lengthening. All three exponents of the diminutive may occur in the same word, as shown in an example with C₁C₂ reduplication in (654a) and one with with C₁ reduplication in (654b).

- (654) a. s-x^wpa<p>ni<:>ɬ-i (cf. *sx^wpaniɬ* ‘deer’)
 NMLZ-deer<DIM><DIM>-DIM
 ‘small deer’ (Nater, 1990, p.107, Bella Coola)
- b. x^wŋ<x^wn>a<:>ɬ-i (cf. *x^wnaɬ* ‘spring of water’)
 spring<DIM><DIM>-DIM
 ‘small spring of water’ (Bagemihl, 1991, p.615, Bella Coola)

The three exponents of the diminutive need not surface concurrently in a single word. The diminutive suffix *-i* or reduplication are dropped if the word already ends in an /i/ or /y/ or has lexicalised reduplication, respectively. It is therefore possible to get diminutives without diminutive reduplication if the word already has reduplication, as shown in (655).⁵³ Vowel lengthening is often absent, though the conditioning factors are unclear at present. I focus only on reduplication in the remainder of this section, but vowel length is discussed further in Chapter 6 (Section 6.7.1.2).

- (655) a. naɣnx-i (cf. *naxnx* ‘mallard duck’)
 mallard.duck-DIM
 ‘small mallard duck’ (Nater, 1990, p.78, Bella Coola)
- b. mami<:>s-i (cf. *mamis* ‘black fly’)
 black.fly<DIM>-DIM
 ‘small black fly’ (Bagemihl, 1991, p.598, Bella Coola)

The choice between C₁ or C₁C₂ reduplication is determined by whether a moraic vowel or a sonorant comes first in the word. C₁ is observed when a vowel other than [ə] comes before the first sonorant consonant, while C₁C₂ is found if a sonorant consonant comes before the first non-[ə] vowel.⁵⁴ Both types of reduplication are positioned relative to the first vowel or sonorant consonant in the word; like plural reduplication, diminutive reduplication can target suffixal material if the first vowel does not occur in the root. Diminutive C₁ reduplication is discussed in Section A.6.3.1; diminutive C₁C₂ reduplication is discussed in Section A.6.3.2.

⁵³There are some other diminutive forms without reduplication, such as *kasmiwi* from *kasmiw* ‘golden eagle’ (Nater, 1990, p.44)

⁵⁴I have found one example of an obstruent-only word with diminutive marking and it does not undergo reduplication: *ɬxti* ‘stone (dim.)’ from *ɬxt* ‘stone’ in (Newman, 1937, p.10).

Examples in the following sections show that both diminutive C_1 and C_1C_2 reduplication are attested with nouns. Newman (1971, p.34) notes that diminutive reduplication is “used primarily with nominal [roots]”; it marks a small entity. While it is not clear if diminutive reduplication is permitted with verbs, there is one example of diminutive reduplication applied to adjectives, which are given in (656).⁵⁵ The translation of this diminutive adjective suggests a small amount reading, rather than small participants.

- (656) nu-qṇ<ḳṇ>k-i (cf. *qənk* ‘deep’)
 inside-deep<DIM>-DIM
 ‘water is a bit deep’ (Nater, 1990, p.94, Bella Coola)

A.6.3.1 Bella Coola Diminutive C_1 Reduplication

C_1 reduplication marks diminutivity when the first moraic segment in an unreduplicated form is a vowel. Words with C_1 reduplication have a copy of the first CV sequence of the word, as shown in (657).

- (657) a. su<su>m-i<: > (cf. *sum* ‘pants’)
 trousers<DIM>-DIM<DIM>
 ‘small trousers’ (Nater, 1984, p.109, Bella Coola)
- b. qa<ḳa>yt-i (cf. *qayt* ‘hat’)
 hat<DIM>-DIM
 ‘small toadstool’ (Bagemihl, 1991, p.598, Bella Coola)
- c. tqḷa<: ><ḷa>-y (cf. *tqḷa* ‘knife’)
 knife<DIM><DIM>-DIM
 ‘small knife’ (Bagemihl, 1991, p.609, Bella Coola)
- d. qḷpa<: ><ḷpa>-yi (cf. *qḷpa* ‘egg’)
 egg<DIM><DIM>-DIM
 ‘small egg’ (Bagemihl, 1991, p.609, Bella Coola)

⁵⁵Setting aside the vowel length and -i associated with the diminutive, the fact that diminutive reduplication is almost exclusively recorded with nouns means that forms with C_1C_2 reduplication are not ambiguous: a verb with C_1C_2 reduplication is probably pluractional, not diminutive.

The copied vowel in C₁ reduplication may be omitted in words where the copied consonant is an obstruent and the following consonant is also an obstruent (i.e., an obstruent-vowel-obstruent sequence reduplicates as obstruent-vowel-<**obstruent**>-obstruent, while an obstruent-vowel-sonorant sequence reduplicates as obstruent-vowel-<**obstruent-vowel**>-sonorant).

- (658) a. tp-i<: ><p>q^w-i (cf. *tpi:x^w* ‘cover, lid’)
spread-head-DIM<DIM>-DIM
‘small cover/lid’ (Bagemihl, 1991, p.609, Bella Coola)
- b. q^wa<: ><x^w>|s-i (cf. *q^wals* ‘fir tree needles’)
navel<DIM><DIM>-DIM
‘fir tree needles (dim.)’ (Bagemihl, 1991, p.599, Bella Coola)
- c. ka<: ><k>pay (cf. *kapay* ‘humpback salmon’)
humpback.salmon<DIM><DIM>
‘small humpback salmon’ (Bagemihl, 1991, p.598, Bella Coola)

A.6.3.2 Bella Coola Diminutive C₁C₂ Reduplication

C₁C₂ reduplication marks diminutivity when the first moraic segment in an unreduplicated form is coda consonant (i.e., a sonorant consonant). Diminutive C₁C₂ reduplication copies the first coda sonorant and the consonant immediately before, as shown in (659).

- (659) a. s-tq^wl<q^wl>-us-i (cf. *stq^wlus* ‘black bear snare’)
NMLZ-weave<DIM>-face-DIM
‘small black bear snare’ (Bagemihl, 1991, p.615, Bella Coola)
- b. s-kṃ<km>a-y (cf. *skma* ‘moose’)
NMLZ-moose<DIM>-DIM
‘small moose’ (Bagemihl, 1991, p.615, Bella Coola)
- c. s-tx^wṃ<x^wm>-i (cf. *stx^wṃ* ‘floor mat’)
NMLZ-floor.mat<DIM>-DIM
‘small floor mat’ (Bagemihl, 1991, p.609, Bella Coola)

- d. s-tɲ<tn>-i<ɪ> (cf. *stɲ* ‘tree’)
 NMLZ-tree<DIM>-DIM<DIM>
 ‘small tree’ (Bagemihl, 1991, p.609, Bella Coola)

Though C₁C₂ reduplication can mark diminutives or plurals, the two types can be distinguished because the diminutives have the suffix *-i* and only copy two segments; plural C₁C₂ reduplication copies a vowel in addition to two consonants.

A.6.4 Phonological Patterns Shared between Different Types of Reduplication in Bella Coola

Plural and diminutive reduplication pattern together with respect to how ejectives and /ʔ/ are reduplicated. When an ejective undergoes fission, the first of the two output segments is [ʔ], as shown in (660). In previous work (e.g., K. Carlson, 1997), reduplication with the form in (660b) is labeled as V₁ reduplication.

- (660) a. **ʔix**~ti:xɬala-m (cf. *tixɬalam* ‘to hop on one leg’)⁵⁶
 PL~hop.on.one.leg-MID
 ‘hop on one leg (cont.)’ (Bagemihl, 1991, p.601, Bella Coola)
- b. **ʔi**~tixɬala-y (cf. *tixɬala* ‘robin’)
 DIM~robin-DIM
 ‘small robin’ (Bagemihl, 1991, p.598, Bella Coola)

When C₁ is a glottal stop, C₁C₂ reduplication has a reduced form (see analysis in Bagemihl, 1991), as shown in (661). These examples appear to have C₂ reduplication.

- (661) ʔix-ʔak^w<**k^w**>a (cf. *ʔak^wa* ‘to buy (something)’)
 INT.DIST-buy<PL>
 ‘to be shopping’ (Nater, 1984, p.111, Bella Coola)
- (662) ʔa<ɪ>c<**c**>a-**yi** (cf. *ʔacaya* ‘fox’)
 fox<DIM><DIM>-DIM
 ‘fox (dim.)’ (Nater, 1984, p.111, Bella Coola)

⁵⁶This form also has a x̣ ~ x alternation (see Nater 1984, p.108).

Appendix B

Survey of Multiple Reduplication Patterns

Multiple reduplication, the combination of more than one reduplicative morpheme in a single word, is attested in 15 of the 23 Salish languages. This section summarises documented combinations of reduplicative morphemes for each language with a discussion of associated meanings where possible.

Table B.1: Multiple Reduplication in Salish Languages by Branch

	Yes	No	Lacks Sufficient Data
Central Salish (n=10)	7	2	1
Tillamook (n=1)	1	0	0
Tsamosan (n=4)	0	3	1
Interior Salish (n=7)	7	0	0
Bella Coola (n=1)	0	1	0
Total (n=23)	15	6	2

Multiple reduplication is absent in Nooksack, Squamish, Cowlitz, Lower Chehalis, Upper Chehalis, and Bella Coola. There is not enough available data to conclude whether multiple reduplication was present in Quinault and Pentlatch, though comparison with closely related languages suggests that it is likely to have been absent in Quinault and present in Pentlatch.

For languages with attested multiple reduplication, the appendix is sorted by branch: Interior Salish, Central Salish, and Tillamook. The languages are organised alphabetically within branch. The amount of detail and description given for each language and each reduplicative process varies based on available documentation. However, multiple reduplication has not been elicited systematically in many languages; some missing combinations may reflect accidental gaps.

Transcription decisions and the source of non-reduplicated forms given for comparison are the same as those detailed in Appendix A. Attested combinations are marked in tables for each language with ✓; × marks ungrammaticality for languages where negative data is available (i.e., Comox-Sliammon and Lillooet). The absence of × in any given cell does not indicate ungrammaticality. Given that the comparisons are pairwise, redundant cells in the table are marked in gray (i.e., DIM-COS is identical to COS-DIM).

Given the additional complexity of forms with multiple reduplication, I have chosen to largely follow the transcription in the original sources with respect to [ə]. Readers should be particularly careful where there appear to be clusters with sonorants as there is likely to be a [ə] that is present in the surface form but absent in the abstract phonemic transcription.

B.1 Interior Salish Multiple Reduplication

B.1.1 Coeur d’Alene Multiple Reduplication

Attested multiple reduplication in Coeur d’Alene is summarised in Table B.2.

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	COS C ₂	PL V ₁
PL	C ₁ C ₂			✓	✓	
PDEN	C ₁ C ₂			✓		
DIM	C ₁				✓	
COS	C ₂					
PL	V ₁					

Table B.2: Multiple Reduplication in Coeur d’Alene

Examples come from Reichard (1938), Barthmaier (1996), Lyon (2010), and Doak (2023).

B.1.1.1 Diminutive Plural

In diminutive plural forms, diminutive C₁ reduplication is leftmost (excluding prefixes); plural C₁C₂ reduplication follows. Examples are provided in (663).

- (663) a. *šə~šiw~šewtəm* (cf. *ššiwətəm* ‘girl’)
 DIM~PL~girl
 ‘little girls’ (Barthmaier, 1996, p.37, Coeur d’Alene)
- b. *?en-g^wə~g^wəx^w~g^wəx^w-ínč* (cf. *?etg^wéx^w* ‘be hanging’)
 LOC-DIM~PL~hang-interior
 ‘Little ones hung here and there on the wall.’ (Reichard, 1938, p.836, Coeur d’Alene)
- c. *čə~čəs~čəs-qín-ən* (cf. *česp* ‘annihilate, consume, destroy, use up’)
 DIM~PL~?-head-?
 ‘catbird’
 Literal: ‘a little fellow who repeatedly picks things at head’ (Lyon, 2010, p.25, Coeur d’Alene)

- d. hən-**ḥə**~**ḥə**~ḥə-əm (cf. *ḥə*ʔč ‘string breaks’)
 LOC-DIM~PL~break.string-mid
 ‘The little one sobbed incessantly.’
 Literal: ‘He (little one) broke string inside again and again.’ (Lyon, 2010, p.48, Coeur d’Alene)
- e. **ʔe**~**ʔe**~ʔeńis (cf. *ʔe*ńis ‘leave’)
 DIM~PL~leave
 ‘little ones went off one by one’ (Reichard, 1938, p.836, Coeur d’Alene)

With a noun, this combination marks a plurality of small entities, as in (663a). The combination marks a small argument (or arguments) participating in multiple events when applied to a verb; events may be distributed in space or time, as shown in (663b) and (663c–e), respectively.

B.1.1.2 Diminutive and Property-Denoting

In words with diminutive and property-denoting reduplication, diminutive C₁ reduplication is positioned before the root; C₁C₂ reduplication follows the root. Examples are given in (664).

- (664) a. sə~siy~**siy**-ús (root: *siy*- ‘smart’)
 DIM~PDEN~smart-face
 ‘little one is smart’ (Doak, 2023, Coeur d’Alene)
- b. sya-**q**^w~q^waʔ<q^weʔ>eʔl (cf. *q^waʔq^weʔlúl* ‘talkative’)
 agentive-DIM~talk~PDEN
 ‘lawyer, little speaker’ (Lyon, 2010, p.66, Coeur d’Alene)

Property-denoting reduplication applies to adjectives and marks a state or property; diminutive reduplication marks that a small argument possesses this state or property.

B.1.1.3 Plural Change of State

In words with plural and change-of-state reduplication, plural C₁C₂ reduplication surfaces as a prefix before the root; change-of-state C₂ reduplication follows the

second consonant of the root. Examples are given in (665).

- (665) a. **g^wár~g^war~ər** (root: *g^war-* ‘scrape’)
 PL~scrape~COS
 ‘They got scraped.’ (Reichard, 1938, p.836, Coeur d’Alene)
- b. **t-čəl~čəl~əl-m-íce?** (cf. *čéliš* ‘S/he stood up.’)
 LOC-PL~upright~COS-MID-cover
 ‘set up on end all over their hides’ (Doak, 2023, Coeur d’Alene)

The combination of C₁C₂ and C₂ reduplication is found with verbs: it is used to mark a change-of-state with multiple participants.

B.1.1.4 Diminutive Change of State

In words with diminutive and change-of-state reduplication, diminutive C₁ reduplication surfaces as a prefix before the root and change-of-state C₂ reduplication follows the second consonant of the root. Examples are given in (666).

- (666) a. **t̥~t̥k^w~k^w-ílup** (root: *t̥k^w-* ‘begin’)
 DIM~begin~COS-foundation
 ‘made a little camp’ (Doak, 2023, Coeur d’Alene)
- b. **t~təx^w~əx^w-í:::** (cf. *táx^w* ‘quiet’)
 DIM-stop~COS-EXAG
 ‘small one lay dead’ (Doak, 2023, Coeur d’Alene)
- c. **k^wə~k^wél~əl** (cf. *k^welt* ‘warm’)
 DIM~hot~COS
 ‘it became clear, pleasant warm weather came on’ (Reichard, 1938, p.836, Coeur d’Alene)
- d. **t̥ə~t̥ék^w~ək^w** (cf. *t̥ék^wncut* ‘He lay down’)
 DIM~fall~COS
 ‘He fell to the ground.’ (Barthmaier, 1996, p.18, Coeur d’Alene)

The combination of C₁ and C₂ reduplication is applied to verb and adjective roots; the word with multiple reduplication is a verb. This combination is used to mark a

change-of-state with a small argument, as shown in (666a–b). The meaning is less clear in (666c–d).

B.1.2 Moses Columbia Multiple Reduplication

Attested multiple reduplication in Moses Columbia is summarised in Table B.3.

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	REP C ₁	COS C ₂
PL	C ₁ C ₂			✓		✓
PDEN	C ₁ C ₂					
DIM	C ₁					✓
REP	C ₁					
COS	C ₂					

Table B.3: Multiple Reduplication in Moses Columbia

Examples come from Kinkade (1982) and Willett (2003).

B.1.2.1 Diminutive Plural

In diminutive plural forms, diminutive C₁ reduplication is leftmost (excluding prefixes); plural C₁C₂ reduplication follows. Examples are provided in (667).

- (667) a. s-ḡ~ḡaʔ~ḡáʔxn̩-s (cf. *sḡaʔxn̩* ‘shoe’)
 NMLZ-DIM~PL-shoe-3POSS
 ‘her little shoes’ (Willett, 2003, p.280, Moses Columbia)
- b. ḡ~ḡəx~ḡəxn̩ (cf. *kn ḡḡəxn̩áʔ* ‘I’m short.’)
 DIM~PL~short
 ‘short pieces’ (Willett, 2003, p.280, Moses Columbia)

The diminutive plural examples in (667) include a nominal and an adjectival root. These examples refer to a plural entity made up of small individuals. While both types of reduplication allow event readings on the own (i.e., pluractional or diminutive event), it is not clear if these readings are available in multiple reduplication.

B.1.2.2 Plural Change of State

In words with plural and change-of-state reduplication, plural C₁C₂ reduplication surfaces as a prefix before the root and change-of-state C₂ reduplication follows

the second consonant of the root. Examples are given in (668).

- (668) a. **ḥac**~ḥác~**əc** (cf. *ḥac-n* ‘I tied it.’)
 PL~tie~COS
 ‘several knots tangled up’ (Kinkade, 1982, p.70, Moses Columbia)
- b. **kʰ-wəkʷ**~wakʷ~**kʷ**-ɬx (cf. *kʰwákʷukʷ* ‘It disappeared.’)
 POS-PL~disappear~COS-AUT
 ‘They went out of sight.’ (Kinkade, 1982, p.69, Moses Columbia)
- c. **ḥəkʷ**~ḥákʷ~**kʷ** (cf. *ḥákʷ-n* ‘I scratched it.’)
 PL~scratch~COS
 ‘go through bushes and get scratched’ (Kinkade, 1982, p.70, Moses Columbia)
- d. **qʷəḡʷ**~qʷáḡʷ~**ḡʷ** (root: *qʷaḡʷ*- ‘drunk’)
 PL~drunk~COS
 ‘rabid’ (Kinkade, 1982, p.70, Moses Columbia)
- e. **kəs**~kás~**əs** (cf. *kássn* ‘I argue with him.’)
 PL~argue~COS
 ‘argue’ (Kinkade, 1982, p.70, Moses Columbia)

Words with C₁C₂ and C₂ reduplication are verbs. This combination marks a change of state that involves distribution across participants, as shown in (668a–b), or across time and space, as shown in (668c). There are other cases where the contribution of the reduplicative morphemes is unclear, such as in (668d–e).

B.1.2.3 Diminutive Change of State

In words with diminutive and change-of-state reduplication, diminutive C₁ reduplication surfaces as a prefix before the root and change-of-state C₂ reduplication follows the second consonant of the root. Examples are given in (669).

- (669) **ɬ**~ɬút~**ət** (cf. *ɬut-nt-xʷ* ‘you inhaled it’)
 DIM~sob~COS
 ‘half crying’ (Kinkade, 1982, p.67, Moses Columbia)

(670) $\dot{\mathbf{m}} \sim \dot{\mathbf{m}}\dot{\mathbf{i}}\dot{\mathbf{n}} \sim \dot{\mathbf{n}}$ (root: $\dot{\mathbf{m}}\dot{\mathbf{i}}\dot{\mathbf{n}}\text{-}$)

DIM~?~COS

‘wiggle’ (Kinkade, 1982, p.68, Moses Columbia)

It is not possible to discern the meaning of a diminutive change-of-state from available documentation. However, note that both examples in (669) may refer to a change of state that is diminished or small in some way, rather than involving a small participant.

B.1.3 Okanagan Multiple Reduplication

Attested multiple reduplication in Okanagan is summarised in Table B.4.

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	COS C ₂	PL V ₁
PL	C ₁ C ₂			✓	✓	
PDEN	C ₁ C ₂			✓		
DIM	C ₁				✓	
COS	C ₂					
PL	V ₁					

Table B.4: Multiple Reduplication in Okanagan

Examples come from A. Mattina (1973), A. Mattina (1987), and A. Mattina and Peterson (1997).

B.1.3.1 Diminutive Plural

Diminutive plural forms either surface as C₁~C₁C₂, with diminutive before plural, or C₁C₂<C₁>, with diminutive infix within the prefixed plural reduplication. Examples of the C₁~C₁C₂ pattern are given in (671a–d); an example of the C₁C₂<C₁> pattern is given in (671e).

- (671) a. **k^wə~k^wəɾ~k^wɾ-ísxən** (cf. *k^wɾi?* ‘yellow, golden’)
 DIM~PL~yellow/gold-round
 ‘golden coins, round objects, pennies’ (A. Mattina, 1987, p.53, Okanagan)
- b. **pə~pək^w~púk^wla?** (cf. *púk^wla?* ‘ball’)
 DIM~PL~ball
 ‘little balls’ (A. Mattina & Peterson, 1997, p.318, Okanagan)

- c. s-tə<t>ɬ~táɬəm (cf. *stáɬəm* ‘boat’)
 NMLZ-PL<DIM>~boat
 ‘They are little boats.’ (A. Mattina, 1973, p.65, Okanagan)

Nouns are able to undergo plural C₁C₂ and diminutive C₁ reduplication; diminutive plural nouns refer to multiple entities that are small. It is not clear from the available documentation whether this type of multiple reduplication occurs with verbs.

B.1.3.2 Diminutive and Property-Denoting

In words with diminutive and property-denoting reduplication, diminutive C₁ is positioned before the root; C₁C₂ reduplication follows the root. Examples are given in (672).

- (672) a. q^w~q^wáɬ~q^waɬ-t (cf. *q^waɬ* ‘silly, crazy, drunk’)
 DIM~PDEN~silly/crazy/drunk-CHAR
 ‘ornery children’ (A. Mattina, 1987, p.164, Okanagan)
- b. x^wə~x^wiɬ~x^wəɬ-t (cf. *x^wiɬ-nt* ‘break something’)
 DIM~PDEN~break-CHAR
 ‘weak, brittle objects’ (A. Mattina, 1987, p.259, Okanagan)
- c. q^w~q^wil~q^wɪ-t (root: *q^wil-*)
 DIM~waste.time~PDEN-CHAR
 ‘waste time’ (A. Mattina, 1987, p.163, Okanagan)

Though the infixed position of C₁C₂ reduplication in (672) above and the presence of the adjectival marker -t are consistent with analysing the C₁C₂ reduplication as property-denoting, this is not clear from the meaning. The translations associated with (672a–b) suggest plural arguments, which are also small in (672a). The meaning and status of this combination is unclear at present.

B.1.3.3 Plural Change of State

In words with plural and change-of-state reduplication, plural C₁C₂ reduplication surfaces as a prefix before the root and change-of-state C₂ reduplication follows the second consonant of the root. Examples are given in (673).

- (673) a. $\dot{\lambda}\dot{\text{ə}}\dot{\text{x}} \sim \dot{\lambda}\dot{\text{á}}\dot{\text{x}} \sim \dot{\text{ə}}\dot{\text{x}}$ (cf. $\dot{\lambda}\dot{\text{ə}}\dot{\text{xap}}$ ‘grown’)
 PL~grow~COS
 ‘They are elders.’ (A. Mattina, 1973, p.65, Okanagan)
- b. $\text{n-}\mathbf{c}\mathbf{\text{ə}}\mathbf{k}^{\text{w}} \sim \text{c}\mathbf{\text{ə}}\mathbf{k}^{\text{w}} \sim \mathbf{k}^{\text{w}}\text{-i}\mathbf{\text{ł}}\text{-}\mathbf{x}\mathbf{\text{ı}}\mathbf{\text{̃}}$ (cf. $\text{ck}^{\text{w}}\text{am}$ ‘haul’)
 LOC-PL~pull~COS-CONN-leg
 ‘muscle cramps’ (A. Mattina, 1987, p.5, Okanagan)

The combination of plural C_1C_2 and change-of-state C_2 reduplication marks a change in state that is distributed across participants, as in (673a), or is distributed in space and/or time, as in (673b).

B.1.3.4 Diminutive Change of State

When diminutive and change-of-state reduplication are combined, C_1 reduplication surfaces as a prefix; C_2 reduplication surfaces as an infix. Examples are given in (674).

- (674) a. $\mathbf{k}^{\text{w}}\mathbf{\text{ə}} \sim \mathbf{k}^{\text{w}} < \mathbf{\text{ı}} > \mathbf{\text{a}}\mathbf{\text{ł}}$ (cf. $\text{k}^{\text{w}}\text{alt}$ ‘warm’)
 DIM-warm<COS>
 ‘bright’ (A. Mattina, 1987, p.53, Okanagan)
- b. $\mathbf{\text{c}}\mathbf{\text{ə}} \sim \mathbf{c} < \mathbf{q}^{\text{w}} > \mathbf{\text{á}}\mathbf{q}^{\text{w}}$ (cf. $\text{c}\mathbf{q}^{\text{w}}\text{aq}^{\text{w}}$ ‘cry’)
 DIM-cry<COS>
 ‘small cry, whimper’ (A. Mattina & Peterson, 1997, p.304, Okanagan)

From the limited data, it appears that diminutive change of state reduplication occurs with verbs or adjectives, as shown in (674a) and (674b), respectively. There is not enough data at present to draw conclusions about the meaning associated with a diminutive change-of-state.

B.1.4 Montana Salish-Spokane-Kalispel Multiple Reduplication

Attested multiple reduplication in Montana Salish-Spokane-Kalispel is summarised in Table B.5.

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	REP C ₁	COS C ₂	PL V ₁	INCH V ₁
PL	C ₁ C ₂			✓				
PDEN	C ₁ C ₂							
DIM	C ₁				✓	✓		
REP	C ₁							
COS	C ₂							
PL	V ₁							
INCH	V ₁							
Other:		DIM-PL-COS						

Table B.5: Multiple Reduplication in Montana Salish-Spokane-Kalispel

Examples come from B. Carlson and Thompson (1981), B. Carlson (1989), and Black (1996). I have not added [ə] in Montana Salish-Spokane-Kalispel forms with multiple reduplication.

B.1.4.1 Diminutive Plural

In diminutive plural forms, diminutive C₁ reduplication is leftmost (excluding prefixes); plural C₁C₂ reduplication follows. Examples are given in (675).

- (675) a. $\dot{m} \sim \dot{m}\dot{\lambda} \sim \dot{m}\dot{e}\dot{\lambda}$ (cf. *he-n-méλ* ‘It’s mixed.’)
 DIM~PL~mix
 ‘little things are scrambled’ (B. Carlson, 1989, p.210, Spokane)
- b. $\check{s} \sim \check{s}\check{l} \sim \check{s}\check{l}$ (cf. *šliš* ‘chop’)
 DIM~PL~chop
 ‘small things are cut’ (B. Carlson, 1989, p.210, Spokane)
- c. $\check{c}t\text{-}q \sim q\check{c} \sim q\check{c}\text{-}i\check{l}\check{s}$ (cf. *qéclš* ‘He ran.’)
 LOC-DIM~PL~run-AUT
 ‘He ran about anxiously on the floor.’ (Black, 1996, p.256, Spokane)

I have only been able to find diminutive plural applied to verbs in Montana Salish-Spokane-Kalispel. Diminutive plural marking is used with verbs with plural participants where the participants are small, as shown in (675a–b). It is not clear if this combination is permitted on nouns or if the absence of forms merely reflects a gap in documentation.

B.1.4.2 Diminutive Repetitive

In diminutive plural forms, diminutive C₁ reduplication is leftmost (excluding prefixes); repetitive C₁ reduplication follows. Examples are given in (676).

- (676) a. *hec-m̥~m̥e~m̥éχ-iʔ* (cf. *he-n-méχ* ‘It’s mixed.’)
 PROG-DIM~REP~mix-CONT
 ‘they (sg.) are mixing things over and over’ (B. Carlson, 1989, p.210, Spokane)
- b. *š~še~šíl* (cf. *šliš* ‘chop’)
 DIM~REP~chop
 ‘one small object is cut repeatedly’ (B. Carlson, 1989, p.211, Spokane)

Diminutive repetitive marking is found with verbs; it can mark a repeated event with a small participant, as shown in (676b).

B.1.4.3 Diminutive Change of State

When diminutive and change-of-state reduplication are combined, C₁ reduplication surfaces as a prefix; C₂ reduplication surfaces following the second consonant of the root. Examples are given in (677).

- (677) a. *š~š<í>íl* (cf. *šliš* ‘chop’)
 DIM~chop<COS>
 ‘small things got all cut up’ (B. Carlson, 1989, p.210, Spokane)
- b. *m̥é<m̥><χ>χ* (cf. *he-n-méχ* ‘It’s mixed.’)
 mix<DIM><COS>
 ‘a little thing got mixed by accident’ (B. Carlson, 1989, p.210, Spokane)

- c. $k^w \sim k^w <\acute{l}> \acute{l}$ (cf. $k^w \acute{l}$ ‘red’)
 DIM~sunny<COS>
 ‘sunny’ (B. Carlson & Thompson, 1981, p.113, Spokane)

Multiple reduplication marking a diminutive change of state is found with verb or adjective roots, as shown in (677). With verbs, it refers to a small argument undergoing a change of state, as shown in (677a–b). With adjectives, such as (677c), the meaning is unclear: it is possible that $k^w k^w \acute{l}$ ‘sunny’ means “got a little bit red”.

B.1.4.4 Diminutive, Plural, and Change of State

Diminutive C_1 , plural $C_1 C_2$, and change-of-state C_2 reduplication can be combined to yield multiple reduplication comprised of three reduplicative morphemes, as shown in (678).

- (678) $\acute{x} \sim \acute{x} \acute{a} \acute{l} \sim \acute{x} <e> <\acute{l}> \acute{l}$ (cf. $\acute{x} \acute{l}$ ‘It stopped.’)
 DIM~PL~still<REP><COS>
 ‘They (sg.) keep having little (uncontrolled) seizures (as in epilepsy).’
 (B. Carlson, 1989, p.211, Spokane)

It is not clear what meanings are possible with diminutive plural change-of-state reduplication, though the verb in (678) seems to refer to a change of state (seizure) of a small magnitude that is repeated.

B.1.5 Lillooet Multiple Reduplication

Attested multiple reduplication in Lillooet is marked with ✓ in Table B.6. I use × to mark ungrammatical combinations, which are discussed in Chapter 5 (Section 5.4).

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	PL C ₁	INIT C ₁	COS C ₂	TRIP C ₂ ²
PL	C ₁ C ₂			✓		×	✓	
PDEN	C ₁ C ₂			✓				
DIM	C ₁					✓	×	
PL	C ₁							
INIT	C ₁						✓	
COS	C ₂							✓
TRIP	C ₂ ²							

Table B.6: Multiple Reduplication in Lillooet

Examples come from Van Eijk (1981a), Van Eijk (1997), H. Davis et al. (in prep), and fieldwork conducted by Henry Davis.

B.1.5.1 Diminutive Plurals

When diminutive and plural reduplication are combined, plural C₁C₂ reduplication surfaces as a prefix; diminutive C₁ reduplication surfaces as an infix following the stressed vowel. The stressed vowel is replaced by [ə] in these forms. Examples are given in (679).

- (679) a. s-**qəw**~qə<**qsqawc ‘potato’)
 NMLZ-PL~potato<DIM>
 ‘little potatoes’ (Van Eijk, 1981a, p.57, Lillooet)**
- b. s-**qəx**~qə<**qsqáxa? ‘dog’)
 NMLZ-PL~dog<DIM>
 ‘pups’ (Van Eijk, 1997, p.59, Lillooet)**
- c. **ǰak**~ǰə<**ǰǰak ‘go along’)
 PL~go.along<DIM>=1PL.SUBJ
 ‘We went along a little ways’ (CA, Lillooet)**

- d. $təw \sim t\acute{ə} < t > wən$ (cf. $t\acute{ə}wən$ ‘visit’, from English *town*)
 PL~visit<DIM>
 ‘go visiting’ (CA, Lillooet)
- e. $q^wə\acute{l} \sim q^wə\acute{l} \acute{ə} < \acute{l} > t$ (cf. $q^wə\acute{l}it$ ‘speak’)
 PL~speak<DIM>
 ‘have a conversation’ (CA, Lillooet)

Both nouns and verbs occur with plural diminutive multiple reduplication, as shown in (679a–b) and (679c–e), respectively. With nouns, this combination refers to multiple small individuals (or young animals). With verbs, the contribution of diminutive reduplication is not always clear. However, (679c) shows that it can reflect a small amount of progress for a event with multiple participants.

B.1.5.2 Diminutive and Property-Denoting

When diminutive and property-denoting reduplication are combined, property-denoting C_1C_2 reduplication surfaces either as a prefix or an infix (see Section 4.2.3.2 for further discussion of the position of property-denoting reduplication). Diminutive C_1 reduplication surfaces as an infix following the stressed vowel; the stressed vowel is replaced by [ə] in these forms. Examples are given in (680).

- (680) a. $s-\acute{x}əq^w \sim \acute{x}ə < \acute{x} > q^w$ (cf. $s\acute{x}əq^w$ ‘to have a spot on one’s body’)
 STAT-PDEN~spot<DIM>
 ‘to have little spots, variegated’ (Van Eijk, 1997, p.150, Lillooet)
- b. $k\acute{ə} < k > ə\acute{n} \sim k\acute{n} -ət$ (cf. $k\acute{í}nk\acute{ə}nt$ ‘dangerous’)
 PDEN<DIM>~dangerous-CHAR
 ‘a little bit dangerous’ (CA, Lillooet)
- c. $z\acute{ə} < z > ə\acute{w} \sim z\acute{w} -ət$ (cf. $z\acute{á}w\acute{z}w\acute{ə}t$ ‘annoying’)
 PDEN<DIM>~annoy-CHAR
 ‘a little bit annoying’ (CA, Lillooet)
- d. $q^w\acute{ə} < q^w > ə\acute{l} \sim q^wə\acute{l} -t$ (cf. $q^w\acute{á}lq^wə\acute{l}t$ ‘hurting’)
 PDEN<DIM>~hurt-CHAR
 ‘hurting a little bit’ (CA, Lillooet)

- e. $ká < \dot{k} > \dot{a}w \sim \dot{k}w - \dot{a}t$ (cf. $káwkw\dot{a}t$ ‘gross, filthy’)
 PDEN < DIM > ~ dirty-CHAR
 ‘a little bit gross/filthy’ (CA, Lillooet)

Forms with property-denoting and diminutive reduplication are adjectives; this combination of reduplicative morphemes marks that a state or property holds to a small degree.

B.1.5.3 Plural Change of State

When plural and change-of-state reduplication are combined, change-of-state C_2 reduplication surfaces as an infix following the stressed vowel, while plural C_1C_2 reduplication surfaces as a prefix to the root. Examples are given in (681).

- (681) a. $p\dot{a}t \sim p\acute{u} < \dot{t} \dot{a} > t$ (cf. $p\acute{u}t\dot{a}t$ ‘to be boiling’)
 PL ~ boil < COS >
 ‘rapids’ (Van Eijk, 1997, p.57, Lillooet)
- b. $c\dot{a}q \sim c\acute{í} < q\dot{a} > q$ (cf. $c\acute{í}q\dot{í}n$ ‘to stab s.o.’)
 PL ~ stab < COS >
 ‘to get stabbed all over’ (Van Eijk, 1997, p.57, Lillooet)
- c. $c\dot{a}k^w \sim ck^w - \acute{a} < n > na? = t\dot{k}an$ (cf. $c\acute{a}k^w\dot{a}n$ ‘pull something’)
 PL ~ pull-ear < COS > = 1SG.SUBJ
 ‘Both my ears got pulled.’ (CA, Lillooet)
- d. $\dot{x}\dot{a}p \sim \dot{x}\acute{u} < p\dot{a} > p$ (root: $\dot{x}up$ - ‘twist, wrap’)
 PL ~ twist < COS >
 ‘twist or wrap (pl.)’ (H. Davis et al., in prep., Lillooet)
- e. $t\dot{a}t\dot{\Gamma}^w \sim t < \dot{\Gamma}^w\dot{a} > \dot{\Gamma}^w$ (root: $t\dot{\Gamma}^w$ - ‘jump’)
 PL ~ bounce/hop < COS >
 ‘bound, hop or jump, of animal (pl.)’ (H. Davis et al., in prep., Lillooet)

The combination of plural C_1C_2 and change-of-state $_2$ reduplication applies to verbs. It can refer to a change-of-state that holds of multiple participants, as in (681a,c-e), or arises from events distributed in time and space, as in (681b).

B.1.5.4 Diminutive and Initial

When diminutive and initial reduplication are combined, initial C₁ reduplication surfaces as a prefix to the root; diminutive C₁ reduplication surfaces as an infix following the stressed vowel. Examples are given in (683).

- (682) a. $\dot{\lambda}ak$ $k^w u = k\dot{\alpha} \sim k\acute{\alpha} < k > \dot{\alpha}w$ (cf. *nkáwlax* ‘go far from shore’)
 go.along DET=INIT.C₁~far<DIM>
 ‘Go a little further!’ (CA, Lillooet)
- b. $\dot{x}\dot{\alpha} \sim \dot{x}z\acute{\alpha} < z > \dot{\alpha}m$ $\eta izá$ (cf. *xzum* ‘big’)
 INIT.C₁~big<DIM> DEM
 ‘These ones are a little bigger.’ (CA, Lillooet)
- c. $x^w \eta it$ $\eta i = s - q^w \dot{\alpha} \sim q^w y\acute{\alpha} < \dot{y} > c = a$
 many PL.DET=NMLZ-INIT.C₁~rabbit<DIM>=EXIS
 $\dot{q}^w u \eta - x\acute{it} - an$
 trap-IND-1 SG.ERG
 ‘I trapped a lot of little rabbits.’ (cf. *sq^w [\acute{\alpha}]yíc* ‘rabbit’) (CA, Lillooet)

The combination of C₁ diminutive reduplication and initial C₁ reduplication is found with nouns and adjectives. With adjectives, this combination can mark a small degree or difference between two degrees, as in (682b). With nouns, it indicates a plurality of small entities.

B.1.5.5 Change of State and Initial

When change-of-state and initial reduplication are combined, initial C₁ reduplication surfaces as a prefix to the root; change-of-state C₂ reduplication surfaces as an infix following the stressed vowel. An example is given in (683).

- (683) $\dot{x}\dot{\alpha} \sim \dot{x}z\acute{u} < m\dot{\alpha} > m$ $\eta i = s\acute{it}k - s = a$ $ta = s - t\acute{t}áyn = a$
 INIT.C₁~big<COS> PL.DET=mesh-3POSS=EXIS DET=NMLZ-net=EXIS
 ‘The meshes of the net got bigger.’ (CA, Lillooet)

The combination of change-of-state C₂ reduplication and initial C₁ reduplication is limited to adjectival roots that take initial C₁ reduplication. The form with multiple reduplication is a verb; it is used to describe plural entities that are undergoing a change of state.

B.1.5.6 Triplication + Change-of-State

There is one example, given in (684), which has both triplication and change-of-state C₂ reduplication. The final consonant (/x^w/) is tripled in triplication, while the second consonant (/l/) is doubled in C₂ reduplication.

- (684) qəl<ḷ><əx^wəx^w>əx^w (root: qḷx^w- 'hurry')
hurry<COS><TRIP>
'rushing to get s.t. done, rushing to get ready' (H. Davis et al., in prep,
Lillooet)

It is not possible to say anything about the meaning of this combination at present.

B.1.6 Shuswap Multiple Reduplication

Attested multiple reduplication in Shuswap is marked with ✓ in Table B.7. I use × to mark ungrammatical combinations where I have elicited negative data.¹

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	1SG C ₁	COS C ₂
PL	C ₁ C ₂			✓	✓	×
PDEN	C ₁ C ₂			✓	✓	
DIM	C ₁					×
1SG	C ₁					
COS	C ₂					
Other:	unclear C ₁ x 2 (comparative)					

Table B.7: Multiple Reduplication in Shuswap

Examples come from Kuipers (1974a), FirstVoices, and my fieldwork.²

B.1.6.1 Diminutive Plural

When diminutive and plural reduplication are combined, plural C₁C₂ reduplication surfaces as a prefix; diminutive C₁ reduplication surfaces as an infix following the

¹Diminutive/first person C₁ + plural C₁C₂ reduplication and change-of-state C₂ + plural C₁C₂ reduplication were rejected, as shown in (i) and (ii). There is one exception, however, *q^wacéc* ‘set out’ appears to have C₂ reduplication and I have recorded the form *q^wacq^wacéc* ‘they set out’, which has both change-of-state C₂ and plural C₁C₂ reduplication.

- (i) a. **λ<əx^w>ú<x^w>(ə)x^w* (cf. *λəx^wúx^w* ‘numb’)
numb<COS><C₁>
Intended: ‘numb (1SG)’ or ‘a little numb’ (GD, Shuswap)
- b. **s<əx>é<x>(ə)x* (cf. *səxéx* ‘numb’)
numb<COS><C₁>
Intended: ‘numb (1SG)’ or ‘a little numb’ (GD, Shuswap)
- (ii) a. **təx^w~λ<əx^w>úx^w* (cf. *λəx^wúx^w* ‘numb’)
PL~numb<COS>
Intended ‘They are numb.’ (GD, Shuswap)
- b. **səx~s<əx>éx* (cf. *səxéx* ‘numb’)
PL~numb<COS>
Intended ‘They are numb.’ (GD, Shuswap)

²FirstVoices: <https://www.firstvoices.com/secwepemc/>

stressed vowel. Examples are given in (685).

- (685) a. **cəq**~cqe<**q̇**>**ɬp** (cf. *cqetp* ‘tree’)
 PL~tree<DIM>
 ‘young fir growth’ (Kuipers, 1974a, p.38, Shuswap)
- b. **pəs**~pé<**p**>sətk^we (cf. *pəsətk^we* ‘lake’)
 PL~lake<DIM>
 ‘small lakes’ (Kuipers, 1974a, p.39, Shuswap)
- c. x-**ɬəq^w**~ɬú<**ɬ**>q^w-məñ
 LOC-PL~scoop<DIM-instrument
 ‘small cups’ (GD, Shuswap)
- d. **yəx**~yé<**y**>x-t (cf. *yext* ‘long’)
 PL~long<DIM>-CHAR
 ‘long little things’ (GD, Shuswap)
- e. k^woyíʔsə ʔux^w s-**nək**~ní<**n**>k-ən-s x^wəx^weyt
 small only NMLZ-PL~cut<DIM>-CTR-3ERG all
 ʔə=ʔəyúʔ
 DET=carrot
 ‘Henry cut all of the carrots a little bit.’ (cf. *nikəm* ‘to cut’) (GD, Shuswap)

The combination of diminutive and plural reduplication is attested with nouns, verbs, and adjectives. Diminutive plural nouns refer to multiple entities that are small or young, as shown in (685a–c). A diminutive plural adjective means that a state or property holds of multiple small entities, as shown in (685d). For diminutive plural verbs, the full range of meanings is unclear, though (685e) shows that it is possible to use this to mark a diminutive event with plural participants.

B.1.6.2 Diminutive and Property-Denoting

When diminutive and property-denoting reduplication are combined, property-denoting C₁C₂ reduplication surfaces as a prefix; diminutive C₁ reduplication surfaces as an infix following the stressed vowel. Examples are given in (686).

- (686) Sander ʔəpəkut-s ɬət~ɬá<ɬ>ʰ
 Sander coat-3POSS PROP.D~wet<DIM>
 ‘Sander’s coat is a little wet.’ (GD, Shuswap)
- a. təx^w~tú<t>x^w-t ɣə=qéwtən-s
 PDEN~straight<DIM>-CHAR DET=hair-3POSS
 ‘Her hair is a little bit straight.’ (GD, Shuswap)
- b. yíʔənə tə=x^wə~x^wéɬ s-təx^w~tú<t>x^w-t-s
 this DET=DIM~road NMLZ-PDEN~straight<DIM>-CHAR-3POSS
 tə=s-ʔiɬx
 DET=NMLZ-other
 ‘This trail is a little straighter than the others.’ (GD, Shuswap)

The combination of diminutive and property-denoting reduplication is found with adjectives; it marks a small amount of the state or property denoted by the adjective or small degree of difference, as shown in (686a–b) and (686c), respectively.

B.1.6.3 First Person and Plural

When first person and plural reduplication are combined, plural C₁C₂ reduplication surfaces as a prefix; first person C₁ reduplication surfaces as an infix following the stressed vowel. Examples are given in (687).

- (687) ʔank^w-t ɣə=n-kəl~ké<k>əlɬ (cf. *kelx* ‘hand’)
 twist-CHAR DET=1SG.POSS-PL~hand<1SG>
 ‘My hands are gnarled.’ (FirstVoices, Shuswap)
- a. ʰhéhəʔən lə=n-s-pəq^w~pəq^wé<q^{ww}?
 where DET=1SG.POSS-NMLZ-PL~book<DIM>
 ‘Where are my books?’ (FirstVoices, Shuswap)

Multiple reduplication marking first person and plurality is attested with nouns, further elicitation is required to determine if it is possible with verbs.³

³With a verb, this combination could potentially allow a pluractional reading with a first person argument; it is not clear if a transitive verb would allow for a plural argument as a subject and first person argument as an object, or vice versa.

B.1.6.4 First Person Properties

When first person and property-denoting reduplication are combined, property-denoting C_1C_2 reduplication surfaces as a prefix; first person C_1 reduplication surfaces as an infix following the stressed vowel. An example is provided in (688).

- (688) $xət \sim xé < \dot{x} > t=kən$ (root: *xet-*)
PDEN $\sim$ be.energetic $< 1SG > = 1SG.SUBJ$
'I am active.' (Shuswap)

The example in (688) is an adjective; this type of multiple reduplication denotes a property that holds of a first person argument.

B.1.6.5 Comparatives

Shuswap has a type of multiple reduplication that is a combination of two C_1 reduplication processes, shown in (689).

- (689) a. $tə \sim té < \dot{\lambda} > x-t$ (cf. $\dot{\lambda}ext$ 'tall')
 $C_1 \sim \text{tall} < C_1 > -CHAR$
'taller' (Kuipers, 1974a, p.38, Shuswap)
b. $cə \sim cé < \dot{c} > t-t$ (cf. $\dot{c}et$ 'cold')
 $C_1 \sim \text{cold} < C_1 > -CHAR$
'colder' (Kuipers, 1974a, p.39, Shuswap)
c. $lə \sim lé < \dot{l} > qt$ (cf. $\dot{l}eqt$ 'broad')
 $C_1 \sim \text{broad} < C_1 >$
'broader' (Kuipers, 1974a, p.198, Shuswap)
d. $yə \sim yé < \dot{y} > x-t$ (cf. $yext$ 'long')
 $C_1 \sim \text{long} < C_1 > -CHAR$
'longer' (Kuipers, 1974a, p.270, Shuswap)

This double C_1 reduplication process is associated with a comparative meaning, though this combination is restricted in distribution and likely is not productive.⁴

⁴It is possible this pattern should be treated as a type of single reduplication (i.e., triplication).

B.1.7 Thompson River Salish Multiple Reduplication

Attested multiple reduplication in Thompson River Salish is summarised in Table B.8.

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	AFF C ₁	COS C ₂
PL	C ₁ C ₂		✓	✓		✓
PDEN	C ₁ C ₂			✓		
DIM	C ₁				✓	✓
AFF	C ₁					✓
COS	C ₂					
Other:		PL-DIM-COS, PL-DIM-PDEN, AFF-DIM-PDEN				

Table B.8: Multiple Reduplication in Thompson River Salish

Examples come from B. Carlson and Thompson (1981), Jimmie (1994), and L. C. Thompson and Thompson (1996).

B.1.7.1 Diminutive Plural

In words with plural C₁C₂ and diminutive C₁ reduplication, the diminutive reduplicant is positioned following the stressed syllable and the plural reduplicant is placed on the left edge of the word. Examples are given in (690).

- (690) a. qíqyáy's (cf. qyaš 'barrel')
- //qəy~qyá<y>s//
- PL~barrel<DIM>
- 'little barrels' (L. C. Thompson & Thompson, 1996, p.283, Thompson River Salish)
- b. k^wə^m~k^wə^m<e?>-xə<x>n̩ (cf. k^wmí?me? 'small')
- PL-small<REP+SPZG>-foot<DIM>
- 'have small feet' (Jimmie, 1994, p.87, Thompson River Salish)

- c. $yəq^w \sim yú < y > q^w$ -n-s (cf. $yəq^w etés$ ‘splash s.o.’)
 PL-splash<DIM>-TR-3ERG
 ‘splash something/someone a little repeatedly’ (Jimmie, 1994, p.31, Thompson River Salish)
- d. $tək \sim te < t > kɬ$ (cf. $stekɬ$ ‘rain’)
 PL~rain<DIM>
 ‘a little shower, rain a little’ (L. C. Thompson & Thompson, 1996, p.344, Thompson River Salish)
- e. $ʔəq^w \sim ʔú < ʔ > q^w$ -n-s (root: $ʔuq^w$ - ‘daub’)
 PL~daub<DIM>-TR-3ERG
 ‘daub a little on several things’
 ‘daub s.t. on a little one’ (L. C. Thompson & Thompson, 1996, p.186, Thompson River Salish)

The combination of diminutive and plural reduplication is found with noun, verb, and adjective roots. It can be used to mark multiple small entities with nouns and adjectives, as shown in (690a) and (690b), respectively. With verbs, it can mark a diminutive plural event (690c–d), a diminutive event with plural participants (690e), or plural events with a small participant (690e).

B.1.7.2 Diminutive and Property-Denoting

In words with property-denoting C_1C_2 and diminutive C_1 reduplication, the diminutive reduplicant is positioned following the stressed syllable and the property-denoting reduplicant is placed either on the left edge of the word or following the second consonant of the root. Examples are given in (691).⁵

- (691) a. $cək \sim cé < c > k$ (cf. $ʔescék$ ‘already cool(ed)’)
 PDEN~cool<DIM>
 ‘a little cooler’ (Jimmie, 1994, p.87, Thompson River Salish)

⁵It is likely that the [u] in (691b) comes from a schwa that has undergone rounding.

- b. $k^w\text{əs} \sim k^w\acute{u} < k^w > s\text{-t}$ (cf. $\text{ʔesk}^w\acute{əs}$ ‘very dim’)
 PDEN $\sim$ darken < DIM > -CHAR
 ‘be a little dark’ (L. C. Thompson & Thompson, 1996, p.119, Thompson River Salish)
- c. $\text{ləx}^w \sim \text{l}\acute{a} < \text{l} > \text{x}^w\text{-t}$ (root: $\text{l}\acute{x}^w$ - ‘flow free’)
 PDEN $\sim$ flow.free < DIM > -CHAR
 ‘a little runny’ (L. C. Thompson & Thompson, 1996, p.141, Thompson River Salish)
- d. $\acute{c}\acute{u} < \acute{c} > \text{l} \sim \acute{c}\text{əl}\text{-t}$ (cf. $\acute{c}\text{ʔul}$ ‘get sour’)
 sour < DIM > $\sim$ PDEN-CHAR
 ‘a little, somewhat, kind of, slightly sour’ (L. C. Thompson & Thompson, 1996, p.1244, Thompson River Salish)
- e. $n\text{-}k^w\text{əl} \sim k^w\text{a} < k^w > \text{l}$ (root: k^wel - ‘lukewarm’)
 LOC-PDEN $\sim$ lukewarm < DIM >
 ‘just (barely) warm’ (L. C. Thompson & Thompson, 1996, p.1374, Thompson River Salish)
- f. $\acute{x}\text{o}\Gamma^w\acute{x}\acute{o}\acute{x}\Gamma^w\text{t}$ (root: $\acute{x}\Gamma^w$ - ‘hard’)
 $//\acute{x}\text{ə}\Gamma^w \sim \acute{x}\acute{o} < \acute{x} > \Gamma^w\text{-t} //$
 PDEN $\sim$ hard < DIM > -CHAR
 ‘somewhat hard’ (L. C. Thompson & Thompson, 1996, p.905, Thompson River Salish)

The combination of diminutive and property-denoting reduplication is found with adjectives. It marks a state or property that holds to a small degree. It is not possible to determine if this combination could be used to refer to a state or property that holds of a small individual.

B.1.7.3 Plural Change of State

In words with plural and change-of-state reduplication, plural C_1C_2 reduplication surfaces as a prefix before the root and change-of-state C_2 reduplication follows the second consonant of the root. Examples are given in (692).

- (692) a. **ḥəp̌~ḥp̌~əp̌** (cf. *ḥp̌əm* ‘hang s.t.’)
 PL~hang.over~COS
 ‘(of plants) droop, wither’ (Jimmie, 1994, p.88, Thompson River Salish)
- b. x^wuḡ=kt **nəǩ~néǩ~əǩ** tə=k=cókns
 going.to=1 PL PL~change~COS OBL=DET=chicken
 ‘We’re going to turn into chickens.’ (cf. *néǩes* ‘change, transform’)
 (B. Carlson & Thompson, 1981, p.118, Thompson River Salish)
- c. cucuwustés (cf. *?escúw* ‘already done, made, built’)
 //cəw~cuw-əw-s-t-és//
 PL~work/do~COS-CAUS-TR-3ERG
 ‘raise children or plants’ (L. C. Thompson & Thompson, 1996, p.39, Thompson River Salish)
- d. **wiʔ~wíʔ~iʔ-t** (cf. *wíʔes* ‘disable someone’)
 PDEN~disable~COS-CHAR
 ‘(become) completely disabled’ (L. C. Thompson & Thompson, 1996, p.379, Thompson River Salish)

The combination of plural and change-of-state reduplication is found with verbs. This type of multiple reduplication marks a change of state with plural participants in (692a–c). The plural marking in (692d) is translated as “completely”: I interpret C₁C² reduplication in this example as having a pluractional interpretation where the change of state is understood to have a spatial distribution (i.e., “completely” may mean “all over”).

B.1.7.4 Diminutive Change of State

There are a handful of examples that have both diminutive C₁ and change-of-state C₂ reduplication, as shown in (693).

- (693) a. n-ḥé<**ḥ**>ḡ~ḡ~m̌-tn (cf. *?esntém̌* ‘be inside’)
 LOC-enter<DIM>~COS-instrument
 ‘a small container’ (Jimmie, 1994, p.88, Thompson River Salish)

- b. $\dot{x}^w u < \dot{x}^w > s \sim \text{əs}$ (cf. $\text{?es}\dot{x}^w \acute{u}s$ ‘(of soapberries) already beaten’)
 foam<DIM>~COS
 ‘a little foam’ (Jimmie, 1994, p.88, Thompson River Salish)

The examples in (693) are atypical for C_2 reduplication because they appear to nouns, based on their translation, despite being formed on verbal roots. The status of diminutive and change-of-state reduplication is unclear at present; this may reflect a lexicalised combination. However, these types of reduplication do co-occur in a multiple reduplication pattern with plural reduplication, discussed in Section B.1.7.8.

B.1.7.5 Plural and Property-Denoting

In words with plural and property-denoting reduplication, plural C_1C_2 reduplication surfaces as a prefix before the root and property-denoting C_1C_2 reduplication follows the second consonant of the root. Examples are given in (694).

- (694) a. $si\text{?s}\acute{e}y\acute{s}i\text{?}$ (cf. $s\acute{e}z\acute{l}m\acute{i}ns$ ‘play around with s.t.’)
 //səỵ~séỵ~səỵ//
 PL~play~PDEN
 ‘several play, have a good time’ (L. C. Thompson & Thompson, 1996, p.326, Thompson River Salish)
- b. $q^w i\text{?}q^w \acute{e}y\acute{q}^w i\text{?}t$ (cf. $q^w i\text{?}st\acute{e}s$ ‘have s.t. ache’)
 //q^wəỵ~q^wéỵ~q^wəỵ-t//
 PL~ache~PDEN-CHAR
 ‘be (generally) very painful, have many aches and pains’ (L. C. Thompson & Thompson, 1996, p.294, Thompson River Salish)

The combination of plural and property-denoting reduplication marks a state or property that holds of plural participants or is spatially distributed, as shown in (694a) and (694b) above.

B.1.7.6 Affective Change of State

When affective and change-of-state reduplication are combined, affective C₁ reduplication surfaces as a prefix to the root; change-of-state C₂ reduplication surfaces following the stressed vowel. An example is given in (695).

- (695) a. $\text{tə} \sim \text{t}^{\text{h}} \text{í} \text{í} \sim \text{í}$ (cf. *?eshtí* ‘scattered’)
AFF~disperse~COS
‘scattered far and wide’ (L. C. Thompson & Thompson, 1996, p.155, Thompson River Salish)
- b. $\text{n-} \text{x}^{\text{w}} \text{ə} \sim \text{x}^{\text{w}} \text{s-cín} \sim \text{n}$ (root: *x^ws-*)
LOC-AFF~diverge-mouth~COS
‘forked logs, sticks all over’ (L. C. Thompson & Thompson, 1996, p.435, Thompson River Salish)

It is unclear from the available data which lexical categories permit the combination of affective and change-of-state reduplication. However, affective reduplication in these forms appears to mark plurality and possibly spatial distribution.⁶

B.1.7.7 Affective Diminutive

When affective and diminutive reduplication are combined, affective C₁ reduplication surfaces as a prefix to the root; diminutive C₁ reduplication surfaces following the stressed vowel. Examples are given in (696).

- (696) a. $\text{?es-cə} \sim \text{c}^{\text{h}} \text{ó} < \text{c} > \text{í}^{\text{w}}$ (cf. *?escóí^w* ‘have a stripe’)
STAT-AFF~DIM~stripe
‘striped (baby talk)’ (L. C. Thompson & Thompson, 1996, p.37, Thompson River Salish)
- b. $\text{kə} \sim \text{ké} < \text{k} > \text{m-i} \text{?x}$ (cf. *kémix* ‘sneak along’)
AFF~stealthy<DIM>-AUT
‘sneak up (esp. in hunting game animals)’ (L. C. Thompson & Thompson, 1996, p.103, Thompson River Salish)

⁶Affective reduplication in these forms is likely cognate to the initial C₁ reduplication process in Lillooet that marks plurality with certain nouns and adjectives.

- c. $\dot{\lambda}\dot{\alpha}\sim\dot{\lambda}u<\dot{\lambda}>q^w$ (cf. $\dot{\lambda}u\dot{\lambda}q^w$ ‘a little speckled animal’)
 AFF~daub<DIM>
 ‘a little speckled animal’ (L. C. Thompson & Thompson, 1996, p.173,
 Thompson River Salish)
- d. $n\text{-}\dot{m}\dot{\alpha}\sim\dot{m}\acute{\alpha}<\dot{m}>\dot{\imath}\text{-}\dot{m}$ (cf. $\dot{m}\acute{\alpha}\dot{\imath}\dot{e}s$ ‘illuminate s.t.’)
 LOC-AFF~light.up<DIM>-MID
 ‘a little lightning bolt’ (L. C. Thompson & Thompson, 1996, p.192,
 Thompson River Salish)

Affective diminutive forms include nouns, verbs, and adjectives. The contribution of the affective reduplication is unclear; diminutive reduplication marks the same range of meanings as described in Section A.2.7.

B.1.7.8 Plural, Diminutive, and Change of State

Diminutive C₁, plural C₁C₂, and change-of-state C₂ reduplication can be combined to yield multiple reduplication comprised of three reduplicative morphemes, as shown in (697).

- (697) a. $n\text{-}\dot{\imath}\dot{\alpha}m\text{-}\dot{\imath}\acute{e}<\dot{\imath}>\dot{m}\sim\dot{\alpha}\dot{m}$ (cf. $n\dot{\imath}\dot{e}\dot{m}\dot{\alpha}m$ ‘to carry unannounced’)
 LOC-PL-enter<DIM>~COS
 ‘Small things are carried unnoticed.’ (Jimmie, 1994, p.90, Thompson River Salish)
- b. $n\dot{\alpha}x\sim n\acute{e}<\dot{n}>x\sim\dot{\alpha}x$ (cf. $n\acute{e}xk^wu$ ‘have waves’)
 PL~wave<DIM>~COS
 ‘small, tiny waves, ripples’ (L. C. Thompson & Thompson, 1996, p.217, Thompson River Salish)
- c. $c\dot{\alpha}m\sim cm<e?>\sim m\text{-}\acute{e}<\dot{m}>kst$ (root: $cm\text{-}$ ‘small’)⁷
 PL-small<REP+SPZG>~COS-hand<DIM>
 ‘have small hands’ (L. C. Thompson & Thompson, 1996, p.35, Thompson River Salish)

⁷This example shows an “out of control” use of change-of-state C₂ reduplication. L. C. Thompson and Thompson (1996, p.35) state that “a person has no control over the size of [thier] hands”.

B.1.7.9 Diminutive, Plural, and Property-Denoting

Diminutive C₁, plural C₁C₂, and property-denoting C₁C₂ reduplication can be combined to yield multiple reduplication comprised of three reduplicative morphemes, as shown in (698).

- (698) a. $\text{ʔe s-t-} \mathbf{pa\dot{\varsigma}} \sim \mathbf{p\acute{a}} < \mathbf{p} > \dot{\varsigma} \sim \mathbf{pa\dot{\varsigma}} \text{-t}$
INT STAT-QUAL-PL $\sim$ fade $\sim$ PDEN-CHAR
'little grey faded ones' (cf. *ʔe stpa\dot{\varsigma}p\acute{a}\dot{\varsigma}pa\dot{\varsigma}t* 'several are grey, faded')
(L. C. Thompson & Thompson, 1996, p.227, Thompson River Salish)
- b. $\mathbf{t\acute{a}p} \sim \mathbf{t\acute{a}} < \mathbf{t} > \mathbf{p} \sim \mathbf{t\acute{a}p} \text{-t}$ (cf. *t\acute{a}pt\acute{e}p* 'between black and grey')
PL $\sim$ black $<$ DIM $>$ $\sim$ PDEN-CHAR
'a very little black one' (L. C. Thompson & Thompson, 1996, p.351, Thompson River Salish)

B.1.7.10 Affective, Diminutive, and Property-Denoting

Affective C₁, diminutive C₁, and property-denoting C₁C₂ reduplication can be combined to yield multiple reduplication comprised of three reduplicative morphemes, as shown in (699).

- (699) $\mathbf{q^w\acute{a}q^w oq^w \acute{z}q^w \acute{z}t}$ (cf. *tq^w \acute{a}zq^w \acute{a}zt* 'blue')
 $// \mathbf{q^w\acute{a}} \sim \mathbf{q^w\acute{a}} < \mathbf{q^w} > \acute{z} \sim \mathbf{q^w\acute{a}\acute{z}} \text{-t} //$
AFF $\sim$ blue $<$ DIM $>$ $\sim$ PDEN-CHAR
'bluish' (L. C. Thompson & Thompson, 1996, p.293, Thompson River Salish)

B.2 Central Salish Multiple Reduplication

B.2.1 Comox-Sliammon Multiple Reduplication

Attested multiple reduplication in Comox-Sliammon is marked with ✓ in Table B.9. I use × to mark ungrammatical combinations, which are discussed in Chapter 5 (Section 5.5).

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	PL C ₁	PROG C ₁	COS C ₂	PL V ₁
PL	C ₁ C ₂			×		✓		✓
PDEN	C ₁ C ₂			✓	✓			
DIM	C ₁				✓			✓
PL	C ₁							
PROG	C ₁						✓	✓
COS	C ₂							×
PL	V ₁							
Other:		PROG-V ₁ .PL-DIM, C ₁ C ₂ .PL-V ₁ .PL-DIM						

Table B.9: Multiple Reduplication in Comox-Sliammon

Examples come from Watanabe (1994a), Blake (2000), Watanabe (2021), and my fieldwork.

B.2.1.1 Diminutive and Property-Denoting

When diminutive and property-denoting reduplication co-occur, diminutive C₁ reduplication is leftmost; property-denoting C₁C₂ reduplication follows. The diminutive always surfaces with its Ci- allomorph in this type of multiple reduplication, regardless of whether the -C- allomorph would be used in the context of single reduplication. Examples are given in (700).

- (700) a. **xi~xəɬ~xəɬ** (cf. **xəɬ** ‘get angry’)
 DIM~PDEN~angry
 ‘a little angry’ (FL, Comox-Sliammon)

- b. **ši~šəm**~šəm (cf. *k^wu šəm* ‘It got dry.’)
 DIM~PDEN~dry
 ‘a little dry’ (FL, Comox-Sliammon)
- c. **xi~xəm**~xəm tə=ʔapustən (cf. *šəm* ‘get wet’)
 DIM~PDEN~wet DET=towel
 ‘The towel is a little wet/damp.’ (BW, Comox-Sliammon)
- d. **t^θi~t^θəy**~t^θay (cf. *t^θəyt^θay* ‘loud’)
 DIM~PDEN~loud
 ‘a little loud’ (FL, Comox-Sliammon)
- e. **xi~xəs**~xəsəm (cf. *šəsəm* ‘strong’)
 DIM~PDEN~strong
 ‘kinda strong’ (FL, Comox-Sliammon)

Forms with diminutive and property-denoting reduplication are adjectives; they refer to a small amount of a state or property.

B.2.1.2 C₁ Plural and Property-Denoting

When plural and property-denoting reduplication co-occur, plural C₁ reduplication is leftmost; property-denoting C₁C₂ reduplication follows. The plural morpheme always surfaces with its Ca- allomorph in this type of multiple reduplication, regardless of whether the -C- allomorph would be used in the context of single reduplication. Examples are given in (701).

- (701) a. hihiw **ša~šəs**~šəsəm tə=təq~tiqiw (cf. *šəsəm* ‘strong’)
 really PL~PDEN~strong DET=PL~horse
 ‘Those horses are really strong.’ (EP, Comox-Sliammon)
- b. hihiw **t^θa~t^θəy**~t^θay-mut (cf. *t^θəyt^θay* ‘loud’)
 really PL~PDEN~noisy-INTS
 ‘They’re really noisy.’ (EP, Comox-Sliammon)
- c. hihiw **ša~xəm**~xəm-mut tə=təm~tumiš (cf. *šəm* ‘get wet’)
 really PL~PDEN~wet-INTS DET=PL~man
 ‘Those men are really wet.’ (EP, Comox-Sliammon)

Forms with diminutive and plural C₁ reduplication are adjectives; they refer a state or property which holds of multiple individuals.

B.2.1.3 Diminutive and C₁ Plural

When diminutive C₁ and plural C₁ reduplication co-occur, diminutive C₁ reduplication is realised with its Ci- allomorph; plural C₁ reduplication is realised with its Ca- allomorph. With respect to surface linear order, both DIM-PL and PL-DIM are attested without an apparent change of meaning, as shown in (702a–b). However, note that only the PL-DIM order is permitted for (702c). Further elicitation is required to determine if there is a systematic reason why the DIM-PL form is rejected.

- (702) a. **pi~pa~pɣ-it** (✓ **pa~pi~pɣ-it**)
 DIM~PL~rip-STAT (PL~DIM~rip-STAT)
 ‘They (sleeves of a shirt) are ripped a little.’ (root: *pɣ-* ‘rip’) (FL, Comox-Sliammon)
- b. **ki~ka~kp-it** (✓ **ka~ki~kp-it**)
 DIM~PL~cut-STAT (PL~DIM~cut-STAT)
 ‘It has small little cuts.’ (root: *kɔ-* ‘cut’) (FL, Comox-Sliammon)
- c. **xa~xi~λm-it** (* **xi~xa~λm-it**)
 PL~DIM~wet-STAT (DIM~PL~wet-STAT)
 ‘They are a little wet.’ (root: *λm-* ‘wet’) (FL, Comox-Sliammon)

The combination of diminutive C₁ and plural C₁ reduplication is attested with stative-marked verbs, as shown in (702); it marks a small degree of a state that holds of plural participants, as shown in (702a–b) or a state that is distributed across a single participant (702c.) It is likely that this combination is also possible with adjectives (without stative marking), though further elicitation is required to confirm this prediction.

B.2.1.4 Diminutive and C₁C₂ Plural

When diminutive and plural reduplication co-occur, diminutive C₁ reduplication is leftmost; plural C₁C₂ reduplication follows. The diminutive always surfaces with its Ci- allomorph in this type of multiple reduplication, regardless of whether the -C- allomorph would be used in the context of single reduplication. Examples are given in (703).

- (703) a. **t̥i~t̥əɬ~tuɬaɬ** (cf. *t̥uɬaɬ* ‘bed’)
 DIM~PL~bed
 ‘a bunch of little beds’ (EP/FL, Comox-Sliammon)
- b. **si~səp~supayu** (cf. *supayu* ‘axe’)
 DIM~PL~axe
 ‘lots of little axes’ (EP/FL, Comox-Sliammon)
- c. **qi~qəs~qəsnaɣ** (cf. *qəsnaɣ* ‘shirt/dress’)
 DIM~PL~shirt/dress
 ‘a bunch of little shirts or dresses’ (EP/FL, Comox-Sliammon)
- d. **pi~pəč~pəču** (cf. *pəču* ‘basket’)
 DIM~PL~basket
 ‘small baskets’ (EP/FL, Comox-Sliammon)

This combination is only attested with nouns; it is used to refer to multiple small entities. It is not possible to form a diminutive pluractional verb using C₁C₂ reduplication, V₁ reduplication must be used instead (see following section).

B.2.1.5 Diminutive and V₁ Plural

When diminutive C₁ and plural V₁ reduplication are combined, diminutive C₁ reduplication surfaces either following the first root vowel (-C- allomorph) or as a prefix to the root (Ci- allomorph); plural V₁ reduplication surfaces immediately before the root vowel. Examples are given in (704).

- (704) a. **p<iʔ>i~pčəm** Bailey
 DIM<PL>~wake Bailey
 ‘Bailey wakes up now and then.’ (BW, Comox-Sliammon)
- b. **č<iʔ>i~čtəm** Freddie
 DIM<PL>~dance-MID Freddie
 ‘Freddie does a little jig now and then.’ (BW, Comox-Sliammon)
- c. **namaɬ=ʔut q<iʔ>i~qk^w-əm**
 little.bit=EXCL DIM<PL>~stop-MID
 ‘It kept stopping and starting for a little while.’ (BW, Comox-Sliammon)

- d. $\dot{k}^w < i? > i \sim \dot{k}^w \text{ən} [t] = \check{c} \text{an}$ Daniel
 DIM < PL > ~ see- [CTR] = 1 SG. SUBJ Daniel
 ‘I’m looking at Daniel (several times to see what he’s doing).’ (BW, Comox-Sliammon)

The combination of diminutive C₁ and plural V₁ reduplication is attested with verbs, as shown in (704) above, and nouns, as shown in (705). With verbs, this type of multiple reduplication marks pluractionality: there are a number of small events that are distributed in time. With nouns, diminutive C₁ and plural V₁ mark multiple small entities.⁸

- (705) a. $s < u? > u < s > \text{payu}$ (cf. *supayu* ‘axe’)
 axe < PL > < DIM >
 ‘little axes’ (JF, Comox-Sliammon)
- b. $\dot{t} < u? > u < \dot{t} > \dot{t} \text{a} \dot{t}$ (cf. *tutāt* ‘bed’)
 DIM < PL > ~ bed
 ‘little beds’ (EP, Comox-Sliammon)
- c. $m < i? > i < m > \text{xat}$ (cf. *mixat* ‘black bear’)
 DIM < PL > ~ bear
 ‘baby black bears’ (FL, Comox-Sliammon)
- d. $\check{c} < i? > i \sim \check{c} \text{uy}$ (cf. *čuy* ‘child’)
 DIM < PL > ~ child
 ‘little children’ (FL, Comox-Sliammon)

B.2.1.6 C₁C₁ Plural and V₁ Plural

When plural C₁C₂ and plural V₁ reduplication are combined, plural C₁C₂ reduplication surfaces as a prefix; plural V₁ reduplication surfaces immediately before the vowel in the C₁C₂ reduplicant. The vowel in the C₁C₂ reduplicant is [a], rather than [ə]. Examples are given in (706).

⁸The meaning of diminutive C₁ and plural V₁ reduplication is identical to that of diminutive C₁ and plural C₁V₂ reduplication in the previous section.

- (706) a. hił Bailey ʔə=t^θ<aʔ>ak^w~t^θək^wt tə θək^w~θuk^wnačtən
 CLF Bailey CLF=PL<PL>~wipe-CTR DET PL~chair
 ‘It was Bailey who wiped down the chairs.’ (cf. t^θək^wtčan ‘I wiped it.’)
 (EP, Comox-Sliammon)
- b. t<aʔ>ak^w~t^θuk^w tə=hiyum (cf. t^θuk^w Fred ‘Fred flew.’)
 PL<PL>~fly DET=seagull
 ‘The seagull flies around (go a little distance, stop, and repeat).’ (BW, Comox-Sliammon)
- c. j<aʔ>a^ʔ~jə^ʔ tə=čanu (cf. jə^ʔ ‘run’)
 PL<PL>~run DET=dog
 ‘The dog is running around the yard (starting and stopping, sniffing at things).’ (BW, Comox-Sliammon)

This combination is only found with verbs, where it marks pluractionality and requires a spatio-temporal distribution. Further elicitation is required to determine how the meaning of this type of reduplication differs from pluractional C₁C₂ reduplication on its own.

B.2.1.7 Progressive Change of State

When progressive and change-of-state reduplication co-occur, progressive C₁ reduplication surfaces as prefix; change-of-state C₂ reduplication surfaces following the first consonant of the root. Examples are given in (707).

- (707) a. t^θu~t^θ<uk^w>uk^w (cf. xəqt t^θuk^w ‘The day is long.’)
 PROG~day<COS>
 ‘becoming daylight’ (EP, Comox-Sliammon)
- b. q^wəl ʔa~ʔ<ah>ah t^θ=čayis (cf. ʔah ‘sore’)
 come PROG~sore<COS> 1SG.POSS=hand
 ‘My hand is starting to get sore.’ (BW/FL, vf)
- c. hu=čan k^wa~k^w<as>as (cf. k^was ‘warm/hot’)
 go=1SG.SUBJ PROG~warm/hot<COS>
 ‘I am getting warm.’ (EP, Comox-Sliammon)

- d. **qə~q<xə>x** (cf. *qəx* ‘more’)
 PROG~more<COS>
 ‘increasing, getting to be more’ (EP/FL, sf/vt)
- e. **pə~p<sə>s** **t^θ=čayiš** (cf. *pəx* ‘numb’)
 PROG~more<COS> 1SG.POSS=hand
 ‘My hand is getting numb.’ (BW, Comox-Sliammon)

Progressive change-of-state forms are built on adjectival roots, but the reduplicated forms are always verbs. This type of multiple reduplication refers to a change of state that is in progress.

B.2.1.8 Progressive and V₁ Plural

There are two recorded forms with progressive C₁ reduplication and plural V₁ reduplication, given in (708).⁹

- (708) a. **t^θ<ɛʔ>ɛ~t^θq-aʔam** (cf. *t^θəqʔəm* ‘throw (s.t.)’)
 PROG<PL>~throw-A.INTR
 ‘throwing lots’ (Blake, 2000, p.285)
- b. **ʔut=čx^w=k^wa** **t<aʔ>a~taw-š...**
 if=2SG.SUBJ=QUOT PROG<PL>~leave-TR
 ‘If you keep leaving them (sg.) alone...’ (cf. *tawəx^was* ‘Someone left them (sg.).’) (Watanabe, 2021, p.101, Comox-Sliammon)

The multiple reduplication pattern shown in (708) has only been recorded with two verbs to my knowledge; the examples are progressive and pluractional, involving multiple participants in (708a) and temporal distribution in (708b).

B.2.1.9 Diminutive, Progressive, and V₁ Plural

Diminutive C₁, progressive C₁, and plural V₁ reduplication can combine to yield multiple reduplication comprised of three reduplicative morphemes, as shown in (709).

⁹I leave the vowel in (708a) as given by Blake (2000) because it seems to correspond to an epenthetic /i/, but I have not managed to re-elicite the form to confirm my analysis.

- (709) a. $\acute{\text{č}} < \text{i?} > \text{i} \sim \acute{\text{č}} \text{i} \sim \acute{\text{č}} \text{ə} \text{ł}$ ʔasq̣ (cf. $\acute{\text{č}} \text{ə} \text{ł}$ ‘rain’)
 PROG<PL>~DIM~rain outside
 ‘It is raining a little (throughout the day).’ (BW, Comox-Sliammon)
- b. $\check{\text{j}} < \text{i?} > \text{i} \sim \check{\text{j}} \text{i} \sim \check{\text{j}} \text{ə} \text{ł}$ (cf. $\check{\text{j}} \text{ə} \text{ł}$ ‘run’)
 PROG<PL>~DIM~run
 ‘running around’ (Watanabe, 1994a, p.70, Comox-Sliammon)

B.2.1.10 Diminutive, V₁ Plural, and C₁C₂ Plural

I have found one form with diminutive C₁, plural V₁, and plural C₁C₂ reduplication, which is given in (710).

- (710) $\text{s} < \text{i?} > \text{i} \sim \text{səp} \sim \text{səplin}$ (cf. *səplin* ‘bread’)
 DIM<PL>~PL~bread
 ‘lots of little bits of bread’ (Blake, 2000, p.285, Comox-Sliammon)

B.2.2 Halkomelem Multiple Reduplication

Attested multiple reduplication in Halkomelem is summarised in Table B.10. Multiple reduplication in Halkomelem is discussed in more detail in Chapter 5 (Section 5.3).

		PL C ₁ C ₂	PDEN C ₁ C ₂	IPFV C ₁	DIM C ₁
PL	C ₁ C ₂			✓	
PDEN	C ₁ C ₂				
IPFV	C ₁				✓
DIM	C ₁				

Table B.10: Multiple Reduplication in Halkomelem

Examples come from Hukari (1976), Leslie (1979), and Suttles (2004).

B.2.2.1 Diminutive Imperfective

In diminutive imperfective forms, diminutive C₁ reduplication is leftmost (excluding prefixes); a glottal stop is infixes following the stressed vowel; imperfective C₁ reduplication immediately precedes the root. Examples are given in (711).

- (711) a. **q^wə~q^wá<ʔ>~q^wəl** (cf. *q^wel* ‘speak’)
 DIM~IPFV<DIM>~speak
 ‘little one is speaking’ (Suttles, 2004, p.193, Downriver Halkomelem)
- b. **q̣ə~q̣í<ʔ>~q̣ək^w** (cf. *q̣ɪk^w* ‘get bitten’)
 DIM~IPFV<DIM>~get.bitten
 ‘little one is getting bitten’ (Suttles, 2004, p.191, Downriver Halkomelem)
- c. **p̣ə~p̣é<ʔ>~p̣ət^θ** (cf. *p̣ét^θət* ‘sew it’)
 DIM~IPFV<DIM>~sew
 ‘sewing (dim.)’ (Suttles, 2004, p.172, Downriver Halkomelem)

- d. $\dot{c}\dot{a}\sim\dot{c}\dot{e}<?>\sim\dot{c}\dot{a}w\dot{a}t$ (cf. $\dot{c}\dot{e}w\dot{a}t$ ‘help them (sg.)’)
 DIM~IPFV<DIM>~help
 ‘help him’ (prog. dim.) (Suttles, 2004, p.170, Downriver Halkomelem)
- e. $\dot{t}\dot{a}\sim\dot{t}\dot{i}<?>\sim\dot{t}\dot{a}l\dot{a}m$ (cf. $ni\ \dot{t}\dot{i}l\dot{a}m$ ‘They (sg.) sang.’)
 DIM~IPFV<DIM>~sing-MID
 ‘A little thing (as a child) is singing.’ (Suttles, 2004, p.172, Downriver Halkomelem)
- f. $s\text{-}p\dot{a}\sim p\dot{i}<?>\sim p\dot{a}w$ (cf. $p\dot{i}w\dot{a}t$ ‘freeze it’)
 STAT-DIM~IPFV<DIM>~freeze
 ‘poor little frozen thing’ (Suttles, 2004, p.189, Downriver Halkomelem)

Diminutive imperfective multiple reduplication applies to verb roots. The diminutive imperfective forms in (711a–e) are imperfective verbs with a small participant. The example in (711f) is has reduplication that Suttles (2004, p.189) labels as “resultative”. The combination of *s*- and *C*₁ reduplication in this form marks a state (i.e., $sp\dot{i}p\dot{a}w$ ‘frozen’); it is unclear if the *C*₁ reduplication in these forms is independently meaningful.

B.2.2.2 Plural Imperfective

When plural *C*₁*C*₂ reduplication co-occurs with imperfective *C*₁ reduplication, plural *C*₁*C*₂ reduplication surfaces as a prefix to the root. This combination is found with words that form their imperfective with *C*₁ reduplication and debuccalisation of the initial sonorant to [h] (see Section A.3.2.4): the [h] is copied in *C*₁*C*₂ reduplication. Examples of plural imperfectives formed by multiple reduplication are given in (712–712).

- (712) a. $h\dot{a}m\sim h\acute{o}<\dot{m}>\acute{o}x$ (cf. $m\acute{e}?x$ ‘remove it’)
 PL~remove<IPFV>
 ‘removing them’ (Suttles, 2004, p.169, Downriver Halkomelem)
- b. $h\dot{a}w\sim h\acute{o}<\dot{w}>\acute{q}^w$ (cf. $w\acute{o}\acute{q}^w$ ‘drift away’)
 PL~drift.away<IPFV>
 ‘They are drifting.’ (Suttles, 2004, p.171, Downriver Halkomelem)

- c. **həṁ**~hə<**ṁ**>ḡ-ət
 PL~swallow<IPFV>-CTR
 ‘swallowing them’ (Hukari, 1976, p.187, Island Halkomelem)
- d. **həl**~hə<**l**>k^w-ət (cf. lək^wat ‘break it’)
 PL~break<IPFV>-CTR
 ‘breaking them’ (Leslie, 1979, p.52, Island Halkomelem)

The combination of plural and imperfective reduplication is only found with verbs, where it marks an imperfective verb with plural participants.

B.2.3 Lushootseed Multiple Reduplication

Attested multiple reduplication in Lushootseed is summarised in Table B.11.

		PL C ₁ C ₂	DIM C ₁	C ₂	PL V ₁
PL	C ₁ C ₂		✓	✓	
DIM	C ₁				✓
	C ₂				
PL	V ₁				

Table B.11: Multiple Reduplication in Lushootseed

Examples come from Hess (1967), Bates et al. (1994), Bates (1999), and Urbanczyk (2006).

B.2.3.1 Diminutive and C₁C₂ Plural

Examples with diminutive C₁ and plural C₁C₂ reduplication are given in (713).

- (713) a. **qí~q̣s~q̣is-šəd** (cf. *q̣is* ‘expose’)
 DIM~PL~expose-leg
 ‘legs partly uncovered’ (Urbanczyk, 2006, p.214, from Bates et al. 1994, Lushootseed)
- b. **sáʔ~sx^w~sax^w-əb** (cf. *sáx^wəb* ‘jump, leap, run’)
 DIM~PL~jump-MID
 ‘hopping’ (Bates et al., 1994, p.201, Lushootseed)
- c. [ʔəb]s-**bi~bəd~bədaʔ** ʔə tiʔəʔ qah~a(h) tiʔəʔ sg^wəlub
 POSS-DIM~PL~offspring OBL DET many~C₂ DET pheasant
 ‘Pheasant had children a-plenty.’ (cf. *bədáʔ* ‘child’) (Bates, 1999, p.4, Lushootseed)

C₁ and C₁C₂ reduplication can co-occur in the order in (713), with the C₁ reduplicant on the left edge, or in the reverse order (714), with the C₁C₂ reduplicant on the left edge. The examples in (714) have a C₁C₂ reduplicant, but

there is no copy of the second consonant.¹⁰

- (714) a. **pá~pa~pastəd** (cf. *pástəd* ‘white people’)
PL~DIM~white.person
‘many white children’ (Bates et al., 1994, p.160, Lushootseed)
- b. **yú~yu~yəbil** (cf. *yúbil* ‘starve’)
PL~DIM~starve
‘Children are starving.’ (Bates et al., 1994, p.279, Lushootseed)
- c. **s-dí~di~du:k^w** (cf. *sdúuk^w* ‘knife’)
NMLZ-PL~DIM~knife
‘small knives’ (Bates et al., 1994, p.85, Lushootseed)
- d. **či~či~čǵa?** (cf. *čǵa?* ‘rock’)
PL~DIM~rock
‘gravel’ (Bates et al., 1994, p.74, Lushootseed)

The pair in (715) shows that the two orders are possible with the same root. The translations are different, but there is not enough data to determine if this is systematic.

- (715) a. **bí~bəd~bəda?** (cf. *bədá?* ‘child’)
DIM~PL~child
‘dolls, litter (of animals)’ (Bates et al., 1994, p.36, Lushootseed)
- b. **bí~bi~bəda?**
PL~DIM~child
‘small children’ (Urbanczyk, 2006, p.214, from Bates et al. 1994, Lushootseed)

This type of multiple reduplication is found with nouns and verbs. With verbs, it can mark diminutive plural events (e.g., small events distributed in space or time), as shown in (713), or multiple small arguments, as shown in (714b). With nouns, it marks small entities (e.g., 714c).

¹⁰Though the second consonant is not copied and therefore it is not clear from the surface forms that these have C₁C₂ reduplication from the surface realisation; they are assumed to be there from the plural meanings and I adopt the assumptions about order from previous work (e.g., Broselow, 1983).

B.2.3.2 Diminutive and V₁ Plural

The combination of diminutive C₁ and plural V₁ reduplication is shown in (713). Diminutive reduplication surfaces either as an infix or a prefix; plural reduplication surfaces as an infix following the stressed vowel.

- (716) a. s-**ɬá**<**a**><**ɬ**>ədəyʔ (cf. *sɬádəyʔ* ‘woman’)
NMLZ-woman<PL><DIM>
‘girls’ (Hess, 1967, p.65, Lushootseed)
- b. s-**tú**<**u**>~tubš (cf. *stubš* ‘man’)
NMLZ-DIM<PL>~man
‘boys’ (Hess, 1967, p.65, Lushootseed)

This combination is found with nouns and marks plural entities that are small or young.

B.2.3.3 C₁C₁ Plural and C₂ Reduplication

When plural C₁ C₂ and C₂ reduplication co-occur, plural reduplication surfaces as prefix; change-of-state C₂ reduplication surfaces between the plural reduplicant and the root. Examples are given in (717).

- (717) a. **sáq^w~aḡ^w~saq^w** (cf. *ʔusáq^w* ‘It flew.’)
PL~COS~fly
‘Many wheeling (flying slowly in circles) all about.’ (Hess, 1967, p.67, Lushootseed)
- b. **sáx^w~ax^w~sax^w-əb** (cf. *sáx^wəb* ‘jump, leap, run’)
PL~COS~jump-MID
‘Many scurrying all about’ (Hess, 1967, p.67, Lushootseed)

This type of multiple reduplication is found with verbs. The meaning of C₂ reduplication in Lushootseed is unclear, but this combination is used to indicate that there are multiple participants in the C₂-marked event.

B.2.4 Sechelt Multiple Reduplication

Attested multiple reduplication in Sechelt is summarised in Table B.12.

		PL C ₁ C ₂	PDEN C ₁ C ₂	DIM C ₁	PL C ₁	IPFV C ₁	COS C ₂
PL	C ₁ C ₂			✓			✓
PDEN	C ₁ C ₂						
DIM	C ₁						
PL	C ₁						✓
IPFV	C ₁						✓
COS	C ₂						

Table B.12: Multiple Reduplication in Sechelt

Examples come from Beaumont (2011).

B.2.4.1 Diminutive and C₁C₂ Plural

When diminutive and plural reduplication co-occur, diminutive C₁ reduplication is leftmost; plural C₁C₂ reduplication follows. The diminutive always surfaces with its Ci- allomorph in this type of multiple reduplication, regardless of whether the -C- allomorph would be used in the context of single reduplication. Examples are given in (718).

- (718) a. **sí~səy~síya** (cf. *síya* ‘tree’)
 DIM~PL~tree
 ‘small trees’ (Beaumont, 2011, p.489, Sechelt)
- b. **s-tí~təl~túlu** (cf. *stúlu* ‘river’)
 NMLZ-DIM~PL~river
 ‘creeks, streams’ (Beaumont, 2011, p.787, Sechelt)
- c. **hí~həp~húpit** (cf. *húpit* ‘deer’)
 DIM~PL~deer
 ‘young deer (more than one)’ (Beaumont, 2011, p.115, Sechelt)
- d. **s-lí~ləm~lómay** (cf. *slómay* ‘branch’)
 NMLZ-DIM~PL~branch

‘small branches’ (Beaumont, 2011, p.771, Sechelt)

- e. $\dot{\text{h}}\dot{\text{i}}\sim\dot{\text{h}}\dot{\text{a}}\dot{\text{c}}\sim\dot{\text{h}}\dot{\text{a}}\dot{\text{c}}\text{-t}\dot{\text{a}}\text{n}$ (cf. $\dot{\text{h}}\dot{\text{a}}\dot{\text{c}}\text{t}\dot{\text{a}}\text{n}$ ‘knife’)

DIM~PL~stick-instrument

‘small knives’ (Beaumont, 2011, p.246, Sechelt)

This type of multiple reduplication is found with nouns: it refers to multiple small entities.

B.2.4.2 Imperfective Change of State

When imperfective and change-of-state reduplication co-occur, C₁ reduplication surfaces as prefix; C₂ reduplication surfaces following the second consonant of the root. Examples are given in (719).

- (719) a. $\dot{\text{k}}^{\text{w}}\dot{\text{a}}\sim\dot{\text{k}}^{\text{w}}\text{as}\sim\text{as}$ (cf. $\dot{\text{k}}^{\text{w}}\text{as}$ ‘warm’)

IPFV~warm~COS

‘getting hotter/warmer’ (Beaumont, 2011, p.511, Sechelt)

- b. $\dot{\text{k}}^{\text{w}}\dot{\text{u}}\sim\dot{\text{k}}^{\text{w}}\text{u}\dot{\text{x}}\sim\text{u}\dot{\text{x}}$ (cf. $\dot{\text{k}}^{\text{w}}\dot{\text{u}}\dot{\text{x}}\text{u}\dot{\text{x}}$ ‘get rusty’)

IPFV~rust~COS

‘getting rusty’ (Beaumont, 2011, p.622, Sechelt)

- c. $\dot{\text{q}}\dot{\text{i}}\sim\dot{\text{q}}\text{i}l<\dot{\text{i}}l>\text{t}$ (cf. $\dot{\text{q}}\text{i}l\text{t}$ ‘light (sky)’)

IPFV~light<COS>

‘daybreak, dawn, starting to get light (sky)’ (Beaumont, 2011, p.259, Sechelt)

- d. $\dot{\text{x}}^{\text{w}}\dot{\text{ə}}\sim\dot{\text{x}}^{\text{w}}\text{s}\sim\text{əs}$ (cf. $\dot{\text{x}}\text{əs}$ ‘fat’)

IPFV~fat~COS

‘gaining a lot of weight, getting fat’ (Beaumont, 2011, p.942, Sechelt)

This type of reduplication applies to adjective roots and marks an ongoing change of state.

B.2.4.3 C₁ Plural and Change of State

When plural C₁ and change-of-state C₂ reduplication co-occur, the Ca- allomorph of plural C₁ reduplication surfaces as prefix; C₂ reduplication surfaces following the second consonant of the root. An example is given in (720).

- (720) $\dot{x}^w\acute{a}\sim\dot{x}^w\text{al}\sim\acute{a}l$ (cf. $\dot{x}^w\acute{a}lám$ ‘glowing (not very brightly)’)
 PL~glow~COS
 ‘starting to glow all over the place (lights, sparks, etc.) (not very brightly)’
 ‘lots of sparks glowing’ (Beaumont, 2011, p.418, Sechelt)

B.2.4.4 C₁C₂ Plural and Change of State

When plural C₁C₂ and change-of-state C₂ reduplication co-occur, plural reduplication surfaces as prefix; C₂ reduplication surfaces following the second consonant of the root. An example is given in (721).

- (721) $\dot{x}\acute{o}\dot{x}\sim\dot{x}\acute{a}\dot{x}\sim\text{a}\dot{x}$ (cf. $\dot{x}\acute{a}\dot{x}\dot{x}$ ‘old/getting old’)
 PL~old~COS
 ‘old/getting old (more than one person/thing)’ (Beaumont, 2011, p.865, Sechelt)

This type of reduplication applies to adjective roots and marks a change of state that involves a plural argument.

B.2.5 Northern Straits and Klallam Multiple Reduplication

Attested multiple reduplication in Northern Straits and Klallam is summarised in Table B.13.

		PL C ₁ C ₂	PDEN C ₁ C ₂	IPFV C ₁	DIM C ₁
PL	C ₁ C ₂				
PDEN	C ₁ C ₂				✓
IPFV	C ₁				✓
DIM	C ₁				

Table B.13: Multiple Reduplication in Northern Straits and Klallam

Multiple reduplication is only attested in the Saanich dialect of Northern Straits; Raffo (1972) and Galloway (1990) do not describe patterns of multiple reduplication in Songish and Samish, respectively. While Crawford (2015, p.92) states that there are “quite a few cases of multiple reduplication” in Klallam (from a survey of published data in Montler 2012 and Montler 2015), this equates to a total of twenty tokens; some forms that appear to have multiple reduplication involve lexicalised reduplication, such as *maʔmúʔsmus* ‘a group of calves, small cows’ (Montler, 2012, p.253). As it is not clear that multiple reduplication is productive in Klallam, I only provide examples from Saanich in this section. Examples come from Montler (2018).

B.2.5.1 Diminutive and Property-Denoting

When diminutive C₁ and property-denoting C₁C₂ reduplication co-occur, diminutive reduplication surfaces as a prefix and property-denoting reduplication follows the second consonant of the root. Examples are provided in (722).

- (722) a. **θiʔ~θəɬ~θəɬ** (cf. *θiʔəŋ* ‘stand’)
 DIM~high~PDEN
 ‘a little high’ (Montler, 2018, p.1426, Saanich - Northern Straits)

- b. $n\acute{a}\text{-}k^w\acute{i}\text{?}\sim k^w\acute{a}m\sim k^w\acute{a}m$ (cf. $n\acute{a}k^w\acute{i}m$ ‘red’)
 color-DIM~red~PDEN
 ‘a little red’ (Montler, 2018, p.1236, Saanich - Northern Straits)

Forms with this type of reduplication are adjectives that refer to a small amount of a property.

B.2.5.2 Imperfective Diminutives

When diminutive C_1 and imperfective C_1 reduplication co-occur, diminutive reduplication surfaces as a prefix and imperfective reduplication surfaces between the diminutive and the root. Examples are provided in (723).

- (723) a. $s\text{-}l\acute{a}\sim l\acute{e}\sim l\acute{a}[\text{?}]\text{-}s\acute{a}t$ (cf. $l\acute{e}\text{?}s\acute{u}t$ ‘fix for someone’)
 STAT-DIM~IPFV~fix-REFL
 ‘fixing (dim)’ (Montler, 2018, p.1251, Saanich)
- b. $\text{?}\acute{a}\sim \text{?}e\sim \text{?}e\text{?}\text{-}\acute{a}j\text{-}\acute{e}l\eta\acute{a}n$ (cf. $\text{?}e\text{?}\acute{a}j$ ‘say’)
 DIM~IPFV~say/do-MID-want
 ‘to want to be saying or doing a little’ (Montler, 2018, p.137, Saanich)

The combination of diminutive and imperfective reduplication is found with verbs.

B.2.6 Twana Multiple Reduplication

Attested multiple reduplication in Twana is summarised in Table B.14.

		PL C ₁ C ₂	DIM C ₁	DIM V ₁
PL	C ₁ C ₂			
DIM	C ₁			✓
DIM	V ₁			

Table B.14: Multiple Reduplication in Twana

Examples come from Drachman (1969).

B.2.6.1 Double Diminutive

C₁ and V₁ reduplication are combined in diminutive forms in Twana, which Drachman (1969) labels as “attenuative”.¹¹ Examples are given in (724).

- (724) a. **čá<ʔa>** ~čələš (cf. *čálaš* ‘hand’)
 DIM<DIM> ~hand
 ‘little hand’ (Drachman, 1969, p.137, Twana)
- b. **ṣ^wú<ʔu>** ~ṣ^wəbət (cf. *ṣ^wuʔbat* ‘paddle’)
 DIM<DIM>hand
 ‘paddle (dim.)’ (Drachman, 1969, p.139, Twana)
- c. **s-k^w<aʔ>á** ~k^wtəd (cf. *s-k^wák^wtəd* ‘rat’)
 DIM<DIM> ~rat
 ‘rat (dim.)’ (Drachman, 1969, p.138, Twana)

While most of C₁- and -V₁- forms are not given full English translations in Drachman (1969), this combination appears to mark a small entity with a noun, as in (724) above. Drachman (1969) notes that the reduplication pattern associated with diminutives formed this way can affect the action or the object; both interpretations are shown in (725a).

¹¹Drachman (1969, p.269) describes this as combination of C₁- and -V₁h-, but most of the given forms have glottal stop instead of /h/.

- (725) a. be- $\acute{k}^w$ <**aʔ**> $\acute{a}$ < $\acute{k}^w$ > $\acute{s}$ -ə-čəd (cf. *be $\acute{k}^w$ ašáčəd* ‘I’m cutting things.’)
 CONT-cut<DIM><DIM>-TR-1 SG.SUBJ
 ‘I’m cutting a little.’
 ‘I’m cutting things small.’ (Drachman, 1969, p.137, Twana)
- b. be- $\acute{c}i$ <**ʔi**>~ $\acute{c}x^w$ -ə-čəd (cf. *be $\acute{c}x^w$ əčəd* ‘I’m slaughtering.’)
 CONT-DIM<DIM>~kill-TR-1 SG.SUBJ
 ‘I’m slaughtering little ones.’
 ‘I’m slaughtering unimportant ones.’ (Drachman, 1969, p.137, Twana)

Though the forms in (724) and (725) have multiple reduplication, there is no semantic evidence for the two types of reduplication having their own semantic contributions; this is likely a single diminutive morpheme, which may be expressed with either a C_1 or a /ʔ/ infix, or both, and vowel reduplication is triggered by the infix. Therefore, these may be treated straightforwardly as diminutives in function despite having the shape of multiple reduplication.

B.3 Tillamook Multiple Reduplication

There are not many examples of multiple reduplication available in the documentation of Tillamook; attested multiple reduplication is summarised in Table B.15.

		PL C ₁ C ₂	DIM or PL C ₁	COS C ₂
PL	C ₁ C ₂			✓
DIM or PL	C ₁			✓
COS	C ₂			

Table B.15: Multiple Reduplication in Tillamook

Examples come from Edel (1939) and Egesdal and Thompson (1996).

B.3.1 Plural Change of State

When plural C₁C₂ and change-of-state C₂ reduplication co-occur, plural reduplication surfaces as prefix; C₂ reduplication surfaces following the second consonant of the root. An example is given in (726).

- (726) a. də c-ḡəl~ḡel<əl>s-ən-i (cf. Proto-Salish *ḡəl ‘to steam-cook’)
 DET STAT-PL~cook.pit<COS>-DRV-1 SG.SUBJ
 ‘I am baking them [clams] in the oven.’ (Egesdal & Thompson, 1996, p.4, Tillamook)
- b. c-yəš~yiš~əš-ni-wəš (cf. Thompson River Salish zix ‘go lower slowly’)
 STAT-C₁C₂~loose~C₂-NTR-1 SG.OBJ
 ‘My sleeveholder is getting loose on me.’ (Edel, 1939, p.16, Tillamook)

B.3.2 Plural + Unclear C₁ Reduplication

The combination of C₁C₂ and C₁ reduplication is shown in (727). This has a plural meaning, though the identity of the two reduplicative morphemes is unclear.

- (727) a. s-lə~ləx~ləx-ən-i (cf. *ləx* ‘remember’)
 STAT-C₁~C₁C₂-remember-DRV-1 SG.SUBJ
 ‘I remember you (pl).’ (Edel, 1939, p.16, Tillamook)
- b. gəʔ ʔə-ʔəl-əl-il (cf. Proto-Salish *ʔal- ‘be alive, move the body’)
 FUT C₁~C₁C₂-move-DEV
 ‘The tide comes in slowly.’ (Reichard, 1959, p.244, Tillamook)
- c. c-yə-yəh-yəh-is-əw-i (cf. *yehis* ‘see’)
 STAT-C₁~C₁C₂~cause-eye-MID-1 SG.SUBJ
 ‘I begin to see.’ (Reichard, 1959, p.244, Tillamook)
- d. ci yə~ye(h)~yə(h)-g^wat-(n/t)-i
 DEM C₁~C₁C₂~cause-know-(CTR)-1 SG.SUBJ
 ‘I know them.’ (cf. *cyeg^watíg^wəl* ‘They (sg.) know them (sg.).’) (Edel, 1939, p.31, Tillamook)

Appendix C

Index of Constraints and Summary of Rankings

C.1 Index of Constraints

The following lists the constraints used in this dissertation and their definitions. The constraints are listed in alphabetical order. Constraints that start with *, or another symbol, are alphabetized by the first letter in the name. The section where the constraint was first defined is also provided. The notation CS refers to Comox-Sliammon, Li refers to Lillooet, and Hk refers to Halkomelem. The codes and their corresponding section are summarised in the following table with a section number to which they correspond.

Table C.1: Sections for Constraint Definitions

Code	Section Name	Section Location
ArchG	Architecture of Grammar (Chapter 3)	(§3.5)
CS1-C ₁	Comox-Sliammon Level 1 C ₁ (Chapter 4)	(§4.1.3)
CS1-C ₁ C ₂	Comox-Sliammon Level 1 C ₁ C ₂ (Chapter 4)	(§4.1.3.2)
CS3-C ₂	Comox-Sliammon Level 3 C ₂ (Chapter 4)	(§4.1.5.1)

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Table C.1: Sections for Constraint Definitions (Continued)

Code	Section Name	Section Location
CS3-V ₁	Comox-Sliammon Level 3 V ₁ (Chapter 4)	(§4.1.5.2)
Li1-C ₁ C ₂	Lillooet Level 1 C ₁ C ₂ (Chapter 4)	(§4.2.3.2)
Li2-C ₂	Lillooet Level 2 C ₂ (Chapter 4)	(§4.2.4.1)
Li2-C ₁	Lillooet Level 2 C ₁ (Chapter 4)	(§4.2.4.2)
Li2-G	Lillooet Level 2 Glottalisation (Chapter 4)	(§4.2.4.3)
MR	Multiple Reduplication (Chapter 5)	
CrS	Cross-Salish Chapter (Chapter 6)	

Table C.2: Constraint Definitions

Constraint	First Defined
*- Δ FT $\succ$ Δ FT: Assign a violation for every non-head syllable in a foot with a greater weight than the corresponding head syllable.	Li2-C ₁
*? _σ : Assign a violation mark for any [?] in a coda.	CrS
*əC _[?h] _σ (*ə? _σ): Sequences of ə? or əh within a single syllable are not permitted.	Li2-C ₁
AFFIX _{L_n} : Assign a violation mark for every affix with an L _n diacritic that is parsed into the phonological word (where n correspond to a phonological level).	ArchG
ANCHOR-L: If α is at the left edge of a foot and $\alpha\Re\beta$, then β is at the left edge of a foot.	Li2-C ₂

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Table C.2: Constraint Definitions (Continued)

Constraint	First Defined
ANCHOR-L(Word, Segment): Any segment at the left periphery of a word in the output has a correspondent at the left periphery of an input word.	CS3-V ₁
ANCHOR-R: If α is at the right edge of a foot and $\alpha\Re\beta$, then β is at the right edge of a foot.	Li2-C ₂
*CLASH: Rhythm is alternating; no two adjacent stressed syllables.	Li2-C ₂
*COMPLEXONSET: Syllables do not have complex onsets.	CS1-C ₁
CULMINATIVITY- μ^+ (CULMIN): Every word has exactly one head foot with one head mora.	Li2-C ₁
DEP: Output segments must have input correspondents	CS1-C ₁
DEP-C: Output consonants must have input correspondents	CS1-C ₁
DEP-LINK _[C.G.] (DEPL _[C.G.]): Any association line between a segment and a phonological feature in the output must have an input correspondent.	Li2-G
DEP-LINK- μ (N): When a nucleus is linked to a mora in the output, it is also linked to a mora in the input.	CS1-C ₁ C ₂
DEP-LINK _{OBS} [C.G.]: Any association line between an obstruent and a phonological feature in the output must have an input correspondent.	Li2-G
DEP-V: Output vowels must have input correspondents	CS1-C ₁
DEP-V $_{\mu}$: Every moraic vowel in the output must have a corresponding moraic vowel in the input.	CS1-C ₁ C ₂

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Table C.2: Constraint Definitions (Continued)

Constraint	First Defined
*FLOAT: $\forall p \in O$, where p is a prosodic unit or phonological feature: $\exists s$, where s is a segment (i.e., phonological content on the segmental tier), and p dominates or is associated with s .	CS1-C ₁
HEAD-MATCH: If α is the prosodic head of the word and $\alpha \Re \beta$, then β is the prosodic head of the word.	Li2-C ₁
IDENT-IO[C.G.]: The specification (or lack of specification) for the feature [c.g.] of an input segment must be preserved in its output correspondent.	CS3-V ₁
INT·AGREE-[C.G.] (INT·AGR-[C.G.]): If two output segments correspond to the same input segment, then either both or neither are specified for [c.g.].	Li2-G
INT·EDGE-(σ): If two output segments correspond to the same input segment, then they are in the same σ .	Li2-C ₁
INTEGRITY: No element of the input has multiple correspondents in the output.	CS1-C ₁
INTEGRITY-C: No consonant in the input has multiple correspondents in the output.	CS1-C ₁
INTEGRITY-V: No vowel in the input has multiple correspondents in the output.	CS1-C ₁
INTEGRITY _{WORD-INITIAL} (INT-WI): No word-initial element of the input has multiple correspondents in the output.	CS3-C ₂
LINEARITY: S_1 is consistent with the precedence structure of S_2 , and vice versa.	CrS

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Table C.2: Constraint Definitions (Continued)

Constraint	First Defined
MAX: All segments in the input have a correspondent in the output. Assign a violation mark for every segment in the input that does not have a correspondent in the output.	CS1-C ₁
MAX _[c.g.] : Any [c.g.] feature must have an output correspondent.	Li2-G
MAX-FLOAT (MAXFLT): All prosodic units or features that are floating in the input have output correspondents.	CS1-C ₁
MAX-CONTIGUITY (MAX-CONTIG): If $x, y \in S_1$ and x immediately precedes ($\rightarrow$) y , then $\exists x', y'$ such that $x \mathfrak{P} x'$ and $y \mathfrak{P} y'$ and x' immediately precedes ($\rightarrow$) y'	CS1-C ₁
MAX- μ : All moras in the input have a correspondent in the output. Assign a violation mark for every mora in the input that does not have a correspondent in the output.	Li2-C ₁
MIN=2 σ : The prosodic word should be minimally two syllables.	CS1-C ₁
*MORA[-sonorant]: Assign a violation mark for every obstruent associated with a mora.	CrS
*MORA[-continuant]: Assign a violation mark for every [-continuant] segment associated with a mora.	CrS

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Table C.2: Constraint Definitions (Continued)

Constraint	First Defined
MORPHEME-AFFILIATION (MAFFIL): For any three output segments S (S_1, S_2, S_3) that belong to the same morpheme M_a , if S_1 is associated with a prosodic unit belonging to M_a and S_2 is associated with a prosodic unit belonging to some other morpheme M_b , then S_3 must be associated with a prosodic unit belonging to either M_a or M_b . Assign a violation mark for any three segments where all three segments belong to the same morpheme and each segment is associated with a prosodic unit belonging to a different morpheme ($*S_1-M_a, S_2-M_b, S_3-M_c$).	MR
MPARSE: Given a candidate ($o, \langle i \rangle, \langle o \rangle, \mathfrak{R}$), if $\mathfrak{R}$ is not a total bijective function from $\langle i \rangle$ to $\langle o \rangle$, assign a violation mark.	MR
OCP: Adjacent identical elements are prohibited. Assign a violation for each sequence of identical elements.	Li2-C ₁
ONSET: Syllables have onsets.	CS1-C ₁
*ONSET' _{σ[c.g.]} : A segment in the onset of a stressed syllable is not associated with a [c.g.] feature.	Li2-G
*P/ə: Do not associate schwa with a prosodic head.	Li2-C ₂
PARSE _{AFFIX} : Assign a violation mark for any affix left unparsed.	ArchG
PARSE- μ : Moras are parsed by feet.	Li2-C ₁
PARSE-SYLL: Syllables are parsed by feet.	Li1- C ₁ C ₂

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Table C.2: Constraint Definitions (Continued)

Constraint	First Defined
PREFIX: The output correspondent(s) of a prefix are not preceded by a segment.	CS1-C ₁
PREFIX/ μ : The output correspondent(s) of a prefix is not preceded by a mora.	CS1-C ₁
PREFIX/ σ : The output correspondent(s) of a prefix is not preceded by a syllable.	CS1-C ₁
PRIORITY: Given an input containing allomorphs $m_1, m_2, \dots, m_n$, and a candidate m_i' , where m_i' is in correspondence with m_i , PRIORITY assigns as many violation marks as the depth of ordering between m_i and the highest dominating morph.	CS1-C ₁
PRWD=GRWD: A grammatical word must be a prosodic word.	Li2-C ₂
REALIZEMORPHEME (RM): Let α be a morphological form, β be a morphosyntactic category, and $F(\alpha)$ be the phonological form from which $F(\alpha+\beta)$ is derived to express a morphosyntactic category β . Then RM is satisfied with respect to β iff $F(\alpha+\beta) \neq F(\alpha)$ phonologically.	CS1-C ₁
*SON _[C.G] /– Δ FT: Glottalised sonorants are only permitted in head feet.	Li2-G
STROLE: A segment in the input and its correspondent in the output must have identical syllabic roles.	CrS
STRESSSTRESS (STRESS ²): Assign a violation mark for any prosodic unit with a [stress] feature in the input that does not correspond to a head in the output.	Li2-C ₁

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Table C.2: Constraint Definitions (Continued)

Constraint	First Defined
*STRUC-FT: No feet. Assign a violation mark for every foot.	Li1-C ₁ C ₂
TROCHEE: Feet are L-dominant at the level of the mora. Assign one violation per foot where the head is not the leftmost mora.	Li2-C ₁
*UNSTRESSED $\sigma_{\mu\mu\mu}$: Unstressed syllables are maximally bimoraic. Assign a violation mark for any unstressed syllable associated with three (or more) moras.	Li2-C ₂
WT-IDENT-IO: If $\alpha \in \text{Domain}(f)$, if α is non-moraic, then $f(\alpha)$ is non-moraic. if α is monomoraic, then $f(\alpha)$ is monomoraic. if α is bimoraic, then $f(\alpha)$ is bimoraic.	CrS

C.2 Summary of Linear Rankings

I provide flat linear rankings in (728–732) that were used to generate tableaux at each strata. While these linear rankings represent an order of constraints that allows for the correct outputs, not all pairwise rankings suggested by these lists represent crucial rankings (e.g., PREFIX/ σ does not need to be ranked above PREFIX); the tables given in Chapter 4 provide a more accurate summary of rankings.

(728) *Comox-Sliammon - Flat Linear Ranking at Level 1*

*FLOAT, MAX, MAX-FLOAT, DEPL- μ (N), DEP-C, INT-V, DEP-V $_{\mu}$, PREFIX/ σ , *COMPLEX, ONSET $\gg$ DEP-V $\gg$ MIN=2 σ $\gg$ PRIORITY, MAX-CONTIGUITY, PREFIX/ μ , PREFIX $\gg$ INT-C

(729) *Comox- Sliammon - Flat Linear Ranking at Level 2*

*FLOAT, REALIZE MORPHEME, MAX-FLOAT, PREFIX/ μ , PREFIX/ σ , DEP-C, ONSET, *COMPLEX, MAX, DEP-V $_{\mu}$, DEP-LINK- μ (N) $\gg$ INT-C $\gg$ INT-V $\gg$ DEP-V

(730) *Comox-Sliammon - Flat Linear Ranking at Level 3*

*FLOAT, REALIZEMORPHEME, INTEGRITY-WI, PREFIX/ σ , PREFIX/ μ ,
ONSET, DEP-V, ANCHOR-L, MAX-FLOAT, IDENT[C.G.] $\gg$ DEP-C $\gg$
INT-C, INT-V

(731) *Lillooet - Flat Linear Ranking at Level 1*

*FLOAT, REALIZEMORPHEME, MAX-FLOAT, PREFIX/ σ , INT-V, DEP-
C, SONORITY, TROCHEE, PARSE-SYLL $\gg$ INT-C, DEP-V, *STRUC-FT,
DEP-V $_{\mu}$

(732) *Lillooet - Flat Linear Ranking at Level 2*

PRWD=GRWD, *FLOAT, REALIZEMORPHEME, MAX-FLOAT,
TROCHEE, CULMIN, * ∂] $_{\sigma}$, INT-V, DEP-V $_{\mu}$, DEP-C, OCP, PARSE-
 μ , ANCHOR_{FOOT}, INT·EDGE-(σ), DEP-LINK_O[C.G.], IDENT[C.G.] $\gg$
STRESS² $\gg$ *P/ ∂ , *STRUC-FT, *SON_[C.G.]/- Δ FT, *ONSET' $_{\sigma}$ [C.G.] $\gg$
HEAD-MATCH, PARSE- σ , DEP-V, PREFIX/ σ , MAX- μ $\gg$ MAX_[C.G.],
MAX-FLOAT $\gg$ INT·AGR-[c.g.] $\gg$ DEPL_[C.G.], INT-C

Appendix D

Sources for Salish Reduplication

This appendix summarises which types of reduplication are discussed in different sources. Primary and secondary sources are discussed in Chapter 1 (Section 1.5). These sources include description, analysis, or brief discussion of the language or pattern. Sources are listed and given by language; languages are listed in alphabetical order. I only include dictionaries and similar sources with reduplication data where the patterns are described in a stand-alone section (e.g., where a brief grammatical sketch accompanies the resource). Data sources for these papers are discussed in Chapter 1. Papers considering reduplication across Salish are not included in this appendix (e.g., Haeberlin, 1918).

A ✓ in a cell means a type of reduplication was mentioned in some capacity, which includes everything from a brief summary of reduplicative patterns to more complete analyses. Where possible, I also show whether the reduplicative pattern is described as a prefix or an infix (e.g., C_1C_2 - vs. $-C_1C_2$). I convert the notation from the original sources to my four core types of reduplication for ease of comparison (see Section 2.4); not all authors

This section is intended to provide a descriptive overview of the patterns in the literature *as described* in the original source materials. I have reanalysed some of the patterns listed in this section in my survey; I have also set aside a number of patterns where there was minimal evidence or the patterns appeared to be marginal or lexicalised (see Chapter 2, Chapter 6, and Appendix A for how I have interpreted and reanalysed data).

D.1 Bella Coola

Table D.1: Sources for Bella Coola Reduplication

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₁ -	-C ₂	-V ₁ -
Angermeyer (2003)	✓		✓			✓
Bagemihl (1990, 1991)	✓		✓			✓
Brown (2004)	✓		✓			
K. Carlson (1997)	✓		✓			✓
Cho and King (2003)	✓					
Cook (1994)	✓		✓			
Frampton (2009)			✓			
M. Gibson (1995)	✓		✓			
Ingram (1973)	✓		✓			
Kirchner (2010)			✓			
Marlo (2004)	✓					
Nater (1978, 1989)						
Nater (1984)	✓	✓	✓		✓	✓
Nelson (2003)			✓			
Newman (1937)	✓		✓			
Newman (1947)	✓		✓		✓	✓
Newman (1971)	✓	✓	✓	✓		(✓)
Raimy and Idsardi (1997)	✓		✓		✓	✓
Saunders and Davis (1972)	✓			✓		
Shaw (2005)			✓			
Topintzi (2005)	✓		✓			✓

Continued on next page

Table D.1: Sources for Bella Coola Reduplication (Continued)

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₁ -	-C ₂	-V ₁ -
Topintzi (2006)	✓		✓			✓
Vaux and Wolfe (2009)			✓			

D.2 Coeur d'Alene

Table D.2: Sources for Coeur d'Alene Reduplication

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₂	-V ₁ -
Doak (1997)	✓	✓	✓	✓	
Doak (2004)	✓			✓	✓
Palmer et al. (2003)	✓		✓		
Reichard (1938)	✓		✓	✓	✓
Reichard (1959)	✓		✓	✓	✓
Vincent (2014)				✓	

D.3 Comox-Sliammon

Table D.3: Sources for Comox-Sliammon Reduplication

Source	C ₁ C ₂ -	C ₁ -	-C ₁ -	-C ₂	V ₁
Berg (2022)	✓				
Blake (1992)	✓	✓		✓	
Blake (2000)	✓	✓		✓	✓*
Czaykowska-Higgins (2006)					✓

Continued on next page

Table D.3: Sources for Comox-Sliammon Reduplication (Continued)

Source	C ₁ C ₂ -	C ₁ -	-C ₁ -	-C ₂	V ₁
J. Davis (1970a, 2019)		✓			
J. Davis (1970b)	✓	✓			
J. Davis (1971)	✓	✓			
J. Davis (2020)	✓				✓
Harris (1981)	✓	✓			
Hagège (1981)	✓	✓		✓	
Huijsmans (2023b)	✓	✓	✓	✓	
Huijsmans and McCarthy (2024)	✓		✓		
Huijsmans and Mellesmoen (2021)	✓		✓		
Huijsmans et al. (2018)			✓		
Kroeber (1988)	✓	✓		✓	
Mellesmoen (2017)			✓		
Mellesmoen (2018a)			✓		✓
Mellesmoen (2020, 2021a)		✓	✓		
Mellesmoen (in press)	✓	✓			
Mellesmoen and Huijsmans (2019c)	✓				
Mellesmoen et al. (2020)			✓		✓
Reisinger (2024)	✓				
Sapir (1915)	✓	✓			
Urbanczyk (2005)		✓			
Watanabe (1994a, 1994b, 2003)	✓	✓		✓	✓**
Watanabe (1997)	✓	✓			✓**
Watanabe (2010)	✓	✓		✓	

Continued on next page

Table D.3: Sources for Comox-Sliammon Reduplication (Continued)

Source	C_1C_2-	C_1-	$-C_1-$	$-C_2$	V_1
Wilbur (1973)	✓				

*Analysed as $L'-$.

**Analysed as $CV?V-$ and $CV?VC-$.

D.4 Cowlitz

Table D.4: Sources for Cowlitz Reduplication

Source	C_1C_2
Kinkade (2004)	✓

D.5 Halkomelem

Table D.5: Sources for Halkomelem Reduplication

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_1-$	$-C_2$	$-V_1-$
Downriver Halkomelem						
H. Davis (2024)	✓		✓	✓		
Inkelas (2014)	✓		✓			
Shaw (2002)	✓					
Shaw (2004)	✓	✓				
Shaw (2008)	✓		✓			
Suttles (2004)	✓		✓			
Upriver Halkomelem						
Brown (2004)	✓		✓			

Continued on next page

Table D.5: Sources for Halkomelem Reduplication (Continued)

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_1-$	$-C_2$	$-V_1-$
Galloway (1973, 1977, 1984a)	✓	✓	✓	✓	✓	
Galloway (1991)	✓	✓	✓	✓		
Galloway (1993a)						
Kurisu (2001)			✓			
Marinakis (2004)	✓		✓			
Monahan (2002)			✓	✓		
J. Thompson (2005a, 2009)	✓					
Urbanczyk (1998)			✓			
Wiltschko (2005)	✓		✓			
Wiltschko (2008)	✓					
Wiltschko and Steriopolo (2007)			✓			
Zimmermann (2009, 2013, 2017)			✓			
Island Halkomelem						
Gerdts and Peter (2021)			✓			
Hukari (1976, 1977, 1978, 1981a, 1981b, 1984)	✓		✓			
Jones (1976)	✓		✓			
Leslie (1979)	✓		✓			✓
Mellesmoen (2024b)	✓		✓			
Mellesmoen and Urbanczyk (2020)			✓			
Mellesmoen and Urbanczyk (2021a)	✓		✓			
Mellesmoen and Urbanczyk (2021b)	✓		✓			
Urbanczyk (2004a)			✓			

Continued on next page

Table D.5: Sources for Halkomelem Reduplication (Continued)

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_1-$	$-C_2$	$-V_1-$
Urbanczyk (2010)	✓		✓			
Urbanczyk (2011)			✓			

D.6 Lillooet

Table D.6: Sources for Lillooet Reduplication

Source	C_1C_2-	C_1-	$-C_1-$	$-C_2$
Bird (2000)			✓	
Caldecott (2009)			✓	
H. Davis (2011)	✓	✓	✓	✓
H. Davis (2014)	✓			
H. Davis (in prep)	✓	?	✓	✓
H. Davis and Mellesmoen (2023)	✓	✓	✓	✓
Downing and Inkelas (2015)			✓	
Nelson (1998)				✓
Shaw (2005)	✓		✓	
Van Eijk (1981a, 1997); Van Eijk (2011)	✓	✓	✓	✓
Van Eijk (1993)	✓			
Van Eijk and Hess (1986)	✓			

D.7 Lower Chehalis

Table D.7: Sources for Lower Chehalis Reduplication

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₂	-V ₁ -	Other
Robertson (2014)	✓	✓	✓	✓	✓	CVCVC
Snow (1969)	✓					CVCVC

D.8 Lushootseed

Table D.8: Sources for Lushootseed Reduplication

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₁ -	-C ₂	V ₁ -
Alderete et al. (1999)					✓	
Bates (1986)	✓			✓		
Bates (1990)						✓
Benua (1997)	✓		✓			
Broselow (1983)	✓		✓			
S. Davis (1988)						✓
Demers (1974)	✓					
Downing and Inkelas (2015)	✓		✓			
Fidelholtz (1978)		✓		✓		
Fitzpatrick (2006)	✓		✓		✓	
Fitzpatrick and Nevins (2003)	✓		✓		✓	
Frampton (2009)	✓		✓		✓	
Hess (1966, 1967)			✓	✓	✓	✓
Kim (1992)				✓		

Continued on next page

Table D.8: Sources for Lushootseed Reduplication (Continued)

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_1-$	$-C_2$	V_1-
Kirkham (1991, 1992)	✓		✓		✓	✓
Kye (2023)	✓		✓		✓	
McCarthy and Broselow (1983)			✓		✓	
Mellesmoen and Urbanczyk (2021b)	✓	✓				
Mellesmoen and Urbanczyk (2024)			✓			
Park (1999, 2000)	✓		✓			
Ransom (1945)	✓		✓			✓
Reichard (1959)	✓		✓			
Samuels (2010)	✓				✓	
Snyder (1968)	✓		✓			✓
Struijke (2000)			✓			
Tweddell (1950)	✓	✓	✓	✓	✓	
Urbanczyk (1996b, 2005)			✓			
Urbanczyk (1999)	✓		✓			
Urbanczyk (2000, 2001, 2006)	✓		✓		✓	
Van Eijk and Hess (1986)	✓					
Zimmermann (2021a)			✓			
Zimmermann (2021b)	✓		✓			

D.9 Montana Salish-Spokane-Kalispel

Table D.9: Sources for Montana Salish-Spokane-Kalispel Reduplication

Source	C ₁ C ₂ -	C ₁ -	-C ₁ -	-C ₂	-V ₁ -
Kalispel					
Idris (1981)	✓	✓		✓	
Speck (1977)	✓	✓		✓	
Vogt (1940)	✓	✓	✓	✓	✓
Montana Salish					
Pete (2010) ✓	✓		✓	✓	
Spokane					
Bates (1990)				✓	
Bates and Carlson (1989)	✓	✓	✓	✓	
Bates and Carlson (1998)	✓	✓		✓	
Black (1996)	✓	✓		✓	
B. Carlson (1972a)	✓	✓	✓	✓	
B. Carlson (1989)	✓	✓		✓	
B. Carlson (1996)				✓	
B. Carlson (1972b)	✓				
B. Carlson (1980)		✓			
B. Carlson and Bates (1990)				✓	
B. Carlson and Thompson (1982)				✓	
Inkelas (2008)			✓		
Kirchner (2010)		✓			
Kroeber (1988)				✓	

Continued on next page

Table D.9: Sources for Montana Salish-Spokane-Kalispel Reduplication
(Continued)

Source	C ₁ C ₂ -	C ₁ -	-C ₁ -	-C ₂	-V ₁ -
Reichard (1959)	✓	✓		✓	✓
Vaux and Wolfe (2009)		✓			
Yu (2005)			✓		
Multiple Dialects					
Thomason (1997)	✓				✓
Thomason (1999)	✓ x 1-3			✓ x 2-3	

D.10 Moses Columbia

Table D.10: Sources for Moses Columbia Reduplication

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₂
Bessell (1993a)	✓			✓
Czaykowska-Higgins (1993a, 1998)	✓			✓
Czaykowska-Higgins (1985, 1990)	✓			
Czaykowska-Higgins (1993b)	✓	✓	✓	✓
Czaykowska-Higgins (2006)				✓
Czaykowska-Higgins and Willett (1997)		✓	✓	✓
Czaykowski (1983)	✓		✓	✓
Kinkade (1982, 1989)				✓
Raimy (2009)	✓			✓
Urbanczyk (2005)	✓	✓		
Willett (2003)	✓	✓	✓	✓

Continued on next page

Table D.10: Sources for Moses Columbia Reduplication (Continued)

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_2$
Willett and Czaykowska-Higgins (1995)		✓	✓	✓

D.11 Nooksack

Table D.11: Sources for Nooksack Reduplication

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_1-$	$-C_2$	V_1-
Galloway (1984b, 1993b)	✓	✓	✓	✓	✓	✓

D.12 Okanagan

Table D.12: Sources for Okanagan Reduplication

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_1-$	$-C_2$	$-V_1-$
A. Mattina (1973)	✓	✓	✓		✓	
A. Mattina (1979)			✓			
A. Mattina (1993)					✓	
A. Mattina and Peterson (1997)			✓	✓		
Pattison (1978)	✓		✓			
Watkins (1970)	✓		✓		✓	✓
Watkins (1974)	✓		✓		✓	

D.13 Pentlatch

Table D.13: Sources for Pentlatch Reduplication

Source	C_1C_2-	C_1-
Pulhug (2023)	✓	✓

D.14 Quinault

Table D.14: Sources for Quinault Reduplication

Source	C_1C_2-	C_1-	$-C_2$	$-V_1-$
Modrow (1971)			✓	✓
Modrow (1969)			✓	✓

D.15 Sechelt

Table D.15: Sources for Sechelt Reduplication

Source	C_1C_2-	C_1-
Beaumont (1985)	✓	✓
Hill-Tout (1904)		✓

D.16 Shuswap

Table D.16: Sources for Shuswap Reduplication

Source	C ₁ C ₂ -	C ₁ -	-C ₁ -	-C ₂
Broselow (1983)	✓		✓	
Bell (1983)			✓	
Bjorkman (2006)	✓	✓		
J. Gibson (1973)	✓	✓		
Hendricks (1998, 1999)			✓	
Ignace and Ignace (2017)	✓	✓	✓	✓
Kim (2017b)		✓		
Kim and Gardiner (2016)	✓		✓	
Kuipers (1974a, 1989)	✓	✓	✓	✓
Kuipers (1974b)	✓			
McCarthy and Broselow (1983)		✓	×	
Nakamura (2000)	✓		✓	
Taylor (1996)	✓	✓	✓	✓
Van Eijk (1990a)			✓	

D.17 Squamish

Table D.17: Sources for Squamish Reduplication

Source	C ₁ C ₂ -	-C ₁ -
Bar-el (2000, 2006, 2008)	✓	✓
Bar-el and Watt (2001)	✓	✓

Continued on next page

Table D.17: Sources for Squamish Reduplication (Continued)

Source	C ₁ C ₂ -	-C ₁ -
S. Davis (1984)	✓	✓
Demers and Horn (1978)	✓	✓
Downing (2008)	✓	
Dyck (2004)	✓	✓
Jacobs (2012)	✓	✓
Kuipers (1967)	✓	✓

D.18 Straits (Northern Straits and Klallam)

Table D.18: Sources for Northern Straits and Klallam Reduplication

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₁ -	V ₁ -
Saanich					
S. Davis and Ueda (2006)			✓		
Kirchner (2010)			✓		
Kiyota (2008)	✓		✓		
Kurisu (2001)			✓		
Leonard (2009, 2019)	✓		✓		
Leonard and Turner (2010)			✓		
Montler (1986)	✓		✓		
Montler (1988, 1989)			✓		
Stonham (1994, 2007)			✓		
Svenonius and Bye (2012)			✓		

Continued on next page

Table D.18: Sources for Northern Straits and Klallam Reduplication
(Continued)

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₁ -	V ₁ -
Turner (2005)			✓	✓	
Turner (2006, 2007, 2011)			✓		
Zimmermann and Trommer (2013)			✓		
Samish					
Galloway (1986, 1990)	✓	✓	✓	✓	
Songish					
M. Mitchell (1975)	✓		✓		
Raffo (1970)	✓		✓		✓
Raffo (1972)	✓	✓	✓		✓
Klallam					
Crawford (2015)	✓	✓	✓	✓	
Montler and Bates (1998)			✓		
Montler (2015)	✓		✓	✓	
L. C. Thompson and Thompson (1971)	✓	✓	✓	✓	
L. C. Thompson and Thompson (1968)			✓		
Sooke					
Efrat (1969)	✓		✓		
Lummi					
Demers and Jelinek (1984)	✓		✓		
Demers and Jelinek (1996)	✓		✓		
Jelinek and Demers (1997)	✓		✓		
Haag (1998)	✓				

Continued on next page

Table D.18: Sources for Northern Straits and Klallam Reduplication
(Continued)

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₁ -	V ₁ -
Demers (1974)	✓				
Multiple Dialects					
Mellesmoen (2023)	✓				

D.19 Thompson

Table D.19: Sources for Thompson Reduplication

Source	C ₁ C ₂ -	-C ₁ C ₂	C ₁ -	-C ₁ -	-C ₂
Broselow (1983)	✓			✓	
Brown (2017)	✓			✓	
B. Carlson and Thompson (1982)				✓	
Coelho (1999)				✓	
Howett (1993)					✓
Jimmie (1994)	✓	✓	✓	✓	✓
Kroeber (1988)					✓
Luntzlara (in prep)					
McCarthy and Broselow (1983)			✓	×	
Robertson (2004)				✓*	
Shaw (2005)	✓		✓		
L. C. Thompson (1985)					✓
L. C. Thompson and Thompson (1981)	✓		✓	✓	✓
L. C. Thompson and Thompson (1992)	✓	✓	✓	✓	✓

Continued on next page

Table D.19: Sources for Thompson Reduplication (Continued)

Source	C_1C_2-	$-C_1C_2$	C_1-	$-C_1-$	$-C_2$
L. C. Thompson et al. (1996)	✓		✓	✓	✓
Zimmermann (2021b)	✓		✓		

*Discusses a process of glottalisation related to reduplication.

D.20 Tillamook

Table D.20: Sources for Tillamook Reduplication

Source	C_1C_2-	C_1-	$-C_2$	$-V_1-$
Edel (1939)	✓	✓	✓	✓
Egesdal and Thompson (1996, 1998)	✓*	✓	✓	
Kim and Gardiner (2016)	✓*			
Reichard (1959)	✓	✓	✓	✓
L. C. Thompson and Thompson (1985)	✓	✓		

*Includes a “wrong side” realisation.

D.21 Twana

Table D.21: Sources for Twana Reduplication

Source	C_1C_2-	C_1-	$-C_1-$	$-C_2$	V_1-
Drachman (1969)	✓	✓		✓	✓
Kim (2017a, 2017b)	✓				
N. Thompson (1979)	✓				
N. Thompson (1985)	✓	✓	✓	✓	✓

D.22 Upper Chehalis

Table D.22: Sources for Upper Chehalis Reduplication

Source	C_1C_2-	C_1-	$-C_2$
Haag (1998)			✓
Kinkade (1963)	✓		✓
Kinkade (1985)			✓
Kinkade (1975, 1995)	✓		
Mellesmoen (2022)	✓	×	✓
Reichard (1959)	✓	×	✓

Appendix E

NAPA to IPA Conversion Chart

A conversion chart between the North American Phonetic Alphabet (NAPA) and the International Phonetic Alphabet (IPA) are given in the following table for obstruent and sonorant consonants. Note that /ʔ/ and the voiced obstruents behave phonologically as sonorants in Salish; this is because they have largely developed from Proto-Salish sonorants (see, e.g., Mellesmoen, 2018b). I provide some additional representations (orthographies and transcriptions) in the rightmost column that are found across source materials cited in Appendix D. While /ʔ/ behaves as a sonorant across the family, /h/ does not (Czaykowska-Higgins & Kinkade, 1998, p.8).¹

Table E.1: NAPA Conversion Chart: Obstruents

NAPA	IPA	Other Representations
p	p	
p̰	pʰ	B
t	t	
t̰	tʰ	D
k	k	kʷ

Continued on next page

¹A sonorant analysis of /h/ has been proposed for some languages.

Table E.1: NAPA Conversion Chart: Obstruents
(Continued)

NAPA	IPA	Other Representations
ḳ	kʼ	
kʷ	kʷ	kw, k [◦] , ǂ, ku
ḳʷ	kʷʼ	kwʼ, kʷʷ, ḳ [◦] , ku, Q
q	q	<u>K</u> , <u>k</u>
q̣	qʼ	<u>k</u> , K
qʷ	qʷ	qw, <u>kw</u> , qu, <u>kw</u> , Ḳ, q [◦] , q [◦]
q̣ʷ	qʷʼ	qʷʷ, <u>kw</u> ʼ, Ḳ, q̣ [◦] , q̣u
t ^θ	t̠	tth
ṭ ^θ	t̠ʼ	θʼ, tthʼ, ǂ
ɬ	t̠	tl
ɬ̣	t̠ʼ	tʼ, ṭ, tlʼ
c	t̠s (or t̠ʃ)	ts
c̣	t̠sʼ	cʼ, t̠s, tsʼ
č	t̠ʃ	Č, tc, ch
č̣	t̠ʃʼ	t̠c, chʼ, J
s	s (or ʃ)	
θ	θ	th, T, ɕ
ɬ	ɬ	ɬ, lh, λ, ɬ, L
š	ʃ	sh, c, Š
x	x	c
xʷ	xʷ	cw, xw, <u>W</u> , x [◦]
ɣ̣	ɣ	ǂ, <u>x</u> , x, X

Continued on next page

Table E.1: NAPA Conversion Chart: Obstruents
(Continued)

NAPA	IPA	Other Representations
$\dot{x}^w$	χ^w	$\check{x}^w$, xw , $\underline{x}w$, $\underline{X}$, $\check{x}^\circ$
$\dot{h}$	h	
$\dot{h}^w$	h^w	
h	h	

Table E.2: NAPA Conversion Chart:
Sonorant Consonants

NAPA	IPA	Other Representations
m	m	
$\dot{m}$	$^?m$	m' , $m?$, $?m$, $m7$, $7m$
b	b	
$\dot{b}$	$^?b$	$b?$, $?b$
n	n	
$\dot{n}$	n' , $^?n$	$n?$, $?n$, $n7$, $7n$
d	d	
$\dot{d}$	$^?d$	$?d$, $d?$
z	z^*	
$\dot{z}$	$^?z^*$	z'
d^z	$\widehat{dz}$	j , dj
l	l	
$\dot{l}$	$^?l$	l' , $l?$, $?l$, $l7$, $7l$
r	r	R

Continued on next page

Table E.2: NAPA Sonorant Conversion Chart:
 Sonorant Consonants
 (Continued)

NAPA	IPA	Other Representations
ṛ	ʔṛ	Ṛ
ṣ̌	ḍ̌	j
ṣ̌̇	ʔḍ̌̇	ṣ̌̇
g	g	gʸ
g ^w	g ^w	g ^w
ḡ ^w	ʔg ^w	
ḡ	ʔg	gʻ
ʁ	ʁ	r
ṛ̌	ʔʁ̌	ṛ̌, rʻ
ŋ	ŋ	<u>N</u>
ṅ̌	ʔŋ̌	<u>N</u> , Ň
y	j	i
ṽ	ʔj	yʻ, yʔ, ʔy, yʔ̌, ʔy̌, iʔ̌, iʔ̌̇
ḡ̌	Ǧ	ḡ̌̇
ḡ̌ ^w	Ǧ ^w	ḡ̌ ^w
w	w	u
ṽ̌	ʔw̌	ʔw, wʔ, wʻ, ʔw̌, w̌̇, uʔ̌̇
ʔ̌	ʔ̌	ʔ̌̇, ʻ, ʔ̌̇
ɣ	ɣ	g
ḡ̌̇	ʔɣ̌̇	gʻ
ɣ̌	ɣ̌ ^w	g ^w , ɣ̌̇

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Table E.2: NAPA Conversion Chart:
Sonorant Consonants
(Continued)

NAPA	IPA	Other Representations
ɰʷ	?ɰʷ	gw', ɰ°

*This is not quite the same as a typical [z]; it is a sonorant in Salish.

Appendix F

Fission and Alternate Analyses of V₁ Reduplication

Decisions about vowel reduplication in this dissertation, and across existing descriptions, are very reliant on particular assumptions about phonological processes. A case in point concerns “link vowels”, which are found in combination with transitivity morphology in many Salish languages. Link vowels occur before certain transitivity suffixes and typically match the quality of the root vowel; it is a question of analysis as to whether these link vowels, which are bolded in the Comox-Sliammon examples in (733) below, should be treated as reduplicated.

- (733) a. *ɣiq-**it**-as* (root: *ɣiq-*)
scratch-CTR-3ERG
‘They (sg.) scratched it.’ (Watanabe, 2003, p.439, Comox-Sliammon)
- b. *q^wum-**ut**-as* (root: *q^wum-*)
put.in.mouth-CTR-3ERG
‘They (sg.) put it in their mouth.’ (Watanabe, 2003, p.439, Comox-Sliammon)

Examples such as these allow for different analyses, some of which would be considered reduplication under my definition. In order to count, the link vowel and the root vowel must correspond to the same input vowel and fission must be

motivated by the addition of a morpheme. While the link vowel is associated with the addition of the control transitivity marker *-t* in the examples in (733), it is not clear that it arises via fission, feature agreement, or spreading between two input segments (one of which may be underspecified); alternatively, the lexical entry could contain four allomorphs (*-t*, *-ut*, *-it*, and *-at*) that are selected based on properties of the root.¹

Where a 1:2 input-output mapping is hard to evaluate, as is the case with link vowels, I include only patterns in the survey where it is possible to propose an empty prosodic unit in the input. This means that in order for the link vowels to be considered as reduplication, the transitivity marker *-t* would have to be accompanied by a floating prosodic unit, such as a mora or syllable. The simplest analysis of the transitivity marker does not involve an affixed prosodic unit: the presence of a link vowel depends on root shape and vowel quality (see Watanabe, 2003, pp.214–216). Furthermore, a reduplicative analysis would need to account for the deletion of an affixed prosodic unit in words where the transitivity marker appears without a link vowel, such as those with a CC root in (734).

- (734) a. təq-t (root: tq- ‘close’)
 close-CTR
 ‘Close it!’ (Watanabe, 2003, p.213, Comox-Sliammon)

¹The middle marker *-Vm* in Comox-Sliammon can also surface with any of the four vowels [i], [a], [u], or [ə], but the vowel does not always match that of the root, unlike with the transitivity marker *-t*. Examples with the middle marker are shown in (i).

- (i) a. tʰikʷ-**am**
 left-MID
 ‘(be on) the left side’
 b. tikʷ-**um**
 disembark-MID
 ‘disembark (from boat, car)’
 c. kʷəj-**im**
 shine-MID
 ‘shining’
 d. jat-**əm**
 move-MID
 ‘move’

(Watanabe, 2003, pp.191–192, Comox-Sliammon)

- b. $yəq-t$ (root: $yq-$ ‘buy’)
 buy-CTR
 ‘Buy it!’ (Watanabe, 2003, p.214, Comox-Sliammon)

There is no independent motivation for the deletion of a prosodic unit. Further, the lack of a link vowel in (734) below cannot be reduced to the fact that there is no underlying vowel in the root: the example given in (735) shows that [a] is the link vowel with CCC roots. If the transitivity suffix had a mora or syllable, then [a] would also be expected in (734). Further, the vowel qualities in (735) do not match, which means that there is no 1:2 mapping between input and output.

- (735) $pəlk^w-at$ (root: plk^w- ‘roll’)
 roll-CTR
 ‘Roll it!’ (Watanabe, 2003, p.30, Comox-Sliammon)

There are four potential analyses for vowel reduplication: (i) as a segmental affix (no fission), (ii) as a segmental affix (fission), (iii) as a mixed affix (fission), and (iv) as a prosodic affix (fission). The latter three all involve fission and are therefore classified as reduplication in this dissertation. Tables F.1 and F.2 show these possible analyses for Montana Salish-Spokane-Kalispel inchoatives in (736) repeated from (10) in Chapter 2 (Section 2.2.2.2), and Comox-Sliammon link vowels in (733), respectively. Subscripts are used to show whether vowels correspond to the same input segment (V_1) or are epenthetic (V_x), as well as if the segments are associated with affixed prosodic material from the input (e.g., V_μ). Angled brackets mark epenthetic material.

- (736) a. $q^w<uʔ>úʔ$ (cf. $hi q^w uʔ$ ‘It is dusty.’)²
 dusty<INCH>
 ‘It gets dusty.’ (Vogt, 1940, p.64, Kalispel)
 b. $kən ʔ<aʔ>áʔ$ (cf. $cʔʔat$ ‘be/get wet’)³

²The Spokane form $q^w ʔuʔ$ is given in B. Carlson and Flett (1989, p.176). It is possible that there is idiolectal or dialectal variation with respect to the presence of a doubled vowel with the inchoative. See Mellesmoen (2024a) for further discussion of V_1 reduplication across different Salish languages with multiple sources of data.

³The underlying /ʔ/ in the root $ʔʔat$ ‘wet’ (10b) is absent in forms with $-ʔ-$, based on the examples provided in A. Mattina (1987, p.88).

1SG.SUBJ wet<INCH>

‘I got wet.’ (Watkins, 1970, p.107, Okanagan)

Potential analyses of the inchoative include a segmental affix consisting of /ʔ/ that either (i) triggers epenthesis (with subsequent harmony or feature sharing), or (ii) triggers fission (“copy-repair”). Both options involve a repair; the difference in analysis comes down to the source of the vowel. The remaining options include an allomorph of the inchoative that is (iii) a mixed affix consisting of segment (/ʔ/) and a prosodic unit (μ), or (iv) a prosodic affix (σ). Fission occurs to fill a prosodic unit in options (iii) and (iv), with the additional epenthesis of [ʔ] in (iv) to prevent vowel hiatus. These options are summarised in Table F.1.⁴ Of the possible analyses, only (i) is not considered reduplication.

Table F.1: Defining Reduplication - Inchoatives

	Not Reduplication <i>Epenthesis</i>	Reduplication <i>Fission</i>		
	i. Segmental Affix	ii. Segmental Affix	iii. Mixed Affix	iv. Prosodic Affix
Input	CV ₁ C + ʔ	CV ₁ C + ʔ	CV ₁ C + μ ʔ	CV ₁ C + σ
	CVʔC CVʔ<ə>C CV ₁ ʔV _x C	CVʔC CV ₁ ʔV ₁ C -	CVʔC μ CV ₁ ʔV ₁ μ C -	CV ₁ V ₁ σ C CV ₁ <ʔ>V ₁ C -
Output	CVʔVC			

Certain analyses in Table F.1 are less well-motivated than others; for example, the prosodic affix option given in (iv) renders any resemblance between the inchoative in other environments (-ʔ-) and vowel reduplication (-Vʔ-) purely accidental, despite their similar form; two distinct allomorphs would be needed. However, all four analyses are plausible. As there is no reason to favour an epenthesis analysis over a fission one, I consider the inchoative as a reduplicative morpheme.

Unlike the inchoative example, I do not treat the link vowels in (733) above as a type of vowel reduplication. The epenthesis analysis in (i) in Table F.2 is the

⁴Tables F.1 and F.2 in this section are intended to illustrate possible analyses by breaking them into “steps”. However, this is just an illustration; lines given between input and output here do not correspond to strata.

most viable explanation of the four.⁵

Table F.2: Defining Reduplication - Link Vowels

	Not Reduplication <i>Epenthesis</i>	Reduplication <i>Fission</i>		
	i. Segmental Affix	ii. ?Segmental Affix	iii. ?Mixed Affix	iv. ?Prosodic Affix
Input	CV ₁ C + t	CV ₁ C + t	CV ₁ C + μ t	CV ₁ C + σ
	CVCt CVC< σ >C CV ₁ CV _x t	CVCt CV ₁ CV ₁ t -	CVCt μ CV ₁ CV ₁ μ t -	CV ₁ C(V ₁) σ CV ₁ CV< t > -
Output	CVCVt			

None of the fission analyses of link vowels are promising. The segmental affix analysis in (ii), with fission of a vowel would reflect a grammar that can only copy moraic vowels. However, [ə] can be copied in change-of-state reduplication (see discussion of C₂ reduplication in Section 2.6). Examples given in (737) show that epenthetic [a] is used if change-of-state reduplication is present, regardless of the form of the root (i.e., [i] would be the link vowel for *píq̣-* ‘wide’ otherwise). However, (737b) shows that epenthetic [ə] is present and copied in change-of-state reduplication; there is therefore no explanation for why [ə] not undergo fission to supply a link vowel.

- (737) a. *p̣<íq̣>íq̣-at* (root: *píq̣-*)
wide<COS>-CTR
‘Make it wide.’ (Watanabe, 2003, p.215, Comox-Sliammon)
- b. *ḵ<əṭ>əṭ-at* (root: *ḵṭ-*)
stiff<COS>-CTR
‘Iron it.’ (Watanabe, 2003, p.215, Comox-Sliammon)

The mixed affix analysis in (iii) proposes that the underlying form of the transitivity contains μ and /t/; the underlying mora in the transitivity would be lost in a word like *təqt* ‘Close it!’ in (734a) above.⁶ However, there is no

⁵My preferred analysis is actually that an underspecified vowel in the transitivity takes on the place features of a nearby vowel, though this makes no difference to the present discussion because neither epenthesis nor feature sharing with an underlying unspecified vowel count as reduplication.

⁶Parsing the /q/ as a moraic coda to satisfy the floating mora is ruled out because it is vacuous: Comox-Sliammon does not allow complex onsets.

motivation for an input prosodic unit to be lost.

Finally, the prosodic affix analysis in (iv) is also problematic because it requires an epenthetic [t] which is not attested elsewhere in the language.

None of alternative analyses of link vowels that fit my definition of reduplication are viable. I therefore only consider patterns in Appendix A and Chapter 2 where there is a viable fission analysis that involves the affixation of a morpheme. I therefore set aside link vowels at present as there is no convincing evidence for any of the fission-based analyses (ii–iv).